MASTER MATHEMATICS
VOLUME-BASED RESEARCH-TRAINING CURRICULUM

10 years - 30 hours/week - 40 quarters - 36-volume subject architecture

Zero prerequisites to research-level entry across analysis, number theory, logic, group theory, representation theory, geometry, topology, probability, dynamics, arithmetic geometry, automorphic forms, additive combinatorics, spectral gaps, fractal measures, cryptography and mathematical physics

Expanded version v6: audited and schedule-specific edition. This revision follows a line-by-line structural audit of all 40 quarters and 36 volumes. It replaces every generic/missing quarter schedule with an explicit 12-week reading/result/problem sequence; removes stale cross-references and copied references; distinguishes proof, architecture, and use-level mastery without conflicting letter codes; makes every volume checkpoint theorem-specific; and updates research-frontier entries through August 2026 where they materially changed.

How to use this curriculum

Core scale: Each quarter contains 12 instructional weeks. At 30 hours/week that is 360 focused hours per quarter, 1,440 hours per 48-week year, and 14,400 scheduled hours over ten years. Four buffer weeks per year may be used for exams, travel, catch-up or research, giving up to 15,600 hours if fully used.

Weekly default: 10h deep reading/lectures; 10h problem solving; 4h proof reconstruction; 3h exposition/LaTeX; 2h spaced review; 1h research orientation. Quarter-specific allocations redistribute these hours without changing the 30h total.

Theorem standard: For every major result know: exact statement, examples/counterexamples, intuition, proof architecture, at least one key lemma in full detail, and what later topics use it. For foundational theorems, reproduce the full proof.

Problem standard: Do not optimize for the raw number of exercises. For each active two-week unit, complete the three precision core problems Uj.P1-Uj.P3 and 3-8 carefully chosen source exercises that reinforce the same technique. At least two core problems per unit should receive a full written solution or theorem audit. Preserve failed attempts and corrections.

Exercise tags: [F] foundation/concrete mastery; [T] reusable technique or lemma; [S] synthesis across ideas; [C] counterexample/hypothesis audit; [R] research bridge or precise theorem audit; [X] computational exploration whose mathematical limits must be stated.

Reading standard: Chapter numbering refers to the named editions where possible. If an edition differs, follow the topic title rather than blindly following a number. Primary sources are introduced only after a modern textbook or lecture-note exposition has made the dependency graph clear.

Research standard: From Year 5 onward, every quarter includes paper-reading. Use three passes: first identify statements and dependencies; second reconstruct proofs of selected lemmas; third write your own exposition and formulate questions.

A practical weekly rhythm

Block	Suggested use
Monday	2h new reading + 2h problems + 1h proof reconstruction + 1h review
Tuesday	2h new reading + 3h problems + 1h exposition
Wednesday	2h lecture notes + 2h problems + 1h proof reconstruction + 1h computation/secondary reading
Thursday	2h new reading + 3h problems + 1h exposition
Friday	2h consolidation + 2h problems + 2h theorem reconstruction
Weekend block	3h long problem session + 2h writing/research orientation + 1h spaced review. Rebalance around work as necessary; preserve the weekly totals.

Verified lecture-note and open-course directory

Use these as companions to the books; the quarter pages specify when each enters.

Foundational and graduate lecture-note companions

MIT OCW 18.100B Real Analysis (Spring 2025)

Proof writing, sequences, metric spaces, compactness, differentiation, Riemann integration, uniform convergence, Picard-Lindelof.
https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/pages/lecture-notes/

MIT OCW 18.785 Number Theory I (Fall 2021)

Valuations, Dedekind domains, local fields, geometry of numbers, units, zeta/PNT, Dirichlet L-functions, class field theory, adeles and Chebotarev.
https://ocw.mit.edu/courses/18-785-number-theory-i-fall-2021/pages/lecture-notes/

MIT OCW 18.786 Number Theory II: Class Field Theory (Spring 2016)

Hilbert symbols, Tate cohomology, local and global class field theory.
https://ocw.mit.edu/courses/18-786-number-theory-ii-class-field-theory-spring-2016/pages/lecture-notes/

J.S. Milne Course Notes

Graduate notes in algebraic number theory, class field theory, algebraic geometry, modular forms, elliptic curves, abelian varieties, etale cohomology, algebraic groups and more.
https://www.jmilne.org/math/CourseNotes/index.html

Terence Tao, UCLA Math 254A Arithmetic Combinatorics

Cauchy-Davenport, Plunnecke, Freiman, BSG, Roth/Gowers, sum-product and finite-field incidence.
https://www.math.ucla.edu/~tao/254a.1.03w/

Terence Tao, UCLA Math 254A Topics in Ergodic Theory

Ergodic theorems, recurrence, factors, spectral theory, Szemeredi connections, nilmanifolds and Ratner.
https://www.math.ucla.edu/~tao/254a.1.08w/

Ravi Vakil, Foundations of Algebraic Geometry

Scheme-first algebraic-geometry notes with extensive exercises.
https://math.stanford.edu/~vakil/0910-216/FOAGevolution.html

MIT OCW 18.755 Lie Groups and Lie Algebras

Manifolds, Lie groups/algebras, homogeneous spaces, semisimple structure, invariant measures and representation-theoretic foundations.
https://ocw.mit.edu/courses/18-755-lie-groups-and-lie-algebras-ii-spring-2024/lists/lecture-notes/

MIT OCW RES.18-015 Topics in Fourier Analysis

Fourier series/transforms, distributions, singular integrals, Hilbert transform and interpolation.
https://ocw.mit.edu/courses/res-18-015-topics-in-fourier-analysis-spring-2024/

MIT OCW 18.152 / 18.156 PDE

Harmonic/heat equations and graduate PDE/Fourier-analysis continuation.
https://ocw.mit.edu/search/?q=18.152%2018.156

MIT OCW 18.175 Theory of Probability

Measure-theoretic probability and advanced probability notes.
https://ocw.mit.edu/search/?q=18.175%20Theory%20of%20Probability

Pavel Etingof et al., Introduction to Representation Theory

Open MIT text. Use Chapters 2-5 for basic representation theory, finite groups, characters, induction and examples; later sections provide Schur-Weyl and GL(2,F_q) examples.

https://math.mit.edu/~etingof/reprbook.pdf

MIT OCW 18.757 Representations of Lie Groups (Fall 2023)

Continuous and unitary representations, matrix coefficients, (g,K)-modules, SL(2,R), principal series and Harish-Chandra theory. Use selectively after the Lie-group quarter.

https://ocw.mit.edu/courses/18-757-representations-of-lie-groups-fall-2023/lists/lecture-notes/

Clara Loh, Geometric Group Theory lecture notes

A clean first route through generators and relations, Cayley graphs, word metrics, quasi-isometries, growth, group actions and large-scale geometry.

https://loeh.app.uni-regensburg.de/teaching/ggt_ss22/lecture_notes.pdf

Bekka-de la Harpe-Valette, Kazhdan's Property (T)

Definitive reference for almost-invariant vectors, Kazhdan pairs/constants, property (T), spectral criteria, property (tau), and applications to expander constructions.

https://perso.univ-rennes1.fr/bachir.bekka/KazhdanTotal.pdf

Hoory-Linial-Wigderson / Lubotzky expander surveys

Use for combinatorial expansion, Cheeger inequalities, random walks, eigenvalue gaps, explicit constructions and links to group theory and computer science.

https://www.math.ias.edu/~avi/PUBLICATIONS/MYPAPERS/HLW06/hlw06.pdf

Ben Green, Approximate groups and their applications

Survey bridge from additive combinatorics and sum-product to Helfgott growth, approximate groups, Bourgain-Gamburd expansion and affine sieve ideas.

https://arxiv.org/abs/0911.3354

Kenny Easwaran, A Cheerful Introduction to Forcing and the Continuum Hypothesis

Graduate-level companion for forcing and the independence of CH; use only after the basic set-theory and model-theory material.

https://arxiv.org/abs/0712.2279

Bjorn Poonen, Hilbert's Tenth Problem over rings of number-theoretic interest

Arizona Winter School notes linking computability, Diophantine definability, DPRM, number fields, Q and arithmetic geometry.

https://math.mit.edu/~poonen/papers/aws2003.pdf

Peter Varju, Recent progress on Bernoulli convolutions

Survey of transversality, entropy, algebraic parameters, Hochman-type dimension results and arithmetic aspects of Bernoulli convolutions.

https://arxiv.org/abs/1608.04210

Ten-year map

Year 1

- Q1. Proof, arithmetic language, and precalculus
- Q2. Single-variable calculus, linear algebra I, and elementary number theory II
- Q3. Multivariable calculus, linear algebra II, and discrete/combinatorial methods
- Q4. Mathematical logic I: first-order logic, computability, and Godel incompleteness

Year 2

- Q5. Transition to rigorous analysis and computational arithmetic

- Q6. Real analysis II, metric spaces, and geometry of numbers I
- Q7. Abstract algebra I, complex analysis I, and finite fields I
- Q8. Topology, algebra II, and algebraic number theory entry

Year 3

- Q9. ODE, smooth geometry, complex analysis II, and elementary analytic number theory
- Q10. Measure theory and analytic number theory I
- Q11. Probability I and additive combinatorics I
- Q12. Representation theory I: finite and compact groups

Year 4

- Q13. Functional analysis, Fourier analysis, and exponential/character sums
- Q14. PDE I, Riemannian manifolds, and algebraic number theory I
- Q15. Harmonic analysis I and analytic number theory II
- Q16. Mathematical logic II: axiomatic set theory, constructibility, forcing, and CH

Year 5

- Q17. Lie/algebraic groups, arithmetic lattices, and geometry of numbers II
- Q18. Representation theory II: unitary and Lie-group representations
- Q19. Local fields, heights, and cryptography I
- Q20. Algebraic topology, commutative algebra, and algebraic geometry I

Year 6

- Q21. Geometric group theory, property (T), expander graphs, and spectral gaps
- Q22. Restriction/Kakeya toolkit and sieve methods
- Q23. Probability II: SDE, SLE, percolation, random graphs and random matrices
- Q24. Algebraic geometry II, elliptic curves, and finite-field arithmetic II

Year 7

- Q25. Ergodic/homogeneous foundations, modular forms I, and cryptography II
- Q26. Carleson's theorem, time-frequency analysis, and the circle method
- Q27. Bourgain-Demeter decoupling and analytic number theory III
- Q28. Sum-product, approximate groups, growth, and the Bourgain-Gamburd machine

Year 8

- Q29. Baker, Schmidt Subspace Theorem, and Diophantine synthesis
- Q30. Hilbert's Tenth Problem, Diophantine definability, and undecidability
- Q31. Borel density, Borel-Harish-Chandra, KAM/billiards, and class field theory I
- Q32. Falconer/Furstenberg/Kakeya and additive combinatorics II

Year 9

- Q33. Entropy and Lq inverse theorems, self-similar measures, and Bernoulli convolutions
- Q34. Ratner, effective homogeneous dynamics, EKL/Benoist-Quint, and Diophantine applications
- Q35. Zimmer conjecture and automorphic forms II: adeles, Tate thesis, GL(2)
- Q36. Sato-Tate, Fermat/FLT, modularity, and automorphic L-functions

Year 10

- Q37. Faltings, Diophantine geometry, and global arithmetic synthesis
- Q38. Etale cohomology, Weil conjectures, Deligne/Bombieri, and finite fields III
- Q39. Poincare, Ricci flow, Hauptvermutung, and cryptography III
- Q40. Particle systems, kinetic/field limits, Ising, and final research synthesis

Milestone logic

- End of Year 2: rigorous proof, first-order logic/computability, real analysis foundations, algebra, topology, number fields, complex methods and computational arithmetic are operational.

- End of Year 4: measure/probability, additive combinatorics, finite/compact representation theory, Fourier analysis, PDE, harmonic analysis, and the logic of constructibility/forcing/CH are operational.
- End of Year 6: Lie and unitary representation theory, local fields, scheme foundations, geometric group theory, property (T), expanders, spectral gaps, restriction/Kakeya and advanced probability are operational.
- End of Year 8: Carleson, decoupling, sum-product, approximate groups, the Bourgain-Gamburd method, Baker/Subspace, Hilbert's Tenth Problem, class field theory and Falconer/Furstenberg methods are at research-entry level.
- End of Year 10: entropy/Lq inverse theorems and Bernoulli convolutions, homogeneous rigidity, automorphic forms, FLT/Sato-Tate, Faltings, Deligne/Weil, Ricci flow, lattice cryptography and particle/kinetic limits have been integrated into selectable research programs.

How the inserted tracks fit the existing programme

- Logic track: Q4 gives syntax/semantics, completeness, computability and the incompleteness theorems. Q16 develops ZFC, ordinals/cardinals, the constructible universe and forcing to the independence of CH. Q30 returns to logic from the arithmetic side through Diophantine definability and the DPRM theorem.
- Group/spectral track: Q12 builds finite and compact representation theory; Q18 develops unitary representations and SL(2,R); Q21 turns these tools into geometric group theory, amenability, property (T)/(tau), expanders and spectral gaps; Q28 develops noncommutative growth and the Bourgain-Gamburd expansion machine.
- Fractal-measure track: Q11/Q13 provide probability and Fourier foundations, Q32 provides advanced additive/Falconer tools, and Q33 studies Hochman entropy inverse theory, Shmerkin Lq inverse theory and Bernoulli convolutions including Varju's arithmetic work.
- The new quarters do not replace the original research targets. They extend the curriculum from 32 to 40 quarters so that the original number theory, dynamics, geometry, topology, PDE, probability, arithmetic geometry and mathematical-physics tracks retain their intended depth.

Q1. Proof, arithmetic language, and precalculus

Quarter overview and reading route

Objective: Build rigorous proof habits while making elementary arithmetic and functions completely fluent. The emphasis is not speed but the ability to unpack quantifiers, choose a proof strategy, and write a proof that another mathematician can audit line by line.

Prerequisites: None beyond school mathematics.

30-hour allocation: Proof/writing 12h; elementary number theory 10h; functions/algebra/precalculus 5h; review and exposition 3h.

Core books and chapters/sections

- Velleman, How to Prove It, Chs. 1-6.
- Hammack, Book of Proof, Parts I-II and selected Part III.
- Niven-Zuckerman-Montgomery, An Introduction to the Theory of Numbers, Chs. 1-3.
- Apostol, Introduction to Analytic Number Theory, Chs. 1-6; Ireland-Rosen, early reciprocity/continued-fraction chapters.

Lecture notes and companions

- MIT OCW 18.100B Real Analysis (2025), Lectures 3-7 for proof writing, Archimedean property and sequence arguments.
- Use MIT 18.785 Number Theory I only as a later-facing preview; the elementary material should be solved from textbooks.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Logic, sets, quantifiers, functions; divisibility and Euclidean algorithm	Velleman Chs. 1-2; Hammack Part I; Niven-Zuckerman-Montgomery Ch. 1. MIT 18.100B Lectures 3-4 as proof-writing models. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Logic, sets, quantifiers, functions; divisibility and Euclidean algorithm	Velleman Chs. 1-2; Hammack Part I; Niven-Zuckerman-Montgomery Ch. 1. MIT 18.100B Lectures 3-4 as proof-writing models. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Direct proof, contrapositive, contradiction; gcd, Bezout, primes	Velleman Chs. 3-4; Hammack proof-method chapters; Apostol Chs. 1-2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Direct proof, contrapositive, contradiction; gcd, Bezout, primes	Velleman Chs. 3-4; Hammack proof-method chapters; Apostol Chs. 1-2. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Induction and recursion; congruences and modular arithmetic	Velleman induction chapter; Hammack induction sections; Niven-Zuckerman-Montgomery congruence chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Induction and recursion; congruences and modular arithmetic	Velleman induction chapter; Hammack induction sections; Niven-Zuckerman-Montgomery	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		congruence chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	
7	Learn: Relations and countability; CRT, Fermat-Euler, arithmetic functions	Velleman relations/functions chapters; Apostol Chs. 2-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Relations and countability; CRT, Fermat-Euler, arithmetic functions	Velleman relations/functions chapters; Apostol Chs. 2-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Möbius inversion; quadratic residues and first reciprocity examples	Apostol Chs. 2, 5; Ireland-Rosen sections on quadratic residues. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Möbius inversion; quadratic residues and first reciprocity examples	Apostol Chs. 2, 5; Ireland-Rosen sections on quadratic residues. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Continued fractions, Pell equations, cumulative proof clinic	Ireland-Rosen continued-fraction/Pell sections; Hardy-Wright selected elementary Diophantine examples; rework MIT 18.100B proof lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Continued fractions, Pell equations, cumulative proof clinic	Ireland-Rosen continued-fraction/Pell sections; Hardy-Wright selected elementary Diophantine examples; rework MIT 18.100B proof lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Logic and proof forms, objects and examples. Reading: Velleman/Hammack sections on statements, quantifiers and direct/contrapositive/contradiction proof. Definition/result focus: quantifier negation; implication vs converse; rigorous counterexamples. Work: Complete V1U1 plus 8 short translation/proof drills.

Week 2 - Logic and proof forms, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete V1U1 plus 8 short translation/proof drills; write a one-page error taxonomy. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Sets, functions and relations, objects and examples. Reading: Hammack/Velleman sections on sets, functions, equivalence relations. Definition/result focus: well-defined maps on quotients; injection/surjection/bijection; Cantor-Schroeder-Bernstein orientation. Work: Complete V1U2.

Week 4 - Sets, functions and relations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete V1U2; construct three quotient maps and audit well-definedness. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Induction, recursion and invariants, objects and examples. Reading: Hammack/Velleman induction and recursion sections. Definition/result focus: ordinary/strong induction; well-ordering principle; recursive correctness. Work: Complete V1U3.

Week 6 - Induction, recursion and invariants, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete V1U3; prove Euclidean-algorithm termination from a decreasing invariant. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Divisibility and Euclid, objects and examples. Reading: Niven-Zuckerman-Montgomery/Apostol elementary divisibility sections. Definition/result focus: Bezout; gcd properties; fundamental theorem of arithmetic. Work: Run extended Euclid on several pairs and give full proofs of Bezout and infinitude of primes.

Week 8 - Divisibility and Euclid, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Run extended Euclid on several pairs and give full proofs of Bezout and infinitude of primes. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Congruences and arithmetic functions, objects and examples. Reading: Apostol sections on congruences, CRT, Mobius inversion. Definition/result focus: CRT; Fermat/Euler; Dirichlet convolution; Mobius inversion. Work: Complete assigned arithmetic problems.

Week 10 - Congruences and arithmetic functions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete assigned arithmetic problems; derive inversion in both pointwise and summatory forms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Continued fractions and Pell, objects and examples. Reading: Ireland-Rosen or Hardy-Wright sections on continued fractions/Pell. Definition/result focus: convergents; best approximation; Pell recurrence. Work: Compute two Pell systems.

Week 12 - Continued fractions and Pell, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute two Pell systems; reconstruct one best-approximation proof; sit Q1 exit assessment. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- statement/quantifier
- direct proof
- contrapositive
- contradiction
- induction
- equivalence relation
- injection/surjection/bijection
- countability
- divisibility
- gcd
- prime
- congruence
- Dirichlet convolution
- Mobius function
- Legendre symbol
- continued fraction
- Pell equation

Key results to know and use

- induction principle
- Cantor diagonal argument
- countability of Z and Q
- uncountability of R
- Bezout identity
- fundamental theorem of arithmetic
- CRT
- Fermat/Euler theorem
- Mobius inversion
- quadratic reciprocity
- continued-fraction best approximation

Quarter-specific mastery targets

M1 - Exact language: state and negate quantified statements; distinguish implication/converse/contrapositive; define injection, surjection, bijection, equivalence relation, gcd, congruence, continued-fraction convergent and Pell equation.

M2 - Proofs to reproduce in full: irrationality of sqrt(2); Euclidean algorithm termination and Bezout; Chinese remainder theorem for pairwise coprime moduli; infinitude of primes; Mobius inversion; the basic best-approximation property of convergents.

M3 - Computation: execute extended Euclid, modular inversion, CRT reconstruction, continued-fraction expansion and Pell recurrence by hand on nontrivial examples, and justify each algorithmic step.

M4 - Proof judgement: diagnose quantifier reversal, converse errors, circularity and hidden induction assumptions; construct a counterexample when a proposed universal statement is false.

M5 - Exit use: given an unfamiliar elementary divisibility or recurrence problem, choose between contradiction, induction, invariant, Euclidean algorithm, congruence or continued fractions without the method being named.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V1U1: Propositions, quantifiers and proof forms

V1U1P1. [F] Formalize five statements with nested quantifiers, including continuity at a point, boundedness of a function, and existence of infinitely many primes. Negate each statement by pushing negations all the way to atomic predicates, then check the negation against examples.

V1U1P2. [T] Prove the irrationality of sqrt(2) by contradiction, prove that n^2 even implies n even by contrapositive, and prove the sum of the first n odd integers by induction. For each proof, identify why that proof form is natural and rewrite one proof in a less natural form to see the difference.

V1U1P3. [C] Diagnose three false arguments: affirming the consequent, illicit reversal of quantifiers, and circular reasoning. Repair each argument if a true statement is nearby, and state exactly which logical step failed.

Assigned block V1U3: Induction, recursion and invariants

V1U3P1. [T] Solve the recurrence a_{n+2}=3a_{n+1}-2a_n twice: once by induction from a guessed closed form and once by linear-algebra/characteristic-polynomial reasoning. Compare which information each method reveals.

V1U3P2. [T] Prove correctness and termination of the Euclidean algorithm using a strictly decreasing nonnegative invariant. Derive Bezout coefficients by back substitution for one nontrivial pair such as (414,662).

V1U3P3. [S] Design an invariant for the following process: replace a pair of integers (a,b) by (a-b,b) or (a,b-a) whenever entries remain positive. Determine which quantity is preserved and use it to characterize possible terminal states.

Assigned block V22U1: Congruences, reciprocity and continued fractions

V22U1P1. [F] Use quadratic reciprocity to determine whether 2,3,5,7 are quadratic residues modulo a curated set of odd primes. Organize the calculations so reciprocity reduces rather than expands the work.

V22U1P2. [T] Compute continued fractions of sqrt(2), sqrt(3), sqrt(5), and sqrt(13), derive Pell fundamental solutions, and prove the recurrence generates all positive solutions in one case.

V22U1P3. [S] Prove continued-fraction convergents are best approximants in a standard sense and apply the result to bound ||q alpha|| for a quadratic irrational.

Assigned block V1U6: Exposition and proof checking

V1U6P1. [T] Take a one-page proof from an introductory text, remove all phrases such as 'clearly' and 'obviously', and expand every hidden inference. Then compress it again without losing logical completeness. Compare the two versions.

V1U6P2. [C] Write a deliberately flawed proof of a true theorem containing exactly three nontrivial errors. One week later, audit it as if it were someone else's work and produce a line-by-line correction.

V1U6P3. [R] Prepare a 15-minute blackboard proof of one theorem with no notes. Afterward, write a dependency graph listing definitions, lemmas, and the theorem. Mark which nodes you could reproduce immediately and which required reconstruction.

Quarter-level integration problems

I1. [S]. Prove that every solution of x^2-Dy^2=1 generated by powers of a fundamental unit satisfies a second-order linear recurrence; identify the characteristic roots and relate exponential growth to continued fractions.

I2. [C]. Write three plausible but false elementary-number-theory claims involving divisibility, congruence and irrationality; for each, find the smallest counterexample and then formulate the strongest nearby true statement you can prove.

Full written solutions required for: V1U1P2, V1U3P2, V22U1P2, V1U6P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 10-15 page note entitled 'Five proof patterns in elementary number theory', including original examples and a complete Pell-equation worked example.

Exit checkpoint

Without notes, prove Bezout, CRT, Möbius inversion, irrationality of sqrt(2), and one best-approximation statement for continued fractions.

Research bridge

This quarter supplies the proof language used everywhere and begins the arithmetic spine leading eventually to Diophantine approximation, Baker theory, the Subspace Theorem and analytic number theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q2. Single-variable calculus, linear algebra I, and elementary number theory II

Quarter overview and reading route

Objective: Acquire computational and conceptual fluency in calculus and finite-dimensional linear algebra while deepening classical arithmetic through reciprocity, continued fractions and basic Diophantine equations.

Prerequisites: Q1.

30-hour allocation: Calculus 11h; linear algebra 10h; number theory 6h; proof/review 3h.

Core books and chapters/sections

- Apostol, Calculus Vol. I, Chs. 1-6 and selected integration/series chapters; or Spivak, corresponding chapters.
- MIT OCW 18.01SC problem sets for computational fluency.
- Axler, Linear Algebra Done Right, 4th ed., Chs. 1-8.
- Artin, Algebra, linear-algebra chapters as a geometric companion.
- Niven-Zuckerman-Montgomery, An Introduction to the Theory of Numbers, Chs. 1-3.
- Apostol, Introduction to Analytic Number Theory, Chs. 1-6; Ireland-Rosen, early reciprocity/continued-fraction chapters.

Lecture notes and companions

- MIT OCW 18.01SC Single Variable Calculus.
- MIT OCW RES.18-011 Algebra I Student Notes, linear algebra/bilinear forms sections.
- Use MIT 18.785 Number Theory I only as a later-facing preview; the elementary material should be solved from textbooks.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Limits, derivatives, vector spaces and linear maps	Apostol Calculus I Chs. 4-5 (or Spivak derivative chapters); Axler Chs. 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Limits, derivatives, vector spaces and linear maps	Apostol Calculus I Chs. 4-5 (or Spivak derivative chapters); Axler Chs. 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Mean-value/Taylor theorems; matrices, kernels, ranges and quotient intuition	Apostol Calculus I derivative/Taylor chapters; Axler Ch. 3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Mean-value/Taylor theorems; matrices, kernels, ranges and quotient intuition	Apostol Calculus I derivative/Taylor chapters; Axler Ch. 3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Riemann integration and FTC; eigenvalues/eigenvectors	Apostol Calculus I integration chapters; Axler Ch. 5. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Riemann integration and FTC; eigenvalues/eigenvectors	Apostol Calculus I integration chapters; Axler Ch. 5. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Sequences and series;	Apostol/Spivak series chapters;	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	invariant subspaces and diagonalization	Axler Chs. 5-6. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
8	Prove/use: Sequences and series; invariant subspaces and diagonalization	Apostol/Spivak series chapters; Axler Chs. 5-6. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Inner products and orthogonality; quadratic reciprocity	Axler Ch. 6; Ireland-Rosen quadratic reciprocity chapter; Apostol relevant chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Inner products and orthogonality; quadratic reciprocity	Axler Ch. 6; Ireland-Rosen quadratic reciprocity chapter; Apostol relevant chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Spectral theorem; continued fractions and Pell recurrences	Axler Chs. 7-8; Ireland-Rosen continued-fraction sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Spectral theorem; continued fractions and Pell recurrences	Axler Chs. 7-8; Ireland-Rosen continued-fraction sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - One-variable limits and derivatives, objects and examples. Reading: standard calculus text plus rigorous companion sections. Definition/result focus: derivative rules; MVT; Taylor theorem with remainder. Work: Derive MVT consequences.

Week 2 - One-variable limits and derivatives, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive MVT consequences; solve nonroutine limit/optimization problems with hypotheses stated. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Integration and series, objects and examples. Reading: calculus chapters on Riemann integration, FTC, Taylor/power series. Definition/result focus: FTC; convergence tests; uniform caution as preview. Work: Compute integrals/series and prove one convergence criterion from definitions available at this stage.

Week 4 - Integration and series, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute integrals/series and prove one convergence criterion from definitions available at this stage. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Vector spaces and linear maps, objects and examples. Reading: Axler opening chapters. Definition/result focus: basis; linear map; kernel/image; rank-nullity. Work: Complete V2U1-style work.

Week 6 - Vector spaces and linear maps, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete V2U1-style work; solve basis-change and quotient-space examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Eigenvalues and inner products, objects and examples. Reading: Axler operator/inner-product chapters. Definition/result focus: diagonalizability; orthogonality; finite-dimensional spectral theorem. Work: Diagonalize symmetric examples and construct defective counterexamples.

Week 8 - Eigenvalues and inner products, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Diagonalize symmetric examples and construct defective counterexamples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Quadratic residues, objects and examples. Reading: Ireland-Rosen/Apostol reciprocity sections. Definition/result focus: Legendre/Jacobi symbols; Euler criterion; Gauss lemma. Work: Compute symbols by reciprocity and prove Euler criterion/Gauss lemma in full.

Week 10 - Quadratic residues, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute symbols by reciprocity and prove Euler criterion/Gauss lemma in full. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Quadratic reciprocity and Pell synthesis, objects and examples. Reading: Ireland-Rosen reciprocity and Pell sections. Definition/result focus: quadratic reciprocity; Pell matrix recurrence. Work: Give one complete reciprocity proof route.

Week 12 - Quadratic reciprocity and Pell synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Give one complete reciprocity proof route; translate Pell recurrence into eigenvalue/linear-algebra language. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- limit
- derivative
- Riemann integral
- Taylor polynomial
- series
- vector space
- basis
- dimension
- linear map
- kernel/image
- eigenvalue
- inner product
- self-adjoint operator
- quadratic form
- divisibility
- gcd
- prime
- congruence
- Dirichlet convolution
- Mobius function
- Legendre symbol
- continued fraction
- Pell equation

Key results to know and use

- mean value theorem
- Taylor theorem
- fundamental theorem of calculus
- rank-nullity
- spectral theorem
- Gram-Schmidt
- classification of real quadratic forms
- Bezout identity
- fundamental theorem of arithmetic
- CRT
- Fermat/Euler theorem
- Mobius inversion
- quadratic reciprocity
- continued-fraction best approximation

Quarter-specific mastery targets

M1 - Exact language: derivative, Riemann integral, Taylor remainder, vector space, span, linear independence, linear map, kernel/image, eigenvalue/eigenspace, inner product, self-adjoint operator, Legendre symbol and quadratic residue.

M2 - Proofs to reproduce: mean value theorem and Taylor theorem in the form used; rank-nullity; diagonalizability with distinct eigenvalues; finite-dimensional spectral theorem for real symmetric/self-adjoint operators; one complete proof of quadratic reciprocity.

M3 - Computation: solve basis-change, kernel/image, eigenvalue and orthogonal diagonalization problems; evaluate Legendre/Jacobi symbols using reciprocity; derive and solve Pell recurrences via a 2x2 matrix.

M4 - Failure modes: give matrices that split but are not diagonalizable and operators diagonalizable over C but not R; explain where differentiability, compactness or symmetry is essential in the analytic/linear-algebra results used.

M5 - Exit use: translate a recurrence or quadratic-form problem into linear algebra and an arithmetic congruence problem into reciprocity, then complete both without a template.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V6U3: Differentiation and Riemann integration

V6U3P1. [T] Derive the fundamental theorem of calculus from the mean value theorem/Riemann integration hypotheses you have established. Identify the exact continuity assumptions in each direction.

V6U3P2. [C] Construct: a differentiable function with discontinuous derivative; a continuous nowhere differentiable function at theorem-statement level; and a Riemann-integrable discontinuous function. Explain which intuitive implications fail.

V6U3P3. [S] Prove Taylor's theorem with remainder and use it to derive a controlled approximation to log(1+x) or exp(x), including an explicit numerical error estimate.

Assigned block V2U1: Vector spaces and linear maps

V2U1P1. [F] For three concrete linear maps R^3->R^2 given by formulas, compute kernel, image, rank, nullity, and matrices in two different bases. Verify rank-nullity numerically and conceptually.

V2U1P2. [T] Prove that a linear map is determined by its values on a basis and derive the universal formula for extending an assignment on a basis. Use it to classify all projections R^3->R^3 with image span(e1,e2).

V2U1P3. [S] Let T be differentiation on polynomials of degree at most n. Find invariant subspaces, kernel, image, nilpotency index, and a convenient basis. Explain why this example anticipates Jordan theory.

Assigned block V2U4: Inner products and spectral theory

V2U4P1. [T] Run Gram-Schmidt on a badly conditioned basis by hand and explain geometrically what each subtraction accomplishes. Use the resulting QR factorization to solve a least-squares problem.

V2U4P2. [T] Prove the finite-dimensional spectral theorem for a real symmetric operator from the existence of a maximizing unit vector for the Rayleigh quotient. Track exactly where compactness and symmetry are used.

V2U4P3. [S] For a quadratic form in three variables, diagonalize it orthogonally, classify its signature, and solve a constrained extremum problem using the eigenvalues. Relate the answer to the min-max principle.

Assigned block V22U1: Congruences, reciprocity and continued fractions

V22U1P1. [F] Use quadratic reciprocity to determine whether 2,3,5,7 are quadratic residues modulo a curated set of odd primes. Organize the calculations so reciprocity reduces rather than expands the work.

V22U1P2. [T] Compute continued fractions of sqrt(2), sqrt(3), sqrt(5), and sqrt(13), derive Pell fundamental solutions, and prove the recurrence generates all positive solutions in one case.

V22U1P3. [S] Prove continued-fraction convergents are best approximants in a standard sense and apply the result to bound ||q alpha|| for a quadratic irrational.

Quarter-level integration problems

I1. [S]. For a symmetric 2x2 integer matrix arising from a Pell recurrence, diagonalize it over R, derive the closed form of its powers, and use the formula to prove an asymptotic statement about the recurrence.

I2. [T]. Starting from Taylor's theorem, derive a rigorous error bound for Newton iteration on sqrt(a); then express one Newton step as a rational approximation and compare it with continued-fraction convergents.

Full written solutions required for: V6U3P2, V2U1P2, V2U4P2, V22U1P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Implement continued fractions and Pell recurrences; prove the correctness of the algorithm and analyze the dominant eigenvalue of the Pell recurrence matrix.

Exit checkpoint

Be able to derive Taylor's theorem in the form used, prove rank-nullity, state/prove a finite-dimensional spectral theorem, and prove quadratic reciprocity at least by one chosen route.

Research bridge

Linear algebra becomes the language of differential equations, representation theory, lattices and finite fields; continued fractions prepare geometry of numbers and Diophantine approximation.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q3. Multivariable calculus, linear algebra II, and discrete/combinatorial methods

Quarter overview and reading route

Objective: Move from one-dimensional calculus to differential forms and multivariable geometry, strengthen linear algebra, and acquire the discrete tools used later in additive combinatorics, probability and graph theory.

Prerequisites: Q2.

30-hour allocation: Multivariable calculus 12h; linear algebra 8h; discrete/combinatorics 7h; review 3h.

Core books and chapters/sections

- Apostol, Calculus Vol. II, multivariable differentiation/integration/vector-calculus chapters.
- Hubbard-Hubbard, Vector Calculus, Linear Algebra, and Differential Forms, corresponding chapters.
- Axler, Linear Algebra Done Right, 4th ed., Chs. 1-8.
- Artin, Algebra, linear-algebra chapters as a geometric companion.
- Graham-Knuth-Patashnik, Concrete Mathematics, Chs. 1-7 selectively.
- van Lint-Wilson, A Course in Combinatorics, introductory counting/graph chapters.

Lecture notes and companions

- MIT OCW 18.02SC Multivariable Calculus.
- MIT OCW RES.18-011 Algebra I Student Notes, linear algebra/bilinear forms sections.
- Use problem-heavy lecture notes only as supplements; prioritize solving recurrences/counting problems.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Differentiability in R^n, Jacobians, inverse/implicit function intuition	Apostol Calculus II multivariable differentiation chapters; Hubbard-Hubbard differentiation chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Differentiability in R^n, Jacobians, inverse/implicit function intuition	Apostol Calculus II multivariable differentiation chapters; Hubbard-Hubbard differentiation chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Multiple integrals, change of variables, determinants and orientation	Apostol Calculus II multiple-integration chapters; Axler determinant chapter/review. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Multiple integrals, change of variables, determinants and orientation	Apostol Calculus II multiple-integration chapters; Axler determinant chapter/review. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Vector fields, line/surface integrals, differential forms	Hubbard-Hubbard differential-forms chapters; Apostol vector-calculus chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Vector fields, line/surface integrals, differential forms	Hubbard-Hubbard differential-forms chapters; Apostol vector-calculus chapters. Second pass: theorem proofs, harder exercises,	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		and a one-page proof reconstruction.	
7	Learn: Green, Stokes and divergence theorems	Hubbard-Hubbard Stokes chapters; MIT 18.02SC corresponding units. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Green, Stokes and divergence theorems	Hubbard-Hubbard Stokes chapters; MIT 18.02SC corresponding units. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Counting, recurrences, generating functions	Concrete Mathematics Chs. 1-2, 7 selectively. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Counting, recurrences, generating functions	Concrete Mathematics Chs. 1-2, 7 selectively. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Graphs, extremal examples, probabilistic counting preview	van Lint-Wilson introductory graph/counting chapters; Concrete Mathematics finite-sums/binomial chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Graphs, extremal examples, probabilistic counting preview	van Lint-Wilson introductory graph/counting chapters; Concrete Mathematics finite-sums/binomial chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Multivariable differentiation, objects and examples. Reading: rigorous multivariable calculus chapters. Definition/result focus: total derivative; chain rule; inverse/implicit function theorem hypotheses. Work: Compute Jacobians and prove chain rule.

Week 2 - Multivariable differentiation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Jacobians and prove chain rule; audit singular examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Multiple integration and change of variables, objects and examples. Reading: multivariable integration chapters. Definition/result focus: Fubini in elementary setting; Jacobian change of variables. Work: Evaluate integrals in nontrivial coordinates.

Week 4 - Multiple integration and change of variables, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Evaluate integrals in nontrivial coordinates; explain orientation/absolute determinant. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Vector calculus and differential forms preview, objects and examples. Reading: Spivak/Lee-compatible sections on forms and Stokes. Definition/result focus: differential forms; pullback; Green/Stokes/divergence statements. Work: Translate the same integral identity into vector-calculus and differential-form notation.

Week 6 - Vector calculus and differential forms preview, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Translate the same integral identity into vector-calculus and differential-form notation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Canonical forms, objects and examples. Reading: Axler/Roman/Hoffman-Kunze operator structure sections. Definition/result focus: minimal polynomial; generalized eigenspace; Jordan/rational canonical form. Work: Compute canonical forms and build examples separating diagonalizability criteria.

Week 8 - Canonical forms, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute canonical forms and build examples separating diagonalizability criteria. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Tensor and exterior algebra, objects and examples. Reading: Roman or multilinear algebra sections. Definition/result focus: tensor universal property; exterior power; determinant as top wedge. Work: Compute wedge products and derive determinant/oriented-volume interpretation.

Week 10 - Tensor and exterior algebra, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute wedge products and derive determinant/oriented-volume interpretation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Discrete mathematics and generating functions, objects and examples. Reading: standard discrete-math/generating-function chapters. Definition/result focus: recurrences; generating functions; basic counting identities. Work: Solve recurrences three ways.

Week 12 - Discrete mathematics and generating functions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve recurrences three ways; write one synthesis linking matrices, generating functions and asymptotics. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Frechet derivative
- Jacobian
- gradient
- Hessian
- line integral
- surface integral
- vector space
- basis
- dimension
- linear map
- kernel/image
- eigenvalue
- inner product
- self-adjoint operator
- quadratic form
- recurrence
- generating function
- graph
- matching
- inclusion-exclusion

Key results to know and use

- multivariable chain rule
- inverse/implicit function theorem
- change of variables
- Green/Stokes/divergence
- rank-nullity
- spectral theorem
- Gram-Schmidt
- classification of real quadratic forms
- inclusion-exclusion
- basic generating-function solution of recurrences
- pigeonhole principle

Quarter-specific mastery targets

M1 - Exact language: total derivative/Jacobian, differential form, orientation, line/surface integral, tensor product, exterior power, Jordan chain, generating function and recurrence relation.

M2 - Proofs/derivations: multivariable chain rule; change-of-variables formula in standard smooth-bijection settings; Green/Stokes/divergence theorem statements with correct orientations; Jordan-chain criteria; tensor universal property.

M3 - Computation: change coordinates and evaluate multidimensional integrals; compute Jordan/rational forms; manipulate wedge products; solve linear recurrences by generating functions and characteristic polynomials.

M4 - Structural understanding: explain determinant and oriented volume through exterior algebra; identify which hypotheses fail in characteristic 2 or at singular coordinate changes; distinguish local differential identities from global Stokes-type statements.

M5 - Exit use: solve a problem that requires moving among coordinates, forms, matrices and a discrete recurrence, making every convention explicit.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V2U5: Canonical forms

V2U5P1. [T] Compute Jordan forms and chains for three matrices exhibiting: one Jordan block, two blocks with the same eigenvalue, and mixed eigenvalues. Recover block sizes from dimensions of ker(T-lambda I)^k.

V2U5P2. [S] Prove the rational canonical form classification over a field at the level of cyclic modules over F[x]. Work out one 4x4 example where rational and Jordan forms differ because the characteristic polynomial does not split.

V2U5P3. [C] Construct two non-similar nilpotent 6x6 matrices with the same rank. Find the smallest list of nullities ker(T^k) that distinguishes their Jordan types.

Assigned block V2U6: Tensor, symmetric and exterior algebra

V2U6P1. [F] Compute V tensor W for V=R^2,W=R^3 in bases, then verify the universal property by factorizing an explicit bilinear map through V tensor W.

V2U6P2. [T] Prove the decompositions V tensor V = Sym^2 V direct-sum Lambda^2 V in characteristic not 2, and calculate dimensions. Explain what fails in characteristic 2.

V2U6P3. [S] Express cross products and oriented volume in R^3 using exterior powers and the Hodge-star intuition. Then derive the Cauchy-Binet formula from minors/exterior powers in one nontrivial case.

Assigned block V1U3: Induction, recursion and invariants

V1U3P1. [T] Solve the recurrence a_{n+2}=3a_{n+1}-2a_n twice: once by induction from a guessed closed form and once by linear-algebra/characteristic-polynomial reasoning. Compare which information each method reveals.

V1U3P2. [T] Prove correctness and termination of the Euclidean algorithm using a strictly decreasing nonnegative invariant. Derive Bezout coefficients by back substitution for one nontrivial pair such as (414,662).

V1U3P3. [S] Design an invariant for the following process: replace a pair of integers (a,b) by (a-b,b) or (a,b-a) whenever entries remain positive. Determine which quantity is preserved and use it to characterize possible terminal states.

Assigned block V12U3: Differential forms and integration

V12U3P1. [F] Compute pullbacks, wedge products, exterior derivatives, and integrals of differential forms on spheres/torus/coordinate domains.

V12U3P2. [T] Prove Stokes' theorem for a rectangle/simplex and map the local proof ingredients to the manifold theorem. Recover Green and divergence theorems as cases.

V12U3P3. [C] Produce closed nonexact forms on S^1 or T^n and explain why local exactness does not imply global exactness.

Quarter-level integration problems

I1. [S]. Parameterize an oriented surface patch, compute its area form using exterior algebra, and verify Stokes' theorem for one nontrivial 1-form by both boundary and surface calculations.

I2. [S]. Encode a coupled linear recurrence as a matrix iteration, compute its Jordan form, and derive the generating function of one coordinate; reconcile the poles of the generating function with eigenvalues.

Full written solutions required for: V2U5P2, V2U6P2, V1U3P2, V12U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Produce a computational notebook that visualizes change of variables and Stokes' theorem on several explicit regions, accompanied by rigorous derivations.

Exit checkpoint

Derive the multivariable chain rule, change-of-variables formula in standard examples, Stokes' theorem statement with orientation conventions, and solve representative recurrence/generating-function problems.

Research bridge

Differential forms lead naturally to smooth manifolds and de Rham theory; combinatorial fluency is prerequisite for random graphs and additive combinatorics.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q4. Mathematical logic I: first-order logic, computability, and Godel incompleteness

Quarter overview and reading route

Objective: Build enough mathematical logic to understand first-order completeness, computability, undecidability and the two Godel incompleteness theorems as mathematical results rather than slogans.

Prerequisites: Q1-Q3. No advanced algebra or analysis is required.

30-hour allocation: formal logic/model theory 9h; computability 7h; incompleteness/provability 8h; exercises and exposition 6h.

Core books and chapters/sections

- Enderton, A Mathematical Introduction to Logic, 2nd ed.: Chs. 1-3, with emphasis on Ch. 2 completeness and Ch. 3 undecidability/incompleteness.
- Boolos, Burgess, Jeffrey, Computability and Logic, 5th ed.: Chs. 1-18; Ch. 25 and Ch. 27 selectively.

- Sipser, Introduction to the Theory of Computation: computability/undecidability chapters as an algorithmic companion.

Lecture notes and primary-paper companions

- Enderton Ch. 2 contains the Henkin-style completeness route; Ch. 3 develops recursive functions, Godel coding and incompleteness.
- BBJ provides a particularly clean path from Turing computability through arithmetization to the diagonal lemma and Godel I/II.
- Use Feferman's historical Hilbert-Godel lectures only after the technical proofs, to understand the foundational context.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Propositional syntax and semantics	Enderton Ch. 1, opening sections; Boolos-Burgess-Jeffrey (BBJ) Chs. 9-10 as a preview of formal syntax/semantics.	Write truth-table and structural-induction notes; 8 exercises.
2	First-order languages, structures, satisfaction	Enderton Ch. 2: languages, terms, formulas, structures, assignments.	Define satisfaction recursively and work 6 model examples.
3	Deduction systems, soundness, completeness	Enderton Ch. 2 through the Soundness and Completeness Theorems.	Reconstruct soundness fully; write Henkin completeness proof architecture.
4	Compactness and Lowenheim-Skolem	Enderton Ch. 2; BBJ Chs. 12-14.	Prove compactness from completeness; give 3 nontrivial applications.
5	Turing machines and recursive/enumerable sets	BBJ Chs. 1-8; Sipser computability lectures as a secondary companion.	Implement/describe 4 machines; prove one closure theorem for r.e. sets.
6	Undecidability and reductions	BBJ Chs. 4, 7-8; Enderton Ch. 3 opening.	Prove the halting problem undecidable; complete 5 many-one reductions.
7	Arithmetization and representability	Enderton Ch. 3, Godel-number and representability sections; BBJ Chs. 15-16.	Construct a concrete Godel coding and represent two primitive recursive functions.
8	Diagonal lemma and Tarski undefinability	BBJ Ch. 17; Enderton Ch. 3.	Prove the diagonal lemma; derive a Tarski-style undefinability statement.
9	Godel first incompleteness theorem	Enderton Ch. 3; BBJ Ch. 17.	Give both computability and self-reference proof architectures; fully prove one version.
10	Godel second incompleteness theorem and provability	Enderton Ch. 3 later sections; BBJ Chs. 18 and 27 selectively.	State derivability conditions and reconstruct the proof skeleton of Godel II.

Week	Main focus	Reading / lecture notes	Required output
11	Nonstandard models and limits of formal systems	BBJ Ch. 25; Enderton model-theory remarks.	Construct a compactness argument yielding a nonstandard model of arithmetic.
12	Synthesis: logic as a mathematical tool	Review all theorem sheets; preview Q16 forcing and Q30 Hilbert X.	Closed-book theorem map + 20-minute oral on completeness vs incompleteness.

Week 1 - First-order syntax and semantics, objects and examples. Reading: Enderton/Boolos-Burgess-Jeffrey sections on languages, structures and satisfaction. Definition/result focus: terms/formulas; assignments; models; semantic consequence. Work: Formalize arithmetic statements and prove substitution/satisfaction lemmas for a toy language.

Week 2 - First-order syntax and semantics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Formalize arithmetic statements and prove substitution/satisfaction lemmas for a toy language. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Proof systems, soundness, completeness and compactness, objects and examples. Reading: Enderton completeness/compactness sections. Definition/result focus: soundness in full; completeness architecture; compactness consequences. Work: Prove soundness.

Week 4 - Proof systems, soundness, completeness and compactness, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove soundness; derive compactness from completeness; construct a nonstandard-model application. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Computability, objects and examples. Reading: Sipser or computability notes on Turing machines/c.e. sets. Definition/result focus: computable vs c.e.; universal machine; coding. Work: Implement/trace toy machines and prove closure properties of computable functions.

Week 6 - Computability, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Implement/trace toy machines and prove closure properties of computable functions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Undecidability and reductions, objects and examples. Reading: computability notes on halting problem and many-one reductions. Definition/result focus: halting undecidability; reduction methodology. Work: Carry out two reductions and explicitly distinguish recognizability from decidability.

Week 8 - Undecidability and reductions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out two reductions and explicitly distinguish recognizability from decidability. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Arithmetization and diagonal lemma, objects and examples. Reading: logic text sections on Godel numbering/representability/fixed points. Definition/result focus: representability; diagonal lemma; provability predicate. Work: Build a toy Godel numbering.

Week 10 - Arithmetization and diagonal lemma, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build a toy Godel numbering; prove the diagonal lemma assuming representability. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Godel I/II and semantic distinctions, objects and examples. Reading: logic text incompleteness sections. Definition/result focus: Godel I derivation; Godel II architecture; truth vs proof vs decidability. Work: Reconstruct Godel I.

Week 12 - Godel I/II and semantic distinctions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reconstruct Godel I; give a dependency audit for Godel II and explain why completeness is not contradicted. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Formal language; term; formula; sentence; free/bound variable
- Structure/model; variable assignment; satisfaction; semantic consequence
- Formal proof; theory; consistency; completeness of a theory
- Recursive/decidable set; recursively enumerable (c.e.) set
- Turing machine; universal machine; halting set; many-one reduction
- Primitive recursive and partial recursive function
- Representability/definability in arithmetic
- Godel numbering; provability predicate
- Diagonal/fixed-point lemma
- Omega-consistency, syntactic consistency, soundness (distinguish them)
- Standard vs nonstandard model of arithmetic
- Decidable theory; complete theory; recursively axiomatizable theory

Key results to know and use

- Soundness theorem for first-order logic (full proof)

- Godel-Henkin completeness theorem (proof architecture, with Henkin construction understood)
- Compactness theorem and at least three applications
- Downward Lowenheim-Skolem theorem (working proof)
- Undecidability of the halting problem (full proof)
- Equivalence/robustness of standard computability models (conceptual)
- Representability of recursive functions in weak arithmetic (architecture)
- Diagonal lemma (full proof)
- Tarski undefinability of arithmetical truth (derive from diagonalization)
- Godel first incompleteness theorem (one full proof, second proof architecture)
- Godel second incompleteness theorem (proof architecture and exact hypotheses)
- Existence of nonstandard models of arithmetic by compactness

Quarter-specific mastery targets

M1 - Exact language: first-order language, structure, assignment, satisfaction, theory, syntactic derivation, computable/decidable/c.e. set, Godel numbering, provability predicate, consistency, soundness and omega-consistency.

M2 - Proofs to reproduce: soundness for the chosen proof system; compactness from completeness and finite proofs; undecidability of the halting problem; diagonal lemma assuming representability; the standard derivation of Godel I from the fixed-point sentence.

M3 - Architecture-only but precise: completeness theorem and Godel II - state hypotheses and conclusions exactly, list the indispensable lemmas, and identify where arithmetization and derivability conditions enter.

M4 - Distinctions: explain why first-order completeness does not contradict incompleteness of sufficiently strong recursively axiomatized arithmetic theories; distinguish semantic truth, provability and decidability.

M5 - Exit use: encode a small syntactic operation arithmetically, carry out one many-one reduction, and present a dependency diagram from computability to Godel incompleteness.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V5U1: Formal languages, structures and deduction

V5U1P1. [F] Formalize a fragment of group theory in a first-order language: write axioms, a sentence asserting commutativity, and a formula defining the center. Distinguish syntax from semantics at every step.

V5U1P2. [T] Build a formal derivation of a simple theorem from logical axioms/rules, then prove soundness for the rules actually used. Identify which part is meta-mathematics rather than a formula in the object language.

V5U1P3. [C] Give properties of structures that are and are not first-order expressible in a chosen language. Explain why finiteness is not captured by a single first-order sentence over arbitrary structures.

Assigned block V5U2: Completeness and compactness

V5U2P1. [T] Use compactness to construct a nonstandard model of arithmetic containing an element larger than every standard numeral. Write the expanded language and finite satisfiability argument explicitly.

V5U2P2. [S] Use compactness to prove the graph-theoretic statement: if every finite subgraph of a graph is k-colorable, then the entire graph is k-colorable. Encode the coloring constraints as first-order/propositional clauses.

V5U2P3. [C] Compare completeness and compactness: derive compactness from completeness plus finite proof length, then explain why semantic completeness is not the same as categorical uniqueness of models.

Assigned block V5U3: Computability and undecidability

V5U3P1. [F] Give an explicit coding of finite binary strings by natural numbers and show concatenation/length are computable. Use the encoding to illustrate how syntax can become arithmetic data.

V5U3P2. [T] Prove undecidability of the halting problem by diagonalization. Then reduce the halting problem to another undecidable decision problem of your choice, writing the reduction as an algorithmic transformation.

V5U3P3. [S] Compare Cantor diagonalization, the halting diagonal argument, and the later Godel diagonal lemma. Write a table specifying the object diagonalized against and the contradiction/fixed-point produced.

Assigned block V5U4: Godel incompleteness

V5U4P1. [T] Construct Godel numbering for a toy formal language and explicitly code substitution for a few formulas. Identify which syntactic operations must be primitive recursive in a full proof.

V5U4P2. [T] Assuming representability and the diagonal lemma, derive a sentence G with G <-> not Prov(G). State exactly what consistency or soundness assumptions are required for each conclusion about G.

V5U4P3. [S] Build a proof-architecture map for the second incompleteness theorem: derivability conditions, formalized first incompleteness, and the obstruction to proving Con(T). Mark which ingredients are internalized in T.

Quarter-level integration problems

I1. [S]. Give a single comparative diagram for Cantor diagonalization, Turing's halting diagonalization and Godel's fixed-point construction: specify domain, coding map, diagonal object, and resulting impossibility statement.

I2. [C]. Construct a recursively axiomatized consistent toy theory too weak for the usual incompleteness conclusion, and explain exactly which representability/arithmetization hypothesis fails.

Full written solutions required for: V5U1P2, V5U2P2, V5U3P2, V5U4P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 15-20 page expository note proving soundness, compactness (via completeness), the halting problem, the diagonal lemma and one form of Godel I. Include a dependency graph that makes clear which theorem is semantic, syntactic or computability-theoretic.

Exit checkpoint

Without notes: define satisfaction, c.e. set and provability predicate; prove the halting problem and diagonal lemma; state completeness, compactness, Godel I and Godel II with hypotheses; answer “why no contradiction?” between completeness and incompleteness.

Research bridge

This quarter feeds Q16 (forcing/CH), Q30 (Hilbert's Tenth Problem), computability aspects of cryptography, and the model-theoretic viewpoint on arithmetic and algebra.

Q5. Transition to rigorous analysis and computational arithmetic

Quarter overview and reading route

Objective: Replace calculus intuition by theorem-proof analysis, while learning to treat computation as an experimental arithmetic laboratory whose outputs are always paired with proofs of correctness.

Prerequisites: Q1-Q3.

30-hour allocation: Real analysis 18h; computational arithmetic 7h; arithmetic review 2h; exposition 3h.

Core books and chapters/sections

- Abbott, Understanding Analysis, 2nd ed., Chs. 1-8.
- Rudin, Principles of Mathematical Analysis, Chs. 1-7 and 9-11 selectively.
- Crandall-Pomerance, Prime Numbers: A Computational Perspective, opening chapters selectively.
- Hoffstein-Pipher-Silverman, computational number-theory preliminaries.
- Niven-Zuckerman-Montgomery, An Introduction to the Theory of Numbers, Chs. 1-3.
- Apostol, Introduction to Analytic Number Theory, Chs. 1-6; Ireland-Rosen, early reciprocity/continued-fraction chapters.

Lecture notes and companions

- MIT OCW 18.100B Real Analysis (2025), Lectures 1-23.
- Implement algorithms in Python/Sage, but write a correctness proof beside each implementation.
- Use MIT 18.785 Number Theory I only as a later-facing preview; the elementary material should be solved from textbooks.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Real-number completeness, supremum, sequences and Cauchy criteria	Abbott Chs. 1-2; Rudin Chs. 1-3; MIT 18.100B Lectures 1-7. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Real-number completeness, supremum, sequences and Cauchy criteria	Abbott Chs. 1-2; Rudin Chs. 1-3; MIT 18.100B Lectures 1-7. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Series, limsup/liminf, compactness on R	Abbott Chs. 2-3; MIT 18.100B Lectures 7-10. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Series, limsup/liminf, compactness on R	Abbott Chs. 2-3; MIT 18.100B Lectures 7-10. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Continuity, uniform continuity, IVT/EVT	Abbott Ch. 4; Rudin Ch. 4; MIT 18.100B Lectures 9-14. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Continuity, uniform continuity, IVT/EVT	Abbott Ch. 4; Rudin Ch. 4; MIT 18.100B Lectures 9-14. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Differentiation and rigorous Taylor theory	Abbott Ch. 5; Rudin Ch. 5; MIT 18.100B Lectures 15-17. First pass: definitions, examples, and 2-3	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
		basic exercises.	
8	Prove/use: Differentiation and rigorous Taylor theory	Abbott Ch. 5; Rudin Ch. 5; MIT 18.100B Lectures 15-17. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Riemann integration, FTC, uniform convergence	Abbott Chs. 7-8; Rudin Chs. 6-7; MIT 18.100B Lectures 17-21. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Riemann integration, FTC, uniform convergence	Abbott Chs. 7-8; Rudin Chs. 6-7; MIT 18.100B Lectures 17-21. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Primality, modular exponentiation, Euclidean algorithms and experimental arithmetic	Crandall-Pomerance opening chapters; Hoffstein-Pipher-Silverman computational preliminaries; implement and prove each algorithm. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Primality, modular exponentiation, Euclidean algorithms and experimental arithmetic	Crandall-Pomerance opening chapters; Hoffstein-Pipher-Silverman computational preliminaries; implement and prove each algorithm. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Real numbers, sequences and completeness, objects and examples. Reading: Abbott/Tao rigorous-analysis opening chapters. Definition/result focus: supremum property; Cauchy convergence; Bolzano-Weierstrass. Work: Prove sequence theorems from the least-upper-bound property.

Week 2 - Real numbers, sequences and completeness, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove sequence theorems from the least-upper-bound property. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Continuity and compactness on intervals, objects and examples. Reading: Abbott continuity/compactness chapters. Definition/result focus: Heine-Borel; extreme/intermediate value; uniform continuity on compacta. Work: Construct counterexamples when compactness is removed.

Week 4 - Continuity and compactness on intervals, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct counterexamples when compactness is removed. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Differentiation, objects and examples. Reading: Abbott differentiation chapter. Definition/result focus: Rolle/MVT; Taylor-type consequences. Work: Prove MVT and use it to derive monotonicity, inverse estimates and error bounds.

Week 6 - Differentiation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove MVT and use it to derive monotonicity, inverse estimates and error bounds. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Riemann integration and uniform convergence, objects and examples. Reading: Abbott integration/sequences-of-functions sections. Definition/result focus: Riemann integrability criteria; uniform-limit/integral interchange in valid settings. Work: Separate pointwise/uniform convergence with explicit examples.

Week 8 - Riemann integration and uniform convergence, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Separate pointwise/uniform convergence with explicit examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Computational arithmetic, objects and examples. Reading: algorithmic number theory notes on Euclid, modular exponentiation, primality. Definition/result focus: algorithm invariants; operation/bit complexity distinction. Work: Implement Euclid, fast powering, continued fractions and toy Miller-Rabin.

Week 10 - Computational arithmetic, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Implement Euclid, fast powering, continued fractions and toy Miller-Rabin; prove correctness. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Cumulative analysis/proof audit, objects and examples. Reading: MIT 18.100B selected problems and prior notes. Definition/result focus: proof selection without prompts. Work: Closed-notes cumulative set.

Week 12 - Cumulative analysis/proof audit, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Closed-notes cumulative set; rewrite two weakest proofs after error analysis. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- supremum
- Cauchy sequence
- compactness
- uniform continuity
- uniform convergence
- Riemann integrability
- bit complexity
- modular exponentiation
- probable prime
- Euclidean algorithm
- divisibility
- gcd
- prime
- congruence
- Dirichlet convolution
- Mobius function
- Legendre symbol
- continued fraction
- Pell equation

Key results to know and use

- Bolzano-Weierstrass
- Heine-Borel
- extreme/intermediate value theorems
- uniform-limit theorem
- Weierstrass M-test
- Arzela-Ascoli (later pass)
- correctness of repeated squaring
- extended Euclidean algorithm
- Miller-Rabin guarantee in its standard randomized form
- Bezout identity
- fundamental theorem of arithmetic
- CRT
- Fermat/Euler theorem
- Mobius inversion
- quadratic reciprocity
- continued-fraction best approximation

Quarter-specific mastery targets

M1 - Exact language: epsilon-delta limit/continuity, Cauchy sequence, compactness in R, uniform convergence, Riemann integrability, arithmetic algorithm correctness and bit/operation complexity at an elementary level.

M2 - Proofs to reproduce: Bolzano-Weierstrass/Heine-Borel in R, uniform-limit theorem, a standard criterion for Riemann integrability, and correctness/termination of Euclid and modular exponentiation.

M3 - Computation: implement gcd, modular exponentiation, primality trial/Miller-Rabin toy tests and continued fractions; give invariants proving correctness and estimate operation counts.

M4 - Counterexamples: distinguish pointwise from uniform convergence, bounded from uniformly continuous, and convergence from absolute convergence using explicit examples.

M5 - Exit use: pass from experimental computation to a rigorous claim - state exactly what the code verifies and what still requires proof.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V6U1: Real numbers, sequences and series

V6U1P1. [T] Prove Cauchy completeness of R from the least-upper-bound property and conversely derive a completeness formulation from Cauchy completeness. Track which construction of R is being assumed.

V6U1P2. [F] Determine convergence/absolute convergence/conditional convergence of a curated list of ten series using comparison, ratio/root, condensation, and Dirichlet tests. For each, justify why the chosen test is informative.

V6U1P3. [C] Construct sequences showing the converses of common convergence implications fail: bounded does not imply convergent; every subsequence has a further convergent subsequence characterizes compactness but not arbitrary boundedness in general metric spaces.

Assigned block V6U2: Metric spaces and compactness

V6U2P1. [T] Prove the equivalence of compactness, sequential compactness, and complete+totally bounded for metric spaces. Give a counterexample to each implication if the metric-space hypothesis is removed or altered.

V6U2P2. [F] Analyze compactness and connectedness of explicit subsets of R^n and of function spaces with different metrics. Explain why the metric, not only the underlying set, matters.

V6U2P3. [S] Prove the contraction mapping theorem and apply it to a nonlinear equation x=cos x with an explicit invariant interval and quantitative error bound.

Assigned block V22U2: Arithmetic functions and convolution

V22U2P1. [F] Compute Dirichlet convolutions among 1, mu, phi, tau and selected multiplicative functions; verify multiplicativity from prime-power data.

V22U2P2. [T] Prove Mobius inversion in arithmetic-function and summatory forms. Use it to count coprime pairs in [1,N]^2 up to a controllable error.

V22U2P3. [S] Derive the Dirichlet hyperbola estimate sum_{n<=x} d(n)=x log x+(2gamma-1)x+O(sqrt x), including the geometric lattice-point interpretation.

Assigned block V36U2: RSA, factoring and primality

V36U2P1. [F] Implement modular exponentiation, extended Euclid, CRT, Miller-Rabin, and RSA on toy parameters. Verify decryption both by Euler/Carmichael and CRT decomposition.

V36U2P2. [T] Prove RSA correctness with all gcd cases handled and analyze textbook RSA's deterministic-security failure. Explain exactly what randomized padding/security notion must change.

V36U2P3. [R] Compare primality testing and factoring: state AKS polynomial-time primality, probabilistic Miller-Rabin, and best-known factoring status without conflating them.

Quarter-level integration problems

I1. [T]. Implement binary modular exponentiation and prove its loop invariant; compare arithmetic-operation count with naive exponentiation and state why this is not yet a bit-complexity proof.

I2. [C]. Produce one sequence of continuous functions converging pointwise but not uniformly and one uniformly convergent series whose termwise derivative cannot be justified under naive hypotheses; state the missing theorem hypotheses.

Full written solutions required for: V6U1P2, V6U2P2, V22U2P2, V36U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Create a theorem notebook for one-variable analysis and an arithmetic code repository containing gcd, modular exponentiation, primality testing and continued fractions, each with correctness proof and complexity discussion.

Exit checkpoint

Pass a closed-book two-hour exam: compactness/continuity proof, uniform convergence problem, Riemann integrability proof, and one algorithm-correctness proof.

Research bridge

This is the key rigor transition. Metric-space analysis next quarter will make compactness and convergence portable to geometry, function spaces and probability.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q6. Real analysis II, metric spaces, and geometry of numbers I

Quarter overview and reading route

Objective: Generalize analysis to metric spaces and simultaneously learn the geometry-of-numbers viewpoint that turns arithmetic approximation problems into lattice geometry.

Prerequisites: Q5 plus Q2 linear algebra.

30-hour allocation: Metric-space analysis 16h; geometry of numbers 10h; proof/exposition 4h.

Core books and chapters/sections

- Abbott, Understanding Analysis, 2nd ed., Chs. 1-8.
- Rudin, Principles of Mathematical Analysis, Chs. 1-7 and 9-11 selectively.
- Cassels, An Introduction to the Geometry of Numbers: first pass through lattices, convex bodies and Minkowski theorems; return later to successive minima, transference and Diophantine applications when those topics enter.
- Gruber-Lekkerkerker, Geometry of Numbers, introductory chapters as reference.

Lecture notes and companions

- MIT OCW 18.100B Real Analysis (2025), Lectures 1-23.
- MIT OCW 18.785 Number Theory I, Lecture 14 (Geometry of Numbers).

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says "selected" or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Metric spaces, open/closed sets, convergence and completeness	Rudin Ch. 2; Abbott metric-space sections; MIT 18.100B Lectures 11-14. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Metric spaces, open/closed sets, convergence and completeness	Rudin Ch. 2; Abbott metric-space sections; MIT 18.100B Lectures 11-14. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Compactness, connectedness and function spaces	Rudin Chs. 2, 7; MIT 18.100B Lectures 13-21. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Compactness, connectedness and function spaces	Rudin Chs. 2, 7; MIT 18.100B Lectures 13-21. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Contraction mapping and Picard-Lindelof; norms as geometry	MIT 18.100B Lectures 22-23; Rudin relevant fixed-point exercises. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Contraction mapping and Picard-Lindelof; norms as geometry	MIT 18.100B Lectures 22-23; Rudin relevant fixed-point exercises. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Lattices, fundamental parallelepipeds, covolume, primitive vectors	Cassels Ch. I; MIT 18.785 Lecture 14. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Lattices, fundamental parallelepipeds, covolume, primitive vectors	Cassels Ch. I; MIT 18.785 Lecture 14. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Blichfeldt and Minkowski	Cassels Chs. II-III; MIT 18.785	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	convex-body theorem	Lecture 14. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
10	Prove/use: Blichfeldt and Minkowski convex-body theorem	Cassels Chs. II-III; MIT 18.785 Lecture 14. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Successive minima and Diophantine applications	Cassels Chs. IV-V; derive Dirichlet approximation from Minkowski. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Successive minima and Diophantine applications	Cassels Chs. IV-V; derive Dirichlet approximation from Minkowski. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Metric spaces and completeness, objects and examples. Reading: Abbott/Folland or MIT 18.100B metric-space sections. Definition/result focus: metric topology; Cauchy; completeness; contraction principle. Work: Prove Banach fixed-point theorem and solve one ODE/fixed-point application.

Week 2 - Metric spaces and completeness, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Banach fixed-point theorem and solve one ODE/fixed-point application. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Compactness and total boundedness, objects and examples. Reading: metric-space compactness sections. Definition/result focus: sequential compactness; total boundedness; complete+totally bounded criterion. Work: Construct examples separating bounded/total bounded/compact.

Week 4 - Compactness and total boundedness, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct examples separating bounded/total bounded/compact. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Lattices and covolume, objects and examples. Reading: Cassels opening sections. Definition/result focus: lattice; fundamental domain; determinant/covolume; primitive vector. Work: Compute explicit lattices and verify covolume under basis changes.

Week 6 - Lattices and covolume, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute explicit lattices and verify covolume under basis changes. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Blichfeldt and Minkowski I, objects and examples. Reading: Cassels convex-body theorem sections. Definition/result focus: Blichfeldt; Minkowski convex-body theorem. Work: Prove both and use Minkowski to solve a nontrivial arithmetic existence problem.

Week 8 - Blichfeldt and Minkowski I, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove both and use Minkowski to solve a nontrivial arithmetic existence problem. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Successive minima and Minkowski II, objects and examples. Reading: Cassels successive-minima sections. Definition/result focus: successive minima; Minkowski second theorem exact inequalities. Work: Calculate minima for planar examples.

Week 10 - Successive minima and Minkowski II, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Calculate minima for planar examples; audit equality/near-equality configurations. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Diophantine approximation via lattices, objects and examples. Reading: Cassels Diophantine applications. Definition/result focus: Dirichlet simultaneous approximation; dual/transference viewpoint. Work: Design lattice+body from an approximation target.

Week 12 - Diophantine approximation via lattices, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Design lattice+body from an approximation target; explain later Dani correspondence preview. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- supremum
- Cauchy sequence
- compactness
- uniform continuity
- uniform convergence
- Riemann integrability
- lattice
- covolume
- fundamental domain
- convex body
- successive minima

- dual lattice

Key results to know and use

- Bolzano-Weierstrass
- Heine-Borel
- extreme/intermediate value theorems
- uniform-limit theorem
- Weierstrass M-test
- Arzela-Ascoli (later pass)
- Blichfeldt lemma
- Minkowski convex body theorem
- Minkowski second theorem
- Dirichlet simultaneous approximation via lattices
- Mahler compactness (later)

Quarter-specific mastery targets

M1 - Exact language: metric, completeness, compactness, total boundedness, lattice, covolume, convex/symmetric body, successive minima and dual lattice.

M2 - Proofs to reproduce: compact iff complete + totally bounded in the chosen setting; Blichfeldt; Minkowski convex-body theorem; Dirichlet simultaneous approximation via a lattice/convex body.

M3 - Computation: calculate bases, determinants and successive minima for explicit 2D/3D lattices; construct the lattice encoding a simultaneous-approximation problem.

M4 - Architecture: state Minkowski's second theorem and basic transference inequalities precisely, and explain how they sharpen first-minimum existence statements.

M5 - Exit use: given a Diophantine approximation claim, design the ambient lattice and convex body rather than being told them.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V6U2: Metric spaces and compactness

V6U2P1. [T] Prove the equivalence of compactness, sequential compactness, and complete+totally bounded for metric spaces. Give a counterexample to each implication if the metric-space hypothesis is removed or altered.

V6U2P2. [F] Analyze compactness and connectedness of explicit subsets of R^n and of function spaces with different metrics. Explain why the metric, not only the underlying set, matters.

V6U2P3. [S] Prove the contraction mapping theorem and apply it to a nonlinear equation x=cos x with an explicit invariant interval and quantitative error bound.

Assigned block V25U1: Lattices and convex bodies

V25U1P1. [F] For lattices generated by explicit 2x2 and 3x3 matrices, compute covolume, shortest vectors, fundamental parallelepipeds, and dual lattices. Compare Euclidean length with covolume under shearing.

V25U1P2. [T] Prove Blichfeldt's lemma and deduce Minkowski's convex body theorem. Apply it to prove existence of a nonzero lattice point in a carefully chosen ellipse/box with an explicit numerical bound.

V25U1P3. [S] Use Minkowski to prove Dirichlet's simultaneous approximation theorem by constructing the appropriate lattice/convex body. Make the dictionary between lattice coordinates and approximation inequalities explicit.

Assigned block V25U2: Successive minima, duality and transference

V25U2P1. [F] Compute successive minima for Z^2 under several convex bodies and for a highly skew lattice. Verify Minkowski's second theorem numerically in dimension 2.

V25U2P2. [T] Prove a selected transference inequality between a lattice and its dual in low dimension, and apply it to simultaneous versus linear-form approximation.

V25U2P3. [S] Analyze how an LLL-reduced basis controls successive minima only approximately. Work a 2D/3D example by hand to prepare the cryptographic use later.

Assigned block V25U3: Classical Diophantine approximation

V25U3P1. [F] Compute convergents and approximation constants for sqrt(2), golden ratio, and one nonquadratic irrational. Verify the sharpness phenomenon behind Hurwitz's theorem for the golden ratio.

V25U3P2. [T] Prove Dirichlet's theorem, Hurwitz's theorem in dimension one, and characterize badly approximable reals via bounded continued-fraction partial quotients.

V25U3P3. [S] For a quadratic irrational alpha, derive a uniform lower bound q||q alpha|| >= c(alpha)>0 using periodic continued fractions or algebraic arguments.

Quarter-level integration problems

I1. [S]. Derive Dirichlet's simultaneous approximation theorem in dimension 2 by an explicit lattice in R^3 and a symmetric box; optimize the parameter choice and compare with pigeonhole proof constants.

I2. [X]. Enumerate lattice points in several planar convex bodies for two lattices of equal covolume but different shape; compare observations with Minkowski's theorem and state clearly what the experiment cannot prove.

Full written solutions required for: V6U2P2, V25U1P2, V25U2P2, V25U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 12-page exposition deriving Dirichlet's simultaneous approximation theorem from a lattice formulation, with diagrams and explicit low-dimensional examples.

Exit checkpoint

Prove completeness/compactness results in metric spaces and Minkowski's convex-body theorem; compute successive minima for at least five explicit lattices.

Research bridge

Geometry of numbers will return in algebraic number theory, homogeneous dynamics, lattice minima, Schmidt's theorem and lattice cryptography.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q7. Abstract algebra I, complex analysis I, and finite fields I

Quarter overview and reading route

Objective: Build the algebraic structures and complex-analytic tools that power modern number theory; finish the quarter able to construct and calculate explicitly in finite fields.

Prerequisites: Q2-Q6.

30-hour allocation: Abstract algebra 12h; complex analysis 10h; finite fields 5h; review 3h.

Core books and chapters/sections

- Dummit-Foote, Abstract Algebra, Chs. 1-14 selectively.
- Artin, Algebra, selected chapters for a second viewpoint.
- Stein-Shakarchi, Complex Analysis, Chs. 1-8 selectively.
- Conway, Functions of One Complex Variable I, corresponding core chapters.
- Lidl-Niederreiter, Finite Fields, Chs. 1-3 first pass; later character/exponential-sum chapters.
- Ireland-Rosen, finite-field and character-sum sections as a number-theoretic companion.

Lecture notes and companions

- MIT OCW 18.701 Algebra I, student notes/problem sets.
- MIT OCW 18.04 Complex Variables with Applications for computations; use a proof-oriented text for theory.
- J.S. Milne field/Galois notes may be used for structural review.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Groups, homomorphisms, actions, quotient groups	Dummit-Foote Chs. 1-4; MIT 18.701 group-theory notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Groups, homomorphisms, actions, quotient groups	Dummit-Foote Chs. 1-4; MIT 18.701 group-theory notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Rings, ideals, quotient rings, polynomial rings	Dummit-Foote Chs. 7-9. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Rings, ideals, quotient rings, polynomial rings	Dummit-Foote Chs. 7-9. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Holomorphic functions, Cauchy-Riemann, contour integration	Stein-Shakarchi Complex Analysis Chs. 1-2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Holomorphic functions, Cauchy-Riemann, contour integration	Stein-Shakarchi Complex Analysis Chs. 1-2. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Cauchy theorem/formula, power series, zeros and residues	Stein-Shakarchi Chs. 2-3; Conway corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Cauchy theorem/formula, power series, zeros and residues	Stein-Shakarchi Chs. 2-3; Conway corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
9	Learn: Fields, extensions, irreducible polynomials, finite-field construction	Dummit-Foote Chs. 13-14 selectively; Lidl-Niederreiter Ch. 1. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Fields, extensions, irreducible polynomials, finite-field construction	Dummit-Foote Chs. 13-14 selectively; Lidl-Niederreiter Ch. 1. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Frobenius, multiplicative groups, trace/norm and explicit computations in F_{p^n}	Lidl-Niederreiter Chs. 1-2; Ireland-Rosen finite-field sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Frobenius, multiplicative groups, trace/norm and explicit computations in F_{p^n}	Lidl-Niederreiter Chs. 1-2; Ireland-Rosen finite-field sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Groups and quotients, objects and examples. Reading: Dummit-Foote group theory opening chapters. Definition/result focus: normality; quotient; isomorphism theorems. Work: Compute quotient groups and prove first isomorphism theorem.

Week 2 - Groups and quotients, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute quotient groups and prove first isomorphism theorem. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Rings, ideals and factorization, objects and examples. Reading: Dummit-Foote ring chapters. Definition/result focus: ideal; quotient; PID/UFD/Euclidean implications. Work: Construct examples/counterexamples among factorization domains.

Week 4 - Rings, ideals and factorization, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct examples/counterexamples among factorization domains. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Fields and polynomial extensions, objects and examples. Reading: Dummit-Foote field-extension sections. Definition/result focus: irreducibility; simple extension; minimal polynomial. Work: Construct finite-degree extensions and compute bases/norms in examples.

Week 6 - Fields and polynomial extensions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct finite-degree extensions and compute bases/norms in examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Complex analysis I, objects and examples. Reading: Ahlfors/Stein-Shakarchi opening complex chapters. Definition/result focus: holomorphicity; Cauchy theorem/formula; residues. Work: Prove Cauchy formula in chosen route.

Week 8 - Complex analysis I, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Cauchy formula in chosen route; compute residues by several methods. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Finite fields, objects and examples. Reading: Lidl-Niederreiter opening chapters. Definition/result focus: existence/uniqueness F_q; Frobenius; cyclic F_q^*. Work: Construct F_9/F_16 examples and compute trace/norm.

Week 10 - Finite fields, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct F_9/F_16 examples and compute trace/norm. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Finite-field polynomial arithmetic, objects and examples. Reading: Lidl-Niederreiter factorization sections. Definition/result focus: irreducible polynomials; subfields; factorization algorithms. Work: Factor explicit polynomials and connect Frobenius orbits to irreducible factors.

Week 12 - Finite-field polynomial arithmetic, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Factor explicit polynomials and connect Frobenius orbits to irreducible factors. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- normal subgroup
- group action
- ideal
- PID/UFD
- irreducible polynomial
- field extension
- Galois group
- holomorphic
- contour integral
- winding number

- meromorphic function
- residue
- argument principle
- finite field
- Frobenius
- trace
- norm
- additive character
- multiplicative character
- Gauss sum
- Jacobi sum

Key results to know and use

- isomorphism theorems
- Sylow theorems
- Gauss lemma
- fundamental theorem of Galois theory
- Cauchy theorem/formula
- Liouville
- maximum modulus
- residue theorem
- Rouche theorem
- existence/uniqueness of F_{p^n}
- cyclicity of F_q^*
- trace/norm formulas
- basic Gauss/Jacobi identities
- Weil bounds (later)

Quarter-specific mastery targets

M1 - Exact language: normal subgroup, quotient, ideal, quotient ring, PID/UFD, field extension, irreducible polynomial, holomorphic function, contour integral, residue, finite field, Frobenius, trace and norm.

M2 - Proofs to reproduce: group/ring isomorphism theorems; quotient-ring universal property; Cauchy integral formula and residue theorem; existence/uniqueness of F_q and cyclicity of F_q^* at textbook level.

M3 - Computation: construct F_{p^n} from irreducible polynomials, compute Frobenius/trace/norm, factor polynomials over finite fields and evaluate residues by several methods.

M4 - Counterexamples: distinguish UFD/PID/Euclidean domain implications; show why holomorphic hypotheses in Cauchy theory cannot simply be removed; identify inseparable phenomena in positive characteristic.

M5 - Exit use: move fluently between quotient algebra, polynomial factorization and finite-field arithmetic in one integrated calculation.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V3U1: Groups, homomorphisms and actions

V3U1P1. [F] Compute kernels, images, stabilizers, and orbits for the natural actions of S4 on {1,2,3,4}, on 2-element subsets, and by conjugation on itself. Check orbit-stabilizer in each case.

V3U1P2. [T] Prove the first isomorphism theorem and use it to classify quotients of Z and kernels of homomorphisms Z^2->Z. Then interpret normal subgroups as exactly kernels.

V3U1P3. [S] Use group actions to prove Cauchy's theorem for finite groups in a concrete prime-order counting argument. Compare this proof to one using Sylow theory later.

Assigned block V3U3: Rings, ideals and factorization

V3U3P1. [F] In Z[x], F_p[x], and k[x,y], compute explicit ideals, quotient rings, units, zero divisors, and prime/maximal behavior. Use quotient calculations rather than only theorem names.

V3U3P2. [T] Prove Euclidean domain => PID => UFD and exhibit rings showing the converses fail. For Z[sqrt(-5)], verify concretely the failure of unique factorization of elements.

V3U3P3. [S] Apply the Chinese remainder theorem to rings: decompose Z/60Z and F[x]/((x-1)(x+1)) when factors are coprime. Explain the idempotents that realize the product decomposition.

Assigned block V8U1: Holomorphic functions and Cauchy theory

V8U1P1. [T] Starting from Cauchy's integral theorem, derive Cauchy's integral formula and the derivative estimates. Use them to prove Liouville and the Fundamental Theorem of Algebra.

V8U1P2. [F] Evaluate contour integrals around circles and rectangles for rational functions with poles inside/outside. For each calculation, do it once directly when possible and once by Cauchy theory.

V8U1P3. [C] Give functions that are complex differentiable at one point but not holomorphic nearby, and contrast with real differentiability. Explain why the Cauchy-Riemann equations need regularity hypotheses for a converse.

Assigned block V27U1: Structure of finite fields

V27U1P1. [F] Construct F_4, F_8, F_9, and F_25 explicitly as polynomial quotients. List subfields, Frobenius automorphisms, multiplicative-group orders, traces, and norms.

V27U1P2. [T] Prove every finite subgroup of a field's multiplicative group is cyclic and deduce F_q^* is cyclic. Use it to count elements of each possible order.

V27U1P3. [S] Prove existence/uniqueness of F_{p^n} up to isomorphism and determine when F_{p^m} embeds in F_{p^n}.

Quarter-level integration problems

I1. [S]. Construct F_9 in two different irreducible-polynomial models and exhibit an explicit field isomorphism; compute trace/norm of each element and verify invariance under the isomorphism.

I2. [S]. Use residues to evaluate a real integral, then reinterpret the algebraic poles as roots in a splitting field; explain precisely which parts are analytic and which purely algebraic.

Full written solutions required for: V3U1P2, V3U3P2, V8U1P2, V27U1P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Construct F_{p^n} for several small (p,n), write addition/multiplication/Frobenius/trace/norm tables, and prove the structural facts your program uses.

Exit checkpoint

Prove the isomorphism theorems, quotient-ring universal property, Cauchy integral formula, residue theorem, and uniqueness/classification of finite fields.

Research bridge

These subjects feed directly into Galois theory, algebraic number theory, L-functions, modular forms and finite-field arithmetic.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q8. Topology, algebra II, and algebraic number theory entry

Quarter overview and reading route

Objective: Acquire point-set topology and Galois/commutative algebra while beginning number fields in a concrete way.

Prerequisites: Q7.

30-hour allocation: Topology 10h; algebra/Galois theory 10h; algebraic number theory 7h; exposition 3h.

Core books and chapters/sections

- Munkres, Topology, Chs. 2-4 and Ch. 9 selectively.
- Lee, Introduction to Topological Manifolds, selected sections as a geometric companion.
- Dummit-Foote, Abstract Algebra, Chs. 1-14 selectively.
- Artin, Algebra, selected chapters for a second viewpoint.
- Marcus, Number Fields, early chapters through ideals, class group and units.
- Neukirch, Algebraic Number Theory, Ch. I; Milne, Algebraic Number Theory notes in parallel.

Lecture notes and companions

- Use standard university topology notes only to reinforce Munkres problem solving.
- MIT OCW 18.701 Algebra I, student notes/problem sets.
- MIT OCW 18.785 Number Theory I, Lectures 1-19.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says "selected" or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Bases, product/quotient topologies, continuity	Munkres Ch. 2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Bases, product/quotient topologies, continuity	Munkres Ch. 2. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Compactness, connectedness, separation and countability axioms	Munkres Chs. 3-4. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Compactness, connectedness, separation and countability axioms	Munkres Chs. 3-4. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Field extensions and Galois correspondence	Dummit-Foote Chs. 13-14; Milne Fields and Galois Theory parallel notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Field extensions and Galois correspondence	Dummit-Foote Chs. 13-14; Milne Fields and Galois Theory parallel notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Algebraic integers, number fields, norm and trace	Marcus Chs. 1-2; Milne Algebraic Number Theory opening chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Algebraic integers, number fields, norm and trace	Marcus Chs. 1-2; Milne Algebraic Number Theory opening chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Dedekind domains, ideals,	Marcus ideal chapters; Neukirch	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	ideal factorization	Ch. I; MIT 18.785 Lectures 1-6. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
10	Prove/use: Dedekind domains, ideals, ideal factorization	Marcus ideal chapters; Neukirch Ch. I; MIT 18.785 Lectures 1-6. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Prime decomposition and Frobenius preview	MIT 18.785 Lectures 6-7; Marcus prime-factorization chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Prime decomposition and Frobenius preview	MIT 18.785 Lectures 6-7; Marcus prime-factorization chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Point-set topology, objects and examples. Reading: Munkres opening topology chapters. Definition/result focus: basis/subbasis; continuity; product/quotient topology. Work: Prove basic universal properties and construct quotient examples.

Week 2 - Point-set topology, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove basic universal properties and construct quotient examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Compactness and connectedness, objects and examples. Reading: Munkres compactness/connectedness chapters. Definition/result focus: Tychonoff finite/product cases as appropriate; connected-image principles. Work: Solve separation/compactness counterexamples.

Week 4 - Compactness and connectedness, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve separation/compactness counterexamples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Galois theory, objects and examples. Reading: Dummit-Foote Galois chapters. Definition/result focus: normal/separable; Galois correspondence; fixed fields. Work: Compute Galois groups of explicit polynomials/cyclotomic examples.

Week 6 - Galois theory, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Galois groups of explicit polynomials/cyclotomic examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Algebraic integers, objects and examples. Reading: Marcus/Neukirch introductory number-field sections. Definition/result focus: integrality; O_K; discriminant; trace/norm. Work: Find integral bases in quadratic/cubic examples.

Week 8 - Algebraic integers, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Find integral bases in quadratic/cubic examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Dedekind domains and ideals, objects and examples. Reading: Marcus/Neukirch ideal-factorization sections. Definition/result focus: Dedekind domain; fractional ideal; ideal factorization. Work: Factor primes in quadratic fields and restore unique factorization via ideals.

Week 10 - Dedekind domains and ideals, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Factor primes in quadratic fields and restore unique factorization via ideals. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Splitting and ramification, objects and examples. Reading: Marcus/Neukirch prime decomposition sections. Definition/result focus: e,f,g; ramification; discriminant criterion with hypotheses. Work: Produce complete arithmetic dossiers for two small number fields.

Week 12 - Splitting and ramification, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Produce complete arithmetic dossiers for two small number fields. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- basis/subbasis
- product topology
- quotient topology
- connectedness
- compactness
- fundamental group
- normal subgroup
- group action
- ideal
- PID/UFD

- irreducible polynomial
- field extension
- Galois group
- number field
- ring of integers
- norm/trace
- Dedekind domain
- fractional ideal
- class group
- ramification
- discriminant
- regulator

Key results to know and use

- compact subsets of Hausdorff spaces are closed
- Tychonoff for finite products
- basic quotient constructions
- isomorphism theorems
- Sylow theorems
- Gauss lemma
- fundamental theorem of Galois theory
- unique factorization of ideals
- Dedekind-Kummer
- Minkowski class-group finiteness
- Dirichlet unit theorem
- analytic class number formula (statement)

Quarter-specific mastery targets

M1 - Exact language: topological basis, compact/connected space, field extension, normal/separable/Galois extension, algebraic integer, ring of integers, prime ideal and Dedekind domain.

M2 - Proofs to reproduce: compact-image and connected-image theorems; fundamental theorem of Galois theory; integral-closure facts used to define O_K; unique factorization of nonzero ideals in a Dedekind domain.

M3 - Computation: determine O_K for quadratic examples, compute discriminants, use Dedekind-style criteria where valid, and factor rational primes in quadratic/cyclotomic fields.

M4 - Structural understanding: connect splitting/ramification/inertia with polynomial factorization mod p, clearly recording hypotheses and exceptional primes.

M5 - Exit use: given a small number field, produce a complete arithmetic dossier: integral basis, discriminant, ramified primes and sample ideal factorizations.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V11U1: Topological spaces and continuity

V11U1P1. [F] For several topologies on the same set, identify open/closed sets, bases, closures, interiors, and convergence. Include standard, discrete, indiscrete, and product examples.

V11U1P2. [T] Prove that continuity via inverse images is equivalent to the epsilon-delta notion in metric spaces and derive the universal property of product topology for maps into products.

V11U1P3. [C] Construct examples showing sequence convergence does not characterize closure in arbitrary topological spaces, while it does in first-countable spaces.

Assigned block V3U6: Galois theory

V3U6P1. [T] Compute Galois groups of x^3-2, x^4-2, and cyclotomic polynomials Phi_5 and Phi_8. Make the correspondence between subgroups and intermediate fields explicit.

V3U6P2. [T] Prove the fundamental theorem of Galois theory for finite Galois extensions at proof-architecture level, then verify every correspondence in Q(zeta_8)/Q.

V3U6P3. [S] Use Galois groups to determine solvability by radicals for selected polynomials. Work through one irreducible quintic with Galois group S5 at the level of the group-theoretic obstruction.

Assigned block V24U1: Number fields and algebraic integers

V24U1P1. [F] For Q(sqrt(5)), Q(sqrt(-5)), Q(cuberoot(2)), and Q(zeta_5), compute candidate rings of integers/integral bases where accessible, field discriminants, traces, and norms of selected elements.

V24U1P2. [T] Prove the trace/norm/discriminant relations from embeddings and multiplication matrices. Use discriminant congruences to identify the quadratic ring of integers.

V24U1P3. [S] Determine splitting of several rational primes in a quadratic field using polynomial factorization modulo p and the discriminant/Legendre symbol.

Assigned block V24U2: Dedekind domains and ideal factorization

V24U2P1. [F] Factor ideals generated by small rational primes in Z[sqrt(-5)] or another quadratic integer ring. Contrast element factorization with unique ideal factorization.

V24U2P2. [T] Prove a selected route that the ring of integers is Dedekind and hence nonzero ideals factor uniquely into prime ideals. Track Noetherian/integrally closed/dimension-one inputs.

V24U2P3. [S] Compute the ideal class group of Q(sqrt(-5)) or Q(sqrt(-23)) using a Minkowski bound and explicit prime ideals.

Quarter-level integration problems

I1. [S]. For K=Q(sqrt(-5)), compute O_K, discriminant and factorizations of (2),(3),(6); use ideals to explain the failure and restoration of unique factorization.

I2. [S]. For a quadratic extension K/Q, relate factorization of a prime p not dividing the discriminant to the Frobenius element in the Galois group and to the factorization of a defining polynomial mod p.

Full written solutions required for: V11U1P2, V3U6P2, V24U1P2, V24U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Compute rings of integers and prime decompositions in at least five quadratic/cyclotomic examples; accompany every computation with the theorem used.

Exit checkpoint

State and prove the Galois correspondence, verify key compactness/connectedness examples, and prove unique ideal factorization in the Dedekind-domain framework at the level of the chosen text.

Research bridge

This establishes the algebraic-number-theory branch leading to local fields, class field theory, elliptic curves, automorphic forms and arithmetic geometry.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q9. ODE, smooth geometry, complex analysis II, and elementary analytic number theory

Quarter overview and reading route

Objective: Develop differential-equation and smooth-manifold language, deepen complex analysis, and use complex methods to enter zeta functions and prime distribution.

Prerequisites: Q6-Q8.

30-hour allocation: ODE/smooth manifolds 10h; complex analysis 9h; analytic number theory 8h; review 3h.

Core books and chapters/sections

- Teschl, Ordinary Differential Equations and Dynamical Systems, Chs. 1-6 selectively.
- Lee, Introduction to Smooth Manifolds, Chs. 1-5 plus vector-fields/forms chapters later.
- Stein-Shakarchi, Complex Analysis, Chs. 1-8 selectively.
- Conway, Functions of One Complex Variable I, corresponding core chapters.
- Davenport, Multiplicative Number Theory, chapters on Dirichlet characters, L-functions, PNT and primes in AP.
- Montgomery-Vaughan, Multiplicative Number Theory I; Iwaniec-Kowalski, corresponding analytic chapters as references.

Lecture notes and companions

- MIT OCW 18.100B Lecture 23 for Picard-Lindelof as a bridge.
- MIT OCW 18.04 Complex Variables with Applications for computations; use a proof-oriented text for theory.
- MIT OCW 18.785 Number Theory I, Lectures 16-19.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says "selected" or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Existence/uniqueness, flows and phase portraits	Teschl Chs. 1-3; MIT 18.100B Lecture 23. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Existence/uniqueness, flows and phase portraits	Teschl Chs. 1-3; MIT 18.100B Lecture 23. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Smooth manifolds, tangent spaces, submersions/immersions	Lee Introduction to Smooth Manifolds Chs. 1-5. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Smooth manifolds, tangent spaces, submersions/immersions	Lee Introduction to Smooth Manifolds Chs. 1-5. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Meromorphic functions, argument principle, analytic continuation	Stein-Shakarchi Chs. 3-5. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Meromorphic functions, argument principle, analytic continuation	Stein-Shakarchi Chs. 3-5. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Gamma/zeta, Dirichlet series and Euler products	Davenport opening chapters; MIT 18.785 Lecture 16. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Gamma/zeta, Dirichlet series and Euler products	Davenport opening chapters; MIT 18.785 Lecture 16. Second pass:	2 theorem reconstructions + 5 harder problems + one 10-minute

Week	Main focus	Reading / lecture notes	Required output
		theorem proofs, harder exercises, and a one-page proof reconstruction.	oral explanation.
9	Learn: PNT architecture and complex-analytic ingredients	MIT 18.785 Lecture 16; Davenport PNT chapter. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: PNT architecture and complex-analytic ingredients	MIT 18.785 Lecture 16; Davenport PNT chapter. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Dirichlet characters/L-functions and primes in arithmetic progressions	Davenport character/L-function chapters; MIT 18.785 Lectures 17-18. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Dirichlet characters/L-functions and primes in arithmetic progressions	Davenport character/L-function chapters; MIT 18.785 Lectures 17-18. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - ODE existence and flows, objects and examples. Reading: standard ODE text/Picard-Lindelof sections. Definition/result focus: local existence/uniqueness; flow property. Work: Prove contraction-based existence and compute nonlinear flows.

Week 2 - ODE existence and flows, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove contraction-based existence and compute nonlinear flows. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Smooth manifolds foundations, objects and examples. Reading: Lee smooth manifolds opening chapters. Definition/result focus: charts; tangent space; differential; immersion/submersion. Work: Compute tangent maps and verify coordinate independence.

Week 4 - Smooth manifolds foundations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute tangent maps and verify coordinate independence. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Complex analysis II, objects and examples. Reading: Ahlfors/Stein-Shakarchi meromorphic/Gamma/argument chapters. Definition/result focus: argument principle; residues; Gamma basics. Work: Use contour methods for sums/integrals and count zeros.

Week 6 - Complex analysis II, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Use contour methods for sums/integrals and count zeros. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Arithmetic functions and Dirichlet series, objects and examples. Reading: Apostol analytic-NT chapters. Definition/result focus: Dirichlet convolution; partial summation; Dirichlet series. Work: Derive average-order formulas via hyperbola/partial summation.

Week 8 - Arithmetic functions and Dirichlet series, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive average-order formulas via hyperbola/partial summation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Zeta and Euler products, objects and examples. Reading: Apostol/Davenport zeta opening sections. Definition/result focus: Euler product; analytic continuation/functional equation architecture. Work: Derive Euler product and map analytic continuation dependencies.

Week 10 - Zeta and Euler products, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive Euler product and map analytic continuation dependencies. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Dirichlet characters preview, objects and examples. Reading: Davenport character sections. Definition/result focus: orthogonality; character decomposition of AP indicators. Work: Build characters mod q and derive the progression indicator formula.

Week 12 - Dirichlet characters preview, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build characters mod q and derive the progression indicator formula. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- flow
- equilibrium
- stability
- smooth manifold
- chart
- tangent space
- submanifold

- holomorphic
- contour integral
- winding number
- meromorphic function
- residue
- argument principle
- Dirichlet series
- Euler product
- Dirichlet character
- Dirichlet L-function
- von Mangoldt function
- zero-free region

Key results to know and use

- Picard-Lindelof
- matrix exponential solution
- regular value/submanifold theorem (statement)
- Cauchy theorem/formula
- Liouville
- maximum modulus
- residue theorem
- Rouche theorem
- Euler product for zeta
- PNT
- Dirichlet theorem on primes in AP
- functional equation for primitive Dirichlet L-functions (architecture)
- PNT in AP (statement)

Quarter-specific mastery targets

M1 - Exact language: local flow, tangent vector, chart/atlas, meromorphic function, Gamma function, Dirichlet series, Euler product, arithmetic function and Dirichlet character.

M2 - Proofs/derivations: Picard-Lindelof in a standard Banach/contraction formulation; tangent-space coordinate invariance; Euler product for zeta in Re(s)>1; character orthogonality.

M3 - Architecture: state analytic continuation/functional equation of zeta, PNT and Dirichlet's theorem on primes in progressions; identify the exact complex-analysis inputs needed later.

M4 - Computation: solve representative nonlinear ODEs/flows, compute tangent maps and residues, and build character tables modulo small q.

M5 - Exit use: explain how a question about primes becomes a statement about zeros/nonvanishing of analytic functions.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V12U1: Smooth manifolds and tangent geometry

V12U1P1. [F] Write atlases and transition maps for S^n, RP^n, the torus, and a matrix Lie group. Compute tangent spaces at selected points using charts and derivations.

V12U1P2. [T] Prove the regular value theorem in the finite-dimensional setting and use it to show spheres, orthogonal groups, and level sets are submanifolds.

V12U1P3. [S] For a smooth map between manifolds, compute its differential in coordinates and intrinsically. Verify chain rule and coordinate independence on a nontrivial example.

Assigned block V8U5: Special functions and zeta preparation

V8U5P1. [F] Derive the Gamma functional equation and evaluate Gamma(n+1), Beta integrals, and one reflection/duplication identity from standard integral representations.

V8U5P2. [T] Starting from theta-function Poisson summation, derive the functional equation of zeta at proof-architecture level, marking the Mellin-transform and analytic-continuation steps.

V8U5P3. [S] Prove Euler's product for zeta on Re(s)>1 and compare its analytic meaning with the later zero-free line/PNT route. Explain exactly why absolute convergence matters.

Assigned block V22U3: Dirichlet series and zeta

V22U3P1. [T] Starting from absolute convergence for Re(s)>1, derive the Euler product for zeta and logarithmic derivative -zeta'/zeta=sum Lambda(n)n^{-s}. State every interchange justification.

V22U3P2. [F] Prove Euler's divergence of sum_p 1/p using the Euler product/finite products and compare it with divergence of sum 1/n.

V22U3P3. [S] Use Mellin transforms/partial summation to translate between summatory arithmetic functions and Dirichlet-series behavior in at least two explicit examples.

Assigned block V22U5: Characters and Dirichlet L-functions

V22U5P1. [F] Construct all Dirichlet characters modulo 5, 8, and 12; separate primitive/imprimitive characters and verify orthogonality relations.

V22U5P2. [T] Compute quadratic Gauss sums and prove |tau(chi)|=sqrt(p) for the Legendre character. Track the role of character orthogonality.

V22U5P3. [S] Derive the Euler product and functional-equation ingredients for L(s,chi) in one primitive parity case, making clear which complex-analysis results are imported.

Quarter-level integration problems

I1. [S]. Starting from character orthogonality, derive the exact formula extracting the n congruent a mod q terms of a Dirichlet series; identify how this produces L(s,chi) in the study of primes in progressions.

I2. [T]. Derive the Euler product for zeta twice - from multiplicativity/unique factorization and from a finite Euler product followed by a justified limit - and audit convergence in both arguments.

Full written solutions required for: V12U1P2, V8U5P2, V22U3P2, V22U5P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Give a 45-minute blackboard lecture proving the Euler product and explaining the proof architecture of the PNT and Dirichlet's theorem, explicitly marking every complex-analysis input.

Exit checkpoint

Solve one nontrivial ODE existence/flow problem, compute tangent spaces via charts, derive the zeta Euler product rigorously, and prove character orthogonality.

Research bridge

Smooth geometry prepares Lie groups and dynamics; analytic number theory will later connect to sieve, circle method, modular/automorphic L-functions and decoupling.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q10. Measure theory and analytic number theory I

Quarter overview and reading route

Objective: Make Lebesgue measure/integration and L^p spaces routine, then use them to strengthen the analytic-number-theory foundation around zeta and Dirichlet L-functions.

Prerequisites: Q5-Q9.

30-hour allocation: Measure/integration 16h; analytic number theory 10h; proof/review 4h.

Core books and chapters/sections

- Folland, Real Analysis, Chs. 1-3.
- Stein-Shakarchi, Real Analysis, Chs. 1-3 as an intuitive companion.
- Davenport, Multiplicative Number Theory, chapters on Dirichlet characters, L-functions, PNT and primes in AP.
- Montgomery-Vaughan, Multiplicative Number Theory I; Iwaniec-Kowalski, corresponding analytic chapters as references.

Lecture notes and companions

- MIT OCW 18.103 Fourier Analysis, Lectures 2-7 for measure/integration/Lp.
- MIT OCW 18.785 Number Theory I, Lectures 16-19.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Sigma-algebras, outer measure and Lebesgue measure	Folland Ch. 1; Stein-Shakarchi Real Analysis Ch. 1; MIT 18.103 Lectures 2-4. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Sigma-algebras, outer measure and Lebesgue measure	Folland Ch. 1; Stein-Shakarchi Real Analysis Ch. 1; MIT 18.103 Lectures 2-4. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Measurable functions and convergence theorems	Folland Ch. 2; MIT 18.103 Lectures 4-6. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Measurable functions and convergence theorems	Folland Ch. 2; MIT 18.103 Lectures 4-6. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: L^p spaces, Holder/Minkowski, completeness	Folland Ch. 2; MIT 18.103 Lectures 6-7. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: L^p spaces, Holder/Minkowski, completeness	Folland Ch. 2; MIT 18.103 Lectures 6-7. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Product measures, Fubini/Tonelli and differentiation ideas	Folland Ch. 2-3 selectively. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Product measures, Fubini/Tonelli and differentiation ideas	Folland Ch. 2-3 selectively. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Zeta/L-functions: analytic continuation, functional equation architecture	Davenport relevant chapters; MIT 18.785 Lectures 16-18. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
10	Prove/use: Zeta/L-functions: analytic continuation, functional equation architecture	Davenport relevant chapters; MIT 18.785 Lectures 16-18. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: PNT and primes in arithmetic progressions: full proof reconstruction	Davenport PNT/Dirichlet chapters; MIT 18.785 Lectures 16-19. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: PNT and primes in arithmetic progressions: full proof reconstruction	Davenport PNT/Dirichlet chapters; MIT 18.785 Lectures 16-19. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Measures and measurable functions, objects and examples. Reading: Folland/Stein-Shakarchi measure chapters. Definition/result focus: sigma-algebra; measure; measurable function; convergence a.e. Work: Construct nontrivial measurable sets/functions and prove basic measure properties.

Week 2 - Measures and measurable functions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct nontrivial measurable sets/functions and prove basic measure properties. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Lebesgue integration, objects and examples. Reading: Folland integration sections. Definition/result focus: MCT; Fatou; DCT. Work: Prove convergence theorems and build examples showing hypotheses matter.

Week 4 - Lebesgue integration, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove convergence theorems and build examples showing hypotheses matter. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Lp spaces, objects and examples. Reading: Folland Lp sections. Definition/result focus: Holder; Minkowski; completeness of Lp. Work: Prove inequalities and solve convergence/interpolation examples.

Week 6 - Lp spaces, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove inequalities and solve convergence/interpolation examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Dirichlet L-functions, objects and examples. Reading: Davenport early L-function chapters. Definition/result focus: characters; Gauss sums; Euler products; L(s,chi). Work: Compute Gauss sums and primitive/imprimitive examples.

Week 8 - Dirichlet L-functions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Gauss sums and primitive/imprimitive examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Nonvanishing and primes in progressions, objects and examples. Reading: Davenport Dirichlet theorem chapters. Definition/result focus: L(1,chi) nonvanishing architecture; Dirichlet theorem. Work: Write a complete dependency chain from orthogonality to primes in AP.

Week 10 - Nonvanishing and primes in progressions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Write a complete dependency chain from orthogonality to primes in AP. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - PNT architecture, objects and examples. Reading: Davenport/Apostol PNT sections. Definition/result focus: psi/theta/pi equivalence; zero-free line strategy. Work: Prove Chebyshev estimates and a chosen Tauberian/complex proof architecture.

Week 12 - PNT architecture, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Chebyshev estimates and a chosen Tauberian/complex proof architecture. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- sigma-algebra
- measure
- measurable function
- almost everywhere
- Lebesgue integral
- L^p space
- product measure
- Dirichlet series
- Euler product
- Dirichlet character
- Dirichlet L-function
- von Mangoldt function

- zero-free region

Key results to know and use

- Caratheodory construction
- MCT
- Fatou
- DCT
- Fubini-Tonelli
- Holder/Minkowski
- Euler product for zeta
- PNT
- Dirichlet theorem on primes in AP
- functional equation for primitive Dirichlet L-functions (architecture)
- PNT in AP (statement)

Quarter-specific mastery targets

M1 - Exact language: sigma-algebra, measurable function, measure, a.e. convergence, L^p, product/signed measure, Dirichlet L-function and zero-free/nonvanishing statement at s=1.

M2 - Proofs to reproduce: monotone convergence, Fatou, dominated convergence, Fubini/Tonelli under correct hypotheses, Holder/Minkowski, and completeness of L^p for 1<=p<=infinity in the chosen text.

M3 - Number-theory chain: reconstruct the logical route zeta/L-function analytic properties -> nonvanishing on Re(s)=1 or at 1 -> PNT/Dirichlet PNT in progressions; know exactly which steps are black boxes.

M4 - Applications: justify exchanges of limit/integral/sum in analytic-number-theory examples and derive partial summation consequences without hidden convergence assumptions.

M5 - Exit use: audit a proof containing an interchange of limits, sums, products and integrals, accepting only steps licensed by named convergence theorems.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V6U4: Measure and Lebesgue integration

V6U4P1. [F] Compute Lebesgue measures/integrals for step functions, Cantor-set indicators, and functions with countably many discontinuities. Compare with their Riemann integrability.

V6U4P2. [T] Prove monotone convergence, Fatou, and dominated convergence in a dependency chain, then construct examples showing why domination or monotonicity hypotheses cannot simply be dropped.

V6U4P3. [S] Derive the Borel-Cantelli lemma's measure-theoretic core from continuity of measure and apply it to a limsup set before probability is formally introduced.

Assigned block V6U6: Lp spaces and convergence modes

V6U6P1. [F] For L^p[0,1], determine membership of x^{-alpha} and log x for ranges of p and alpha. Compute norms where possible.

V6U6P2. [T] Prove Holder and Minkowski inequalities and identify equality cases in a finite-dimensional model before passing to integrals.

V6U6P3. [C] Build examples separating almost-everywhere, in-measure, L^1, and L^p convergence. Draw a one-page implication diagram and annotate each failed converse with your example.

Assigned block V22U4: Prime number theorem

V22U4P1. [F] Prove Chebyshev upper/lower bounds for pi(x) using binomial coefficients or theta/psi estimates. Make constants explicit enough to see pi(x) is of order x/log x.

V22U4P2. [T] Prove equivalence psi(x)~x iff pi(x)~x/log x by partial summation and prime-power error estimates.

V22U4P3. [R] Reconstruct the complex-analytic PNT chain: meromorphic continuation/zero-free line for zeta, Wiener-Ikehara or classical contour method, psi(x)~x. Prove the elementary steps and audit the analytic input precisely.

Assigned block V22U6: Primes in arithmetic progressions

V22U6P1. [T] Derive the character identity for the indicator n=a mod q and use it to reduce prime counting in progressions to estimates for sums chi(n)Lambda(n).

V22U6P2. [S] Prove Dirichlet's theorem on primes in arithmetic progressions following a selected proof route through L(1,chi) != 0; identify separately the real and complex character cases.

V22U6P3. [R] Compare qualitative Dirichlet, PNT in arithmetic progressions, Siegel-Walfisz, and Bombieri-Vinogradov by writing exact quantifier ranges in q and error strengths.

Quarter-level integration problems

I1. [S]. Take a standard proof of PNT or Dirichlet's theorem and annotate every use of Fubini/Tonelli, dominated convergence, uniform convergence or analytic continuation; replace any unjustified interchange with a valid lemma.

I2. [C]. Construct examples separating a.e., in-measure, L1 and L2 convergence, then explain which convergence mode is actually needed in a selected analytic-number-theory limiting argument.

Full written solutions required for: V6U4P2, V6U6P2, V22U4P2, V22U6P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Prepare a 15-page proof dossier containing DCT, Fubini, completeness of L^p, the PNT proof architecture, and Dirichlet's theorem with every analytic lemma cross-referenced.

Exit checkpoint

Closed-book: prove MCT/DCT in a chosen formulation, derive Holder/Minkowski, and reconstruct the main logical chain from nonvanishing of L(1,chi) to primes in arithmetic progressions.

Research bridge

Measure theory is the common foundation for probability, harmonic analysis, ergodic theory and spectral automorphic theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q11. Probability I and additive combinatorics I

Quarter overview and reading route

Objective: Develop measure-theoretic probability and the structure-vs-randomness language of sumsets, with enough finite Fourier intuition to see their future interaction.

Prerequisites: Q10 and Q3 combinatorics.

30-hour allocation: Probability 16h; additive combinatorics 10h; review/exposition 4h.

Core books and chapters/sections

- Durrett, Probability: Theory and Examples, Chs. 1-5 selectively.
- Williams, Probability with Martingales, early chapters.
- Tao-Vu, Additive Combinatorics, Chs. 1-3 first pass; later Fourier/Gowers chapters.
- Nathanson, Additive Number Theory: Inverse Problems and the Geometry of Sumsets, early chapters.

Lecture notes and companions

- MIT OCW 18.440 Probability and Random Variables, Lectures 4-35; MIT 18.175 later.
- Tao UCLA Math 254A arithmetic combinatorics notes, Lecture Notes 1-6.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Probability spaces, independence, Borel-Cantelli	Durrett Chs. 1-2; Williams opening chapters; MIT 18.440 early probability lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Probability spaces, independence, Borel-Cantelli	Durrett Chs. 1-2; Williams opening chapters; MIT 18.440 early probability lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Random variables, expectation, convergence modes	Durrett Chs. 2-3; MIT 18.440 corresponding lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Random variables, expectation, convergence modes	Durrett Chs. 2-3; MIT 18.440 corresponding lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: LLN, CLT and conditional expectation preview	Durrett Chs. 2-3; MIT 18.440 LLN/CLT units. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: LLN, CLT and conditional expectation preview	Durrett Chs. 2-3; MIT 18.440 LLN/CLT units. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Sumsets, Cauchy-Davenport, Ruzsa calculus	Tao-Vu Ch. 1; Tao UCLA Math 254A Lecture Note 1. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Sumsets, Cauchy-Davenport, Ruzsa calculus	Tao-Vu Ch. 1; Tao UCLA Math 254A Lecture Note 1. Second pass: theorem proofs, harder exercises,	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		and a one-page proof reconstruction.	
9	Learn: Plunnecke, Ruzsa covering and additive energy	Tao-Vu Chs. 1-2; Tao 254A Notes 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Plunnecke, Ruzsa covering and additive energy	Tao-Vu Chs. 1-2; Tao 254A Notes 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Balog-Szemeredi-Gowers and Freiman theory	Tao-Vu Chs. 2-3; Tao 254A Notes 2-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Balog-Szemeredi-Gowers and Freiman theory	Tao-Vu Chs. 2-3; Tao 254A Notes 2-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Probability spaces and expectation, objects and examples. Reading: Durrett/Billingsley foundations. Definition/result focus: independence; expectation; distribution; conditional expectation preview. Work: Construct pairwise-not-joint independence and compute distributions.

Week 2 - Probability spaces and expectation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct pairwise-not-joint independence and compute distributions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Convergence and limit laws, objects and examples. Reading: Durrett convergence chapters. Definition/result focus: Borel-Cantelli; weak/strong laws; CLT statement and mechanism. Work: Build examples separating convergence modes.

Week 4 - Convergence and limit laws, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build examples separating convergence modes; prove Borel-Cantelli. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Conditional expectation and martingales, objects and examples. Reading: Durrett/Williams martingale sections. Definition/result focus: conditional expectation; martingale; stopping time. Work: Prove elementary martingale properties and gambler’s ruin.

Week 6 - Conditional expectation and martingales, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove elementary martingale properties and gambler’s ruin. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Sumsets and Ruzsa calculus, objects and examples. Reading: Tao-Vu early chapters. Definition/result focus: sumset; doubling; Ruzsa triangle/covering. Work: Complete additive-energy calculations and Ruzsa inequalities.

Week 8 - Sumsets and Ruzsa calculus, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete additive-energy calculations and Ruzsa inequalities. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Energy, BSG and Freiman orientation, objects and examples. Reading: Tao-Vu inverse-theory chapters. Definition/result focus: additive energy; BSG; Freiman homomorphism/GAP. Work: Prove energy lower bound.

Week 10 - Energy, BSG and Freiman orientation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove energy lower bound; reconstruct simplified BSG mechanism. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Roth/Fourier preview, objects and examples. Reading: Tao-Vu finite Fourier sections. Definition/result focus: Fourier transform on finite groups; density increment concept. Work: Prove finite-group Parseval/convolution identities and outline Roth’s argument.

Week 12 - Roth/Fourier preview, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove finite-group Parseval/convolution identities and outline Roth’s argument. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- random variable
- independence
- conditional expectation
- filtration
- martingale
- mode of convergence
- sumset
- doubling constant
- additive energy
- Ruzsa distance
- Freiman homomorphism

- Gowers U^k norm

Key results to know and use

- Borel-Cantelli
- weak/strong laws
- Jensen
- optional stopping in standard settings
- martingale convergence (later)
- Cauchy-Davenport
- Ruzsa triangle/covering
- Plunnecke
- Balog-Szemeredi-Gowers
- Freiman theorem
- Roth theorem (later)

Quarter-specific mastery targets

M1 - Exact language: independence, expectation, variance, convergence modes, additive energy, doubling constant, Ruzsa distance, Freiman homomorphism and approximate progression.

M2 - Proofs to reproduce: Borel-Cantelli I/II in their standard hypotheses; a LLN proof route; Cauchy-Davenport; Ruzsa triangle inequality/covering lemma; energy lower bound from Cauchy-Schwarz.

M3 - Architecture: state a quantitative BSG theorem and a finite Freiman theorem in one standard formulation; identify the energy-to-small-doubling-to-structure pipeline.

M4 - Computation: compare random sets, arithmetic progressions, subgroups and unions of progressions via |A+A| and E(A); verify probabilistic convergence examples/counterexamples.

M5 - Exit use: when presented with unexpectedly large additive energy, derive a structurally meaningful conclusion rather than merely estimating the energy.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V14U1: Probability and independence

V14U1P1. [F] Construct examples of independent, pairwise-independent-but-not-jointly-independent, and conditionally independent random variables. Verify each from sigma-algebras/probabilities, not intuition.

V14U1P2. [T] Prove Borel-Cantelli I and II and apply them to independent events with probabilities 1/n^alpha. Determine the almost-sure limsup behavior for all alpha.

V14U1P3. [C] Build examples showing uncorrelated does not imply independent, and independence can be destroyed by conditioning. Explain the structural reason in each example.

Assigned block V14U2: Convergence and laws of large numbers

V14U2P1. [F] Construct sequences separating almost sure, in probability, in L^1, and in distribution convergence. Reuse one sequence in more than one comparison where possible.

V14U2P2. [T] Prove the weak and strong laws for a standard iid setting by two routes available in your text (Chebyshev/subsequence versus martingale/truncation where appropriate).

V14U2P3. [S] Analyze sums of independent but non-identically distributed variables and verify a Kolmogorov-type criterion in a concrete example.

Assigned block V28U1: Sumsets and Ruzsa calculus

V28U1P1. [F] For arithmetic progressions, random-looking sets, subgroups, and geometric progressions in Z/pZ, compute |A+A|, |A-A|, and additive energy. Compare structure signatures.

V28U1P2. [T] Prove Ruzsa triangle inequality, covering lemma, and a selected Plunnecke inequality. Starting from |A+A|<=K|A|, derive explicit polynomial-in-K bounds for mA-nA.

V28U1P3. [S] Derive E(A)>=|A|^4/|A+A| by Cauchy-Schwarz and explain why large energy is the bridge from small sumsets to inverse structure.

Assigned block V28U2: Inverse sumset theory

V28U2P1. [F] Classify exact/small-doubling examples in Z or Z/pZ for small sizes and compare arithmetic progressions with unions/cosets.

V28U2P2. [T] Work through a simplified Balog-Szemeredi-Gowers implication: large additive energy gives a large subset with small doubling. Track losses in all parameters.

V28U2P3. [R] State Freiman's theorem in Z and one finite-field/model-group version precisely. For sample sets, identify the generalized arithmetic progression predicted by the theorem.

Quarter-level integration problems

I1. [S]. For A subset Z/NZ, derive E(A)>=|A|^4/|A+A| and use it to explain why small doubling forces high additive energy; compare numerically with random sets and arithmetic progressions.

I2. [R]. Starting from a simplified high-energy hypothesis, carry out the first extraction step toward Balog-Szemeredi-Gowers and state precisely what additional lemma is needed to obtain a large small-doubling subset.

Full written solutions required for: V14U1P2, V14U2P2, V28U1P2, V28U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a mini-survey 'Small doubling means structure' and include a computational experiment comparing random sets, arithmetic progressions and unions of progressions in Z/NZ.

Exit checkpoint

Prove Borel-Cantelli, one version of the LLN/CLT assumptions carefully, Cauchy-Davenport, Ruzsa triangle inequality, and state BSG/Freiman with a clear dependency graph.

Research bridge

Probability will branch to SDE/SLE/random structures; additive combinatorics will later interact with Roth/Szemeredi, primes, Kakeya and finite-field incidence geometry.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q12. Representation theory I: finite and compact groups

Quarter overview and reading route

Objective: Develop the representation theory of finite and compact groups to the point where characters, induction, Peter-Weyl, SU(2), and representation-theoretic spectral methods are routine.

Prerequisites: Q7 algebra and Q11 probability/additive combinatorics; Q13 Fourier will later deepen the analytic side, but is not required to start.

30-hour allocation: finite-group representations 11h; characters/induction 8h; compact groups/Peter-Weyl 6h; examples and exercises 5h.

Core books and chapters/sections

- Etingof et al., Introduction to Representation Theory: Chs. 2-5, especially Ch. 4 and the finite-group sections of Ch. 5.
- Serre, Linear Representations of Finite Groups: Parts I-II, focusing on characters, induction and examples.
- Fulton-Harris, Representation Theory: early chapters on sl2/SU(2), symmetric groups and tensor constructions, selectively.
- Folland, A Course in Abstract Harmonic Analysis: compact-group/Peter-Weyl chapter for Weeks 8-10.

Lecture notes and primary-paper companions

- MIT Etingof open text provides the main problem-rich route.
- Use the GL2(Fq) sections as a bridge to quasirandomness and Bourgain-Gamburd later.
- For Peter-Weyl, compare finite-group regular representation with the compact-group theorem before reading the general proof.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Representations as modules; Schur lemma	Etingof et al. Chs. 2-3 through Schur lemma, semisimplicity preliminaries.	Compute subrepresentations and intertwiners for cyclic/dihedral examples.
2	Maschke and regular representation	Etingof Ch. 4.1-4.3; Serre, Linear Representations of Finite Groups, opening	Full proof of Maschke + regular representation

Week	Main focus	Reading / lecture notes	Required output
		chapters.	decompositions.
3	Characters and orthogonality	Etingof Ch. 4; Serre character chapters.	Derive row/column orthogonality and compute S3, D8 character tables.
4	Tensor products, duals, character ring	Etingof Ch. 4 and 5 selections.	Decompose 6 tensor products using characters; build representation ring examples.
5	Induction/restriction and Frobenius reciprocity	Etingof Ch. 5 induction sections; Serre induced representation chapters.	Prove Frobenius reciprocity and work Mackey-style finite examples.
6	Symmetric groups and Specht modules	Etingof Ch. 5 symmetric-group sections; Fulton-Harris selections.	Construct Specht modules for n<=4 and verify dimensions.
7	Representations over finite fields / GL2(Fq) preview	Etingof Ch. 5 late sections, especially GL2(Fq).	Classify conjugacy types in GL2(Fq); compute selected induced characters.
8	Compact groups and Haar averaging	Folland Abstract Harmonic Analysis: locally compact/compact group chapters; Peter-Weyl setup.	Prove existence consequences from Haar measure assuming Haar theorem.
9	Peter-Weyl theorem	Folland compact-group chapter; standard Peter-Weyl notes.	Proof architecture + full proof for compact abelian groups / circle.
10	Fourier analysis on compact groups	Folland compact groups; compare finite group Fourier transform.	Write inversion/Plancherel formulas for finite groups and S1.
11	SU(2), highest weights, Clebsch-Gordan	Fulton-Harris early chapters; Etingof sl2 sections.	Derive irreducibles of SU(2) and Clebsch-Gordan formula.
12	Synthesis toward property T and automorphic forms	Review; preview Q18 and Q21.	Closed-book character exam + oral on Peter-Weyl and induction.

Week 1 - Finite-group representations, objects and examples. Reading: Serre/Fulton-Harris opening chapters. Definition/result focus: representation; subrepresentation; irreducible; intertwiner. Work: Decompose explicit cyclic/dihedral representations.

Week 2 - Finite-group representations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Decompose explicit cyclic/dihedral representations. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Maschke and Schur, objects and examples. Reading: Serre character theory chapters. Definition/result focus: Maschke; Schur lemma; complete reducibility. Work: Prove both in full and diagnose characteristic-dividing-order failure.

Week 4 - Maschke and Schur, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove both in full and diagnose characteristic-dividing-order failure. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Characters and orthogonality, objects and examples. Reading: Serre chapters on characters. Definition/result focus: character table; class function; orthogonality. Work: Compute S3/D4 character tables and tensor decompositions.

Week 6 - Characters and orthogonality, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute S3/D4 character tables and tensor decompositions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Induction/restriction, objects and examples. Reading: Serre/Fulton-Harris induction sections. Definition/result focus: induced representation; Frobenius reciprocity; Mackey preview. Work: Compute induced characters in explicit subgroup pairs.

Week 8 - Induction/restriction, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute induced characters in explicit subgroup pairs. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Compact groups, objects and examples. Reading: Folland compact harmonic analysis/Peter-Weyl companion. Definition/result focus: Haar measure; unitary representation; matrix coefficient. Work: Prove Haar averaging consequences and understand Peter-Weyl statement.

Week 10 - Compact groups, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Haar averaging consequences and understand Peter-Weyl statement. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Representation-theoretic synthesis, objects and examples. Reading: selected examples SU(2), finite groups. Definition/result focus: Clebsch-Gordan for SU(2) orientation; regular representation. Work: Derive SU(2) low-weight tensor products and prepare bridge to Lie/unitary theory.

Week 12 - Representation-theoretic synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive SU(2) low-weight tensor products and prepare bridge to Lie/unitary theory. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Group representation; subrepresentation; quotient representation; irreducible representation
- Intertwining operator; equivalence/isomorphism of representations
- Group algebra and module viewpoint
- Character; class function; character inner product
- Regular representation; permutation representation
- Tensor product, dual and representation ring
- Induced and restricted representation
- Frobenius reciprocity
- Specht module/partition (working definition)
- Compact group; Haar probability measure
- Unitary representation of a compact group
- Matrix coefficient
- Peter-Weyl decomposition
- Highest weight for SU(2); symmetric-power model

Key results to know and use

- Schur lemma (full proof)
- Maschke theorem (full proof)
- Sum of squares formula for dimensions of irreducibles
- Character orthogonality relations (full derivation)
- Characters determine finite-dimensional complex representations of finite groups
- Frobenius reciprocity (full proof in finite-group setting)
- Basic structure/classification of irreducible S_n representations via partitions (working knowledge)
- Peter-Weyl theorem (proof architecture; full proof in abelian/finite cases)
- Fourier inversion and Plancherel on finite/compact groups
- Classification of irreducible representations of SU(2) and the Clebsch-Gordan formula

Quarter-specific mastery targets

M1 - Exact language: representation, subrepresentation, irreducible, character, intertwiner, induced/restricted representation, compact group, Haar probability and matrix coefficient.

M2 - Proofs to reproduce: Maschke, Schur, character orthogonality and Frobenius reciprocity for finite groups; the finite-group decomposition of the regular representation.

M3 - Computation: produce full character tables for S3 and D8 and substantial data for S4; decompose tensor/induced representations; classify irreducibles of finite abelian groups.

M4 - Architecture: state Peter-Weyl precisely and explain classification of irreducible SU(2) representations via highest weights/characters; no full Peter-Weyl proof required yet.

M5 - Exit use: translate an operator on a finite Cayley graph or class function into representation-theoretic blocks and recover spectral information.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V18U1: Finite-group representations

V18U1P1. [F] Decompose the regular representations of S3, D4, and a cyclic group into irreducibles using characters. Verify the sum-of-squares dimension formula.

V18U1P2. [T] Prove Maschke's theorem by averaging a projection and identify exactly why characteristic dividing |G| breaks the proof.

V18U1P3. [S] Derive character orthogonality and use it to reconstruct the full character table of S3 or D4 from minimal information.

Assigned block V18U2: Induction/restriction and symmetric groups

V18U2P1. [T] Construct induced representations from a subgroup in a finite example and verify Frobenius reciprocity by explicit Hom-space dimensions.

V18U2P2. [F] Use Young diagrams/tableaux to compute dimensions of selected irreducible S_n representations for n<=5 and verify branching in small cases.

V18U2P3. [S] Compare restriction/induction along S_{n-1}<S_n with permutation representations on cosets; identify the combinatorial meaning of branching.

Assigned block V18U3: Compact groups and Peter-Weyl

V18U3P1. [F] For S^1 and SU(2), list/construct irreducible unitary representations and compute matrix coefficients in low-dimensional cases.

V18U3P2. [T] Prove Peter-Weyl at proof-architecture level, highlighting compactness, Haar measure, and density of matrix coefficients. Verify Fourier series on S^1 as the abelian case.

V18U3P3. [S] Derive Schur orthogonality for compact groups and use it to compute a convolution projection onto one isotypic component.

Assigned block V2U6: Tensor, symmetric and exterior algebra

V2U6P1. [F] Compute V tensor W for V=R^2,W=R^3 in bases, then verify the universal property by factorizing an explicit bilinear map through V tensor W.

V2U6P2. [T] Prove the decompositions V tensor V = Sym^2 V direct-sum Lambda^2 V in characteristic not 2, and calculate dimensions. Explain what fails in characteristic 2.

V2U6P3. [S] Express cross products and oriented volume in R^3 using exterior powers and the Hodge-star intuition. Then derive the Cauchy-Binet formula from minors/exterior powers in one nontrivial case.

Quarter-level integration problems

I1. [S]. Decompose the adjacency operator of a Cayley graph of S3 with a symmetric generating set into irreducible representation blocks and recover all graph eigenvalues from character/representation data.

I2. [T]. Compute Ind_H^G 1 for a nontrivial subgroup H of S4, decompose it by Frobenius reciprocity, and interpret multiplicities as fixed-vector dimensions.

Full written solutions required for: V18U1P2, V18U2P2, V18U3P2, V2U6P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Prepare a character/representation atlas for S3, S4, D8, a finite abelian group, SU(2), and one GL2(Fq). Include decomposition rules, character tables where feasible, and one page on how each example reappears later.

Exit checkpoint

Closed book: prove Maschke, Schur and Frobenius reciprocity; derive character orthogonality; compute a character table; state Peter-Weyl and classify SU(2) irreducibles.

Research bridge

This is the algebraic backbone for Q18 unitary representations, Q21 property (T)/spectral gaps, Q28 quasirandomness and Q35 automorphic representation theory.

Q13. Functional analysis, Fourier analysis, and exponential/character sums

Quarter overview and reading route

Objective: Learn Banach/Hilbert-space methods and Fourier analysis as a single toolkit, then apply them to character and exponential sums.

Prerequisites: Q10-Q11.

30-hour allocation: Functional analysis 10h; Fourier analysis 12h; exponential/character sums 5h; review 3h.

Core books and chapters/sections

- Brezis, Functional Analysis, Sobolev Spaces and PDE, Chs. 1-4.
- Stein-Shakarchi, Fourier Analysis, Chs. 1-5; Grafakos, Classical Fourier Analysis, opening chapters.
- Iwaniec-Kowalski, Analytic Number Theory, chapters/sections on character and exponential sums.
- Vaughan, The Hardy-Littlewood Method, preliminary exponential-sum sections.
- Lidl-Niederreiter, Finite Fields, Chs. 1-3 first pass; later character/exponential-sum chapters.
- Ireland-Rosen, finite-field and character-sum sections as a number-theoretic companion.

Lecture notes and companions

- MIT OCW 18.103 Lectures 6-13; MIT OCW RES.18-015 Lectures 1-24.
- Use finite Fourier analysis notes plus MIT Fourier notes for analytic prerequisites.
- J.S. Milne field/Galois notes may be used for structural review.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Normed/Banach spaces, bounded operators, Hahn-Banach	Brezis Chs. 1-2; MIT 18.103 Hilbert/Banach lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Normed/Banach spaces, bounded operators, Hahn-Banach	Brezis Chs. 1-2; MIT 18.103 Hilbert/Banach lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Hilbert spaces, orthogonality, Riesz representation	Brezis Ch. 4 selectively; MIT 18.103 Hilbert-space units. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Hilbert spaces, orthogonality, Riesz representation	Brezis Ch. 4 selectively; MIT 18.103 Hilbert-space units. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Fourier series, convolution, Parseval/Plancherel	Stein-Shakarchi Fourier Analysis Chs. 1-3; MIT RES.18-015 Lectures 1-9. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Fourier series, convolution, Parseval/Plancherel	Stein-Shakarchi Fourier Analysis Chs. 1-3; MIT RES.18-015 Lectures 1-9. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Fourier transform, Schwartz space and distributions	Stein-Shakarchi Chs. 2-3; MIT RES.18-015 Lectures 10-18. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
8	Prove/use: Fourier transform, Schwartz space and distributions	Stein-Shakarchi Chs. 2-3; MIT RES.18-015 Lectures 10-18. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Finite Fourier transform, multiplicative/additive characters, Gauss sums	Lidl-Niederreiter character-sum sections; Ireland-Rosen corresponding sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Finite Fourier transform, multiplicative/additive characters, Gauss sums	Lidl-Niederreiter character-sum sections; Ireland-Rosen corresponding sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Weyl differencing and first nontrivial exponential-sum bounds	Iwaniec-Kowalski exponential-sum sections; Vaughan preliminary chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Weyl differencing and first nontrivial exponential-sum bounds	Iwaniec-Kowalski exponential-sum sections; Vaughan preliminary chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Banach spaces and bounded operators, objects and examples. Reading: Rudin/Conway functional-analysis opening chapters. Definition/result focus: Banach; bounded operator; dual; uniform boundedness. Work: Prove Banach-Steinhaus and solve operator-norm examples.

Week 2 - Banach spaces and bounded operators, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Banach-Steinhaus and solve operator-norm examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Hahn-Banach/open mapping/closed graph, objects and examples. Reading: functional-analysis core chapters. Definition/result focus: Hahn-Banach; open mapping; closed graph. Work: Prove standard versions and construct use cases.

Week 4 - Hahn-Banach/open mapping/closed graph, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove standard versions and construct use cases. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Hilbert spaces, objects and examples. Reading: functional-analysis Hilbert chapters. Definition/result focus: orthogonal projection; Riesz representation; compact operator. Work: Prove projection/Riesz theorems.

Week 6 - Hilbert spaces, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove projection/Riesz theorems; compute orthogonal expansions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Fourier series and transform, objects and examples. Reading: Stein-Shakarchi Fourier Analysis. Definition/result focus: Fourier coefficients; convolution; Plancherel. Work: Prove Parseval/Plancherel in selected settings and run explicit transforms.

Week 8 - Fourier series and transform, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Parseval/Plancherel in selected settings and run explicit transforms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Poisson summation and uncertainty, objects and examples. Reading: Stein-Shakarchi Fourier chapters. Definition/result focus: Poisson summation; approximate identity; uncertainty principle. Work: Apply Poisson to theta/Gaussian and lattice sums.

Week 10 - Poisson summation and uncertainty, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Apply Poisson to theta/Gaussian and lattice sums. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Exponential sums entry, objects and examples. Reading: Iwaniec-Kowalski/Vaughan introductory sections. Definition/result focus: Weyl sums; differencing; completion. Work: Prove a basic Weyl inequality and test numerically on polynomial phases.

Week 12 - Exponential sums entry, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a basic Weyl inequality and test numerically on polynomial phases. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Banach space
- dual space

- weak convergence
- Hilbert space
- Fourier transform
- Schwartz function
- tempered distribution
- exponential sum
- Kloosterman sum
- Weyl differencing
- van der Corput process
- finite field
- Frobenius
- trace
- norm
- additive character
- multiplicative character
- Gauss sum
- Jacobi sum

Key results to know and use

- Hahn-Banach
- uniform boundedness
- open mapping/closed graph
- Riesz representation
- Parseval/Plancherel
- Fourier inversion
- Poisson summation
- quadratic Gauss sum evaluation
- Weyl inequality
- van der Corput inequality (basic form)
- existence/uniqueness of F_{p^n}
- cyclicity of F_q^*
- trace/norm formulas
- basic Gauss/Jacobi identities
- Weil bounds (later)

Quarter-specific mastery targets

M1 - Exact language: Banach/Hilbert space, bounded functional/operator, weak convergence, compact operator, Fourier transform, convolution, tempered distribution, additive/multiplicative character and Gauss sum.

M2 - Proofs to reproduce: Hahn-Banach in one standard form; Riesz representation in Hilbert space; uniform boundedness/open mapping as assigned; Plancherel; Poisson summation under Schwartz hypotheses; |tau(chi)|=sqrt(p) for quadratic Gauss sums.

M3 - Computation: compute Fourier transforms on R/T/Z_N and finite fields, diagonalize convolution operators, evaluate Gauss/Jacobi sums in examples.

M4 - Failure modes: distinguish norm/weak convergence, bounded/compact operators, pointwise/L2 Fourier convergence; track normalization constants consistently across Fourier settings.

M5 - Exit use: recognize Fourier diagonalization as the common mechanism behind convolution, characters and spectral calculations.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V7U2: Hahn-Banach and duality

V7U2P1. [T] Use Hahn-Banach to extend a functional from a one-dimensional subspace of a normed space, first explicitly and then abstractly. Derive separation of a point from a closed subspace.

V7U2P2. [F] Identify duals of finite-dimensional spaces and of l^1/l^p in the standard ranges covered by your text. Check norm equality for explicit functionals.

V7U2P3. [S] Prove that if x is not in a closed subspace M of a normed space, there is a continuous functional vanishing on M but not on x. Then explain how this becomes a separation tool in convexity and PDE.

Assigned block V7U5: Compact operators and spectral theory

V7U5P1. [F] On l^2, analyze a diagonal compact operator with entries tending to zero: determine spectrum, eigenvalues, norm, and compactness directly.

V7U5P2. [T] Prove the spectral theorem for compact self-adjoint operators at full proof-architecture level, including existence of an extremal eigenvector and orthogonal decomposition.

V7U5P3. [C] Compare compact and bounded operators by giving examples of bounded noncompact operators and compact operators with nonclosed range. Explain which finite-dimensional intuitions survive.

Assigned block V9U2: Fourier transform on R^n

V9U2P1. [T] Compute Fourier transforms of Gaussians, indicators of intervals, exponentials times Gaussians, and derivatives. Verify translation, modulation, scaling, and differentiation identities.

V9U2P2. [T] Prove Fourier inversion first for Schwartz functions and derive Plancherel. Identify where dominated convergence/Fubini enter.

V9U2P3. [S] Derive Poisson summation for a Schwartz function and apply it to the theta function or a lattice Gaussian sum.

Assigned block V27U3: Characters, Gauss and Jacobi sums

V27U3P1. [F] Construct additive and multiplicative characters of F_p and F_{p^n} using trace. Verify orthogonality identities explicitly.

V27U3P2. [T] Prove the quadratic Gauss-sum magnitude and derive relations between Gauss and Jacobi sums. Evaluate a Jacobi sum in a small field.

V27U3P3. [S] Use character orthogonality to count solutions of one polynomial equation over F_q, separating the main term and character-sum error.

Quarter-level integration problems

I1. [S]. Derive Poisson summation for a Schwartz Gaussian and use it to prove the theta transformation formula; identify the functional-analytic and Fourier-normalization steps explicitly.

I2. [S]. Diagonalize convolution by a multiplicative/additive character basis on a finite field and use this to derive the norm of a Gauss-sum operator.

Full written solutions required for: V7U2P2, V7U5P2, V9U2P2, V27U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Build a 'Fourier dictionary' across R, T, Z/NZ and finite fields, proving each transform/inversion/Plancherel identity and computing Gauss sums in examples.

Exit checkpoint

Prove Hahn-Banach in a chosen form, Riesz representation for Hilbert spaces, Plancherel, Poisson summation under suitable hypotheses, and the quadratic Gauss-sum evaluation.

Research bridge

This quarter is a central junction: PDE, harmonic analysis, analytic number theory, additive combinatorics, automorphic forms and spectral theory all reuse these tools.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q14. PDE I, Riemannian manifolds, and algebraic number theory I

Quarter overview and reading route

Objective: Establish first graduate-level PDE and Riemannian geometry while completing the classical global theory of number fields through class groups and units.

Prerequisites: Q6-Q13.

30-hour allocation: PDE 10h; Riemannian geometry 8h; algebraic number theory 9h; review 3h.

Core books and chapters/sections

- Evans, Partial Differential Equations, Chs. 1-2 and 5-7 selectively.
- Brezis, Functional Analysis, Sobolev Spaces and PDE, Sobolev/PDE chapters.
- Lee, Riemannian Manifolds: An Introduction to Curvature, Chs. 1-7 selectively.
- do Carmo, Riemannian Geometry, corresponding classical chapters.
- Marcus, Number Fields, early chapters through ideals, class group and units.
- Neukirch, Algebraic Number Theory, Ch. I; Milne, Algebraic Number Theory notes in parallel.
- Cassels, An Introduction to the Geometry of Numbers: first pass through lattices, convex bodies and Minkowski theorems; return later to successive minima, transference and Diophantine applications when those topics enter.
- Gruber-Lekkerkerker, Geometry of Numbers, introductory chapters as reference.

Lecture notes and companions

- MIT OCW 18.152 Lectures 1-23; MIT 18.156 Lectures 7-30 for graduate continuation.
- MIT 18.755 Chapter I differential-geometry notes are a useful companion.
- MIT OCW 18.785 Number Theory I, Lectures 1-19.
- MIT OCW 18.785 Number Theory I, Lecture 14 (Geometry of Numbers).

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Laplace equation, harmonic functions and maximum principles	Evans Ch. 2 elliptic sections; MIT 18.152 Lectures 1-7. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Laplace equation, harmonic functions and maximum principles	Evans Ch. 2 elliptic sections; MIT 18.152 Lectures 1-7. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Heat/wave equations and energy methods	Evans Ch. 2 parabolic/hyperbolic sections; MIT 18.152 corresponding lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Heat/wave equations and energy methods	Evans Ch. 2 parabolic/hyperbolic sections; MIT 18.152 corresponding lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Riemannian metrics, Levi-Civita connection, geodesics	Lee Riemannian Manifolds Chs. 1-5. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Riemannian metrics, Levi-Civita connection, geodesics	Lee Riemannian Manifolds Chs. 1-5. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Curvature and comparison	Lee Chs. 6-7; do Carmo	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	intuition	corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
8	Prove/use: Curvature and comparison intuition	Lee Chs. 6-7; do Carmo corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Minkowski embedding, class group finiteness	Marcus class-group chapters; MIT 18.785 Lecture 14; Neukirch Ch. I. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Minkowski embedding, class group finiteness	Marcus class-group chapters; MIT 18.785 Lecture 14; Neukirch Ch. I. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Dirichlet unit theorem and analytic class-number preview	Marcus units chapter; MIT 18.785 Lectures 15 and 19. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Dirichlet unit theorem and analytic class-number preview	Marcus units chapter; MIT 18.785 Lectures 15 and 19. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Distributions and weak derivatives, objects and examples. Reading: Evans/Brezis distribution chapters. Definition/result focus: distribution; weak derivative; Sobolev W^{k,p}. Work: Compute weak derivatives and solve approximation exercises.

Week 2 - Distributions and weak derivatives, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute weak derivatives and solve approximation exercises. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Sobolev inequalities, objects and examples. Reading: Evans/Brezis Sobolev sections. Definition/result focus: Poincare; Sobolev embedding; Rellich orientation. Work: Derive scaling exponents.

Week 4 - Sobolev inequalities, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive scaling exponents; prove one embedding and construct endpoint failures. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Elliptic PDE/variational methods, objects and examples. Reading: Evans elliptic chapters. Definition/result focus: weak solution; Lax-Milgram; energy estimates. Work: Solve Poisson variationally and prove uniqueness/energy bounds.

Week 6 - Elliptic PDE/variational methods, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve Poisson variationally and prove uniqueness/energy bounds. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Riemannian geometry foundations, objects and examples. Reading: Lee/do Carmo curvature chapters. Definition/result focus: Levi-Civita; geodesic; curvature; exponential map. Work: Compute connections/curvature on sphere/hyperbolic models.

Week 8 - Riemannian geometry foundations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute connections/curvature on sphere/hyperbolic models. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Geometry of numbers II, objects and examples. Reading: Cassels successive minima/duality/transference sections. Definition/result focus: dual lattice; transference; Mahler compactness orientation. Work: Prove/derive selected transference bounds and compactness criterion.

Week 10 - Geometry of numbers II, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove/derive selected transference bounds and compactness criterion. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Analytic NT I synthesis, objects and examples. Reading: Iwaniec-Kowalski/Davenport selected sieve/exponential-sum sections. Definition/result focus: partial summation; character sums; large-sieve preview. Work: Complete a mixed problem linking lattice geometry, Fourier methods and arithmetic estimates.

Week 12 - Analytic NT I synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete a mixed problem linking lattice geometry, Fourier methods and arithmetic estimates. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- classical/weak solution
- fundamental solution

- maximum principle
- Sobolev space
- energy estimate
- Riemannian metric
- Levi-Civita connection
- geodesic
- curvature tensor
- sectional curvature
- number field
- ring of integers
- norm/trace
- Dedekind domain
- fractional ideal
- class group
- ramification
- discriminant
- regulator
- lattice
- covolume
- fundamental domain
- convex body
- successive minima
- dual lattice

Key results to know and use

- mean-value/max principle
- heat kernel
- wave energy identity
- Sobolev embedding
- Lax-Milgram (later)
- Calderon-Zygmund estimate (later)
- Levi-Civita existence/uniqueness
- Hopf-Rinow (statement)
- basic curvature identities
- unique factorization of ideals
- Dedekind-Kummer
- Minkowski class-group finiteness
- Dirichlet unit theorem
- analytic class number formula (statement)
- Blichfeldt lemma
- Minkowski convex body theorem
- Minkowski second theorem
- Dirichlet simultaneous approximation via lattices
- Mahler compactness (later)

Quarter-specific mastery targets

M1 - Exact language: distribution/weak derivative, Sobolev space, weak solution, elliptic bilinear form, Riemannian metric, Levi-Civita connection, curvature, ideal class group, unit rank and Minkowski embedding.

M2 - Proofs to reproduce: a model energy estimate/Lax-Milgram application; basic Sobolev embedding in the range used; existence/uniqueness of Levi-Civita connection; class-group finiteness via Minkowski; Dirichlet unit theorem proof architecture with the lattice step explicit.

M3 - Computation: solve one weak elliptic problem, compute Christoffel symbols/curvature for standard metrics, and compute Minkowski bounds/class groups/units for small fields.

M4 - Architecture: distinguish PDE regularity input from functional-analysis existence input; distinguish algebraic ideal structure from geometry-of-numbers finiteness input.

M5 - Exit use: explain two different appearances of coercivity/compactness - in PDE and in arithmetic lattices - without conflating the underlying structures.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V10U2: Sobolev spaces

V10U2P1. [F] Determine H^1 membership for x^alpha, |x|, and piecewise functions on intervals. Compute weak derivatives explicitly.

V10U2P2. [T] Prove Poincare inequality on an interval and then a bounded domain in a standard formulation. Use scaling to predict how the constant transforms.

V10U2P3. [S] Derive one Sobolev embedding by scaling and a standard proof. Construct a concentrating sequence showing an endpoint or supercritical embedding fails.

Assigned block V10U3: Elliptic weak theory

V10U3P1. [T] Set up the weak formulation of -Delta u=f with Dirichlet data and verify boundedness/coercivity of the bilinear form. Apply Lax-Milgram to obtain existence/uniqueness.

V10U3P2. [F] Solve the same Poisson problem on an interval both classically and weakly, and verify the solutions agree.

V10U3P3. [S] Derive an energy estimate and use it to prove continuous dependence on f. Identify the exact norm in which the estimate closes.

Assigned block V13U1: Metrics, connections and geodesics

V13U1P1. [F] Compute Levi-Civita connections and geodesic equations for Euclidean space, the sphere in standard coordinates, and a conformally flat metric in dimension 2. Check metric compatibility and zero torsion directly.

V13U1P2. [T] Derive the first variation of energy/length and show geodesics satisfy the Euler-Lagrange equation. Explain parametrization issues and why constant-speed geodesics are natural.

V13U1P3. [S] Use the exponential map on S^2 to identify conjugate/cut phenomena explicitly and compare local normal coordinates with global failure of injectivity.

Assigned block V24U3: Minkowski theory, units and class finiteness

V24U3P1. [T] Derive Minkowski's embedding and covolume formula for O_K in a number field. Verify it numerically for a quadratic field.

V24U3P2. [F] Use Minkowski's bound to prove finiteness of the class group and execute the proof in one explicit field far enough to enumerate ideal-class representatives.

V24U3P3. [S] Compute units in Q(sqrt(2)) or Q(sqrt(5)) via Pell equations and relate the logarithmic embedding to Dirichlet's unit theorem lattice.

Quarter-level integration problems

I1. [S]. Prove class-group finiteness for one imaginary quadratic field using Minkowski, compute the bound numerically, and then determine the class group by checking only the necessary ideals.

I2. [S]. For the Laplace-Beltrami operator on a simple compact surface/metric, derive a weak formulation and energy identity; compare geometric curvature data with analytic coercivity data without conflating them.

Full written solutions required for: V10U2P2, V10U3P2, V13U1P2, V24U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write two linked notes: an energy-method solution of a model PDE, and a geometry-of-numbers proof of class-group finiteness plus Dirichlet's unit theorem architecture.

Exit checkpoint

Derive maximum/energy estimates, compute Levi-Civita connections and curvature in standard examples, and prove class-group finiteness and the unit theorem at textbook level.

Research bridge

PDE/geometry will feed Ricci flow and mathematical physics; algebraic number theory now connects directly to local fields, heights and class field theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q15. Harmonic analysis I and analytic number theory II

Quarter overview and reading route

Objective: Learn maximal functions, singular integrals, interpolation and Littlewood-Paley theory, while advancing analytic number theory toward large-sieve and mean-value techniques.

Prerequisites: Q13-Q14.

30-hour allocation: Harmonic analysis 16h; analytic number theory 10h; review 4h.

Core books and chapters/sections

- Stein, Singular Integrals and Differentiability Properties of Functions, Chs. I-III selectively.
- Grafakos, Classical Fourier Analysis, chapters on maximal operators, singular integrals, interpolation, Littlewood-Paley.
- Davenport, Multiplicative Number Theory, chapters on Dirichlet characters, L-functions, PNT and primes in AP.
- Montgomery-Vaughan, Multiplicative Number Theory I; Iwaniec-Kowalski, corresponding analytic chapters as references.
- Iwaniec-Kowalski, Analytic Number Theory, chapters/sections on character and exponential sums.
- Vaughan, The Hardy-Littlewood Method, preliminary exponential-sum sections.

Lecture notes and companions

- MIT RES.18-015 Lectures 20-24; MIT 18.156 Lectures 22-30.
- MIT OCW 18.785 Number Theory I, Lectures 16-19.
- Use finite Fourier analysis notes plus MIT Fourier notes for analytic prerequisites.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Hardy-Littlewood maximal operator and differentiation	Grafakos maximal-function chapters; Stein Singular Integrals Ch. I. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Hardy-Littlewood maximal operator and differentiation	Grafakos maximal-function chapters; Stein Singular Integrals Ch. I. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Interpolation: Riesz-Thorin and Marcinkiewicz	Grafakos interpolation chapters; MIT RES.18-015 interpolation lectures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Interpolation: Riesz-Thorin and Marcinkiewicz	Grafakos interpolation chapters; MIT RES.18-015 interpolation lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Hilbert/Riesz transforms and Calderon-Zygmund kernels	Stein Chs. II-III; MIT RES.18-015 Lectures 20-24. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Hilbert/Riesz transforms and Calderon-Zygmund kernels	Stein Chs. II-III; MIT RES.18-015 Lectures 20-24. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Littlewood-Paley decompositions and square functions	Grafakos Littlewood-Paley chapters; MIT 18.156 LP lectures. First pass: definitions, examples,	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
		and 2-3 basic exercises.	
8	Prove/use: Littlewood-Paley decompositions and square functions	Grafakos Littlewood-Paley chapters; MIT 18.156 LP lectures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Large sieve and mean values of Dirichlet polynomials	Iwaniec-Kowalski large-sieve sections; Montgomery-Vaughan corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Large sieve and mean values of Dirichlet polynomials	Iwaniec-Kowalski large-sieve sections; Montgomery-Vaughan corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Zero density, character sums and analytic-number-theory synthesis	Iwaniec-Kowalski selected chapters; reconstruct key estimates rather than survey-reading only. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Zero density, character sums and analytic-number-theory synthesis	Iwaniec-Kowalski selected chapters; reconstruct key estimates rather than survey-reading only. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Calderon-Zygmund foundations, objects and examples. Reading: Stein Singular Integrals/Harmonic Analysis opening CZ sections. Definition/result focus: CZ kernel; maximal function; weak type. Work: Prove Hardy-Littlewood maximal weak (1,1) and interpolation consequences.

Week 2 - Calderon-Zygmund foundations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Hardy-Littlewood maximal weak (1,1) and interpolation consequences. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Singular integrals, objects and examples. Reading: Stein CZ chapters. Definition/result focus: Hilbert/Riesz transforms; Lp boundedness architecture. Work: Work through one Calderon-Zygmund decomposition proof.

Week 4 - Singular integrals, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work through one Calderon-Zygmund decomposition proof. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Littlewood-Paley, objects and examples. Reading: Stein harmonic analysis LP chapters. Definition/result focus: dyadic decomposition; square function. Work: Prove basic L2 square-function estimates and use frequency localization.

Week 6 - Littlewood-Paley, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove basic L2 square-function estimates and use frequency localization. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Large sieve and character sums, objects and examples. Reading: Iwaniec-Kowalski large-sieve sections. Definition/result focus: large sieve inequality; duality principle. Work: Derive a large-sieve form and apply it to characters.

Week 8 - Large sieve and character sums, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive a large-sieve form and apply it to characters. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Sieve methods, objects and examples. Reading: Iwaniec-Kowalski sieve chapters. Definition/result focus: upper-bound/Selberg sieve; sieve dimension; parity issue. Work: Build a toy Selberg sieve and identify parity obstruction.

Week 10 - Sieve methods, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build a toy Selberg sieve and identify parity obstruction. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Zero density/Bombieri-Vinogradov orientation, objects and examples. Reading: Iwaniec-Kowalski distribution of primes sections. Definition/result focus: level of distribution; Bombieri-Vinogradov exact statement. Work: Map the proof architecture and solve a simplified average-over-moduli estimate.

Week 12 - Zero density/Bombieri-Vinogradov orientation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Map the proof architecture and solve a simplified average-over-moduli estimate. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- maximal operator
- weak type
- CZ kernel
- principal value
- Littlewood-Paley projection
- Dirichlet series
- Euler product
- Dirichlet character
- Dirichlet L-function
- von Mangoldt function
- zero-free region
- exponential sum
- Kloosterman sum
- Weyl differencing
- van der Corput process

Key results to know and use

- Hardy-Littlewood maximal theorem
- Lebesgue differentiation
- Hilbert transform L2 boundedness
- CZ weak (1,1)
- Riesz-Thorin
- Marcinkiewicz interpolation
- Euler product for zeta
- PNT
- Dirichlet theorem on primes in AP
- functional equation for primitive Dirichlet L-functions (architecture)
- PNT in AP (statement)
- quadratic Gauss sum evaluation
- Weyl inequality
- van der Corput inequality (basic form)

Quarter-specific mastery targets

M1 - Exact language: Hardy-Littlewood maximal operator, weak type, interpolation, Calderon-Zygmund kernel/operator, Littlewood-Paley square function, large-sieve constant and level of distribution.

M2 - Proofs to reproduce: weak (1,1) maximal inequality; Riesz-Thorin; one model CZ L2/weak-type argument; LP square-function estimate in a tractable setting; large sieve in one standard formulation.

M3 - Applications: use dyadic decomposition and maximal estimates in concrete Fourier problems; derive a large-sieve bound for characters or well-spaced frequencies.

M4 - Architecture: state Bombieri-Vinogradov precisely and identify where large sieve replaces GRH-strength pointwise control on average; know the role of LP/CZ tools in later time-frequency/restriction work.

M5 - Exit use: decide whether a new operator problem calls for maximal, interpolation, CZ, LP or orthogonality machinery, and justify the choice by scale/kernel structure.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V9U3: Maximal functions and differentiation

V9U3P1. [F] Compute the Hardy-Littlewood maximal function for simple indicators in R and estimate the level sets explicitly.

V9U3P2. [T] Prove the weak-(1,1) maximal inequality using a covering argument and derive the Lebesgue differentiation theorem. Track constants and the role of Vitali-type selection.

V9U3P3. [C] Construct an L^1 function whose maximal function is not integrable, showing weak-(1,1) is generally the right endpoint statement.

Assigned block V9U4: Interpolation and singular integrals

V9U4P1. [T] Prove Riesz-Thorin interpolation for a finite-dimensional/simple-function model before reading the full complex interpolation proof. Apply it to interpolate operator norms between two endpoints.

V9U4P2. [T] Verify the Hilbert transform kernel satisfies cancellation and calculate its Fourier multiplier. Prove L^2 boundedness directly by Plancherel.

V9U4P3. [S] Work through a Calderon-Zygmund decomposition of a concrete L^1 function at height lambda and use it to outline weak-(1,1) boundedness of a singular integral.

Assigned block V9U5: Littlewood-Paley and multipliers

V9U5P1. [F] Construct a dyadic Littlewood-Paley partition and compute frequency pieces of a function with separated Fourier support. Verify square-function orthogonality in L^2.

V9U5P2. [T] Prove a basic Mikhlin multiplier theorem in a one-dimensional or radial simplified setting, identifying derivative conditions and kernel decay.

V9U5P3. [S] Use Littlewood-Paley decomposition to derive a Sobolev norm equivalence and explain why frequency localization is useful in nonlinear PDE.

Assigned block V23U2: Large sieve and distribution of primes

V23U2P1. [T] Prove the large sieve inequality in a standard finite form or reconstruct its duality proof. Apply it to a family of character/exponential sums.

V23U2P2. [S] Starting from a large-sieve/Bombieri-Vinogradov statement, derive a nontrivial average result for primes in arithmetic progressions over q<=Q. Track the level-of-distribution exponent.

V23U2P3. [R] Compare Brun-Titchmarsh, Siegel-Walfisz, Bombieri-Vinogradov, and Elliott-Halberstam by exact ranges and averaging. Mark theorem versus conjecture.

Quarter-level integration problems

I1. [S]. Starting from a well-spaced set of frequencies, derive a large-sieve inequality and rewrite it as an operator-norm estimate; compare its logic with an L2 singular-integral estimate.

I2. [C]. Construct a kernel violating one Calderon-Zygmund hypothesis and show exactly which part of the standard proof breaks; do the analogous hypothesis audit for a large-sieve spacing assumption.

Full written solutions required for: V9U3P2, V9U4P2, V9U5P2, V23U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Create a theorem map showing how maximal, interpolation, CZ and LP tools enter later Carleson/restriction/decoupling arguments; pair it with a worked large-sieve application.

Exit checkpoint

Prove weak (1,1) maximal inequality, Riesz-Thorin, a model CZ estimate, LP square-function estimate in a tractable setting, and the large sieve in one standard formulation.

Research bridge

This is the analytic toolkit required for Carleson's theorem, restriction/Kakeya, decoupling and spectral analytic number theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q16. Mathematical logic II: axiomatic set theory, constructibility, forcing, and CH

Quarter overview and reading route

Objective: Understand the formal set-theoretic foundations behind the continuum hypothesis: ZFC, ordinals/cardinals, constructibility, forcing, and the precise meaning of independence.

Prerequisites: Q4 logic I plus mature proof technique from Q5-Q15.

30-hour allocation: axiomatic set theory 7h; ordinals/cardinals 5h; constructibility 6h; forcing/CH 9h; exercises/exposition 3h.

Core books and chapters/sections

- Jech, Set Theory (3rd millennium ed.): Chs. 1-6 selectively; Ch. 12 (models), Ch. 13 (constructible sets), Ch. 14 (forcing), and selected Chs. 15-16.
- Kunen, Set Theory / independence-proof material, selected as a second source if Jech is too compressed.

- Enderton, Elements of Set Theory, for a gentler route through ordinals/cardinals before Jech.

Lecture notes and primary-paper companions

- Easwaran, A Cheerful Introduction to Forcing and the Continuum Hypothesis, as the first forcing exposition.
- Feferman's Bernays/Hilbert/CH lectures for historical and foundational context after the technical proof architecture is understood.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	ZF/ZFC axioms and cumulative hierarchy	Jech Chs. 1-2; review formal set theory from Q4.	Axiom sheet; build V_alpha examples and transfinite-recursion proofs.
2	Ordinals, transfinite induction and recursion	Jech Ch. 2.	Full proofs of transfinite induction/recursion; ordinal arithmetic exercises.
3	Cardinals, cofinality, AC and Zorn	Jech Chs. 3-6 selectively.	Equivalence map for major forms of AC; cardinal arithmetic exercises.
4	Models of set theory and relative consistency	Jech Ch. 12.	Write 4-page note on inner/transitive models and absoluteness.
5	Constructible universe L	Jech Ch. 13.	Define L_alpha carefully and prove key closure/transitivity lemmas.
6	Godel relative consistency of AC/GCH	Jech Ch. 13; Feferman CH historical companion.	Proof architecture of L models ZFC+GCH; reconstruct definable well-order idea.
7	Partial orders, dense sets, generic filters	Jech Ch. 14 opening; Easwaran forcing notes.	Construct generics over countable transitive models in toy examples.
8	Names, forcing relation, forcing theorem	Jech Ch. 14; Easwaran middle sections.	Prove truth lemma in a simplified forcing setup; build valuation of names.
9	Cohen forcing and adding a real	Jech Chs. 14-15.	Show Cohen generic is new; prove ccc for basic Cohen poset.
10	Independence of CH	Jech Ch. 14 CH consistency proof; Easwaran.	Write complete architecture of Cohen independence proof with preservation steps.
11	Applications: MA, forcing axioms, descriptive-set preview	Jech Chs. 15-16 selectively.	State MA and derive two basic consequences; distinguish CH/GCH/MA.
12	Synthesis and foundations perspective	Review Godel L versus Cohen forcing.	Oral: Con(ZFC)->Con(ZFC+CH) and Con(ZFC)->Con(ZFC+not CH), with caveats.

Week 1 - ZF/ZFC and ordinals, objects and examples. Reading: Jech/Kunen introductory set-theory sections. Definition/result focus: ZF axioms; ordinal; transfinite recursion. Work: Construct ordinal arithmetic and prove transfinite induction.

Week 2 - ZF/ZFC and ordinals, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct ordinal arithmetic and prove transfinite induction. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Cardinals and choice, objects and examples. Reading: set-theory chapters on cardinals/AC. Definition/result focus: cardinal comparison; AC/Zorn/well-ordering equivalences. Work: Prove selected equivalences and cardinal arithmetic facts.

Week 4 - Cardinals and choice, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove selected equivalences and cardinal arithmetic facts. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Constructible universe, objects and examples. Reading: Jech/Kunen sections on L. Definition/result focus: definability; L_alpha; constructibility. Work: Compute early L stages and map the proof that L satisfies ZFC.

Week 6 - Constructible universe, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute early L stages and map the proof that L satisfies ZFC. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - CH in L, objects and examples. Reading: constructibility/GCH sections. Definition/result focus: Godel relative consistency of AC/GCH architecture. Work: State condensation-related inputs and write exact consistency implication.

Week 8 - CH in L, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State condensation-related inputs and write exact consistency implication. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Forcing basics, objects and examples. Reading: forcing notes/Jech introductory forcing. Definition/result focus: poset; dense set; generic filter; name. Work: Work with Cohen forcing finite partial functions and dense sets.

Week 10 - Forcing basics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work with Cohen forcing finite partial functions and dense sets. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Cohen independence of CH, objects and examples. Reading: forcing theorem/CH sections. Definition/result focus: forcing relation; generic extension; CH independence architecture. Work: Trace M -> M[G], distinguish relative consistency from truth, and sit oral audit.

Week 12 - Cohen independence of CH, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Trace M -> M[G], distinguish relative consistency from truth, and sit oral audit. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- ZF and ZFC axioms
- Ordinal, transitive set, successor/limit ordinal
- Transfinite induction and recursion
- Cardinal, aleph, cofinality, regular/singular cardinal
- Axiom of Choice; Zorn lemma; well-ordering theorem
- Cumulative hierarchy V_alpha
- Inner model and transitive model
- Definability and absoluteness
- Constructible hierarchy L_alpha and constructible universe L
- Forcing poset; compatibility; dense set
- Filter and M-generic filter
- P-name; valuation of a name
- Forcing relation p forces phi
- Cohen forcing
- Countable chain condition (ccc)
- Continuum hypothesis (CH) and generalized CH (GCH)
- Martin axiom (orientation only)

Key results to know and use

- Transfinite induction and recursion theorems (full proof)
- Basic ordinal/cardinal arithmetic and Hartogs-style arguments
- Equivalences among standard formulations of Choice (selected full proofs)
- Construction and basic properties of L
- Godel theorem: relative consistency of AC and GCH via L (proof architecture)
- Rasiowa-Sikorski lemma for countable families of dense sets
- Forcing theorem/truth lemma (proof architecture)
- Cohen forcing adds a new real

- ccc and preservation of cardinals in the basic CH-independence construction (working proof)
- Cohen independence theorem for CH: relative consistency of CH and of not-CH (architecture with exact metamathematical interpretation)
- Basic Martin axiom consequences (orientation)

Quarter-specific mastery targets

M1 - Exact language: ZF/ZFC, ordinal, cardinal, transfinite recursion, constructible hierarchy L_alpha, inner model, forcing poset, dense set, generic filter, forcing relation and Cohen real.

M2 - Proofs to reproduce: basic transfinite recursion/cardinal arguments used; construction/properties of early L_alpha; elementary forcing calculations for Cohen forcing and the generic real.

M3 - Architecture-only but exact: Godel's relative consistency of CH via L and Cohen's relative consistency of not-CH via forcing; state metatheoretic assumptions and do not turn relative consistency into absolute consistency.

M4 - Conceptual distinctions: internal vs external model language, countable transitive-model heuristic vs formal forcing theorem, and CH vs GCH.

M5 - Exit use: trace a single sentence through M -> M[G], identifying what is preserved, what is added and where the forcing theorem is invoked.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V5U5: ZF/ZFC, ordinals and cardinals

V5U5P1. [F] Construct the first several finite and transfinite ordinals, compute simple ordinal sums/products, and exhibit noncommutativity such as 1+omega != omega+1.

V5U5P2. [T] Prove Cantor's theorem |X|<|P(X)| and use Hartogs' theorem/choice assumptions to distinguish cardinal comparability issues. Work several cardinal-arithmetic examples under and without CH assumptions.

V5U5P3. [S] Prove transfinite recursion in a simplified setting and use it to define the cumulative hierarchy V_alpha for initial stages. Explain how Replacement enters the full ZF construction.

Assigned block V5U6: Constructibility, forcing and CH

V5U6P1. [F] Compute the first stages L_0,L_1,... of the constructible hierarchy and contrast 'all subsets' with 'definable subsets' at successor stages.

V5U6P2. [T] For Cohen forcing Fn(omega,2,<omega), decide compatibility of explicit conditions, construct dense sets ensuring a total binary sequence, and show how a generic filter determines a new real.

V5U6P3. [R] Write a two-column proof architecture for CH independence: Godel's L gives relative consistency of CH; Cohen forcing gives relative consistency of not-CH. For each column state the model transformation and exactly what metatheoretic assumption is being made.

Assigned block V1U4: Cardinality and countability

V1U4P1. [F] Construct explicit enumerations of Z, Q, the set of finite binary strings, and Z^2. Explain why the same strategy fails for infinite binary sequences.

V1U4P2. [T] Reproduce Cantor's diagonal proof that {0,1}^N is uncountable, then prove P(N) is uncountable by a direct diagonal argument. State exactly where self-reference enters.

V1U4P3. [S] Prove that the set of algebraic numbers is countable and hence transcendental numbers exist. Strengthen the conclusion by showing that almost every real number is transcendental once measure theory becomes available; note clearly what is not yet proved at this stage.

Assigned block V5U2: Completeness and compactness

V5U2P1. [T] Use compactness to construct a nonstandard model of arithmetic containing an element larger than every standard numeral. Write the expanded language and finite satisfiability argument explicitly.

V5U2P2. [S] Use compactness to prove the graph-theoretic statement: if every finite subgraph of a graph is k-colorable, then the entire graph is k-colorable. Encode the coloring constraints as first-order/propositional clauses.

V5U2P3. [C] Compare completeness and compactness: derive compactness from completeness plus finite proof length, then explain why semantic completeness is not the same as categorical uniqueness of models.

Quarter-level integration problems

I1. [S]. In Cohen forcing with finite partial functions omega->2, write five explicit dense sets, prove density, and show how a filter meeting them determines a total new real.

I2. [R]. Produce a two-column proof architecture: L satisfies CH versus a forcing extension satisfies not-CH. Every row must name a lemma/theorem and state whether it is proved this quarter or used as a black box.

Full written solutions required for: V5U5P2, V5U6P2, V1U4P2, V5U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 20-page “CH independence dossier”: formal statement of CH, the role of L, the role of Cohen forcing, the two relative-consistency directions, a glossary of models/forcing terminology, and a dependency diagram. Do not claim an absolute consistency proof of ZFC.

Exit checkpoint

Explain, without notes, why Godel gives the CH-consistent side and Cohen the not-CH side; define L, generic filter, forcing relation and Cohen real; state exactly what assumptions are being made in the relative-consistency statements.

Research bridge

The forcing/model-theory language is useful beyond CH: descriptive set theory, ergodic theory, set-theoretic topology, and the model-theoretic aspects of Diophantine definability in Q30.

Q17. Lie/algebraic groups, arithmetic lattices, and geometry of numbers II

Quarter overview and reading route

Objective: Develop Lie group/algebra language, Haar measure and homogeneous spaces; connect arithmetic groups and lattices back to geometry of numbers.

Prerequisites: Q8 topology/algebra, Q14 geometry, Q6 geometry of numbers.

30-hour allocation: Lie/algebraic groups 16h; arithmetic groups/lattices 8h; geometry of numbers 3h; review 3h.

Core books and chapters/sections

- Hall, Lie Groups, Lie Algebras, and Representations, Chs. 1-8 selectively.
- Witte Morris, Introduction to Arithmetic Groups, early Lie/Haar/lattice chapters; Milne, Algebraic Groups notes.
- Cassels, An Introduction to the Geometry of Numbers: first pass through lattices, convex bodies and Minkowski theorems; return later to successive minima, transference and Diophantine applications when those topics enter.
- Gruber-Lekkerkerker, Geometry of Numbers, introductory chapters as reference.
- Marcus, Number Fields, early chapters through ideals, class group and units.
- Neukirch, Algebraic Number Theory, Ch. I; Milne, Algebraic Number Theory notes in parallel.

Lecture notes and companions

- MIT OCW 18.755 Introduction to Lie Groups, Chapters I-II.
- MIT OCW 18.785 Number Theory I, Lecture 14 (Geometry of Numbers).
- MIT OCW 18.785 Number Theory I, Lectures 1-19.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Manifolds, Lie groups, Lie algebras, exponential map	Hall Chs. 1-4; MIT 18.755 Sections 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Manifolds, Lie groups, Lie algebras, exponential map	Hall Chs. 1-4; MIT 18.755 Sections 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Homogeneous spaces and Lie group actions	MIT 18.755 Sections 4 onward; Hall corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Homogeneous spaces and Lie group actions	MIT 18.755 Sections 4 onward; Hall corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Haar measure, modular functions and unimodularity	Witte Morris introductory chapters; MIT 18.755 invariant-measure sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Haar measure, modular functions and unimodularity	Witte Morris introductory chapters; MIT 18.755 invariant-measure sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Semisimple groups, SL2(R), SO(2,1), Iwasawa decomposition	MIT 18.755 semisimple/Iwasawa sections; Hall semisimple chapters.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
		First pass: definitions, examples, and 2-3 basic exercises.	
8	Prove/use: Semisimple groups, SL2(R), SO(2,1), Iwasawa decomposition	MIT 18.755 semisimple/Iwasawa sections; Hall semisimple chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Arithmetic subgroups, lattices and Borel-Harish-Chandra preview	Witte Morris arithmetic-group chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Arithmetic subgroups, lattices and Borel-Harish-Chandra preview	Witte Morris arithmetic-group chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Successive minima, reduction and homogeneous-space dictionary	Cassels later chapters; identify unimodular lattices with SL_n(R)/SL_n(Z). First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Successive minima, reduction and homogeneous-space dictionary	Cassels later chapters; identify unimodular lattices with SL_n(R)/SL_n(Z). Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Lie groups and Lie algebras, objects and examples. Reading: Hall/Knapp/MIT 18.755 opening sections. Definition/result focus: Lie group/algebra; exponential; adjoint representation. Work: Compute Lie algebras/exponentials for matrix groups.

Week 2 - Lie groups and Lie algebras, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Lie algebras/exponentials for matrix groups. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Semisimple structure, objects and examples. Reading: Knapp/Hall semisimple sections. Definition/result focus: Killing form; root system; Cartan decomposition orientation. Work: Compute sl2 root data and representations.

Week 4 - Semisimple structure, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute sl2 root data and representations. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Algebraic groups, objects and examples. Reading: Borel/Springer introductory sections. Definition/result focus: Zariski closure; torus; unipotent; reductive group. Work: Analyze SL2, SO(q), Borel/unipotent subgroups algebraically.

Week 6 - Algebraic groups, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Analyze SL2, SO(q), Borel/unipotent subgroups algebraically. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Haar measure and lattices, objects and examples. Reading: Raghunathan/Morris arithmetic groups sections. Definition/result focus: Haar measure; lattice; unimodularity. Work: Prove quotient-measure criteria and finite-covolume examples.

Week 8 - Haar measure and lattices, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove quotient-measure criteria and finite-covolume examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Borel density, objects and examples. Reading: Morris/Raghunathan Borel density chapters. Definition/result focus: Borel density exact statement; Zariski density of lattices. Work: Reconstruct proof architecture using invariant measures/representation theory.

Week 10 - Borel density, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reconstruct proof architecture using invariant measures/representation theory. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Borel-Harish-Chandra and arithmetic lattices, objects and examples. Reading: Morris arithmetic groups chapters. Definition/result focus: arithmetic subgroup; reduction theory; BHC theorem. Work: Explain why SO(Q,Z) is a lattice under hypotheses.

Week 12 - Borel-Harish-Chandra and arithmetic lattices, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Explain why SO(Q,Z) is a lattice under hypotheses; connect to quadratic forms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Lie group
- Lie algebra
- exponential map
- adjoint representation

- Haar measure
- homogeneous space
- unimodular
- unipotent
- lattice
- covolume
- fundamental domain
- convex body
- successive minima
- dual lattice
- number field
- ring of integers
- norm/trace
- Dedekind domain
- fractional ideal
- class group
- ramification
- discriminant
- regulator

Key results to know and use

- Iwasawa decomposition for SL2
- existence/uniqueness of Haar measure
- invariant-measure criterion on G/H
- Mahler compactness (with lattice theory)
- Blichfeldt lemma
- Minkowski convex body theorem
- Minkowski second theorem
- Dirichlet simultaneous approximation via lattices
- Mahler compactness (later)
- unique factorization of ideals
- Dedekind-Kummer
- Minkowski class-group finiteness
- Dirichlet unit theorem
- analytic class number formula (statement)

Quarter-specific mastery targets

M1 - Exact language: Lie group/algebra, exponential map, algebraic group, lattice, arithmetic subgroup, Haar measure, modular function, homogeneous space G/H and invariant measure criterion.

M2 - Proofs to reproduce: tangent Lie algebra construction and basic exponential facts; Haar uniqueness at theorem level; unimodularity of connected reductive groups via Ad/determinant; existence criterion for G-invariant measure on G/H in the standard modular-function form.

M3 - Computation: identify SL2(R)/N and SL2(R)/P explicitly; compute stabilizers/orbits and modular functions in standard solvable examples; identify SL_n(R)/SL_n(Z) with unimodular lattices.

M4 - Architecture: state Borel-Harish-Chandra and Borel density accurately, but reserve their detailed use for Q31; know which hypotheses are arithmetic/reductive/lattice hypotheses.

M5 - Exit use: convert a geometric/arithmetic object into a homogeneous-space statement and verify the measure/lattice hypotheses rather than assuming them.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V18U4: Lie groups and Lie algebras

V18U4P1. [F] Compute Lie algebras of GL_n, SL_n, SO_n, SU_n and verify brackets from matrix commutators. Calculate exponential maps for selected one-parameter subgroups.

V18U4P2. [T] Derive the adjoint representation and compute ad_X for sl_2 basis elements. Verify the sl_2 commutation relations and Killing form.

V18U4P3. [S] Prove the Lie correspondence between Lie algebra homomorphisms and local/simply-connected group homomorphisms at theorem-architecture level; work one explicit integration.

Assigned block V20U2: Lie groups, lattices and arithmeticity background

V20U2P1. [F] Show SL_2(R)/SL_2(Z) parametrizes unimodular lattices in R^2 and identify the stabilizer of Z^2. Compute diagonal and unipotent actions on lattice bases.

V20U2P2. [T] Prove unimodularity of a reductive Lie group in a concrete adjoint-determinant formulation and derive the criterion for a G-invariant measure on G/H in examples.

V20U2P3. [S] Work through Borel density/Borel-Harish-Chandra in one arithmetic lattice example at theorem-application level: identify algebraic group, lattice, and conclusion.

Assigned block V25U1: Lattices and convex bodies

V25U1P1. [F] For lattices generated by explicit 2x2 and 3x3 matrices, compute covolume, shortest vectors, fundamental parallelepipeds, and dual lattices. Compare Euclidean length with covolume under shearing.

V25U1P2. [T] Prove Blichfeldt's lemma and deduce Minkowski's convex body theorem. Apply it to prove existence of a nonzero lattice point in a carefully chosen ellipse/box with an explicit numerical bound.

V25U1P3. [S] Use Minkowski to prove Dirichlet's simultaneous approximation theorem by constructing the appropriate lattice/convex body. Make the dictionary between lattice coordinates and approximation inequalities explicit.

Assigned block V24U3: Minkowski theory, units and class finiteness

V24U3P1. [T] Derive Minkowski's embedding and covolume formula for O_K in a number field. Verify it numerically for a quadratic field.

V24U3P2. [F] Use Minkowski's bound to prove finiteness of the class group and execute the proof in one explicit field far enough to enumerate ideal-class representatives.

V24U3P3. [S] Compute units in Q(sqrt(2)) or Q(sqrt(5)) via Pell equations and relate the logarithmic embedding to Dirichlet's unit theorem lattice.

Quarter-level integration problems

I1. [S]. Prove directly that the stabilizer of a nonzero vector in the natural SL2(R) action is conjugate to N, then derive the identification SL2(R)/N with R^2 minus {0} and check topology carefully.

I2. [S]. Starting from a quadratic form of signature (2,1), identify SO(Q) as an algebraic subgroup, write the corresponding orbit/lattice problem, and explain what arithmeticity data would be needed for a finite-volume quotient.

Full written solutions required for: V18U4P2, V20U2P2, V25U1P2, V24U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a rigorous exposition of SL2(R)/N and SL2(R)/P as homogeneous spaces, and of unimodular lattices as SL_n(R)/SL_n(Z), including invariant-measure conditions.

Exit checkpoint

Prove basic Lie subgroup/homogeneous-space statements, existence/uniqueness of Haar measure at theorem level, unimodularity for reductive groups, and the lattice-homogeneous-space identification.

Research bridge

This is the entrance to Borel density, Ratner, EKL, Zimmer and the dynamical form of Diophantine approximation.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q18. Representation theory II: unitary and Lie-group representations

Quarter overview and reading route

Objective: Move from finite/compact representation theory to unitary representations of locally compact and Lie groups, with enough SL(2,R) and spectral language for property (T), homogeneous dynamics, and automorphic forms.

Prerequisites: Q12 representation theory I, Q13 functional/Fourier analysis, and Q17 Lie groups in parallel or immediately before.

30-hour allocation: locally compact harmonic analysis 6h; unitary representation theory 8h; induction/Mackey 6h; SL(2,R)/semisimple examples 7h; exercises/exposition 3h.

Core books and chapters/sections

- Folland, A Course in Abstract Harmonic Analysis, 2nd ed.: chapters on locally compact groups, basic representation theory, compact groups, induced representations and further representation theory.
- MIT OCW 18.757 Representations of Lie Groups: Lectures 1-9 and 19 as the main Lie-group companion.

- Knapp, Representation Theory of Semisimple Groups: opening SL(2,R) and principal/discrete-series chapters selectively.

Lecture notes and primary-paper companions

- MIT 18.757 gives continuous representations, K-finite vectors, (g,K)-modules and a worked SL(2,R) classification route.
- Bekka-de la Harpe-Valette introductory chapters provide the spectral-gap vocabulary that will be formalized in Q21.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Locally compact groups and Haar measure	Folland, Locally Compact Groups chapter; review Q17 Lie groups as needed.	Prove left/right translation identities; compute modular functions in examples.
2	Unitary representations and matrix coefficients	Folland Basic Representation Theory; MIT 18.757 Lectures 1-2.	Classify basic unitary reps of R, S1 and finite groups.
3	Group algebras, positive-definite functions, GNS	Folland Basic Representation Theory; MIT 18.757 Lectures 3-4 selectively.	Construct GNS for 3 positive-definite functions.
4	Weak containment, regular representation, amenability preview	Folland further topics; Bekka-Valette introductory sections.	Prove basic almost-invariant vector criteria in examples.
5	Induced representations and Frobenius reciprocity	Folland Induced Representations chapter.	Construct induced reps for semidirect products and homogeneous spaces.
6	Mackey machine and imprimitivity	Folland induced-representation chapter.	Work affine group / Euclidean motion examples.
7	K-finite and smooth vectors; (g,K)-modules	MIT 18.757 Lectures 1-7.	Translate a continuous representation into its infinitesimal data.
8	SL(2,R): structure and principal series	MIT 18.757 Lectures 9 and 19; Knapp SL2(R) chapters.	Construct principal series and compute K-types.
9	Discrete and complementary series	MIT 18.757 Lecture 9; Knapp.	Classify irreducible unitary SL2(R) representations at the theorem-statement level.
10	Matrix coefficient decay and spectral viewpoint	Bekka-Valette introductory representation chapters; automorphic preview.	Estimate matrix coefficients in explicit rank-one examples.
11	Spherical representations / harmonic analysis on G/K	Folland/Knapp selections.	Compute spherical functions in a basic rank-one model.
12	Synthesis toward automorphic forms and property T	Review; preview Q21, Q25, Q35.	Oral: regular rep, induction, SL2(R), almost-invariant vectors.

Week 1 - Unitary representations, objects and examples. Reading: Folland/Knapp unitary representation sections. Definition/result focus: unitary rep; cyclic vector; matrix coefficient. Work: Construct regular reps and invariant subspaces.

Week 2 - Unitary representations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct regular reps and invariant subspaces. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Positive-definite functions and GNS, objects and examples. Reading: Bekka-de la Harpe-Valette/Folland sections. Definition/result focus: positive definite; GNS construction. Work: Carry out GNS for explicit functions/groups.

Week 4 - Positive-definite functions and GNS, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out GNS for explicit functions/groups. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Induced representations/Mackey, objects and examples. Reading: Folland/Knapp induction chapters. Definition/result focus: induction; systems of imprimitivity; Mackey machine orientation. Work: Compute induction for simple semidirect products.

Week 6 - Induced representations/Mackey, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute induction for simple semidirect products. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - SL2(R) structure, objects and examples. Reading: Knapp/Bump SL2 chapters. Definition/result focus: Iwasawa KAN; Casimir; K-types. Work: Compute Iwasawa coordinates and Lie algebra action.

Week 8 - SL2(R) structure, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Iwasawa coordinates and Lie algebra action. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Principal/discrete/complementary series, objects and examples. Reading: Knapp/Bump classification sections. Definition/result focus: principal/discrete/complementary series exact models. Work: Work through low-weight examples and unitarity ranges.

Week 10 - Principal/discrete/complementary series, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work through low-weight examples and unitarity ranges. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Spectral gap preview, objects and examples. Reading: Bekka-de la Harpe-Valette property T opening sections. Definition/result focus: almost invariant vectors; trivial rep isolation. Work: Compute almost-invariant examples for Z and prepare property-(T) quarter.

Week 12 - Spectral gap preview, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute almost-invariant examples for Z and prepare property-(T) quarter. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Locally compact group; Haar measure; modular function
- Strongly continuous unitary representation
- Matrix coefficient; cyclic vector
- Positive-definite function; GNS construction
- Left/right regular representation
- Weak containment (working definition)
- Almost invariant vector; invariant vector
- Induced representation; system of imprimitivity
- Smooth vector; K-finite vector
- (g,K)-module / Harish-Chandra module
- Principal, complementary and discrete series for SL(2,R)
- Spherical vector/function
- Tempered representation (orientation)
- Spectral gap of a unitary representation

Key results to know and use

- Haar measure uniqueness and modular-function consequences (working proof)
- GNS construction (full proof)
- Basic equivalence between positive-definite functions and cyclic unitary representations
- Frobenius reciprocity for induced unitary representations (working theorem)
- Mackey imprimitivity theorem (architecture and examples)
- Density of smooth/K-finite vectors in the relevant Lie-group setting (working use)
- Classification architecture of irreducible unitary representations of SL(2,R)
- How trivial-representation isolation is expressed in the unitary dual
- Relation between almost invariant vectors and spectral gap for averaging operators

Quarter-specific mastery targets

M1 - Exact language: unitary representation, cyclic vector, positive-definite function, GNS construction, induced representation, almost invariant vector, weak containment and spectral gap.

M2 - Proofs to reproduce: GNS; basic induction construction; equivalence between absence of almost-invariant vectors on the orthogonal complement and a representation-theoretic spectral gap in the standard setting.

M3 - Computation: decompose regular/unitary examples, calculate matrix coefficients, and work principal/discrete/complementary-series parameters for SL2(R) at an introductory but correct level.

M4 - Architecture: state the unitary-dual picture of SL2(R) and identify how complementary series approaches the trivial representation; explain why this matters for property (T) and spectral gaps.

M5 - Exit use: translate between a random-walk/operator gap, almost-invariant vectors and isolation of the trivial representation.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V18U3: Compact groups and Peter-Weyl

V18U3P1. [F] For S^1 and SU(2), list/construct irreducible unitary representations and compute matrix coefficients in low-dimensional cases.

V18U3P2. [T] Prove Peter-Weyl at proof-architecture level, highlighting compactness, Haar measure, and density of matrix coefficients. Verify Fourier series on S^1 as the abelian case.

V18U3P3. [S] Derive Schur orthogonality for compact groups and use it to compute a convolution projection onto one isotypic component.

Assigned block V18U6: Unitary reps of real groups

V18U6P1. [F] Write the principal series representation of SL_2(R) in one standard model and calculate the action of basic matrices/one-parameter subgroups.

V18U6P2. [T] Compute K-types and Casimir eigenvalues in a selected SL_2(R) representation. Relate the Casimir to the hyperbolic Laplacian on automorphic functions.

V18U6P3. [R] Compare principal, complementary, and discrete series: parameter ranges, unitarity, K-types, and where spectral-gap questions occur. Prepare this as a one-page table for later property (T)/automorphic use.

Assigned block V21U5: Kazhdan property (T)

V21U5P1. [F] Show Z does not have property (T) by exhibiting almost-invariant vectors in suitable one-dimensional/Fourier representations. Show finite groups have property (T).

V21U5P2. [T] Prove one standard equivalence for property (T): almost invariant vectors imply invariant vectors versus isolation of the trivial representation. Work through Kazhdan constants in a finite example.

V21U5P3. [S] Prove that a discrete amenable group with property (T) is finite. Explain the more general locally compact statement (amenable + property (T) forces compactness under the standard hypotheses) and identify which representation/mean argument encodes amenability.

Assigned block V29U1: Spectral graph theory

V29U1P1. [F] Compute spectra, spectral gaps, conductance, and mixing times for cycles, complete graphs, hypercubes, and small Cayley graphs. Compare normalized versus unnormalized adjacency conventions.

V29U1P2. [T] Prove the variational characterization of lambda_2 and one side of Cheeger's inequality. Use it to convert a combinatorial expansion bound into a spectral bound and vice versa.

V29U1P3. [S] Derive an L^2 and total-variation mixing estimate for a reversible regular graph from the spectral gap, including dependence on the smallest stationary mass.

Quarter-level integration problems

I1. [S]. For a symmetric probability measure on a finite group, prove that a spectral gap of its convolution operator on constants-perp is equivalent to absence of almost invariant vectors in the regular representation, with normalization explicit.

I2. [C]. Analyze the regular characters e^{itx} of Z (or R) to construct almost-invariant vectors/characters near the trivial representation and explain why this prevents property (T).

Full written solutions required for: V18U3P2, V18U6P2, V21U5P2, V29U1P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 15-page “unitary representation toolkit” note: Haar measure, regular representation, GNS, induction, almost invariant vectors, SL(2,R) principal/discrete/complementary series, and a diagram showing how these notions enter property (T) and automorphic forms.

Exit checkpoint

Compute modular functions in examples; carry out GNS; construct one induced representation; state the SL(2,R) unitary dual architecture; translate “spectral gap” into an operator/representation statement.

Research bridge

Q21 uses almost-invariant vectors and spectral gaps; Q25/Q34 use regular representations on homogeneous spaces; Q35 uses admissible and automorphic representation theory.

Q19. Local fields, heights, and cryptography I

Quarter overview and reading route

Objective: Learn p-adic/local arithmetic and height functions while beginning mathematically serious public-key cryptography.

Prerequisites: Q8-Q17, especially algebraic number theory and finite fields.

30-hour allocation: Local fields 13h; heights/Diophantine 7h; cryptography 7h; review 3h.

Core books and chapters/sections

- Serre, Local Fields, Chs. I-V selectively.
- Bombieri-Gubler, Heights in Diophantine Geometry, Ch. 1 and early Ch. 2; Neukirch local chapters.
- Hoffstein-Pipher-Silverman, An Introduction to Mathematical Cryptography, Chs. 1-4 and elliptic-curve chapters later.
- Katz-Lindell, Introduction to Modern Cryptography, security-definition chapters.
- Lidl-Niederreiter, Finite Fields, Chs. 1-3 first pass; later character/exponential-sum chapters.
- Ireland-Rosen, finite-field and character-sum sections as a number-theoretic companion.

Lecture notes and companions

- MIT OCW 18.785 Lectures 8-13.
- Use implementations only with toy parameters; treat security reductions separately from arithmetic correctness.
- J.S. Milne field/Galois notes may be used for structural review.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Valuations, completions, Q_p and Hensel's lemma	Serre Local Fields Chs. I-II; MIT 18.785 Lectures 8-9. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Valuations, completions, Q_p and Hensel's lemma	Serre Local Fields Chs. I-II; MIT 18.785 Lectures 8-9. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Extensions of local fields, ramification and residue fields	Serre Chs. II-IV; MIT 18.785 Lectures 10-12. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Extensions of local fields, ramification and residue fields	Serre Chs. II-IV; MIT 18.785 Lectures 10-12. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Different/discriminant and product formula	Serre/Neukirch relevant sections; MIT 18.785 Lectures 12-13. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Different/discriminant and product formula	Serre/Neukirch relevant sections; MIT 18.785 Lectures 12-13. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Absolute/projective heights and Northcott philosophy	Bombieri-Gubler Ch. 1 and early Ch. 2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Absolute/projective heights and Northcott philosophy	Bombieri-Gubler Ch. 1 and early Ch. 2. Second pass: theorem	2 theorem reconstructions + 5 harder problems + one 10-minute

Week	Main focus	Reading / lecture notes	Required output
		proofs, harder exercises, and a one-page proof reconstruction.	oral explanation.
9	Learn: RSA, Diffie-Hellman, ElGamal and arithmetic correctness	Hoffstein-Pipher-Silverman Chs. 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: RSA, Diffie-Hellman, ElGamal and arithmetic correctness	Hoffstein-Pipher-Silverman Chs. 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Security definitions/reductions; discrete log/factoring algorithms	Katz-Lindell introductory security chapters; HPS relevant algorithm chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Security definitions/reductions; discrete log/factoring algorithms	Katz-Lindell introductory security chapters; HPS relevant algorithm chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Valuations and completions, objects and examples. Reading: Serre Local Fields/Neukirch local chapters. Definition/result focus: valuation; completion; Q_p. Work: Construct p-adic expansions and prove Hensel in a standard form.

Week 2 - Valuations and completions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct p-adic expansions and prove Hensel in a standard form. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Local fields and ramification, objects and examples. Reading: Serre Local Fields. Definition/result focus: DVR; ramification index; residue degree; higher ramification orientation. Work: Compute extensions/ramification in quadratic p-adic examples.

Week 4 - Local fields and ramification, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute extensions/ramification in quadratic p-adic examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Heights, objects and examples. Reading: Bombieri-Gubler/Silverman height opening sections. Definition/result focus: absolute logarithmic height; product formula; Northcott. Work: Prove height identities and Northcott in tractable setting.

Week 6 - Heights, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove height identities and Northcott in tractable setting. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Local-global arithmetic, objects and examples. Reading: Neukirch/Serre sections. Definition/result focus: Hilbert symbols/Hasse principle orientation. Work: Solve local solvability tests for quadratic forms.

Week 8 - Local-global arithmetic, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve local solvability tests for quadratic forms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Classical public-key cryptography, objects and examples. Reading: Hoffstein-Pipher-Silverman. Definition/result focus: RSA; discrete log; Diffie-Hellman; semantic-security preview. Work: Implement toy schemes and distinguish correctness from security.

Week 10 - Classical public-key cryptography, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Implement toy schemes and distinguish correctness from security. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Elliptic-curve cryptography entry, objects and examples. Reading: Washington/HPS elliptic-curve chapters. Definition/result focus: EC group law; scalar multiplication; ECDLP. Work: Compute curves over F_p, point groups and toy ECDH.

Week 12 - Elliptic-curve cryptography entry, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute curves over F_p, point groups and toy ECDH. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- valuation
- completion
- uniformizer
- ramification
- absolute/projective height
- Northcott property
- one-way function
- discrete log problem
- RSA problem
- semantic security

- IND-CPA
- finite field
- Frobenius
- trace
- norm
- additive character
- multiplicative character
- Gauss sum
- Jacobi sum

Key results to know and use

- Hensel lemma
- structure of finite extensions of Q_p
- product formula
- Northcott theorem
- RSA correctness
- Diffie-Hellman correctness
- baby-step giant-step complexity
- semantic security/IND equivalence (statement)
- existence/uniqueness of F_{p^n}
- cyclicity of F_q^*
- trace/norm formulas
- basic Gauss/Jacobi identities
- Weil bounds (later)

Quarter-specific mastery targets

M1 - Exact language: nonarchimedean valuation, completion, valuation ring, uniformizer, ramification index/residue degree, local/global height, RSA/DH/ElGamal correctness and IND-style security notion.

M2 - Proofs to reproduce: Hensel lemma; basic local-field structure results used; product formula in representative number fields; elementary height identities/Northcott statement; RSA correctness.

M3 - Computation: p-adic expansions and Hensel lifts; local norms in examples; RSA/DH/ElGamal toy implementations and discrete-log/factoring experiments.

M4 - Security discipline: distinguish correctness from computational hardness and from reduction-based security; state the adversary experiment before making a security claim.

M5 - Exit use: connect local arithmetic and global height estimates to an explicit Diophantine/cryptographic computation without confusing exact mathematics with complexity assumptions.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V24U4: Valuations and local fields

V24U4P1. [F] Construct Q_p from a p-adic absolute value/completion and compute p-adic expansions of several rationals. Compare archimedean and nonarchimedean triangle inequalities.

V24U4P2. [T] Apply Hensel's lemma to lift roots modulo p to Q_p; determine solvability of selected equations such as x^2=a over Q_p.

V24U4P3. [S] For a finite extension of Q_p, compute ramification index/residue degree in a simple unramified or Eisenstein extension and verify e f <= [L:K] with equality in the separable local setting.

Assigned block V26U1: Heights and finiteness principles

V26U1P1. [F] Compute absolute logarithmic heights of rational numbers, quadratic algebraic numbers, roots of unity, and points in P^n(Q) from local/global definitions.

V26U1P2. [T] Prove basic height inequalities h(alpha beta), h(alpha+beta), h(alpha^n), and Northcott finiteness for bounded degree and height in a selected formulation.

V26U1P3. [S] Use heights to turn a naive infinite Diophantine search into a finite search once a height bound is given. Work a simple Thue/Mordell-type toy example.

Assigned block V36U2: RSA, factoring and primality

V36U2P1. [F] Implement modular exponentiation, extended Euclid, CRT, Miller-Rabin, and RSA on toy parameters. Verify decryption both by Euler/Carmichael and CRT decomposition.

V36U2P2. [T] Prove RSA correctness with all gcd cases handled and analyze textbook RSA's deterministic-security failure. Explain exactly what randomized padding/security notion must change.

V36U2P3. [R] Compare primality testing and factoring: state AKS polynomial-time primality, probabilistic Miller-Rabin, and best-known factoring status without conflating them.

Assigned block V36U3: Discrete logarithms in finite fields

V36U3P1. [F] Compute discrete logarithms in small F_p^* by brute force, baby-step/giant-step, and Pollard-rho conceptually. Compare time-memory tradeoffs.

V36U3P2. [T] Derive Diffie-Hellman/ElGamal correctness and formulate CDH, DDH, and DLP assumptions. Give a group where DDH is easy although DLP may remain hard, explaining the distinction.

V36U3P3. [S] Analyze subgroup/confinement attacks in a composite-order multiplicative group and show how group-order validation/cofactor handling changes security.

Quarter-level integration problems

I1. [S]. Choose a quadratic number field and a rational prime p; compute one p-adic completion/local factor explicitly, then verify the local contribution to the global product formula on a selected algebraic number.

I2. [C]. Write a correct RSA encryption/decryption proof and then list at least four security claims that do not follow from correctness; formulate one as a precise adversarial experiment.

Full written solutions required for: V24U4P2, V26U1P2, V36U2P2, V36U3P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Implement toy RSA/DH/ElGamal and write separate documents for (a) mathematical correctness and (b) security assumptions/reductions; also compute p-adic expansions and Hensel lifts.

Exit checkpoint

Prove Hensel's lemma, product formula in representative global fields, basic height identities, RSA correctness, and formulate IND-style security without conflating it with arithmetic correctness.

Research bridge

Local fields and heights are indispensable for Baker/Subspace/Faltings; cryptography will later branch to elliptic curves and lattices.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q20. Algebraic topology, commutative algebra, and algebraic geometry I

Quarter overview and reading route

Objective: Build the topological and commutative-algebra infrastructure needed for modern geometry, schemes, cohomology and arithmetic geometry.

Prerequisites: Q8 topology/algebra and Q17 manifolds.

30-hour allocation: Algebraic topology 10h; commutative algebra 9h; algebraic geometry 8h; review 3h.

Core books and chapters/sections

- Hatcher, Algebraic Topology, Chs. 0-3 selectively.
- Atiyah-Macdonald, Introduction to Commutative Algebra, Chs. 1-5,7,9 selectively; Vakil FOAG early chapters; Hartshorne Ch. I.
- Dummit-Foote, Abstract Algebra, Chs. 1-14 selectively.
- Artin, Algebra, selected chapters for a second viewpoint.

Lecture notes and companions

- Ravi Vakil FOAG course-note portal; J.S. Milne Algebraic Geometry notes.
- MIT OCW 18.701 Algebra I, student notes/problem sets.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Fundamental group, covering spaces	Hatcher Ch. 1. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Fundamental group, covering spaces	Hatcher Ch. 1. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Homology and exact sequences	Hatcher Ch. 2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Homology and exact sequences	Hatcher Ch. 2. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Modules, localization, prime/maximal ideals, Nakayama	Atiyah-Macdonald Chs. 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Modules, localization, prime/maximal ideals, Nakayama	Atiyah-Macdonald Chs. 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Integral dependence, dimension/Noetherian ideas	Atiyah-Macdonald Chs. 5,7,9 selectively. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Integral dependence, dimension/Noetherian ideas	Atiyah-Macdonald Chs. 5,7,9 selectively. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Affine varieties/schemes, Spec, structure sheaf	Vakil FOAG opening chapters; Hartshorne Ch. I Sections 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
10	Prove/use: Affine varieties/schemes, Spec, structure sheaf	Vakil FOAG opening chapters; Hartshorne Ch. I Sections 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Morphisms, projective space, divisors/line bundles preview	Vakil corresponding chapters; Hartshorne Ch. I remaining selected sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Morphisms, projective space, divisors/line bundles preview	Vakil corresponding chapters; Hartshorne Ch. I remaining selected sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Fundamental group and covering spaces, objects and examples. Reading: Hatcher Ch. 1. Definition/result focus: pi_1; covering space; van Kampen. Work: Compute fundamental groups of graphs/surfaces and use van Kampen.

Week 2 - Fundamental group and covering spaces, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute fundamental groups of graphs/surfaces and use van Kampen. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Homology, objects and examples. Reading: Hatcher homology chapters. Definition/result focus: chain complex; homology; exact sequence. Work: Compute cellular/simplicial homology and long exact sequences.

Week 4 - Homology, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute cellular/simplicial homology and long exact sequences. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Cohomology orientation, objects and examples. Reading: Hatcher cohomology sections. Definition/result focus: cochain; cup product; duality orientation. Work: Compute cohomology rings in basic spaces.

Week 6 - Cohomology orientation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute cohomology rings in basic spaces. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Commutative algebra I, objects and examples. Reading: Atiyah-Macdonald opening chapters. Definition/result focus: prime/maximal; localization; Noetherian; Nakayama. Work: Compute spectra/localizations and prove Nakayama.

Week 8 - Commutative algebra I, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute spectra/localizations and prove Nakayama. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Affine schemes, objects and examples. Reading: Vakil/Hartshorne I.1-I.3 style material. Definition/result focus: Spec; structure sheaf; morphism; fiber product basics. Work: Compute Spec Z, Spec k[x]/I, and explicit morphisms.

Week 10 - Affine schemes, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Spec Z, Spec k[x]/I, and explicit morphisms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Schemes and geometry bridge, objects and examples. Reading: Vakil early scheme properties. Definition/result focus: dimension; irreducibility; generic point; sheaf gluing. Work: Relate arithmetic prime splitting to scheme fibers and prepare AG II.

Week 12 - Schemes and geometry bridge, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Relate arithmetic prime splitting to scheme fibers and prepare AG II. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- fundamental group
- homology
- chain complex
- localization
- Noetherian ring
- Spec(A)
- scheme
- structure sheaf
- normal subgroup
- group action
- ideal
- PID/UFD

- irreducible polynomial
- field extension
- Galois group

Key results to know and use

- van Kampen
- long exact homology sequence
- Nakayama lemma
- Hilbert basis theorem
- affine-scheme/ring dictionary
- isomorphism theorems
- Sylow theorems
- Gauss lemma
- fundamental theorem of Galois theory

Quarter-specific mastery targets

M1 - Exact language: fundamental group, covering, chain complex/homology/cohomology, localization, local ring, Noetherian ring, Spec, prime spectrum, structure sheaf and fiber product.

M2 - Proofs to reproduce: van Kampen in usable form; long exact sequence/excision calculations as assigned; Nakayama and localization universal properties; affine-scheme equivalence and Spec functoriality for rings.

M3 - Computation: pi_1/homology of circles, tori, projective spaces or CW examples; localize explicit rings; draw/compute Spec Z, k[x], quotient rings and fibers.

M4 - Structural understanding: explain contravariance of Spec and how prime ideals encode geometric points/specializations; distinguish classical varieties from schemes by a concrete nilpotent/arithmetic example.

M5 - Exit use: pass from a ring map to its geometric morphism, fiber and local-ring data, and from a cell decomposition to homological invariants.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V11U3: Fundamental group and coverings

V11U3P1. [F] Compute pi_1 of S^1, torus, wedge of circles, and a simple punctured plane using van Kampen. Make generators and relations explicit.

V11U3P2. [T] Classify connected covering spaces of S^1 and a bouquet of circles via subgroups of the fundamental group in concrete cases.

V11U3P3. [S] Use covering spaces to prove a fixed-point/nonexistence statement, such as no retraction D^2->S^1 or Brouwer in dimension 2 via a standard route.

Assigned block V11U4: Homology

V11U4P1. [F] Compute simplicial/cellular homology of S^n, torus, projective plane, and one lens-space/simple CW example. Write all boundary matrices.

V11U4P2. [T] Prove homotopy invariance and exactness via the long exact sequence of a pair at proof-architecture level; then use exactness in a calculation.

V11U4P3. [S] Derive Euler characteristic from chain complexes and verify multiplicative/additive behavior in selected CW constructions.

Assigned block V4U1: Localization and local rings

V4U1P1. [F] Compute S^{-1}R for R=Z and S generated by a prime p; for R=k[x,y] localized at (x,y); and for localization at a prime ideal. Identify units and maximal ideals explicitly.

V4U1P2. [T] Prove the universal property of localization and use it to factor a homomorphism that sends every element of S to a unit. Check uniqueness in a concrete ring map.

V4U1P3. [S] Show that localizing Z at (p) detects p-adic divisibility but forgets all other primes. Translate this into the geometric statement that localization focuses on a neighborhood of a prime.

Assigned block V31U2: Schemes, Spec and fiber products

V31U2P1. [F] Draw/describe Spec Z, Spec k[x], Spec Z[i], and Spec k[x,y]/(xy), listing generic/closed points and specialization relations.

V31U2P2. [T] Compute fiber products on affine schemes by tensor products in examples such as Spec Z[i] x_{Spec Z} Spec F_p. Interpret split/inert/ramified fibers.

V31U2P3. [S] Glue two affine lines along G_m to construct P^1 and verify the structure sheaf on overlaps. Explain why schemes retain nilpotent/infinitesimal data lost by point sets.

Quarter-level integration problems

I1. [S]. Compute Spec Z[i] -> Spec Z over primes p=2, p congruent 1 mod 4 and p congruent 3 mod 4; interpret ramification/splitting/inertia simultaneously in ring-theoretic and geometric language.

I2. [S]. Build a CW decomposition of a torus, compute homology, then compare the topological quotient/gluing data with a scheme-theoretic fiber product - identifying analogy and non-analogy.

Full written solutions required for: V11U3P2, V11U4P2, V4U1P2, V31U2P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Construct Spec examples for Z, k[x], k[x,y]/I and number rings; compute fundamental groups/homology of standard spaces; write a dictionary between algebra and affine geometry.

Exit checkpoint

Prove Seifert-van Kampen in selected examples, compute homology by exact sequences, prove Nakayama/localization properties, and manipulate Spec/morphisms/sheaves in explicit cases.

Research bridge

This quarter is the gateway to elliptic curves, schemes, etale cohomology, Faltings, Deligne and the topology side of Poincare/Hauptvermutung.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q21. Geometric group theory, property (T), expander graphs, and spectral gaps

Quarter overview and reading route

Objective: Build the geometric and spectral theory of finitely generated groups, then understand Kazhdan property (T), property (tau), expander graphs and the equivalence between combinatorial, probabilistic and spectral expansion.

Prerequisites: Q12 and Q18 representation theory; Q17 Lie/arithmetic groups; Q13 functional/Fourier analysis.

30-hour allocation: geometric group theory 9h; amenability/random walks 5h; property (T)/(tau) 7h; expander/spectral graph theory 7h; synthesis 2h.

Core books and chapters/sections

- Loh, Geometric Group Theory: chapters on generators/relations, Cayley graphs, word metrics, quasi-isometries, actions, growth and hyperbolicity.
- Drutu-Kapovich, Geometric Group Theory: selected chapters on Cayley geometry, amenability, hyperbolic groups and property (T).
- Bekka-de la Harpe-Valette, Kazhdan's Property (T): Chs. 1-6 selectively, especially spectral criteria and expander applications.
- Hoory-Linial-Wigderson, Expander Graphs and Their Applications; Lubotzky, Expander Graphs in Pure and Applied Mathematics.

Lecture notes and primary-paper companions

- Calegari's discrete-groups notes are an optional companion on quasi-isometry, amenability and property T.
- Use Lubotzky's survey to connect property T/tau with arithmetic expanders and Ramanujan graphs.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Cayley graphs, word metrics, quasi-isometries	Loh GGT Chs. 1-3; Drutu-Kapovich Cayley/word-metric chapters.	Compute Cayley graphs/growth for 6 groups; prove generating-set invariance up to quasi-isometry.
2	Group actions and Milnor-	Loh action/quasi-isometry	Full proof of Milnor-Svarc;

Week	Main focus	Reading / lecture notes	Required output
	Svarc	chapters.	applications to fundamental groups.
3	Growth, ends, trees and Bass-Serre preview	Loh growth material; Serre Trees selections.	Compute growth/ends for free, abelian and surface-group examples.
4	Hyperbolic groups and boundaries	Loh / Drutu-Kapovich hyperbolic-group chapters.	Prove thin-triangle properties in trees and hyperbolic plane; boundary examples.
5	Amenability, Folner sets and random walks	Drutu-Kapovich amenability chapter; Bekka-Valette background.	Prove amenability of abelian groups and nonamenability of F2.
6	Property (T): definitions and elementary criteria	Bekka-de la Harpe-Valette Chs. 1-2.	Equivalence sheet: almost-invariant vectors, Kazhdan pairs, isolation of trivial rep.
7	Property (T) for higher-rank examples	Bekka-de la Harpe-Valette Chs. 3-5 selectively.	Study SL3(R)/SL3(Z) route; reconstruct one proof or spectral criterion.
8	Spectral graph theory and expansion	HLW survey Sections 2-4; Lubotzky expander survey.	Prove Cheeger-type inequalities for regular graphs in a selected normalization.
9	Property (tau), quotients and Margulis expanders	Bekka-Valette Ch. 6; Lubotzky.	Derive expander family from property T/tau for finite quotients.
10	Ramanujan graphs and optimal spectral gaps	Lubotzky survey; Alon-Boppana and Ramanujan sections.	Prove Alon-Boppana in a simplified form; compute spectra of examples.
11	Mixing times, flattening preview, expander applications	HLW random walk/applications sections.	Derive L2 mixing estimate from spectral gap; preview Bourgain-Gamburd.
12	Synthesis: geometry-representation-spectrum triangle	Review; read Green approximate-groups survey introduction.	30-minute oral connecting quasi-isometry, T, tau, Cheeger, mixing and expanders.

Week 1 - Cayley graphs and word metrics, objects and examples. Reading: de la Harpe/Bridson-Haefliger introductory GGT. Definition/result focus: word metric; Cayley graph; quasi-isometry. Work: Draw/compute growth for Z^d/free groups.

Week 2 - Cayley graphs and word metrics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Draw/compute growth for Z^d/free groups; prove generator independence up to quasi-isometry. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Hyperbolic groups and boundaries, objects and examples. Reading: Bridson-Haefliger/Ghys-de la Harpe sections. Definition/result focus: delta-hyperbolic; Gromov boundary. Work: Prove trees are 0-hyperbolic and Z^2 is not hyperbolic.

Week 4 - Hyperbolic groups and boundaries, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove trees are 0-hyperbolic and Z^2 is not hyperbolic. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Amenability, objects and examples. Reading: Paterson/Bekka-de la Harpe-Valette sections. Definition/result focus: Folner sequence; invariant mean; nonamenability. Work: Build Folner sequences for Z^d and prove free-group nonamenability.

Week 6 - Amenability, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build Folner sequences for Z^d and prove free-group nonamenability. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Property (T), objects and examples. Reading: Bekka-de la Harpe-Valette core chapters. Definition/result focus: Kazhdan pair; almost invariant vector; fixed-point formulations. Work: Prove finite groups have T and Z does not.

Week 8 - Property (T), theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove finite groups have T and Z does not; understand standard equivalences. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Spectral graph theory and expanders, objects and examples. Reading: Hoory-Linial-Wigderson/Lubotzky surveys. Definition/result focus: Cheeger constant; adjacency/Markov spectrum; expander. Work: Prove one Cheeger inequality direction.

Week 10 - Spectral graph theory and expanders, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove one Cheeger inequality direction; compute mixing from spectral gap. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Property tau and group-expander bridge, objects and examples. Reading: Lubotzky/Bekka-de la Harpe-Valette. Definition/result focus: property (tau); finite quotients; uniform spectral gap. Work: Derive expander families from tau and prepare Bourgain-Gamburd mechanism.

Week 12 - Property tau and group-expander bridge, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive expander families from tau and prepare Bourgain-Gamburd mechanism. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Cayley graph; word metric; finitely generated/presented group
- Quasi-isometry; coarse equivalence; quasi-geodesic
- Geometric action; proper/cocompact action
- Growth function and growth type
- End of a group
- Gromov hyperbolic metric space/group; boundary
- Amenable group; Folner sequence; invariant mean
- Kazhdan property (T); Kazhdan pair; Kazhdan constant
- Property (tau)
- Spectral gap of a Markov/adjacency operator
- Edge/vertex expansion; Cheeger constant
- Expander family
- Normalized graph Laplacian
- Random-walk mixing time
- Ramanujan graph
- Quasirandom finite group (representation-theoretic sense)

Key results to know and use

- Milnor-Svarc lemma (full proof)
- Quasi-isometry invariance of growth type
- Basic hyperbolicity of trees and H^n; Morse stability as working theorem
- Folner criterion for amenability; amenability of abelian/solvable examples
- Nonamenability of free groups (full combinatorial proof)
- Equivalent formulations of property (T) via almost invariant vectors / isolation of trivial representation
- Property (T) for standard higher-rank lattices/groups: know one proof architecture
- Cheeger inequalities relating expansion and spectral gap (one normalization fully proved)
- Property (T) or property (tau) yields uniform expansion in finite quotients
- Alon-Boppana bound (architecture)
- Ramanujan optimality definition and examples
- Spectral-gap mixing estimate for random walks on regular graphs

Quarter-specific mastery targets

M1 - Exact language: word metric, Cayley graph, quasi-isometry, growth, amenability/Folner set, Kazhdan property (T), property (tau), Cheeger constant, expander and spectral gap.

M2 - Proofs to reproduce: Milnor-Svarc; nonamenability of a free group by expansion/Folner obstruction; a Kazhdan-pair criterion in a standard formulation; one direction of Cheeger and spectral mixing from lambda_2.

M3 - Computation: compare growth/quasi-isometry examples; compute spectra/conductance/mixing for cycles, complete graphs and small Cayley graphs; test almost-invariant vectors in elementary representations.

M4 - Architecture: explain property (T) -> uniform spectral gaps on finite quotients and property (tau) relative to a family -> expander family, with normalization and generating-set dependence stated correctly.

M5 - Exit use: move through the dictionary unitary representation <-> Markov operator <-> spectrum <-> isoperimetry <-> mixing on a concrete quotient family.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V21U1: Cayley geometry and quasi-isometry

V21U1P1. [F] Draw/describe Cayley graphs and compute word metrics/growth for Z, Z^2, F_2, D_infinity, and the discrete Heisenberg group with standard generators.

V21U1P2. [T] Prove that word metrics from two finite generating sets are quasi-isometric. Then use the Milnor-Svarc lemma to compare a cocompact group action with the ambient metric space in a concrete example.

V21U1P3. [S] Use growth to prove Z^2 is not quasi-isometric to F_2. Identify a second coarse invariant that distinguishes another pair of groups.

Assigned block V21U4: Amenability

V21U4P1. [F] Construct Folner sequences for Z^d and prove nonamenability of F_2 via boundary expansion/paradoxical decomposition in a selected formulation.

V21U4P2. [T] Prove equivalence of two standard amenability criteria in the discrete finitely generated setting, such as Folner and invariant mean at theorem-architecture level.

V21U4P3. [S] Analyze simple random walk on Z^d versus F_2: return probabilities/spectral radius qualitatively, and explain how nonamenability manifests spectrally.

Assigned block V21U5: Kazhdan property (T)

V21U5P1. [F] Show Z does not have property (T) by exhibiting almost-invariant vectors in suitable one-dimensional/Fourier representations. Show finite groups have property (T).

V21U5P2. [T] Prove one standard equivalence for property (T): almost invariant vectors imply invariant vectors versus isolation of the trivial representation. Work through Kazhdan constants in a finite example.

V21U5P3. [S] Prove that a discrete amenable group with property (T) is finite. Explain the more general locally compact statement (amenable + property (T) forces compactness under the standard hypotheses) and identify which representation/mean argument encodes amenability.

Assigned block V21U6: Property (tau), spectral gaps and expanders

V21U6P1. [F] For a finite d-regular Cayley graph, derive the normalized adjacency/random-walk operator and relate eigenvalues to L^2 mixing of the walk.

V21U6P2. [T] Prove one direction of Cheeger's inequality for regular graphs and apply it to cycles, complete graphs, and a candidate expander family.

V21U6P3. [S] Show how property (tau) for a family of finite-index subgroups produces a uniform spectral gap/expander family of finite quotients. Make every identification between representation spaces and graph eigenfunctions explicit.

Assigned block V29U1: Spectral graph theory

V29U1P1. [F] Compute spectra, spectral gaps, conductance, and mixing times for cycles, complete graphs, hypercubes, and small Cayley graphs. Compare normalized versus unnormalized adjacency conventions.

V29U1P2. [T] Prove the variational characterization of lambda_2 and one side of Cheeger's inequality. Use it to convert a combinatorial expansion bound into a spectral bound and vice versa.

V29U1P3. [S] Derive an L^2 and total-variation mixing estimate for a reversible regular graph from the spectral gap, including dependence on the smallest stationary mass.

Quarter-level integration problems

I1. [S]. For a finite quotient family of a fixed finitely generated group, derive the Markov operator and prove carefully how a uniform spectral gap yields a uniform Cheeger/expansion constant and exponential mixing.

I2. [C]. Compare Z^d and F_2: construct a Folner sequence for the former, prove a boundary-expansion obstruction for the latter, and explain why this difference is quasi-isometry invariant at the level studied.

Full written solutions required for: V21U1P2, V21U1P3, V21U4P2, V21U4P3, V21U5P2, V21U5P3, V21U6P2, V21U6P3, V29U1P2, V29U1P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Build a “spectral-gap dictionary” with five columns: group action/unitary representation, Cayley graph, Markov operator, isoperimetry, random walk. Prove at least one implication between every adjacent pair and work one property-T expander family.

Exit checkpoint

Prove Milnor-Svarc and one Cheeger inequality; define amenability, T, tau and expander family exactly; explain property T -> expander quotients and derive a mixing estimate.

Research bridge

This quarter is the conceptual prerequisite for Q28 Bourgain-Gamburd, arithmetic super-approximation, expander-based sieve, and spectral-gap arguments throughout homogeneous dynamics.

Q22. Restriction/Kakeya toolkit and sieve methods

Quarter overview and reading route

Objective: Begin modern Fourier restriction and Kakeya theory while learning classical and large-sieve methods in analytic number theory. The unifying theme is converting geometric or arithmetic sparsity into norm estimates.

Prerequisites: Q13-Q15 harmonic/Fourier analysis; Q9-Q15 analytic number theory.

30-hour allocation: Restriction/Kakeya 14h; sieve methods 10h; review/problem synthesis 6h.

Core books and chapters/sections

- Stein, Harmonic Analysis, restriction/oscillatory-integral chapters selectively.
- Demeter, Fourier Restriction, Decoupling, and Applications, Chs. 1-2.
- Iwaniec-Kowalski, Analytic Number Theory, sieve/large-sieve chapters.
- Friedlander-Iwaniec, Opera de Cribro, introductory chapters.
- Stein, Singular Integrals and Differentiability Properties of Functions, Chs. I-III selectively.
- Grafakos, Classical Fourier Analysis, chapters on maximal operators, singular integrals, interpolation, Littlewood-Paley.

Lecture notes and companions

- Use Tao/Guth restriction-Kakeya lecture notes/surveys; revisit MIT 18.156 Strichartz lectures.
- Use graduate analytic-number-theory course notes only as a companion to the monographs.
- MIT RES.18-015 Lectures 20-24; MIT 18.156 Lectures 22-30.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says "selected" or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Oscillatory integrals, stationary phase and extension operators	Stein Harmonic Analysis oscillatory-integral/restriction sections; Demeter Ch. 1. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Oscillatory integrals, stationary phase and extension operators	Stein Harmonic Analysis oscillatory-integral/restriction sections; Demeter Ch. 1. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Tomas-Stein restriction and Knapp examples	Demeter Chs. 1-2; standard restriction lecture notes; revisit Plancherel/interpolation. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Tomas-Stein restriction and Knapp examples	Demeter Chs. 1-2; standard restriction lecture notes; revisit Plancherel/interpolation. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Kakeya maximal viewpoint and wave-packet heuristic	Demeter opening Kakeya sections; Guth/Tao restriction-Kakeya expositions. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Kakeya maximal viewpoint and wave-packet heuristic	Demeter opening Kakeya sections; Guth/Tao restriction-Kakeya expositions. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Brun/Selberg sieve and	Iwaniec-Kowalski sieve chapters;	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	sieve dimension	Opera de Cribro introductory chapters. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
8	Prove/use: Brun/Selberg sieve and sieve dimension	Iwaniec-Kowalski sieve chapters; Opera de Cribro introductory chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Large sieve, fundamental lemma and parity obstruction	Iwaniec-Kowalski large-sieve/sieve chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Large sieve, fundamental lemma and parity obstruction	Iwaniec-Kowalski large-sieve/sieve chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Applications: almost primes, distribution in progressions, restriction-sieve comparison	Work one full sieve application and one full restriction estimate from source to conclusion. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Applications: almost primes, distribution in progressions, restriction-sieve comparison	Work one full sieve application and one full restriction estimate from source to conclusion. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Oscillatory integrals and stationary phase, objects and examples. Reading: Stein Harmonic Analysis oscillatory-integral sections. Definition/result focus: stationary phase; van der Corput. Work: Prove 1D van der Corput and compute model oscillatory integrals.

Week 2 - Oscillatory integrals and stationary phase, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove 1D van der Corput and compute model oscillatory integrals. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Restriction theorem foundations, objects and examples. Reading: Stein/Tao restriction notes. Definition/result focus: extension operator; Tomas-Stein; Knapp example. Work: Derive Knapp necessary conditions and prove L2-based Tomas-Stein route.

Week 4 - Restriction theorem foundations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive Knapp necessary conditions and prove L2-based Tomas-Stein route. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Wave packets and Kakeya connection, objects and examples. Reading: Tao/Guth restriction notes. Definition/result focus: wave-packet decomposition; tube geometry. Work: Decompose a paraboloid wave packet model and quantify transversality.

Week 6 - Wave packets and Kakeya connection, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Decompose a paraboloid wave packet model and quantify transversality. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Sieve foundations, objects and examples. Reading: Iwaniec-Kowalski sieve sections. Definition/result focus: sifting function; fundamental lemma orientation. Work: Run Brun/Selberg sieve on a toy sequence.

Week 8 - Sieve foundations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Run Brun/Selberg sieve on a toy sequence. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Large sieve and distribution, objects and examples. Reading: Iwaniec-Kowalski. Definition/result focus: large sieve; Bombieri-Vinogradov exact statement. Work: Prove a duality version and track level-of-distribution parameters.

Week 10 - Large sieve and distribution, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a duality version and track level-of-distribution parameters. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Restriction/Kakeya/sieve synthesis, objects and examples. Reading: selected survey problems. Definition/result focus: scaling; incidence vs Fourier cancellation. Work: Write two tool-selection essays: one geometric-analytic, one arithmetic-sieve.

Week 12 - Restriction/Kakeya/sieve synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Write two tool-selection essays: one geometric-analytic, one arithmetic-sieve. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- restriction operator
- extension operator
- Knapp example

- oscillatory integral
- stationary phase
- Kakeya set
- sieve weight
- sieve dimension
- level of distribution
- large sieve
- parity problem
- maximal operator
- weak type
- CZ kernel
- principal value
- Littlewood-Paley projection

Key results to know and use

- Tomas-Stein restriction theorem
- stationary-phase lemma
- basic Kakeya-restriction implications
- Selberg sieve
- fundamental lemma of sieve
- large-sieve inequality
- Bombieri-Vinogradov
- Chen theorem (statement/architecture)
- Hardy-Littlewood maximal theorem
- Lebesgue differentiation
- Hilbert transform L2 boundedness
- CZ weak (1,1)
- Riesz-Thorin
- Marcinkiewicz interpolation

Quarter-specific mastery targets

M1 - Exact language: extension/restriction operator, Fourier decay of surface measure, Knapp example, Kakeya maximal function, sieve weight, sifting density, sieve dimension and parity barrier.

M2 - Proofs to reproduce: stationary phase in a model nondegenerate case; Stein-Tomas by TT*; Knapp necessary condition; Selberg upper-bound sieve in one standard finite-dimensional formulation.

M3 - Computation: determine scaling exponents for parabola/sphere restriction; construct cap/tube examples; optimize simple Selberg weights and derive an almost-prime upper bound.

M4 - Architecture: state a representative Kakeya/restriction implication and fundamental lemma/sieve limitation; explain exactly why the parity problem prevents elementary sieve detection of primes.

M5 - Exit use: identify whether an estimate is obstructed by scaling/Knapp geometry or by arithmetic parity/distribution, and choose the appropriate analytic/sieve response.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V35U2: Restriction theory

V35U2P1. [F] Determine scaling and Knapp necessary conditions for restriction to the parabola/sphere. Construct the cap example and associated physical-space tube explicitly.

V35U2P2. [T] Prove the Stein-Tomas theorem by TT* and Fourier decay of surface measure, tracking exponents.

V35U2P3. [S] Perform a wave-packet decomposition for a cap-localized function at scale R in a model parabola setting and derive tube dimensions/orientations from uncertainty principle.

Assigned block V35U3: Kakeya, incidence geometry and polynomial method

V35U3P1. [F] Construct epsilon-neighborhoods of line sets in R^2/R^3 and compute volume lower bounds from elementary direction counting. Contrast true Kakeya sets with finite-scale Kakeya configurations.

V35U3P2. [T] Work through a polynomial partitioning toy example for incidences in R^2 or points/lines. Count contributions in cells versus on the zero set.

V35U3P3. [R] State the Wang-Zahl 2025 theorem that every Kakeya set in R^3 has Hausdorff and Minkowski dimension 3. Build a proof-architecture map through the non-clustering/sticky reduction, multiscale analysis and volume estimates, using Guth's later outline as a companion; explicitly note that the general Kakeya conjecture in dimensions >=4 is not settled by this theorem.

Assigned block V23U1: Combinatorial and Selberg sieves

V23U1P1. [F] Build the Eratosthenes/Legendre sieve for integers avoiding a finite set of prime divisors and compute exact upper/lower counts for a small z.

V23U1P2. [T] Derive Selberg's upper-bound sieve weights for a simplified dimension-1 problem and use them to bound twin-prime-like sifted sets at the correct order of magnitude up to constants/log factors.

V23U1P3. [C] Explain the parity problem by constructing a sieve situation where primes and semiprimes are indistinguishable to purely combinatorial inclusion-exclusion data.

Assigned block V23U2: Large sieve and distribution of primes

V23U2P1. [T] Prove the large sieve inequality in a standard finite form or reconstruct its duality proof. Apply it to a family of character/exponential sums.

V23U2P2. [S] Starting from a large-sieve/Bombieri-Vinogradov statement, derive a nontrivial average result for primes in arithmetic progressions over q<=Q. Track the level-of-distribution exponent.

V23U2P3. [R] Compare Brun-Titchmarsh, Siegel-Walfisz, Bombieri-Vinogradov, and Elliott-Halberstam by exact ranges and averaging. Mark theorem versus conjecture.

Quarter-level integration problems

I1. [S]. Derive the Stein-Tomas exponent by TT* and surface-measure decay, then construct a Knapp example and verify that the exponents are consistent; every scaling power must be checked.

I2. [C]. Work a Selberg-sieve upper bound for a simple sequence and then identify precisely why the same framework cannot distinguish integers with an even versus odd number of prime factors strongly enough to isolate primes.

Full written solutions required for: V35U2P2, V35U2P3, V35U3P2, V35U3P3, V23U1P2, V23U2P2, V23U2P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Prepare two blackboard lectures: Tomas-Stein with sharpness tests, and the Selberg sieve with a concrete almost-prime application.

Exit checkpoint

Derive a stationary-phase estimate, prove Tomas-Stein in the standard range, explain Knapp obstruction, construct Selberg sieve weights, and state the parity barrier precisely.

Research bridge

Restriction/Kakeya leads to decoupling and modern Falconer/Furstenberg/Kakeya work; sieve theory feeds prime-distribution and circle-method problems.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q23. Probability II: SDE, SLE, percolation, random graphs and random matrices

Quarter overview and reading route

Objective: Build a coherent advanced-probability toolkit and survey four major research directions deeply enough to choose one or two for later specialization.

Prerequisites: Q10-Q11 probability/measure; Q7-Q9 complex analysis useful for SLE.

30-hour allocation: Stochastic calculus/SDE 10h; SLE 5h; percolation 5h; random graphs 4h; random matrices 4h; synthesis 2h.

Core books and chapters/sections

- Karatzas-Shreve, Brownian Motion and Stochastic Calculus, Chs. 1-5 selectively.
- Lawler, Conformally Invariant Processes in the Plane; Grimmett, Percolation; Bollobas, Random Graphs; Tao, Topics in Random Matrix Theory, early chapters.
- Durrett, Probability: Theory and Examples, Chs. 1-5 selectively.
- Williams, Probability with Martingales, early chapters.

Lecture notes and companions

- MIT OCW 18.175 Theory of Probability lecture notes.
- MIT OCW 18.440 Probability and Random Variables, Lectures 4-35; MIT 18.175 later.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says "selected" or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Conditional expectation, martingales, stopping and Brownian motion	Durrett/Williams martingale chapters; MIT 18.175 opening probability notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Conditional expectation, martingales, stopping and Brownian motion	Durrett/Williams martingale chapters; MIT 18.175 opening probability notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Ito integral, Ito formula and SDE existence/uniqueness	Karatzas-Shreve Chs. 2-5 selectively. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Ito integral, Ito formula and SDE existence/uniqueness	Karatzas-Shreve Chs. 2-5 selectively. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Planar Brownian motion, conformal maps and Loewner evolution	Lawler opening SLE/conformal-invariance chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Planar Brownian motion, conformal maps and Loewner evolution	Lawler opening SLE/conformal-invariance chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Percolation: criticality, crossings, RSW and scaling philosophy	Grimmett opening/planar percolation chapters; selected modern notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
8	Prove/use: Percolation: criticality, crossings, RSW and scaling philosophy	Grimmett opening/planar percolation chapters; selected modern notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Random graphs: Erdos-Renyi phase transition and concentration	Bollobas introductory random-graph chapters; probabilistic-method review. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Random graphs: Erdos-Renyi phase transition and concentration	Bollobas introductory random-graph chapters; probabilistic-method review. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Random matrices: semicircle law, moments and concentration	Tao Topics in Random Matrix Theory opening chapters; prove the moment method in a model ensemble. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Random matrices: semicircle law, moments and concentration	Tao Topics in Random Matrix Theory opening chapters; prove the moment method in a model ensemble. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Conditional expectation and martingale convergence, objects and examples. Reading: Durrett/Williams advanced probability sections. Definition/result focus: martingale convergence; optional stopping hypotheses. Work: Prove bounded martingale convergence case and audit optional-stopping failures.

Week 2 - Conditional expectation and martingale convergence, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove bounded martingale convergence case and audit optional-stopping failures. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Brownian motion, objects and examples. Reading: Karatzas-Shreve opening chapters. Definition/result focus: Brownian motion; stopping times; hitting probabilities. Work: Derive reflection principle/hitting probabilities in standard forms.

Week 4 - Brownian motion, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive reflection principle/hitting probabilities in standard forms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Ito calculus and SDE, objects and examples. Reading: Karatzas-Shreve. Definition/result focus: Ito integral/formula; strong solution; Lipschitz existence/uniqueness. Work: Derive Ito for polynomials then solve geometric Brownian/OU examples.

Week 6 - Ito calculus and SDE, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive Ito for polynomials then solve geometric Brownian/OU examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Percolation and SLE, objects and examples. Reading: Grimmett/Lawler selected foundational chapters. Definition/result focus: critical probability; RSW orientation; SLE Loewner equation. Work: Prove basic percolation inequalities and compute simple Loewner evolutions.

Week 8 - Percolation and SLE, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove basic percolation inequalities and compute simple Loewner evolutions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Random graphs, objects and examples. Reading: Bollobas/Janson-Luczak-Rucinski. Definition/result focus: Erdos-Renyi; branching-process heuristic; giant component. Work: Derive threshold heuristics and prove first/second moment results.

Week 10 - Random graphs, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive threshold heuristics and prove first/second moment results. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Random matrices, objects and examples. Reading: Tao/Anderson-Guionnet-Zeitouni opening material. Definition/result focus: Wigner ensemble; moments; semicircle law. Work: Run moment method through Catalan leading terms.

Week 12 - Random matrices, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Run moment method through Catalan leading terms; simulate eigenvalue histograms. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Brownian motion

- Ito integral
- quadratic variation
- SDE
- Loewner chain
- SLE_kappa
- percolation threshold
- Wigner matrix
- random variable
- independence
- conditional expectation
- filtration
- martingale
- mode of convergence

Key results to know and use

- Ito formula
- SDE existence/uniqueness under Lipschitz conditions
- Donsker principle (statement)
- giant-component threshold
- Wigner semicircle law
- Borel-Cantelli
- weak/strong laws
- Jensen
- optional stopping in standard settings
- martingale convergence (later)

Quarter-specific mastery targets

M1 - Exact language: conditional expectation, stopping time, martingale, Ito integral, SDE/generator, Loewner chain/SLE_kappa, percolation crossing, Erdos-Renyi threshold, Wigner matrix and empirical spectral law.

M2 - Proofs to reproduce: optional stopping under a clearly stated valid hypothesis set; Ito formula; existence/uniqueness for a Lipschitz SDE in the chosen text; one Erdos-Renyi threshold calculation; semicircle moment calculation at introductory level.

M3 - Computation/simulation: Brownian/SDE sample paths, simple Loewner maps, percolation crossings, random-graph components and Wigner spectra, with numerical claims kept separate from proofs.

M4 - Architecture: state SLE conformal-invariance role and one percolation/random-matrix universality result at theorem level; identify which pieces are not proved in this quarter.

M5 - Exit use: move between stochastic process, martingale/generator and geometric/combinatorial observable in a new example.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V14U4: Conditional expectation and martingales

V14U4P1. [F] Compute conditional expectations in finite partitions, discrete filtrations, and one continuous random-variable example. Verify the defining integral property.

V14U4P2. [T] Show simple random walk and compensated counting processes are martingales in appropriate filtrations. Derive Doob decomposition in a finite discrete-time example.

V14U4P3. [S] Solve gambler's ruin by optional stopping, then list all hypotheses needed and modify the problem to show what can go wrong without boundedness/uniform integrability conditions.

Assigned block V15U1: Ito calculus

V15U1P1. [F] Compute quadratic variation of Brownian motion along dyadic partitions and use it to motivate Ito's correction term. Contrast with bounded-variation paths.

V15U1P2. [T] Derive Ito's formula first for f(x)=x^2,x^3 and then in general. Use it to compute d(exp(B_t-t/2)) and verify the martingale property.

V15U1P3. [S] Apply Ito to log of geometric Brownian motion and to the Ornstein-Uhlenbeck process. Extract mean, variance, and long-time behavior.

Assigned block V16U5: Loewner evolution and SLE

V16U5P1. [F] Solve the chordal Loewner equation for constant driving function and identify the growing slit explicitly. Track half-plane capacity under scaling.

V16U5P2. [T] Use Brownian scaling to verify SLE_kappa scale invariance and explain the domain Markov property. Determine which kappa regimes correspond to simple/non-simple/space-filling paths at theorem-statement level.

V16U5P3. [S] Derive one SLE martingale/PDE in a simple boundary-event setting and explain how solving it gives a crossing probability or exponent.

Assigned block V15U4: Concentration and random graphs

V15U4P1. [F] In G(n,p), compute expected isolated vertices, triangles, and degrees; locate heuristic thresholds from first moments and test numerically.

V15U4P2. [T] Prove a bounded-difference/McDiarmid inequality in a standard form and apply it to a Lipschitz graph statistic or product-space observable.

V15U4P3. [S] Use first and second moment methods on a simple random-graph property, clearly distinguishing what each method proves and where dependencies matter.

Assigned block V15U5: Wigner matrices and global laws

V15U5P1. [F] For a Wigner matrix, compute expected normalized trace of powers k=2,4,6 and identify Catalan pairings in the leading terms.

V15U5P2. [T] Derive the semicircle law by the moment method at proof-architecture level, including why non-pairing index patterns are lower order.

V15U5P3. [X] Simulate eigenvalue histograms for increasing matrix size and compare with the semicircle density. Record finite-size effects and what the simulation cannot prove.

Quarter-level integration problems

I1. [S]. Starting from Brownian motion, use Ito's formula to derive the generator and Feynman-Kac equation for one solvable SDE; verify the formula by an independent expectation calculation.

I2. [X]. Simulate Erdos-Renyi component sizes and Wigner spectra over increasing n; state the theoretical threshold/semicircle predictions and quantify finite-size deviations without treating simulation as proof.

Full written solutions required for: V14U4P2, V14U4P3, V15U1P2, V15U1P3, V16U5P2, V16U5P3, V15U4P2, V15U4P3, V15U5P2, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Choose one of SLE/percolation/random graphs/random matrices and write a 20-page mini-survey with one complete central proof and a computational experiment.

Exit checkpoint

Prove optional stopping under valid hypotheses, Ito formula, solve basic SDEs, explain Loewner encoding, derive an Erdos-Renyi threshold, and prove the semicircle moment calculation at introductory level.

Research bridge

This quarter supplies the probability foundation for percolation/SLE and statistical-mechanics/particle-system targets in the final year.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q24. Algebraic geometry II, elliptic curves, and finite-field arithmetic II

Quarter overview and reading route

Objective: Move from affine/scheme foundations to curves, divisors and elliptic curves; deepen finite-field arithmetic enough to connect point counting, Frobenius and cryptography.

Prerequisites: Q20 algebraic geometry; Q7 finite fields; Q19 local arithmetic.

30-hour allocation: Algebraic geometry/curves 11h; elliptic curves 11h; finite fields 5h; review 3h.

Core books and chapters/sections

- Silverman, The Arithmetic of Elliptic Curves, Chs. I-III first; later reduction/heights chapters selectively.
- Hartshorne/Vakil chapters on curves/divisors; Washington, Elliptic Curves: Number Theory and Cryptography, early computational chapters.
- Lidl-Niederreiter, Finite Fields, Chs. 1-3 first pass; later character/exponential-sum chapters.
- Ireland-Rosen, finite-field and character-sum sections as a number-theoretic companion.
- Hatcher, Algebraic Topology, Chs. 0-3 selectively.
- Atiyah-Macdonald, Introduction to Commutative Algebra, Chs. 1-5,7,9 selectively; Vakil FOAG early chapters; Hartshorne Ch. I.

Lecture notes and companions

- J.S. Milne Elliptic Curves notes; use MIT 18.783 materials if available.
- J.S. Milne field/Galois notes may be used for structural review.
- Ravi Vakil FOAG course-note portal; J.S. Milne Algebraic Geometry notes.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Sheaves, divisors, line bundles and curves	Vakil FOAG relevant curve/divisor chapters; Hartshorne Ch. I Sections 5-6 and Ch. II selected foundations. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Sheaves, divisors, line bundles and curves	Vakil FOAG relevant curve/divisor chapters; Hartshorne Ch. I Sections 5-6 and Ch. II selected foundations. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Riemann-Roch for curves and genus calculations	Hartshorne Ch. IV selected or a curves text; Vakil corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Riemann-Roch for curves and genus calculations	Hartshorne Ch. IV selected or a curves text; Vakil corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Elliptic curves: Weierstrass equations, group law, isogenies	Silverman Arithmetic of Elliptic Curves Chs. I-II; Milne Elliptic Curves notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Elliptic curves: Weierstrass equations, group law, isogenies	Silverman Arithmetic of Elliptic Curves Chs. I-II; Milne Elliptic Curves notes. Second pass:	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		theorem proofs, harder exercises, and a one-page proof reconstruction.	
7	Learn: Elliptic curves over finite fields, Frobenius and Hasse bound	Silverman Ch. V selected; Washington finite-field/ECC chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Elliptic curves over finite fields, Frobenius and Hasse bound	Silverman Ch. V selected; Washington finite-field/ECC chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Finite-field characters, Gauss/Jacobi sums and point counting	Lidl-Niederreiter character-sum chapters; Ireland-Rosen companion sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Finite-field characters, Gauss/Jacobi sums and point counting	Lidl-Niederreiter character-sum chapters; Ireland-Rosen companion sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Reduction, local/global invariants and arithmetic of rational points	Silverman Chs. VII-VIII selectively as preview; consolidate explicit computations. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Reduction, local/global invariants and arithmetic of rational points	Silverman Chs. VII-VIII selectively as preview; consolidate explicit computations. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Projective varieties and divisors, objects and examples. Reading: Hartshorne I/Vakil divisors sections. Definition/result focus: projective variety; divisor; line bundle. Work: Compute divisors on P1 and plane curves.

Week 2 - Projective varieties and divisors, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute divisors on P1 and plane curves. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Sheaves/cohomology entry, objects and examples. Reading: Hartshorne III/Vakil cohomology introduction. Definition/result focus: sheaf cohomology; Cech model; exact sequences. Work: Compute H^i(P1,O(n)) in basic cases.

Week 4 - Sheaves/cohomology entry, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute H^i(P1,O(n)) in basic cases. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Algebraic curves and Riemann-Roch, objects and examples. Reading: Hartshorne IV/standard curve notes. Definition/result focus: genus; canonical divisor; Riemann-Roch. Work: Use RR on P1 and elliptic curves.

Week 6 - Algebraic curves and Riemann-Roch, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Use RR on P1 and elliptic curves; compute dimensions of L(D). Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Elliptic curves arithmetic, objects and examples. Reading: Silverman I-III. Definition/result focus: Weierstrass model; group law; reduction. Work: Compute torsion/group law and reduction mod primes.

Week 8 - Elliptic curves arithmetic, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute torsion/group law and reduction mod primes. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Finite fields II: curves and point counts, objects and examples. Reading: Lidl-Niederreiter/finite-field curve notes. Definition/result focus: zeta function; Frobenius; Hasse-Weil for curves. Work: Compute #E(F_q) and zeta functions for small examples.

Week 10 - Finite fields II: curves and point counts, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute #E(F_q) and zeta functions for small examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Mordell-Weil/descent orientation, objects and examples. Reading: Silverman height/descent sections. Definition/result focus: Mordell-Weil theorem; descent; canonical height preview. Work: Carry out 2-descent/toy height calculations and map prerequisites for later Faltings/FLT.

Week 12 - Mordell-Weil/descent orientation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out 2-descent/toy height calculations and map prerequisites for later Faltings/FLT. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- divisor
- genus
- Riemann-Roch space
- elliptic curve
- isogeny
- Frobenius endomorphism
- zeta function of a curve
- finite field
- Frobenius
- trace
- norm
- additive character
- multiplicative character
- Gauss sum
- Jacobi sum
- fundamental group
- homology
- chain complex
- localization
- Noetherian ring
- Spec(A)
- scheme
- structure sheaf

Key results to know and use

- Riemann-Roch for curves (statement)
- elliptic-curve group law
- Hasse bound
- Frobenius characteristic polynomial
- existence/uniqueness of F_{p^n}
- cyclicity of F_q^*
- trace/norm formulas
- basic Gauss/Jacobi identities
- Weil bounds (later)
- van Kampen
- long exact homology sequence
- Nakayama lemma
- Hilbert basis theorem
- affine-scheme/ring dictionary

Quarter-specific mastery targets

M1 - Exact language: divisor, line bundle, Riemann-Roch space, elliptic curve, isogeny, good/bad reduction, Frobenius trace, zeta function of a curve and Hasse-Weil bound.

M2 - Proofs to reproduce: algebraic chord-tangent group law and associativity route at the level assigned; Riemann-Roch for curves at proof-architecture or proof level per source; Hasse bound via the chosen elementary/isogeny argument if feasible.

M3 - Computation: point counts and group structures of E(F_q), Frobenius traces, low-degree isogenies, divisors on P1/elliptic curves and zeta functions of simple curves.

M4 - Structural understanding: connect Frobenius characteristic polynomial, point counts over F_{q^n} and local zeta factors; distinguish geometric morphisms from arithmetic reduction data.

M5 - Exit use: produce a rigorous Sage-assisted arithmetic dossier for an elliptic curve, with every computational invariant tied to a theorem.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V31U4: Divisors, line bundles and curves

V31U4P1. [F] Compute Weil/Cartier divisors and Picard groups on P^1, and divisors of rational functions on a smooth projective curve.

V31U4P2. [T] Prove Riemann-Roch for P^1 directly and verify the general curve formula on several low-degree divisors where dimensions are known.

V31U4P3. [S] Relate line bundles, Cartier divisors, and invertible sheaves by converting one explicit object through all three descriptions.

Assigned block V32U1: Elliptic curves and isogenies

V32U1P1. [F] Put several cubic curves into Weierstrass form, compute discriminant/j-invariant, add points by chord-tangent law, and identify torsion points in small examples.

V32U1P2. [T] Prove the group law is algebraic/associative at theorem-architecture level via divisor/Picard interpretation rather than a brute-force coordinate proof.

V32U1P3. [S] Compute degrees/kernels of multiplication-by-n and one explicit isogeny. Verify dual-isogeny relations in a small example where formulas are available.

Assigned block V32U2: Finite/local arithmetic and reduction

V32U2P1. [F] Count E(F_p) for several small elliptic curves and verify Hasse's bound. Compute Frobenius trace a_p and local Euler factors.

V32U2P2. [T] Apply good/multiplicative/additive reduction criteria at several primes for a curve over Q; compute minimal-model data where manageable.

V32U2P3. [S] For a local field Q_p and good reduction, describe the exact reduction sequence E_1(Q_p)->E_0(Q_p)->E_tilde(F_p) and use it to analyze local points in a simple case.

Assigned block V27U6: Zeta functions and Frobenius

V27U6P1. [F] Count #P^1(F_{q^n}) and derive its zeta function directly from the exponential generating definition.

V27U6P2. [T] For an elliptic curve over a small finite field, count points over F_q and F_{q^2}, infer Frobenius trace, and check the local zeta numerator relation.

V27U6P3. [R] Derive the rationality/formal determinant expression for curve zeta functions at theorem-architecture level, preparing the cohomological Frobenius formulation in Volume 33.

Quarter-level integration problems

I1. [S]. For one elliptic curve over Q with good reduction at several primes, compute #E(F_p), Frobenius traces a_p and local zeta factors; verify the recurrence for #E(F_{p^n}) in small n.

I2. [S]. On an elliptic curve, express a low-degree divisor class using the group law and Riemann-Roch spaces; explain how divisor arithmetic and point-group arithmetic encode the same geometry.

Full written solutions required for: V31U4P2, V31U4P3, V32U1P2, V32U1P3, V32U2P2, V32U2P3, V27U6P2, V27U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Create an elliptic-curve laboratory in Sage: group law, point counts, Frobenius traces and small isogenies; write a rigorous mathematical companion explaining every invariant.

Exit checkpoint

Prove the chord-tangent group law algebraically, state Riemann-Roch precisely, derive Hasse's bound at an expository level, and calculate curves over F_q by hand and machine.

Research bridge

This quarter is central to Sato-Tate, modularity/FLT, Faltings, finite-field arithmetic and elliptic-curve cryptography.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q25. Ergodic/homogeneous foundations, modular forms I, and cryptography II

Quarter overview and reading route

Objective: Learn foundational ergodic theory and classical modular forms while moving cryptography from finite groups to elliptic curves.

Prerequisites: Q10 measure, Q17 Lie groups, Q24 elliptic curves, Q9 complex analysis.

30-hour allocation: Ergodic theory 11h; modular forms 11h; elliptic-curve cryptography 5h; review 3h.

Core books and chapters/sections

- Einsiedler-Ward, Ergodic Theory with a View Towards Number Theory, Chs. 1-6 selectively.
- Diamond-Shurman, A First Course in Modular Forms, Chs. 1-4; Serre, A Course in Arithmetic, modular-form chapters.
- Hoffstein-Pipher-Silverman, An Introduction to Mathematical Cryptography, Chs. 1-4 and elliptic-curve chapters later.
- Katz-Lindell, Introduction to Modern Cryptography, security-definition chapters.
- Silverman, The Arithmetic of Elliptic Curves, Chs. I-III first; later reduction/heights chapters selectively.
- Hartshorne/Vakil chapters on curves/divisors; Washington, Elliptic Curves: Number Theory and Cryptography, early computational chapters.

Lecture notes and companions

- Tao UCLA ergodic-theory course notes; Milne Modular Functions and Modular Forms notes.
- Use implementations only with toy parameters; treat security reductions separately from arithmetic correctness.
- J.S. Milne Elliptic Curves notes; use MIT 18.783 materials if available.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Measure-preserving systems, recurrence and ergodicity	Einsiedler-Ward Chs. 1-2; Tao UCLA ergodic notes Lectures 1-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Measure-preserving systems, recurrence and ergodicity	Einsiedler-Ward Chs. 1-2; Tao UCLA ergodic notes Lectures 1-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Mean/Birkhoff ergodic theorems, spectral viewpoint	Einsiedler-Ward Chs. 2-4; Tao ergodic notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Mean/Birkhoff ergodic theorems, spectral viewpoint	Einsiedler-Ward Chs. 2-4; Tao ergodic notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Modular group, upper half-plane, fundamental domains	Diamond-Shurman Ch. 1; Serre Course in Arithmetic modular-form section; Milne Modular Functions notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Modular group, upper half-plane, fundamental domains	Diamond-Shurman Ch. 1; Serre Course in Arithmetic modular-form section; Milne Modular Functions notes. Second pass: theorem proofs, harder exercises, and a one-page proof	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		reconstruction.	
7	Learn: Modular forms, Eisenstein series, cusp forms and q-expansions	Diamond-Shurman Chs. 2-3. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Modular forms, Eisenstein series, cusp forms and q-expansions	Diamond-Shurman Chs. 2-3. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Hecke operators, eigenforms and modular L-functions	Diamond-Shurman Ch. 5 selected; Milne modular-form notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Hecke operators, eigenforms and modular L-functions	Diamond-Shurman Ch. 5 selected; Milne modular-form notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: ECC: ECDH, signatures, discrete logarithm and implementation issues	Hoffstein-Pipher-Silverman elliptic-curve crypto chapters; Washington corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: ECC: ECDH, signatures, discrete logarithm and implementation issues	Hoffstein-Pipher-Silverman elliptic-curve crypto chapters; Washington corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Measure-preserving dynamics, objects and examples. Reading: Einsiedler-Ward/Walters opening ergodic chapters. Definition/result focus: ergodic; invariant measure; recurrence. Work: Prove Poincare recurrence and analyze rotations/Bernoulli shifts.

Week 2 - Measure-preserving dynamics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove Poincare recurrence and analyze rotations/Bernoulli shifts. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Ergodic theorems, objects and examples. Reading: Einsiedler-Ward/Walters. Definition/result focus: Birkhoff/von Neumann exact statements. Work: Prove mean ergodic theorem in Hilbert setting and apply Birkhoff as theorem.

Week 4 - Ergodic theorems, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove mean ergodic theorem in Hilbert setting and apply Birkhoff as theorem. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Homogeneous spaces, objects and examples. Reading: Witte Morris/Einsiedler-Ward selected chapters. Definition/result focus: G/Gamma; unipotent/diagonal flow; Haar probability. Work: Identify lattice-space interpretations and explicit orbits.

Week 6 - Homogeneous spaces, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Identify lattice-space interpretations and explicit orbits. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Dani correspondence/nondivergence preview, objects and examples. Reading: Kleinbock-Margulis/Dani expositions. Definition/result focus: short vectors; cusp excursion; Diophantine correspondence. Work: Derive 1D/2D correspondence for approximation.

Week 8 - Dani correspondence/nondivergence preview, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive 1D/2D correspondence for approximation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Modular forms I, objects and examples. Reading: Diamond-Shurman/Serre Course in Arithmetic. Definition/result focus: modular form; cusp form; q-expansion; Eisenstein series. Work: Compute E4,E6,Delta transformations/Fourier coefficients.

Week 10 - Modular forms I, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute E4,E6,Delta transformations/Fourier coefficients. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Hecke operators and crypto II, objects and examples. Reading: Diamond-Shurman + Washington/HPS. Definition/result focus: Hecke eigenform; Euler factor; ECC security notions. Work: Compute low Hecke actions and an elliptic-curve crypto implementation.

Week 12 - Hecke operators and crypto II, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute low Hecke actions and an elliptic-curve crypto implementation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- ergodic

- mixing
- Koopman operator
- modular form
- cusp form
- Hecke operator
- eigenform
- one-way function
- discrete log problem
- RSA problem
- semantic security
- IND-CPA
- divisor
- genus
- Riemann-Roch space
- elliptic curve
- isogeny
- Frobenius endomorphism
- zeta function of a curve

Key results to know and use

- Poincare recurrence
- mean ergodic theorem
- Birkhoff theorem (architecture)
- ergodicity of irrational rotations
- Hecke multiplicativity
- Euler product of eigenform L-function
- RSA correctness
- Diffie-Hellman correctness
- baby-step giant-step complexity
- semantic security/IND equivalence (statement)
- Riemann-Roch for curves (statement)
- elliptic-curve group law
- Hasse bound
- Frobenius characteristic polynomial

Quarter-specific mastery targets

M1 - Exact language: measure-preserving system, ergodic, invariant sigma-algebra, modular form/cusp form, q-expansion, Hecke operator/eigenform, elliptic-curve discrete-log problem and ECDH.

M2 - Proofs to reproduce: Poincare recurrence; mean ergodic theorem; basic dimension/generation results for M_k(SL2(Z)) used; Hecke multiplicativity for normalized eigenforms in a standard formulation.

M3 - Computation: q-expansions of E4,E6,Delta; Hecke eigenvalues; modular-form bases in small weights; toy ECDH with subgroup/order checks.

M4 - Architecture: explain how Hecke eigenvalues generate Euler factors and why ergodic recurrence differs from mixing/equidistribution; state ECC hardness assumption separately from correctness.

M5 - Exit use: translate a classical modular form into spectral/arithmetic data and independently analyze an elliptic-curve protocol's algebraic group assumptions.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V20U1: Ergodic-theory core

V20U1P1. [F] Prove mean ergodic theorem in a Hilbert-space formulation and Birkhoff ergodic theorem at proof-architecture level. Apply both to irrational rotation and a Bernoulli shift.

V20U1P2. [T] Use Fourier series to prove ergodicity/unique ergodicity of an irrational rotation and compute time averages of trigonometric polynomials.

V20U1P3. [C] Construct measure-preserving systems that are ergodic but not mixing and mixing systems that illustrate stronger correlation decay; state the distinctions precisely.

Assigned block V30U1: Classical modular forms

V30U1P1. [F] Compute q-expansions of E4, E6, Delta and verify modular weights/products to enough order to see identities such as E4^3-E6^2 proportional to Delta.

V30U1P2. [T] Prove the valence/dimension formula in a selected level-one setting and use it to determine spaces M_k and S_k for several small weights.

V30U1P3. [S] Derive Mellin transform of a cusp form and the associated completed L-function functional equation from modular transformation, with all normalizing powers recorded.

Assigned block V30U2: Hecke operators, eigenforms and newforms

V30U2P1. [F] Compute T_p on q-expansions and verify Hecke relations T_m T_n for coprime m,n and prime powers on selected forms.

V30U2P2. [T] Prove simultaneous diagonalization of commuting self-adjoint Hecke operators under the Petersson inner product in a finite-dimensional cusp space.

V30U2P3. [S] From Hecke eigenform relations derive multiplicativity of Fourier coefficients and the Euler product of L(s,f), including the quadratic local factor at unramified primes.

Assigned block V36U4: Elliptic curves and pairings

V36U4P1. [F] Implement elliptic-curve addition/doubling over a small finite field and compute group order, scalar multiples, and a toy ECDH exchange.

V36U4P2. [T] Explain why generic discrete-log algorithms cost about sqrt(|G|) and compare field versus elliptic-curve parameter sizes. Avoid relying on empirical 'security bits' without assumptions.

V36U4P3. [R] For pairings, state bilinearity/nondegeneracy and work a toy algebraic consequence such as DDH testing or identity-based-encryption structure; do not treat insecure tiny examples as deployable crypto.

Quarter-level integration problems

I1. [S]. Compute T_p on a low-weight modular-form basis, diagonalize it, and derive the corresponding Euler factor 1-a_p p^{-s}+p^{k-1-2s}; state every normalization used.

I2. [C]. Give examples of measure-preserving transformations that are recurrent but not mixing and of modular-form computations that produce correct q-expansions but no security claim for ECC; explain the distinct notions being tested.

Full written solutions required for: V20U1P2, V30U1P2, V30U1P3, V30U2P2, V30U2P3, V36U4P2, V36U4P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Compute a basis of low-weight modular forms and Hecke eigenvalues in Sage; separately implement toy ECDH and analyze its underlying group problem.

Exit checkpoint

Prove Poincare recurrence and mean ergodic theorem, derive q-expansions of E4/E6/Delta, prove basic Hecke relations, and explain ECC security assumptions without hand-waving.

Research bridge

Ergodic theory is the base for Ratner/EKL/Zimmer; modular forms are the classical gateway to automorphic representations, Sato-Tate and FLT.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q26. Carleson's theorem, time-frequency analysis, and the circle method

Quarter overview and reading route

Objective: Develop enough time-frequency analysis to understand a modern proof of Carleson's theorem and, in parallel, master the major/minor arc architecture of the Hardy-Littlewood circle method.

Prerequisites: Q13-Q15 harmonic analysis; Q22 oscillatory methods; Q9-Q15 analytic number theory.

30-hour allocation: Carleson/time-frequency 16h; circle method 10h; review/exposition 4h.

Core books and chapters/sections

- Muscalu-Schlag, Classical and Multilinear Harmonic Analysis Vol. II, time-frequency/Carleson chapters selectively.
- Vaughan, The Hardy-Littlewood Method, Chs. 1-4 and selected applications.
- Stein, Singular Integrals and Differentiability Properties of Functions, Chs. I-III selectively.
- Grafakos, Classical Fourier Analysis, chapters on maximal operators, singular integrals, interpolation, Littlewood-Paley.
- Iwaniec-Kowalski, Analytic Number Theory, chapters/sections on character and exponential sums.
- Vaughan, The Hardy-Littlewood Method, preliminary exponential-sum sections.

Lecture notes and companions

- Use Lacey/Tao time-frequency lecture notes; revisit MIT Fourier/Hilbert-transform notes.
- MIT RES.18-015 Lectures 20-24; MIT 18.156 Lectures 22-30.
- Use finite Fourier analysis notes plus MIT Fourier notes for analytic prerequisites.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Carleson operator, maximal partial Fourier sums and model reductions	Muscalu-Schlag Vol. II Carleson/time-frequency chapters; Lacey/Tao lecture notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Carleson operator, maximal partial Fourier sums and model reductions	Muscalu-Schlag Vol. II Carleson/time-frequency chapters; Lacey/Tao lecture notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Tiles, trees, phase plane and size/energy	Continue Muscalu-Schlag/Lacey notes; reconstruct tree lemma. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Tiles, trees, phase plane and size/energy	Continue Muscalu-Schlag/Lacey notes; reconstruct tree lemma. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Tree selection, almost orthogonality and proof assembly	Complete a modern Carleson proof at architecture-to-detail level. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Tree selection, almost orthogonality and proof assembly	Complete a modern Carleson proof at architecture-to-detail level. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
7	Learn: Circle method: generating integrals, major/minor arcs	Vaughan Chs. 1-2. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Circle method: generating integrals, major/minor arcs	Vaughan Chs. 1-2. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Weyl sums, singular series/integral, Waring prototypes	Vaughan Chs. 2-4; Iwaniec-Kowalski companion sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Weyl sums, singular series/integral, Waring prototypes	Vaughan Chs. 2-4; Iwaniec-Kowalski companion sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Vinogradov mean values preview and comparative Fourier architecture	Vaughan selected later sections; prepare bridge to decoupling. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Vinogradov mean values preview and comparative Fourier architecture	Vaughan selected later sections; prepare bridge to decoupling. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Maximal Fourier sums, objects and examples. Reading: Carleson/Hunt expositions, Lacey-Thiele notes introduction. Definition/result focus: Carleson operator; maximal partial sum. Work: Derive why pointwise Fourier convergence reduces to maximal bounds.

Week 2 - Maximal Fourier sums, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive why pointwise Fourier convergence reduces to maximal bounds. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Tiles and phase plane, objects and examples. Reading: Lacey-Thiele exposition. Definition/result focus: tile; partial order; tree; size/density. Work: Construct explicit tile collections and orthogonality estimates.

Week 4 - Tiles and phase plane, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Construct explicit tile collections and orthogonality estimates. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Tree estimates and decomposition, objects and examples. Reading: Lacey-Thiele proof sections. Definition/result focus: tree lemma; size lemma architecture. Work: Prove a model tree estimate and reconstruct the decomposition logic.

Week 6 - Tree estimates and decomposition, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a model tree estimate and reconstruct the decomposition logic. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Circle method basics, objects and examples. Reading: Vaughan Hardy-Littlewood Method opening chapters. Definition/result focus: major/minor arcs; singular series/integral. Work: Derive circle-method identity for Waring-type count.

Week 8 - Circle method basics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive circle-method identity for Waring-type count. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Weyl estimates/minor arcs, objects and examples. Reading: Vaughan/Iwaniec-Kowalski. Definition/result focus: Weyl differencing; minor-arc control. Work: Prove a polynomial exponential-sum bound and use it on minor arcs.

Week 10 - Weyl estimates/minor arcs, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a polynomial exponential-sum bound and use it on minor arcs. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Carleson-circle synthesis, objects and examples. Reading: selected expositions. Definition/result focus: time-frequency vs arithmetic frequency decomposition. Work: Write proof maps for Carleson and one circle-method theorem, emphasizing analogous decompositions/differences.

Week 12 - Carleson-circle synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Write proof maps for Carleson and one circle-method theorem, emphasizing analogous decompositions/differences. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Carleson operator
- tile
- tree

- size
- energy
- major arc
- minor arc
- singular series
- maximal operator
- weak type
- CZ kernel
- principal value
- Littlewood-Paley projection
- exponential sum
- Kloosterman sum
- Weyl differencing
- van der Corput process

Key results to know and use

- Carleson-Hunt theorem
- tree lemma
- size/energy decomposition
- circle-method major/minor-arc decomposition
- Weyl inequality
- Hardy-Littlewood maximal theorem
- Lebesgue differentiation
- Hilbert transform L2 boundedness
- CZ weak (1,1)
- Riesz-Thorin
- Marcinkiewicz interpolation
- quadratic Gauss sum evaluation
- van der Corput inequality (basic form)

Quarter-specific mastery targets

M1 - Exact language: Carleson maximal operator, tile, tree, size/density, major/minor arc, Weyl differencing, singular series and singular integral.

M2 - Proofs to reproduce: a finite/model tree estimate and size decomposition sufficient to expose Carleson architecture; Weyl differencing and a standard Weyl-type bound; one major-arc asymptotic calculation.

M3 - Architecture-only but exact: Carleson's theorem on a.e. convergence of Fourier series and one representative circle-method theorem; identify every reduction and the role of orthogonality versus arithmetic approximation.

M4 - Computation: discretize a toy Carleson operator into tiles; compute major/minor arcs for a small Waring-type model and local factors of a singular series.

M5 - Exit use: compare time-frequency localization with circle-method frequency localization and articulate why the decompositions solve different obstructions.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V35U1: Time-frequency analysis and Carleson

V35U1P1. [F] Starting from Fourier partial sums, derive the maximal partial-sum operator whose L^2 boundedness implies almost-everywhere convergence. Explain why uniform boundedness of individual partial-sum operators is insufficient.

V35U1P2. [T] Discretize a model Carleson operator into tiles; define time/frequency intervals, partial order, trees, size, and density. Work a finite toy tile collection completely.

V35U1P3. [R] Prove a simplified tree estimate and size decomposition, then write a dependency map showing how orthogonality + tree control + packing yield the Carleson theorem architecture.

Assigned block V23U3: Exponential and character sums

V23U3P1. [F] Prove Weyl differencing for a polynomial exponential sum and derive a nontrivial bound for a quadratic phase. Compare with exact Gauss-sum evaluation when available.

V23U3P2. [T] Derive van der Corput's A/B process in a selected form and apply it to one incomplete exponential sum.

V23U3P3. [S] Prove a Polya-Vinogradov bound for character sums or study Burgess at proof-architecture level; identify what extra idea improves the exponent/range.

Assigned block V23U4: Circle method and Vinogradov methods

V23U4P1. [F] Write the Hardy-Littlewood circle-method representation for the number of solutions of a Waring-type equation. Separate major/minor arcs and compute the formal singular series/integral in a toy case.

V23U4P2. [T] Prove a minor-arc bound using Weyl's inequality sufficient for one classical Waring result in a simplified exponent range.

V23U4P3. [R] State Vinogradov mean value theorem in modern sharp form and show algebraically how it improves Weyl-type bounds. Then connect to decoupling in Volume 35.

Assigned block V35U6: Decoupling and Vinogradov mean values

V35U6P1. [F] State l^2 decoupling for the parabola and moment curve with the exact scaling exponent in a standard range. Verify the trivial orthogonality and triangle-inequality bounds for comparison.

V35U6P2. [T] Work through one induction-on-scales step in a simplified decoupling proof, separating transverse from nearly parallel pieces and recording the recurrence for the decoupling constant.

V35U6P3. [R] Derive the Vinogradov mean-value consequence from decoupling by expressing the relevant moment integral as a count of Diophantine solutions. Check normalization/exponent arithmetic carefully.

Quarter-level integration problems

I1. [S]. Build a finite tile model for a Carleson operator, perform a tree decomposition and prove the model tree estimate; record exactly where orthogonality and packing are used.

I2. [S]. For a cubic Weyl sum, carry out one differencing step, derive a minor-arc estimate and compare it to the major-arc rational approximation; explain why the same frequency split would not solve Carleson.

Full written solutions required for: V35U1P2, V35U1P3, V23U3P2, V23U3P3, V23U4P2, V23U4P3, V35U6P2, V35U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write two detailed proof maps: Carleson's theorem and a circle-method theorem. Each map must identify the role of every lemma and include at least one lemma proved independently.

Exit checkpoint

State Carleson's theorem precisely, prove the core model tree estimate, carry out a major-arc computation, derive a minor-arc bound via Weyl differencing, and explain the singular series.

Research bridge

The next quarter uses decoupling to revisit restriction and Vinogradov mean values; this also supports the user's dedicated Carleson and Bourgain-decoupling targets.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q27. Bourgain-Demeter decoupling and analytic number theory III

Quarter overview and reading route

Objective: Understand l^2 decoupling for the paraboloid/moment curve, its induction-on-scales mechanism and its arithmetic applications, especially modern Vinogradov mean value theory.

Prerequisites: Q15 harmonic analysis, Q22 restriction, Q26 circle method.

30-hour allocation: Decoupling 18h; analytic-number-theory applications 8h; synthesis 4h.

Core books and chapters/sections

- Demeter, Fourier Restriction, Decoupling, and Applications, core decoupling chapters.
- Bourgain-Demeter, The proof of the l^2 decoupling conjecture (Annals); modern VMVT expositions.
- Stein, Harmonic Analysis, restriction/oscillatory-integral chapters selectively.
- Demeter, Fourier Restriction, Decoupling, and Applications, Chs. 1-2.
- Iwaniec-Kowalski, Analytic Number Theory, chapters/sections on character and exponential sums.
- Vaughan, The Hardy-Littlewood Method, preliminary exponential-sum sections.
- Davenport, Multiplicative Number Theory, chapters on Dirichlet characters, L-functions, PNT and primes in AP.
- Montgomery-Vaughan, Multiplicative Number Theory I; Iwaniec-Kowalski, corresponding analytic chapters as references.

Lecture notes and companions

- Use Demeter's book as lecture notes; revisit MIT 18.156 Littlewood-Paley/Strichartz prerequisites.
- Use Tao/Guth restriction-Kakeya lecture notes/surveys; revisit MIT 18.156 Strichartz lectures.
- Use finite Fourier analysis notes plus MIT Fourier notes for analytic prerequisites.
- MIT OCW 18.785 Number Theory I, Lectures 16-19.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Decoupling constants, parabolic rescaling and uncertainty-scale geometry	Demeter, Fourier Restriction, Decoupling, and Applications, opening decoupling chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Decoupling constants, parabolic rescaling and uncertainty-scale geometry	Demeter, Fourier Restriction, Decoupling, and Applications, opening decoupling chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Multilinear Kakeya/restriction input and broad-narrow philosophy	Demeter core chapters; Bourgain-Demeter paper introductory sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Multilinear Kakeya/restriction input and broad-narrow philosophy	Demeter core chapters; Bourgain-Demeter paper introductory sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Induction on scales and l^2 decoupling theorem	Demeter proof chapters; read Bourgain-Demeter in parallel after textbook exposition. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Induction on scales and l^2 decoupling theorem	Demeter proof chapters; read Bourgain-Demeter in parallel after textbook exposition. Second pass:	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		theorem proofs, harder exercises, and a one-page proof reconstruction.	
7	Learn: Moment curve decoupling and Vinogradov mean values	Demeter arithmetic applications; modern VMVT expositions. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Moment curve decoupling and Vinogradov mean values	Demeter arithmetic applications; modern VMVT expositions. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Applications to exponential sums and Diophantine counting	Work through at least one full quantitative application. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Applications to exponential sums and Diophantine counting	Work through at least one full quantitative application. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Compare decoupling, efficient congruencing and classical circle-method inputs	Survey-level comparison followed by proof reconstruction of chosen decoupling lemmas. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Compare decoupling, efficient congruencing and classical circle-method inputs	Survey-level comparison followed by proof reconstruction of chosen decoupling lemmas. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Decoupling setup and scaling, objects and examples. Reading: Demeter book opening chapters. Definition/result focus: decoupling constant; caps; critical exponent. Work: Compute scaling and Knapp-type necessary exponents.

Week 2 - Decoupling setup and scaling, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute scaling and Knapp-type necessary exponents. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Wave packets and induction on scales, objects and examples. Reading: Demeter/Bourgain-Demeter expositions. Definition/result focus: wave packets; parabolic rescaling. Work: Carry out rescaling identities and a toy induction recurrence.

Week 4 - Wave packets and induction on scales, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out rescaling identities and a toy induction recurrence. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Multilinear input, objects and examples. Reading: Bennett-Carbery-Tao/Demeter expositions. Definition/result focus: transversality; multilinear restriction/decoupling. Work: Prove a low-dimensional toy multilinear estimate.

Week 6 - Multilinear input, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a low-dimensional toy multilinear estimate. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Bourgain-Demeter theorem architecture, objects and examples. Reading: Demeter theorem chapters/original paper selectively. Definition/result focus: l2 decoupling for paraboloid/moment curve. Work: State exact theorem and map induction/multilinear components.

Week 8 - Bourgain-Demeter theorem architecture, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State exact theorem and map induction/multilinear components. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Vinogradov mean value theorem, objects and examples. Reading: Wooley/Bourgain-Demeter-Guth expositions. Definition/result focus: mean value J_{s,k}(N); main conjecture. Work: Translate decoupling to VMVT and verify exponents.

Week 10 - Vinogradov mean value theorem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Translate decoupling to VMVT and verify exponents. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Analytic NT III applications, objects and examples. Reading: Iwaniec-Kowalski/current notes. Definition/result focus: efficient congruencing/decoupling comparison; exponential sums. Work: Solve one application to Waring/Weyl sums and produce research-paper dependency map.

Week 12 - Analytic NT III applications, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve one application to Waring/Weyl sums and produce research-paper dependency map. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- decoupling constant
- frequency cap
- parabolic rescaling
- induction on scales
- Vinogradov mean value integral
- restriction operator
- extension operator
- Knapp example
- oscillatory integral
- stationary phase
- Kakeya set
- exponential sum
- Kloosterman sum
- Weyl differencing
- van der Corput process
- Dirichlet series
- Euler product
- Dirichlet character
- Dirichlet L-function
- von Mangoldt function
- zero-free region

Key results to know and use

- l2 decoupling theorem for paraboloid/moment curve
- parabolic rescaling lemma
- VMVT main conjecture range
- Weyl/Waring consequences
- Tomas-Stein restriction theorem
- stationary-phase lemma
- basic Kakeya-restriction implications
- quadratic Gauss sum evaluation
- Weyl inequality
- van der Corput inequality (basic form)
- Euler product for zeta
- PNT
- Dirichlet theorem on primes in AP
- functional equation for primitive Dirichlet L-functions (architecture)
- PNT in AP (statement)

Quarter-specific mastery targets

M1 - Exact language: decoupling constant, extension operator, parabolic rescaling, wave packet, transverse multilinear estimate, Vinogradov mean value $J_{s,k}(N)$, approximate functional equation and spectral summation formula.

M2 - Proofs/derivations: parabolic rescaling; trivial/orthogonality decoupling bounds; one complete induction-on-scales step; conversion of a moment integral to a Diophantine solution count.

M3 - Architecture: state Bourgain-Demeter l2 decoupling for the moment curve/parabola in the chosen normalization and the main VMVT consequence; identify multilinear and induction inputs.

M4 - Analytic-number-theory use: connect decoupling-derived VMVT to Weyl sums/circle-method estimates; state where mean-value or spectral L-function methods enter beyond decoupling.

M5 - Exit use: check every exponent/scaling normalization in the decoupling -> VMVT deduction without relying on memorized formulas.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V35U6: Decoupling and Vinogradov mean values

V35U6P1. [F] State l^2 decoupling for the parabola and moment curve with the exact scaling exponent in a standard range. Verify the trivial orthogonality and triangle-inequality bounds for comparison.

V35U6P2. [T] Work through one induction-on-scales step in a simplified decoupling proof, separating transverse from nearly parallel pieces and recording the recurrence for the decoupling constant.

V35U6P3. [R] Derive the Vinogradov mean-value consequence from decoupling by expressing the relevant moment integral as a count of Diophantine solutions. Check normalization/exponent arithmetic carefully.

Assigned block V23U4: Circle method and Vinogradov methods

V23U4P1. [F] Write the Hardy-Littlewood circle-method representation for the number of solutions of a Waring-type equation. Separate major/minor arcs and compute the formal singular series/integral in a toy case.

V23U4P2. [T] Prove a minor-arc bound using Weyl's inequality sufficient for one classical Waring result in a simplified exponent range.

V23U4P3. [R] State Vinogradov mean value theorem in modern sharp form and show algebraically how it improves Weyl-type bounds. Then connect to decoupling in Volume 35.

Assigned block V23U5: Zeta and L-function mean values

V23U5P1. [F] Derive an approximate functional equation for zeta or a primitive Dirichlet L-function in a simplified symmetric form, explaining the truncation length near the square root of conductor.

V23U5P2. [T] Compute the second moment of zeta on the critical line to the main-term level using Dirichlet polynomials and diagonal/off-diagonal separation in a standard route.

V23U5P3. [R] Write the hierarchy convexity -> subconvexity -> Lindelof for a family of L-functions with conductor C. Give exact exponents in one GL(1) or GL(2) example and name the method behind a known subconvex result.

Assigned block V23U6: Spectral methods and trace formulae

V23U6P1. [F] Derive the Petersson trace formula in a finite-dimensional holomorphic cusp-form setting at theorem-application level and compute the diagonal term for a small weight/level example.

V23U6P2. [T] State the Kuznetsov formula with the roles of Kloosterman sums and Bessel transforms clearly identified. Use a simplified form to see spectral/geometric duality.

V23U6P3. [R] Compare Petersson, Kuznetsov, Selberg trace, and Poisson/Voronoi summation: for each, list the spectral side, arithmetic/geometric side, and a typical analytic-number-theory application.

Quarter-level integration problems

I1. [S]. Starting from the l2 decoupling statement for the moment curve, derive the standard VMVT moment bound, writing the Fourier expansion that converts the integral to the Vinogradov system of equations.

I2. [C]. Check parabolic rescaling and critical exponents under a deliberate wrong normalization, locate the contradiction, then redo the argument correctly; include dimensions of wave packets at each scale.

Full written solutions required for: V35U6P2, V35U6P3, V23U4P2, V23U4P3, V23U5P2, V23U5P3, V23U6P2, V23U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Give a 60-minute seminar on the l^2 decoupling theorem and a second seminar deriving the Vinogradov mean-value consequence.

Exit checkpoint

Derive parabolic rescaling, state multilinear input accurately, reconstruct the induction-on-scales skeleton, and carry out the deduction from decoupling to a VMVT estimate.

Research bridge

This quarter is the key analytic bridge among restriction/Kakeya, exponential sums, circle method and modern analytic number theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q28. Sum-product, approximate groups, growth, and the Bourgain-Gamburd machine

Quarter overview and reading route

Objective: Understand the modern chain from sum-product estimates and approximate groups to growth in linear groups and the Bourgain-Gamburd spectral-gap/expansion method, including super-approximation and affine-sieve bridges.

Prerequisites: Q11 additive combinatorics I, Q12/Q18 representation theory, Q21 expanders/property T, Q7 finite fields, Q15 analytic number theory.

30-hour allocation: sum-product/incidence 7h; approximate groups/noncommutative growth 7h; Helfgott/product theorems 5h; Bourgain-Gamburd/flattening 8h; super-approximation/sieve synthesis 3h.

Core books and chapters/sections

- Tao-Vu, Additive Combinatorics: Ch. 2 (especially 2.7-2.8), Ch. 8 (especially 8.3 and 8.5).
- Green, Approximate Groups and Their Applications: full survey as the main expository spine.
- Helfgott, Growth and generation in SL2(Z/pZ), Annals 2008.
- Breuillard-Green-Tao, The structure of approximate groups: introduction, main theorem and selected proof architecture.
- Bourgain-Gamburd, Uniform expansion bounds for Cayley graphs of SL2(Fp), Annals 2008.
- Bourgain-Gamburd-Sarnak, Affine linear sieve, expanders, and sum-product; Salehi Golsefidy-Varju, Expansion in perfect groups, selectively.

Lecture notes and primary-paper companions

- Bourgain-Katz-Tao, A sum-product estimate in finite fields, for the finite-field input.
- Kowalski's expository notes on growth/expansion in SL2 can be used before the original Bourgain-Gamburd proof.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Ruzsa calculus and energy refresh	Tao-Vu Ch. 2, especially 2.3-2.7.	Reprove triangle/covering/BSG variants in multiplicative notation.
2	Sum-product over R	Tao-Vu Ch. 8.1-8.3; Szemeredi-Trotter proof route.	Prove a classical real sum-product bound from incidence geometry.
3	Finite-field sum-product: Bourgain-Katz-Tao	Tao-Vu Ch. 8.5; Bourgain-Katz-Tao paper introduction and key lemmas.	Write theorem dependency tree and reconstruct one energy/incidence lemma.
4	Approximate groups and noncommutative product sets	Tao-Vu 2.7; Green survey; BGT introduction.	Prove basic Ruzsa covering/product-set consequences in groups.
5	Helfgott growth in SL2(Fp)	Helfgott Annals paper; Green survey.	Understand escape from subvarieties/centralizers; reconstruct pivot argument architecture.
6	Product theorems and BGT structure	Breuillard-Green-Tao approximate groups paper introduction/theorem; selected expository notes.	State BGT theorem precisely; work nilpotent progression examples.
7	Quasirandomness and L2 flattening	Bourgain-Gamburd 2008 Section 2 architecture; representation theory review Q12/Q18.	Prove a flattening lemma in a finite-group toy setting.
8	Escape from proper subgroups and nonconcentration	Bourgain-Gamburd; Helfgott.	Establish nonconcentration for a model random walk and classify obstruction

Week	Main focus	Reading / lecture notes	Required output
			subgroups for SL2(Fp).
9	Bourgain-Gamburd expansion machine	Bourgain-Gamburd Annals 2008, theorem/proof architecture.	Write the three-input machine: growth, quasirandomness, escape -> spectral gap.
10	Spectral gap in compact groups	Bourgain-Gamburd SU(2) paper introduction and selected sections.	Compare finite-quotient and compact-group flattening arguments.
11	Super-approximation and affine sieve	Bourgain-Gamburd-Sarnak affine sieve; Salehi Golsefidy-Varju expansion in perfect groups.	Map expansion -> equidistribution -> sieve; work one thin-orbit example.
12	Synthesis and modern expansion	Green/Breuillard-Lubotzky surveys.	30-minute oral proving the Bourgain-Gamburd machine conceptually and tracking every prerequisite.

Week 1 - Finite-field sum-product, objects and examples. Reading: Bourgain-Katz-Tao paper + Tao-Vu background. Definition/result focus: additive/multiplicative energy; sum-product theorem. Work: Work through toy incidence formulation and exact obstruction from subfields.

Week 2 - Finite-field sum-product, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work through toy incidence formulation and exact obstruction from subfields. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Product growth in SL2, objects and examples. Reading: Helfgott product-growth paper/expositions. Definition/result focus: tripling; escape from subvarieties; tori/centralizers. Work: Compute growth for small SL2(Fp) sets and map Helfgott proof.

Week 4 - Product growth in SL2, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute growth for small SL2(Fp) sets and map Helfgott proof. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Approximate groups, objects and examples. Reading: Breuillard-Green-Tao theorem expositions. Definition/result focus: K-approximate group; control; nilprogression. Work: Prove elementary small-doubling consequences and state BGT structure theorem precisely.

Week 6 - Approximate groups, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove elementary small-doubling consequences and state BGT structure theorem precisely. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Quasirandomness and flattening, objects and examples. Reading: Gowers/Bourgain-Gamburd expositions. Definition/result focus: minimal representation dimension; L2 flattening. Work: Derive convolution/spectral identities and a toy flattening lemma.

Week 8 - Quasirandomness and flattening, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive convolution/spectral identities and a toy flattening lemma. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Bourgain-Gamburd expansion machine, objects and examples. Reading: Bourgain-Gamburd Annals 2008. Definition/result focus: escape; product growth; quasirandomness; spectral gap. Work: Build a box-by-box proof architecture for SL2(Fp) expansion.

Week 10 - Bourgain-Gamburd expansion machine, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build a box-by-box proof architecture for SL2(Fp) expansion. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Super-approximation, objects and examples. Reading: Salehi Golsefidy-Varju expositions/papers. Definition/result focus: perfect connected component criterion in square-free quotient setting. Work: State hypotheses exactly.

Week 12 - Super-approximation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State hypotheses exactly; compare property T/tau and combinatorial expansion routes. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Additive/multiplicative energy
- Ruzsa distance and covering in noncommutative groups
- Sum-product phenomenon
- Approximate subgroup / K-approximate group
- Control of one approximate group by another
- Product-set growth
- Escape from subvarieties/proper subgroups
- Quasirandom finite group; minimal representation degree

- Convolution probability measure and L2 flattening
- Nonconcentration
- Cayley graph expansion / spectral gap
- Helfgott product theorem
- Bourgain-Gamburd expansion machine
- Super-approximation
- Thin subgroup/orbit
- Affine sieve (conceptual definition)

Key results to know and use

- Ruzsa triangle/covering and Balog-Szemeredi-Gowers tools in multiplicative form
- Szemeredi-Trotter implication for a real sum-product estimate
- Bourgain-Katz-Tao finite-field sum-product theorem (proof architecture)
- Helfgott growth theorem for SL2(Fp) (proof architecture with pivot/escape mechanism)
- Breuillard-Green-Tao structure theorem for approximate groups (statement and conceptual architecture)
- Quasirandomness lower bound for nontrivial representations of SL2(Fp) (working use)
- L2 flattening mechanism: failure to flatten implies approximate-group structure
- Bourgain-Gamburd theorem: uniform expansion for suitable Cayley graphs of SL2(Fp)
- Bourgain-Gamburd spectral gap for algebraic generators in SU(2) (architecture)
- Salehi Golsefidy-Varju super-approximation/expansion in perfect groups (statement and place in theory)
- Expansion input to affine sieve (working architecture)

Quarter-specific mastery targets

M1 - Exact language: additive/multiplicative energy, sum-product, K-approximate group, control, quasirandom group, nonconcentration, L2 flattening, escape from subgroups, super-approximation and affine sieve.

M2 - Proofs to reproduce: Ruzsa/covering/energy lemmas used in noncommutative form; one toy BSG/flattening implication; spectral-gap extraction from L2 mixing once the norm is sufficiently flat.

M3 - Research theorems, exact statement + architecture: Bourgain-Katz-Tao finite-field sum-product, Helfgott growth in SL2(F_p), BGT approximate-group structure, Bourgain-Gamburd expansion, and one super-approximation theorem.

M4 - Dependency mastery: for Bourgain-Gamburd, identify separately nonconcentration/escape, product growth, flattening, quasirandomness and final operator-norm contraction; know which theorem supplies each input.

M5 - Exit use: given a family of generating measures on finite groups, diagnose which Bourgain-Gamburd input is missing before claiming expansion.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V28U5: Sum-product phenomena

V28U5P1. [F] Compare A+A and A*A for arithmetic and geometric progressions in R/F_p. Quantify why simultaneous smallness is exceptional.

V28U5P2. [T] Prove an elementary sum-product lower bound via incidences/energy in a toy regime, or reconstruct the Bourgain-Katz-Tao finite-field proof architecture with exact non-concentration assumptions.

V28U5P3. [S] Convert a sum-product estimate into an expansion statement for a simple set of matrices/Mobius transformations, showing explicitly how additive and multiplicative growth interact.

Assigned block V28U6: Growth in groups and approximate groups

V28U6P1. [F] Compute A^2 and A^3 for small subsets of S_3, Heisenberg groups, and SL_2(F_p); identify examples with small tripling caused by subgroup/coset structure.

V28U6P2. [T] Prove noncommutative Ruzsa covering/triangle lemmas in a standard form and derive a small-tripling consequence.

V28U6P3. [R] Compare Helfgott product growth in SL_2(F_p) with the Breuillard-Green-Tao approximate-group theorem: state hypotheses/conclusions and explain the structural obstruction to growth in each.

Assigned block V29U3: Quasirandomness and flattening

V29U3P1. [F] For probability measures mu on a finite group, compute L^1/L^2 norms of convolution powers in subgroup-supported, uniform, and generating examples.

V29U3P2. [T] Prove a finite-group L^2-flattening lemma in a simplified setting from high multiplicative energy plus BSG/product-growth input. Make the contrapositive structure explicit.

V29U3P3. [S] Explain quasirandomness through minimal dimensions of nontrivial representations and prove how this forces a nearly flat measure to mix once its L^2 norm crosses a threshold.

Assigned block V29U4: Escape from subgroups and product growth

V29U4P1. [F] In SL_2(F_p), classify/inspect standard proper subgroup types needed in an expansion proof and compute intersections of a sample generating set with them for small p.

V29U4P2. [T] Prove a toy escape-from-subgroups statement for a free/non-elementary generating set using word maps or random-walk counting in a simplified model.

V29U4P3. [R] Audit Helfgott growth: identify trace amplification, tori/centralizers, escape from subvarieties, and the conclusion |A^3| >= min(|A|^{1+delta},|G|). Mark which ingredients feed Bourgain-Gamburd.

Assigned block V29U5: The Bourgain-Gamburd expansion machine

V29U5P1. [S] Starting from (i) non-concentration in proper subgroups, (ii) product growth/flattening, and (iii) quasirandomness, write and prove the abstract three-phase route to a spectral gap for convolution by a generating measure.

V29U5P2. [T] Translate the operator spectral gap into expansion of the Cayley graph and exponential random-walk mixing. Keep normalization constants consistent throughout.

V29U5P3. [R] Reconstruct one Bourgain-Gamburd theorem for SL_2(F_p): state the generating-set assumptions uniformly in p, the expansion conclusion, and where each of the three machine inputs is established.

Quarter-level integration problems

I1. [R]. Assume a probability measure mu on SL2(F_p) satisfies a quantified nonconcentration condition. Write the precise chain product growth -> L2 flattening -> quasirandom mixing -> operator spectral gap, including one numerical exponent placeholder at each step and where uniformity in p is needed.

I2. [S]. Compare the BKT sum-product obstruction (subfield/additive-multiplicative structure) with Helfgott's group-growth obstruction (proper subgroup/torus concentration); explain how each becomes an approximate-structure obstruction in the Bourgain-Gamburd machine.

Full written solutions required for: V28U5P2, V28U5P3, V28U6P2, V28U6P3, V29U3P2, V29U3P3, V29U4P2, V29U4P3, V29U5P2, V29U5P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 20-page proof map of the Bourgain-Gamburd method. It must isolate: sum-product/growth; escape/nonconcentration; approximate-group obstruction; quasirandomness; L2 flattening; spectral-gap extraction; and one arithmetic application such as affine sieve or super-approximation.

Exit checkpoint

State BKT, Helfgott and Bourgain-Gamburd accurately; prove Ruzsa/BSG tools and one flattening toy lemma; explain the Bourgain-Gamburd machine from memory in under 25 minutes.

Research bridge

This module connects additive combinatorics to group theory, arithmetic dynamics and expander methods. It also clarifies the role of spectral gap in Benoist-Quint/effective dynamics and modern sieve on thin groups.

Q29. Baker, Schmidt Subspace Theorem, and Diophantine synthesis

Quarter overview and reading route

Objective: Reach research-entry level in transcendence/Diophantine approximation via heights, linear forms in logarithms and the Subspace Theorem, and learn when each method is the right tool.

Prerequisites: Q6 geometry of numbers, Q14 algebraic number theory, Q19 local fields/heights.

30-hour allocation: Baker theory 11h; Subspace Theorem 11h; heights/Diophantine applications 5h; exposition 3h.

Core books and chapters/sections

- Baker, Transcendental Number Theory, logarithmic-form chapters.
- Bombieri-Gubler, Heights in Diophantine Geometry, Chs. 1-2; Schmidt, Diophantine Approximation, Subspace-Theorem chapters; Waldschmidt/Evertse surveys.
- Serre, Local Fields, Chs. I-V selectively.
- Bombieri-Gubler, Heights in Diophantine Geometry, Ch. 1 and early Ch. 2; Neukirch local chapters.
- Cassels, An Introduction to the Geometry of Numbers: first pass through lattices, convex bodies and Minkowski theorems; return later to successive minima, transference and Diophantine applications when those topics enter.
- Gruber-Lekkerkerker, Geometry of Numbers, introductory chapters as reference.

Lecture notes and companions

- Use Waldschmidt/Evertse lecture notes as the main expository bridge.
- MIT OCW 18.785 Lectures 8-13.
- MIT OCW 18.785 Number Theory I, Lecture 14 (Geometry of Numbers).

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Absolute values, product formula and heights review	Bombieri-Gubler Chs. 1-2; local-field notes review. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Absolute values, product formula and heights review	Bombieri-Gubler Chs. 1-2; local-field notes review. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Linear forms in logarithms: statements and transcendence inputs	Baker Transcendental Number Theory logarithmic-form chapters; Waldschmidt exposition. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Linear forms in logarithms: statements and transcendence inputs	Baker Transcendental Number Theory logarithmic-form chapters; Waldschmidt exposition. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Effective Diophantine applications: S-unit/Pillai-type equations	Work selected Baker-method applications from Baker/Waldschmidt. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Effective Diophantine applications: S-unit/Pillai-type equations	Work selected Baker-method applications from Baker/Waldschmidt. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
7	Learn: Schmidt Subspace Theorem: formulation and geometry	Schmidt Diophantine Approximation Subspace-Theorem chapters; Evertse surveys. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Schmidt Subspace Theorem: formulation and geometry	Schmidt Diophantine Approximation Subspace-Theorem chapters; Evertse surveys. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: S-unit equations and applications of the Subspace Theorem	Evertse/Bombieri-Gubler expository treatments. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: S-unit equations and applications of the Subspace Theorem	Evertse/Bombieri-Gubler expository treatments. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Comparison: Roth, Baker, Subspace theorem; formulate research-level examples	Build a method-selection chart and study one recent paper using each family of methods. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Comparison: Roth, Baker, Subspace theorem; formulate research-level examples	Build a method-selection chart and study one recent paper using each family of methods. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Heights and Roth, objects and examples. Reading: Bombieri-Gubler/Schmidt preliminaries. Definition/result focus: height; product formula; Roth theorem. Work: Prove elementary height inequalities and state Roth exactly.

Week 2 - Heights and Roth, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove elementary height inequalities and state Roth exactly. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Linear forms in logarithms setup, objects and examples. Reading: Baker/Waldschmidt introductions. Definition/result focus: logarithmic form; algebraic logarithm; height data. Work: Reduce a concrete exponential Diophantine equation to a nonzero linear form.

Week 4 - Linear forms in logarithms setup, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reduce a concrete exponential Diophantine equation to a nonzero linear form. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Baker lower bounds, objects and examples. Reading: Baker/Matveev-style expositions. Definition/result focus: effective lower bound theorem with parameter dependence. Work: Apply a stated theorem to derive an explicit upper bound in a toy equation.

Week 6 - Baker lower bounds, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Apply a stated theorem to derive an explicit upper bound in a toy equation. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - S-unit equations, objects and examples. Reading: Evertse/Schmidt notes. Definition/result focus: S-unit; height; finiteness. Work: Derive standard reductions and compare effective/non-effective inputs.

Week 8 - S-unit equations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive standard reductions and compare effective/non-effective inputs. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Subspace Theorem, objects and examples. Reading: Schmidt/Evertse expositions. Definition/result focus: multiplicative inequalities; exceptional subspaces. Work: State theorem precisely and work through a low-dimensional consequence.

Week 10 - Subspace Theorem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State theorem precisely and work through a low-dimensional consequence. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Applications and comparison, objects and examples. Reading: survey papers. Definition/result focus: S-unit finiteness; Roth relation; recurrences. Work: Complete one Baker-method case study and one Subspace-Theorem application ledger.

Week 12 - Applications and comparison, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Complete one Baker-method case study and one Subspace-Theorem application ledger. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- linear form in logarithms
- S-unit
- auxiliary polynomial
- exceptional subspace
- Subspace Theorem
- valuation
- completion
- uniformizer
- ramification
- absolute/projective height
- Northcott property
- lattice
- covolume
- fundamental domain
- convex body
- successive minima
- dual lattice

Key results to know and use

- Baker lower-bound theorem
- Subspace Theorem
- S-unit equation finiteness
- Roth as related consequence
- Hensel lemma
- structure of finite extensions of Q_p
- product formula
- Northcott theorem
- Blichfeldt lemma
- Minkowski convex body theorem
- Minkowski second theorem
- Dirichlet simultaneous approximation via lattices
- Mahler compactness (later)

Quarter-specific mastery targets

M1 - Exact language: absolute/projective height, logarithmic height, S-unit, linear form in logarithms, exceptional subspace and Subspace Theorem formulation.

M2 - Proofs to reproduce: Northcott-type finiteness in standard projective setting; elementary height identities; one complete Baker-style reduction after accepting a quantitative logarithmic-form bound; derivation of an S-unit finiteness consequence from the Subspace Theorem.

M3 - Research theorems, exact statement + architecture: a quantitative Baker/Matveev-type lower bound with all degree/height/nonzero hypotheses; Schmidt Subspace Theorem in a standard linear-forms formulation.

M4 - Method choice: distinguish when logarithmic forms give effective bounds from when the Subspace Theorem gives qualitative finiteness/structure; identify ineffectivity/constant dependence honestly.

M5 - Exit use: solve one explicit exponential Diophantine equation by a Baker-style chain and one unit-equation/finiteness problem by a Subspace-Theorem chain.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V25U3: Classical Diophantine approximation

V25U3P1. [F] Compute convergents and approximation constants for sqrt(2), golden ratio, and one nonquadratic irrational. Verify the sharpness phenomenon behind Hurwitz's theorem for the golden ratio.

V25U3P2. [T] Prove Dirichlet's theorem, Hurwitz's theorem in dimension one, and characterize badly approximable reals via bounded continued-fraction partial quotients.

V25U3P3. [S] For a quadratic irrational alpha, derive a uniform lower bound q||q alpha|| >= c(alpha)>0 using periodic continued fractions or algebraic arguments.

Assigned block V26U1: Heights and finiteness principles

V26U1P1. [F] Compute absolute logarithmic heights of rational numbers, quadratic algebraic numbers, roots of unity, and points in P^n(Q) from local/global definitions.

V26U1P2. [T] Prove basic height inequalities h(alpha beta), h(alpha+beta), h(alpha^n), and Northcott finiteness for bounded degree and height in a selected formulation.

V26U1P3. [S] Use heights to turn a naive infinite Diophantine search into a finite search once a height bound is given. Work a simple Thue/Mordell-type toy example.

Assigned block V26U3: Baker theory and linear forms in logarithms

V26U3P1. [F] For equations such as 2^m-3^n=1 or x^2-Dy^2=1, isolate a candidate nonzero linear form in logarithms whose smallness follows from a hypothetical large solution.

V26U3P2. [T] Assuming a standard Baker-Wustholz/Matveev lower bound, combine it with an elementary upper bound to derive an explicit upper bound for an exponent in one exponential Diophantine equation.

V26U3P3. [R] Audit one modern explicit linear-form theorem: exact degree/height/logarithm hypotheses, dependence of constants, and which parameters dominate applications. Then compare with a qualitative transcendence statement.

Assigned block V26U4: Schmidt Subspace Theorem and unit equations

V26U4P1. [F] Rewrite a unit equation x+y=1 in S-units as a projective linear-form problem and compute solutions in one small S for Q if feasible.

V26U4P2. [T] State Schmidt's Subspace Theorem in a standard absolute-value/height formulation and derive the finiteness of nondegenerate solutions to a simple S-unit equation at proof-architecture level.

V26U4P3. [S] Compare Roth and the Subspace Theorem by embedding rational approximation into a two-variable linear-form inequality. Identify exactly how the latter generalizes the former.

Assigned block V26U5: Integral points and effective equations

V26U5P1. [F] Solve several classical Thue/Pell/Thue-Mahler equations computationally after deriving a rigorous finite search region from elementary or Baker-type estimates.

V26U5P2. [T] State Siegel's theorem on integral points and contrast qualitative finiteness with effective bounds from Baker theory. Give examples where each applies naturally.

V26U5P3. [R] Choose one effective Diophantine theorem (Catalan/Mihailescu, S-unit bounds, linear recurrence equation). Trace the reduction from the original equation to heights/logarithmic forms.

Quarter-level integration problems

I1. [S]. Solve one concrete equation such as x^2-Dy^2=k or a^x-b^y=c by reducing it to a nonzero linear form in logarithms; carry the argument through to an explicit finite search bound after accepting the chosen Baker/Matveev estimate.

I2. [S]. Starting from the Subspace Theorem, derive finiteness of solutions to an S-unit equation x+y=1 outside a finite set of exceptional subspaces, and identify exactly where the hypotheses on valuations/linear forms enter.

Full written solutions required for: V25U3P2, V25U3P3, V26U1P2, V26U1P3, V26U3P2, V26U3P3, V26U4P2, V26U4P3, V26U5P2, V26U5P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 25-page comparison of Baker's method and the Subspace Theorem, with one complete worked Diophantine equation for each method.

Exit checkpoint

State a quantitative linear-forms theorem with its hypotheses, execute a Baker-style reduction in an example, state the Subspace Theorem in projective/linear-form language, and derive an S-unit finiteness consequence.

Research bridge

This is a direct endpoint of the arithmetic spine and supports future work on Diophantine approximation, recurrence sequences and digit phenomena.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q30. Hilbert's Tenth Problem, Diophantine definability, and undecidability

Quarter overview and reading route

Objective: Study Hilbert's Tenth Problem as the meeting point of logic and Diophantine equations: Diophantine definability, MRDP, recurrence sequences, undecidability, and arithmetic variants over Q, number fields and subrings.

Prerequisites: Q4 mathematical logic I, Q8/Q14/Q19 algebraic number theory/local fields, and Q24 elliptic curves are useful; Q29 Baker/Subspace provides further Diophantine maturity.

30-hour allocation: computability review 4h; Diophantine representation/MRDP 12h; model theory/decidability 6h; arithmetic variants 6h; exposition 2h.

Core books and chapters/sections

- Matiyasevich, Hilbert's Tenth Problem: use the chapters on Diophantine representation, Pell/Fibonacci recurrences, exponentiation and the MRDP conclusion.
- Poonen, Hilbert's Tenth Problem over rings of number-theoretic interest: survey sections on Diophantine models, number fields/subrings and H10(Q).
- Koymans-Pagano, Hilbert's tenth problem via additive combinatorics (2025), followed by their 2026 expository article Hilbert's tenth problem for finitely generated rings: read theorem statements and the elliptic-curve rank-stability/2-descent/additive-combinatorics architecture.
- Optional research bridge: Poonen and Shlapentokh sources on Diophantine definability over number fields and large subrings; keep H10(Q) logically separate.

Lecture notes and primary-paper companions

- Use Q4 notes to review Turing machines, c.e. sets and reductions rather than relearning them.
- A modern local-global H10 survey may be read in Weeks 10-12 to see how definability, quadratic forms and arithmetic geometry interact.
- Optional research update: recent expository work on H10 for finitely generated rings, read only after the classical DPRM and Poonen routes are secure.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Decision problems, recursive and c.e. sets review	Poonen AWS Sections 2-5; Q4 notes.	Reprove halting undecidability and normalize definitions over rings.
2	Diophantine sets and closures	Matiyasevich Ch. 1; Poonen Sections 5-6.	Prove closure of Diophantine sets under basic logical operations where valid.
3	Davis-Putnam-Robinson strategy	Poonen outline of DPRM; Matiyasevich Chs. 1-2.	Write a dependency graph for “c.e. -> exponential Diophantine -> Diophantine.”
4	Pell/Fibonacci-type recurrences and exponentiation	Matiyasevich Ch. 2.	Work through the recurrence mechanism making exponentiation Diophantine.
5	Coding and universal Diophantine equations	Matiyasevich Chs. 3-4.	Construct explicit coding steps and a small universal-family toy model.
6	MRDP / Hilbert X unsolvability	Matiyasevich Ch. 5; Poonen Section 6.	Give proof architecture of MRDP and derive undecidability of H10(Z).
7	First-order definability and decidable fields	Poonen Sections 7-11.	Compare Diophantine vs first-order definability; study algebraically closed/real closed/p-adic

Week	Main focus	Reading / lecture notes	Required output
			cases.
8	H10 over Q and number fields	Poonen Sections 12-14.	Map open/known cases, elliptic-curve inputs and definability strategies.
9	Large subrings of Q and arithmetic geometry	Poonen Section 15 and JAMS paper.	Understand elliptic-divisibility/Diophantine model mechanism at architecture level.
10	Local-global/model-theoretic methods	Modern survey on local-global methods; selected Koenigsmann/Park context.	Write a comparison of existential and first-order definability routes.
11	Connections to Godel, Turing, undecidability in number theory	Review Q4; Matiyasevich Ch. 7 selections.	Build one unified reduction diagram but keep theorem hypotheses distinct.
12	Research frontier and synthesis	Poonen survey plus a recent expository update on finitely generated rings.	Oral exam + 8-page research map of H10 variants.

Week 1 - Diophantine definability, objects and examples. Reading: Matiyasevich opening chapters; Poonen H10 survey. Definition/result focus: Diophantine set/model; closure operations. Work: Encode divisibility, conjunction and finite unions by polynomial equations.

Week 2 - Diophantine definability, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Encode divisibility, conjunction and finite unions by polynomial equations. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Computability-to-Diophantine bridge, objects and examples. Reading: Matiyasevich/Davis-Putnam-Robinson history sections. Definition/result focus: c.e. sets; exponential Diophantine representation. Work: Write the DPR dependency chain and compare with Godel/Turing reductions.

Week 4 - Computability-to-Diophantine bridge, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Write the DPR dependency chain and compare with Godel/Turing reductions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Pell/Fibonacci recurrence mechanisms, objects and examples. Reading: Matiyasevich chapters on exponentiation. Definition/result focus: recurrence growth; Diophantine coding. Work: Work explicit Pell/Fibonacci identities and identify the exponential-growth ingredient.

Week 6 - Pell/Fibonacci recurrence mechanisms, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work explicit Pell/Fibonacci identities and identify the exponential-growth ingredient. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - MRDP and H10(Z), objects and examples. Reading: Matiyasevich proof overview. Definition/result focus: MRDP theorem; H10(Z) undecidability. Work: Reconstruct the deduction from MRDP to undecidability and audit quantifiers.

Week 8 - MRDP and H10(Z), theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reconstruct the deduction from MRDP to undecidability and audit quantifiers. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Other rings and H10(Q), objects and examples. Reading: Poonen survey/number-field H10 literature. Definition/result focus: Diophantine models; ring of integers; rational-field obstruction. Work: Separate known transfer mechanisms from the still-open H10(Q) problem.

Week 10 - Other rings and H10(Q), theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Separate known transfer mechanisms from the still-open H10(Q) problem. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - 2025-26 finitely generated ring theorem, objects and examples. Reading: Koymans-Pagano 2025 paper and 2026 exposition. Definition/result focus: H10 undecidable for every infinite ring finitely generated over Z; elliptic-curve rank stability + 2-descent/additive combinatorics architecture. Work: State theorem exactly, explain why Q is not covered, and give a dependency audit of the elliptic-curve/additive-combinatorics input.

Week 12 - 2025-26 finitely generated ring theorem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State theorem exactly, explain why Q is not covered, and give a dependency audit of the elliptic-curve/additive-combinatorics input. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Diophantine equation over a ring R
- Diophantine/existentially definable subset of R^n
- Recursive/decidable and c.e./listable set
- Diophantine representation of a relation/function
- Exponential Diophantine equation

- Universal Diophantine equation
- Diophantine model of one ring in another
- Hilbert's Tenth Problem over R
- First-order theory of a ring/field
- Existential theory
- Quantifier elimination (working notion)
- Elliptic divisibility sequence / elliptic-curve Diophantine model (orientation)
- Local-global definability strategy

Key results to know and use

- DPRM/MRDP theorem: every computably enumerable subset of Z is Diophantine (exact statement; proof architecture).
- Matiyasevich breakthrough: exponential growth/recurrence is Diophantinely representable in the form needed to complete MRDP (mechanism understood, not a one-line black box).
- Undecidability of Hilbert's Tenth Problem over Z (full deduction from MRDP).
- Closure/coding properties of Diophantine sets and existence of universal Diophantine equations (working results).
- Comparison cases: decidability for algebraically closed and real closed fields via quantifier elimination; p-adic decidability perspective stated with the chosen language/theorem formulation.
- Koymans-Pagano theorem (2025): Hilbert's tenth problem is undecidable for every infinite ring finitely generated over Z; understand the reduction to rank-stable elliptic curves and the role of full rational 2-torsion, 2-descent and additive combinatorics.
- H10(Q) remains open: understand why Q is not a finitely generated ring over Z and why the finitely-generated-ring theorem does not settle the rational case.
- Logical distinction among undecidability, independence from a formal theory, and unprovability of an individual sentence.

Quarter-specific mastery targets

M1 - Exact language: Diophantine set, existential polynomial definition, recursively enumerable set, MRDP theorem, universal Diophantine representation and Hilbert's tenth decision problem.

M2 - Proofs to reproduce: closure of Diophantine sets under elementary logical constructions; halting/r.e. reduction logic; derivation H10(Z) undecidable from MRDP; selected Pell/recurrence identities used to encode exponential growth.

M3 - Architecture-only but precise: Davis-Putnam-Robinson reduction and Matiyasevich's exponential/recurrence breakthrough; state what each stage contributed instead of treating MRDP as one black box.

M4 - Boundary of knowledge: distinguish H10(Z), the now-settled case of infinite rings finitely generated over Z, and H10(Q); explain explicitly why the 2025 finitely-generated-ring theorem does not include Q.

M5 - Exit use: take a simple arithmetic relation and construct an explicit Diophantine encoding, then explain how such encodings interact with computability reductions.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V5U3: Computability and undecidability

V5U3P1. [F] Give an explicit coding of finite binary strings by natural numbers and show concatenation/length are computable. Use the encoding to illustrate how syntax can become arithmetic data.

V5U3P2. [T] Prove undecidability of the halting problem by diagonalization. Then reduce the halting problem to another undecidable decision problem of your choice, writing the reduction as an algorithmic transformation.

V5U3P3. [S] Compare Cantor diagonalization, the halting diagonal argument, and the later Godel diagonal lemma. Write a table specifying the object diagonalized against and the contradiction/fixed-point produced.

Assigned block V26U6: Hilbert's tenth problem and Diophantine undecidability

V26U6P1. [F] Show that conjunction and finite disjunction of Diophantine conditions can be encoded by polynomial equations over Z. Encode divisibility and selected bounded arithmetic relations explicitly.

V26U6P2. [T] Work out Pell/Fibonacci recurrence identities that exhibit exponential growth through polynomial relations, and explain why such growth was the missing ingredient in the Davis-Putnam-Robinson route.

V26U6P3. [R] Reconstruct MRDP as the chain c.e. -> exponential-Diophantine -> Diophantine -> undecidability of H10(Z). Then state the Koymans-Pagano 2025 theorem for every infinite ring finitely generated over Z, explain the elliptic-curve/2-descent/additive-combinatorics architecture, and prove from definitions why this theorem does not include Q; finish with the exact open status of H10(Q).

Assigned block V1U4: Cardinality and countability

V1U4P1. [F] Construct explicit enumerations of Z, Q, the set of finite binary strings, and Z^2. Explain why the same strategy fails for infinite binary sequences.

V1U4P2. [T] Reproduce Cantor's diagonal proof that $\{0,1\}^N$ is uncountable, then prove P(N) is uncountable by a direct diagonal argument. State exactly where self-reference enters.

V1U4P3. [S] Prove that the set of algebraic numbers is countable and hence transcendental numbers exist. Strengthen the conclusion by showing that almost every real number is transcendental once measure theory becomes available; note clearly what is not yet proved at this stage.

Assigned block V5U4: Godel incompleteness

V5U4P1. [T] Construct Godel numbering for a toy formal language and explicitly code substitution for a few formulas. Identify which syntactic operations must be primitive recursive in a full proof.

V5U4P2. [T] Assuming representability and the diagonal lemma, derive a sentence G with G <-> not Prov(G). State exactly what consistency or soundness assumptions are required for each conclusion about G.

V5U4P3. [S] Build a proof-architecture map for the second incompleteness theorem: derivability conditions, formalized first incompleteness, and the obstruction to proving Con(T). Mark which ingredients are internalized in T.

Quarter-level integration problems

I1. [S]. Encode simultaneously x|y, z=x+y and w=xy as a single polynomial equation over Z using sums of squares/auxiliary variables; distinguish logical equivalence over Z from over N.

I2. [R]. Compare two undecidability architectures: MRDP over Z and Koymans-Pagano over arbitrary infinite finitely generated Z-algebras. For each, give the exact theorem statement, the encoding/model used, the genuinely new arithmetic input, and the remaining reason H10(Q) is outside the theorem.

Full written solutions required for: V5U3P2, V5U3P3, V26U6P2, V26U6P3, V1U4P2, V1U4P3, V5U4P2, V5U4P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Prepare a 20-page seminar manuscript beginning with Hilbert's 1900 question, proving the reduction from MRDP to undecidability, explaining the exponentiation-via-recurrence breakthrough, and ending with a carefully sourced map of H10 over Q and arithmetic rings.

Exit checkpoint

From memory: define Diophantine set, state MRDP, explain the recurrence/exponentiation step, derive H10(Z) undecidable, and distinguish H10(Q) from H10(Z). Give a precise comparison with Godel incompleteness.

Research bridge

This is the main logic-number-theory bridge in the curriculum. It also links Pell-type recurrences, elliptic curves, definability, model theory and arithmetic geometry.

Q31. Borel density, Borel-Harish-Chandra, KAM/billiards, and class field theory I

Quarter overview and reading route

Objective: Study two separate advanced branches in parallel: arithmetic lattices/Borel theory plus local-global class field theory, and Hamiltonian/KAM/billiard dynamics.

Prerequisites: Q17 Lie/arithmetic groups, Q19 local fields, Q25 ergodic theory; Q9 ODE/smooth geometry.

30-hour allocation: Borel/BHC and arithmetic groups 8h; class field theory 10h; KAM/billiards 9h; synthesis 3h.

Core books and chapters/sections

- Witte Morris, Introduction to Arithmetic Groups, Borel density/BHC chapters.
- Arnold, Mathematical Methods of Classical Mechanics, Hamiltonian chapters; Tabachnikov, Geometry and Billiards; Poeschel/de la Llave KAM notes.
- Neukirch, Algebraic Number Theory, class-field-theory chapters; Milne, Class Field Theory notes.
- Hall, Lie Groups, Lie Algebras, and Representations, Chs. 1-8 selectively.
- Witte Morris, Introduction to Arithmetic Groups, early Lie/Haar/lattice chapters; Milne, Algebraic Groups notes.
- Serre, Local Fields, Chs. I-V selectively.
- Bombieri-Gubler, Heights in Diophantine Geometry, Ch. 1 and early Ch. 2; Neukirch local chapters.

Lecture notes and companions

- MIT OCW 18.786 Class Field Theory, Lectures 1-23; J.S. Milne CFT notes.
- MIT OCW 18.755 Introduction to Lie Groups, Chapters I-II.
- MIT OCW 18.785 Lectures 8-13.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Borel density: statement, Zariski closure and representation-theoretic proof ideas	Witte Morris Borel-density chapters; complementary arithmetic-group notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Borel density: statement, Zariski closure and representation-theoretic proof ideas	Witte Morris Borel-density chapters; complementary arithmetic-group notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Borel-Harish-Chandra and arithmetic lattice construction	Witte Morris BHC/arithmetic-lattice chapters; revisit reduction theory/geometry of numbers. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Borel-Harish-Chandra and arithmetic lattice construction	Witte Morris BHC/arithmetic-lattice chapters; revisit reduction theory/geometry of numbers. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Hilbert symbols, local reciprocity and Tate cohomology entry	MIT 18.786 Lectures 1-13; Milne Class Field Theory opening chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Hilbert symbols, local reciprocity and Tate cohomology entry	MIT 18.786 Lectures 1-13; Milne Class Field Theory opening chapters. Second pass: theorem	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		proofs, harder exercises, and a one-page proof reconstruction.	
7	Learn: Local class field theory and norm groups	MIT 18.786 Lectures 14-18; Milne corresponding chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Local class field theory and norm groups	MIT 18.786 Lectures 14-18; Milne corresponding chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Hamiltonian systems, action-angle variables, small divisors	Arnold Mathematical Methods of Classical Mechanics Hamiltonian/action-angle chapters; KAM lecture notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Hamiltonian systems, action-angle variables, small divisors	Arnold Mathematical Methods of Classical Mechanics Hamiltonian/action-angle chapters; KAM lecture notes. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: KAM theorem architecture and billiard maps	Poeschel/de la Llave KAM notes; Tabachnikov Geometry and Billiards selected chapters; preview global CFT. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: KAM theorem architecture and billiard maps	Poeschel/de la Llave KAM notes; Tabachnikov Geometry and Billiards selected chapters; preview global CFT. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Borel density review, objects and examples. Reading: Witte Morris/Raghunathan. Definition/result focus: Zariski density of lattices; invariant vectors. Work: Reprove key representation-theoretic lemma and apply to SO(2,1) examples.

Week 2 - Borel density review, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reprove key representation-theoretic lemma and apply to SO(2,1) examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Borel-Harish-Chandra/reduction theory, objects and examples. Reading: Witte Morris/Raghunathan. Definition/result focus: arithmetic lattice criterion; finite covolume. Work: Trace one Q-group arithmetic lattice example in full.

Week 4 - Borel-Harish-Chandra/reduction theory, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Trace one Q-group arithmetic lattice example in full. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Hamiltonian systems and KAM setup, objects and examples. Reading: Arnold classical mechanics/KAM notes. Definition/result focus: symplectic form; integrable system; action-angle variables. Work: Compute action-angle coordinates in low-dimensional examples.

Week 6 - Hamiltonian systems and KAM setup, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute action-angle coordinates in low-dimensional examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - KAM theorem mechanism, objects and examples. Reading: Poeschel/de la Llave notes. Definition/result focus: Diophantine frequency; small divisors; twist/nondegeneracy. Work: Carry out one homological equation estimate and theorem hypothesis audit.

Week 8 - KAM theorem mechanism, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out one homological equation estimate and theorem hypothesis audit. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Billiards, objects and examples. Reading: Tabachnikov billiards chapters. Definition/result focus: billiard map; caustic; invariant curve. Work: Analyze circle/ellipse billiards and connect to twist maps.

Week 10 - Billiards, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Analyze circle/ellipse billiards and connect to twist maps. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Class field theory I, objects and examples. Reading: Neukirch/Milne/Serre local-global CFT entry. Definition/result focus: Artin map; ray class group; local reciprocity orientation. Work: Compute ray/class groups in examples and map local-to-global reciprocity prerequisites.

Week 12 - Class field theory I, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute ray/class groups in examples and map local-to-global reciprocity prerequisites. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Zariski-dense subgroup
- arithmetic subgroup/lattice
- Haar measure and homogeneous space
- Hamiltonian/symplectic system
- action-angle variable
- Diophantine frequency
- twist/nondegeneracy condition
- billiard map and caustic
- local field and Hilbert symbol
- idele and idele class group
- Artin reciprocity map
- ray class group

Key results to know and use

- Borel density theorem (precise hypotheses and conclusion).
- Borel-Harish-Chandra theorem/reduction-theory route to arithmetic lattices (architecture).
- Iwasawa decomposition and invariant-measure criterion on G/H as tools used in the density/lattice discussion.
- A precise KAM theorem in the chosen finite-dimensional nearly-integrable setting, including Diophantine and nondegeneracy hypotheses.
- Area/symplectic preservation of the billiard map; integrability/caustics for circle or ellipse as explicit models.
- Local reciprocity and global Artin reciprocity in the selected class-field-theory formulation.
- Kronecker-Weber as the Q example; compute at least one local norm/Hilbert-symbol example.

Quarter-specific mastery targets

M1 - Exact language: arithmetic lattice, Zariski density, Borel-Harish-Chandra hypothesis, local reciprocity/Artin map, Hilbert symbol, Hamiltonian action-angle coordinate, Diophantine frequency and KAM nondegeneracy.

M2 - Proofs to reproduce: one standard consequence of Borel density and one arithmetic-lattice example from BHC; explicit local norm/Hilbert-symbol calculations; area-preserving billiard map derivation; a toy small-divisor/KAM iterative estimate.

M3 - Architecture: state Borel density, Borel-Harish-Chandra and local Artin reciprocity exactly; state a standard KAM theorem with regularity/nondegeneracy/smallness hypotheses.

M4 - Computation: verify lattice covolume/arithmetic subgroup examples, compute local symbols/norm groups, and calculate twist maps/rotation numbers for simple billiards.

M5 - Exit use: separate rigidity/arithmetic-lattice inputs, local-class-field inputs and small-divisor dynamics rather than merging them into a generic 'advanced structure' toolkit.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V20U2: Lie groups, lattices and arithmeticity background

V20U2P1. [F] Show SL_2(R)/SL_2(Z) parametrizes unimodular lattices in R^2 and identify the stabilizer of Z^2. Compute diagonal and unipotent actions on lattice bases.

V20U2P2. [T] Prove unimodularity of a reductive Lie group in a concrete adjoint-determinant formulation and derive the criterion for a G-invariant measure on G/H in examples.

V20U2P3. [S] Work through Borel density/Borel-Harish-Chandra in one arithmetic lattice example at theorem-application level: identify algebraic group, lattice, and conclusion.

Assigned block V24U5: Adeles, ideles and class field theory

V24U5P1. [F] Write explicit finite adeles/ideles for Q with only a few nonunit components and verify restricted-product conditions. Embed Q diagonally and test discreteness/cocompact quotient ideas.

V24U5P2. [T] Prove strong approximation for Q in a selected elementary form (simultaneous approximation at finitely many places) using CRT/weak approximation.

V24U5P3. [R] State local and global Artin reciprocity precisely for one standard formulation. Work out the cyclotomic/Q example that recovers Kronecker-Weber/class-field-theoretic reciprocity data.

Assigned block V19U5: Small divisors and KAM

V19U5P1. [F] Solve a cohomological equation u(x+alpha)-u(x)=f(x) by Fourier series and identify the small divisors |e^{2pi i k alpha}-1|. Compare Diophantine and Liouville alpha.

V19U5P2. [T] Prove a quantitative bound for the cohomological equation assuming a Diophantine condition on alpha and smooth/analytic decay of f-hat.

V19U5P3. [R] Build the KAM iteration architecture for a near-integrable Hamiltonian: linearized equation, loss of derivatives/small divisors, parameter exclusion, quadratic convergence, surviving Cantor set.

Assigned block V19U6: Billiards and hyperbolic examples

V19U6P1. [F] Derive the billiard map for a circular/elliptical billiard and identify conserved quantities/invariant curves in integrable cases.

V19U6P2. [T] For dispersing Sinai billiards, compute how curvature of wave fronts evolves across free flight/reflection in a model setup and relate this to hyperbolicity.

V19U6P3. [R] Compare smooth hyperbolic systems and billiards with singularities: identify which standard Anosov arguments need modification because of grazing collisions/discontinuity sets.

Quarter-level integration problems

I1. [S]. For SL2(Z) inside SL2(R), state the Borel-Harish-Chandra theorem that yields lattice status and separately state Borel density for the quotient; explain why neither theorem is a consequence of the other.

I2. [S]. In a twist-map toy model, solve the cohomological equation Fourier coefficient by Fourier coefficient, derive the small-divisor denominator, and impose a Diophantine condition sufficient for convergence at the chosen regularity level.

Full written solutions required for: V20U2P2, V20U2P3, V24U5P2, V24U5P3, V19U5P2, V19U5P3, V19U6P2, V19U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Produce three short expositions: why arithmetic groups are lattices, a local reciprocity example, and the small-divisor mechanism behind a KAM iteration.

Exit checkpoint

Explain Borel density and BHC hypotheses/conclusions, compute Hilbert symbols/norm groups in examples, state local reciprocity precisely, derive an area-preserving billiard map, and explain a KAM nondegeneracy condition.

Research bridge

Borel theory leads into Ratner/Zimmer; CFT supplies the abelian local-global template for automorphic forms; KAM/billiards forms the smooth-dynamics specialization.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q32. Falconer/Furstenberg/Kakeya and additive combinatorics II

Quarter overview and reading route

Objective: Study modern geometric-measure and incidence methods surrounding Falconer distance, Furstenberg sets and Kakeya, while returning to higher additive combinatorics and sum-product phenomena.

Prerequisites: Q15 harmonic analysis, Q22 restriction/Kakeya, Q11 additive combinatorics, Q27 decoupling.

30-hour allocation: Geometric measure/Falconer/Furstenberg 15h; additive combinatorics 10h; synthesis 5h.

Core books and chapters/sections

- Mattila, Fourier Analysis and Hausdorff Dimension, energy/projection/distance chapters.
- Guth, Polynomial Methods in Combinatorics, incidence/polynomial-partitioning chapters; Tao-Vu later Gowers/Fourier chapters.
- Tao-Vu, Additive Combinatorics, Chs. 1-3 first pass; later Fourier/Gowers chapters.
- Nathanson, Additive Number Theory: Inverse Problems and the Geometry of Sumsets, early chapters.
- Stein, Harmonic Analysis, restriction/oscillatory-integral chapters selectively.
- Demeter, Fourier Restriction, Decoupling, and Applications, Chs. 1-2.

Lecture notes and companions

- Tao arithmetic-combinatorics Notes 4-6; use author surveys on modern Falconer/Furstenberg/Kakeya results.
- Tao UCLA Math 254A arithmetic combinatorics notes, Lecture Notes 1-6.
- Use Tao/Guth restriction-Kakeya lecture notes/surveys; revisit MIT 18.156 Strichartz lectures.

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Hausdorff dimension, Frostman measures and energy integrals	Mattila energy/Hausdorff-dimension chapters; revisit Fourier transforms of measures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Hausdorff dimension, Frostman measures and energy integrals	Mattila energy/Hausdorff-dimension chapters; revisit Fourier transforms of measures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Projection theorems and Falconer distance formulation	Mattila projection/distance chapters; selected modern Falconer surveys. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Projection theorems and Falconer distance formulation	Mattila projection/distance chapters; selected modern Falconer surveys. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Kakeya/Furstenberg sets, incidence bounds and polynomial partitioning	Guth Polynomial Methods incidence/partitioning chapters; restriction-Kakeya notes. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Kakeya/Furstenberg sets, incidence bounds and polynomial partitioning	Guth Polynomial Methods incidence/partitioning chapters; restriction-Kakeya notes. Second pass: theorem proofs, harder exercises, and a one-page proof	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		reconstruction.	
7	Learn: Finite-field sum-product, Szemeredi-Trotter analogues	Tao UCLA 254A Note 6; Tao-Vu relevant chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Finite-field sum-product, Szemeredi-Trotter analogues	Tao UCLA 254A Note 6; Tao-Vu relevant chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Gowers norms, Roth/Szemeredi and inverse-theorem philosophy	Tao-Vu Fourier/Gowers chapters; Tao 254A Notes 4-5. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Gowers norms, Roth/Szemeredi and inverse-theorem philosophy	Tao-Vu Fourier/Gowers chapters; Tao 254A Notes 4-5. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Modern geometric-combinatorial synthesis: read selected Wang/Guth/Zahl-type papers via surveys first	Choose one Falconer, one Furstenberg and one Kakeya paper; produce dependency maps rather than attempting all details immediately. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Modern geometric-combinatorial synthesis: read selected Wang/Guth/Zahl-type papers via surveys first	Choose one Falconer, one Furstenberg and one Kakeya paper; produce dependency maps rather than attempting all details immediately. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Frostman energy and Fourier dimension, objects and examples. Reading: Mattila opening energy/Fourier chapters. Definition/result focus: Frostman lemma; s-energy; Fourier-energy identity. Work: Prove energy identity in the chosen normalization and compute model energies.

Week 2 - Frostman energy and Fourier dimension, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove energy identity in the chosen normalization and compute model energies. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Falconer distance problem, objects and examples. Reading: Mattila distance chapters + Guth-Iosevich-Ou-Wang paper. Definition/result focus: Mattila integral; pinned distance set; planar 5/4 pinned positive-measure theorem. Work: Derive Mattila criterion and audit the 5/4 theorem’s main inputs.

Week 4 - Falconer distance problem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive Mattila criterion and audit the 5/4 theorem’s main inputs. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Furstenberg sets, objects and examples. Reading: Ren-Wang paper plus expository background. Definition/result focus: (s,t)-Furstenberg set; sharp planar bound min(s+t,(3s+t)/2,s+1). Work: Check parameter regimes/equality cases and map incidence/multiscale mechanism.

Week 6 - Furstenberg sets, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Check parameter regimes/equality cases and map incidence/multiscale mechanism. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Kakeya in three dimensions, objects and examples. Reading: Wang-Zahl 2025 + Guth 2025 outline. Definition/result focus: Kakeya set; Minkowski/Hausdorff dimension; non-clustering/sticky structures. Work: State the R3 theorem exactly and build a proof architecture without treating higher-dimensional Kakeya as solved.

Week 8 - Kakeya in three dimensions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State the R3 theorem exactly and build a proof architecture without treating higher-dimensional Kakeya as solved. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Polynomial partitioning/incidences, objects and examples. Reading: Guth polynomial methods and related notes. Definition/result focus: polynomial partitioning; cells/wall; broad-narrow orientation. Work: Prove a toy incidence bound and track scale induction.

Week 10 - Polynomial partitioning/incidences, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove a toy incidence bound and track scale induction. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Additive-combinatorial synthesis, objects and examples. Reading: Tao-Vu later Fourier/Gowers sections + Ren-Wang sum-product corollary. Definition/result focus: Gowers norms; inverse/counting mechanisms; discretized sum-product bridge. Work: Discretize one geometric set problem and translate the incidence estimate back to dimension.

Week 12 - Additive-combinatorial synthesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Discretize one geometric set problem and translate the incidence estimate back to dimension. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Hausdorff dimension
- Frostman measure
- energy integral
- distance set
- Mattila integral
- polynomial partitioning
- Gowers U^k norm
- sumset
- doubling constant
- additive energy
- Ruzsa distance
- Freiman homomorphism
- restriction operator
- extension operator
- Knapp example
- oscillatory integral
- stationary phase
- Kakeya set

Key results to know and use

- Frostman lemma and the Fourier-energy identity (full working proofs in the form used).
- Marstrand projection theorem and Mattila integral criterion (exact statements; proof architecture).
- Guth-Iosevich-Ou-Wang planar pinned-distance theorem: for compact E subset R^2 with dim_H(E)>5/4, there is a point with pinned distance set of positive Lebesgue measure (state the precise source formulation used).
- Ren-Wang planar (s,t)-Furstenberg theorem: dim_H E >= min(s+t,(3s+t)/2,s+1), with the parameter ranges stated correctly.
- Wang-Zahl theorem (2025): every Kakeya set in R^3 has Hausdorff and Minkowski dimension 3. Keep the general Kakeya conjecture in dimensions >=4 separate.
- Polynomial partitioning and wave-packet/incidence mechanisms used in modern geometric analysis (working toy versions + research architecture).
- Cauchy-Davenport, Ruzsa triangle/covering, Plunnecke inequalities and Balog-Szemeredi-Gowers (proof/use level).
- Freiman/inverse-additive theory and Gowers-norm counting/inverse principles at the level needed for applications.

Quarter-specific mastery targets

M1 - Exact language: Frostman measure, s-energy, distance measure, Mattila integral, Furstenberg/Kakeya set at finite scale, Gowers U^k norm and additive energy at multiple scales.

M2 - Proofs to reproduce: Frostman's lemma in the chosen direction(s); Fourier-energy identity sufficient for the Mattila criterion; one incidence/polynomial-partitioning toy estimate; Gowers norm identities in finite abelian groups.

M3 - Research theorems, exact statements + architectures: the planar >5/4 pinned-distance theorem of Guth-Iosevich-Ou-Wang, the sharp planar Ren-Wang (s,t)-Furstenberg theorem, the Wang-Zahl R^3 Kakeya theorem, and one additive-combinatorial inverse/counting result actually used in a selected proof route.

M4 - Scaling/hypothesis audit: calculate critical dimensional exponents and construct model configurations showing why naive energy/incidence estimates fail at endpoints.

M5 - Exit use: take a geometric problem, discretize it at scale delta, identify the resulting incidence/additive structure, and translate the quantitative bound back to dimension.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V35U4: Falconer distance problem

V35U4P1. [F] For a Frostman measure mu, derive the Fourier representation of the distance measure and identify the Mattila integral controlling L^2 density.

V35U4P2. [T] Reproduce the classical Falconer threshold from a chosen Fourier/energy estimate in the dimension range covered by your source.

V35U4P3. [R] State the planar pinned-distance result of Guth-Iosevich-Ou-Wang in the source formulation (in particular the dim_H(E)>5/4 threshold giving a pinned distance set of positive measure). Then compare the Fourier/decoupling, incidence/polynomial and radial-projection mechanisms that appear in modern Falconer work, marking which conclusions are theorems and which Falconer thresholds remain open.

Assigned block V35U5: Furstenberg sets and modern geometric combinatorics

V35U5P1. [F] Construct standard Furstenberg set examples and derive basic lower bounds from counting/slicing. Work both continuous and discretized formulations.

V35U5P2. [T] Translate a discretized Furstenberg problem into an incidence problem between points and tubes/lines, keeping scale and multiplicity parameters explicit.

V35U5P3. [R] State the Ren-Wang sharp planar (s,t)-Furstenberg theorem, including s,t ranges and the bound dim_H(E)>=min(s+t,(3s+t)/2,s+1). Analyze which term is dominant in each parameter region, then reconstruct the multiscale/incidence architecture and one sum-product/projection consequence.

Assigned block V28U3: Fourier analysis and Roth

V28U3P1. [F] Compute Fourier transforms of indicators in Z/NZ for intervals, subgroups, and random subsets. Verify Parseval and convolution identities.

V28U3P2. [T] Prove Roth's theorem in Z/NZ via a Fourier density-increment argument in a simplified quantitative form. Identify the large Fourier coefficient and progression/subspace increment step.

V28U3P3. [S] Compare physical-space BSG/Freiman and Fourier-space Roth methods: given three additive problems, decide which framework is more natural and explain why.

Assigned block V28U4: Gowers norms and higher-order structure

V28U4P1. [F] Compute U^2 and U^3 norms for characters, random functions, and polynomial phases on finite vector spaces. Verify U^2 controls Fourier coefficients.

V28U4P2. [T] Prove the generalized von Neumann inequality for 3-term progressions and formulate the analogous role of U^{k-1} for k-term patterns.

V28U4P3. [R] State one inverse theorem for Gowers norms precisely in the setting you study (finite fields or integers), and work several examples of structured functions matching the conclusion.

Assigned block V34U2: Projections, Fourier decay and distance methods

V34U2P1. [F] Compute projections of product/Cantor sets in special directions and identify exceptional directions caused by arithmetic structure.

V34U2P2. [T] Prove the Mattila projection theorem/Marstrand theorem at proof-architecture level via energies/Fourier transform in the formulation used by your text.

V34U2P3. [S] Derive the Mattila integral criterion for the Falconer distance problem conceptually and test the scaling threshold on a Frostman measure.

Quarter-level integration problems

I1. [S]. Starting from a Frostman measure, derive the distance-measure Fourier formula and Mattila integral; then identify the exact Fourier-decay/energy estimate whose improvement would lower the Falconer dimension threshold.

I2. [S]. Discretize a Furstenberg-set problem at scale delta into a point-tube incidence statement and formulate the additive-combinatorial quantity (energy/Gowers norm/sum-product) that could detect concentration; track all powers of delta.

Full written solutions required for: V35U4P2, V35U4P3, V35U5P2, V35U5P3, V28U3P2, V28U3P3, V28U4P2, V28U4P3, V34U2P2, V34U2P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 30-page survey connecting energy methods, polynomial partitioning and additive-combinatorial structure. Include a detailed exposition of one modern result close to the Falconer/Furstenberg/Kakeya frontier.

Exit checkpoint

Prove Frostman's lemma in an appropriate formulation or master its proof, derive the Mattila integral criterion, prove a sum-product estimate in a model setting, and work a Gowers-norm calculation.

Research bridge

This quarter directly supports research-level reading of modern Falconer, Furstenberg and Kakeya work and explains why additive combinatorics repeatedly appears in geometric incidence problems.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q33. Entropy and Lq inverse theorems, self-similar measures, and Bernoulli convolutions

Quarter overview and reading route

Objective: Learn the multiscale inverse-theorem approach to self-similar measures: Hochman's entropy inverse theory, Shmerkin's Lq convolution inverse theory, and arithmetic applications to Bernoulli convolutions including Varju's work.

Prerequisites: Q10 measure theory, Q11 probability/additive combinatorics, Q13 Fourier analysis, Q24 finite-field/arithmetic geometry maturity, Q32 Falconer/Furstenberg/additive combinatorics II.

30-hour allocation: fractal/self-similar foundations 5h; entropy methods 9h; Lq inverse methods 7h; Bernoulli convolutions/arithmetic 7h; exposition 2h.

Core books and chapters/sections

- Falconer, Fractal Geometry: self-similar sets/measures and dimension chapters, selectively.
- Hochman, On self-similar sets with overlaps and inverse theorems for entropy, Annals 2014: Sections 1-5 in detail, applications selectively.
- Shmerkin, On Furstenberg's intersection conjecture, self-similar measures, and the Lq norms of convolutions, Annals 2019: inverse-theorem and self-similar sections.
- Shmerkin, Lq dimensions of self-similar measures, and applications: survey, as an easier second pass.
- Varju, Recent progress on Bernoulli convolutions; Breuillard-Varju, On the dimension of Bernoulli convolutions; Varju, On the dimension of Bernoulli convolutions for all transcendental parameters.

Lecture notes and primary-paper companions

- Use Hochman first to learn entropy at scale and local entropy averages; only then read the inverse theorem.
- Use Shmerkin's survey before the full 2019 Annals paper if the Lq multifractal formalism is unfamiliar.
- Track the number-theoretic inputs - algebraic separation, Mahler measure and Lehmer-type issues - separately from the entropy/additive-combinatorial inputs.

Reading rule: Read the expository route first. For the research papers, begin with the introduction and theorem statements, then reconstruct only the lemmas named in the weekly schedule. Do not attempt a line-by-line first pass of the hardest papers.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Self-similar sets/measures and Bernoulli convolutions	Falconer Fractal Geometry self-similar chapters; Varju survey Sections 1-2.	Compute similarity dimensions and exact-overlap examples.
2	Entropy at scale and dimension	Hochman paper Sections 1-2; Varju entropy preliminaries.	Prove basic entropy chain/scaling inequalities for dyadic partitions.
3	Local entropy averages	Hochman preliminaries and entropy machinery.	Derive local entropy average formula in a simplified dyadic setting.
4	Entropy growth under convolution	Hochman inverse-theorem setup.	Prove basic monotonicity/subadditivity examples; identify no-growth obstructions.
5	Hochman inverse theorem for entropy	Hochman 2014 central inverse theorem sections.	Write structural statement carefully; reconstruct one multiscale lemma.
6	Self-similar dimension and superexponential concentration	Hochman applications sections.	Derive dimension-drop -> superexponentially close cylinders architecture.
7	Algebraic parameters and exponential separation	Hochman/Varju; Mahler-measure background from ANT.	Explain why algebraicity converts separation information into rigidity.
8	Lq dimensions and convolution norms	Shmerkin 2019 Sections 1-4; Shmerkin survey.	Compute Lq dimensions of model Cantor measures and convolution examples.
9	Shmerkin inverse Lq theorem	Shmerkin central inverse theorem sections.	State the inverse theorem precisely and compare

Week	Main focus	Reading / lecture notes	Required output
			every component with Hochman entropy.
10	Furstenberg intersections and Lq applications	Shmerkin applications sections.	Map inverse theorem -> self-similar Lq dimension -> intersection theorem.
11	Bernoulli convolutions: Hochman-Shmerkin-Varju	Varju survey; Breuillard-Varju 2016; Varju 2019.	Trace dimension/entropy/transcendence/Mahler-measure implications.
12	Absolute continuity, open problems, synthesis	Shmerkin survey + Varju survey; optional later papers.	30-minute oral + capstone proof map of modern Bernoulli-convolution theory.

Week 1 - Self-similar measures and dimensions, objects and examples. Reading: Falconer fractal geometry/self-similar chapters. Definition/result focus: IFS; OSC/SSC; exact dimensionality orientation. Work: Compute dimensions for separated Bernoulli/Cantor measures.

Week 2 - Self-similar measures and dimensions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute dimensions for separated Bernoulli/Cantor measures. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Entropy at scale, objects and examples. Reading: Hochman 2014 sections on entropy/local entropy averages. Definition/result focus: dyadic entropy; entropy dimension; components. Work: Compute H_n for atomic/uniform/Cantor measures and prove elementary entropy inequalities.

Week 4 - Entropy at scale, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute H_n for atomic/uniform/Cantor measures and prove elementary entropy inequalities. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Hochman inverse theorem, objects and examples. Reading: Hochman 2014 central inverse-theorem sections. Definition/result focus: small entropy growth under convolution => multiscale structure. Work: State theorem precisely and map how it yields superexponential concentration under dimension drop.

Week 6 - Hochman inverse theorem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State theorem precisely and map how it yields superexponential concentration under dimension drop. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Shmerkin inverse Lq, objects and examples. Reading: Shmerkin 2019 inverse-theorem sections. Definition/result focus: Lq norm decay; Lq dimension; inverse convolution theorem. Work: Compute finite-scale Lq quantities and compare conclusion with entropy inverse theory.

Week 8 - Shmerkin inverse Lq, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute finite-scale Lq quantities and compare conclusion with entropy inverse theory. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Bernoulli convolutions: classical and modern, objects and examples. Reading: Erdos/Garsia/Solomyak expositions; Hochman/Shmerkin results. Definition/result focus: inverse Pisot singular examples; Garsia-number absolute continuity; a.e. Solomyak; zero-dimensional exceptional sets. Work: Prepare a theorem table with hypotheses carefully separated - especially Pisot vs Garsia parameters.

Week 10 - Bernoulli convolutions: classical and modern, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prepare a theorem table with hypotheses carefully separated - especially Pisot vs Garsia parameters. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Varju/Breuillard-Varju arithmetic input, objects and examples. Reading: Breuillard-Varju 2016; Varju 2019. Definition/result focus: algebraic approximation/Mahler measure; dim nu_lambda=1 for every transcendental lambda in (1/2,1). Work: Reconstruct the arithmetic-separation logic and compare dimension-one vs absolute-continuity conclusions.

Week 12 - Varju/Breuillard-Varju arithmetic input, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Reconstruct the arithmetic-separation logic and compare dimension-one vs absolute-continuity conclusions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Iterated function system (IFS); self-similar set and measure
- Open set condition; exact overlap; exponential separation
- Similarity dimension; Hausdorff dimension; exact dimension
- Bernoulli convolution mu_lambda
- Shannon entropy; entropy with respect to a partition
- Entropy at scale H(mu;r) and conditional entropy
- Entropy dimension / local entropy average
- Convolution of measures and entropy growth
- Inverse theorem for entropy growth

- Lq norm of a discretized measure
- Lq dimension and Frostman exponent
- Inverse theorem for Lq convolution
- Superexponential concentration of cylinders
- Mahler measure; Lehmer-type separation input
- Transversality (in Bernoulli-convolution context)

Key results to know and use

- Basic dimension formula for separated self-similar measures and entropy/Lq dimension calculations in model cases.
- Hochman inverse theorem for entropy growth under convolution (precise finite-scale statement and proof architecture).
- Hochman 2014: dimension drop for one-dimensional self-similar measures forces superexponentially close cylinders; for algebraic defining data this confirms the expected exact-overlap consequence.
- Shmerkin 2019: inverse theorem for decay of discrete Lq norms under convolution and the resulting Lq-dimension machinery; know the Furstenberg-intersection application.
- Bernoulli convolutions - Erdos: reciprocal-Pisot parameters give singular/non-Rajchman examples; do not confuse these with Garsia parameters.
- Bernoulli convolutions - Garsia: classical absolute-continuity/bounded-density theorem for reciprocals of Garsia numbers (the algebraic-integer norm/conjugate hypothesis must be stated from the chosen source, not replaced by “Pisot-like”).
- Solomyak: absolute continuity for Lebesgue-a.e. lambda in (1/2,1); Hochman/Shmerkin: much thinner exceptional sets for dimension/regularity statements, with conclusions kept distinct.
- Breuillard-Varju algebraic-approximation theorem for dimension-drop parameters and Varju 2019: dim nu_lambda=1 for every transcendental lambda in (1/2,1). Dimension one is not, by itself, the same as absolute continuity.

Quarter-specific mastery targets

M1 - Exact language: self-similar measure, exact/exponential separation, entropy at scale, local entropy average, L^q norm/dimension, Bernoulli convolution and Mahler measure.

M2 - Proofs to reproduce: elementary entropy and convolution inequalities; a finite-scale local-entropy identity; discrete L^q convolution inequalities; polynomial/root-separation estimates in representative algebraic examples.

M3 - Research theorems, exact statement + architecture: Hochman's inverse entropy theorem in one finite-scale form; Hochman's self-similar dimension consequence; Shmerkin inverse L^q theorem in the chosen formulation; one Varju/Breuillard-Varju Bernoulli-convolution theorem.

M4 - Mechanism mastery: explain how 'failure of entropy/Lq growth' forces multiscale structure and how arithmetic separation rules out the bad alternative in Bernoulli-convolution applications.

M5 - Exit use: for a parameter lambda, determine which facts are known from separation/algebraicity/transcendence assumptions and which regularity conclusions cannot be inferred.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V34U3: Self-similar measures, overlaps and Bernoulli convolutions

V34U3P1. [F] For Bernoulli convolution mu_lambda, derive the self-similarity equation and Fourier product formula. Work explicitly with lambda=1/2 and one reciprocal Pisot parameter.

V34U3P2. [X] Compute finite Bernoulli-convolution approximations for lambda=0.45,0.5,0.6,1/phi and inspect overlaps/Fourier coefficients. Record phenomena without inferring unproved regularity.

V34U3P3. [S] Translate an exact/near overlap into a small value of a polynomial with coefficients in {-1,0,1}. Explain why algebraic properties of lambda become relevant.

Assigned block V34U4: Entropy at scale and Hochman inverse theory

V34U4P1. [F] Compute dyadic entropy H_n(mu) for a Dirac mass, uniform interval measure, finite uniform sets, and finite Cantor approximations. Normalize by n and compare with dimension.

V34U4P2. [T] Prove elementary entropy inequalities under convolution/conditioning and a local-entropy-averages identity in a finite dyadic model.

V34U4P3. [R] State Hochman's inverse theorem for entropy growth in one precise finite-scale formulation. Build a toy example for each structural alternative, then trace how the theorem yields the dimension formula for self-similar measures absent superexponential concentration.

Assigned block V34U5: Lq dimensions and Shmerkin inverse theory

V34U5P1. [F] Compute dyadic L^q sums sum_I mu(I)^q and L^q dimensions for uniform/self-similar measures under separation. Compare q=2 with additive energy intuition.

V34U5P2. [T] Prove basic convolution inequalities for discrete L^q norms and construct examples where convolution flattens strongly versus barely changes the norm.

V34U5P3. [R] State Shmerkin's inverse theorem for L^q norms of convolutions in the exact formulation chosen from the reference. For a toy multiscale measure, identify the porous/concentrated alternatives and explain the implication for L^q dimension growth.

Assigned block V34U6: Varju, algebraic parameters and modern Bernoulli convolutions

V34U6P1. [F] For algebraic lambda, compute minimal polynomial, degree, and Mahler measure in several examples; relate polynomial evaluations at lambda to root-separation bounds.

V34U6P2. [R] Select one Varju or Breuillard-Varju theorem on Bernoulli convolutions and write its hypotheses exactly, including algebraicity/Mahler-measure or approximation conditions. Trace how the arithmetic estimate feeds entropy/dimension growth.

V34U6P3. [S] Prepare a chronological theorem map Erdos -> Garsia -> Solomyak -> Hochman -> Shmerkin -> Breuillard-Varju -> Varju. For each stage give one exact statement and conclusion type (singularity/Fourier decay, absolute continuity, dimension, or Lq regularity). Explicitly separate reciprocal Pisot parameters (Erdos singularity) from reciprocal Garsia-number parameters (Garsia absolute continuity), and distinguish dimension one from absolute continuity.

Assigned block V28U1: Sumsets and Ruzsa calculus

V28U1P1. [F] For arithmetic progressions, random-looking sets, subgroups, and geometric progressions in Z/pZ, compute |A+A|, |A-A|, and additive energy. Compare structure signatures.

V28U1P2. [T] Prove Ruzsa triangle inequality, covering lemma, and a selected Plunnecke inequality. Starting from |A+A|<=K|A|, derive explicit polynomial-in-K bounds for mA-nA.

V28U1P3. [S] Derive E(A)>=|A|^4/|A+A| by Cauchy-Schwarz and explain why large energy is the bridge from small sumsets to inverse structure.

Quarter-level integration problems

I1. [S]. For Bernoulli convolution mu_lambda, derive the Fourier product and finite-scale entropy. Analyze lambda=1/2 and one reciprocal Pisot parameter, and explain why Fourier decay and entropy dimension answer different regularity questions.

I2. [R]. Build one complete application diagram: algebraic/root separation -> exclusion of super-exponential concentration -> Hochman entropy theorem -> dimension conclusion, then indicate where a Shmerkin/Varju input is needed to upgrade dimension to stronger regularity.

Full written solutions required for: V34U3P2, V34U3P3, V34U4P2, V34U4P3, V34U5P2, V34U5P3, V34U6P2, V34U6P3, V28U1P2, V28U1P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 25-page survey with three proof diagrams: (A) entropy inverse theorem -> self-similar dimension; (B) Lq inverse theorem -> Furstenberg/self-similar consequences; (C) arithmetic separation + entropy -> Bernoulli-convolution results of Hochman/Breuillard-Varju/Varju. Add a final section connecting these methods to Cantor measures and Fourier questions.

Exit checkpoint

Define entropy at scale, exponential separation, Lq dimension and Bernoulli convolution; state Hochman and Shmerkin inverse theorems at a usable level; explain the proof architecture of Varju's transcendental-parameter dimension theorem.

Research bridge

This quarter directly supports research on Cantor measures, Furstenberg sets/intersections, fractal Fourier analysis and Diophantine separation. It also shows how additive-combinatorial inverse theorems migrate into fractal geometry.

Q34. Ratner, effective homogeneous dynamics, EKL/Benoist-Quint, and Diophantine applications

Quarter overview and reading route

Objective: Move from foundational ergodic/Lie theory to measure/orbit rigidity for unipotent and higher-rank actions, then connect homogeneous dynamics to Diophantine approximation and equidistribution.

Prerequisites: Q17 Lie groups/lattices, Q25 ergodic theory, Q31 Borel density/BHC.

30-hour allocation: Ratner/unipotent dynamics 13h; higher-rank/EKL 7h; effective dynamics/Marklof-Strombergsson 5h; Benoist-Quint and Diophantine applications 5h.

Core books and chapters/sections

- Witte Morris, Ratner's Theorems on Unipotent Flows / introductory homogeneous-dynamics notes; use for precise statements and examples.
- Dani-Margulis and Kleinbock-Margulis papers/notes on nondivergence and Diophantine applications; read theorem statements before proofs.
- Einsiedler-Katok-Lindenstrauss papers/expositions on higher-rank diagonal actions, entropy and Littlewood-type applications.
- Marklof-Strömbergsson and Strömbergsson papers/expositions for effective/equidistribution questions in rank-one homogeneous dynamics; record the exact spaces, norms and rates.
- Benoist-Quint, Random Walks on Reductive Groups / related papers: stationary measures, orbit closures and equidistribution for random walks.
- Use Raghunathan or Witte Morris for algebraic subgroups/lattices whenever Ratner hypotheses need structural verification.

Lecture notes and companions

- Tao UCLA ergodic-theory notes for recurrence/entropy background; Witte Morris/Ratner lecture notes for unipotent-flow statements.
- Kleinbock-Margulis/Dani expositions for quantitative nondivergence and Diophantine correspondences.
- Author surveys/lecture notes by Einsiedler-Lindenstrauss and Benoist-Quint for higher-rank measure rigidity and random walks; use original papers for exact frontier statements.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Unipotent flows, horocycles, recurrence/nondivergence	Witte Morris Ratner's Theorems opening chapters; Tao ergodic notes homogeneous sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Unipotent flows, horocycles, recurrence/nondivergence	Witte Morris Ratner's Theorems opening chapters; Tao ergodic notes homogeneous sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Ratner measure classification: statement and proof architecture	Witte Morris central Ratner chapters; read Ratner originals only after exposition. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Ratner measure classification: statement and proof architecture	Witte Morris central Ratner chapters; read Ratner originals only after exposition. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Orbit closure/equidistribution	Witte Morris applications; Margulis/Oppenheim expositions.	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
	consequences and Oppenheim	First pass: definitions, examples, and 2-3 basic exercises.	
6	Prove/use: Orbit closure/equidistribution consequences and Oppenheim	Witte Morris applications; Margulis/Oppenheim expositions. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Higher-rank diagonal actions and EKL measure rigidity	Einsiedler-Lindenstrauss/Katok expository papers, then the theorem paper selectively. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Higher-rank diagonal actions and EKL measure rigidity	Einsiedler-Lindenstrauss/Katok expository papers, then the theorem paper selectively. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Effective homogeneous dynamics in rank one/two, Marklof-Strombergsson	Strombergsson/Marklof surveys and selected theorem proofs; track quantitative nondivergence/mixing inputs. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Effective homogeneous dynamics in rank one/two, Marklof-Strombergsson	Strombergsson/Marklof surveys and selected theorem proofs; track quantitative nondivergence/mixing inputs. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Benoist-Quint random walks and Diophantine/fractal applications	Benoist-Quint expositions/selected chapters and contemporary Diophantine applications. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Benoist-Quint random walks and Diophantine/fractal applications	Benoist-Quint expositions/selected chapters and contemporary Diophantine applications. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Unipotent dynamics and nondivergence, objects and examples. Reading: Ratner/Dani-Margulis expositions. Definition/result focus: unipotent flow; recurrence; quantitative nondivergence. Work: Compute horocycle action and prove a simple nondivergence estimate.

Week 2 - Unipotent dynamics and nondivergence, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute horocycle action and prove a simple nondivergence estimate. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Ratner orbit closures, objects and examples. Reading: Ratner/Witte Morris notes. Definition/result focus: orbit-closure theorem exact statement; homogeneous closures. Work: Test theorem on SL2/SL3 examples and classify candidate intermediate subgroups.

Week 4 - Ratner orbit closures, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Test theorem on SL2/SL3 examples and classify candidate intermediate subgroups. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Ratner measure classification, objects and examples. Reading: Ratner expositions. Definition/result focus: ergodic invariant measure classification. Work: Build proof architecture and distinguish from orbit closure/equidistribution.

Week 6 - Ratner measure classification, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build proof architecture and distinguish from orbit closure/equidistribution. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Effective homogeneous dynamics, objects and examples. Reading: Strömbergsson/Marklof-Strömbergsson/Kleinbock-Margulis selected papers. Definition/result focus: effective equidistribution; Sobolev norms; spectral gap/mixing inputs. Work: Audit one effective theorem with all norms/rates and identify rank-one limitations.

Week 8 - Effective homogeneous dynamics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Audit one effective theorem with all norms/rates and identify rank-one limitations. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Higher-rank measure rigidity, objects and examples. Reading: Einsiedler-Katok-Lindenstrauss. Definition/result focus: entropy contribution; positive entropy; algebraic measure. Work: State EKL theorem used for Littlewood-type applications and map entropy mechanism.

Week 10 - Higher-rank measure rigidity, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State EKL theorem used for Littlewood-type applications and map entropy mechanism. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Benoist-Quint/random walks and Diophantine applications, objects and examples. Reading: Benoist-Quint plus relevant Diophantine papers. Definition/result focus: stationary measure; orbit closure/equidistribution; random walk. Work: Compare unipotent-rigidity and random-walk rigidity methods in a concrete arithmetic application.

Week 12 - Benoist-Quint/random walks and Diophantine applications, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compare unipotent-rigidity and random-walk rigidity methods in a concrete arithmetic application. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- unipotent one-parameter subgroup/flow
- homogeneous probability measure
- nondivergence and cusp excursion
- linearization method
- additional invariance
- ergodic component and entropy contribution
- stationary measure
- random walk on a homogeneous/algebraic space
- spectral gap/mixing rate in an effective theorem
- Diophantine correspondence/orbit closure

Key results to know and use

- Ratner orbit-closure theorem, measure-classification theorem and equidistribution theorem: keep the three statements distinct.
- Margulis deduction of the Oppenheim conjecture from unipotent orbit closure/measure rigidity.
- Dani-Margulis/Kleinbock-Margulis quantitative nondivergence in the specific formulation used for Diophantine applications.
- One Strömbergsson or Marklof-Strömbergsson effective equidistribution theorem with space, test-function norm and error rate stated exactly.
- Einsiedler-Katok-Lindenstrauss higher-rank measure-rigidity theorem used in the Littlewood-conjecture direction, including the positive-entropy hypothesis/conclusion.
- Benoist-Quint stationary-measure/orbit-closure principle for a selected random-walk setting, with irreducibility/Zariski-density hypotheses recorded.

Quarter-specific mastery targets

M1 - Exact language: unipotent flow, homogeneous measure/orbit, Ratner measure classification/orbit closure/equidistribution, linearization, entropy contribution, higher-rank diagonal action, stationary measure and effective equidistribution.

M2 - Proofs to reproduce: Dani correspondence in a basic Diophantine example; nondivergence/linearization lemmas in tractable model cases; explicit Oppenheim-to-orbit-closure reduction.

M3 - Research theorems, exact statements + architecture: Ratner's principal theorems; Margulis Oppenheim deduction; EKL measure rigidity; a representative effective Ratner/equidistribution theorem; a Benoist-Quint stationary-measure theorem.

M4 - Hypothesis audit: identify unipotence, higher rank, arithmeticity/noncompactness and spectral-gap assumptions separately; give examples showing why diagonal and unipotent dynamics behave differently.

M5 - Exit use: translate a new Diophantine problem into a homogeneous orbit problem and state exactly which rigidity/effectivity theorem would be needed.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V20U3: Ratner theory

V20U3P1. [F] Analyze horocycle orbits on SL_2(R)/SL_2(Z): identify periodic versus nonperiodic examples and compare with geodesic-flow behavior.

V20U3P2. [T] State Ratner orbit-closure and measure-classification theorems precisely for unipotent flows. Test the conclusion against at least five homogeneous-space examples.

V20U3P3. [R] Reconstruct the shearing principle behind Ratner's method in a toy SL_2 setting by comparing nearby unipotent orbits; identify how polynomial divergence replaces exponential hyperbolicity.

Assigned block V20U4: Oppenheim and linearization

V20U4P1. [T] For an indefinite ternary quadratic form Q, identify SO(Q) inside SL_3(R), show its standard representation is the 3-dimensional irreducible representation of sl_2 in a model basis, and locate unipotent one-parameter subgroups.

V20U4P2. [S] Translate Oppenheim into density of an H=SO(Q)-orbit in SL_3(R)/SL_3(Z): make the correspondence between lattice points, values Q(v), and orbit closure explicit.

V20U4P3. [R] Build the linearization/nondivergence proof architecture used to exclude improper orbit closures. State exactly where arithmeticity/rational forms enter.

Assigned block V20U5: Higher-rank measure rigidity

V20U5P1. [F] Compute entropy contributions for commuting diagonal maps on a torus or homogeneous space in a toy algebraic example.

V20U5P2. [T] State a representative Einsiedler-Katok-Lindenstrauss measure-rigidity theorem with all rank/entropy/invariance assumptions. Produce examples showing why higher rank and positive entropy are substantive.

V20U5P3. [R] Trace one entropy-to-unipotent-invariance mechanism at proof-architecture level: coarse Lyapunov foliations, conditional measures, entropy contribution, extra invariance.

Assigned block V20U6: Effective equidistribution and random walks

V20U6P1. [F] For a random walk on a compact or finite homogeneous quotient, compute convolution powers and stationary measures in an explicit example.

V20U6P2. [T] Derive how a spectral gap gives exponential mixing/equidistribution for mean-zero L^2 observables. Quantify the rate in terms of the second eigenvalue/operator norm.

V20U6P3. [R] Compare Strombergsson-type effective equidistribution, Marklof-Strombergsson limit theorems, Lei Yang higher-rank effectiveness, and Benoist-Quint random-walk rigidity by writing for each: acting object, space, qualitative theorem, quantitative input.

Assigned block V25U6: Homogeneous dynamics correspondence

V25U6P1. [F] Show SL_2(R)/SL_2(Z) parametrizes unimodular lattices and write the lattice associated with a real alpha so that short vectors encode good rational approximations.

V25U6P2. [T] Prove the Dani correspondence for badly approximable numbers: boundedness of a diagonal orbit iff alpha is badly approximable, with all norm comparisons explicit.

V25U6P3. [R] Extend the dictionary to simultaneous approximation in R^n and formulate one higher-dimensional/fractal Diophantine problem as a shrinking-target or nondivergence problem on a homogeneous space.

Quarter-level integration problems

I1. [S]. Write the Oppenheim problem for an indefinite ternary quadratic form as a statement about an SO(2,1)-orbit in SL3(R)/SL3(Z); prove the elementary equivalence before invoking Ratner/Margulis.

I2. [R]. Compare Ratner, EKL and Benoist-Quint in a table whose rows are acting object, invariant/stationary measure, rigidity hypothesis, conclusion and quantitative/effective status; add one example where applying the wrong theorem would fail.

Full written solutions required for: V20U3P2, V20U3P3, V20U4P2, V20U4P3, V20U5P2, V20U5P3, V20U6P2, V20U6P3, V25U6P2, V25U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Create a 35-page 'rigidity atlas': Ratner, EKL, effective Ratner-type results and Benoist-Quint, with a table of acting groups, invariant measures, hypotheses, conclusions and Diophantine consequences.

Exit checkpoint

Be able to state Ratner's three principal theorems precisely, explain the Oppenheim deduction, outline EKL entropy rigidity, identify where effectiveness enters quantitative results, and work one homogeneous reformulation of a Diophantine problem.

Research bridge

This is the main endpoint for the homogeneous-dynamics program and prepares Zimmer rigidity and current Diophantine applications on fractals/homogeneous spaces.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q35. Zimmer conjecture and automorphic forms II: adeles, Tate thesis, GL(2)

Quarter overview and reading route

Objective: Run two advanced representation/rigidity tracks in parallel: Zimmer-type actions of lattices and the adelic/representation-theoretic formulation of automorphic forms on GL(1) and GL(2).

Prerequisites: Q17 Lie groups, Q25 ergodic/modular forms, Q31 Borel theory/CFT, Q34 rigidity.

30-hour allocation: Zimmer/superrigidity 12h; adeles/Tate thesis 8h; GL(2) automorphic representations 8h; synthesis 2h.

Core books and chapters/sections

- Brown-Fisher-Hurtado, Zimmer's conjecture: Subexponential growth, measure rigidity, and strong property (T), Annals 2022: read Introduction and main theorem statements first, then the derivative-growth and strong-(T) architecture.
- Fisher/Zimmer-program surveys for cocycle superrigidity, derivative cocycles and dimension thresholds; use original theorem statements when quoting a case.
- Bekka-de la Harpe-Valette plus a strong-property-(T) survey/source for the representation-theoretic distinction relevant to BFH.
- Tate thesis (with a modern exposition such as Bump, Gelbart or Weil): adeles/ideles, local zeta integrals, Fourier transform and functional equation.
- Bump, Automorphic Forms and Representations: GL(2), local components, spherical vectors and Satake parameters; Casselman/Gelbart notes as companions.
- Milne class field/adeles notes as background only where notation or restricted products need reinforcement.

Lecture notes and companions

- Fisher/Zimmer-program lecture notes or surveys for cocycles and superrigidity; read BFH 2022 alongside an exposition of strong property (T).
- MIT/graduate notes on adeles and Tate's thesis, with Milne as notation/background support.
- Bump/Gelbart/Casselman-style automorphic-representation notes for GL(2), spherical representations and Satake parameters.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Cocycles, superrigidity and algebraic hulls	Zimmer Ergodic Theory and Semisimple Groups cocycle/superrigidity chapters; Fisher surveys. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Cocycles, superrigidity and algebraic hulls	Zimmer Ergodic Theory and Semisimple Groups cocycle/superrigidity chapters; Fisher surveys. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Zimmer conjecture statement, dimension bounds and Brown-Fisher-Hurtado architecture	Modern surveys first; then selected BFH sections with proof-dependency map. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Zimmer conjecture statement, dimension bounds and Brown-Fisher-Hurtado architecture	Modern surveys first; then selected BFH sections with proof-dependency map. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Adeles/ideles, strong approximation and Haar analysis	MIT 18.785 Lectures 25-27; Bump introductory adelic chapters. First	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
		pass: definitions, examples, and 2-3 basic exercises.	
6	Prove/use: Adeles/ideles, strong approximation and Haar analysis	MIT 18.785 Lectures 25-27; Bump introductory adelic chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Tate's thesis: local zeta integrals, Fourier transform, global functional equation	Tate thesis with an expository companion; calculate the R and Q_p local factors. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Tate's thesis: local zeta integrals, Fourier transform, global functional equation	Tate thesis with an expository companion; calculate the R and Q_p local factors. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Automorphic forms/representations on GL(2), local components	Bump introductory GL(2)/automorphic representation chapters; Gelbart as companion. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Automorphic forms/representations on GL(2), local components	Bump introductory GL(2)/automorphic representation chapters; Gelbart as companion. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Hecke eigenforms as automorphic representations; Euler products and L-functions	Translate classical modular forms from Q25 into adelic representation language. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Hecke eigenforms as automorphic representations; Euler products and L-functions	Translate classical modular forms from Q25 into adelic representation language. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Zimmer programme and cocycles, objects and examples. Reading: Zimmer/Fisher surveys; BFH introduction. Definition/result focus: cocycle; derivative cocycle; superrigidity; low-dimensional action. Work: Translate a smooth action into derivative cocycle data and state the dimension thresholds carefully.

Week 2 - Zimmer programme and cocycles, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Translate a smooth action into derivative cocycle data and state the dimension thresholds carefully. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Subexponential derivative growth, objects and examples. Reading: Brown-Fisher-Hurtado 2022. Definition/result focus: uniform subexponential derivative growth; Lyapunov/measure-rigidity input. Work: Map the contradiction architecture leading to derivative control.

Week 4 - Subexponential derivative growth, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Map the contradiction architecture leading to derivative control. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Strong property T and invariant metrics, objects and examples. Reading: BFH + Lafforgue strong-T background. Definition/result focus: strong property T; averaging; invariant Riemannian metric. Work: Explain why ordinary property T is not the same input and trace the final finite-image step.

Week 6 - Strong property T and invariant metrics, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Explain why ordinary property T is not the same input and trace the final finite-image step. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Adeles and ideles, objects and examples. Reading: Tate thesis/Weil/Milne notes. Definition/result focus: restricted product; adele/ideles; strong approximation. Work: Compute Q-adeles examples and factor test functions locally.

Week 8 - Adeles and ideles, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Q-adeles examples and factor test functions locally. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Tate thesis, objects and examples. Reading: Tate thesis expositions. Definition/result focus: zeta integral; Fourier transform; local factors; functional equation. Work: Compute local unramified zeta integrals and recover zeta/L factors.

Week 10 - Tate thesis, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute local unramified zeta integrals and recover zeta/L factors. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - GL2 automorphic representations, objects and examples. Reading: Bump/Gelbart/Casselman notes. Definition/result focus: automorphic form/representation; local component; spherical vector; Satake parameter. Work: Adelize a classical eigenform and recover Hecke eigenvalues from local data.

Week 12 - GL2 automorphic representations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Adelize a classical eigenform and recover Hecke eigenvalues from local data. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- measurable cocycle
- derivative cocycle
- Lyapunov exponent
- cocycle superrigidity
- uniform subexponential derivative growth
- Kazhdan property (T) versus strong property (T)
- invariant Riemannian metric
- adele and idele
- restricted tensor product
- automorphic representation/local component
- spherical vector and Satake parameter
- local/global zeta integral

Key results to know and use

- Zimmer cocycle-superrigidity theorem in the precise higher-rank setting used as background.
- Brown-Fisher-Hurtado 2022 finite-image theorem for cocompact lattices in SL(n,R): dim M<n-1, or dim M<n in the volume-preserving case, with smoothness/hypotheses stated from the source.
- Uniform subexponential growth of derivatives as BFH’s key intermediate theorem.
- Strong property (T) input producing an invariant Riemannian metric after derivative control; ordinary property (T) is not an interchangeable substitute.
- Restricted-product construction of adeles/ideles and basic strong-approximation examples.
- Tate thesis: local zeta integrals, unramified local factors, Poisson/Fourier input and global functional equation.
- Classical Hecke eigenform to irreducible automorphic representation of GL(2,A); spherical local vectors, Satake parameters and Euler factors.

Quarter-specific mastery targets

M1 - Exact language: measurable cocycle, superrigidity, Zimmer dimension threshold framework, adele/idele, automorphic form/representation, restricted tensor product, local zeta integral and Satake parameter.

M2 - Proofs/derivations: basic adelic quotient/restricted-product calculations; Tate local zeta integrals in simple archimedean and unramified nonarchimedean cases; functional equation architecture via Poisson summation.

M3 - Research theorems, exact statement + architecture: Zimmer cocycle superrigidity and the relevant Brown-Fisher-Hurtado Zimmer-conjecture theorem; Tate thesis global functional equation; local-global factorization for GL2 automorphic representations.

M4 - Translation mastery: convert a classical modular eigenform into an adelic automorphic representation and recover Hecke/Satake data at unramified places.

M5 - Exit use: keep rigidity dynamics and automorphic representation theory conceptually distinct while recognizing their shared Lie/arithmetic-group infrastructure.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V20U5: Higher-rank measure rigidity

V20U5P1. [F] Compute entropy contributions for commuting diagonal maps on a torus or homogeneous space in a toy algebraic example.

V20U5P2. [T] State a representative Einsiedler-Katok-Lindenstrauss measure-rigidity theorem with all rank/entropy/invariance assumptions. Produce examples showing why higher rank and positive entropy are substantive.

V20U5P3. [R] Trace one entropy-to-unipotent-invariance mechanism at proof-architecture level: coarse Lyapunov foliations, conditional measures, entropy contribution, extra invariance.

Assigned block V21U5: Kazhdan property (T)

V21U5P1. [F] Show Z does not have property (T) by exhibiting almost-invariant vectors in suitable one-dimensional/Fourier representations. Show finite groups have property (T).

V21U5P2. [T] Prove one standard equivalence for property (T): almost invariant vectors imply invariant vectors versus isolation of the trivial representation. Work through Kazhdan constants in a finite example.

V21U5P3. [S] Prove that a discrete amenable group with property (T) is finite. Explain the more general locally compact statement (amenable + property (T) forces compactness under the standard hypotheses) and identify which representation/mean argument encodes amenability.

Assigned block V30U4: Maass forms and spectral theory

V30U4P1. [F] Write the hyperbolic Laplacian, verify invariance under SL_2(R), and compute Fourier-Bessel form of a Maass cusp eigenfunction from periodicity plus eigen-equation.

V30U4P2. [T] Derive the spectral decomposition of L^2(SL_2(Z)\H) at theorem-architecture level: constants, cusp spectrum, Eisenstein continuous spectrum.

V30U4P3. [S] Use unfolding on an Eisenstein/Poincare series to derive one Rankin-Selberg integral identity or Petersson-type formula.

Assigned block V30U5: Adeles and automorphic representations of GL2

V30U5P1. [F] Construct the adele ring A_Q and the embedding of GL_2(Q) into GL_2(A). Rewrite a classical modular form as an adelic function satisfying left rational invariance and right compact/K-type conditions.

V30U5P2. [T] Factor an automorphic representation pi as a restricted tensor product of local pi_v in the unramified almost-everywhere setting and match T_p eigenvalues with Satake parameters.

V30U5P3. [S] Starting from an unramified local GL_2 representation, write its local L-factor and recover the classical Euler factor of an eigenform.

Assigned block V30U6: Rankin-Selberg, trace formula and Langlands bridges

V30U6P1. [T] Unfold a Rankin-Selberg convolution integral and identify the Euler product/local zeta integrals in a standard GL_2 example.

V30U6P2. [R] Compare Selberg/Petersson/Kuznetsov trace formulas with the Arthur trace-formula philosophy: list geometric conjugacy/orbital data versus spectral automorphic data without pretending to prove the full formula.

V30U6P3. [R] Build a Langlands bridge diagram for GL_1 class field theory and GL_2 modular forms: local parameters, global automorphic representation, L-function, and Galois representation where available.

Quarter-level integration problems

I1. [S]. Starting from a normalized Hecke eigenform f, construct its adelic lift and identify unramified local components; recover a_p from Satake parameters and the local Euler factor, with all normalizations stated.

I2. [R]. For one Zimmer-conjecture theorem, write a complete hypothesis audit (acting lattice/group, rank, target-manifold dimension, regularity, volume preservation if relevant) and identify exactly where cocycle superrigidity enters the proof architecture.

Full written solutions required for: V20U5P2, V20U5P3, V21U5P2, V21U5P3, V30U4P2, V30U4P3, V30U5P2, V30U5P3, V30U6P2, V30U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write one expository note on Zimmer conjecture proof architecture and another deriving the functional equation in Tate's thesis and translating a modular eigenform into automorphic-representation language.

Exit checkpoint

State cocycle superrigidity and Zimmer's dimension threshold framework, compute adelic quotient examples, derive local zeta integrals at archimedean/nonarchimedean places, and explain the restricted tensor product of local representations.

Research bridge

The automorphic track now supplies the representation/L-function language used in Sato-Tate, modularity and Langlands-type questions; the rigidity track reaches modern Zimmer results.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q36. Sato-Tate, Fermat/FLT, modularity, and automorphic L-functions

Quarter overview and reading route

Objective: Synthesize elliptic curves, modular forms, Galois representations and automorphic L-functions to understand the architecture of FLT and Sato-Tate at a serious graduate/research-entry level.

Prerequisites: Q24 elliptic curves, Q25 modular forms, Q35 automorphic forms, Q19 local fields.

30-hour allocation: Modularity/FLT 15h; Sato-Tate 8h; automorphic/Galois L-functions 5h; review 2h.

Core books and chapters/sections

- Silverman, The Arithmetic of Elliptic Curves: elliptic curves, reduction, Tate modules/heights as needed.
- Diamond-Shurman, A First Course in Modular Forms: newforms, Hecke operators and modularity background.
- Cornell-Silverman-Stevens (eds.), Modular Forms and Fermat's Last Theorem, plus a careful Wiles/Taylor-Wiles exposition: Frey-Ribet, deformation theory and R=T architecture.
- Wiles and Taylor-Wiles original papers only after the expository route; extract exact modularity-lifting hypotheses rather than attempting a first-pass line-by-line proof.
- Serre/Sato-Tate expositions and modern potential-automorphy surveys for the analytic criterion via symmetric powers and the theorem for non-CM elliptic curves over Q.
- Bump/Gelbart or automorphic L-function notes for local Euler factors, symmetric powers and analytic continuation/nonvanishing vocabulary.

Lecture notes and companions

- Milne elliptic-curve/modular-form notes and a graduate modularity-lifting course as preparation for Wiles/Taylor-Wiles.
- A careful FLT seminar source (e.g. Modular Forms and Fermat's Last Theorem volume) before original papers.
- Serre/Sato-Tate and potential-automorphy lecture notes for the symmetric-power/equidistribution route.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Elliptic curves, modular forms and l-adic Galois representations review	Silverman/Diamond-Shurman review; Cornell-Silverman-Stevens preparatory chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Elliptic curves, modular forms and l-adic Galois representations review	Silverman/Diamond-Shurman review; Cornell-Silverman-Stevens preparatory chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Modularity theorem statement; deformation/representation framework	Expository modularity-lifting notes before Wiles/Taylor-Wiles; identify local deformation conditions. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Modularity theorem statement; deformation/representation framework	Expository modularity-lifting notes before Wiles/Taylor-Wiles; identify local deformation conditions. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Wiles/Taylor-Wiles proof architecture and FLT deduction	Modular Forms and Fermat's Last Theorem selected chapters; Wiles paper only after dependency map. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Wiles/Taylor-Wiles proof architecture and FLT deduction	Modular Forms and Fermat's Last Theorem selected chapters; Wiles paper only after dependency map. Second pass: theorem proofs,	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
		harder exercises, and a one-page proof reconstruction.	
7	Learn: Sato-Tate statement, Frobenius angles and equidistribution	Sato-Tate surveys; compute numerical Frobenius-angle histograms. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Sato-Tate statement, Frobenius angles and equidistribution	Sato-Tate surveys; compute numerical Frobenius-angle histograms. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Symmetric-power L-functions and automorphy strategy behind Sato-Tate	Proof surveys/papers selectively; connect to analytic equidistribution criteria. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Symmetric-power L-functions and automorphy strategy behind Sato-Tate	Proof surveys/papers selectively; connect to analytic equidistribution criteria. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Automorphic L-functions, conductors/local factors and synthesis	Bump/modern notes on GL(2) L-functions; compare classical and adelic normalizations. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Automorphic L-functions, conductors/local factors and synthesis	Bump/modern notes on GL(2) L-functions; compare classical and adelic normalizations. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Elliptic curves and Galois representations, objects and examples. Reading: Silverman + modularity expositions. Definition/result focus: Tate module; l-adic representation; conductor. Work: Compute Frobenius traces and local Euler factors for examples.

Week 2 - Elliptic curves and Galois representations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute Frobenius traces and local Euler factors for examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Modular forms/Galois correspondence, objects and examples. Reading: Diamond-Shurman + Deligne representation expositions. Definition/result focus: newform; Hecke eigenvalue; attached Galois representation. Work: Match a_p and Frobenius characteristic polynomials away from bad primes.

Week 4 - Modular forms/Galois correspondence, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Match a_p and Frobenius characteristic polynomials away from bad primes. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Frey curve and Ribet, objects and examples. Reading: FLT expositions. Definition/result focus: Frey curve; level lowering; Ribet theorem. Work: Write the exact contradiction chain conditional on modularity.

Week 6 - Frey curve and Ribet, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Write the exact contradiction chain conditional on modularity. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Wiles/Taylor-Wiles modularity, objects and examples. Reading: Wiles/Taylor-Wiles expositions. Definition/result focus: deformation ring; Hecke algebra; R=T architecture. Work: State a representative modularity-lifting theorem and map minimal/nonminimal steps.

Week 8 - Wiles/Taylor-Wiles modularity, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State a representative modularity-lifting theorem and map minimal/nonminimal steps. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Sato-Tate criterion, objects and examples. Reading: Serre/Sato-Tate expositions. Definition/result focus: equidistribution; SU(2); symmetric-power L-functions; Weyl criterion. Work: Derive how analytic properties of symmetric powers imply moment/equidistribution statements.

Week 10 - Sato-Tate criterion, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive how analytic properties of symmetric powers imply moment/equidistribution statements. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Potential automorphy and Sato-Tate theorem, objects and examples. Reading: BLGHT-style expositions/current surveys. Definition/result focus: potential automorphy; analytic continuation/nonvanishing consequences. Work: Give exact scope for non-CM elliptic curves over Q and separate theorem statement from proof machinery.

Week 12 - Potential automorphy and Sato-Tate theorem, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Give exact scope for non-CM elliptic curves over Q and separate theorem statement from proof machinery. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- elliptic curve and good/bad reduction
- Tate module and l-adic Galois representation
- conductor and Frobenius trace a_p
- normalized newform and Hecke eigenvalue
- Frey curve
- level lowering
- deformation ring and Hecke algebra
- modularity lifting and R=T
- Sato-Tate measure
- symmetric-power L-function
- potential automorphy
- Weyl equidistribution criterion

Key results to know and use

- Modularity theorem for elliptic curves over Q in the formulation sufficient for FLT.
- Ribet level-lowering/nonmodularity consequence for the Frey curve, with conductor hypotheses tracked.
- Frey-Ribet-modularity contradiction giving Fermat’s Last Theorem (full logical deduction once research theorems are granted).
- A representative Wiles/Taylor-Wiles modularity-lifting theorem: exact residual/deformation hypotheses and R=T architecture.
- Deligne/standard relation between weight-2 eigenforms and two-dimensional l-adic Galois representations at good primes.
- Sato-Tate theorem for non-CM elliptic curves over Q, with normalized trace distribution stated exactly.
- Weyl/Serre analytic criterion: suitable analytic continuation and nonvanishing of symmetric-power L-functions imply Sato-Tate equidistribution; distinguish criterion from the automorphy theorem supplying it.

Quarter-specific mastery targets

M1 - Exact language: elliptic-curve Galois representation, residual/deformation representation, modularity lifting, Frey curve, Ribet level lowering, Sato-Tate measure and symmetric-power L-function.

M2 - Proofs to reproduce: elementary Frey-curve discriminant/conductor computations in the standard setup; the logical Frey + Ribet + modularity implication to FLT; Weyl criterion/equidistribution deductions from suitable analytic L-function hypotheses.

M3 - Research theorems, exact statement + architecture: Wiles/Taylor-Wiles modularity-lifting theorem in a representative formulation, modularity theorem sufficient for FLT, Sato-Tate for non-CM elliptic curves over Q, and the automorphy/analytic input used for equidistribution.

M4 - Dependency mastery: distinguish residual modularity, deformation theory, R=T, level lowering and automorphic L-function analytic continuation; do not collapse them into 'modularity'.

M5 - Exit use: present the full FLT and Sato-Tate dependency graphs, marking which classical lemmas you can prove and which research theorems are used as precise black boxes.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V30U2: Hecke operators, eigenforms and newforms

V30U2P1. [F] Compute T_p on q-expansions and verify Hecke relations T_m T_n for coprime m,n and prime powers on selected forms.

V30U2P2. [T] Prove simultaneous diagonalization of commuting self-adjoint Hecke operators under the Petersson inner product in a finite-dimensional cusp space.

V30U2P3. [S] From Hecke eigenform relations derive multiplicativity of Fourier coefficients and the Euler product of L(s,f), including the quadratic local factor at unramified primes.

Assigned block V30U5: Adeles and automorphic representations of GL2

V30U5P1. [F] Construct the adele ring A_Q and the embedding of GL_2(Q) into GL_2(A). Rewrite a classical modular form as an adelic function satisfying left rational invariance and right compact/K-type conditions.

V30U5P2. [T] Factor an automorphic representation pi as a restricted tensor product of local pi_v in the unramified almost-everywhere setting and match T_p eigenvalues with Satake parameters.

V30U5P3. [S] Starting from an unramified local GL_2 representation, write its local L-factor and recover the classical Euler factor of an eigenform.

Assigned block V32U4: Tate modules and Galois representations

V32U4P1. [F] Compute E[l] over an algebraic closure for l not equal to p and describe the Galois action matrix for simple reductions/examples conceptually.

V32U4P2. [T] Define the Tate module T_l(E) and prove its rank-2 Z_l structure in characteristic 0/good settings. Relate determinant of Frobenius to p and trace to a_p.

V32U4P3. [S] State the l-adic representation rho_{E,l}:G_Q->GL_2(Z_l), identify ramification properties, and connect its characteristic polynomial at good primes to the Hasse-Weil L-function.

Assigned block V32U5: Modular curves, modularity and Fermat

V32U5P1. [F] Study X_0(N) moduli interpretation for small N: identify points as elliptic curves plus cyclic subgroup and understand cusps at a basic level.

V32U5P2. [T] Translate a normalized newform of weight 2 into an isogeny class of elliptic curves over Q at modularity-theorem statement level; compare Fourier coefficients a_p with point counts.

V32U5P3. [R] Build the FLT proof map: hypothetical Fermat solution -> Frey curve -> semistability -> Ribet level lowering/nonmodularity -> modularity theorem -> contradiction. State exactly what each theorem contributes.

Assigned block V32U6: Faltings and Sato-Tate roadmaps

V32U6P1. [R] State Faltings' theorem (Mordell conjecture) and the isogeny theorem in precise standard formulations. Build a dependency map through abelian varieties, heights, moduli, and finiteness without attempting the full proof.

V32U6P2. [F] For an elliptic curve without CM, compute normalized Frobenius traces a_p/(2sqrt p) for many small good primes by code and compare with the Sato-Tate density; for a CM example compare the different behavior.

V32U6P3. [R] State a modern Sato-Tate theorem for elliptic curves over Q in a precise non-CM formulation and map the automorphy/potential automorphy plus equidistribution inputs.

Quarter-level integration problems

I1. [S]. Write the Frey-Ribet-modularity contradiction as a formal implication chain with hypotheses at each arrow; for the Frey curve attached to a hypothetical Fermat solution, compute discriminant/conductor features used by the chain.

I2. [R]. Use Weyl's criterion to show how analytic continuation/nonvanishing of symmetric-power L-functions would imply Sato-Tate equidistribution; identify which symmetric powers are needed in the approximation argument.

Full written solutions required for: V30U2P2, V30U2P3, V30U5P2, V30U5P3, V32U4P2, V32U4P3, V32U5P2, V32U5P3, V32U6P2, V32U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Produce a 40-page capstone exposition: 'From elliptic curves to FLT and Sato-Tate'. Include a dependency graph, one modularity-lifting lemma studied in detail, and numerical Sato-Tate experiments.

Exit checkpoint

Explain the Frey curve-Ribet-modularity implication to FLT without gaps in logical dependencies, formulate residual/deformation representations accurately, state Sato-Tate precisely, and show how symmetric-power analytic properties imply equidistribution.

Research bridge

This quarter is the central synthesis of algebraic, analytic and automorphic number theory.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q37. Faltings, Diophantine geometry, and global arithmetic synthesis

Quarter overview and reading route

Objective: Develop the height/abelian-variety machinery needed to understand Faltings' proof of Mordell at the level of architecture and selected technical components, and place it in the wider Diophantine-geometric toolkit.

Prerequisites: Q20/Q24 algebraic geometry and elliptic curves; Q19/Q29 heights and Diophantine approximation; Q35 automorphic/global arithmetic helpful.

30-hour allocation: Diophantine geometry/heights 10h; abelian varieties 10h; Faltings/Mordell 8h; synthesis 2h.

Core books and chapters/sections

- Hindry-Silverman, Diophantine Geometry: heights, abelian varieties and finiteness framework.
- Bombieri-Gubler, Heights in Diophantine Geometry: Northcott, height machinery and global height comparisons.
- Milne, Abelian Varieties notes; Silverman, The Arithmetic of Elliptic Curves, for Jacobians/Tate modules/isogeny background.
- Faltings original papers only after an expository account; build a theorem-dependency ledger rather than trying to reproduce all technical proofs on first pass.
- Hartshorne/Vakil sections on curves/divisors/Jacobians as geometric background.
- Optional bridge only: Baker/Subspace-Theorem material from Q29 when comparing effective Diophantine methods with Faltings-style non-effective finiteness.

Lecture notes and companions

- Milne Abelian Varieties notes and a modern Faltings/Mordell exposition as the principal companions.
- Silverman/Hindry-Silverman problem sets for heights, descent, Jacobians and abelian varieties.
- Baker/Subspace-Theorem notes are optional comparison material only, not a replacement for the Faltings/abelian-variety route.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Heights on projective varieties and height machine	Hindry-Silverman Chs. A-B; Bombieri-Gubler height chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Heights on projective varieties and height machine	Hindry-Silverman Chs. A-B; Bombieri-Gubler height chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Abelian varieties, isogenies, polarizations, duality	Milne Abelian Varieties opening/middle chapters; Hindry-Silverman companion sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Abelian varieties, isogenies, polarizations, duality	Milne Abelian Varieties opening/middle chapters; Hindry-Silverman companion sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Mordell-Weil, canonical heights and finiteness mechanisms	Hindry-Silverman relevant chapters; Silverman height chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Mordell-Weil, canonical heights and finiteness mechanisms	Hindry-Silverman relevant chapters; Silverman height chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Faltings theorem(s): finiteness of rational	Milne Abelian Varieties Faltings discussion; strong expository	Definition sheet + 5 basic problems + 1 worked example.

Week	Main focus	Reading / lecture notes	Required output
	points/isogeny theorem	sources before original paper. First pass: definitions, examples, and 2-3 basic exercises.	
8	Prove/use: Faltings theorem(s): finiteness of rational points/isogeny theorem	Milne Abelian Varieties Faltings discussion; strong expository sources before original paper. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Proof architecture: moduli, semistable reduction, Arakelov/height ingredients	Read a modern detailed exposition; mark which ingredients are black boxes at first pass. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Proof architecture: moduli, semistable reduction, Arakelov/height ingredients	Read a modern detailed exposition; mark which ingredients are black boxes at first pass. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Compare Faltings, Chabauty, Baker and Subspace methods	Solve examples where different finiteness/effectivity methods apply. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Compare Faltings, Chabauty, Baker and Subspace methods	Solve examples where different finiteness/effectivity methods apply. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Heights and Jacobians, objects and examples. Reading: Hindry-Silverman/Bombieri-Gubler. Definition/result focus: Weil height; canonical height; Jacobian; Abel-Jacobi. Work: Compute divisors/Jacobian classes on genus-2 examples.

Week 2 - Heights and Jacobians, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute divisors/Jacobian classes on genus-2 examples. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Mordell-Weil and descent review, objects and examples. Reading: Silverman/Hindry-Silverman. Definition/result focus: Mordell-Weil; descent; height pairing. Work: Carry out a descent/height finite-generation toy proof.

Week 4 - Mordell-Weil and descent review, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Carry out a descent/height finite-generation toy proof. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Abelian varieties and isogenies, objects and examples. Reading: Milne Abelian Varieties. Definition/result focus: abelian variety; isogeny; Tate module; polarization. Work: Compute basic isogenies and state Tate/Faltings isogeny theorem precisely.

Week 6 - Abelian varieties and isogenies, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute basic isogenies and state Tate/Faltings isogeny theorem precisely. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Faltings finiteness machinery, objects and examples. Reading: Faltings expositions/Hindry-Silverman. Definition/result focus: Shafarevich finiteness; Faltings height; moduli/Arakelov inputs. Work: Create a black-box ledger with exact hypotheses/conclusions.

Week 8 - Faltings finiteness machinery, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Create a black-box ledger with exact hypotheses/conclusions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Mordell-Faltings deduction, objects and examples. Reading: standard Faltings expositions. Definition/result focus: finite rational points on genus >1 curves. Work: Explain Abel-Jacobi/Jacobian route and why MW alone is insufficient.

Week 10 - Mordell-Faltings deduction, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Explain Abel-Jacobi/Jacobian route and why MW alone is insufficient. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Effectivity and modern Diophantine geometry, objects and examples. Reading: Bombieri-Gubler/surveys. Definition/result focus: effectivity gap; Chabauty/Faltings comparison orientation. Work: For a genus-2 curve, distinguish what theorem guarantees from what an algorithm can compute.

Week 12 - Effectivity and modern Diophantine geometry, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: For a genus-2 curve, distinguish what theorem guarantees from what an algorithm can compute. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Weil/projective height
- Northcott property
- canonical/Neron-Tate height
- abelian variety
- polarization
- Jacobian and Abel-Jacobi map
- isogeny and Tate module
- Shafarevich finiteness condition
- Faltings height (working definition/orientation)
- Mordell-Faltings finiteness statement
- effectivity versus ineffectivity in Diophantine finiteness

Key results to know and use

- Northcott finiteness and standard height functoriality/triangle estimates.
- Mordell-Weil theorem and quadraticity/positivity properties of the Neron-Tate height.
- Faltings isogeny theorem (Tate conjecture for homomorphisms of abelian varieties over number fields) in a precise standard formulation.
- Shafarevich/Faltings finiteness theorem for abelian varieties with prescribed dimension/polarization/good reduction data in the formulation used by the exposition.
- Mordell-Faltings theorem: a smooth projective curve of genus >1 over a number field has finitely many rational points.
- Abel-Jacobi/Jacobian embedding mechanism explaining how curve arithmetic is connected to abelian varieties.
- Effectivity audit: identify which finiteness statements do not automatically provide computable bounds or an algorithm for all rational points.

Quarter-specific mastery targets

M1 - Exact language: Weil/canonical height, Jacobian, Abel-Jacobi map, isogeny class, Mordell-Weil group, Faltings height and finiteness of rational points on genus >=2 curves.

M2 - Proofs to reproduce: Northcott and canonical-height properties in tractable cases; Abel-Jacobi constructions for explicit curves; descent/height arguments already available from earlier quarters.

M3 - Research theorems, exact statement + architecture: Faltings isogeny theorem, finiteness theorem for abelian varieties of bounded data, and Mordell-Faltings; identify the Tate-conjecture/isogeny and Arakelov/moduli inputs at architecture level.

M4 - Effectivity audit: explain precisely what finiteness is obtained and where effective height bounds/rational-point algorithms are absent or incomplete.

M5 - Exit use: given a genus-2 example, explain the Jacobian/height route and which Faltings-level theorem would convert arithmetic control into finiteness.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V26U1: Heights and finiteness principles

V26U1P1. [F] Compute absolute logarithmic heights of rational numbers, quadratic algebraic numbers, roots of unity, and points in P^n(Q) from local/global definitions.

V26U1P2. [T] Prove basic height inequalities h(alpha beta), h(alpha+beta), h(alpha^n), and Northcott finiteness for bounded degree and height in a selected formulation.

V26U1P3. [S] Use heights to turn a naive infinite Diophantine search into a finite search once a height bound is given. Work a simple Thue/Mordell-type toy example.

Assigned block V31U4: Divisors, line bundles and curves

V31U4P1. [F] Compute Weil/Cartier divisors and Picard groups on P^1, and divisors of rational functions on a smooth projective curve.

V31U4P2. [T] Prove Riemann-Roch for P^1 directly and verify the general curve formula on several low-degree divisors where dimensions are known.

V31U4P3. [S] Relate line bundles, Cartier divisors, and invertible sheaves by converting one explicit object through all three descriptions.

Assigned block V32U3: Heights, descent and Mordell-Weil

V32U3P1. [F] Compute naive/canonical heights numerically for multiples of a rational point on a simple curve and observe quadratic growth hhat(nP)=n^2 hhat(P).

V32U3P2. [T] Work through a 2-descent on a curve with accessible rational 2-torsion or use a standard worked example to bound $E(Q)/2E(Q)$ and the rank.

V32U3P3. [R] Reconstruct Mordell-Weil theorem architecture: weak Mordell-Weil + height descent + Northcott. Identify precisely where each earlier number-theory result is used.

Assigned block V32U6: Faltings and Sato-Tate roadmaps

V32U6P1. [R] State Faltings' theorem (Mordell conjecture) and the isogeny theorem in precise standard formulations. Build a dependency map through abelian varieties, heights, moduli, and finiteness without attempting the full proof.

V32U6P2. [F] For an elliptic curve without CM, compute normalized Frobenius traces $a_p/(2\sqrt{p})$ for many small good primes by code and compare with the Sato-Tate density; for a CM example compare the different behavior.

V32U6P3. [R] State a modern Sato-Tate theorem for elliptic curves over Q in a precise non-CM formulation and map the automorphy/potential automorphy plus equidistribution inputs.

Quarter-level integration problems

I1. [S]. For a genus-2 hyperelliptic curve with a rational base point, write its Abel-Jacobi embedding into the Jacobian, compute divisor classes in examples, and explain why Mordell-Weil finiteness alone does not imply finiteness of $C(Q)$.

I2. [R]. Build a Faltings proof-dependency ledger separating: heights/Northcott, abelian varieties and isogenies, moduli/Arakelov finiteness, and the final Mordell deduction; for each black box give a precise theorem statement.

Full written solutions required for: V26U1P2, V26U1P3, V31U4P2, V31U4P3, V32U3P2, V32U3P3, V32U6P2, V32U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write a 35-45 page exposition of Mordell-Faltings with a 'proof dependency ledger' separating fully mastered lemmas, understood black boxes and future prerequisites.

Exit checkpoint

Prove the Northcott/canonical-height statements used in simple cases, compute Jacobians/Abel-Jacobi maps for tractable curves, state Faltings isogeny/Mordell results precisely, and explain why the proof is ineffective in the relevant sense.

Research bridge

This quarter realizes the Faltings/Mordell target and provides global arithmetic context for arithmetic geometry and Langlands-adjacent research.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q38. Etale cohomology, Weil conjectures, Deligne/Bombieri, and finite fields III

Quarter overview and reading route

Objective: Build enough etale/l-adic cohomology to understand the cohomological formulation of zeta functions and the Weil conjectures, then study Deligne's purity/RH result and Bombieri-type consequences for exponential sums.

Prerequisites: Q20/Q24 algebraic geometry, Q7/Q24 finite fields, Q13 Fourier/character sums.

30-hour allocation: Etale/l-adic cohomology 16h; Weil/Deligne 9h; exponential-sum consequences 3h; review 2h.

Core books and chapters/sections

- Milne, Lectures on Etale Cohomology: etale sites, l-adic cohomology, Poincare duality, trace formula and Weil-conjecture applications.
- Deligne, La conjecture de Weil I (use Milne’s English translation) and selected Weil II/weight expositions after Milne.
- Milne/Katz expositions on l-adic sheaves, trace functions and exponential sums; use Katz for conductor/monodromy examples selectively.
- Bombieri, Counting points on curves over finite fields [after Stepanov], Bourbaki 1972/73: study the alternative curve-level auxiliary-function proof and keep its scope distinct from Deligne’s higher-dimensional weight theorem.
- Lidl-Niederreiter / Ireland-Rosen finite-field and character-sum sections only as computational support, not as replacement for the cohomological material.
- Background refresh (only if needed): the scheme/cohomology prerequisites from Q20/Q24 rather than rereading unrelated introductory topology chapters.

Lecture notes and companions

- J.S. Milne, Lectures on Etale Cohomology, as the main lecture-note spine.
- Milne’s Deligne Weil I translation/notes and Katz-style sheaf/exponential-sum notes for weights and trace functions.
- Bombieri’s Bourbaki exposition of Stepanov’s curve method as a separate non-cohomological comparison route.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Etale morphisms, sites, sheaves and finite etale covers	Milne Lectures on Etale Cohomology opening chapters; Vakil/Milne geometry review. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Etale morphisms, sites, sheaves and finite etale covers	Milne Lectures on Etale Cohomology opening chapters; Vakil/Milne geometry review. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Etale cohomology, fundamental group and comparison intuition	Milne core cohomology chapters; work examples G_m, curves. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Etale cohomology, fundamental group and comparison intuition	Milne core cohomology chapters; work examples G_m, curves. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: l-adic cohomology, Frobenius and Grothendieck-Lefschetz trace formula	Milne trace-formula chapters. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: l-adic cohomology, Frobenius and Grothendieck-Lefschetz trace formula	Milne trace-formula chapters. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Zeta functions and Weil	Milne Weil-conjecture chapters;	Definition sheet + 5 basic problems

Week	Main focus	Reading / lecture notes	Required output
	conjectures: rationality, functional equation, Betti numbers	read historical/expository overview. First pass: definitions, examples, and 2-3 basic exercises.	+ 1 worked example.
8	Prove/use: Zeta functions and Weil conjectures: rationality, functional equation, Betti numbers	Milne Weil-conjecture chapters; read historical/expository overview. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Deligne purity and Riemann hypothesis over finite fields	Milne's proof exposition; Deligne Weil I/II selected sections after the map is clear. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Deligne purity and Riemann hypothesis over finite fields	Milne's proof exposition; Deligne Weil I/II selected sections after the map is clear. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Bombieri/Weil bounds, character/exponential sums and arithmetic applications	Katz/Milne expositions; derive square-root cancellation consequences in concrete sums. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Bombieri/Weil bounds, character/exponential sums and arithmetic applications	Katz/Milne expositions; derive square-root cancellation consequences in concrete sums. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - Etale topology and l-adic cohomology, objects and examples. Reading: Milne Lectures on Etale Cohomology opening chapters. Definition/result focus: etale morphism/site; constructible sheaf; l-adic cohomology. Work: Compute standard examples such as A1/P1/G_m using cited theorems.

Week 2 - Etale topology and l-adic cohomology, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute standard examples such as A1/P1/G_m using cited theorems. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Trace formula and zeta functions, objects and examples. Reading: Milne trace-formula chapters. Definition/result focus: Grothendieck-Lefschetz trace formula; zeta factorization. Work: Derive point counts from Frobenius traces for P1 and elliptic curves.

Week 4 - Trace formula and zeta functions, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive point counts from Frobenius traces for P1 and elliptic curves. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Weil conjectures: structure, objects and examples. Reading: Milne/Deligne Weil I introduction. Definition/result focus: rationality; functional equation; Betti numbers; RH/weights. Work: State each conjectural component separately and identify historical/cohomological proof inputs.

Week 6 - Weil conjectures: structure, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State each conjectural component separately and identify historical/cohomological proof inputs. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Deligne weights/RH, objects and examples. Reading: Deligne Weil I (Milne translation) + Weil II orientation. Definition/result focus: purity; eigenvalue absolute value q^{i/2}. Work: Map the proof architecture and derive square-root cancellation from weight bounds.

Week 8 - Deligne weights/RH, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Map the proof architecture and derive square-root cancellation from weight bounds. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Curves: Weil and Stepanov-Bombieri, objects and examples. Reading: Bombieri Bourbaki 1973 after Stepanov; standard curve texts. Definition/result focus: Hasse-Weil bound/RH for curves; Stepanov-Bombieri function-field method. Work: Work through the central auxiliary-function/point-count mechanism.

Week 10 - Curves: Weil and Stepanov-Bombieri, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Work through the central auxiliary-function/point-count mechanism; do not conflate it with Deligne for higher-dimensional varieties. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Trace functions and exponential sums, objects and examples. Reading: Katz sheaf/exponential-sum expositions. Definition/result focus: conductor; trace function; Deligne/Katz square-root bounds. Work: Represent a character/exponential sum cohomologically and identify singularity/conductor contributions.

Week 12 - Trace functions and exponential sums, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Represent a character/exponential sum cohomologically and identify singularity/conductor contributions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- etale morphism and etale site
- constructible/l-adic sheaf
- l-adic etale cohomology
- compactly supported cohomology
- geometric/arithmetic Frobenius
- Grothendieck-Lefschetz trace formula
- zeta function over F_q
- weight and purity
- trace function and conductor
- additive/multiplicative character
- Gauss/Jacobi/Kloosterman-type sum
- Stepanov auxiliary-function method for curves

Key results to know and use

- Grothendieck-Lefschetz trace formula expressing finite-field point counts as alternating Frobenius traces.
- Weil conjectures separated into rationality, functional equation/duality, Betti-number interpretation and Riemann-hypothesis/weight statement.
- Deligne Weil I purity/RH: Frobenius eigenvalues on H^i of a smooth projective variety have complex absolute value $q^{i/2}$.
- Hasse-Weil/Riemann-hypothesis bound for smooth projective curves and its zeta-function formulation.
- Stepanov-Bombieri auxiliary-function proof route for the curve-level RH/Hasse-Weil bound; keep its scope distinct from Deligne's higher-dimensional theorem.
- Deligne/Katz trace-function square-root cancellation in a selected sheaf-theoretic special case with conductor/singularity hypotheses.
- Explicit finite-field trace/norm and Gauss/Jacobi-sum identities used as computational test cases.

Quarter-specific mastery targets

M1 - Exact language: etale morphism/site, etale/l-adic cohomology, Frobenius action, compact support, Grothendieck-Lefschetz trace formula, weight, purity and zeta function of a variety over F_q.

M2 - Proofs/computations: compute etale cohomology in basic examples using standard results; derive point counts from known Frobenius eigenvalues; prove elementary character-sum bounds available before Deligne.

M3 - Research theorems, exact statement + architecture: all Weil conjectures; Grothendieck trace formula; Deligne purity/Riemann-hypothesis theorem; a Bombieri/Katz application to exponential sums.

M4 - Mechanism mastery: explain why cohomological eigenvalue weights imply square-root cancellation and how local geometric singularities affect conductor/cohomology terms.

M5 - Exit use: turn a concrete finite-field sum into a trace function/sheaf-style point-count problem and identify the exact Deligne-type input needed.

Quarter exercise assignment - exact core set

Required set: 15 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V33U1: Etale morphisms and sites

V33U1P1. [F] Verify examples of etale morphisms: Spec $k[t,t^{-1}]$->Spec $k[s,s^{-1}]$ with $s=t^n$ when char(k) does not divide n, and identify failure in bad characteristic.

V33U1P2. [T] Compare Zariski and etale covers on a scheme and construct the etale site of a very simple scheme at the level of objects/covers. Explain why ordinary topology is too coarse for finite-cover/Galois information.

V33U1P3. [S] Compute the etale fundamental group of Spec F_q and explain its relation to profinite completion of Z generated by Frobenius.

Assigned block V33U2: Etale and l-adic cohomology

V33U2P1. [F] Compute $H^i_{et}(\mathrm{Spec}\ F_q, Z/lZ)$ in low degrees using continuous Galois cohomology in a simple coefficient module.

V33U2P2. [T] For P^1 over an algebraically closed field, state and verify the expected l-adic cohomology groups and Tate twist in degree 2.

V33U2P3. [S] Compare singular cohomology of a complex variety with l-adic etale cohomology via the comparison theorem in an example. Identify which formal properties are being transported.

Assigned block V33U4: Frobenius and trace formula

V33U4P1. [F] For P^1/F_q and an elliptic curve/F_q, express $\#X(F_{q^n})$ in terms of Frobenius eigenvalues and recover the zeta function.

V33U4P2. [T] Derive the Grothendieck-Lefschetz trace formula in these examples from known cohomology groups, checking signs and Tate twists.

V33U4P3. [S] Show how rationality and functional-equation structure of zeta follows formally from finite-dimensional Frobenius actions plus duality, leaving the Riemann-hypothesis bound as the deep input.

Assigned block V33U5: Weil conjectures and Deligne

V33U5P1. [F] State all Weil conjectures for a smooth projective variety over F_q and verify them for P^n and curves using known formulas.

V33U5P2. [R] Reconstruct Deligne's proof architecture for the Riemann hypothesis over finite fields: weights, Lefschetz pencils/monodromy or the chosen exposition, tensor-power trick, purity conclusion. Mark black-box inputs.

V33U5P3. [S] Translate the weight bound |alpha|=q^{i/2} into a square-root cancellation estimate for a point count. Work one explicit error term.

Assigned block V33U6: Bombieri, Katz and exponential sums

V33U6P1. [T] Use an l-adic sheaf/trace-function viewpoint to reinterpret a classical character or Kloosterman sum. Identify conductor/ramification qualitatively in a simple example.

V33U6P2. [R] State a precise Deligne/Katz square-root bound for a selected trace-function or character-sum class, including geometric irreducibility/conductor hypotheses, and derive the numerical O(q^{1/2})-type estimate. Do not call this the Stepanov-Bombieri proof of the curve RH; that is a different method.

V33U6P3. [R] Compare two routes side by side: (A) the Stepanov-Bombieri auxiliary-function proof of the Hasse-Weil/RH bound for curves; (B) a Deligne-Katz sheaf/trace-function square-root estimate. State the object, hypotheses, mechanism and scope of each, and explain why route (A) is not a proof of the higher-dimensional Weil conjectures.

Quarter-level integration problems

I1. [S]. Compute Z(P^1/F_q,t) and Z(E/F_q,t) from point counts, then recover the expected cohomological factors and Frobenius eigenvalue weights; check the functional form explicitly.

I2. [R]. Choose a nontrivial character/exponential sum that is controlled by a curve or l-adic sheaf; write the trace formula representation and show exactly how Deligne's weight bound produces square-root cancellation.

Full written solutions required for: V33U1P2, V33U1P3, V33U2P2, V33U2P3, V33U4P2, V33U4P3, V33U5P2, V33U5P3, V33U6P2, V33U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Prepare a 45-page set of notes 'From point counts to eigenvalues of Frobenius', including explicit zeta functions, the trace formula, a theorem map for Deligne and several exponential-sum applications.

Exit checkpoint

Compute zeta functions of basic varieties, use the trace formula in examples, state all Weil conjectures precisely, explain weights/purity, and derive a Weil/Deligne square-root bound for a selected sum.

Research bridge

This quarter realizes the finite-field RH/Weil/Deligne/Bombieri target and creates the cohomological language behind modern arithmetic geometry.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q39. Poincare, Ricci flow, Hauptvermutung, and cryptography III

Quarter overview and reading route

Objective: Complete two independent capstone branches: geometric/topological understanding of the Poincare conjecture and Hauptvermutung history, and modern lattice-based cryptography grounded in geometry of numbers.

Prerequisites: Q14 Riemannian geometry/PDE, Q20 algebraic topology, Q6/Q17 geometry of numbers, Q19 cryptography.

30-hour allocation: Ricci flow/Poincare 15h; topology/Hauptvermutung 5h; lattice cryptography 8h; synthesis 2h.

Core books and chapters/sections

- Kleiner-Lott, Notes on Perelman’s Papers; Morgan-Tian, Ricci Flow and the Poincare Conjecture: use for the geometrization/Poincare proof architecture.
- Lee/do Carmo Riemannian geometry only as a targeted curvature/Ricci-flow refresher.
- A standard PL-topology/triangulation source for the Hauptvermutung: record category and dimension in every statement.
- Lenstra-Lenstra-Lovasz original theorem or a modern lattice-algorithms text for LLL; connect to the geometry-of-numbers material already mastered rather than rereading Cassels from the beginning.
- Regev’s LWE paper plus Peikert/Micciancio-Regev lecture notes for SIS/LWE and worst-case-to-average-case reductions; state parameter regimes carefully.
- Optional computational lab: Sage/Python implementation of LLL and toy LWE to test correctness, never as evidence of security.

Lecture notes and companions

- Kleiner-Lott notes are the primary lecture-note companion for Poincare; use modern lattice-crypto course notes for LWE.
- MIT 18.755 Chapter I differential-geometry notes are a useful companion.
- MIT OCW 18.785 Number Theory I, Lecture 14 (Geometry of Numbers).

Reading rule: Do a fast first pass before lecture/problem work, then a proof-level second pass. Where a reference says “selected” or names a topic rather than a chapter number, follow the topic heading in your edition; numbering varies across editions.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Ricci curvature and Ricci-flow equation; evolution identities	Kleiner-Lott opening notes; Morgan-Tian foundational chapters; review Riemannian curvature. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Ricci curvature and Ricci-flow equation; evolution identities	Kleiner-Lott opening notes; Morgan-Tian foundational chapters; review Riemannian curvature. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Singularities, monotonic quantities and noncollapsing	Kleiner-Lott corresponding Perelman-note sections. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Singularities, monotonic quantities and noncollapsing	Kleiner-Lott corresponding Perelman-note sections. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
5	Learn: Surgery, geometrization architecture and Poincare deduction	Kleiner-Lott/Morgan-Tian later chapters; focus on theorem dependencies. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Surgery,	Kleiner-Lott/Morgan-Tian later	2 theorem reconstructions + 5

Week	Main focus	Reading / lecture notes	Required output
	geometrization architecture and Poincare deduction	chapters; focus on theorem dependencies. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	harder problems + one 10-minute oral explanation.
7	Learn: Triangulations, PL topology and Hauptvermutung: statements/counterexamples	A standard PL-topology source and historical expositions; connect to manifold topology. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Triangulations, PL topology and Hauptvermutung: statements/counterexamples	A standard PL-topology source and historical expositions; connect to manifold topology. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: LLL, SVP/CVP, SIS and lattice hardness	Micciancio-Regev/Peikert lattice-crypto notes; revisit geometry of numbers. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: LLL, SVP/CVP, SIS and lattice hardness	Micciancio-Regev/Peikert lattice-crypto notes; revisit geometry of numbers. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: LWE, reductions and post-quantum constructions	Regev LWE paper after lecture-note exposition; modern lattice-crypto notes on encryption/signatures. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: LWE, reductions and post-quantum constructions	Regev LWE paper after lecture-note exposition; modern lattice-crypto notes on encryption/signatures. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - 3-manifold topology and geometrization setup, objects and examples. Reading: Morgan-Tian/Kleiner-Lott introductory chapters. Definition/result focus: prime decomposition; geometrization; Ricci flow. Work: Review curvature/3-manifold language and simple Ricci-flow solutions.

Week 2 - 3-manifold topology and geometrization setup, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Review curvature/3-manifold language and simple Ricci-flow solutions. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Perelman functionals/noncollapsing, objects and examples. Reading: Kleiner-Lott Perelman notes. Definition/result focus: W/entropy; reduced volume; no local collapsing. Work: Compute monotonicity in model settings and state no-local-collapsing precisely.

Week 4 - Perelman functionals/noncollapsing, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compute monotonicity in model settings and state no-local-collapsing precisely. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Surgery and Poincare, objects and examples. Reading: Kleiner-Lott/Morgan-Tian. Definition/result focus: canonical neighborhoods; surgery; extinction. Work: Build the geometrization/Poincare proof architecture with each black box named.

Week 6 - Surgery and Poincare, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Build the geometrization/Poincare proof architecture with each black box named. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Hauptvermutung and triangulations, objects and examples. Reading: PL-topology sources. Definition/result focus: PL homeomorphism; triangulation; Hauptvermutung status by dimension/category. Work: Prepare a precise status table and examples of failure.

Week 8 - Hauptvermutung and triangulations, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prepare a precise status table and examples of failure. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Lattice algorithms, objects and examples. Reading: LLL sources/Micciancio-Regev notes. Definition/result focus: Gram-Schmidt; LLL; SVP/CVP approximation. Work: Prove LLL guarantee and implement small-dimensional reduction.

Week 10 - Lattice algorithms, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Prove LLL guarantee and implement small-dimensional reduction. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - LWE/SIS, objects and examples. Reading: Regev/Peikert notes. Definition/result focus: SIS; LWE; worst-case-to-average-case reduction. Work: State a representative reduction with parameters and identify classical/quantum components.

Week 12 - LWE/SIS, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State a representative reduction with parameters and identify classical/quantum components. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- Ricci flow and Ricci soliton
- Perelman W-entropy/reduced volume
- no-local-collapsing
- canonical neighborhood and surgery
- geometrization/Poincare statement
- triangulation, PL homeomorphism and Hauptvermutung
- lattice basis and Gram-Schmidt data
- LLL-reduced basis
- SVP/CVP
- SIS
- LWE
- discrete Gaussian and smoothing parameter

Key results to know and use

- Basic Ricci-flow evolution equations in tractable settings; Perelman monotonicity and no-local-collapsing statements used in the 3-manifold argument.
- Perelman/Kleiner-Lott/Morgan-Tian geometrization-Poincare architecture: canonical neighborhoods, surgery/extinction and the exact theorem conclusion.
- Hauptvermutung status by dimension/category: know precise positive and negative results from the chosen PL-topology source rather than one blanket sentence.
- LLL theorem: polynomial-time lattice-basis reduction and its quantitative first-vector/orthogonality guarantees in the chosen normalization.
- Representative SIS worst-case/average-case hardness theorem and representative LWE worst-case/average-case reduction with dimensions, approximation factors, modulus/noise regime and classical/quantum status recorded.

Quarter-specific mastery targets

M1 - Exact language: Ricci flow, Ricci soliton, Perelman entropy/reduced volume/noncollapsing, surgery, triangulation/Hauptvermutung, lattice successive minima, LLL, SIS and LWE.

M2 - Proofs to reproduce: basic Ricci-flow evolution calculations for metric/volume/curvature in simple settings; LLL approximation/orthogonality-defect bounds; correctness reductions for toy SIS/LWE constructions.

M3 - Research theorems, exact statement + architecture: Perelman no-local-collapsing and geometrization/Poincare route; status/failure of Hauptvermutung by category/dimension; worst-case-to-average-case LWE/SIS reduction in a representative formulation.

M4 - Separation of evidence: distinguish geometric-flow proof architecture from topological classification and from complexity-theoretic cryptographic reduction; record assumptions in each.

M5 - Exit use: explain one deep proof architecture (Poincare) and one modern hardness architecture (LWE) with no unnamed 'standard arguments'.

Quarter exercise assignment - exact core set

Required set: 12 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V13U5: Cobordism and Hauptvermutung context

V13U5P1. [F] Compare simplicial, PL, smooth, and topological manifolds on low-dimensional examples; triangulate basic surfaces and compute links of simplices.

V13U5P2. [T] Reconstruct the exact statement of the Hauptvermutung and build a chronology of its positive low-dimensional cases and counterexamples. Distinguish 'triangulable' from 'unique PL structure'.

V13U5P3. [R] Choose one obstruction/counterexample theorem (Milnor or later developments) and write a precise theorem audit: hypotheses, category, conclusion, and what it disproves.

Assigned block V13U6: Ricci flow and Poincare/Geometrization

V13U6P1. [F] Derive Ricci flow on constant-curvature metrics and solve the resulting ODE for the scale factor. Interpret finite-time shrinking/expansion.

V13U6P2. [T] Derive evolution of volume and scalar curvature in a selected simple setting, and compute one monotone quantity in a model case.

V13U6P3. [R] Build a proof-dependency map for Perelman's route to Poincare/Geometrization: short-time existence, singularities, entropy/noncollapsing, surgery or modern replacement, long-time/topological conclusion. For each box identify the reference chapter to read.

Assigned block V36U5: Lattices, LLL, SIS and LWE

V36U5P1. [F] Run LLL by hand on a small integer basis, recording Gram-Schmidt coefficients and Lovasz checks. Compare output vector lengths with successive minima bounds.

V36U5P2. [T] Define SVP, CVP, SIS, and LWE precisely. Construct a toy LWE instance and show why solving exact linear equations fails because of noise while a secret still exists.

V36U5P3. [R] Trace the worst-case/average-case reduction philosophy for LWE at theorem-architecture level: lattice problem, Gaussian/smoothing parameter, average LWE distribution, cryptographic hardness conclusion.

Assigned block V36U6: Post-quantum constructions and proof frameworks

V36U6P1. [T] Prove IND-CPA security of a simple LWE/Regev-style encryption scheme from decisional LWE in a clean hybrid sequence, writing each hybrid difference and reduction.

V36U6P2. [S] Compare module/ring lattice assumptions with plain LWE: what algebraic structure buys efficiency and what additional assumption/attack surface it introduces.

V36U6P3. [R] Choose one standard post-quantum primitive family (lattice KEM/signature, code-based, hash-based). Produce a proof-framework audit: hardness assumption, reduction model, correctness failure probability, and implementation-side issues that are outside the mathematical proof.

Quarter-level integration problems

I1. [S]. Derive the Ricci-flow evolution of volume in a simple homogeneous/constant-curvature setting, identify the singularity time, and explain which additional Perelman estimates are needed to continue the global geometrization strategy.

I2. [R]. State a representative LWE worst-case-to-average-case reduction with parameter regime; trace where lattice smoothing/Gaussian/dual-lattice facts enter and separate classical from quantum assumptions if the source does.

Full written solutions required for: V13U5P2, V13U5P3, V13U6P2, V13U6P3, V36U5P2, V36U5P3, V36U6P2, V36U6P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Write two capstone notes: a proof architecture of Poincare via Ricci flow, and a mathematical analysis of LWE from geometry-of-numbers concepts to security reduction.

Exit checkpoint

Derive basic Ricci-flow evolution equations in simple cases, explain Perelman's entropy/noncollapsing role, state Hauptvermutung and its status in relevant categories, prove LLL approximation bounds, and formulate LWE/SIS hardness reductions accurately.

Research bridge

This closes the topology/geometric-analysis and cryptography branches while reinforcing how geometry of numbers reaches both pure Diophantine theory and modern cryptography.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Q40. Particle systems, kinetic/field limits, Ising, and final research synthesis

Quarter overview and reading route

Objective: Use analysis, probability, PDE and statistical mechanics to understand microscopic-to-macroscopic limits, hard-particle kinetic theory and Ising-type models, then convert the ten-year / 40-quarter curriculum into active research programs.

Prerequisites: Q10-Q15 analysis/PDE, Q23 advanced probability, plus mature proof skills from the entire program.

30-hour allocation: Kinetic/particle limits 14h; Ising/statistical mechanics 8h; selected research specialization 6h; final synthesis 2h.

Core books and chapters/sections

- Spohn, Large Scale Dynamics of Interacting Particles: interacting systems, kinetic/hydrodynamic viewpoints and propagation-of-chaos orientation.
- Cercignani-Illner-Pulvirenti, The Mathematical Theory of Dilute Gases: BBGKY, Boltzmann-Grad scaling, collision trees and Lanford theorem background.
- Deng-Hani-Ma, Long time derivation of the Boltzmann equation from hard sphere dynamics (v3, 2025): read Introduction, theorem statements and detailed proof overview; use the 2026 expository literature as a companion if useful.
- Bodineau-Gallagher-Saint-Raymond sources on the Brownian-motion limit of a tagged hard sphere; Saint-Raymond/hydrodynamic-limit sources for fluid limits.
- Friedli-Velenik, Statistical Mechanics of Lattice Systems: Ising model, Gibbs measures, correlation inequalities and phase transition.
- Use Karatzas-Shreve/Evans/Brezis only as prerequisite refreshers for stochastic/PDE tools already covered; do not allocate a fresh foundational block here.

Lecture notes and companions

- Cercignani-Illner-Pulvirenti/Saint-Raymond lecture-style treatments for BBGKY, Lanford and hydrodynamic limits.
- Deng-Hani-Ma theorem overview and subsequent expository accounts for the long-time hard-sphere result; record version/date because this is frontier material.
- Bodineau-Gallagher-Saint-Raymond notes/papers for the tagged-particle Brownian limit; Friedli-Velenik for Ising/Gibbs/phase-transition problems.

12-week schedule

Week	Main focus	Reading / lecture notes	Required output
1	Learn: Hamiltonian particle systems, empirical measures and BBGKY hierarchy	Spohn opening chapters; Cercignani-Illner-Pulvirenti kinetic preliminaries. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
2	Prove/use: Hamiltonian particle systems, empirical measures and BBGKY hierarchy	Spohn opening chapters; Cercignani-Illner-Pulvirenti kinetic preliminaries. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
3	Learn: Boltzmann equation and Lanford theorem architecture	Cercignani-Illner-Pulvirenti; modern Lanford lecture notes before primary sources. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
4	Prove/use: Boltzmann equation and Lanford theorem architecture	Cercignani-Illner-Pulvirenti; modern Lanford lecture notes before primary sources. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week	Main focus	Reading / lecture notes	Required output
5	Learn: Hydrodynamic limits and Hilbert's sixth problem	Saint-Raymond selected chapters; recent survey literature on derivations from particles. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
6	Prove/use: Hydrodynamic limits and Hilbert's sixth problem	Saint-Raymond selected chapters; recent survey literature on derivations from particles. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
7	Learn: Hard-sphere diffusion/Brownian limits and fluctuation ideas	Author surveys and recent papers; identify assumptions, scaling and topology of convergence. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
8	Prove/use: Hard-sphere diffusion/Brownian limits and fluctuation ideas	Author surveys and recent papers; identify assumptions, scaling and topology of convergence. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
9	Learn: Ising model, Gibbs measures, phase transitions and correlation inequalities	Friedli-Velenik introductory/Ising chapters; work exact 1D/2D structural examples. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
10	Prove/use: Ising model, Gibbs measures, phase transitions and correlation inequalities	Friedli-Velenik introductory/Ising chapters; work exact 1D/2D structural examples. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.
11	Learn: Research synthesis: choose 2 primary and 2 secondary tracks and write concrete 12-month research-reading plans	Revisit all quarter exit checkpoints; select current survey and 3-5 seminal/recent papers per chosen track. First pass: definitions, examples, and 2-3 basic exercises.	Definition sheet + 5 basic problems + 1 worked example.
12	Prove/use: Research synthesis: choose 2 primary and 2 secondary tracks and write concrete 12-month research-reading plans	Revisit all quarter exit checkpoints; select current survey and 3-5 seminal/recent papers per chosen track. Second pass: theorem proofs, harder exercises, and a one-page proof reconstruction.	2 theorem reconstructions + 5 harder problems + one 10-minute oral explanation.

Week 1 - BBGKY and kinetic scalings, objects and examples. Reading: Cercignani-Illner-Pulvirenti/Spohn. Definition/result focus: BBGKY; empirical measures; mean-field vs Boltzmann-Grad. Work: Derive first hierarchy equations and formal closures.

Week 2 - BBGKY and kinetic scalings, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Derive first hierarchy equations and formal closures. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 3 - Lanford short-time Boltzmann limit, objects and examples. Reading: Lanford expositions/CIP. Definition/result focus: collision trees; recollisions; short-time theorem. Work: State scaling, marginals/topology and time horizon.

Week 4 - Lanford short-time Boltzmann limit, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: State scaling, marginals/topology and time horizon; trace where short time enters. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 5 - Long-time hard-sphere Boltzmann derivation, objects and examples. Reading: Deng-Hani-Ma 2025 v3 + 2026 expository follow-up if desired. Definition/result focus: arbitrarily long times while the regular Boltzmann solution exists; cumulant/diagram expansion. Work: Compare exactly with Lanford.

Week 6 - Long-time hard-sphere Boltzmann derivation, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Compare exactly with Lanford; treat the authors’ Hilbert-VI claim as the rarefied hard-sphere branch, not every formulation of Hilbert VI. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 7 - Hard-sphere Brownian/hydrodynamic limits, objects and examples. Reading: Bodineau-Gallagher-Saint-Raymond and Saint-Raymond sources. Definition/result focus: tagged-particle Brownian limit via linear Boltzmann; representative hydrodynamic scaling. Work: Create a scaling/topology/time-horizon comparison table.

Week 8 - Hard-sphere Brownian/hydrodynamic limits, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Create a scaling/topology/time-horizon comparison table. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 9 - Ising and interacting systems, objects and examples. Reading: Friedli-Velenik/Spohn. Definition/result focus: Gibbs measure; Peierls argument; FKG; phase transition. Work: Solve 1D Ising exactly and prove Peierls/association results in chosen setting.

Week 10 - Ising and interacting systems, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Solve 1D Ising exactly and prove Peierls/association results in chosen setting. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Week 11 - Final synthesis and research selection, objects and examples. Reading: all theorem/problem notebooks. Definition/result focus: dependency graph; theorem barrier; feasible entry problem. Work: Defend two research directions with precise known frontier, toy model, prerequisites and six-month reading/problem plan.

Week 12 - Final synthesis and research selection, theorem/technique pass. Finish the same reading at proof/architecture level. Required output: Defend two research directions with precise known frontier, toy model, prerequisites and six-month reading/problem plan. Reattempt the block's principal technique from a blank page 5-7 days later; record any failed hypothesis or normalization explicitly.

Definitions and theorem targets

Definitions you should be able to state instantly

- N-particle marginal and BBGKY hierarchy
- propagation of chaos
- mean-field scaling
- Boltzmann-Grad scaling
- Boltzmann collision operator
- collision tree and recollision
- cumulant/correlation expansion
- hydrodynamic scaling/local equilibrium
- tagged particle and linear Boltzmann equation
- Brownian scaling limit
- Gibbs/DLR measure
- Ising model, Peierls contour and FKG inequality

Key results to know and use

- BBGKY hierarchy derivation and formal mean-field/kinetic closures for representative N-particle models.
- Boltzmann H-theorem and hard-sphere collision-operator structure in the chosen normalization.
- Lanford theorem: Boltzmann-Grad limit of hard spheres for a sufficiently short time, with scaling, marginals/topology and time horizon stated exactly.
- Deng-Hani-Ma long-time hard-sphere-to-Boltzmann theorem (v3 2025): arbitrary long times while the regular Boltzmann solution exists; cumulant/diagram/cutting architecture understood at research-map level.
- Scope discipline: this long-time result concerns the rarefied hard-sphere/Boltzmann branch described by the authors in relation to Hilbert VI; it does not by itself settle every fluid/kinetic formulation of Hilbert's sixth problem.
- Bodineau-Gallagher-Saint-Raymond tagged-hard-sphere Brownian-motion limit via a linear Boltzmann regime, with exact scaling/time assumptions from the selected paper.
- One representative hydrodynamic limit theorem with microscopic/kinetic scaling, topology and regularity assumptions stated.
- 1D Ising exact solution; Peierls phase-transition argument and FKG/positive-association theorem in a standard finite/infinite-volume setting.

Quarter-specific mastery targets

M1 - Exact language: interacting particle system, empirical measure, BBGKY hierarchy, propagation of chaos, mean-field/Boltzmann-Grad/hydrodynamic scaling, Gibbs measure, Ising model and kinetic equation.

M2 - Proofs to reproduce: derivation of BBGKY for a finite N model; a simple mean-field/Vlasov limit estimate under strong regularity; exact 1D Ising partition function/correlations; one hydrodynamic-limit toy calculation.

M3 - Research theorems, exact statement + architecture: Lanford's short-time theorem; the Deng-Hani-Ma long-time extension (with the regular-Boltzmann-solution condition); a representative tagged-hard-sphere Brownian theorem; and one hydrodynamic limit, each with scaling, topology, time horizon and assumptions explicitly compared.

M4 - Method comparison: distinguish deterministic chaos/BBGKY, martingale/probability, entropy and PDE compactness approaches; explain which scale limit each controls.

M5 - Exit use: formulate two credible research-entry problems, each with a dependency chain, known theorem barrier, toy model and a first six-month reading/problem programme.

Quarter exercise assignment - exact core set

Required set: 18 precision problems from the named volume units below, plus I1-I2. The full wording is repeated here so this quarter can be followed without hunting through the volume section.

Assigned block V17U1: Interacting particle systems

V17U1P1. [F] For the simple exclusion/contact/zero-range process on a finite graph, write the generator and compute its action on basic observables. Identify invariant measures in one model.

V17U1P2. [T] Derive the graphical construction/coupling for an attractive interacting particle system and use it to prove monotonicity with respect to initial data.

V17U1P3. [S] Compute a one- or two-point evolution equation and identify exactly where closure fails, motivating correlation hierarchies.

Assigned block V17U2: Hydrodynamic limits

V17U2P1. [F] For independent random walks or exclusion on a discrete torus, derive the empirical density and state the expected hydrodynamic PDE.

V17U2P2. [T] Prove a law-of-large-numbers hydrodynamic limit in a toy independent-particle model via test functions and martingale estimates.

V17U2P3. [R] Read the entropy-method architecture for exclusion/zero-range: replacement lemma, local equilibrium, entropy inequality, PDE uniqueness. Explain the role of each.

Assigned block V17U4: BBGKY and mean-field limits

V17U4P1. [F] Derive the BBGKY hierarchy for an N-particle mean-field Hamiltonian or stochastic system through the first two marginals.

V17U4P2. [T] Assuming propagation of chaos, formally derive the Vlasov/McKean-Vlasov equation. Then prove a stability estimate in a Lipschitz mean-field model sufficient for a simple chaos result.

V17U4P3. [S] Compare mean-field scaling 1/N with Boltzmann-Grad scaling. Write a table of density, interaction range/strength, collision frequency, and limiting equation.

Assigned block V17U5: Boltzmann-Grad and dilute gases

V17U5P1. [F] Derive the Boltzmann collision operator formally from binary-collision geometry for hard spheres, including pre/post-collisional velocity parametrization.

V17U5P2. [T] Work through the first collision-tree terms of Lanford's expansion and identify recollisions as the main obstruction. Quantify the combinatorial growth at a schematic level.

V17U5P3. [R] State Lanford's theorem precisely enough to include time scale, topology/marginals, and Boltzmann-Grad regime. Build a proof architecture showing where short time enters.

Assigned block V17U6: Brownian/kinetic limits and Hilbert VI frontier

V17U6P1. [S] For a tagged particle in a dilute gas, derive formally how repeated weakly correlated collisions could yield Brownian motion; identify centering and variance scaling.

V17U6P2. [R] Make a four-row theorem table: Lanford short-time hard spheres -> Boltzmann; Deng-Hani-Ma long-time hard spheres -> Boltzmann; tagged hard sphere -> Brownian motion via linear Boltzmann; kinetic -> hydrodynamic PDE. For each record scaling, topology/marginals, time horizon, regularity assumptions and the obstruction overcome/not overcome. Add a scope note on what is meant by the hard-sphere branch of Hilbert VI.

V17U6P3. [X] Implement a small event-driven hard-sphere or random-collision simulation and compare empirical velocity/position statistics to kinetic/Brownian predictions. State clearly which correlations the simulation does not control.

Assigned block V16U4: Ising and random-cluster models

V16U4P1. [F] Derive the high-temperature expansion of the Ising model on a small graph and the Fortuin-Kasteleyn random-cluster representation in a finite example.

V16U4P2. [T] Prove FKG for the random-cluster measure in the relevant q>=1 setting at theorem-architecture level and translate an Ising spin correlation into a connectivity statement under Edwards-Sokal coupling.

V16U4P3. [S] Compare Ising, FK, and percolation observables on the same planar graph; identify which monotonicity/duality tools survive across models.

Quarter-level integration problems

I1. [S]. Derive the first two equations of the BBGKY hierarchy for a symmetric N-particle system and show formally how factorization leads to a mean-field/kinetic closure; list the estimates required to make the limit rigorous.

I2. [R]. Compare Lanford, Deng-Hani-Ma, a tagged-hard-sphere Brownian theorem and one hydrodynamic limit in a single matrix: microscopic scaling, limiting equation/process, convergence object/topology, time horizon, regularity/data assumptions and principal remaining obstruction. Then formulate one research question that changes exactly one row/column feature without claiming a broader Hilbert-VI resolution than the cited theorem states.

Full written solutions required for: V17U1P2, V17U1P3, V17U2P2, V17U2P3, V17U4P2, V17U4P3, V17U5P2, V17U5P3, V17U6P2, V16U4P2, V16U4P3, I1 and I2. For the remaining assigned problems, keep a checkable solution outline/computation log; a one-line answer is not sufficient.

Reattempt rule: one week after finishing each assigned block, redo its P2 from a blank page without notes. A quarter is not passed until at least three of these delayed P2 reattempts are essentially correct.

Quarter project

Final portfolio: a 50-page research prospectus containing two primary research directions, literature maps, prerequisites already mastered, remaining gaps, three candidate problems per direction and a one-year seminar/problem schedule.

Exit checkpoint

Explain BBGKY-to-Boltzmann architecture; state Lanford and the Deng-Hani-Ma long-time extension with their different time assumptions; compare a tagged-hard-sphere Brownian theorem and one hydrodynamic limit by scaling/topology; analyze the 1D Ising model exactly; and defend two research directions without overstating the scope of any Hilbert-VI result.

Research bridge

The curriculum ends by converting broad mastery into selective depth. The correct next step is no longer another survey course, but repeated cycles of paper reading, seminar exposition, lemma reconstruction and problem formulation.

Pass criterion: complete the project, pass the closed-notes checkpoint at roughly 70% quality or better, and have no undefined symbols in your theorem notebook. If not, use the buffer week before proceeding.

Appendix A. Problem-volume targets and exam policy

- Years 1-2: 40-70 serious problems/quarter (30 listed here plus textbook exercises). Favor proof-writing volume and computational fluency.
- Years 3-4: 35-55 problems/quarter. Longer proofs replace some short exercises.
- Years 5-6: 25-45 problems/quarter plus at least one substantial proof reconstruction from lecture notes or a paper.
- Years 7-8: 20-35 substantial problems/quarter plus paper reading and proof reconstruction. Years 9-10: 15-30 substantial problems/quarter plus seminars and capstone expositions; one research-level lemma can count as several routine exercises.
- Every quarter ends with a 2-3 hour closed-book written exam and a 30-60 minute oral exam. Every year ends with a cumulative oral that deliberately crosses subject boundaries.

Problem log template

- Problem/source and date
- First approach and why it seemed plausible
- Where the approach failed
- Key idea that unlocked the problem
- Clean final proof
- Generalization/counterexample
- Spaced-review dates: +2 days, +2 weeks, +2 months

When to read solutions

For routine exercises, make at least two genuine attempts separated by time. For hard proof problems, spend 60-120 focused minutes and record partial lemmas before consulting a hint. For research-level material, reading a proof is itself part of training, but you must subsequently close the source and reconstruct the argument.

Appendix B. Research-paper reading protocol

1. Pass 0 - context: read abstract/introduction/conclusion and identify the exact theorem, predecessors and claimed new ingredient.
2. Pass 1 - dependency map: write every definition and cited theorem needed to parse the main statement. Do not yet chase all proofs.
3. Pass 2 - architecture: divide the proof into 5-15 modules; for each module write input, output and mechanism.
4. Pass 3 - lemma reconstruction: choose the two or three lemmas carrying the genuinely new idea and reproduce them line by line.
5. Pass 4 - stress test: vary hypotheses, dimensions or parameters. Identify what breaks and whether the obstruction is conceptual or technical.
6. Pass 5 - exposition: give a seminar without following the paper page by page. The seminar should be organized by ideas, not chronology.
7. Pass 6 - research conversion: write three questions: one clarification, one plausible extension, and one deliberately ambitious problem. Track which is actually new by literature search.

Paper notebook fields

- Full citation and version/date
- Main theorem in your own notation
- Why it matters
- Prerequisite dependency graph
- Key new idea
- Technical bottleneck
- One lemma reconstructed in full
- Potential generalizations
- Questions for an expert
- Follow-up papers

Appendix C. Computational mathematics as a proof laboratory

Recommended tools: Python for basic experimentation; SageMath for algebra/number theory/elliptic curves/modular forms; PARI/GP for computational number theory; Mathematica/Maple only when helpful for symbolic checks. Computation is evidence and exploration unless a formally verified argument is supplied.

- Elementary/analytic NT: sieve experiments, prime counts, character sums, exponential sums, continued fractions and Pell recurrences.
- Algebraic NT/finite fields: factor primes in number fields, unit/class-group experiments, finite-field construction and Frobenius calculations.
- Additive combinatorics: enumerate sumsets, energies, Fourier spectra and small Gowers norms.
- Automorphic/modular: q-expansions, Hecke operators, Frobenius traces and L-function coefficient experiments.
- Geometry of numbers/crypto: lattice bases, Gram-Schmidt, LLL, toy SVP/CVP and LWE demonstrations.
- Probability/PDE/dynamics: simulations only after the model, scaling and convergence notion are stated mathematically.
- Geometric analysis: visualize flows/curvature for intuition; never substitute plots for estimates.

Correctness checklist for code

- Specify mathematical input/output types
- Prove termination when nontrivial
- Prove correctness/invariants
- Estimate complexity at least roughly
- Test boundary cases and random examples
- Separate floating-point error from mathematical error
- Record seeds and versions for reproducibility

Appendix D. The theorem notebook and dependency graph

Maintain one living document rather than isolated course notebooks. Each theorem receives a compact card and explicit links to prerequisites and descendants.

- Name / exact statement / standard notation.
- Definitions used in the statement.
- Two examples and one counterexample or sharpness example.
- Proof skeleton in 5-15 lines.
- Hardest lemma and its proof status: mastered / understood / black box.
- Where the theorem is used later in this curriculum.
- Equivalent formulations and common normalization differences.
- One problem that forces you to use the theorem without naming it.

High-value cross-field bridges to record explicitly

- Geometry of numbers <-> Diophantine approximation <-> homogeneous spaces.
- Fourier/harmonic analysis <-> exponential sums <-> circle method <-> decoupling.
- Finite fields <-> character sums <-> algebraic curves <-> Frobenius/etale cohomology.
- Number fields/local fields <-> adeles <-> automorphic representations/L-functions.
- Additive combinatorics <-> ergodic theory <-> incidence geometry/Kakeya.
- Elliptic curves <-> modular forms <-> Galois representations <-> FLT/Sato-Tate.
- Heights <-> Baker/Subspace <-> Mordell/Faltings.
- PDE/Riemannian geometry <-> Ricci flow; probability/PDE <-> kinetic and particle limits.
- Geometry of numbers <-> LLL/SVP/CVP <-> lattice cryptography.

Appendix E. Final specialization rule

The breadth of this ten-year plan is deliberately extreme. By the final two years, the goal is not equal mastery of every frontier. Use the following rule to convert breadth into research depth:

1. Choose two PRIMARY tracks in which you will read current papers and try problems.

2. Choose two SECONDARY tracks in which you maintain seminar-level literacy.

3. Maintain the remaining tracks at reference level through theorem notebooks and annual review.

4. For each primary track, identify a canonical monograph, 3-5 survey/expository papers, 5-10 foundational papers and 5-10 recent papers.

5. Run a weekly research seminar for yourself: one theorem statement, one lemma proof, one example/computation, one open question.

6. Every six months, write a 5-page state-of-understanding memo and delete low-value reading commitments.

Suggested primary-track combinations

- Analytic/additive: restriction-decoupling + Falconer/Furstenberg/Kakeya + analytic number theory.
- Arithmetic/automorphic: algebraic number theory + automorphic forms + Sato-Tate/FLT.
- Diophantine/dynamics: Baker/Subspace + homogeneous dynamics + Diophantine approximation.
- Arithmetic geometry: elliptic/abelian varieties + Faltings + etale cohomology/finite fields.
- Probability/statistical physics: SLE/percolation + particle/kinetic limits + Ising.
- Geometry/dynamics: Ricci flow/topology + KAM/billiards + rigidity theory.
- Computational arithmetic: finite fields + geometry of numbers + elliptic/lattice cryptography.

The curriculum is complete when you can move from a new paper to its dependency graph, reconstruct its central argument at the appropriate level, explain why the theorem matters, and formulate a meaningful next question. Finishing pages is not the criterion; mathematical independence is.

PART V. THE 36-VOLUME MASTER SYLLABUS

This part reorganizes the preceding 40-quarter programme as a coherent 36-volume mathematical series. Each volume is a 360-hour quarter at 30 hours/week. It tells you what to learn, which major results to master, where to read them, and what problems to solve; it does not reproduce the proofs.

Result-mastery code used only in the 36 volume syllabi: [PROOF] = exact statement plus a complete standard proof and one application; [ARCH] = exact hypotheses/conclusion plus a lemma-by-lemma proof architecture and at least two applications; [USE] = exact statement, normalization and dependency chain, with the ability to recognize when the theorem applies. Compound labels such as [PROOF/ARCH] mean: prove the standard core form and know the stronger/research form architecturally. These codes are intentionally distinct from exercise tags [F,T,S,C,R,X].

Reference convention: Chapter numbering varies by edition, so topic titles are authoritative. Where exact lecture numbers are given, follow the named online course version in the verified lecture-note directory earlier in this document.

Ten-year chronological map using the 36 volumes

Year	Q1	Q2	Q3	Q4
Year 1	V1 Proof	V2 Linear algebra	V3 Abstract algebra	V6 Real analysis
Year 2	V5 Logic	V4 Comm./homological algebra	V8 Complex analysis	V11 Topology
Year 3	V14 Probability	V7 Functional analysis	V12 Smooth manifolds	V22 Elementary/analytic NT
Year 4	V9 Harmonic analysis	V10 PDE	V18 Lie/representation	V24 Algebraic NT
Year 5	V25 Geometry of numbers/DA	V15 SDE/random structures	V19 Dynamics/KAM/billiards	V21 GGT/property (T)
Year 6	V28 Additive combinatorics	V29 Expanders/Bourgain-Gamburd	V23 Advanced analytic NT	V27 Finite fields
Year 7	V26 Diophantine/H10	V30 Automorphic forms	V31 Algebraic geometry	V32 Arithmetic geometry
Year 8	V36 Cryptography	V16 Percolation/SLE	V17 Particle systems	V20 Homogeneous dynamics
Year 9	V34 Fractal/entropy	V35 Modern harmonic/geometric	V13 Geometric topology/Ricci	V33 Etale/Weil
Year 10	Arithmetic synthesis	Groups/dynamics synthesis	Analysis/probability synthesis	Research capstone

Year 10 synthesis quarters

- Q37 Arithmetic synthesis: integrate V23-V33 around L-functions, automorphy, Galois representations, Diophantine geometry and finite-field Frobenius; produce theorem roadmaps for FLT, Faltings, Sato-Tate and Deligne/Weil.
- Q38 Groups/dynamics synthesis: integrate V18, V20-V21, V28-V29 and V34 around property (T), spectral gaps, approximate groups, Bourgain-Gamburd, Ratner/EKL/Benoist-Quint and entropy inverse theorems.
- Q39 Analysis/probability synthesis: integrate V9-V10, V14-V17 and V35 around Carleson, restriction/Kakeya/Falconer/Furstenberg, decoupling, SLE and particle/kinetic limits.
- Q40 Research capstone: one primary and one secondary field; four seminar talks; a 40-60 page expository manuscript; one genuinely open-ended project or computational experiment.

VOLUME 1. Proof, Sets, Functions and the Language of Mathematics

Purpose: Build the proof-writing, set-theoretic and structural language used everywhere else. The aim is reliable mathematical reasoning from zero prerequisites.

Prerequisites: None beyond school algebra.

Primary and secondary references: Hammack, Book of Proof; Velleman, How to Prove It; Halmos, Naive Set Theory; MIT OCW 18.100B proof-writing notes

Quarter output: Produce a polished proof portfolio of 20 results, each with strategy note and example/counterexample.

Research bridges: Feeds every later volume; cardinality prepares logic/set theory, and quotient/well-definedness skills prepare algebra/topology.

Six two-week study units

Unit 1 (Weeks 1-2): Propositions, quantifiers and proof forms

Topics: propositions; predicates; quantifiers; negation; implication; biconditional; direct proof; contrapositive; contradiction

Key definitions: truth value; predicate; free/bound variable; necessary/sufficient condition; counterexample

Key results to learn:

- De Morgan laws for quantified statements [PROOF]
- equivalence of implication and contrapositive [PROOF]
- standard contradiction templates [PROOF]

Reading route: Velleman sections on propositional/quantified logic; Hammack chapters on statements and proof techniques

Warm-up/source exercise focus (secondary): Translate 25 statements into quantified form and negate them; prove at least ten implications by two methods.

Week 1: Propositions, quantifiers and proof forms - definitions and examples. Read: Velleman sections on propositional/quantified logic; Hammack chapters on statements and proof techniques. Make explicit examples/nonexamples for truth value; predicate; free/bound variable; necessary/sufficient condition; counterexample. Work through propositions; predicates; quantifiers; negation; implication; biconditional; direct proof; contrapositive; contradiction. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Propositions, quantifiers and proof forms - theorem and technique pass. Master/audit: De Morgan laws for quantified statements; equivalence of implication and contrapositive; standard contradiction templates. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Sets, relations and functions

Topics: set operations; Cartesian products; relations; equivalence relations; partial orders; functions; inverse images

Key definitions: equivalence relation; partition; partial order; injection; surjection; bijection; image; preimage

Key results to learn:

- equivalence relations correspond to partitions [PROOF]
- composition laws for injective/surjective maps [PROOF]
- Cantor-Schroeder-Bernstein [ARCH]

Reading route: Hammack/Velleman chapters on sets, relations and functions; Halmos selections

Warm-up/source exercise focus (secondary): Construct quotient sets; prove maps on quotients well-defined; solve 15 bijection/injection/surjection problems.

Week 3: Sets, relations and functions - definitions and examples. Read: Hammack/Velleman chapters on sets, relations and functions; Halmos selections. Make explicit examples/nonexamples for equivalence relation; partition; partial order; injection; surjection; bijection; image; preimage. Work through set operations; Cartesian products; relations; equivalence relations; partial orders; functions; inverse images. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Sets, relations and functions - theorem and technique pass. Master/audit: equivalence relations correspond to partitions; composition laws for injective/surjective maps; Cantor-Schroeder-Bernstein. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Induction, recursion and invariants

Topics: ordinary/strong induction; recursive definitions; minimal counterexample; invariants

Key definitions: inductive hypothesis; recursion; invariant; well-founded order

Key results to learn:

- strong induction [PROOF]
- well-ordering/induction equivalence on N [ARCH]
- uniqueness of recursively defined objects under standard hypotheses [ARCH]

Reading route: Hammack induction chapter; Velleman induction/recursion sections

Warm-up/source exercise focus (secondary): Prove formulas, inequalities and divisibility statements; prove Euclidean algorithm termination using an invariant.

Week 5: Induction, recursion and invariants - definitions and examples. Read: Hammack induction chapter; Velleman induction/recursion sections. Make explicit examples/nonexamples for inductive hypothesis; recursion; invariant; well-founded order. Work through ordinary/strong induction; recursive definitions; minimal counterexample; invariants. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Induction, recursion and invariants - theorem and technique pass. Master/audit: strong induction; well-ordering/induction equivalence on N; uniqueness of recursively defined objects under standard hypotheses. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Cardinality and countability

Topics: finite/infinite sets; countable unions; diagonal arguments; cardinal comparison

Key definitions: countable; denumerable; cardinality; power set

Key results to learn:

- countability of Z,Q and finite products [PROOF]
- countable union of countable sets [PROOF]
- uncountability of R [PROOF]
- Cantor theorem |P(X)|>|X| [ARCH]

Reading route: Halmos cardinality sections; Hammack cardinality chapters

Warm-up/source exercise focus (secondary): Enumerate Q and algebraic numbers; construct diagonal arguments for sequence/function spaces.

Week 7: Cardinality and countability - definitions and examples. Read: Halmos cardinality sections; Hammack cardinality chapters. Make explicit examples/nonexamples for countable; denumerable; cardinality; power set. Work through finite/infinite sets; countable unions; diagonal arguments; cardinal comparison. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Cardinality and countability - theorem and technique pass. Master/audit: countability of Z,Q and finite products; countable union of countable sets; uncountability of R. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Construction and counterexamples

Topics: existence by construction; extremal examples; pigeonhole; counterexample design

Key definitions: witness; extremal element; minimal/maximal counterexample

Key results to learn:

- pigeonhole and infinite pigeonhole principles [PROOF]
- standard extremal-principle arguments [PROOF]

Reading route: Velleman problem-solving chapters; selected proof exercises

Warm-up/source exercise focus (secondary): Build counterexamples to false converses; identify hidden hypotheses in 20 statements.

Week 9: Construction and counterexamples - definitions and examples. Read: Velleman problem-solving chapters; selected proof exercises. Make explicit examples/nonexamples for witness; extremal element; minimal/maximal counterexample. Work through existence by construction; extremal examples; pigeonhole; counterexample design. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Construction and counterexamples - theorem and technique pass. Master/audit: pigeonhole and infinite pigeonhole principles; standard extremal-principle arguments. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Exposition and proof checking

Topics: definition-first writing; lemma structure; notation; dependency tracking; proof auditing

Key definitions: lemma; theorem; corollary; iff proof; dependency graph

Key results to learn:

- distinguish statement/proof/example/heuristic [PROOF]
- detect circular reasoning and unjustified existence [PROOF]

Reading route: Model proofs from Abbott/Tao/MIT 18.100B

Warm-up/source exercise focus (secondary): Write a 6-8 page mini-note with definitions, examples, counterexamples and a dependency diagram.

Week 11: Exposition and proof checking - definitions and examples. Read: Model proofs from Abbott/Tao/MIT 18.100B. Make explicit examples/nonexamples for lemma; theorem; corollary; iff proof; dependency graph. Work through definition-first writing; lemma structure; notation; dependency tracking; proof auditing. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Exposition and proof checking - theorem and technique pass. Master/audit: distinguish statement/proof/example/heuristic; detect circular reasoning and unjustified existence. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Propositions, quantifiers and proof forms

U1.P1. [F] Formalize five statements with nested quantifiers, including continuity at a point, boundedness of a function, and existence of infinitely many primes. Negate each statement by pushing negations all the way to atomic predicates, then check the negation against examples.

U1.P2. [T] Prove the irrationality of $\sqrt{2}$ by contradiction, prove that n^2 even implies n even by contrapositive, and prove the sum of the first n odd integers by induction. For each proof, identify why that proof form is natural and rewrite one proof in a less natural form to see the difference.

U1.P3. [C] Diagnose three false arguments: affirming the consequent, illicit reversal of quantifiers, and circular reasoning. Repair each argument if a true statement is nearby, and state exactly which logical step failed.

Programme U2: Sets, relations and functions

U2.P1. [F] On Z define $a\sim b$ iff $a-b$ is divisible by 6. Prove this is an equivalence relation, describe $Z/\sim$ explicitly, and identify the quotient map. Then give a relation that is reflexive and symmetric but not transitive.

U2.P2. [T] For maps $f:A\to B$ and $g:B\to C$ prove carefully: $g \circ f$ injective implies f injective; $g \circ f$ surjective implies g surjective. Construct counterexamples to both converses.

U2.P3. [S] Prove the Cantor-Schroeder-Bernstein theorem from an explicit decomposition of A into forward/backward chains. Apply it to produce a bijection between (0,1) and R without writing a closed-form bijection first.

Programme U3: Induction, recursion and invariants

U3.P1. [T] Solve the recurrence $a_{n+2}=3a_{n+1}-2a_n$ twice: once by induction from a guessed closed form and once by linear-algebra/characteristic-polynomial reasoning. Compare which information each method reveals.

U3.P2. [T] Prove correctness and termination of the Euclidean algorithm using a strictly decreasing nonnegative invariant. Derive Bezout coefficients by back substitution for one nontrivial pair such as (414,662).

U3.P3. [S] Design an invariant for the following process: replace a pair of integers (a,b) by (a-b,b) or (a,b-a) whenever entries remain positive. Determine which quantity is preserved and use it to characterize possible terminal states.

Programme U4: Cardinality and countability

U4.P1. [F] Construct explicit enumerations of Z, Q, the set of finite binary strings, and Z^2. Explain why the same strategy fails for infinite binary sequences.

U4.P2. [T] Reproduce Cantor's diagonal proof that $\{0,1\}^N$ is uncountable, then prove P(N) is uncountable by a direct diagonal argument. State exactly where self-reference enters.

U4.P3. [S] Prove that the set of algebraic numbers is countable and hence transcendental numbers exist. Strengthen the conclusion by showing that almost every real number is transcendental once measure theory becomes available; note clearly what is not yet proved at this stage.

Programme U5: Construction and counterexamples

U5.P1. [C] Construct: an injective non-surjective map $N\to N$, a surjective non-injective map $N\to N$, a bijection $Q\to Q$ with infinitely many fixed points, and a function $R\to R$ discontinuous at every point. Verify each claim from definitions.

U5.P2. [C] For each false statement, give the simplest counterexample you can: union preserves intersections under arbitrary maps; image preserves complements; every infinite subset of a countable set is cofinite; every relation is comparable to an equivalence relation by inclusion.

U5.P3. [S] Starting from the claim 'every infinite set has a countably infinite subset', identify the choice principle hidden in common proofs. Distinguish what can be proved in elementary settings from what depends on stronger set-theoretic assumptions.

Programme U6: Exposition and proof checking

U6.P1. [T] Take a one-page proof from an introductory text, remove all phrases such as 'clearly' and 'obviously', and expand every hidden inference. Then compress it again without losing logical completeness. Compare the two versions.

U6.P2. [C] Write a deliberately flawed proof of a true theorem containing exactly three nontrivial errors. One week later, audit it as if it were someone else's work and produce a line-by-line correction.

U6.P3. [R] Prepare a 15-minute blackboard proof of one theorem with no notes. Afterward, write a dependency graph listing definitions, lemmas, and the theorem. Mark which nodes you could reproduce immediately and which required reconstruction.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Proof, Sets, Functions and the Language of Mathematics: start from “Propositions, quantifiers and proof forms”, pass through “Cardinality and countability”, and finish at “Exposition and proof checking”. Use the named results “De Morgan laws for quantified statements” and “detect circular reasoning and unjustified existence” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds every later volume; cardinality prepares logic/set theory, and quotient/well-definedness skills prepare algebra/topology. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: De Morgan laws for quantified statements; equivalence of implication and contrapositive; standard contradiction templates; equivalence relations correspond to partitions; composition laws for injective/surjective maps. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.

- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Cantor-Schroeder-Bernstein; well-ordering/induction equivalence on N; uniqueness of recursively defined objects under standard hypotheses; Cantor theorem $|P(X)|>|X|$.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Propositions, quantifiers and proof forms” develops into “Exposition and proof checking”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 2. Linear and Multilinear Algebra

Purpose: Develop linear algebra at a level adequate for functional analysis, representation theory, differential geometry, PDE, probability and arithmetic.

Prerequisites: Volume I.

Primary and secondary references: Axler, Linear Algebra Done Right; Hoffman-Kunze, Linear Algebra; Roman, Advanced Linear Algebra

Quarter output: Write a 10-page structural summary of one operator using invariant subspaces, polynomial modules, inner products and tensor constructions.

Research bridges: Direct prerequisite for functional analysis, Lie/representation theory, differential geometry and random matrices.

Six two-week study units

Unit 1 (Weeks 1-2): Vector spaces and linear maps

Topics: subspaces; span; bases; dimension; quotient spaces; linear maps

Key definitions: linear independence; basis; kernel; image; quotient space

Key results to learn:

- basis extension [PROOF]
- rank-nullity [PROOF]
- first isomorphism theorem for vector spaces [PROOF]

Reading route: Axler chapters on vector spaces and linear maps

Warm-up/source exercise focus (secondary): Prove rank-nullity; compute kernels/images/quotients; construct maps with prescribed invariant data.

Week 1: Vector spaces and linear maps - definitions and examples. Read: Axler chapters on vector spaces and linear maps. Make explicit examples/nonexamples for linear independence; basis; kernel; image; quotient space. Work through subspaces; span; bases; dimension; quotient spaces; linear maps. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Vector spaces and linear maps - theorem and technique pass. Master/audit: basis extension; rank-nullity; first isomorphism theorem for vector spaces. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Matrices, determinants and duality

Topics: matrix representation; change of basis; determinants; dual spaces; annihilators

Key definitions: determinant; dual map; annihilator; trace

Key results to learn:

- determinant multiplicativity [ARCH]
- determinant/eigenvalue relation [PROOF]
- finite-dimensional double-dual identification [PROOF]

Reading route: Axler determinant/dual chapters; Hoffman-Kunze determinant chapters

Warm-up/source exercise focus (secondary): Prove basis-independence of trace/determinant; solve duality/annihilator problems.

Week 3: Matrices, determinants and duality - definitions and examples. Read: Axler determinant/dual chapters; Hoffman-Kunze determinant chapters. Make explicit examples/nonexamples for determinant; dual map; annihilator; trace. Work through matrix representation; change of basis; determinants; dual spaces; annihilators. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Matrices, determinants and duality - theorem and technique pass. Master/audit: determinant multiplicativity; determinant/eigenvalue relation; finite-dimensional double-dual identification. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Eigenvalues and polynomial calculus

Topics: invariant subspaces; characteristic/minimal polynomials; generalized eigenspaces

Key definitions: eigenvalue; minimal polynomial; generalized eigenspace

Key results to learn:

- Cayley-Hamilton [PROOF/ARCH]
- primary decomposition [ARCH]
- diagonalizability criteria [PROOF]

Reading route: Axler operator chapters; Roman polynomial-module treatment

Warm-up/source exercise focus (secondary): Determine minimal polynomials/decompositions; construct examples separating diagonalizability criteria.

Week 5: Eigenvalues and polynomial calculus - definitions and examples. Read: Axler operator chapters; Roman polynomial-module treatment. Make explicit examples/nonexamples for eigenvalue; minimal polynomial; generalized eigenspace. Work through invariant subspaces;

characteristic/minimal polynomials; generalized eigenspaces. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Eigenvalues and polynomial calculus - theorem and technique pass. Master/audit: Cayley-Hamilton; primary decomposition; diagonalizability criteria. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Inner products and spectral theory

Topics: orthogonality; projections; adjoints; normal/self-adjoint/unitary operators

Key definitions: inner product; adjoint; normal operator; positive operator

Key results to learn:

- Gram-Schmidt [PROOF]
- projection theorem [PROOF]
- finite-dimensional spectral theorem [PROOF]
- SVD [ARCH]

Reading route: Axler inner-product chapters

Warm-up/source exercise focus (secondary): Diagonalize normal operators; derive least squares; compute SVDs and condition numbers.

Week 7: Inner products and spectral theory - definitions and examples. Read: Axler inner-product chapters. Make explicit examples/nonexamples for inner product; adjoint; normal operator; positive operator. Work through orthogonality; projections; adjoints; normal/self-adjoint/unitary operators. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Inner products and spectral theory - theorem and technique pass. Master/audit: Gram-Schmidt; projection theorem; finite-dimensional spectral theorem. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Canonical forms

Topics: Jordan form; rational canonical form; cyclic decomposition

Key definitions: Jordan block; invariant factor; elementary divisor; cyclic vector

Key results to learn:

- Jordan canonical form [ARCH]
- rational canonical form [ARCH]
- uniqueness of canonical invariants [ARCH]

Reading route: Roman/Hoffman-Kunze canonical forms

Warm-up/source exercise focus (secondary): Compute Jordan/rational forms; infer blocks from ranks of powers; compare matrices with same characteristic polynomial.

Week 9: Canonical forms - definitions and examples. Read: Roman/Hoffman-Kunze canonical forms. Make explicit examples/nonexamples for Jordan block; invariant factor; elementary divisor; cyclic vector. Work through Jordan form; rational canonical form; cyclic decomposition. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Canonical forms - theorem and technique pass. Master/audit: Jordan canonical form; rational canonical form; uniqueness of canonical invariants. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Tensor, symmetric and exterior algebra

Topics: tensor products; universal properties; multilinear maps; symmetric/exterior powers

Key definitions: tensor product; exterior power; wedge product

Key results to learn:

- universal property of tensor product [PROOF]
- basis theorem for tensor products [PROOF]
- determinant as alternating top form [ARCH]

Reading route: Roman tensor chapters; Fulton-Harris preliminaries

Warm-up/source exercise focus (secondary): Compute tensor/exterior powers; derive determinant identities; prepare forms/representation examples.

Week 11: Tensor, symmetric and exterior algebra - definitions and examples. Read: Roman tensor chapters; Fulton-Harris preliminaries. Make explicit examples/nonexamples for tensor product; exterior power; wedge product. Work through tensor products; universal properties; multilinear maps; symmetric/exterior powers. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Tensor, symmetric and exterior algebra - theorem and technique pass. Master/audit: universal property of tensor product; basis theorem for tensor products; determinant as alternating top form. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Vector spaces and linear maps

U1.P1. [F] For three concrete linear maps R^3->R^2 given by formulas, compute kernel, image, rank, nullity, and matrices in two different bases. Verify rank-nullity numerically and conceptually.

U1.P2. [T] Prove that a linear map is determined by its values on a basis and derive the universal formula for extending an assignment on a basis. Use it to classify all projections R^3->R^3 with image span(e1,e2).

U1.P3. [S] Let T be differentiation on polynomials of degree at most n. Find invariant subspaces, kernel, image, nilpotency index, and a convenient basis. Explain why this example anticipates Jordan theory.

Programme U2: Matrices, determinants and duality

U2.P1. [T] Derive determinant multilinearity and alternating behavior from elementary row operations, then prove det(AB)=det(A)det(B) without using eigenvalues. Apply it to determine when a parameterized 4x4 matrix is invertible.

U2.P2. [F] Compute dual bases for two non-orthonormal bases of R^3 and express a linear functional in both primal and dual coordinates. Verify the naturality of the double-dual embedding in finite dimension.

U2.P3. [S] Use exterior algebra to rederive the determinant as the scalar by which a linear map acts on the top exterior power. Explain precisely which determinant properties become immediate in this formulation.

Programme U3: Eigenvalues and polynomial calculus

U3.P1. [T] For a matrix with distinct eigenvalues, prove diagonalizability. Then construct a matrix whose characteristic polynomial splits but which is not diagonalizable, and determine its minimal polynomial.

U3.P2. [T] Compute p(T) for several polynomials p using both direct matrix powers and reduction modulo the minimal polynomial. Derive Cayley-Hamilton as a computational shortcut in a concrete 3x3 example.

U3.P3. [C] Give examples showing: diagonalizable over C but not R; minimal polynomial with repeated root but characteristic polynomial not squarefree; same characteristic polynomial but non-similar matrices.

Programme U4: Inner products and spectral theory

U4.P1. [T] Run Gram-Schmidt on a badly conditioned basis by hand and explain geometrically what each subtraction accomplishes. Use the resulting QR factorization to solve a least-squares problem.

U4.P2. [T] Prove the finite-dimensional spectral theorem for a real symmetric operator from the existence of a maximizing unit vector for the Rayleigh quotient. Track exactly where compactness and symmetry are used.

U4.P3. [S] For a quadratic form in three variables, diagonalize it orthogonally, classify its signature, and solve a constrained extremum problem using the eigenvalues. Relate the answer to the min-max principle.

Programme U5: Canonical forms

U5.P1. [T] Compute Jordan forms and chains for three matrices exhibiting: one Jordan block, two blocks with the same eigenvalue, and mixed eigenvalues. Recover block sizes from dimensions of ker(T-lambda I)^k.

U5.P2. [S] Prove the rational canonical form classification over a field at the level of cyclic modules over F[x]. Work out one 4x4 example where rational and Jordan forms differ because the characteristic polynomial does not split.

U5.P3. [C] Construct two non-similar nilpotent 6x6 matrices with the same rank. Find the smallest list of nullities ker(T^k) that distinguishes their Jordan types.

Programme U6: Tensor, symmetric and exterior algebra

U6.P1. [F] Compute V tensor W for V=R^2,W=R^3 in bases, then verify the universal property by factorizing an explicit bilinear map through V tensor W.

U6.P2. [T] Prove the decompositions V tensor V = Sym^2 V direct-sum Lambda^2 V in characteristic not 2, and calculate dimensions. Explain what fails in characteristic 2.

U6.P3. [S] Express cross products and oriented volume in R^3 using exterior powers and the Hodge-star intuition. Then derive the Cauchy-Binet formula from minors/exterior powers in one nontrivial case.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Linear and Multilinear Algebra: start from “Vector spaces and linear maps”, pass through “Inner products and spectral theory”, and finish at “Tensor, symmetric and exterior algebra”. Use the named results “basis extension” and “determinant as alternating top form” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct prerequisite for functional analysis, Lie/representation theory, differential geometry and random matrices. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: basis extension; rank-nullity; first isomorphism theorem for vector spaces; determinant/eigenvalue relation; finite-dimensional double-dual identification. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.

- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: SVD; Jordan canonical form; rational canonical form; uniqueness of canonical invariants; determinant as alternating top form.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Vector spaces and linear maps” develops into “Tensor, symmetric and exterior algebra”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 3. Groups, Rings, Fields and Galois Theory

Purpose: Acquire the algebraic language underlying number theory, topology, Lie theory, algebraic geometry, finite fields and cryptography.

Prerequisites: Volumes I-II.

Primary and secondary references: Dummit-Foote, Abstract Algebra; Artin, Algebra; Lang, Algebra (reference)

Quarter output: Compute and explain five complete Galois correspondences and one nontrivial finite-group classification.

Research bridges: Feeds algebraic number theory, finite fields, representation theory, topology and algebraic geometry.

Six two-week study units

Unit 1 (Weeks 1-2): Groups, homomorphisms and actions

Topics: subgroups; normality; quotients; group actions; orbits/stabilizers

Key definitions: group action; stabilizer; orbit; kernel; normal subgroup

Key results to learn:

- isomorphism theorems [PROOF]
- orbit-stabilizer [PROOF]
- Cayley theorem [PROOF]
- class equation [PROOF]

Reading route: Dummit-Foote/Artin group chapters

Warm-up/source exercise focus (secondary): Classify actions in small examples; compute quotients; prove statements through group actions.

Week 1: Groups, homomorphisms and actions - definitions and examples. Read: Dummit-Foote/Artin group chapters. Make explicit examples/nonexamples for group action; stabilizer; orbit; kernel; normal subgroup. Work through subgroups; normality; quotients; group actions; orbits/stabilizers. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Groups, homomorphisms and actions - theorem and technique pass. Master/audit: isomorphism theorems; orbit-stabilizer; Cayley theorem. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Finite groups and Sylow theory

Topics: p-groups; Sylow subgroups; semidirect products; solvability

Key definitions: Sylow subgroup; semidirect product; composition series; solvable group

Key results to learn:

- Sylow theorems [PROOF]
- finite abelian group classification [PROOF/ARCH]
- Jordan-Holder [ARCH]

Reading route: Dummit-Foote finite-group chapters

Warm-up/source exercise focus (secondary): Classify groups of several small orders; build semidirect products; compute composition factors.

Week 3: Finite groups and Sylow theory - definitions and examples. Read: Dummit-Foote finite-group chapters. Make explicit examples/nonexamples for Sylow subgroup; semidirect product; composition series; solvable group. Work through p-groups; Sylow subgroups; semidirect products; solvability. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Finite groups and Sylow theory - theorem and technique pass. Master/audit: Sylow theorems; finite abelian group classification; Jordan-Holder. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Rings, ideals and factorization

Topics: integral domains; ideals; quotient rings; polynomial rings; divisibility

Key definitions: prime/maximal ideal; PID; UFD; Euclidean domain

Key results to learn:

- CRT for rings [PROOF]
- PID implies UFD [PROOF/ARCH]
- Gauss lemma [PROOF]

Reading route: Dummit-Foote ring chapters

Warm-up/source exercise focus (secondary): Compute ideals/quotients; use Eisenstein/modular reduction; exhibit PID/UFD distinctions.

Week 5: Rings, ideals and factorization - definitions and examples. Read: Dummit-Foote ring chapters. Make explicit examples/nonexamples for prime/maximal ideal; PID; UFD; Euclidean domain. Work through integral domains; ideals; quotient rings; polynomial rings; divisibility. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Rings, ideals and factorization - theorem and technique pass. Master/audit: CRT for rings; PID implies UFD; Gauss lemma. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Modules and structure theorems

Topics: modules; torsion; free modules; modules over PID

Key definitions: module; torsion; annihilator; invariant factor

Key results to learn:

- structure theorem for finitely generated modules over a PID [ARCH]
- finite abelian classification via modules [ARCH]

Reading route: Dummit-Foote module chapters

Warm-up/source exercise focus (secondary): Compute Smith normal form and module decompositions.

Week 7: Modules and structure theorems - definitions and examples. Read: Dummit-Foote module chapters. Make explicit examples/nonexamples for module; torsion; annihilator; invariant factor. Work through modules; torsion; free modules; modules over PID. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Modules and structure theorems - theorem and technique pass. Master/audit: structure theorem for finitely generated modules over a PID; finite abelian classification via modules. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Field extensions

Topics: algebraic/transcendental extensions; splitting fields; separability/normality

Key definitions: minimal polynomial; degree; splitting field; separable extension

Key results to learn:

- tower law [PROOF]
- existence/uniqueness of splitting fields [ARCH]
- primitive element theorem in standard settings [ARCH]

Reading route: Dummit-Foote/Artin field chapters

Warm-up/source exercise focus (secondary): Construct splitting fields; compute degrees; test separability/normality.

Week 9: Field extensions - definitions and examples. Read: Dummit-Foote/Artin field chapters. Make explicit examples/nonexamples for minimal polynomial; degree; splitting field; separable extension. Work through algebraic/transcendental extensions; splitting fields; separability/normality. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Field extensions - theorem and technique pass. Master/audit: tower law; existence/uniqueness of splitting fields; primitive element theorem in standard settings. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Galois theory

Topics: Galois groups; fixed fields; cyclotomic extensions; radicals

Key definitions: Galois extension; fixed field; radical extension

Key results to learn:

- fundamental theorem of Galois theory [PROOF/ARCH]
- Galois groups of finite/cyclotomic fields [PROOF]
- solvability by radicals criterion [ARCH]

Reading route: Dummit-Foote Galois chapters; Artin

Warm-up/source exercise focus (secondary): Compute five Galois correspondences; determine groups of low-degree polynomials; derive constructibility results.

Week 11: Galois theory - definitions and examples. Read: Dummit-Foote Galois chapters; Artin. Make explicit examples/nonexamples for Galois extension; fixed field; radical extension. Work through Galois groups; fixed fields; cyclotomic extensions; radicals. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Galois theory - theorem and technique pass. Master/audit: fundamental theorem of Galois theory; Galois groups of finite/cyclotomic fields; solvability by radicals criterion. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Groups, homomorphisms and actions

U1.P1. [F] Compute kernels, images, stabilizers, and orbits for the natural actions of S4 on {1,2,3,4}, on 2-element subsets, and by conjugation on itself. Check orbit-stabilizer in each case.

U1.P2. [T] Prove the first isomorphism theorem and use it to classify quotients of Z and kernels of homomorphisms Z^2->Z. Then interpret normal subgroups as exactly kernels.

U1.P3. [S] Use group actions to prove Cauchy's theorem for finite groups in a concrete prime-order counting argument. Compare this proof to one using Sylow theory later.

Programme U2: Finite groups and Sylow theory

U2.P1. [T] Determine all Sylow subgroups and possible group structures for groups of orders 15, 21, and 45 as far as Sylow theory alone permits. State where extra arguments are needed.

U2.P2. [C] Find examples showing that a Sylow p-subgroup need not be normal, that normal Sylow subgroups need not make a group abelian, and that groups with the same order can be nonisomorphic.

U2.P3. [S] Classify groups of order pq for primes p<q under the condition p does not divide q-1. Then analyze what changes when p divides q-1, using order 6 or 21 as a model.

Programme U3: Rings, ideals and factorization

U3.P1. [F] In Z[x], F_p[x], and k[x,y], compute explicit ideals, quotient rings, units, zero divisors, and prime/maximal behavior. Use quotient calculations rather than only theorem names.

U3.P2. [T] Prove Euclidean domain => PID => UFD and exhibit rings showing the converses fail. For Z[sqrt(-5)], verify concretely the failure of unique factorization of elements.

U3.P3. [S] Apply the Chinese remainder theorem to rings: decompose Z/60Z and F[x]/((x-1)(x+1)) when factors are coprime. Explain the idempotents that realize the product decomposition.

Programme U4: Modules and structure theorems

U4.P1. [T] Treat a finitely generated abelian group as a Z-module and compute Smith normal form for three integer matrices. Read off cokernels and invariant factors.

U4.P2. [T] Prove the structure theorem for finitely generated modules over a PID in the cyclic decomposition form, identifying the reduction steps where PID is essential.

U4.P3. [C] Give a finitely generated module over a non-PID for which the familiar invariant-factor classification fails or is no longer applicable; explain precisely which proof step breaks.

Programme U5: Field extensions

U5.P1. [F] For Q(sqrt(2)), Q(cuberoot(2)), and F_p(t), compute degrees, minimal polynomials, bases, traces, and norms for selected elements.

U5.P2. [T] Prove the tower law and use it to rule out classical straightedge-and-compass constructions such as trisecting a generic angle or doubling the cube.

U5.P3. [S] Determine splitting fields and degrees for x^4-2 and x^3-2 over Q. Identify which roots must be adjoined and which automorphisms are forced.

Programme U6: Galois theory

U6.P1. [T] Compute Galois groups of x^3-2, x^4-2, and cyclotomic polynomials Phi_5 and Phi_8. Make the correspondence between subgroups and intermediate fields explicit.

U6.P2. [T] Prove the fundamental theorem of Galois theory for finite Galois extensions at proof-architecture level, then verify every correspondence in Q(zeta_8)/Q.

U6.P3. [S] Use Galois groups to determine solvability by radicals for selected polynomials. Work through one irreducible quintic with Galois group S5 at the level of the group-theoretic obstruction.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Groups, Rings, Fields and Galois Theory: start from “Groups, homomorphisms and actions”, pass through “Modules and structure theorems”, and finish at “Galois theory”. Use the named results “isomorphism theorems” and “solvability by radicals criterion” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds algebraic number theory, finite fields, representation theory, topology and algebraic geometry. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: isomorphism theorems; orbit-stabilizer; Cayley theorem; class equation; Sylow theorems. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: structure theorem for finitely generated modules over a PID; finite abelian classification via modules; existence/uniqueness of splitting fields; primitive element theorem in standard settings; solvability by radicals criterion.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Groups, homomorphisms and actions” develops into “Galois theory”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 4. Commutative and Homological Algebra

Purpose: Build the algebraic machinery needed for schemes, local algebra, cohomology, arithmetic geometry and modern representation theory.

Prerequisites: Volumes II-III.

Primary and secondary references: Atiyah-Macdonald, Introduction to Commutative Algebra; Eisenbud, Commutative Algebra with a View Toward Algebraic Geometry; Weibel, An Introduction to Homological Algebra; Matsumura, Commutative Ring Theory (reference)

Quarter output: Prepare a 12-page local-to-global algebra toolkit explaining localization, dimension, resolutions and Ext/Tor.

Research bridges: Essential for schemes, sheaf/etale cohomology and advanced representation theory.

Six two-week study units

Unit 1 (Weeks 1-2): Localization and local rings

Topics: localization of rings/modules; local properties; spectra preview

Key definitions: multiplicative set; localization; local ring; residue field; support

Key results to learn:

- universal property of localization [PROOF]
- primes under localization [PROOF]
- Nakayama lemma [PROOF]

Reading route: Atiyah-Macdonald localization/local-ring sections; Eisenbud

Warm-up/source exercise focus (secondary): Localize explicit rings/modules; compute at primes; use Nakayama on minimal generators.

Week 1: Localization and local rings - definitions and examples. Read: Atiyah-Macdonald localization/local-ring sections; Eisenbud. Make explicit examples/nonexamples for multiplicative set; localization; local ring; residue field; support. Work through localization of rings/modules; local properties; spectra preview. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Localization and local rings - theorem and technique pass. Master/audit: universal property of localization; primes under localization; Nakayama lemma. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Noetherian rings and dimension

Topics: ACC; finitely generated ideals/modules; chains of primes

Key definitions: Noetherian ring/module; Krull dimension; height; regular sequence

Key results to learn:

- Hilbert basis theorem [PROOF]
- Krull principal ideal theorem [ARCH]
- standard dimension inequalities [ARCH]

Reading route: Atiyah-Macdonald Noetherian/dimension chapters; Eisenbud

Warm-up/source exercise focus (secondary): Compute dimensions of polynomial quotients; bound dimensions through prime chains.

Week 3: Noetherian rings and dimension - definitions and examples. Read: Atiyah-Macdonald Noetherian/dimension chapters; Eisenbud. Make explicit examples/nonexamples for Noetherian ring/module; Krull dimension; height; regular sequence. Work through ACC; finitely generated ideals/modules; chains of primes. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Noetherian rings and dimension - theorem and technique pass. Master/audit: Hilbert basis theorem; Krull principal ideal theorem; standard dimension inequalities. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Integral dependence and normalization

Topics: integral elements; finite extensions; going-up/down; normalization

Key definitions: integral extension; integral closure; normalization

Key results to learn:

- lying-over and going-up [ARCH]
- incomparability [ARCH]
- going-down under standard hypotheses [USE]

Reading route: Atiyah-Macdonald integral-dependence chapter

Warm-up/source exercise focus (secondary): Compute integral closures; use characteristic-polynomial trick; analyse examples relevant to ramification.

Week 5: Integral dependence and normalization - definitions and examples. Read: Atiyah-Macdonald integral-dependence chapter. Make explicit examples/nonexamples for integral extension; integral closure; normalization. Work through integral elements; finite extensions; going-up/down; normalization. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Integral dependence and normalization - theorem and technique pass. Master/audit: lying-over and going-up; incomparability; going-down under standard hypotheses. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Primary decomposition

Topics: primary ideals; radicals; zero divisors; associated primes

Key definitions: primary ideal; associated prime; support

Key results to learn:

- Lasker-Noether primary decomposition [ARCH]
- finiteness of associated primes in Noetherian settings [ARCH]

Reading route: Atiyah-Macdonald primary decomposition; Eisenbud

Warm-up/source exercise focus (secondary): Compute primary decompositions of simple/monomial ideals; interpret embedded components.

Week 7: Primary decomposition - definitions and examples. Read: Atiyah-Macdonald primary decomposition; Eisenbud. Make explicit examples/nonexamples for primary ideal; associated prime; support. Work through primary ideals; radicals; zero divisors; associated primes. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Primary decomposition - theorem and technique pass. Master/audit: Lasker-Noether primary decomposition; finiteness of associated primes in Noetherian settings. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Homological algebra I

Topics: complexes; exact sequences; projective/injective/flat modules; resolutions

Key definitions: chain complex; homology; exact functor; resolution

Key results to learn:

- snake lemma [PROOF]
- five lemma [PROOF]
- projective resolutions [ARCH]
- tensor-Hom adjunction [ARCH]

Reading route: Weibel Chapters 1-3 by topic

Warm-up/source exercise focus (secondary): Chase diagrams; construct resolutions over Z; test exactness after tensoring.

Week 9: Homological algebra I - definitions and examples. Read: Weibel Chapters 1-3 by topic. Make explicit examples/nonexamples for chain complex; homology; exact functor; resolution. Work through complexes; exact sequences; projective/injective/flat modules; resolutions. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Homological algebra I - theorem and technique pass. Master/audit: snake lemma; five lemma; projective resolutions. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Derived functors, Ext/Tor and spectral sequences

Topics: Ext; Tor; derived functors; long exact sequences; spectral-sequence literacy

Key definitions: Ext; Tor; derived functor; filtration; E_r page

Key results to learn:

- long exact Ext/Tor sequences [PROOF/ARCH]
- universal-coefficient type consequences [ARCH]
- Grothendieck spectral-sequence philosophy [USE]

Reading route: Weibel derived-functor/spectral-sequence chapters

Warm-up/source exercise focus (secondary): Compute Ext/Tor for cyclic groups; read simple spectral sequences; map to sheaf cohomology.

Week 11: Derived functors, Ext/Tor and spectral sequences - definitions and examples. Read: Weibel derived-functor/spectral-sequence chapters. Make explicit examples/nonexamples for Ext; Tor; derived functor; filtration; E_r page. Work through Ext; Tor; derived functors; long exact sequences; spectral-sequence literacy. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Derived functors, Ext/Tor and spectral sequences - theorem and technique pass. Master/audit: long exact Ext/Tor sequences; universal-coefficient type consequences; Grothendieck spectral-sequence philosophy. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Localization and local rings

U1.P1. [F] Compute S^{-1}R for R=Z and S generated by a prime p; for R=k[x,y] localized at (x,y); and for localization at a prime ideal. Identify units and maximal ideals explicitly.

U1.P2. [T] Prove the universal property of localization and use it to factor a homomorphism that sends every element of S to a unit. Check uniqueness in a concrete ring map.

U1.P3. [S] Show that localizing Z at (p) detects p-adic divisibility but forgets all other primes. Translate this into the geometric statement that localization focuses on a neighborhood of a prime.

Programme U2: Noetherian rings and dimension

U2.P1. [T] Prove Hilbert basis theorem and use it to establish that k[x1,...,xn] is Noetherian. Identify why the leading-coefficient ideal argument terminates.

U2.P2. [F] Compute chains of prime ideals in k[x,y], Z[x], and quotient rings such as k[x,y]/(xy). Estimate or determine Krull dimensions.

U2.P3. [C] Construct an infinite ascending chain of ideals in k[x1,x2,...] to show non-Noetherian behavior, and locate the step of a Noetherian proof that fails.

Programme U3: Integral dependence and normalization

U3.P1. [F] Determine elements integral over Z inside Q(sqrt(5)) and compute the integral closure of Z in several quadratic fields from minimal polynomials.

U3.P2. [T] Prove transitivity of integrality and the finite-module criterion for integral extensions in a usable form. Apply it to a normalization calculation for a cusp ring k[t^2,t^3].

U3.P3. [S] Normalize k[x,y]/(y^2-x^3) by identifying it with k[t^2,t^3] inside k[t]. Explain geometrically which singularity is repaired.

Programme U4: Primary decomposition

U4.P1. [F] Compute radicals and associated primes in examples such as (x^2,xy) in k[x,y] and (12) in Z. Produce explicit primary decompositions where feasible.

U4.P2. [T] Prove existence of primary decomposition in the Noetherian case at theorem-architecture level, emphasizing the role of maximal counterexamples/Noetherian induction.

U4.P3. [S] Compare two non-unique primary decompositions of an ideal with embedded components and identify which associated primes are invariant.

Programme U5: Homological algebra I

U5.P1. [F] For short exact sequences of abelian groups, test exactness and construct splitting maps. Classify when 0->Z->Z->Z/nZ->0 splits.

U5.P2. [T] Prove the snake lemma in a concrete diagram and use it to derive a connecting homomorphism. Then apply the five lemma to an explicit diagram.

U5.P3. [S] Build a projective resolution of Z/nZ as a Z-module and compute Tor_1^Z(Z/nZ,Z/mZ) directly. Compare with the gcd formula.

Programme U6: Derived functors, Ext/Tor and spectral sequences

U6.P1. [T] Compute Ext^1_Z(Z/nZ,Z) and Tor_1^Z(Z/nZ,Z/mZ) from resolutions. Interpret Ext^1 as extension classes in this example.

U6.P2. [F] Construct the double complex associated with a short bicomplex and calculate the first pages of both spectral sequences. Verify that both converge to the same total homology in the toy example.

U6.P3. [R] Take one spectral sequence used later (Serre, Hochschild-Serre, or Grothendieck). Write its E2-page, convergence target, and one low-degree exact sequence, but do not attempt the full theory yet.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Commutative and Homological Algebra: start from “Localization and local rings”, pass through “Primary decomposition”, and finish at “Derived functors, Ext/Tor and spectral sequences”. Use the named results “universal property of localization” and “Grothendieck spectral-sequence philosophy” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Essential for schemes, sheaf/etale cohomology and advanced representation theory. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: universal property of localization; primes under localization; Nakayama lemma; Hilbert basis theorem; snake lemma. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: finiteness of associated primes in Noetherian settings; projective resolutions; tensor-Hom adjunction; universal-coefficient type consequences; Grothendieck spectral-sequence philosophy.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Localization and local rings” develops into “Derived functors, Ext/Tor and spectral sequences”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 5. Mathematical Logic, Computability and Foundations

Purpose: Develop first-order logic, computability, incompleteness, axiomatic set theory, forcing and the independence of the continuum hypothesis, with a bridge to Diophantine undecidability.

Prerequisites: Volume I; Volume III helps for arithmetic examples.

Primary and secondary references: Enderton, A Mathematical Introduction to Logic; Boolos-Burgess-Jeffrey, Computability and Logic; Kunen, Set Theory; Jech, Set Theory; Cohen, Set Theory and the Continuum Hypothesis; Easwaran, A Cheerful Introduction to Forcing and CH

Quarter output: Give a seminar exposition connecting completeness, incompleteness and forcing while clearly separating the three phenomena.

Research bridges: Prepares Hilbert's tenth problem and foundational reading across mathematics.

Six two-week study units

Unit 1 (Weeks 1-2): Formal languages, structures and deduction

Topics: first-order syntax; structures; satisfaction; theories; proof systems

Key definitions: language; term; formula; sentence; structure; model; consistency

Key results to learn:

- soundness theorem [PROOF]
- syntactic induction/substitution lemmas [PROOF]

Reading route: Enderton syntax/semantics chapters

Warm-up/source exercise focus (secondary): Formalize arithmetic/group/order statements; build models/nonmodels; prove soundness for a deductive calculus.

Week 1: Formal languages, structures and deduction - definitions and examples. Read: Enderton syntax/semantics chapters. Make explicit examples/nonexamples for language; term; formula; sentence; structure; model; consistency. Work through first-order syntax; structures; satisfaction; theories; proof systems. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Formal languages, structures and deduction - theorem and technique pass. Master/audit: soundness theorem; syntactic induction/substitution lemmas. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Completeness and compactness

Topics: Henkin constructions; compactness; elementary substructures; LS phenomena

Key definitions: complete theory; elementary equivalence; elementary embedding

Key results to learn:

- Godel completeness theorem [ARCH]
- compactness theorem [PROOF/ARCH]
- upward/downward Lowenheim-Skolem [ARCH]

Reading route: Enderton completeness chapters

Warm-up/source exercise focus (secondary): Use compactness to build nonstandard models; explain the Skolem paradox precisely.

Week 3: Completeness and compactness - definitions and examples. Read: Enderton completeness chapters. Make explicit examples/nonexamples for complete theory; elementary equivalence; elementary embedding. Work through Henkin constructions; compactness; elementary substructures; LS phenomena. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Completeness and compactness - theorem and technique pass. Master/audit: Godel completeness theorem; compactness theorem; upward/downward Lowenheim-Skolem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Computability and undecidability

Topics: Turing machines; recursive functions; r.e. sets; reductions; halting

Key definitions: computable function; decidable/r.e. set; many-one reduction

Key results to learn:

- equivalence of standard computability models [USE]
- halting undecidable [PROOF]
- Church theorem for first-order validity [ARCH]

Reading route: Boolos-Burgess-Jeffrey computability chapters

Warm-up/source exercise focus (secondary): Construct reductions; encode finite computations; prove at least three undecidability results.

Week 5: Computability and undecidability - definitions and examples. Read: Boolos-Burgess-Jeffrey computability chapters. Make explicit examples/nonexamples for computable function; decidable/r.e. set; many-one reduction. Work through Turing machines; recursive functions; r.e. sets; reductions; halting. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Computability and undecidability - theorem and technique pass. Master/audit: equivalence of standard computability models; halting undecidable; Church theorem for first-order validity. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Godel incompleteness

Topics: arithmetization; representability; fixed-point lemma; provability predicates

Key definitions: Godel numbering; representability; provability predicate

Key results to learn:

- diagonal lemma [PROOF/ARCH]
- Godel I [ARCH]
- Godel II [ARCH]
- Tarski undefinability [ARCH]

Reading route: Enderton/Boolos incompleteness sections

Warm-up/source exercise focus (secondary): Write dependency chains; distinguish truth, provability, consistency and soundness.

Week 7: Godel incompleteness - definitions and examples. Read: Enderton/Boolos incompleteness sections. Make explicit examples/nonexamples for Godel numbering; representability; provability predicate. Work through arithmetization; representability; fixed-point lemma; provability predicates. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Godel incompleteness - theorem and technique pass. Master/audit: diagonal lemma; Godel I; Godel II. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): ZF/ZFC, ordinals and cardinals

Topics: choice; transfinite induction/recursion; ordinal/cardinal arithmetic

Key definitions: ordinal; cardinal; well-order; cofinality; AC

Key results to learn:

- well-ordering theorem equivalent to AC [ARCH]
- transfinite recursion [PROOF/ARCH]
- Cantor theorem [PROOF]
- Hartogs theorem [ARCH]

Reading route: Kunen/Jech introductory chapters

Warm-up/source exercise focus (secondary): Compute ordinal arithmetic; prove transfinite induction results; compare cardinal arithmetic under/without CH.

Week 9: ZF/ZFC, ordinals and cardinals - definitions and examples. Read: Kunen/Jech introductory chapters. Make explicit examples/nonexamples for ordinal; cardinal; well-order; cofinality; AC. Work through choice; transfinite induction/recursion; ordinal/cardinal arithmetic. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: ZF/ZFC, ordinals and cardinals - theorem and technique pass. Master/audit: well-ordering theorem equivalent to AC; transfinite recursion; Cantor theorem. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Constructibility, forcing and CH

Topics: constructible universe; forcing posets; generics; names; forcing relation

Key definitions: L; forcing notion; dense set; generic filter; CH/GCH

Key results to learn:

- Godel relative consistency of ZFC+GCH via L [USE]
- forcing theorem architecture [USE]
- Cohen relative consistency of not-CH [USE]
- CH independent of ZFC assuming consistency [ARCH/USE]

Reading route: Kunen/Jech forcing chapters; Cohen; Easwaran notes

Warm-up/source exercise focus (secondary): Work toy forcing posets; trace the independence strategy in a 10-page exposition.

Week 11: Constructibility, forcing and CH - definitions and examples. Read: Kunen/Jech forcing chapters; Cohen; Easwaran notes. Make explicit examples/nonexamples for L; forcing notion; dense set; generic filter; CH/GCH. Work through constructible universe; forcing posets; generics; names; forcing relation. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Constructibility, forcing and CH - theorem and technique pass. Master/audit: Godel relative consistency of ZFC+GCH via L; forcing theorem architecture; Cohen relative consistency of not-CH. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Formal languages, structures and deduction

U1.P1. [F] Formalize a fragment of group theory in a first-order language: write axioms, a sentence asserting commutativity, and a formula defining the center. Distinguish syntax from semantics at every step.

U1.P2. [T] Build a formal derivation of a simple theorem from logical axioms/rules, then prove soundness for the rules actually used. Identify which part is meta-mathematics rather than a formula in the object language.

U1.P3. [C] Give properties of structures that are and are not first-order expressible in a chosen language. Explain why finiteness is not captured by a single first-order sentence over arbitrary structures.

Programme U2: Completeness and compactness

U2.P1. [T] Use compactness to construct a nonstandard model of arithmetic containing an element larger than every standard numeral. Write the expanded language and finite satisfiability argument explicitly.

U2.P2. [S] Use compactness to prove the graph-theoretic statement: if every finite subgraph of a graph is k-colorable, then the entire graph is k-colorable. Encode the coloring constraints as first-order/propositional clauses.

U2.P3. [C] Compare completeness and compactness: derive compactness from completeness plus finite proof length, then explain why semantic completeness is not the same as categorical uniqueness of models.

Programme U3: Computability and undecidability

U3.P1. [F] Give an explicit coding of finite binary strings by natural numbers and show concatenation/length are computable. Use the encoding to illustrate how syntax can become arithmetic data.

U3.P2. [T] Prove undecidability of the halting problem by diagonalization. Then reduce the halting problem to another undecidable decision problem of your choice, writing the reduction as an algorithmic transformation.

U3.P3. [S] Compare Cantor diagonalization, the halting diagonal argument, and the later Godel diagonal lemma. Write a table specifying the object diagonalized against and the contradiction/fixed-point produced.

Programme U4: Godel incompleteness

U4.P1. [T] Construct Godel numbering for a toy formal language and explicitly code substitution for a few formulas. Identify which syntactic operations must be primitive recursive in a full proof.

U4.P2. [T] Assuming representability and the diagonal lemma, derive a sentence G with G <-> not Prov(G). State exactly what consistency or soundness assumptions are required for each conclusion about G.

U4.P3. [S] Build a proof-architecture map for the second incompleteness theorem: derivability conditions, formalized first incompleteness, and the obstruction to proving Con(T). Mark which ingredients are internalized in T.

Programme U5: ZF/ZFC, ordinals and cardinals

U5.P1. [F] Construct the first several finite and transfinite ordinals, compute simple ordinal sums/products, and exhibit noncommutativity such as 1+omega != omega+1.

U5.P2. [T] Prove Cantor's theorem |X|<|P(X)| and use Hartogs' theorem/choice assumptions to distinguish cardinal comparability issues. Work several cardinal-arithmetic examples under and without CH assumptions.

U5.P3. [S] Prove transfinite recursion in a simplified setting and use it to define the cumulative hierarchy V_alpha for initial stages. Explain how Replacement enters the full ZF construction.

Programme U6: Constructibility, forcing and CH

U6.P1. [F] Compute the first stages L_0,L_1,... of the constructible hierarchy and contrast 'all subsets' with 'definable subsets' at successor stages.

U6.P2. [T] For Cohen forcing Fn(omega,2,<omega), decide compatibility of explicit conditions, construct dense sets ensuring a total binary sequence, and show how a generic filter determines a new real.

U6.P3. [R] Write a two-column proof architecture for CH independence: Godel's L gives relative consistency of CH; Cohen forcing gives relative consistency of not-CH. For each column state the model transformation and exactly what metatheoretic assumption is being made.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Mathematical Logic, Computability and Foundations: start from “Formal languages, structures and deduction”, pass through “Godel incompleteness”, and finish at “Constructibility, forcing and CH”. Use the named results “soundness theorem” and “CH independent of ZFC assuming consistency” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Prepares Hilbert’s tenth problem and foundational reading across mathematics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: soundness theorem; syntactic induction/substitution lemmas; compactness theorem; halting undecidable; diagonal lemma. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Hartogs theorem; Godel relative consistency of ZFC+GCH via L; forcing theorem architecture; Cohen relative consistency of not-CH; CH independent of ZFC assuming consistency.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.

- Oral synthesis: in 30-45 minutes explain how “Formal languages, structures and deduction” develops into “Constructibility, forcing and CH”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 6. Real Analysis and Measure Theory

Purpose: Build rigorous real analysis and modern integration theory sufficient for probability, functional/harmonic analysis, PDE and dynamics.

Prerequisites: Volumes I-II.

Primary and secondary references: Abbott, Understanding Analysis; Rudin, Principles of Mathematical Analysis; Folland, Real Analysis; Stein-Shakarchi, Real Analysis; MIT OCW 18.100B

Quarter output: Create a theorem dependency chart from completeness of R through Lebesgue integration, Lp and Radon-Nikodym.

Research bridges: Direct prerequisite for probability, harmonic/functional analysis, PDE and ergodic theory.

Six two-week study units

Unit 1 (Weeks 1-2): Real numbers, sequences and series

Topics: completeness; limits; limsup/liminf; series; rearrangements

Key definitions: Cauchy sequence; completeness; limsup; absolute/conditional convergence

Key results to learn:

- monotone sequence theorem [PROOF]
- Cauchy criterion [PROOF]
- Bolzano-Weierstrass [PROOF]
- absolute convergence implies convergence [PROOF]

Reading route: Abbott/Rudin early chapters; MIT notes

Warm-up/source exercise focus (secondary): Construct pathological sequences; prove convergence criteria; analyse rearrangements and power series.

Week 1: Real numbers, sequences and series - definitions and examples. Read: Abbott/Rudin early chapters; MIT notes. Make explicit examples/nonexamples for Cauchy sequence; completeness; limsup; absolute/conditional convergence. Work through completeness; limits; limsup/liminf; series; rearrangements. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Real numbers, sequences and series - theorem and technique pass. Master/audit: monotone sequence theorem; Cauchy criterion; Bolzano-Weierstrass. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Metric spaces and compactness

Topics: continuity; compactness; completeness; contractions

Key definitions: metric; compactness; total boundedness; uniform continuity; contraction

Key results to learn:

- Heine-Borel in R^n [PROOF]
- metric compactness equivalences [PROOF]
- Banach contraction theorem [PROOF]

Reading route: Rudin metric chapters; MIT notes

Warm-up/source exercise focus (secondary): Prove compactness criteria; apply contractions to nonlinear equations/ODE toy problems.

Week 3: Metric spaces and compactness - definitions and examples. Read: Rudin metric chapters; MIT notes. Make explicit examples/nonexamples for metric; compactness; total boundedness; uniform continuity; contraction. Work through continuity; compactness; completeness; contractions. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Metric spaces and compactness - theorem and technique pass. Master/audit: Heine-Borel in R^n; metric compactness equivalences; Banach contraction theorem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Differentiation and Riemann integration

Topics: mean value; Taylor; Riemann/Stieltjes integration; uniform convergence

Key definitions: bounded variation; Riemann integrable; equicontinuity

Key results to learn:

- mean value theorem [PROOF]
- Taylor theorem [PROOF]
- Arzela-Ascoli [ARCH]
- uniform-limit interchange principles [PROOF]

Reading route: Rudin differentiation/integration/function sequence chapters

Warm-up/source exercise focus (secondary): Build examples where pointwise limits lose properties; prove equicontinuity compactness cases.

Week 5: Differentiation and Riemann integration - definitions and examples. Read: Rudin differentiation/integration/function sequence chapters. Make explicit examples/nonexamples for bounded variation; Riemann integrable; equicontinuity. Work through mean value; Taylor; Riemann/Stieltjes integration; uniform convergence. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Differentiation and Riemann integration - theorem and technique pass. Master/audit: mean value theorem; Taylor theorem; Arzela-Ascoli. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Measure and Lebesgue integration

Topics: outer measure; measurable sets/functions; integration

Key definitions: sigma-algebra; measure; outer measure; simple function

Key results to learn:

- Caratheodory construction [ARCH]
- monotone convergence [PROOF]
- Fatou [PROOF]
- dominated convergence [PROOF]

Reading route: Folland Chapters 1-2 by topic; Stein-Shakarchi

Warm-up/source exercise focus (secondary): Construct Lebesgue measure; justify limit-integral exchanges with the convergence theorems.

Week 7: Measure and Lebesgue integration - definitions and examples. Read: Folland Chapters 1-2 by topic; Stein-Shakarchi. Make explicit examples/nonexamples for sigma-algebra; measure; outer measure; simple function. Work through outer measure; measurable sets/functions; integration. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Measure and Lebesgue integration - theorem and technique pass. Master/audit: Caratheodory construction; monotone convergence; Fatou. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Product and signed measures

Topics: Fubini/Tonelli; signed measures; Radon-Nikodym; differentiation

Key definitions: product measure; signed measure; absolute continuity; RN derivative

Key results to learn:

- Tonelli/Fubini [PROOF]
- Hahn/Jordan decomposition [ARCH]
- Radon-Nikodym [ARCH]
- Lebesgue differentiation [ARCH]

Reading route: Folland product/signed-measure chapters

Warm-up/source exercise focus (secondary): Compute product integrals; derive densities; apply differentiation theorem.

Week 9: Product and signed measures - definitions and examples. Read: Folland product/signed-measure chapters. Make explicit examples/nonexamples for product measure; signed measure; absolute continuity; RN derivative. Work through Fubini/Tonelli; signed measures; Radon-Nikodym; differentiation. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Product and signed measures - theorem and technique pass. Master/audit: Tonelli/Fubini; Hahn/Jordan decomposition; Radon-Nikodym. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Lp spaces and convergence modes

Topics: Holder/Minkowski; Lp completeness; convergence in measure/a.e./Lp

Key definitions: Lp norm; essential supremum; convergence in measure; uniform integrability

Key results to learn:

- Holder/Minkowski [PROOF]
- completeness of Lp [ARCH]
- standard subsequence principles [ARCH]

Reading route: Folland Lp chapters; Brezis preliminaries

Warm-up/source exercise focus (secondary): Prove inequalities; separate convergence modes by counterexamples; approximate functions by simple/smooth ones.

Week 11: Lp spaces and convergence modes - definitions and examples. Read: Folland Lp chapters; Brezis preliminaries. Make explicit examples/nonexamples for Lp norm; essential supremum; convergence in measure; uniform integrability. Work through Holder/Minkowski; Lp completeness; convergence in measure/a.e./Lp. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Lp spaces and convergence modes - theorem and technique pass. Master/audit: Holder/Minkowski; completeness of Lp; standard subsequence principles. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Real numbers, sequences and series

U1.P1. [T] Prove Cauchy completeness of R from the least-upper-bound property and conversely derive a completeness formulation from Cauchy completeness. Track which construction of R is being assumed.

U1.P2. [F] Determine convergence/absolute convergence/conditional convergence of a curated list of ten series using comparison, ratio/root, condensation, and Dirichlet tests. For each, justify why the chosen test is informative.

U1.P3. [C] Construct sequences showing the converses of common convergence implications fail: bounded does not imply convergent; every subsequence has a further convergent subsequence characterizes compactness but not arbitrary boundedness in general metric spaces.

Programme U2: Metric spaces and compactness

U2.P1. [T] Prove the equivalence of compactness, sequential compactness, and complete+totally bounded for metric spaces. Give a counterexample to each implication if the metric-space hypothesis is removed or altered.

U2.P2. [F] Analyze compactness and connectedness of explicit subsets of R^n and of function spaces with different metrics. Explain why the metric, not only the underlying set, matters.

U2.P3. [S] Prove the contraction mapping theorem and apply it to a nonlinear equation x=cos x with an explicit invariant interval and quantitative error bound.

Programme U3: Differentiation and Riemann integration

U3.P1. [T] Derive the fundamental theorem of calculus from the mean value theorem/Riemann integration hypotheses you have established. Identify the exact continuity assumptions in each direction.

U3.P2. [C] Construct: a differentiable function with discontinuous derivative; a continuous nowhere differentiable function at theorem-statement level; and a Riemann-integrable discontinuous function. Explain which intuitive implications fail.

U3.P3. [S] Prove Taylor's theorem with remainder and use it to derive a controlled approximation to log(1+x) or exp(x), including an explicit numerical error estimate.

Programme U4: Measure and Lebesgue integration

U4.P1. [F] Compute Lebesgue measures/integrals for step functions, Cantor-set indicators, and functions with countably many discontinuities. Compare with their Riemann integrability.

U4.P2. [T] Prove monotone convergence, Fatou, and dominated convergence in a dependency chain, then construct examples showing why domination or monotonicity hypotheses cannot simply be dropped.

U4.P3. [S] Derive the Borel-Cantelli lemma's measure-theoretic core from continuity of measure and apply it to a limsup set before probability is formally introduced.

Programme U5: Product and signed measures

U5.P1. [T] Prove Tonelli and Fubini in a selected standard formulation and apply them to evaluate a nontrivial double integral in two orders. Construct a conditionally integrable example where naive interchange fails.

U5.P2. [F] Compute Jordan decompositions and total variation for explicit signed measures with densities and finite atomic parts.

U5.P3. [S] Derive Radon-Nikodym in a simple absolutely continuous setting and use it to explain conditional density/change-of-measure intuition needed later in probability.

Programme U6: Lp spaces and convergence modes

U6.P1. [F] For L^p[0,1], determine membership of x^{-alpha} and log x for ranges of p and alpha. Compute norms where possible.

U6.P2. [T] Prove Holder and Minkowski inequalities and identify equality cases in a finite-dimensional model before passing to integrals.

U6.P3. [C] Build examples separating almost-everywhere, in-measure, L^1, and L^p convergence. Draw a one-page implication diagram and annotate each failed converse with your example.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Real Analysis and Measure Theory: start from “Real numbers, sequences and series”, pass through “Measure and Lebesgue integration”, and finish at “Lp spaces and convergence modes”. Use the named results “monotone sequence theorem” and “standard subsequence principles” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct prerequisite for probability, harmonic/functional analysis, PDE and ergodic theory. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: monotone sequence theorem; Cauchy criterion; Bolzano-Weierstrass; absolute convergence implies convergence; Heine-Borel in R^n. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Hahn/Jordan decomposition; Radon-Nikodym; Lebesgue differentiation; completeness of Lp; standard subsequence principles.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.

- Oral synthesis: in 30-45 minutes explain how “Real numbers, sequences and series” develops into “Lp spaces and convergence modes”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 7. Functional Analysis and Operator Theory

Purpose: Develop Banach/Hilbert-space methods, duality and operator theory as a common language for PDE, spectral theory, representation theory and probability.

Prerequisites: Volumes II and VI.

Primary and secondary references: Brezis, Functional Analysis, Sobolev Spaces and PDEs; Conway, A Course in Functional Analysis; Reed-Simon I, Functional Analysis; Rudin, Functional Analysis

Quarter output: Solve one spectral problem from PDE and one from representation theory, identifying where each functional-analytic theorem enters.

Research bridges: Feeds PDE, unitary representations, spectral gaps, automorphic analysis and stochastic processes.

Six two-week study units

Unit 1 (Weeks 1-2): Banach and Hilbert spaces

Topics: normed spaces; completion; Hilbert geometry; orthogonal projection

Key definitions: Banach space; Hilbert space; completion; orthonormal basis

Key results to learn:

- completion theorem [ARCH]
- projection theorem [PROOF]
- Bessel/Parseval [PROOF]

Reading route: Brezis/Conway opening chapters

Warm-up/source exercise focus (secondary): Work in l^p,L^p,C(K),Sobolev examples; identify complete/incomplete spaces; compute projections.

Week 1: Banach and Hilbert spaces - definitions and examples. Read: Brezis/Conway opening chapters. Make explicit examples/nonexamples for Banach space; Hilbert space; completion; orthonormal basis. Work through normed spaces; completion; Hilbert geometry; orthogonal projection. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Banach and Hilbert spaces - theorem and technique pass. Master/audit: completion theorem; projection theorem; Bessel/Parseval. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Hahn-Banach and duality

Topics: linear functionals; separation; dual spaces; weak topologies

Key definitions: dual; annihilator; weak/weak-* topology; polar set

Key results to learn:

- Hahn-Banach [PROOF/ARCH]
- separation theorems [ARCH]
- canonical double-dual embedding [PROOF]

Reading route: Brezis/Conway Hahn-Banach/duality chapters

Warm-up/source exercise focus (secondary): Extend functionals explicitly; separate convex sets; compute standard duals.

Week 3: Hahn-Banach and duality - definitions and examples. Read: Brezis/Conway Hahn-Banach/duality chapters. Make explicit examples/nonexamples for dual; annihilator; weak/weak-* topology; polar set. Work through linear functionals; separation; dual spaces; weak topologies. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Hahn-Banach and duality - theorem and technique pass. Master/audit: Hahn-Banach; separation theorems; canonical double-dual embedding. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Uniform boundedness/open mapping/closed graph

Topics: the three basic Banach-space principles

Key definitions: bounded operator; graph; pointwise bounded family

Key results to learn:

- Banach-Steinhaus [PROOF/ARCH]
- open mapping [ARCH]
- closed graph [ARCH]

Reading route: Brezis/Conway foundational theorems

Warm-up/source exercise focus (secondary): Construct counterexamples without completeness; use each theorem in several applications.

Week 5: Uniform boundedness/open mapping/closed graph - definitions and examples. Read: Brezis/Conway foundational theorems. Make explicit examples/nonexamples for bounded operator; graph; pointwise bounded family. Work through the three basic Banach-space principles. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Uniform boundedness/open mapping/closed graph - theorem and technique pass. Master/audit: Banach-Steinhaus; open mapping; closed graph. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Weak compactness and convexity

Topics: reflexivity; Banach-Alaoglu; weak compactness; lower semicontinuity

Key definitions: reflexive space; weak compactness; weak-* compactness

Key results to learn:

- Banach-Alaoglu [ARCH]
- weak compactness in reflexive spaces [ARCH]
- Mazur lemma [ARCH]

Reading route: Brezis weak-topology chapters

Warm-up/source exercise focus (secondary): Solve compactness/existence problems for minimizing sequences; contrast weak and norm convergence.

Week 7: Weak compactness and convexity - definitions and examples. Read: Brezis weak-topology chapters. Make explicit examples/nonexamples for reflexive space; weak compactness; weak-* compactness. Work through reflexivity; Banach-Alaoglu; weak compactness; lower semicontinuity. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Weak compactness and convexity - theorem and technique pass. Master/audit: Banach-Alaoglu; weak compactness in reflexive spaces; Mazur lemma. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Compact operators and spectral theory

Topics: compact operators; Fredholm alternative; compact self-adjoint spectrum

Key definitions: compact operator; spectrum; resolvent; eigenprojection

Key results to learn:

- Riesz-Schauder theory [ARCH]
- Fredholm alternative [PROOF/ARCH]
- compact self-adjoint spectral theorem [PROOF/ARCH]

Reading route: Conway/Reed-Simon compact/spectral chapters

Warm-up/source exercise focus (secondary): Compute spectra of integral operators; derive eigenfunction expansions; solve Fredholm equations.

Week 9: Compact operators and spectral theory - definitions and examples. Read: Conway/Reed-Simon compact/spectral chapters. Make explicit examples/nonexamples for compact operator; spectrum; resolvent; eigenprojection. Work through compact operators; Fredholm alternative; compact self-adjoint spectrum. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Compact operators and spectral theory - theorem and technique pass. Master/audit: Riesz-Schauder theory; Fredholm alternative; compact self-adjoint spectral theorem. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Unbounded operators and C*-viewpoint

Topics: closed densely defined operators; self-adjointness; functional calculus; C*-algebras

Key definitions: unbounded operator; symmetric/self-adjoint; C*-algebra; positive functional

Key results to learn:

- spectral theorem for self-adjoint operators (architecture) [USE]
- Stone theorem [ARCH/USE]
- GNS construction [ARCH]

Reading route: Reed-Simon I; Conway C*-chapters

Warm-up/source exercise focus (secondary): Analyse derivative/Laplacian domains; distinguish symmetric/self-adjoint; construct GNS in simple examples.

Week 11: Unbounded operators and C*-viewpoint - definitions and examples. Read: Reed-Simon I; Conway C*-chapters. Make explicit examples/nonexamples for unbounded operator; symmetric/self-adjoint; C*-algebra; positive functional. Work through closed densely defined operators; self-adjointness; functional calculus; C*-algebras. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Unbounded operators and C*-viewpoint - theorem and technique pass. Master/audit: spectral theorem for self-adjoint operators (architecture); Stone theorem; GNS construction. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Banach and Hilbert spaces

U1.P1. [F] Compare l^1, l^2, l^infinity, C[0,1], and finite-dimensional R^n: determine completeness, separability, dual behavior known at this stage, and canonical dense subsets.

U1.P2. [T] Prove the projection theorem in a Hilbert space and use it to solve a least-squares/closest-point problem in L^2[0,1] for a finite-dimensional polynomial subspace.

U1.P3. [C] Exhibit a normed space that is not complete and a closed subspace of a Banach space that is complete. Explain why completeness is not inherited by arbitrary subspaces.

Programme U2: Hahn-Banach and duality

U2.P1. [T] Use Hahn-Banach to extend a functional from a one-dimensional subspace of a normed space, first explicitly and then abstractly. Derive separation of a point from a closed subspace.

U2.P2. [F] Identify duals of finite-dimensional spaces and of l^1/l^p in the standard ranges covered by your text. Check norm equality for explicit functionals.

U2.P3. [S] Prove that if x is not in a closed subspace M of a normed space, there is a continuous functional vanishing on M but not on x. Then explain how this becomes a separation tool in convexity and PDE.

Programme U3: Uniform boundedness/open mapping/closed graph

U3.P1. [T] Prove the Uniform Boundedness Principle via Baire category and then derive a concrete corollary for a pointwise-bounded family of operators on a Banach space.

U3.P2. [T] Derive the Open Mapping and Closed Graph theorems from one another using the standard Banach-space framework. Record exactly where completeness is needed.

U3.P3. [C] Construct examples showing each theorem can fail on incomplete normed spaces or if linearity/closedness assumptions are weakened.

Programme U4: Weak compactness and convexity

U4.P1. [F] Work out weak and weak-* convergence for standard sequences in l^p and L^p. Compare with norm convergence and identify bounded sequences with no norm-convergent subsequence.

U4.P2. [T] Prove Banach-Alaoglu in the separable/metric special case first, then state the general product-topology proof architecture. Use it to extract a weak-* convergent subsequence of bounded measures in a toy setting.

U4.P3. [S] Prove the Hahn-Banach separation theorem for a closed convex set in a locally convex/normed setting sufficient for later use. Apply it to justify a simple dual variational principle.

Programme U5: Compact operators and spectral theory

U5.P1. [F] On l^2, analyze a diagonal compact operator with entries tending to zero: determine spectrum, eigenvalues, norm, and compactness directly.

U5.P2. [T] Prove the spectral theorem for compact self-adjoint operators at full proof-architecture level, including existence of an extremal eigenvector and orthogonal decomposition.

U5.P3. [C] Compare compact and bounded operators by giving examples of bounded noncompact operators and compact operators with nonclosed range. Explain which finite-dimensional intuitions survive.

Programme U6: Unbounded operators and C*-viewpoint

U6.P1. [T] Analyze the differentiation operator on L^2 or C[0,1] with a specified domain: determine whether it is densely defined, closed, symmetric, or self-adjoint in simple boundary-condition cases.

U6.P2. [F] For C(X), verify the C*-identity ||f*f||=||f||^2 and identify maximal ideals/evaluation functionals in a simple compact space.

U6.P3. [R] State the spectral theorem for an unbounded self-adjoint operator and for a commutative C*-algebra in precise standard forms. For each, give one later theorem in quantum mechanics/PDE/representation theory that uses it.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Functional Analysis and Operator Theory: start from “Banach and Hilbert spaces”, pass through “Weak compactness and convexity”, and finish at “Unbounded operators and C*-viewpoint”. Use the named results “completion theorem” and “GNS construction” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds PDE, unitary representations, spectral gaps, automorphic analysis and stochastic processes. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: projection theorem; Bessel/Parseval; Hahn-Banach; canonical double-dual embedding; Banach-Steinhaus. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Mazur lemma; Riesz-Schauder theory; spectral theorem for self-adjoint operators (architecture); Stone theorem; GNS construction.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.

- Oral synthesis: in 30-45 minutes explain how “Banach and Hilbert spaces” develops into “Unbounded operators and C*-viewpoint”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 8. Complex Analysis and Riemann Surfaces

Purpose: Master one-complex-variable analysis and its geometric extension to Riemann surfaces, supporting analytic number theory, conformal probability and algebraic geometry.

Prerequisites: Volumes II and VI.

Primary and secondary references: Ahlfors, Complex Analysis; Conway, Functions of One Complex Variable I; Stein-Shakarchi, Complex Analysis; Forster, Lectures on Riemann Surfaces

Quarter output: Write a zeta functional-equation strategy note from theta/Mellin methods plus a complete genus-1 worked example.

Research bridges: Feeds analytic number theory, modular forms, SLE and algebraic curves.

Six two-week study units

Unit 1 (Weeks 1-2): Holomorphic functions and Cauchy theory

Topics: complex differentiability; power series; contour integrals; homotopy Cauchy theorem

Key definitions: holomorphic; analytic; contour; winding number

Key results to learn:

- Cauchy theorem [PROOF/ARCH]
- Cauchy integral formula [PROOF]
- Morera [PROOF]
- identity theorem [PROOF]

Reading route: Ahlfors/Conway opening chapters; Stein-Shakarchi

Warm-up/source exercise focus (secondary): Derive standard consequences from Cauchy formula; compute integrals by deformation; construct continuations.

Week 1: Holomorphic functions and Cauchy theory - definitions and examples. Read: Ahlfors/Conway opening chapters; Stein-Shakarchi. Make explicit examples/nonexamples for holomorphic; analytic; contour; winding number. Work through complex differentiability; power series; contour integrals; homotopy Cauchy theorem. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Holomorphic functions and Cauchy theory - theorem and technique pass. Master/audit: Cauchy theorem; Cauchy integral formula; Morera. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Zeros, poles and residues

Topics: isolated singularities; Laurent series; residues; argument principle

Key definitions: zero order; pole; essential singularity; residue

Key results to learn:

- residue theorem [PROOF]
- argument principle [PROOF]
- Rouche [PROOF]
- Liouville and FTA [PROOF]

Reading route: Ahlfors singularity/residue chapters

Warm-up/source exercise focus (secondary): Evaluate contour integrals; count zeros; derive real integrals/asymptotics.

Week 3: Zeros, poles and residues - definitions and examples. Read: Ahlfors singularity/residue chapters. Make explicit examples/nonexamples for zero order; pole; essential singularity; residue. Work through isolated singularities; Laurent series; residues; argument principle. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Zeros, poles and residues - theorem and technique pass. Master/audit: residue theorem; argument principle; Rouche. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Conformal maps and normal families

Topics: Mobius maps; Schwarz lemma; harmonic functions; normal families

Key definitions: conformal map; harmonic conjugate; normal family

Key results to learn:

- Schwarz lemma [PROOF]
- maximum principle [PROOF]
- Montel [ARCH]
- Riemann mapping [ARCH/USE]

Reading route: Ahlfors conformal chapters; Conway normal families

Warm-up/source exercise focus (secondary): Construct explicit maps; solve extremal problems; use normal-family arguments.

Week 5: Conformal maps and normal families - definitions and examples. Read: Ahlfors conformal chapters; Conway normal families. Make explicit examples/nonexamples for conformal map; harmonic conjugate; normal family. Work through Mobius maps; Schwarz lemma; harmonic functions; normal families. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Conformal maps and normal families - theorem and technique pass. Master/audit: Schwarz lemma; maximum principle; Montel. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Meromorphic construction and approximation

Topics: Weierstrass products; Mittag-Leffler; Runge; analytic continuation

Key definitions: canonical product; divisor; meromorphic function

Key results to learn:

- Weierstrass factorization [ARCH]
- Mittag-Leffler [ARCH]
- Runge [ARCH/USE]

Reading route: Conway advanced chapters

Warm-up/source exercise focus (secondary): Build entire/meromorphic functions with prescribed zeros/poles; solve approximation problems.

Week 7: Meromorphic construction and approximation - definitions and examples. Read: Conway advanced chapters. Make explicit examples/nonexamples for canonical product; divisor; meromorphic function. Work through Weierstrass products; Mittag-Leffler; Runge; analytic continuation. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Meromorphic construction and approximation - theorem and technique pass. Master/audit: Weierstrass factorization; Mittag-Leffler; Runge. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Special functions and zeta preparation

Topics: Gamma/Beta; Mellin transform; theta functions; zeta continuation prototypes

Key definitions: Gamma function; Mellin transform; theta series

Key results to learn:

- Gamma functional equation [PROOF]
- theta transformation via Poisson summation [ARCH]
- zeta functional-equation architecture [USE]

Reading route: Stein-Shakarchi; analytic-number-theory preliminaries

Warm-up/source exercise focus (secondary): Derive Gamma identities; compute Mellin transforms; work through theta transformation.

Week 9: Special functions and zeta preparation - definitions and examples. Read: Stein-Shakarchi; analytic-number-theory preliminaries. Make explicit examples/nonexamples for Gamma function; Mellin transform; theta series. Work through Gamma/Beta; Mellin transform; theta functions; zeta continuation prototypes. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Special functions and zeta preparation - theorem and technique pass. Master/audit: Gamma functional equation; theta transformation via Poisson summation; zeta functional-equation architecture. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Riemann surfaces

Topics: charts; holomorphic maps; divisors; differentials; covering spaces; genus

Key definitions: Riemann surface; meromorphic differential; divisor; genus

Key results to learn:

- uniformization theorem (statement) [USE]
- Riemann-Roch for compact Riemann surfaces [ARCH/USE]
- Abel-type ideas [USE]

Reading route: Forster; algebraic-curve supplements

Warm-up/source exercise focus (secondary): Construct sphere/torus/hyperelliptic examples; compute divisors/differentials; connect polygon gluing to complex structures.

Week 11: Riemann surfaces - definitions and examples. Read: Forster; algebraic-curve supplements. Make explicit examples/nonexamples for Riemann surface; meromorphic differential; divisor; genus. Work through charts; holomorphic maps; divisors; differentials; covering spaces; genus. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Riemann surfaces - theorem and technique pass. Master/audit: uniformization theorem (statement); Riemann-Roch for compact Riemann surfaces; Abel-type ideas. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Holomorphic functions and Cauchy theory

U1.P1. [T] Starting from Cauchy's integral theorem, derive Cauchy's integral formula and the derivative estimates. Use them to prove Liouville and the Fundamental Theorem of Algebra.

U1.P2. [F] Evaluate contour integrals around circles and rectangles for rational functions with poles inside/outside. For each calculation, do it once directly when possible and once by Cauchy theory.

U1.P3. [C] Give functions that are complex differentiable at one point but not holomorphic nearby, and contrast with real differentiability. Explain why the Cauchy-Riemann equations need regularity hypotheses for a converse.

Programme U2: Zeros, poles and residues

U2.P1. [T] Prove the argument principle from the residue theorem and apply it to count zeros of a parameterized polynomial in a disk. Then derive Rouche's theorem as a consequence.

U2.P2. [F] Compute Laurent expansions and residues for ten representative functions, including higher-order poles and essential singularities. Classify the isolated singularities.

U2.P3. [S] Use residues to evaluate one real integral and one infinite series. Explicitly justify contour choice, decay on auxiliary arcs, and all limiting steps.

Programme U3: Conformal maps and normal families

U3.P1. [T] Construct Mobius maps sending three prescribed boundary/interior points to standard positions and use them to map half-planes to disks. Track cross-ratio invariance.

U3.P2. [T] Prove Schwarz lemma and derive a rigidity statement for automorphisms of the disk. Then classify disk automorphisms explicitly.

U3.P3. [S] Use Montel's theorem/normal families to prove a compactness result for bounded holomorphic families and explain how this supports the Riemann mapping theorem proof architecture.

Programme U4: Meromorphic construction and approximation

U4.P1. [T] Construct an entire function with prescribed zeros using a simple Weierstrass product and check convergence in a concrete sequence such as the integers.

U4.P2. [F] Use Mittag-Leffler to build a meromorphic function with specified simple poles/residues on Z in a controlled example.

U4.P3. [R] Compare Runge, Weierstrass, and Mittag-Leffler approximation/construction theorems by giving one problem each that only one of the three naturally solves.

Programme U5: Special functions and zeta preparation

U5.P1. [F] Derive the Gamma functional equation and evaluate Gamma(n+1), Beta integrals, and one reflection/duplication identity from standard integral representations.

U5.P2. [T] Starting from theta-function Poisson summation, derive the functional equation of zeta at proof-architecture level, marking the Mellin-transform and analytic-continuation steps.

U5.P3. [S] Prove Euler's product for zeta on Re(s)>1 and compare its analytic meaning with the later zero-free line/PNT route. Explain exactly why absolute convergence matters.

Programme U6: Riemann surfaces

U6.P1. [F] Build Riemann surfaces for sqrt(z) and log z by gluing sheets/cutting planes. Identify branch points and monodromy explicitly.

U6.P2. [T] Compute divisors of meromorphic functions/differentials on the Riemann sphere and verify degree constraints. Work out one map P^1->P^1 and its ramification data.

U6.P3. [S] State Riemann-Roch for compact Riemann surfaces precisely and verify it in genus 0 on several divisors. Record what new geometry appears in genus 1.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Complex Analysis and Riemann Surfaces: start from “Holomorphic functions and Cauchy theory”, pass through “Meromorphic construction and approximation”, and finish at “Riemann surfaces”. Use the named results “Cauchy theorem” and “Abel-type ideas” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds analytic number theory, modular forms, SLE and algebraic curves. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Cauchy theorem; Cauchy integral formula; Morera; identity theorem; residue theorem. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: theta transformation via Poisson summation; zeta functional-equation architecture; uniformization theorem (statement); Riemann-Roch for compact Riemann surfaces; Abel-type ideas.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.

- Oral synthesis: in 30-45 minutes explain how “Holomorphic functions and Cauchy theory” develops into “Riemann surfaces”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 9. Fourier and Harmonic Analysis

Purpose: Build the classical and modern harmonic-analysis toolkit used in PDE, analytic number theory, ergodic theory, decoupling and fractal geometry.

Prerequisites: Volumes VI-VIII; Volume VII helpful.

Primary and secondary references: Stein-Shakarchi, Fourier Analysis; Grafakos, Classical Fourier Analysis; Grafakos, Modern Fourier Analysis; Stein, Harmonic Analysis

Quarter output: Solve one coherent operator-estimate chain and one curved-surface restriction problem.

Research bridges: Prerequisite for Carleson, decoupling, dispersive PDE, analytic NT and fractal Fourier methods.

Six two-week study units

Unit 1 (Weeks 1-2): Fourier series

Topics: Fourier coefficients; convergence; kernels; L2 theory

Key definitions: Fourier coefficient; Dirichlet/Fejer kernel

Key results to learn:

- Bessel [PROOF]
- Parseval/Plancherel for series [PROOF]
- Fejer theorem [PROOF]

Reading route: Stein-Shakarchi Fourier-series chapters

Warm-up/source exercise focus (secondary): Compute series/convergence; analyse Gibbs; prove Fejer convergence.

Week 1: Fourier series - definitions and examples. Read: Stein-Shakarchi Fourier-series chapters. Make explicit examples/nonexamples for Fourier coefficient; Dirichlet/Fejer kernel. Work through Fourier coefficients; convergence; kernels; L2 theory. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Fourier series - theorem and technique pass. Master/audit: Bessel; Parseval/Plancherel for series; Fejer theorem. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Fourier transform on R^n

Topics: Schwartz space; inversion; Plancherel; convolution; scaling

Key definitions: Schwartz function; tempered-distribution preview; convolution

Key results to learn:

- Fourier inversion [PROOF/ARCH]
- Plancherel [PROOF/ARCH]
- convolution theorem [PROOF]
- Poisson summation [ARCH]

Reading route: Stein-Shakarchi; Grafakos classical chapters

Warm-up/source exercise focus (secondary): Compute transforms; prove uncertainty-style inequalities; use Poisson summation.

Week 3: Fourier transform on R^n - definitions and examples. Read: Stein-Shakarchi; Grafakos classical chapters. Make explicit examples/nonexamples for Schwartz function; tempered-distribution preview; convolution. Work through Schwartz space; inversion; Plancherel; convolution; scaling. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Fourier transform on R^n - theorem and technique pass. Master/audit: Fourier inversion; Plancherel; convolution theorem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Maximal functions and differentiation

Topics: Hardy-Littlewood maximal operator; covering lemmas; differentiation

Key definitions: maximal function; weak/strong type

Key results to learn:

- Vitali covering [ARCH]
- weak (1,1) maximal inequality [PROOF/ARCH]
- Lebesgue differentiation [ARCH]

Reading route: Grafakos/Stein maximal chapters

Warm-up/source exercise focus (secondary): Prove weak bounds; deduce differentiation; construct endpoint examples.

Week 5: Maximal functions and differentiation - definitions and examples. Read: Grafakos/Stein maximal chapters. Make explicit examples/nonexamples for maximal function; weak/strong type. Work through Hardy-Littlewood maximal operator; covering lemmas; differentiation. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Maximal functions and differentiation - theorem and technique pass. Master/audit: Vitali covering; weak (1,1) maximal inequality; Lebesgue differentiation. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Interpolation and singular integrals

Topics: Riesz-Thorin; Marcinkiewicz; Hilbert/Riesz transforms; CZ kernels

Key definitions: interpolation; CZ kernel; principal value

Key results to learn:

- Riesz-Thorin [PROOF/ARCH]
- Marcinkiewicz [ARCH]
- Lp Hilbert/Riesz transform bounds [ARCH]
- CZ theorem [ARCH]

Reading route: Grafakos; Stein singular-integral chapters

Warm-up/source exercise focus (secondary): Interpolate operator bounds; verify CZ hypotheses; solve cancellation estimates.

Week 7: Interpolation and singular integrals - definitions and examples. Read: Grafakos; Stein singular-integral chapters. Make explicit examples/nonexamples for interpolation; CZ kernel; principal value. Work through Riesz-Thorin; Marcinkiewicz; Hilbert/Riesz transforms; CZ kernels. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Interpolation and singular integrals - theorem and technique pass. Master/audit: Riesz-Thorin; Marcinkiewicz; Lp Hilbert/Riesz transform bounds. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Littlewood-Paley and multipliers

Topics: dyadic decompositions; square functions; multipliers

Key definitions: LP projection; square function; Fourier multiplier

Key results to learn:

- LP norm equivalence [ARCH]
- Mikhlin multiplier theorem [ARCH]
- Bernstein inequalities [PROOF/ARCH]

Reading route: Grafakos Modern; Stein

Warm-up/source exercise focus (secondary): Run dyadic decompositions; prove multiplier estimates; connect to Sobolev spaces.

Week 9: Littlewood-Paley and multipliers - definitions and examples. Read: Grafakos Modern; Stein. Make explicit examples/nonexamples for LP projection; square function; Fourier multiplier. Work through dyadic decompositions; square functions; multipliers. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Littlewood-Paley and multipliers - theorem and technique pass. Master/audit: LP norm equivalence; Mikhlin multiplier theorem; Bernstein inequalities. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Restriction and oscillatory entry

Topics: extension operator; stationary phase; curvature; wave packets preview

Key definitions: restriction estimate; oscillatory integral; nondegenerate phase

Key results to learn:

- Tomas-Stein [ARCH]
- stationary phase [ARCH]
- van der Corput oscillatory lemma [PROOF/ARCH]

Reading route: Stein Harmonic Analysis; restriction notes

Warm-up/source exercise focus (secondary): Prove model oscillatory estimates; derive sphere restriction at Tomas-Stein level; prepare wave packets.

Week 11: Restriction and oscillatory entry - definitions and examples. Read: Stein Harmonic Analysis; restriction notes. Make explicit examples/nonexamples for restriction estimate; oscillatory integral; nondegenerate phase. Work through extension operator; stationary phase; curvature; wave packets preview. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Restriction and oscillatory entry - theorem and technique pass. Master/audit: Tomas-Stein; stationary phase; van der Corput oscillatory lemma. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Fourier series

U1.P1. [F] Compute Fourier coefficients for a square wave, sawtooth, and |x| on the circle. Determine pointwise convergence at continuity and jump points using Dirichlet-type results.

U1.P2. [T] Prove Parseval/Plancherel for trigonometric polynomials and extend by density in the formulation used by your text. Use it to evaluate one classical series such as sum 1/n^2.

U1.P3. [C] Examine Gibbs phenomenon numerically and analytically near a jump. Explain why convergence in L^2 does not imply uniform convergence.

Programme U2: Fourier transform on R^n

U2.P1. [T] Compute Fourier transforms of Gaussians, indicators of intervals, exponentials times Gaussians, and derivatives. Verify translation, modulation, scaling, and differentiation identities.

U2.P2. [T] Prove Fourier inversion first for Schwartz functions and derive Plancherel. Identify where dominated convergence/Fubini enter.

U2.P3. [S] Derive Poisson summation for a Schwartz function and apply it to the theta function or a lattice Gaussian sum.

Programme U3: Maximal functions and differentiation

U3.P1. [F] Compute the Hardy-Littlewood maximal function for simple indicators in R and estimate the level sets explicitly.

U3.P2. [T] Prove the weak-(1,1) maximal inequality using a covering argument and derive the Lebesgue differentiation theorem. Track constants and the role of Vitali-type selection.

U3.P3. [C] Construct an L^1 function whose maximal function is not integrable, showing weak-(1,1) is generally the right endpoint statement.

Programme U4: Interpolation and singular integrals

U4.P1. [T] Prove Riesz-Thorin interpolation for a finite-dimensional/simple-function model before reading the full complex interpolation proof. Apply it to interpolate operator norms between two endpoints.

U4.P2. [T] Verify the Hilbert transform kernel satisfies cancellation and calculate its Fourier multiplier. Prove L^2 boundedness directly by Plancherel.

U4.P3. [S] Work through a Calderon-Zygmund decomposition of a concrete L^1 function at height lambda and use it to outline weak-(1,1) boundedness of a singular integral.

Programme U5: Littlewood-Paley and multipliers

U5.P1. [F] Construct a dyadic Littlewood-Paley partition and compute frequency pieces of a function with separated Fourier support. Verify square-function orthogonality in L^2.

U5.P2. [T] Prove a basic Mikhlin multiplier theorem in a one-dimensional or radial simplified setting, identifying derivative conditions and kernel decay.

U5.P3. [S] Use Littlewood-Paley decomposition to derive a Sobolev norm equivalence and explain why frequency localization is useful in nonlinear PDE.

Programme U6: Restriction and oscillatory entry

U6.P1. [F] For the parabola or sphere, compute the adjoint restriction/extension operator and determine its scaling under dilation.

U6.P2. [T] Derive the Knapp necessary condition for a model restriction estimate by testing a cap-supported function. Explain the geometry of the associated wave packet.

U6.P3. [R] Prove the Stein-Tomas restriction theorem at proof-architecture level, isolating the Fourier-decay and TT* steps. Then state one later decoupling/Kakeya question that it motivates.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Fourier and Harmonic Analysis: start from “Fourier series”, pass through “Interpolation and singular integrals”, and finish at “Restriction and oscillatory entry”. Use the named results “Bessel” and “van der Corput oscillatory lemma” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Prerequisite for Carleson, decoupling, dispersive PDE, analytic NT and fractal Fourier methods. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Bessel; Parseval/Plancherel for series; Fejer theorem; Fourier inversion; Plancherel. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: CZ theorem; LP norm equivalence; Mikhlin multiplier theorem; Tomas-Stein; stationary phase.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Fourier series” develops into “Restriction and oscillatory entry”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 10. Partial Differential Equations, Distributions and Sobolev Spaces

Purpose: Acquire foundational linear/nonlinear PDE tools needed for mathematical physics, geometric analysis, hydrodynamic limits and harmonic analysis.

Prerequisites: Volumes VI-VII and IX.

Primary and secondary references: Evans, Partial Differential Equations; Brezis, Functional Analysis, Sobolev Spaces and PDEs; Gilbarg-Trudinger, Elliptic PDE; Dafermos, Hyperbolic Conservation Laws

Quarter output: Produce a PDE methods atlas comparing variational, energy, maximum-principle, semigroup, characteristic and compactness methods.

Research bridges: Feeds kinetic limits, Ricci flow, dispersive analysis and mathematical physics.

Six two-week study units

Unit 1 (Weeks 1-2): Distributions and weak derivatives

Topics: test functions; distributions; weak derivatives; mollification

Key definitions: distribution; weak derivative; fundamental solution

Key results to learn:

- mollification/density principles [PROOF/ARCH]
- distributional derivative identities [PROOF]

Reading route: Evans distribution sections; Brezis

Warm-up/source exercise focus (secondary): Compute weak derivatives; approximate rough functions; derive fundamental solutions.

Week 1: Distributions and weak derivatives - definitions and examples. Read: Evans distribution sections; Brezis. Make explicit examples/nonexamples for distribution; weak derivative; fundamental solution. Work through test functions; distributions; weak derivatives; mollification. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Distributions and weak derivatives - theorem and technique pass. Master/audit: mollification/density principles; distributional derivative identities. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Sobolev spaces

Topics: W^{k,p}; traces; extension; compactness; embeddings

Key definitions: Sobolev space; weak gradient; trace

Key results to learn:

- Sobolev embedding [PROOF/ARCH]
- Poincare [PROOF]
- Rellich-Kondrachov [ARCH]
- trace theorem [ARCH]

Reading route: Evans Sobolev chapters; Brezis

Warm-up/source exercise focus (secondary): Prove inequalities on simple domains; test sharpness by scaling; solve compactness problems.

Week 3: Sobolev spaces - definitions and examples. Read: Evans Sobolev chapters; Brezis. Make explicit examples/nonexamples for Sobolev space; weak gradient; trace. Work through W^{k,p}; traces; extension; compactness; embeddings. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Sobolev spaces - theorem and technique pass. Master/audit: Sobolev embedding; Poincare; Rellich-Kondrachov. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Elliptic weak theory

Topics: variational formulation; coercivity; weak solutions; regularity

Key definitions: weak solution; bilinear form; coercivity

Key results to learn:

- Lax-Milgram [PROOF/ARCH]
- energy estimates [PROOF]
- weak maximum principle [PROOF/ARCH]
- elliptic regularity architecture [ARCH/USE]

Reading route: Evans elliptic chapters; Gilbarg-Trudinger

Warm-up/source exercise focus (secondary): Solve elliptic BVPs variationally; derive a priori estimates; compare weak/classical solutions.

Week 5: Elliptic weak theory - definitions and examples. Read: Evans elliptic chapters; Gilbarg-Trudinger. Make explicit examples/nonexamples for weak solution; bilinear form; coercivity. Work through variational formulation; coercivity; weak solutions; regularity. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Elliptic weak theory - theorem and technique pass. Master/audit: Lax-Milgram; energy estimates; weak maximum principle. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Parabolic equations

Topics: heat equation; semigroups; energy; maximum principle

Key definitions: heat kernel; parabolic weak solution

Key results to learn:

- heat-kernel representation [PROOF]
- parabolic maximum principle [PROOF/ARCH]
- basic regularity/smoothing [ARCH]

Reading route: Evans parabolic chapters

Warm-up/source exercise focus (secondary): Solve heat equation in several geometries; derive smoothing/decay estimates.

Week 7: Parabolic equations - definitions and examples. Read: Evans parabolic chapters. Make explicit examples/nonexamples for heat kernel; parabolic weak solution. Work through heat equation; semigroups; energy; maximum principle. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Parabolic equations - theorem and technique pass. Master/audit: heat-kernel representation; parabolic maximum principle; basic regularity/smoothing. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Hyperbolic and conservation laws

Topics: wave equation; finite propagation; characteristics; scalar conservation

Key definitions: domain of dependence; characteristic; entropy solution

Key results to learn:

- wave energy conservation [PROOF]
- finite propagation [ARCH]
- characteristics [PROOF]
- Kruzkov entropy uniqueness statement [USE]

Reading route: Evans wave/first-order; Dafermos

Warm-up/source exercise focus (secondary): Compute characteristic solutions; derive Rankine-Hugoniot; analyse shock formation.

Week 9: Hyperbolic and conservation laws - definitions and examples. Read: Evans wave/first-order; Dafermos. Make explicit examples/nonexamples for domain of dependence; characteristic; entropy solution. Work through wave equation; finite propagation; characteristics; scalar conservation. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Hyperbolic and conservation laws - theorem and technique pass. Master/audit: wave energy conservation; finite propagation; characteristics. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Nonlinear/variational PDE entry

Topics: calculus of variations; fixed point; monotonicity; viscosity intro

Key definitions: weak lower semicontinuity; viscosity solution; monotone operator

Key results to learn:

- direct method [ARCH]
- standard fixed-point existence schemes [ARCH]
- viscosity comparison philosophy [USE]

Reading route: Evans nonlinear/variational selections; Brezis

Warm-up/source exercise focus (secondary): Solve one variational and one nonlinear fixed-point problem; map tools to Navier-Stokes/Boltzmann/Ricci flow.

Week 11: Nonlinear/variational PDE entry - definitions and examples. Read: Evans nonlinear/variational selections; Brezis. Make explicit examples/nonexamples for weak lower semicontinuity; viscosity solution; monotone operator. Work through calculus of variations; fixed point; monotonicity; viscosity intro. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Nonlinear/variational PDE entry - theorem and technique pass. Master/audit: direct method; standard fixed-point existence schemes; viscosity comparison philosophy. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Distributions and weak derivatives

U1.P1. [F] Compute distributional derivatives of the Heaviside function, |x|, and a piecewise smooth function. Verify each by integration against a test function.

U1.P2. [T] Prove that distributional differentiation is continuous in D' and derive convolution identities with a smooth compactly supported kernel.

U1.P3. [S] Construct a fundamental solution for the one-dimensional derivative/Laplacian and check it distributionally. Explain how this viewpoint generalizes to PDE Green functions.

Programme U2: Sobolev spaces

U2.P1. [F] Determine H^1 membership for x^alpha, |x|, and piecewise functions on intervals. Compute weak derivatives explicitly.

U2.P2. [T] Prove Poincare inequality on an interval and then a bounded domain in a standard formulation. Use scaling to predict how the constant transforms.

U2.P3. [S] Derive one Sobolev embedding by scaling and a standard proof. Construct a concentrating sequence showing an endpoint or supercritical embedding fails.

Programme U3: Elliptic weak theory

U3.P1. [T] Set up the weak formulation of -Delta u=f with Dirichlet data and verify boundedness/coercivity of the bilinear form. Apply Lax-Milgram to obtain existence/uniqueness.

U3.P2. [F] Solve the same Poisson problem on an interval both classically and weakly, and verify the solutions agree.

U3.P3. [S] Derive an energy estimate and use it to prove continuous dependence on f. Identify the exact norm in which the estimate closes.

Programme U4: Parabolic equations

U4.P1. [T] Derive the heat equation energy identity and maximum principle for smooth solutions. Then formulate the weak analog used for existence.

U4.P2. [F] Solve the heat equation on a circle/interval by Fourier series and verify smoothing by estimating high-frequency modes.

U4.P3. [S] Derive the heat semigroup contraction in L^2 and explain how semigroup language anticipates Markov processes and functional calculus.

Programme U5: Hyperbolic and conservation laws

U5.P1. [F] Solve the one-dimensional wave equation by d'Alembert and verify finite propagation speed directly.

U5.P2. [T] Derive the conserved energy for the wave equation and use it to prove uniqueness. State exactly which boundary terms vanish under the chosen conditions.

U5.P3. [R] For a scalar conservation law, solve a Riemann problem with one shock and one rarefaction. Check Rankine-Hugoniot and entropy admissibility.

Programme U6: Nonlinear/variational PDE entry

U6.P1. [T] Minimize a coercive convex functional on H_0^1 using the direct method: bounded minimizing sequence, weak compactness, lower semicontinuity, Euler-Lagrange equation.

U6.P2. [C] Give a variational problem where coercivity or lower semicontinuity fails and explain how minimizing sequences can escape or oscillate.

U6.P3. [R] Choose one nonlinear PDE (semilinear elliptic, Navier-Stokes, or nonlinear wave). Write a one-page map of the a priori estimate needed before any fixed-point/compactness scheme can work.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Partial Differential Equations, Distributions and Sobolev Spaces: start from “Distributions and weak derivatives”, pass through “Parabolic equations”, and finish at “Nonlinear/variational PDE entry”. Use the named results “mollification/density principles” and “viscosity comparison philosophy” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds kinetic limits, Ricci flow, dispersive analysis and mathematical physics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: mollification/density principles; distributional derivative identities; Sobolev embedding; Poincare; Lax-Milgram. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: finite propagation; Kruzkov entropy uniqueness statement; direct method; standard fixed-point existence schemes; viscosity comparison philosophy.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Distributions and weak derivatives” develops into “Nonlinear/variational PDE entry”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 11. Topology and Algebraic Topology

Purpose: Develop point-set topology and the basic homotopy/homology/cohomology toolkit supporting manifolds, geometric topology, bundles and algebraic geometry.

Prerequisites: Volumes I-III.

Primary and secondary references: Munkres, Topology; Hatcher, Algebraic Topology; May, A Concise Course in Algebraic Topology

Quarter output: Compute and compare pi1, homology and cohomology ring for at least six spaces, including surfaces and projective/lens examples.

Research bridges: Feeds manifolds, geometric topology, characteristic classes, Riemann surfaces and cohomological analogies.

Six two-week study units

Unit 1 (Weeks 1-2): Topological spaces and continuity

Topics: bases/subbases; continuity; products/quotients; universal constructions

Key definitions: topology; basis; quotient topology; homeomorphism

Key results to learn:

- universal properties of product/quotient topologies [PROOF]
- pasting lemma [PROOF]

Reading route: Munkres basic topology chapters

Warm-up/source exercise focus (secondary): Build quotient spaces; prove continuity via bases; compare standard/nonstandard topologies.

Week 1: Topological spaces and continuity - definitions and examples. Read: Munkres basic topology chapters. Make explicit examples/nonexamples for topology; basis; quotient topology; homeomorphism. Work through bases/subbases; continuity; products/quotients; universal constructions. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Topological spaces and continuity - theorem and technique pass. Master/audit: universal properties of product/quotient topologies; pasting lemma. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Compactness/connectedness/separation

Topics: compactness; connectedness; Hausdorff/normal spaces

Key definitions: compact; connected; path connected; normal; locally compact

Key results to learn:

- finite-product Tychonoff [PROOF] and general statement [USE]
- Urysohn lemma [ARCH]
- Tietze extension [ARCH]

Reading route: Munkres compactness/separation chapters

Warm-up/source exercise focus (secondary): Construct examples; analyse separation axioms; solve connectedness by continuous images.

Week 3: Compactness/connectedness/separation - definitions and examples. Read: Munkres compactness/separation chapters. Make explicit examples/nonexamples for compact; connected; path connected; normal; locally compact. Work through compactness; connectedness; Hausdorff/normal spaces. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Compactness/connectedness/separation - theorem and technique pass. Master/audit: finite-product Tychonoff [PROOF] and general statement; Urysohn lemma; Tietze extension. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Fundamental group and coverings

Topics: homotopy; pi1; covering spaces; lifting

Key definitions: fundamental group; covering; deck transformation

Key results to learn:

- Seifert-van Kampen [PROOF/ARCH]
- lifting criteria [ARCH]
- covering classification under standard hypotheses [ARCH]

Reading route: Hatcher Ch.1; Munkres supplements

Warm-up/source exercise focus (secondary): Compute pi1 of graphs, surfaces, wedges and quotients; classify covers.

Week 5: Fundamental group and coverings - definitions and examples. Read: Hatcher Ch.1; Munkres supplements. Make explicit examples/nonexamples for fundamental group; covering; deck transformation. Work through homotopy; pi1; covering spaces; lifting. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Fundamental group and coverings - theorem and technique pass. Master/audit: Seifert-van Kampen; lifting criteria; covering classification under standard hypotheses. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Homology

Topics: singular/simplicial homology; chain complexes; exact sequences

Key definitions: cycle; boundary; reduced homology

Key results to learn:

- homotopy invariance [ARCH]
- excision [ARCH]
- LES of pair [ARCH]
- Mayer-Vietoris [PROOF/ARCH]

Reading route: Hatcher homology chapters

Warm-up/source exercise focus (secondary): Compute homology of CW complexes/surfaces/projective spaces; use Mayer-Vietoris repeatedly.

Week 7: Homology - definitions and examples. Read: Hatcher homology chapters. Make explicit examples/nonexamples for cycle; boundary; reduced homology. Work through singular/simplicial homology; chain complexes; exact sequences. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Homology - theorem and technique pass. Master/audit: homotopy invariance; excision; LES of pair. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Cohomology and products

Topics: cochains; cup product; universal coefficient; duality entry

Key definitions: cohomology; cup product; cohomology ring

Key results to learn:

- universal coefficient [ARCH]
- Kunneth core cases [ARCH]
- Poincare duality statement [USE]

Reading route: Hatcher cohomology chapters

Warm-up/source exercise focus (secondary): Compute cohomology rings; use cup products to distinguish spaces.

Week 9: Cohomology and products - definitions and examples. Read: Hatcher cohomology chapters. Make explicit examples/nonexamples for cohomology; cup product; cohomology ring. Work through cochains; cup product; universal coefficient; duality entry. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Cohomology and products - theorem and technique pass. Master/audit: universal coefficient; Kunneth core cases; Poincare duality statement. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): CW complexes and homotopy entry

Topics: cellular approximation; higher homotopy; obstruction viewpoint

Key definitions: CW complex; attaching map; homotopy group

Key results to learn:

- cellular homology [ARCH]
- Whitehead [C/B]
- Hurewicz [C/B]

Reading route: Hatcher later selections; May

Warm-up/source exercise focus (secondary): Build CW structures; compute cellular homology; trace first higher-homotopy examples.

Week 11: CW complexes and homotopy entry - definitions and examples. Read: Hatcher later selections; May. Make explicit examples/nonexamples for CW complex; attaching map; homotopy group. Work through cellular approximation; higher homotopy; obstruction viewpoint. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: CW complexes and homotopy entry - theorem and technique pass. Master/audit: cellular homology; Whitehead [C/B]; Hurewicz [C/B]. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Topological spaces and continuity

U1.P1. [F] For several topologies on the same set, identify open/closed sets, bases, closures, interiors, and convergence. Include standard, discrete, indiscrete, and product examples.

U1.P2. [T] Prove that continuity via inverse images is equivalent to the epsilon-delta notion in metric spaces and derive the universal property of product topology for maps into products.

U1.P3. [C] Construct examples showing sequence convergence does not characterize closure in arbitrary topological spaces, while it does in first-countable spaces.

Programme U2: Compactness/connectedness/separation

U2.P1. [T] Prove compactness of products in the finite case and understand the proof architecture of Tychonoff in the general case. Work explicit compactness arguments on [0,1]^N using a metric.

U2.P2. [F] Determine connected components/path components of the topologist's sine curve, comb space, and simple quotient spaces.

U2.P3. [C] Give examples separating T0,T1,Hausdorff,regular,normal; identify which separation assumptions are needed for uniqueness of limits and Urysohn-type results.

Programme U3: Fundamental group and coverings

U3.P1. [F] Compute pi_1 of S^1, torus, wedge of circles, and a simple punctured plane using van Kampen. Make generators and relations explicit.

U3.P2. [T] Classify connected covering spaces of S^1 and a bouquet of circles via subgroups of the fundamental group in concrete cases.

U3.P3. [S] Use covering spaces to prove a fixed-point/nonexistence statement, such as no retraction D^2->S^1 or Brouwer in dimension 2 via a standard route.

Programme U4: Homology

U4.P1. [F] Compute simplicial/cellular homology of S^n, torus, projective plane, and one lens-space/simple CW example. Write all boundary matrices.

U4.P2. [T] Prove homotopy invariance and exactness via the long exact sequence of a pair at proof-architecture level; then use exactness in a calculation.

U4.P3. [S] Derive Euler characteristic from chain complexes and verify multiplicative/additive behavior in selected CW constructions.

Programme U5: Cohomology and products

U5.P1. [F] Compute cohomology groups and cup products of S^n, T^2, and RP^2 with suitable coefficients. Distinguish spaces with same homology but different ring structure where possible.

U5.P2. [T] Prove universal coefficient theorem in a simplified PID setting or use it carefully to compute torsion examples.

U5.P3. [S] Use Poincare duality to recover homology/cohomology information for a closed oriented surface and explain how orientation enters.

Programme U6: CW complexes and homotopy entry

U6.P1. [F] Build CW structures for projective spaces, surfaces, and a mapping torus. Compute cellular chain complexes where feasible.

U6.P2. [T] Work through the cellular approximation theorem or Whitehead theorem at proof-architecture level and apply it to a concrete homotopy equivalence question.

U6.P3. [R] Construct the Hopf map S^3->S^2 and compute/interpret one invariant (linking number or cohomological obstruction) that shows it is nontrivial.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Topology and Algebraic Topology: start from “Topological spaces and continuity”, pass through “Homology”, and finish at “CW complexes and homotopy entry”. Use the named results “universal properties of product/quotient topologies” and “Hurewicz [C/B]” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds manifolds, geometric topology, characteristic classes, Riemann surfaces and cohomological analogies. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: universal properties of product/quotient topologies; pasting lemma; finite-product Tychonoff; Seifert-van Kampen; Mayer-Vietoris. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: LES of pair; universal coefficient; Kunneth core cases; Poincare duality statement; cellular homology.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Topological spaces and continuity” develops into “CW complexes and homotopy entry”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 12. Smooth Manifolds and Differential Geometry

Purpose: Build smooth-manifold, differential-form, bundle and Morse-theoretic foundations.

Prerequisites: Volumes II, VI and XI.

Primary and secondary references: Lee, Introduction to Smooth Manifolds; Tu, An Introduction to Manifolds; Bott-Tu, Differential Forms in Algebraic Topology; Milnor, Morse Theory

Quarter output: Build a full manifold dossier for a torus or genus-2 surface: atlas, tangent bundle, forms/cohomology, Morse function and bundle data.

Research bridges: Feeds Riemannian geometry, topology, Lie groups and dynamical systems.

Six two-week study units

Unit 1 (Weeks 1-2): Smooth manifolds and tangent geometry

Topics: charts; submanifolds; tangent/cotangent; immersions/submersions

Key definitions: smooth manifold; tangent vector; differential; immersion; submersion

Key results to learn:

- inverse/implicit-function applications [PROOF]
- regular-value theorem [PROOF/ARCH]
- rank theorem [ARCH]

Reading route: Lee opening chapters; Tu

Warm-up/source exercise focus (secondary): Construct atlases; compute tangent maps; prove level sets are submanifolds.

Week 1: Smooth manifolds and tangent geometry - definitions and examples. Read: Lee opening chapters; Tu. Make explicit examples/nonexamples for smooth manifold; tangent vector; differential; immersion; submersion. Work through charts; submanifolds; tangent/cotangent; immersions/submersions. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Smooth manifolds and tangent geometry - theorem and technique pass. Master/audit: inverse/implicit-function applications; regular-value theorem; rank theorem. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Vector fields, flows and brackets

Topics: vector fields; flows; distributions; Frobenius

Key definitions: flow; complete vector field; Lie bracket; distribution

Key results to learn:

- local-flow existence/uniqueness [ARCH]
- Frobenius theorem [ARCH/USE]

Reading route: Lee vector-field chapters

Warm-up/source exercise focus (secondary): Compute flows/brackets; test integrability; connect brackets to commutators.

Week 3: Vector fields, flows and brackets - definitions and examples. Read: Lee vector-field chapters. Make explicit examples/nonexamples for flow; complete vector field; Lie bracket; distribution. Work through vector fields; flows; distributions; Frobenius. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Vector fields, flows and brackets - theorem and technique pass. Master/audit: local-flow existence/uniqueness; Frobenius theorem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Differential forms and integration

Topics: forms; exterior derivative; pullback; orientation

Key definitions: differential form; exterior derivative; orientation

Key results to learn:

- Stokes theorem [PROOF/ARCH]
- Poincare lemma [PROOF/ARCH]

Reading route: Lee forms/Stokes; Bott-Tu

Warm-up/source exercise focus (secondary): Compute de Rham complexes; use Stokes to derive classical integral theorems.

Week 5: Differential forms and integration - definitions and examples. Read: Lee forms/Stokes; Bott-Tu. Make explicit examples/nonexamples for differential form; exterior derivative; orientation. Work through forms; exterior derivative; pullback; orientation. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Differential forms and integration - theorem and technique pass. Master/audit: Stokes theorem; Poincare lemma. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): de Rham cohomology

Topics: closed/exact forms; Mayer-Vietoris; de Rham theorem

Key definitions: de Rham cohomology; partition of unity

Key results to learn:

- de Rham theorem [ARCH/USE]
- homotopy invariance [PROOF/ARCH]

Reading route: Bott-Tu early chapters; Lee

Warm-up/source exercise focus (secondary): Compute de Rham cohomology of spheres/tori and compare with singular cohomology.

Week 7: de Rham cohomology - definitions and examples. Read: Bott-Tu early chapters; Lee. Make explicit examples/nonexamples for de Rham cohomology; partition of unity. Work through closed/exact forms; Mayer-Vietoris; de Rham theorem. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: de Rham cohomology - theorem and technique pass. Master/audit: de Rham theorem; homotopy invariance. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Bundles and characteristic classes

Topics: vector/principal bundles; connections preview; Chern/Stiefel-Whitney classes

Key definitions: vector bundle; section; transition function; characteristic class

Key results to learn:

- Whitney embedding [C/B]
- tubular neighborhood [ARCH]
- clutching classifications in examples [ARCH]

Reading route: Lee bundle chapters; Bott-Tu

Warm-up/source exercise focus (secondary): Construct tangent/normal bundles; compute elementary characteristic classes.

Week 9: Bundles and characteristic classes - definitions and examples. Read: Lee bundle chapters; Bott-Tu. Make explicit examples/nonexamples for vector bundle; section; transition function; characteristic class. Work through vector/principal bundles; connections preview; Chern/Stiefel-Whitney classes. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Bundles and characteristic classes - theorem and technique pass. Master/audit: Whitney embedding [C/B]; tubular neighborhood; clutching classifications in examples. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Morse theory and transversality

Topics: critical points; Morse functions; handles; Sard/transversality

Key definitions: Morse index; regular value; transverse intersection

Key results to learn:

- Sard [ARCH]
- transversality core form [C/B]
- Morse lemma [PROOF/ARCH]
- Morse inequalities [ARCH]

Reading route: Milnor Morse Theory; Lee transversality

Warm-up/source exercise focus (secondary): Compute Morse functions on surfaces/tori; recover homology information; practice transverse perturbations.

Week 11: Morse theory and transversality - definitions and examples. Read: Milnor Morse Theory; Lee transversality. Make explicit examples/nonexamples for Morse index; regular value; transverse intersection. Work through critical points; Morse functions; handles; Sard/transversality. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Morse theory and transversality - theorem and technique pass. Master/audit: Sard; transversality core form [C/B]; Morse lemma. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Smooth manifolds and tangent geometry

U1.P1. [F] Write atlases and transition maps for S^n, RP^n, the torus, and a matrix Lie group. Compute tangent spaces at selected points using charts and derivations.

U1.P2. [T] Prove the regular value theorem in the finite-dimensional setting and use it to show spheres, orthogonal groups, and level sets are submanifolds.

U1.P3. [S] For a smooth map between manifolds, compute its differential in coordinates and intrinsically. Verify chain rule and coordinate independence on a nontrivial example.

Programme U2: Vector fields, flows and brackets

U2.P1. [F] Compute Lie brackets of vector fields on R^2 and left-invariant vector fields on a matrix group. Find examples of commuting/noncommuting flows.

U2.P2. [T] Prove existence/uniqueness of local flows from ODE theory and derive the relation between bracket zero and commuting flows in a standard local formulation.

U2.P3. [S] Verify Frobenius integrability for an explicit distribution, and construct a nonintegrable contact-type distribution to see the obstruction.

Programme U3: Differential forms and integration

U3.P1. [F] Compute pullbacks, wedge products, exterior derivatives, and integrals of differential forms on spheres/torus/coordinate domains.

U3.P2. [T] Prove Stokes' theorem for a rectangle/simplex and map the local proof ingredients to the manifold theorem. Recover Green and divergence theorems as cases.

U3.P3. [C] Produce closed nonexact forms on S^1 or T^n and explain why local exactness does not imply global exactness.

Programme U4: de Rham cohomology

U4.P1. [F] Compute de Rham cohomology of R^n, S^1, S^n, and T^2 using Poincare lemma/Mayer-Vietoris.

U4.P2. [T] Work through the Mayer-Vietoris sequence for forms and use it to calculate H^1(S^1) or H^*(S^n).

U4.P3. [S] Compare singular and de Rham cohomology on one manifold by identifying explicit classes/integrals, preparing the de Rham theorem viewpoint.

Programme U5: Bundles and characteristic classes

U5.P1. [F] Construct tangent/cotangent bundles and transition functions for S^1,S^2, and the Mobius bundle. Decide triviality in simple cases.

U5.P2. [T] Compute the first Chern class of a line bundle over S^2 at an introductory clutching-function level and relate it to winding number.

U5.P3. [R] State the Chern-Weil construction precisely enough to see how curvature produces characteristic classes. Work one U(1) connection example.

Programme U6: Morse theory and transversality

U6.P1. [F] Find critical points and Morse indices of height functions on S^2, torus, and RP^2 where convenient. Relate sublevel-set changes to handles.

U6.P2. [T] Prove a basic transversality consequence: a generic perturbation makes intersections transverse. Use dimension counting to predict intersection dimensions.

U6.P3. [S] Use Morse inequalities on a compact surface to constrain the number of critical points. Compare with its homology and Euler characteristic.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Smooth Manifolds and Differential Geometry: start from “Smooth manifolds and tangent geometry”, pass through “de Rham cohomology”, and finish at “Morse theory and transversality”. Use the named results “inverse/implicit-function applications” and “Morse inequalities” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds Riemannian geometry, topology, Lie groups and dynamical systems. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: inverse/implicit-function applications; regular-value theorem; Stokes theorem; Poincare lemma; homotopy invariance. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: de Rham theorem; tubular neighborhood; clutching classifications in examples; Sard; Morse inequalities.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Smooth manifolds and tangent geometry” develops into “Morse theory and transversality”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 13. Riemannian Geometry, Geometric Topology and Ricci Flow

Purpose: Develop curvature/comparison geometry, topology of manifolds and the architecture leading to Poincare/Geometrization and Hauptvermutung-related topology.

Prerequisites: Volumes VII, X-XII.

Primary and secondary references: do Carmo, Riemannian Geometry; Petersen, Riemannian Geometry; Chow et al., Ricci Flow; Morgan-Tian, Ricci Flow and the Poincare Conjecture; Milnor, Lectures on the h-Cobordism Theorem

Quarter output: Prepare a 20-page guided roadmap to the Poincare proof and a historical theorem chart for Hauptvermutung.

Research bridges: Connects topology, PDE and geometry; supports the Poincare/Hauptvermutung target track.

Six two-week study units

Unit 1 (Weeks 1-2): Metrics, connections and geodesics

Topics: Riemannian metrics; Levi-Civita connection; geodesics; exponential map

Key definitions: Riemannian metric; connection; geodesic; completeness

Key results to learn:

- fundamental theorem of Riemannian geometry [PROOF/ARCH]
- Hopf-Rinow [PROOF/ARCH]

Reading route: do Carmo/Petersen opening chapters

Warm-up/source exercise focus (secondary): Compute connections/geodesics on sphere, hyperbolic plane and Lie-group examples.

Week 1: Metrics, connections and geodesics - definitions and examples. Read: do Carmo/Petersen opening chapters. Make explicit examples/nonexamples for Riemannian metric; connection; geodesic; completeness. Work through Riemannian metrics; Levi-Civita connection; geodesics; exponential map. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Metrics, connections and geodesics - theorem and technique pass. Master/audit: fundamental theorem of Riemannian geometry; Hopf-Rinow. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Curvature and Jacobi fields

Topics: Riemann/Ricci/scalar curvature; sectional curvature; Jacobi fields

Key definitions: curvature tensor; sectional curvature; conjugate point

Key results to learn:

- Gauss equation [ARCH]
- Jacobi equation [PROOF/ARCH]
- Bonnet-Myers [ARCH]
- Cartan-Hadamard [ARCH]

Reading route: Petersen curvature/comparison chapters

Warm-up/source exercise focus (secondary): Compute curvature; analyse geodesic deviation; apply comparison to topology.

Week 3: Curvature and Jacobi fields - definitions and examples. Read: Petersen curvature/comparison chapters. Make explicit examples/nonexamples for curvature tensor; sectional curvature; conjugate point. Work through Riemann/Ricci/scalar curvature; sectional curvature; Jacobi fields. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Curvature and Jacobi fields - theorem and technique pass. Master/audit: Gauss equation; Jacobi equation; Bonnet-Myers. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Comparison and volume

Topics: Rauch; Toponogov; Bishop-Gromov; injectivity radius

Key definitions: comparison triangle; volume growth; injectivity radius

Key results to learn:

- Rauch comparison [C/B]
- Toponogov [USE]
- Bishop-Gromov [ARCH/USE]

Reading route: Petersen comparison chapters

Warm-up/source exercise focus (secondary): Work model-space comparisons; derive volume/topological consequences.

Week 5: Comparison and volume - definitions and examples. Read: Petersen comparison chapters. Make explicit examples/nonexamples for comparison triangle; volume growth; injectivity radius. Work through Rauch; Toponogov; Bishop-Gromov; injectivity radius. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Comparison and volume - theorem and technique pass. Master/audit: Rauch comparison [C/B]; Toponogov; Bishop-Gromov. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Hodge theory

Topics: Laplacian on forms; harmonic representatives; elliptic complexes

Key definitions: Hodge star; Laplace-Beltrami; harmonic form

Key results to learn:

- Hodge decomposition on compact manifolds [ARCH/USE]
- Hodge-de Rham identification [ARCH]

Reading route: Petersen/Bott-Tu supplements

Warm-up/source exercise focus (secondary): Compute harmonic forms on tori/surfaces; relate Betti numbers to geometry.

Week 7: Hodge theory - definitions and examples. Read: Petersen/Bott-Tu supplements. Make explicit examples/nonexamples for Hodge star; Laplace-Beltrami; harmonic form. Work through Laplacian on forms; harmonic representatives; elliptic complexes. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Hodge theory - theorem and technique pass. Master/audit: Hodge decomposition on compact manifolds; Hodge-de Rham identification. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Cobordism and Hauptvermutung context

Topics: handles; h-cobordism; PL/topological/smooth categories; triangulation

Key definitions: cobordism; h-cobordism; simple homotopy; triangulation

Key results to learn:

- h-cobordism theorem in high dimension [USE]
- s-cobordism statement [USE]
- failure/qualified forms of Hauptvermutung [USE]

Reading route: Milnor; Kirby-Siebenmann/Ranicki expositions

Warm-up/source exercise focus (secondary): Work low-dimensional handles; write a category-by-category theorem map for Hauptvermutung.

Week 9: Cobordism and Hauptvermutung context - definitions and examples. Read: Milnor; Kirby-Siebenmann/Ranicki expositions. Make explicit examples/nonexamples for cobordism; h-cobordism; simple homotopy; triangulation. Work through handles; h-cobordism; PL/topological/smooth categories; triangulation. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Cobordism and Hauptvermutung context - theorem and technique pass. Master/audit: h-cobordism theorem in high dimension; s-cobordism statement; failure/qualified forms of Hauptvermutung. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Ricci flow and Poincare/Geometrization

Topics: Ricci flow; singularities; surgery; entropy; reduced volume

Key definitions: Ricci flow; ancient solution; neck; surgery; Perelman entropy

Key results to learn:

- short-time existence architecture [USE]
- Hamilton maximum-principle ideas [USE]
- Perelman no-local-collapsing [USE]
- Poincare/Geometrization via Ricci flow [USE]

Reading route: Chow et al.; Morgan-Tian; Perelman with expository guides

Warm-up/source exercise focus (secondary): Compute homogeneous Ricci flows; build a dependency map of Perelman’s proof; present one monotonicity mechanism.

Week 11: Ricci flow and Poincare/Geometrization - definitions and examples. Read: Chow et al.; Morgan-Tian; Perelman with expository guides. Make explicit examples/nonexamples for Ricci flow; ancient solution; neck; surgery; Perelman entropy. Work through Ricci flow; singularities; surgery; entropy; reduced volume. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Ricci flow and Poincare/Geometrization - theorem and technique pass. Master/audit: short-time existence architecture; Hamilton maximum-principle ideas; Perelman no-local-collapsing. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Metrics, connections and geodesics

U1.P1. [F] Compute Levi-Civita connections and geodesic equations for Euclidean space, the sphere in standard coordinates, and a conformally flat metric in dimension 2. Check metric compatibility and zero torsion directly.

U1.P2. [T] Derive the first variation of energy/length and show geodesics satisfy the Euler-Lagrange equation. Explain parametrization issues and why constant-speed geodesics are natural.

U1.P3. [S] Use the exponential map on S^2 to identify conjugate/cut phenomena explicitly and compare local normal coordinates with global failure of injectivity.

Programme U2: Curvature and Jacobi fields

U2.P1. [F] Compute sectional/Gaussian curvature for the sphere, hyperbolic plane, and a product metric. Verify sign conventions against Jacobi-field behavior.

U2.P2. [T] Derive the Jacobi equation from variation through geodesics and solve it in constant curvature. Relate zeros of Jacobi fields to conjugate points.

U2.P3. [S] Prove a selected version of Bonnet-Myers or Cartan-Hadamard at proof-architecture level, identifying exactly how curvature sign enters global geometry.

Programme U3: Comparison and volume

U3.P1. [T] Prove Bishop-Gromov monotonicity in a model/selected formulation or reconstruct its comparison-Jacobian proof architecture. Verify equality on space forms.

U3.P2. [F] Use comparison theorems to estimate diameters/volumes in explicit constant-curvature examples and test sharpness.

U3.P3. [S] Compare Toponogov, Rauch, and Bishop-Gromov: for three geometric questions, decide which comparison theorem is the appropriate tool and justify the choice.

Programme U4: Hodge theory

U4.P1. [F] Compute the Hodge star and Laplacian on forms on a flat torus or circle. Identify harmonic representatives of de Rham classes.

U4.P2. [T] Prove Hodge decomposition at proof-architecture level from elliptic self-adjoint spectral theory. Track the role of compactness and boundary assumptions.

U4.P3. [S] Use Hodge theory to recover Betti numbers of a closed oriented surface and derive one consequence involving harmonic 1-forms.

Programme U5: Cobordism and Hauptvermutung context

U5.P1. [F] Compare simplicial, PL, smooth, and topological manifolds on low-dimensional examples; triangulate basic surfaces and compute links of simplices.

U5.P2. [T] Reconstruct the exact statement of the Hauptvermutung and build a chronology of its positive low-dimensional cases and counterexamples. Distinguish 'triangulable' from 'unique PL structure'.

U5.P3. [R] Choose one obstruction/counterexample theorem (Milnor or later developments) and write a precise theorem audit: hypotheses, category, conclusion, and what it disproves.

Programme U6: Ricci flow and Poincare/Geometrization

U6.P1. [F] Derive Ricci flow on constant-curvature metrics and solve the resulting ODE for the scale factor. Interpret finite-time shrinking/expansion.

U6.P2. [T] Derive evolution of volume and scalar curvature in a selected simple setting, and compute one monotone quantity in a model case.

U6.P3. [R] Build a proof-dependency map for Perelman's route to Poincare/Geometrization: short-time existence, singularities, entropy/noncollapsing, surgery or modern replacement, long-time/topological conclusion. For each box identify the reference chapter to read.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Riemannian Geometry, Geometric Topology and Ricci Flow: start from “Metrics, connections and geodesics”, pass through “Hodge theory”, and finish at “Ricci flow and Poincare/Geometrization”. Use the named results “fundamental theorem of Riemannian geometry” and “Poincare/Geometrization via Ricci flow” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Connects topology, PDE and geometry; supports the Poincare/Hauptvermutung target track. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: fundamental theorem of Riemannian geometry; Hopf-Rinow; Jacobi equation. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: failure/qualified forms of Hauptvermutung; short-time existence architecture; Hamilton maximum-principle ideas; Perelman no-local-collapsing; Poincare/Geometrization via Ricci flow.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Metrics, connections and geodesics” develops into “Ricci flow and Poincare/Geometrization”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 14. Probability, Martingales and Stochastic Processes

Purpose: Develop measure-theoretic probability through martingales, Brownian motion and Markov processes.

Prerequisites: Volume VI; Volume VII helpful.

Primary and secondary references: Durrett, Probability: Theory and Examples; Williams, Probability with Martingales; Kallenberg, Foundations of Modern Probability; Lawler, Introduction to Stochastic Processes

Quarter output: Complete a portfolio containing one stopping problem, one CLT/invariance problem, one Brownian hitting problem and one Markov mixing analysis.

Research bridges: Feeds SDE, percolation/SLE, random matrices, particle systems and ergodic theory.

Six two-week study units

Unit 1 (Weeks 1-2): Probability and independence

Topics: probability measures; random variables; independence; product spaces

Key definitions: distribution; expectation; independence; generated sigma-field

Key results to learn:

- Borel-Cantelli lemmas [PROOF]
- Kolmogorov 0-1 law [ARCH]

Reading route: Durrett early chapters; Williams

Warm-up/source exercise focus (secondary): Construct independent sequences; use Borel-Cantelli for almost-sure limsup events.

Week 1: Probability and independence - definitions and examples. Read: Durrett early chapters; Williams. Make explicit examples/nonexamples for distribution; expectation; independence; generated sigma-field. Work through probability measures; random variables; independence; product spaces. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Probability and independence - theorem and technique pass. Master/audit: Borel-Cantelli lemmas; Kolmogorov 0-1 law. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Convergence and laws of large numbers

Topics: a.s./probability/Lp/distribution convergence; uniform integrability

Key definitions: tightness; convergence in distribution; characteristic function

Key results to learn:

- weak/strong LLN [PROOF/ARCH]
- Slutsky and continuous-mapping principles [PROOF]

Reading route: Durrett convergence chapters

Warm-up/source exercise focus (secondary): Build counterexamples separating convergence modes; prove LLNs in several settings.

Week 3: Convergence and laws of large numbers - definitions and examples. Read: Durrett convergence chapters. Make explicit examples/nonexamples for tightness; convergence in distribution; characteristic function. Work through a.s./probability/Lp/distribution convergence; uniform integrability. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Convergence and laws of large numbers - theorem and technique pass. Master/audit: weak/strong LLN; Slutsky and continuous-mapping principles. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Central limit theory

Topics: characteristic functions; triangular arrays; Lindeberg conditions

Key definitions: characteristic function; variance; Lindeberg condition

Key results to learn:

- Levy continuity theorem [ARCH]
- classical CLT [PROOF/ARCH]
- Lindeberg-Feller CLT [C/B]

Reading route: Durrett CLT chapters

Warm-up/source exercise focus (secondary): Prove iid CLT via characteristic functions; check triangular-array conditions.

Week 5: Central limit theory - definitions and examples. Read: Durrett CLT chapters. Make explicit examples/nonexamples for characteristic function; variance; Lindeberg condition. Work through characteristic functions; triangular arrays; Lindeberg conditions. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Central limit theory - theorem and technique pass. Master/audit: Levy continuity theorem; classical CLT; Lindeberg-Feller CLT [C/B]. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Conditional expectation and martingales

Topics: filtrations; conditional expectation; stopping times

Key definitions: filtration; martingale; submartingale; stopping time

Key results to learn:

- conditional expectation properties [PROOF/ARCH]
- optional stopping under standard hypotheses [PROOF/ARCH]
- Doob inequalities [ARCH]

Reading route: Williams martingale chapters; Durrett

Warm-up/source exercise focus (secondary): Solve random-walk/gambler martingale problems; derive hitting probabilities.

Week 7: Conditional expectation and martingales - definitions and examples. Read: Williams martingale chapters; Durrett. Make explicit examples/nonexamples for filtration; martingale; submartingale; stopping time. Work through filtrations; conditional expectation; stopping times. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Conditional expectation and martingales - theorem and technique pass. Master/audit: conditional expectation properties; optional stopping under standard hypotheses; Doob inequalities. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Martingale convergence and concentration

Topics: uniform integrability; martingale convergence; Azuma-type bounds

Key definitions: uniformly integrable martingale; discrete quadratic variation

Key results to learn:

- martingale convergence theorems [ARCH]
- Azuma-Hoeffding [PROOF/ARCH]

Reading route: Durrett/Williams

Warm-up/source exercise focus (secondary): Prove convergence in model processes; derive concentration estimates.

Week 9: Martingale convergence and concentration - definitions and examples. Read: Durrett/Williams. Make explicit examples/nonexamples for uniformly integrable martingale; discrete quadratic variation. Work through uniform integrability; martingale convergence; Azuma-type bounds. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Martingale convergence and concentration - theorem and technique pass. Master/audit: martingale convergence theorems; Azuma-Hoeffding. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Brownian motion and Markov chains

Topics: Brownian properties; recurrence; Markov chains; invariant laws

Key definitions: Brownian motion; transition kernel; invariant measure; recurrence

Key results to learn:

- Donsker invariance principle statement [USE]
- reflection principle [PROOF/ARCH]
- finite-state Markov ergodic theorem [PROOF/ARCH]

Reading route: Durrett; Lawler

Warm-up/source exercise focus (secondary): Compute Brownian hitting laws; classify random-walk recurrence; analyse finite-chain mixing/stationarity.

Week 11: Brownian motion and Markov chains - definitions and examples. Read: Durrett; Lawler. Make explicit examples/nonexamples for Brownian motion; transition kernel; invariant measure; recurrence. Work through Brownian properties; recurrence; Markov chains; invariant laws. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Brownian motion and Markov chains - theorem and technique pass. Master/audit: Donsker invariance principle statement; reflection principle; finite-state Markov ergodic theorem. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Probability and independence

U1.P1. [F] Construct examples of independent, pairwise-independent-but-not-jointly-independent, and conditionally independent random variables. Verify each from sigma-algebras/probabilities, not intuition.

U1.P2. [T] Prove Borel-Cantelli I and II and apply them to independent events with probabilities 1/n^alpha. Determine the almost-sure limsup behavior for all alpha.

U1.P3. [C] Build examples showing uncorrelated does not imply independent, and independence can be destroyed by conditioning. Explain the structural reason in each example.

Programme U2: Convergence and laws of large numbers

U2.P1. [F] Construct sequences separating almost sure, in probability, in L^1, and in distribution convergence. Reuse one sequence in more than one comparison where possible.

U2.P2. [T] Prove the weak and strong laws for a standard iid setting by two routes available in your text (Chebyshev/subsequence versus martingale/truncation where appropriate).

U2.P3. [S] Analyze sums of independent but non-identically distributed variables and verify a Kolmogorov-type criterion in a concrete example.

Programme U3: Central limit theory

U3.P1. [T] Prove the classical CLT via characteristic functions for iid variables with finite variance. Carry out the Taylor expansion and limiting argument explicitly.

U3.P2. [F] Compute characteristic functions for Bernoulli, Gaussian, Poisson, and sums; use Levy continuity theorem in one convergence example.

U3.P3. [C] Construct/heavily analyze a heavy-tailed example where variance is infinite and the classical sqrt(n) CLT fails; identify the stable-law direction without overclaiming.

Programme U4: Conditional expectation and martingales

U4.P1. [F] Compute conditional expectations in finite partitions, discrete filtrations, and one continuous random-variable example. Verify the defining integral property.

U4.P2. [T] Show simple random walk and compensated counting processes are martingales in appropriate filtrations. Derive Doob decomposition in a finite discrete-time example.

U4.P3. [S] Solve gambler's ruin by optional stopping, then list all hypotheses needed and modify the problem to show what can go wrong without boundedness/uniform integrability conditions.

Programme U5: Martingale convergence and concentration

U5.P1. [T] Prove a selected martingale convergence theorem and apply it to a bounded martingale arising from conditional expectations of an L^1 variable.

U5.P2. [F] Derive Azuma-Hoeffding or Hoeffding inequality from exponential supermartingales and apply it to concentration of a bounded-difference sum.

U5.P3. [S] Compare martingale concentration, Chernoff bounds, and variance-based Chebyshev on the same binomial problem; quantify the gain in the tail exponent.

Programme U6: Brownian motion and Markov chains

U6.P1. [F] Construct Brownian finite-dimensional distributions and verify covariance min(s,t); compute hitting probabilities and expected exit times in one dimension using martingales.

U6.P2. [T] Derive reflection principle and use it to find the distribution of max_{s<=t} B_s. Explain why path continuity matters.

U6.P3. [S] For a finite reversible Markov chain, compute stationary distribution, eigenvalues, spectral gap, and mixing bound. Connect this explicitly to expander/spectral-gap material later.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Probability, Martingales and Stochastic Processes: start from “Probability and independence”, pass through “Conditional expectation and martingales”, and finish at “Brownian motion and Markov chains”. Use the named results “Borel-Cantelli lemmas” and “finite-state Markov ergodic theorem” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds SDE, percolation/SLE, random matrices, particle systems and ergodic theory. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Borel-Cantelli lemmas; weak/strong LLN; Slutsky and continuous-mapping principles; classical CLT; conditional expectation properties. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Kolmogorov 0-1 law; Levy continuity theorem; Doob inequalities; martingale convergence theorems; Donsker invariance principle statement.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Probability and independence” develops into “Brownian motion and Markov chains”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 15. SDEs, Random Structures and Random Matrices

Purpose: Combine stochastic calculus with random graphs/matrices and concentration tools used in modern probability.

Prerequisites: Volumes VII and XIV; Volume IX useful.

Primary and secondary references: Karatzas-Shreve, Brownian Motion and Stochastic Calculus; Oksendal, Stochastic Differential Equations; Anderson-Guionnet-Zeitouni, Random Matrices; Tao, Topics in Random Matrix Theory; Janson-Luczak-Rucinski, Random Graphs

Quarter output: Create a computational/theoretical project comparing random-graph and Wigner spectra, plus one SDE-to-PDE derivation.

Research bridges: Feeds stochastic dynamics, statistical mechanics and random-operator methods.

Six two-week study units

Unit 1 (Weeks 1-2): Ito calculus

Topics: quadratic variation; stochastic integrals; Ito formula

Key definitions: Ito integral; quadratic variation; semimartingale

Key results to learn:

- Ito isometry [PROOF]
- Ito formula [PROOF/ARCH]

Reading route: Karatzas-Shreve/Oksendal opening chapters

Warm-up/source exercise focus (secondary): Compute stochastic integrals and transformations; derive geometric Brownian/Bessel-type examples.

Week 1: Ito calculus - definitions and examples. Read: Karatzas-Shreve/Oksendal opening chapters. Make explicit examples/nonexamples for Ito integral; quadratic variation; semimartingale. Work through quadratic variation; stochastic integrals; Ito formula. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Ito calculus - theorem and technique pass. Master/audit: Ito isometry; Ito formula. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): SDE existence and generators

Topics: strong/weak solutions; Lipschitz conditions; generators

Key definitions: strong solution; weak solution; generator

Key results to learn:

- existence/uniqueness under Lipschitz growth [PROOF/ARCH]
- strong Markov property statement [ARCH]

Reading route: Karatzas-Shreve/Oksendal SDE chapters

Warm-up/source exercise focus (secondary): Solve linear SDEs; test pathwise uniqueness; derive Kolmogorov equations.

Week 3: SDE existence and generators - definitions and examples. Read: Karatzas-Shreve/Oksendal SDE chapters. Make explicit examples/nonexamples for strong solution; weak solution; generator. Work through strong/weak solutions; Lipschitz conditions; generators. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: SDE existence and generators - theorem and technique pass. Master/audit: existence/uniqueness under Lipschitz growth; strong Markov property statement. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Girsanov and Feynman-Kac

Topics: change of measure; martingale problems; PDE representation

Key definitions: density process; equivalent measure

Key results to learn:

- Girsanov theorem [ARCH]
- Feynman-Kac formula [ARCH]

Reading route: Karatzas-Shreve

Warm-up/source exercise focus (secondary): Perform drift changes; solve PDE expectations through Feynman-Kac.

Week 5: Girsanov and Feynman-Kac - definitions and examples. Read: Karatzas-Shreve. Make explicit examples/nonexamples for density process; equivalent measure. Work through change of measure; martingale problems; PDE representation. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Girsanov and Feynman-Kac - theorem and technique pass. Master/audit: Girsanov theorem; Feynman-Kac formula. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Concentration and random graphs

Topics: Chernoff/Hoeffding; martingale concentration; Erdos-Renyi

Key definitions: threshold; giant component; concentration inequality

Key results to learn:

- Chernoff/Hoeffding [PROOF]
- Erdos-Renyi phase-transition statements [ARCH/USE]

Reading route: Janson-Luczak-Rucinski

Warm-up/source exercise focus (secondary): Derive thresholds for isolated vertices/triangles; simulate and prove concentration.

Week 7: Concentration and random graphs - definitions and examples. Read: Janson-Luczak-Rucinski. Make explicit examples/nonexamples for threshold; giant component; concentration inequality. Work through Chernoff/Hoeffding; martingale concentration; Erdos-Renyi. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Concentration and random graphs - theorem and technique pass. Master/audit: Chernoff/Hoeffding; Erdos-Renyi phase-transition statements. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Wigner matrices and global laws

Topics: Wigner ensembles; resolvents; moments; empirical spectral measures

Key definitions: empirical spectral distribution; Stieltjes transform; Wigner matrix

Key results to learn:

- Wigner semicircle law [ARCH]
- moment/Stieltjes proof architectures [ARCH]

Reading route: Tao; AGZ opening chapters

Warm-up/source exercise focus (secondary): Compute moments by pairings; simulate spectra; prove simplified convergence.

Week 9: Wigner matrices and global laws - definitions and examples. Read: Tao; AGZ opening chapters. Make explicit examples/nonexamples for empirical spectral distribution; Stieltjes transform; Wigner matrix. Work through Wigner ensembles; resolvents; moments; empirical spectral measures. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Wigner matrices and global laws - theorem and technique pass. Master/audit: Wigner semicircle law; moment/Stieltjes proof architectures. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Local laws and universality entry

Topics: operator norm; local semicircle; edge behavior; covariance ensembles

Key definitions: local law; universality; Tracy-Widom concept

Key results to learn:

- Bai-Yin/operator-norm law statement [USE]
- local semicircle and universality landmark statements [USE]

Reading route: Tao/AGZ advanced selections

Warm-up/source exercise focus (secondary): Read a local-law proof outline; distinguish global/local spectral questions; compare ensembles.

Week 11: Local laws and universality entry - definitions and examples. Read: Tao/AGZ advanced selections. Make explicit examples/nonexamples for local law; universality; Tracy-Widom concept. Work through operator norm; local semicircle; edge behavior; covariance ensembles. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Local laws and universality entry - theorem and technique pass. Master/audit: Bai-Yin/operator-norm law statement; local semicircle and universality landmark statements. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Ito calculus

U1.P1. [F] Compute quadratic variation of Brownian motion along dyadic partitions and use it to motivate Ito's correction term. Contrast with bounded-variation paths.

U1.P2. [T] Derive Ito's formula first for f(x)=x^2,x^3 and then in general. Use it to compute d(exp(B_t-t/2)) and verify the martingale property.

U1.P3. [S] Apply Ito to log of geometric Brownian motion and to the Ornstein-Uhlenbeck process. Extract mean, variance, and long-time behavior.

Programme U2: SDE existence and generators

U2.P1. [T] Prove existence/uniqueness for a Lipschitz SDE by Picard iteration in L^2 on a finite time interval. Identify the contraction estimate.

U2.P2. [F] Compute infinitesimal generators for Brownian motion, Ornstein-Uhlenbeck, and geometric Brownian motion. Verify Dynkin's formula in a simple case.

U2.P3. [C] Examine an SDE with non-Lipschitz coefficient such as dX=sqrt(|X|)dB and identify which uniqueness theorem no longer applies; state carefully what is and is not concluded.

Programme U3: Girsanov and Feynman-Kac

U3.P1. [T] Derive Girsanov in the constant-drift Brownian case explicitly via exponential martingales and Radon-Nikodym derivatives. Check Novikov in the example.

U3.P2. [F] Use Feynman-Kac to solve a simple terminal-value PDE with potential or drift and verify the solution directly.

U3.P3. [S] Explain the triangle among generator, semigroup, martingale problem, and PDE by writing all four objects for one diffusion.

Programme U4: Concentration and random graphs

U4.P1. [F] In G(n,p), compute expected isolated vertices, triangles, and degrees; locate heuristic thresholds from first moments and test numerically.

U4.P2. [T] Prove a bounded-difference/McDiarmid inequality in a standard form and apply it to a Lipschitz graph statistic or product-space observable.

U4.P3. [S] Use first and second moment methods on a simple random-graph property, clearly distinguishing what each method proves and where dependencies matter.

Programme U5: Wigner matrices and global laws

U5.P1. [F] For a Wigner matrix, compute expected normalized trace of powers k=2,4,6 and identify Catalan pairings in the leading terms.

U5.P2. [T] Derive the semicircle law by the moment method at proof-architecture level, including why non-pairing index patterns are lower order.

U5.P3. [X] Simulate eigenvalue histograms for increasing matrix size and compare with the semicircle density. Record finite-size effects and what the simulation cannot prove.

Programme U6: Local laws and universality entry

U6.P1. [T] Derive the resolvent identity and basic Stieltjes-transform properties. Compute the Stieltjes transform of a simple empirical measure and interpret imaginary parts as smoothed densities.

U6.P2. [S] Write the self-consistent equation for the semicircle Stieltjes transform and solve the algebraic branch condition. Explain why stability of this equation is central to local laws.

U6.P3. [R] Read one introductory local-semicircle-law proof outline and produce a dependency chart: resolvent identities, concentration, self-consistency, stability, eigenvalue consequences.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for SDEs, Random Structures and Random Matrices: start from “Ito calculus”, pass through “Concentration and random graphs”, and finish at “Local laws and universality entry”. Use the named results “Ito isometry” and “local semicircle and universality landmark statements” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds stochastic dynamics, statistical mechanics and random-operator methods. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Ito isometry; Ito formula; existence/uniqueness under Lipschitz growth; Chernoff/Hoeffding. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Erdos-Renyi phase-transition statements; Wigner semicircle law; moment/Stieltjes proof architectures; Bai-Yin/operator-norm law statement; local semicircle and universality landmark statements.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Ito calculus” develops into “Local laws and universality entry”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 16. Percolation, SLE and Statistical Mechanics

Purpose: Develop rigorous critical models and reach conformal invariance/SLE at research-reading level.

Prerequisites: Volumes VIII-IX and XIV.

Primary and secondary references: Grimmett, Percolation; Bollobas-Riordan, Percolation; Lawler, Conformally Invariant Processes in the Plane; Werner, SLE notes; Friedli-Velenik, Statistical Mechanics of Lattice Systems

Quarter output: Write a 15-20 page survey from planar percolation through RSW and discrete holomorphicity to SLE.

Research bridges: Connects probability, complex analysis, conformal geometry and statistical physics.

Six two-week study units

Unit 1 (Weeks 1-2): Bernoulli percolation

Topics: bond/site percolation; clusters; criticality; duality

Key definitions: open cluster; percolation probability; critical threshold

Key results to learn:

- existence of critical threshold [PROOF/ARCH]
- uniqueness of infinite cluster under standard hypotheses [ARCH]
- planar duality [PROOF/ARCH]

Reading route: Grimmett opening chapters

Warm-up/source exercise focus (secondary): Prove monotonicity/coupling; compute trees/1D; bound critical probabilities.

Week 1: Bernoulli percolation - definitions and examples. Read: Grimmett opening chapters. Make explicit examples/nonexamples for open cluster; percolation probability; critical threshold. Work through bond/site percolation; clusters; criticality; duality. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Bernoulli percolation - theorem and technique pass. Master/audit: existence of critical threshold; uniqueness of infinite cluster under standard hypotheses; planar duality. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Correlation inequalities and RSW

Topics: FKG; BK; Russo formula; crossings; RSW

Key definitions: increasing event; pivotal edge; crossing probability

Key results to learn:

- FKG [ARCH]
- Russo formula [ARCH]
- RSW theorem [ARCH/USE]

Reading route: Grimmett/Bollobas-Riordan planar chapters

Warm-up/source exercise focus (secondary): Use pivotality; prove consequences of box-crossing estimates.

Week 3: Correlation inequalities and RSW - definitions and examples. Read: Grimmett/Bollobas-Riordan planar chapters. Make explicit examples/nonexamples for increasing event; pivotal edge; crossing probability. Work through FKG; BK; Russo formula; crossings; RSW. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Correlation inequalities and RSW - theorem and technique pass. Master/audit: FKG; Russo formula; RSW theorem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Critical exponents/scaling

Topics: arm events; correlation length; near-criticality

Key definitions: critical exponent; arm event; correlation length

Key results to learn:

- Kesten scaling-relation architecture [USE]
- known planar exponent statements [USE]

Reading route: Grimmett plus surveys

Warm-up/source exercise focus (secondary): Compute heuristic exponents; separate rigorous theorems from physics predictions.

Week 5: Critical exponents/scaling - definitions and examples. Read: Grimmett plus surveys. Make explicit examples/nonexamples for critical exponent; arm event; correlation length. Work through arm events; correlation length; near-criticality. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Critical exponents/scaling - theorem and technique pass. Master/audit: Kesten scaling-relation architecture; known planar exponent statements. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Ising and random-cluster models

Topics: Gibbs measures; FK representation; phase transition

Key definitions: Gibbs measure; spontaneous magnetization; random-cluster measure

Key results to learn:

- FKG for Ising/FK [ARCH]
- Edwards-Sokal coupling [ARCH]
- phase-transition statements [ARCH/USE]

Reading route: Friedli-Velenik; Grimmett random-cluster sections

Warm-up/source exercise focus (secondary): Solve 1D Ising; derive expansions; construct FK coupling.

Week 7: Ising and random-cluster models - definitions and examples. Read: Friedli-Velenik; Grimmett random-cluster sections. Make explicit examples/nonexamples for Gibbs measure; spontaneous magnetization; random-cluster measure. Work through Gibbs measures; FK representation; phase transition. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Ising and random-cluster models - theorem and technique pass. Master/audit: FKG for Ising/FK; Edwards-Sokal coupling; phase-transition statements. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Loewner evolution and SLE

Topics: Loewner equation; hulls; capacity; SLE_kappa

Key definitions: half-plane capacity; Loewner chain; SLE

Key results to learn:

- Loewner correspondence [ARCH]
- conformal/Markov characterization [ARCH/USE]
- phase properties vs kappa [USE]

Reading route: Lawler; Werner

Warm-up/source exercise focus (secondary): Solve deterministic Loewner examples; compute conformal covariance; simulate drivers.

Week 9: Loewner evolution and SLE - definitions and examples. Read: Lawler; Werner. Make explicit examples/nonexamples for half-plane capacity; Loewner chain; SLE. Work through Loewner equation; hulls; capacity; SLE_kappa. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Loewner evolution and SLE - theorem and technique pass. Master/audit: Loewner correspondence; conformal/Markov characterization; phase properties vs kappa. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Conformal invariance landmarks

Topics: Cardy formula; percolation/Ising interfaces; CLE entry

Key definitions: exploration path; conformal invariant observable

Key results to learn:

- Smirnov Cardy/conformal-invariance theorem for triangular percolation [USE]
- key SLE identification statements [USE]

Reading route: Lawler/Werner plus original expositions

Warm-up/source exercise focus (secondary): Read one Smirnov proof architecture; map RSW/discrete-holomorphic inputs to SLE convergence.

Week 11: Conformal invariance landmarks - definitions and examples. Read: Lawler/Werner plus original expositions. Make explicit examples/nonexamples for exploration path; conformal invariant observable. Work through Cardy formula; percolation/Ising interfaces; CLE entry. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Conformal invariance landmarks - theorem and technique pass. Master/audit: Smirnov Cardy/conformal-invariance theorem for triangular percolation; key SLE identification statements. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Bernoulli percolation

U1.P1. [F] On a finite box, compute crossing/connectivity probabilities for very small bond/site percolation configurations exactly and verify monotonicity in p.

U1.P2. [T] Prove existence of the critical probability p_c as a monotonic threshold and establish a basic Peierls-type argument for nontriviality in a model lattice/dimension where your text presents it.

U1.P3. [S] Derive a differential/finite-energy comparison for an increasing event and explain how pivotal edges enter Russo's formula.

Programme U2: Correlation inequalities and RSW

U2.P1. [T] Prove Harris/FKG inequality in a finite product-space formulation and apply it to crossing events. Identify precisely where monotonicity is used.

U2.P2. [F] Work through RSW gluing: assuming a square-crossing lower bound, derive a rectangle-crossing bound for a fixed aspect ratio in a simplified planar setup.

U2.P3. [R] Create a dependency map from FKG + RSW + arm estimates to tightness/conformal-invariance results. Mark which inputs are model-specific.

Programme U3: Critical exponents/scaling

U3.P1. [F] Define one-arm, four-arm, susceptibility, and correlation-length exponents and derive scaling relations heuristically from dimensional arguments. Label every heuristic step.

U3.P2. [T] For critical site percolation on the triangular lattice, state exact exponent results and identify which require SLE. Do not reproduce proofs; verify consistency among scaling relations.

U3.P3. [X] Simulate critical percolation clusters/crossings at several scales and estimate one exponent crudely. Compare numerical behavior to theorem values without treating simulation as evidence of proof.

Programme U4: Ising and random-cluster models

U4.P1. [F] Derive the high-temperature expansion of the Ising model on a small graph and the Fortuin-Kasteleyn random-cluster representation in a finite example.

U4.P2. [T] Prove FKG for the random-cluster measure in the relevant q>=1 setting at theorem-architecture level and translate an Ising spin correlation into a connectivity statement under Edwards-Sokal coupling.

U4.P3. [S] Compare Ising, FK, and percolation observables on the same planar graph; identify which monotonicity/duality tools survive across models.

Programme U5: Loewner evolution and SLE

U5.P1. [F] Solve the chordal Loewner equation for constant driving function and identify the growing slit explicitly. Track half-plane capacity under scaling.

U5.P2. [T] Use Brownian scaling to verify SLE_kappa scale invariance and explain the domain Markov property. Determine which kappa regimes correspond to simple/non-simple/space-filling paths at theorem-statement level.

U5.P3. [S] Derive one SLE martingale/PDE in a simple boundary-event setting and explain how solving it gives a crossing probability or exponent.

Programme U6: Conformal invariance landmarks

U6.P1. [R] Reconstruct Smirnov's percolation-to-SLE6 theorem architecture: discrete observable, precompactness, conformal covariance, identification of driving process, convergence of interfaces.

U6.P2. [R] For the planar Ising model, select one conformal-invariance theorem and write a theorem audit separating discrete complex analysis, tightness, and continuum identification inputs.

U6.P3. [S] Compare Cardy's formula, SLE6, and RSW: state exactly which is an estimate, which is a scaling-limit identification, and which is an explicit conformal formula.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Percolation, SLE and Statistical Mechanics: start from “Bernoulli percolation”, pass through “Ising and random-cluster models”, and finish at “Conformal invariance landmarks”. Use the named results “existence of critical threshold” and “key SLE identification statements” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Connects probability, complex analysis, conformal geometry and statistical physics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: existence of critical threshold; planar duality. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Loewner correspondence; conformal/Markov characterization; phase properties vs kappa; Smirnov Cardy/conformal-invariance theorem for triangular percolation; key SLE identification statements.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Bernoulli percolation” develops into “Conformal invariance landmarks”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 17. Particle Systems, Kinetic Limits and Interacting Systems

Purpose: Build a rigorous route from microscopic stochastic/deterministic particles to macroscopic PDE and kinetic equations, including Hilbert VI themes.

Prerequisites: Volumes X, XIV-XV.

Primary and secondary references: Liggett, Interacting Particle Systems; Kipnis-Landim, Scaling Limits of Interacting Particle Systems; Spohn, Large Scale Dynamics of Interacting Particles; Cercignani-Illner-Pulvirenti, Mathematical Theory of Dilute Gases; Saint-Raymond, Hydrodynamic Limits of the Boltzmann Equation

Quarter output: Produce a 20-page Hilbert VI particle-to-PDE roadmap with one formal derivation and one rigorous theorem dependency tree.

Research bridges: Directly supports hard-ball/Brownian and particle-to-kinetic research targets.

Six two-week study units

Unit 1 (Weeks 1-2): Interacting particle systems

Topics: spin systems; exclusion; contact process; generators; invariant measures

Key definitions: Feller process; generator; invariant measure; attractiveness

Key results to learn:

- basic existence/generator framework [ARCH]
- coupling/attractiveness consequences [ARCH]

Reading route: Liggett opening chapters

Warm-up/source exercise focus (secondary): Construct generators; derive invariant product measures; use coupling.

Week 1: Interacting particle systems - definitions and examples. Read: Liggett opening chapters. Make explicit examples/nonexamples for Feller process; generator; invariant measure; attractiveness. Work through spin systems; exclusion; contact process; generators; invariant measures. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Interacting particle systems - theorem and technique pass. Master/audit: basic existence/generator framework; coupling/attractiveness consequences. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Hydrodynamic limits

Topics: empirical measures; LLN scaling; entropy method

Key definitions: empirical density; hydrodynamic limit; local equilibrium

Key results to learn:

- hydrodynamic limit for symmetric exclusion model theorem [ARCH/USE]
- replacement/entropy architecture [USE]

Reading route: Kipnis-Landim; Spohn

Warm-up/source exercise focus (secondary): Derive formal PDE limits; prove toy independent-particle hydrodynamics; trace a replacement lemma.

Week 3: Hydrodynamic limits - definitions and examples. Read: Kipnis-Landim; Spohn. Make explicit examples/nonexamples for empirical density; hydrodynamic limit; local equilibrium. Work through empirical measures; LLN scaling; entropy method. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Hydrodynamic limits - theorem and technique pass. Master/audit: hydrodynamic limit for symmetric exclusion model theorem; replacement/entropy architecture. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Fluctuations and KPZ entry

Topics: fluctuation fields; asymmetric exclusion; universality

Key definitions: fluctuation field; current; KPZ universality

Key results to learn:

- equilibrium OU fluctuation limits in model systems [USE]
- KPZ scaling landmarks [USE]

Reading route: Kipnis-Landim; modern surveys

Warm-up/source exercise focus (secondary): Compute covariance of fluctuation fields; compare diffusive/KPZ scaling.

Week 5: Fluctuations and KPZ entry - definitions and examples. Read: Kipnis-Landim; modern surveys. Make explicit examples/nonexamples for fluctuation field; current; KPZ universality. Work through fluctuation fields; asymmetric exclusion; universality. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Fluctuations and KPZ entry - theorem and technique pass. Master/audit: equilibrium OU fluctuation limits in model systems; KPZ scaling landmarks. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): BBGKY and mean-field limits

Topics: Liouville equation; marginals; Vlasov; propagation of chaos

Key definitions: BBGKY hierarchy; empirical measure; chaotic sequence

Key results to learn:

- mean-field/Vlasov limit under regular interactions [ARCH/USE]
- propagation of chaos [ARCH/USE]

Reading route: Spohn; mean-field notes

Warm-up/source exercise focus (secondary): Derive hierarchy; close it formally; prove a toy Gronwall/Wasserstein estimate.

Week 7: BBGKY and mean-field limits - definitions and examples. Read: Spohn; mean-field notes. Make explicit examples/nonexamples for BBGKY hierarchy; empirical measure; chaotic sequence. Work through Liouville equation; marginals; Vlasov; propagation of chaos. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: BBGKY and mean-field limits - theorem and technique pass. Master/audit: mean-field/Vlasov limit under regular interactions; propagation of chaos. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Boltzmann-Grad and dilute gases

Topics: hard spheres; collision operator; Boltzmann equation; recollisions

Key definitions: Boltzmann-Grad scaling; molecular chaos; collision kernel

Key results to learn:

- Lanford theorem statement/architecture [USE]
- Deng-Hani-Ma long-time hard-sphere-to-Boltzmann theorem (v3 2025): exact statement and architecture; compare its regular-Boltzmann-solution condition with Lanford's short-time restriction [USE]
- H-theorem [ARCH]

Reading route: Cercignani et al.; Saint-Raymond

Warm-up/source exercise focus (secondary): Derive collision operator formally; analyse collision trees and recollision obstruction.

Week 9: Boltzmann-Grad and dilute gases - definitions and examples. Read: Cercignani et al.; Saint-Raymond. Make explicit examples/nonexamples for Boltzmann-Grad scaling; molecular chaos; collision kernel. Work through hard spheres; collision operator; Boltzmann equation; recollisions. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Boltzmann-Grad and dilute gases - theorem and technique pass. Master/audit: Lanford theorem statement/architecture; H-theorem. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Brownian/kinetic limits and Hilbert VI frontier

Topics: Lorentz gas; tagged particles; diffusive limits; kinetic-to-fluid limits

Key definitions: diffusive scaling; kinetic equation; hydrodynamic scaling

Key results to learn:

- selected tagged-particle invariance principles [USE]
- Boltzmann-to-Euler/Navier-Stokes landmarks [USE]
- modern hard-sphere/kinetic research map [USE]

Reading route: Saint-Raymond plus target surveys/papers

Warm-up/source exercise focus (secondary): Build a literature map separating microscopic-to-kinetic, kinetic-to-fluid and direct diffusive limits.

Week 11: Brownian/kinetic limits and Hilbert VI frontier - definitions and examples. Read: Saint-Raymond plus target surveys/papers. Make explicit examples/nonexamples for diffusive scaling; kinetic equation; hydrodynamic scaling. Work through Lorentz gas; tagged particles; diffusive limits; kinetic-to-fluid limits. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Brownian/kinetic limits and Hilbert VI frontier - theorem and technique pass. Master/audit: selected tagged-particle invariance principles; Boltzmann-to-Euler/Navier-Stokes landmarks; modern hard-sphere/kinetic research map. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Interacting particle systems

U1.P1. [F] For the simple exclusion/contact/zero-range process on a finite graph, write the generator and compute its action on basic observables. Identify invariant measures in one model.

U1.P2. [T] Derive the graphical construction/coupling for an attractive interacting particle system and use it to prove monotonicity with respect to initial data.

U1.P3. [S] Compute a one- or two-point evolution equation and identify exactly where closure fails, motivating correlation hierarchies.

Programme U2: Hydrodynamic limits

U2.P1. [F] For independent random walks or exclusion on a discrete torus, derive the empirical density and state the expected hydrodynamic PDE.

U2.P2. [T] Prove a law-of-large-numbers hydrodynamic limit in a toy independent-particle model via test functions and martingale estimates.

U2.P3. [R] Read the entropy-method architecture for exclusion/zero-range: replacement lemma, local equilibrium, entropy inequality, PDE uniqueness. Explain the role of each.

Programme U3: Fluctuations and KPZ entry

U3.P1. [F] Derive central-limit-scale fluctuation fields for independent particles and compute their covariance. Identify the candidate Gaussian limiting field.

U3.P2. [S] Perform dimensional/scaling analysis for KPZ versus Edwards-Wilkinson and explain why the 1:2:3 scaling is special in one dimension.

U3.P3. [R] Choose one rigorous KPZ landmark (ASEP/TASEP, stochastic heat equation, or KPZ equation). Produce a theorem-and-method map without attempting all technical details.

Programme U4: BBGKY and mean-field limits

U4.P1. [F] Derive the BBGKY hierarchy for an N-particle mean-field Hamiltonian or stochastic system through the first two marginals.

U4.P2. [T] Assuming propagation of chaos, formally derive the Vlasov/McKean-Vlasov equation. Then prove a stability estimate in a Lipschitz mean-field model sufficient for a simple chaos result.

U4.P3. [S] Compare mean-field scaling 1/N with Boltzmann-Grad scaling. Write a table of density, interaction range/strength, collision frequency, and limiting equation.

Programme U5: Boltzmann-Grad and dilute gases

U5.P1. [F] Derive the Boltzmann collision operator formally from binary-collision geometry for hard spheres, including pre/post-collisional velocity parametrization.

U5.P2. [T] Work through the first collision-tree terms of Lanford's expansion and identify recollisions as the main obstruction. Quantify the combinatorial growth at a schematic level.

U5.P3. [R] State Lanford's theorem precisely enough to include time scale, topology/marginals, and Boltzmann-Grad regime. Build a proof architecture showing where short time enters.

Programme U6: Brownian/kinetic limits and Hilbert VI frontier

U6.P1. [S] For a tagged particle in a dilute gas, derive formally how repeated weakly correlated collisions could yield Brownian motion; identify centering and variance scaling.

U6.P2. [R] Make a four-row theorem table: Lanford short-time hard spheres -> Boltzmann; Deng-Hani-Ma long-time hard spheres -> Boltzmann; tagged hard sphere -> Brownian motion via linear Boltzmann; kinetic -> hydrodynamic PDE. For each record scaling, topology/marginals, time horizon, regularity assumptions and the obstruction overcome/not overcome. Add a scope note on what is meant by the hard-sphere branch of Hilbert VI.

U6.P3. [X] Implement a small event-driven hard-sphere or random-collision simulation and compare empirical velocity/position statistics to kinetic/Brownian predictions. State clearly which correlations the simulation does not control.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Particle Systems, Kinetic Limits and Interacting Systems: start from “Interacting particle systems”, pass through “BBGKY and mean-field limits”, and finish at “Brownian/kinetic limits and Hilbert VI frontier”. Use the named results “basic existence/generator framework” and “modern hard-sphere/kinetic research map” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Directly supports hard-ball/Brownian and particle-to-kinetic research targets. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: basic existence/generator framework; coupling/attractiveness consequences; hydrodynamic limit for symmetric exclusion model theorem. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Lanford theorem statement/architecture; H-theorem; selected tagged-particle invariance principles; Boltzmann-to-Euler/Navier-Stokes landmarks; modern hard-sphere/kinetic research map.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Interacting particle systems” develops into “Brownian/kinetic limits and Hilbert VI frontier”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 18. Lie Groups, Lie Algebras and Representation Theory

Purpose: Develop finite/compact/Lie-group representation theory through unitary representations of semisimple groups.

Prerequisites: Volumes II-III, VII, XII.

Primary and secondary references: Etingof et al., Introduction to Representation Theory; Fulton-Harris, Representation Theory; Hall, Lie Groups, Lie Algebras and Representations; Knapp, Representation Theory of Semisimple Groups; MIT OCW 18.755 and 18.757

Quarter output: Prepare a representation atlas for sl2/SU2/SL2(R), highlighting links to automorphic forms and homogeneous dynamics.

Research bridges: Direct prerequisite for property (T), automorphic forms, spectral gaps and homogeneous dynamics.

Six two-week study units

Unit 1 (Weeks 1-2): Finite-group representations

Topics: representations; intertwiners; characters

Key definitions: irreducible representation; character; regular representation

Key results to learn:

- Maschke [PROOF]
- Schur lemma [PROOF]
- character orthogonality [PROOF/ARCH]
- irreps vs conjugacy classes [PROOF]

Reading route: Etingof Chs.2-4 by topic; Fulton-Harris

Warm-up/source exercise focus (secondary): Compute character tables of S3,D_n,Q8,A4; decompose regular/permutation representations.

Week 1: Finite-group representations - definitions and examples. Read: Etingof Chs.2-4 by topic; Fulton-Harris. Make explicit examples/nonexamples for irreducible representation; character; regular representation. Work through representations; intertwiners; characters. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Finite-group representations - theorem and technique pass. Master/audit: Maschke; Schur lemma; character orthogonality. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Induction/restriction and symmetric groups

Topics: induced reps; Frobenius reciprocity; Mackey ideas; Specht modules

Key definitions: induced representation; representation ring

Key results to learn:

- Frobenius reciprocity [PROOF/ARCH]
- Mackey formula [ARCH]
- irreps of S_n via partitions [ARCH/USE]

Reading route: Fulton-Harris; Etingof

Warm-up/source exercise focus (secondary): Induce characters; compute branching; work Young diagrams/Schur-Weyl examples.

Week 3: Induction/restriction and symmetric groups - definitions and examples. Read: Fulton-Harris; Etingof. Make explicit examples/nonexamples for induced representation; representation ring. Work through induced reps; Frobenius reciprocity; Mackey ideas; Specht modules. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Induction/restriction and symmetric groups - theorem and technique pass. Master/audit: Frobenius reciprocity; Mackey formula; irreps of S_n via partitions. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Compact groups and Peter-Weyl

Topics: Haar measure; unitary reps; matrix coefficients

Key definitions: compact group; Haar measure; matrix coefficient

Key results to learn:

- Haar existence/uniqueness [ARCH]
- Peter-Weyl [ARCH/USE]

Reading route: Hall; representation notes

Warm-up/source exercise focus (secondary): Do Fourier analysis on compact abelian groups and SU(2); derive coefficient orthogonality.

Week 5: Compact groups and Peter-Weyl - definitions and examples. Read: Hall; representation notes. Make explicit examples/nonexamples for compact group; Haar measure; matrix coefficient. Work through Haar measure; unitary reps; matrix coefficients. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Compact groups and Peter-Weyl - theorem and technique pass. Master/audit: Haar existence/uniqueness; Peter-Weyl. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Lie groups and Lie algebras

Topics: exponential map; adjoint; roots; semisimple algebras

Key definitions: Lie group; Lie algebra; Cartan subalgebra; root

Key results to learn:

- Lie correspondence basics [ARCH]
- Engel/Lie theorems [ARCH]
- Killing criterion [ARCH]
- semisimple classification framework [USE]

Reading route: Hall; MIT 18.755; Fulton-Harris

Warm-up/source exercise focus (secondary): Compute classical Lie algebras and root systems A1,A2,B2; represent sl2 explicitly.

Week 7: Lie groups and Lie algebras - definitions and examples. Read: Hall; MIT 18.755; Fulton-Harris. Make explicit examples/nonexamples for Lie group; Lie algebra; Cartan subalgebra; root. Work through exponential map; adjoint; roots; semisimple algebras. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Lie groups and Lie algebras - theorem and technique pass. Master/audit: Lie correspondence basics; Engel/Lie theorems; Killing criterion. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Highest weights

Topics: weights; Weyl group; highest-weight modules

Key definitions: dominant integral weight; highest weight

Key results to learn:

- Weyl complete reducibility [ARCH]
- highest-weight classification [ARCH/USE]
- Weyl character formula [C/B]

Reading route: Fulton-Harris; Etingof; MIT 18.755

Warm-up/source exercise focus (secondary): Decompose sl2 tensor products; apply Clebsch-Gordan; compute small characters.

Week 9: Highest weights - definitions and examples. Read: Fulton-Harris; Etingof; MIT 18.755. Make explicit examples/nonexamples for dominant integral weight; highest weight. Work through weights; Weyl group; highest-weight modules. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Highest weights - theorem and technique pass. Master/audit: Weyl complete reducibility; highest-weight classification; Weyl character formula [C/B]. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Unitary reps of real groups

Topics: principal/discrete/complementary series; induced reps; (g,K)-modules

Key definitions: unitary representation; K-finite vector; principal series; discrete series

Key results to learn:

- SL2(R) unitary-dual architecture [USE]
- Casimir/Laplacian relation [ARCH/USE]
- matrix-coefficient decay principles [USE]

Reading route: MIT 18.757; Knapp

Warm-up/source exercise focus (secondary): Work SL2(R) Iwasawa/induced reps; identify K-types; connect complementary series to spectral gaps.

Week 11: Unitary reps of real groups - definitions and examples. Read: MIT 18.757; Knapp. Make explicit examples/nonexamples for unitary representation; K-finite vector; principal series; discrete series. Work through principal/discrete/complementary series; induced reps; (g,K)-modules. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Unitary reps of real groups - theorem and technique pass. Master/audit: SL2(R) unitary-dual architecture; Casimir/Laplacian relation; matrix-coefficient decay principles. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Finite-group representations

U1.P1. [F] Decompose the regular representations of S3, D4, and a cyclic group into irreducibles using characters. Verify the sum-of-squares dimension formula.

U1.P2. [T] Prove Maschke's theorem by averaging a projection and identify exactly why characteristic dividing |G| breaks the proof.

U1.P3. [S] Derive character orthogonality and use it to reconstruct the full character table of S3 or D4 from minimal information.

Programme U2: Induction/restriction and symmetric groups

U2.P1. [T] Construct induced representations from a subgroup in a finite example and verify Frobenius reciprocity by explicit Hom-space dimensions.

U2.P2. [F] Use Young diagrams/tableaux to compute dimensions of selected irreducible S_n representations for n<=5 and verify branching in small cases.

U2.P3. [S] Compare restriction/induction along S_{n-1}<S_n with permutation representations on cosets; identify the combinatorial meaning of branching.

Programme U3: Compact groups and Peter-Weyl

U3.P1. [F] For S^1 and SU(2), list/construct irreducible unitary representations and compute matrix coefficients in low-dimensional cases.

U3.P2. [T] Prove Peter-Weyl at proof-architecture level, highlighting compactness, Haar measure, and density of matrix coefficients. Verify Fourier series on S^1 as the abelian case.

U3.P3. [S] Derive Schur orthogonality for compact groups and use it to compute a convolution projection onto one isotypic component.

Programme U4: Lie groups and Lie algebras

U4.P1. [F] Compute Lie algebras of GL_n, SL_n, SO_n, SU_n and verify brackets from matrix commutators. Calculate exponential maps for selected one-parameter subgroups.

U4.P2. [T] Derive the adjoint representation and compute ad_X for sl_2 basis elements. Verify the sl_2 commutation relations and Killing form.

U4.P3. [S] Prove the Lie correspondence between Lie algebra homomorphisms and local/simply-connected group homomorphisms at theorem-architecture level; work one explicit integration.

Programme U5: Highest weights

U5.P1. [F] Work out root systems and weights for sl_2 and sl_3. Draw weight diagrams for several low-dimensional irreducibles.

U5.P2. [T] Prove the classification of finite-dimensional sl_2 representations completely. Use raising/lowering operators to derive dimensions and characters.

U5.P3. [S] State the highest-weight theorem for compact semisimple/Lie algebras and use it to construct representations of SU(2) or SU(3) from dominant weights.

Programme U6: Unitary reps of real groups

U6.P1. [F] Write the principal series representation of SL_2(R) in one standard model and calculate the action of basic matrices/one-parameter subgroups.

U6.P2. [T] Compute K-types and Casimir eigenvalues in a selected SL_2(R) representation. Relate the Casimir to the hyperbolic Laplacian on automorphic functions.

U6.P3. [R] Compare principal, complementary, and discrete series: parameter ranges, unitarity, K-types, and where spectral-gap questions occur. Prepare this as a one-page table for later property (T)/automorphic use.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Lie Groups, Lie Algebras and Representation Theory: start from “Finite-group representations”, pass through “Lie groups and Lie algebras”, and finish at “Unitary reps of real groups”. Use the named results “Maschke” and “matrix-coefficient decay principles” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct prerequisite for property (T), automorphic forms, spectral gaps and homogeneous dynamics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Maschke; Schur lemma; character orthogonality; irreps vs conjugacy classes; Frobenius reciprocity. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Weyl complete reducibility; highest-weight classification; SL2(R) unitary-dual architecture; Casimir/Laplacian relation; matrix-coefficient decay principles.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Finite-group representations” develops into “Unitary reps of real groups”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 19. Smooth Dynamics, KAM Theory and Billiards

Purpose: Build the smooth-dynamical and Hamiltonian toolkit needed for hyperbolic dynamics, KAM theory, billiards and later rigidity theory.

Prerequisites: Volumes 2, 6-7, 12, 14; basic ODEs and measure theory.

Primary and secondary references: Katok-Hasselblatt, Introduction to the Modern Theory of Dynamical Systems, Parts I-II and selected III; Hasselblatt-Katok (eds.), Handbook of Dynamical Systems, selected surveys; Arnold, Mathematical Methods of Classical Mechanics, Chs. 8-10; Tabachnikov, Geometry and Billiards, selected chapters; de la Llave, tutorial/review notes on KAM theory

Quarter output: Write a 20-25 page comparison of an integrable billiard, a KAM perturbation and a Sinai dispersing billiard, emphasizing which invariant structures survive and which statistical properties emerge.

Research bridges: Prepares Volume 20 homogeneous dynamics and supplies the smooth-dynamics background for Zimmer rigidity and entropy methods.

Six two-week study units

Unit 1 (Weeks 1-2): Flows, maps and recurrence

Topics: Discrete maps and flows; invariant sets and measures; conjugacy; recurrence; linearization near fixed/periodic points.

Key definitions: flow; diffeomorphism; invariant measure; recurrent/nonwandering point; topological conjugacy; periodic orbit; derivative cocycle.

Key results to learn:

- Poincare recurrence theorem [PROOF]
- recurrence implies no wandering set of positive finite measure [PROOF]
- Hartman-Grobman theorem [ARCH]
- local linear stability criteria [PROOF].

Reading route: Katok-Hasselblatt Chs. 1-3; Arnold introductory dynamical sections.

Warm-up/source exercise focus (secondary): Classify fixed points of linear and nonlinear planar systems; prove recurrence for rotations; construct conjugacy and nonconjugacy examples.

Week 1: Flows, maps and recurrence - definitions and examples. Read: Katok-Hasselblatt Chs. 1-3; Arnold introductory dynamical sections.. Make explicit examples/nonexamples for flow; diffeomorphism; invariant measure; recurrent/nonwandering point; topological conjugacy; periodic orbit; derivative cocycle.. Work through Discrete maps and flows; invariant sets and measures; conjugacy; recurrence; linearization near fixed/periodic points.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Flows, maps and recurrence - theorem and technique pass. Master/audit: Poincare recurrence theorem; recurrence implies no wandering set of positive finite measure; Hartman-Grobman theorem. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Hyperbolic dynamics

Topics: Uniform hyperbolicity; stable/unstable manifolds; symbolic dynamics; shadowing; Anosov systems.

Key definitions: hyperbolic fixed point/set; stable/unstable bundle; Anosov diffeomorphism/flow; local product structure; Markov partition; shift of finite type.

Key results to learn:

- Stable manifold theorem [ARCH]
- lambda-lemma/inclination lemma [ARCH]
- shadowing lemma [ARCH]
- Anosov closing lemma [ARCH/USE]
- spectral decomposition for Axiom A systems [USE]
- existence of Markov partitions [USE].

Reading route: Katok-Hasselblatt Chs. 5-7, 18 selected.

Warm-up/source exercise focus (secondary): Compute stable/unstable foliations for toral automorphisms; code a horseshoe symbolically; trace the proof dependencies for closing and shadowing.

Week 3: Hyperbolic dynamics - definitions and examples. Read: Katok-Hasselblatt Chs. 5-7, 18 selected.. Make explicit examples/nonexamples for hyperbolic fixed point/set; stable/unstable bundle; Anosov diffeomorphism/flow; local product structure; Markov partition; shift of finite type.. Work through Uniform hyperbolicity; stable/unstable manifolds; symbolic dynamics; shadowing; Anosov systems.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Hyperbolic dynamics - theorem and technique pass. Master/audit: Stable manifold theorem; lambda-lemma/inclination lemma; shadowing lemma. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Entropy and nonuniform hyperbolicity

Topics: Measure/topological entropy; Lyapunov exponents; Oseledets splitting; Pesin theory; SRB measures.

Key definitions: Kolmogorov-Sinai entropy; topological entropy; Lyapunov exponent; Oseledets subspace; Pesin block; SRB measure.

Key results to learn:

• Kolmogorov-Sinai theorem [ARCH]
• variational principle [ARCH/USE]
• Oseledets multiplicative ergodic theorem [ARCH/USE]
• Pesin stable manifold theorem [USE]
• Ruelle inequality [USE]
• Pesin entropy formula under standard hypotheses [USE].

Reading route: Katok-Hasselblatt entropy and nonuniform-hyperbolicity chapters; supplementary Pesin surveys.

Warm-up/source exercise focus (secondary): Compute entropy/Lyapunov exponents for shifts and toral automorphisms; compare metric and topological entropy; prepare a dependency map for Pesin theory.

Week 5: Entropy and nonuniform hyperbolicity - definitions and examples. Read: Katok-Hasselblatt entropy and nonuniform-hyperbolicity chapters; supplementary Pesin surveys.. Make explicit examples/nonexamples for Kolmogorov-Sinai entropy; topological entropy; Lyapunov exponent; Oseledets subspace; Pesin block; SRB measure.. Work through Measure/topological entropy; Lyapunov exponents; Oseledets splitting; Pesin theory; SRB measures.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Entropy and nonuniform hyperbolicity - theorem and technique pass. Master/audit: Kolmogorov-Sinai theorem; variational principle; Oseledets multiplicative ergodic theorem. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Hamiltonian dynamics and integrability

Topics: Symplectic geometry in dynamics; Hamilton equations; canonical transformations; integrability; action-angle variables.

Key definitions: symplectic form; Hamiltonian vector field; Poisson bracket; canonical transformation; first integral; complete integrability; action-angle coordinate.

Key results to learn:

• Hamilton equations from symplectic form [PROOF]
• Liouville theorem on volume preservation [PROOF]
• Poisson commuting integrals criterion [PROOF]
• Liouville-Arnold theorem [ARCH/USE].

Reading route: Arnold, Mathematical Methods of Classical Mechanics, symplectic/Hamiltonian and integrability chapters.

Warm-up/source exercise focus (secondary): Derive Hamilton flows for pendulum/oscillator; compute Poisson brackets; construct action-angle variables in one integrable model.

Week 7: Hamiltonian dynamics and integrability - definitions and examples. Read: Arnold, Mathematical Methods of Classical Mechanics, symplectic/Hamiltonian and integrability chapters.. Make explicit examples/nonexamples for symplectic form; Hamiltonian vector field; Poisson bracket; canonical transformation; first integral; complete integrability; action-angle coordinate.. Work through Symplectic geometry in dynamics; Hamilton equations; canonical transformations; integrability; action-angle variables.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Hamiltonian dynamics and integrability - theorem and technique pass. Master/audit: Hamilton equations from symplectic form; Liouville theorem on volume preservation; Poisson commuting integrals criterion. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Small divisors and KAM

Topics: Perturbation of integrable Hamiltonians; resonances; Diophantine frequencies; invariant tori; twist/nondegeneracy.

Key definitions: Diophantine vector; resonance; small divisor; invariant Lagrangian torus; twist/nondegeneracy condition; Kolmogorov normal form.

Key results to learn:

• Basic cohomological-equation estimates for Diophantine rotations [PROOF/ARCH]
• Kolmogorov-Arnold-Moser theorem in a standard analytic or smooth formulation [ARCH/USE]
• persistence of a positive-measure Cantor family of invariant tori [USE].

Reading route: Arnold KAM sections; de la Llave KAM tutorial; optional Poeschel notes.

Warm-up/source exercise focus (secondary): Solve rotation cohomological equations by Fourier series; locate resonances; state one KAM theorem exactly and annotate every hypothesis and loss of regularity.

Week 9: Small divisors and KAM - definitions and examples. Read: Arnold KAM sections; de la Llave KAM tutorial; optional Poeschel notes.. Make explicit examples/nonexamples for Diophantine vector; resonance; small divisor; invariant Lagrangian torus; twist/nondegeneracy condition; Kolmogorov normal form.. Work through Perturbation of integrable Hamiltonians; resonances; Diophantine frequencies; invariant tori; twist/nondegeneracy.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Small divisors and KAM - theorem and technique pass. Master/audit: Basic cohomological-equation estimates for Diophantine rotations; Kolmogorov-Arnold-Moser theorem in a standard analytic or smooth formulation; persistence of a positive-measure Cantor family of invariant tori [USE].. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Billiards and hyperbolic examples

Topics: Billiard map; convex and dispersing billiards; twist maps; caustics; Sinai billiards; ergodicity/mixing mechanisms.

Key definitions: billiard map; invariant measure; caustic; twist map; dispersing billiard; singularity set; finite/infinite horizon.

Key results to learn:

- Birkhoff billiard map is an area-preserving twist map for smooth strictly convex tables [ARCH]
- existence/geometry of caustics in integrable examples [ARCH]
- hyperbolicity and ergodicity of finite-horizon dispersing billiards [USE]
- exponential/statistical properties under stronger hypotheses [USE].

Reading route: Tabachnikov, Geometry and Billiards, core chapters; Katok-Hasselblatt examples; selected Sinai/Chernov surveys.

Warm-up/source exercise focus (secondary): Compute circle/ellipse billiard maps; compare integrable and dispersing tables; make a theorem map from geometric defocusing/dispersion to ergodicity.

Week 11: Billiards and hyperbolic examples - definitions and examples. Read: Tabachnikov, Geometry and Billiards, core chapters; Katok-Hasselblatt examples; selected Sinai/Chernov surveys.. Make explicit examples/nonexamples for billiard map; invariant measure; caustic; twist map; dispersing billiard; singularity set; finite/infinite horizon.. Work through Billiard map; convex and dispersing billiards; twist maps; caustics; Sinai billiards; ergodicity/mixing mechanisms.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Billiards and hyperbolic examples - theorem and technique pass. Master/audit: Birkhoff billiard map is an area-preserving twist map for smooth strictly convex tables; existence/geometry of caustics in integrable examples; hyperbolicity and ergodicity of finite-horizon dispersing billiards. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Flows, maps and recurrence

U1.P1. [F] For rotations of the circle, toral automorphisms, and the logistic map, identify invariant sets, periodic points, recurrence behavior, and natural invariant measures where known.

U1.P2. [T] Prove Poincare recurrence theorem from measure preservation and finite measure. Construct a dissipative example showing why finite invariant measure matters.

U1.P3. [S] Compare topological transitivity, minimality, ergodicity, and mixing on explicit rotations/shift systems; give implications and counterexamples.

Programme U2: Hyperbolic dynamics

U2.P1. [F] Compute stable/unstable eigenspaces for a hyperbolic toral automorphism and draw how a small rectangle evolves. Verify exponential contraction/expansion rates.

U2.P2. [T] Prove the stable manifold theorem at proof-architecture level from a graph transform/contraction scheme, identifying the norm and cone/invariance estimates.

U2.P3. [S] Construct a Markov partition for a simple hyperbolic map at a schematic level and explain how symbolic dynamics encodes orbits.

Programme U3: Entropy and nonuniform hyperbolicity

U3.P1. [F] Compute Kolmogorov-Sinai entropy for Bernoulli shifts and compare with zero entropy of an irrational rotation. Verify entropy of a finite partition under refinement.

U3.P2. [T] State and apply Oseledets theorem to a constant/random matrix cocycle example; compute Lyapunov exponents explicitly where possible.

U3.P3. [R] Write a theorem audit for Pesin theory: hypotheses, local stable manifolds, entropy formula context, and how nonuniform hyperbolicity differs from Anosov uniformity.

Programme U4: Hamiltonian dynamics and integrability

U4.P1. [F] Put the harmonic oscillator and pendulum in Hamiltonian form, find integrals, action-angle variables in the integrable region, and phase portraits.

U4.P2. [T] Verify symplectic invariance of Hamiltonian flow and derive Poisson brackets for first integrals. Check involution in a simple integrable system.

U4.P3. [S] State Liouville-Arnold precisely and apply it to an explicit integrable two-degree-of-freedom model at the level of invariant tori/actions.

Programme U5: Small divisors and KAM

U5.P1. [F] Solve a cohomological equation u(x+alpha)-u(x)=f(x) by Fourier series and identify the small divisors |e^{2pi i k alpha}-1|. Compare Diophantine and Liouville alpha.

U5.P2. [T] Prove a quantitative bound for the cohomological equation assuming a Diophantine condition on alpha and smooth/analytic decay of f-hat.

U5.P3. [R] Build the KAM iteration architecture for a near-integrable Hamiltonian: linearized equation, loss of derivatives/small divisors, parameter exclusion, quadratic convergence, surviving Cantor set.

Programme U6: Billiards and hyperbolic examples

U6.P1. [F] Derive the billiard map for a circular/elliptical billiard and identify conserved quantities/invariant curves in integrable cases.

U6.P2. [T] For dispersing Sinai billiards, compute how curvature of wave fronts evolves across free flight/reflection in a model setup and relate this to hyperbolicity.

U6.P3. [R] Compare smooth hyperbolic systems and billiards with singularities: identify which standard Anosov arguments need modification because of grazing collisions/discontinuity sets.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Smooth Dynamics, KAM Theory and Billiards: start from “Flows, maps and recurrence”, pass through “Hamiltonian dynamics and integrability”, and finish at “Billiards and hyperbolic examples”. Use the named results “Poincare recurrence theorem” and “exponential/statistical properties under stronger hypotheses.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Prepares Volume 20 homogeneous dynamics and supplies the smooth-dynamics background for Zimmer rigidity and entropy methods. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Poincare recurrence theorem; recurrence implies no wandering set of positive finite measure; local linear stability criteria; Hamilton equations from symplectic form; Liouville theorem on volume preservation. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: persistence of a positive-measure Cantor family of invariant tori; Birkhoff billiard map is an area-preserving twist map for smooth strictly convex tables; existence/geometry of caustics in integrable examples; hyperbolicity and ergodicity of finite-horizon dispersing billiards; exponential/statistical properties under stronger hypotheses.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Flows, maps and recurrence” develops into “Billiards and hyperbolic examples”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 20. Ergodic Theory and Homogeneous Dynamics

Purpose: Develop ergodic theory of Lie-group actions through Ratner, Oppenheim, higher-rank measure rigidity, effective equidistribution and random walks on homogeneous spaces.

Prerequisites: Volumes 12, 14, 18-19; Volume 24 can be studied concurrently for arithmetic lattices.

Primary and secondary references: Einsiedler-Ward, Ergodic Theory with a View Towards Number Theory; Witte Morris, Introduction to Arithmetic Groups; Bekka-Mayer, Ergodic Theory and Topological Dynamics of Group Actions on Homogeneous Spaces; Ratner, selected original papers/surveys; Einsiedler-Katok-Lindenstrauss, selected papers/lecture notes; Benoist-Quint, Random Walks on Reductive Groups; selected Strombergsson/Marklof/Yang papers

Quarter output: Prepare a 25-30 page research roadmap linking Ratner -> Oppenheim -> EKL -> effective unipotent dynamics -> Benoist-Quint, with a dependency graph of every theorem used.

Research bridges: Central preparation for rigidity, Diophantine dynamics, effective homogeneous dynamics and the Zimmer/Benoist-Quint directions.

Six two-week study units

Unit 1 (Weeks 1-2): Ergodic-theory core

Topics: Measure-preserving actions; ergodicity; mixing; conditional expectation; factors; joinings; ergodic decomposition.

Key definitions: measure-preserving action; ergodic/mixing/weak mixing; invariant sigma-algebra; factor; joining; generic point.

Key results to learn:

- von Neumann mean ergodic theorem [PROOF/ARCH]
- Birkhoff pointwise ergodic theorem [ARCH]
- ergodic decomposition [ARCH/USE]
- equivalences of weak mixing [ARCH].

Reading route: Einsiedler-Ward early chapters; supplement with Walters or Petersen.

Warm-up/source exercise focus (secondary): Prove ergodicity/mixing for standard shifts/toral maps; compute invariant functions; analyze simple joinings.

Week 1: Ergodic-theory core - definitions and examples. Read: Einsiedler-Ward early chapters; supplement with Walters or Petersen.. Make explicit examples/nonexamples for measure-preserving action; ergodic/mixing/weak mixing; invariant sigma-algebra; factor; joining; generic point.. Work through Measure-preserving actions; ergodicity; mixing; conditional expectation; factors; joinings; ergodic decomposition.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Ergodic-theory core - theorem and technique pass. Master/audit: von Neumann mean ergodic theorem; Birkhoff pointwise ergodic theorem; ergodic decomposition. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Lie groups, lattices and arithmeticity background

Topics: Homogeneous spaces G/Gamma; Haar measure; lattices; arithmetic groups; recurrence/nondivergence; unipotent and semisimple flows.

Key definitions: lattice; arithmetic lattice; homogeneous measure; unipotent element/subgroup; horospherical subgroup; covolume.

Key results to learn:

- Borel density theorem [ARCH/USE]
- Borel-Harish-Chandra finite-volume theorem for arithmetic groups [ARCH/USE]
- Moore ergodicity theorem [ARCH/USE]
- Mahler compactness criterion [PROOF/ARCH].

Reading route: Witte Morris selected chapters; Einsiedler-Ward homogeneous-space chapters; Borel-Harish-Chandra/Borel density expositions.

Warm-up/source exercise focus (secondary): Verify Mahler criterion in SL2; identify unipotent flows; work examples relating rational structures to closed homogeneous orbits.

Week 3: Lie groups, lattices and arithmeticity background - definitions and examples. Read: Witte Morris selected chapters; Einsiedler-Ward homogeneous-space chapters; Borel-Harish-Chandra/Borel density expositions.. Make explicit examples/nonexamples for lattice; arithmetic lattice; homogeneous measure; unipotent element/subgroup; horospherical subgroup; covolume.. Work through Homogeneous spaces G/Gamma; Haar measure; lattices; arithmetic groups; recurrence/nondivergence; unipotent and semisimple flows.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Lie groups, lattices and arithmeticity background - theorem and technique pass. Master/audit: Borel density theorem; Borel-Harish-Chandra finite-volume theorem for arithmetic groups; Moore ergodicity theorem. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Ratner theory

Topics: Orbit closures, invariant measures and equidistribution for unipotent flows; polynomial divergence; shearing.

Key definitions: algebraic/homogeneous measure; unipotent-generated subgroup; generic point; polynomial divergence; Ratner property.

Key results to learn:

- Ratner measure-classification theorem [USE]
- Ratner orbit-closure theorem [USE]
- Ratner equidistribution theorem [USE]
- Dani-Margulis nondivergence as a key input [USE].

Reading route: Ratner surveys/original theorem statements; Einsiedler-Ward and expository notes on unipotent flows.

Warm-up/source exercise focus (secondary): State all three Ratner theorems precisely and show their logical implications in examples; analyze horocycle flow on SL2(R)/SL2(Z).

Week 5: Ratner theory - definitions and examples. Read: Ratner surveys/original theorem statements; Einsiedler-Ward and expository notes on unipotent flows.. Make explicit examples/nonexamples for algebraic/homogeneous measure; unipotent-generated subgroup; generic point; polynomial divergence; Ratner property.. Work through Orbit closures, invariant measures and equidistribution for unipotent flows; polynomial divergence; shearing.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Ratner theory - theorem and technique pass. Master/audit: Ratner measure-classification theorem; Ratner orbit-closure theorem; Ratner equidistribution theorem. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Oppenheim and linearization

Topics: Indefinite quadratic forms; SO(2,1)-orbits; unipotent subgroups; quantitative nondivergence; linearization.

Key definitions: irrational quadratic form; stabilizer; closed orbit; linearization method; quantitative nondivergence.

Key results to learn:

- Margulis theorem resolving Oppenheim for n>=3 [ARCH/USE]
- Dani-Margulis nondivergence [USE]
- connection between rational forms and closed algebraic orbits [ARCH/USE]
- linearization principle in standard applications [USE].

Reading route: Margulis/Oppenheim expositions; Eskin-Margulis-Mozes selected; Witte Morris background.

Warm-up/source exercise focus (secondary): Work out SO(2,1) inside SL3 explicitly; explain why unipotents generate the connected group; write an Oppenheim-to-dynamics dictionary.

Week 7: Oppenheim and linearization - definitions and examples. Read: Margulis/Oppenheim expositions; Eskin-Margulis-Mozes selected; Witte Morris background.. Make explicit examples/nonexamples for irrational quadratic form; stabilizer; closed orbit; linearization method; quantitative nondivergence.. Work through Indefinite quadratic forms; SO(2,1)-orbits; unipotent subgroups; quantitative nondivergence; linearization.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Oppenheim and linearization - theorem and technique pass. Master/audit: Margulis theorem resolving Oppenheim for n>=3; Dani-Margulis nondivergence; connection between rational forms and closed algebraic orbits. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Higher-rank measure rigidity

Topics: Diagonal actions; entropy; coarse Lyapunov foliations; conditional measures; measure rigidity.

Key definitions: higher rank; Cartan action; coarse Lyapunov exponent/foliation; entropy contribution; leafwise measure.

Key results to learn:

- Measure rigidity theorem of Einsiedler-Katok-Lindenstrauss for higher-rank diagonal actions in its standard arithmetic setting [USE]
- entropy contribution formula/principles [USE]
- low-entropy method architecture [USE].

Reading route: EKL papers and lecture notes; Einsiedler-Ward relevant chapters.

Warm-up/source exercise focus (secondary): Compute Lyapunov weights in SL3; trace entropy contributions; prepare a blackboard proof architecture for the EKL theorem without technical proofs.

Week 9: Higher-rank measure rigidity - definitions and examples. Read: EKL papers and lecture notes; Einsiedler-Ward relevant chapters.. Make explicit examples/nonexamples for higher rank; Cartan action; coarse Lyapunov exponent/foliation; entropy contribution; leafwise measure.. Work through Diagonal actions; entropy; coarse Lyapunov foliations; conditional measures; measure rigidity.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Higher-rank measure rigidity - theorem and technique pass. Master/audit: Measure rigidity theorem of Einsiedler-Katok-Lindenstrauss for higher-rank diagonal actions in its standard arithmetic setting; entropy contribution formula/principles; low-entropy method architecture [USE].. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Effective equidistribution and random walks

Topics: Effective horocycle/unipotent equidistribution; lattice-point applications; Lorentz gas; stationary measures; Benoist-Quint rigidity.

Key definitions: spectral gap; effective equidistribution; stationary measure; proximality; strong irreducibility; renewal theorem.

Key results to learn:

- Selected Strombergsson effective equidistribution theorem(s) [USE]
- Marklof-Strombergsson limit theorems for periodic Lorentz gas [USE]
- Benoist-Quint stationary-measure/equidistribution theorems [USE]
- selected higher-rank effective Ratner-type result of Lei Yang [USE].

Reading route: Selected original papers plus Benoist-Quint book/surveys; Strombergsson-Marklof surveys.

Warm-up/source exercise focus (secondary): Choose one theorem from each of the four strands and write exact hypotheses/conclusions; identify where spectral gaps, nondivergence and arithmetic enter.

Week 11: Effective equidistribution and random walks - definitions and examples. Read: Selected original papers plus Benoist-Quint book/surveys; Strombergsson-Marklof surveys.. Make explicit examples/nonexamples for spectral gap; effective equidistribution; stationary measure; proximality; strong irreducibility; renewal theorem.. Work through Effective horocycle/unipotent equidistribution; lattice-point applications; Lorentz gas; stationary measures; Benoist-Quint rigidity.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Effective equidistribution and random walks - theorem and technique pass. Master/audit: Selected Strombergsson effective equidistribution theorem(s); Marklof-Strombergsson limit theorems for periodic Lorentz gas; Benoist-Quint stationary-measure/equidistribution theorems. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Ergodic-theory core

U1.P1. [F] Prove mean ergodic theorem in a Hilbert-space formulation and Birkhoff ergodic theorem at proof-architecture level. Apply both to irrational rotation and a Bernoulli shift.

U1.P2. [T] Use Fourier series to prove ergodicity/unique ergodicity of an irrational rotation and compute time averages of trigonometric polynomials.

U1.P3. [C] Construct measure-preserving systems that are ergodic but not mixing and mixing systems that illustrate stronger correlation decay; state the distinctions precisely.

Programme U2: Lie groups, lattices and arithmeticity background

U2.P1. [F] Show SL_2(R)/SL_2(Z) parametrizes unimodular lattices in R^2 and identify the stabilizer of Z^2. Compute diagonal and unipotent actions on lattice bases.

U2.P2. [T] Prove unimodularity of a reductive Lie group in a concrete adjoint-determinant formulation and derive the criterion for a G-invariant measure on G/H in examples.

U2.P3. [S] Work through Borel density/Borel-Harish-Chandra in one arithmetic lattice example at theorem-application level: identify algebraic group, lattice, and conclusion.

Programme U3: Ratner theory

U3.P1. [F] Analyze horocycle orbits on SL_2(R)/SL_2(Z): identify periodic versus nonperiodic examples and compare with geodesic-flow behavior.

U3.P2. [T] State Ratner orbit-closure and measure-classification theorems precisely for unipotent flows. Test the conclusion against at least five homogeneous-space examples.

U3.P3. [R] Reconstruct the shearing principle behind Ratner's method in a toy SL_2 setting by comparing nearby unipotent orbits; identify how polynomial divergence replaces exponential hyperbolicity.

Programme U4: Oppenheim and linearization

U4.P1. [T] For an indefinite ternary quadratic form Q, identify SO(Q) inside SL_3(R), show its standard representation is the 3-dimensional irreducible representation of sl_2 in a model basis, and locate unipotent one-parameter subgroups.

U4.P2. [S] Translate Oppenheim into density of an H=SO(Q)-orbit in SL_3(R)/SL_3(Z): make the correspondence between lattice points, values Q(v), and orbit closure explicit.

U4.P3. [R] Build the linearization/nondivergence proof architecture used to exclude improper orbit closures. State exactly where arithmeticity/rational forms enter.

Programme U5: Higher-rank measure rigidity

U5.P1. [F] Compute entropy contributions for commuting diagonal maps on a torus or homogeneous space in a toy algebraic example.

U5.P2. [T] State a representative Einsiedler-Katok-Lindenstrauss measure-rigidity theorem with all rank/entropy/invariance assumptions. Produce examples showing why higher rank and positive entropy are substantive.

U5.P3. [R] Trace one entropy-to-unipotent-invariance mechanism at proof-architecture level: coarse Lyapunov foliations, conditional measures, entropy contribution, extra invariance.

Programme U6: Effective equidistribution and random walks

U6.P1. [F] For a random walk on a compact or finite homogeneous quotient, compute convolution powers and stationary measures in an explicit example.

U6.P2. [T] Derive how a spectral gap gives exponential mixing/equidistribution for mean-zero L^2 observables. Quantify the rate in terms of the second eigenvalue/operator norm.

U6.P3. [R] Compare Strombergsson-type effective equidistribution, Marklof-Strombergsson limit theorems, Lei Yang higher-rank effectiveness, and Benoist-Quint random-walk rigidity by writing for each: acting object, space, qualitative theorem, quantitative input.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Ergodic Theory and Homogeneous Dynamics: start from “Ergodic-theory core”, pass through “Oppenheim and linearization”, and finish at “Effective equidistribution and random walks”. Use the named results “von Neumann mean ergodic theorem” and “selected higher-rank effective Ratner-type result of Lei Yang.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Central preparation for rigidity, Diophantine dynamics, effective homogeneous dynamics and the Zimmer/Benoist-Quint directions. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: von Neumann mean ergodic theorem; Mahler compactness criterion. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: low-entropy method architecture; Selected Strombergsson effective equidistribution theorem(s); Marklof-Strombergsson limit theorems for periodic Lorentz gas; Benoist-Quint stationary-measure/equidistribution theorems; selected higher-rank effective Ratner-type result of Lei Yang.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Ergodic-theory core” develops into “Effective equidistribution and random walks”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 21. Geometric Group Theory, Amenability and Property (T)

Purpose: Learn groups through their large-scale geometry and unitary representations, culminating in amenability, Kazhdan property (T), property (tau) and the geometric origins of expander phenomena.

Prerequisites: Volumes 3, 7, 11, 18; basic probability helpful.

Primary and secondary references: Bridson-Haefliger, Metric Spaces of Non-Positive Curvature, selected parts; de la Harpe, Topics in Geometric Group Theory; Clara Loh, Geometric Group Theory lecture notes; Bekka-de la Harpe-Valette, Kazhdan's Property (T); Lubotzky, Discrete Groups, Expanding Graphs and Invariant Measures

Quarter output: Write a 20-page “from quasi-isometry to spectral rigidity” synthesis, including one worked Cayley-graph family and a theorem map connecting amenability, (T), (tau), random walks and expanders.

Research bridges: Direct prerequisite for Volume 29 Bourgain-Gamburd/super-approximation and useful for Zimmer rigidity, arithmetic groups and random walks.

Six two-week study units

Unit 1 (Weeks 1-2): Cayley geometry and quasi-isometry

Topics: Word metrics; Cayley graphs; quasi-isometries; group actions on metric spaces; coarse invariants.

Key definitions: finite generating set; word metric; Cayley graph; quasi-isometry; proper/cocompact action; coarse invariant.

Key results to learn:

- Independence of quasi-isometry type from finite generating set [PROOF]
- Milnor-Schwarz/Milnor-Svarc lemma [PROOF/ARCH]
- growth type is a quasi-isometry invariant [PROOF/ARCH].

Reading route: de la Harpe Chs. I-IV; Loh introductory notes.

Warm-up/source exercise focus (secondary): Compute Cayley graphs/growth for Z^d, free groups, Heisenberg group; prove quasi-isometry examples and nonexamples.

Week 1: Cayley geometry and quasi-isometry - definitions and examples. Read: de la Harpe Chs. I-IV; Loh introductory notes.. Make explicit examples/nonexamples for finite generating set; word metric; Cayley graph; quasi-isometry; proper/cocompact action; coarse invariant.. Work through Word metrics; Cayley graphs; quasi-isometries; group actions on metric spaces; coarse invariants.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Cayley geometry and quasi-isometry - theorem and technique pass. Master/audit: Independence of quasi-isometry type from finite generating set; Milnor-Schwarz/Milnor-Svarc lemma; growth type is a quasi-isometry invariant [PROOF/ARCH].. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Trees, splittings and growth

Topics: Groups acting on trees; graphs of groups; free products/HNN extensions; ends; polynomial/exponential growth.

Key definitions: Bass-Serre tree; graph of groups; amalgamated free product; HNN extension; end of a group; growth function.

Key results to learn:

- Bass-Serre correspondence [ARCH/USE]
- Stallings theorem on ends, statement and examples [USE]
- Gromov theorem on polynomial growth, exact statement and nilpotent conclusion [USE].

Reading route: Serre, Trees selected; de la Harpe growth chapters; Bridson-Haefliger selected.

Warm-up/source exercise focus (secondary): Construct Bass-Serre trees for free products; compute ends; compare growth in abelian/free/nilpotent groups.

Week 3: Trees, splittings and growth - definitions and examples. Read: Serre, Trees selected; de la Harpe growth chapters; Bridson-Haefliger selected.. Make explicit examples/nonexamples for Bass-Serre tree; graph of groups; amalgamated free product; HNN extension; end of a group; growth function.. Work through Groups acting on trees; graphs of groups; free products/HNN extensions; ends; polynomial/exponential growth.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Trees, splittings and growth - theorem and technique pass. Master/audit: Bass-Serre correspondence; Stallings theorem on ends, statement and examples; Gromov theorem on polynomial growth, exact statement and nilpotent conclusion [USE].. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Hyperbolic and CAT(0) groups

Topics: Gromov hyperbolicity; boundaries; quasi-geodesics; CAT(0) geometry; rank-one behavior.

Key definitions: delta-hyperbolic space; Gromov product; boundary; quasi-geodesic; CAT(0) inequality; visual boundary.

Key results to learn:

- Morse lemma [PROOF/ARCH]
- boundary extension under quasi-isometries [ARCH]
- thin-triangle characterizations [PROOF/ARCH]
- Cartan fixed-point theorem in CAT(0) spaces [ARCH].

Reading route: Bridson-Haefliger hyperbolic/CAT(0) chapters; de la Harpe selected.

Warm-up/source exercise focus (secondary): Prove tree hyperbolicity; calculate boundaries of free groups; work CAT(0) product and fixed-point examples.

Week 5: Hyperbolic and CAT(0) groups - definitions and examples. Read: Bridson-Haefliger hyperbolic/CAT(0) chapters; de la Harpe selected.. Make explicit examples/nonexamples for delta-hyperbolic space; Gromov product; boundary; quasi-geodesic; CAT(0) inequality; visual boundary.. Work through Gromov hyperbolicity; boundaries; quasi-geodesics; CAT(0) geometry; rank-one behavior.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Hyperbolic and CAT(0) groups - theorem and technique pass. Master/audit: Morse lemma; boundary extension under quasi-isometries; thin-triangle characterizations. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Amenability

Topics: Invariant means; Folner sets; paradoxical decompositions; random walks; spectral radius.

Key definitions: amenable group; invariant mean; Folner sequence; paradoxical decomposition; return probability; spectral radius.

Key results to learn:

- Folner criterion [PROOF/ARCH]
- closure properties of amenability [PROOF/ARCH]
- Tarski alternative [USE]
- Kesten amenability/spectral-radius criterion for symmetric random walks [ARCH/USE].

Reading route: Bekka-de la Harpe-Valette preliminaries; de la Harpe amenability sections; Woess selected.

Warm-up/source exercise focus (secondary): Construct Folner sequences in Z^d/solvable examples; prove free group nonamenability; compute random-walk spectral radii in simple cases.

Week 7: Amenability - definitions and examples. Read: Bekka-de la Harpe-Valette preliminaries; de la Harpe amenability sections; Woess selected.. Make explicit examples/nonexamples for amenable group; invariant mean; Folner sequence; paradoxical decomposition; return probability; spectral radius.. Work through Invariant means; Folner sets; paradoxical decompositions; random walks; spectral radius.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Amenability - theorem and technique pass. Master/audit: Folner criterion; closure properties of amenability; Tarski alternative. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Kazhdan property (T)

Topics: Almost invariant vectors; Kazhdan pairs/constants; affine isometric actions; cohomology; permanence.

Key definitions: unitary representation; almost invariant vector; Kazhdan pair; Kazhdan constant; property (FH); 1-cocycle/coboundary.

Key results to learn:

- Equivalent formulations of property (T) [ARCH]
- Delorme-Guichardet theorem (T) <-> (FH) for sigma-compact locally compact groups, standard hypotheses [ARCH/USE]
- property (T) for SL_n(R), n>=3, and inheritance by lattices [USE]
- finite generation/finite abelianization consequences for discrete groups [ARCH].

Reading route: Bekka-de la Harpe-Valette Chs. 1-3 and classical examples.

Warm-up/source exercise focus (secondary): Compute Kazhdan behavior in finite groups; prove basic permanence; explain why Z fails (T); trace a proof strategy for higher-rank groups.

Week 9: Kazhdan property (T) - definitions and examples. Read: Bekka-de la Harpe-Valette Chs. 1-3 and classical examples.. Make explicit examples/nonexamples for unitary representation; almost invariant vector; Kazhdan pair; Kazhdan constant; property (FH); 1-cocycle/coboundary.. Work through Almost invariant vectors; Kazhdan pairs/constants; affine isometric actions; cohomology; permanence.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Kazhdan property (T) - theorem and technique pass. Master/audit: Equivalent formulations of property (T); Delorme-Guichardet theorem (T) <-> (FH) for sigma-compact locally compact groups, standard hypotheses; property (T) for SL_n(R), n>=3, and inheritance by lattices. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Property (tau), spectral gaps and expanders

Topics: Finite quotients; congruence families; spectral gap; Cheeger constants; expander Cayley graphs.

Key definitions: property (tau); Schreier graph; normalized Laplacian; spectral gap; Cheeger constant; expander family.

Key results to learn:

- Property (T) implies property (tau) for quotient families [PROOF/ARCH]
- Cheeger inequalities for regular graphs [PROOF/ARCH]
- uniform spectral gap equivalent to expansion in bounded-degree regular families [PROOF/ARCH]
- Selberg 3/16 theorem as a prototypical arithmetic spectral gap [USE].

Reading route: Lubotzky core chapters; Bekka-de la Harpe-Valette applications; Hoory-Linial-Wigderson survey selected.

Warm-up/source exercise focus (secondary): Derive spectral gaps for toy Cayley graphs; prove the easy Cheeger direction; compare (T), (tau) and expansion in a diagram.

Week 11: Property (tau), spectral gaps and expanders - definitions and examples. Read: Lubotzky core chapters; Bekka-de la Harpe-Valette applications; Hoory-Linial-Wigderson survey selected.. Make explicit examples/nonexamples for property (tau); Schreier graph; normalized Laplacian; spectral gap; Cheeger constant; expander family.. Work through Finite quotients; congruence families; spectral gap; Cheeger constants; expander Cayley graphs.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Property (tau), spectral gaps and expanders - theorem and technique pass. Master/audit: Property (T) implies property (tau) for quotient families; Cheeger inequalities for regular graphs; uniform spectral gap equivalent to expansion in bounded-degree regular families. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Cayley geometry and quasi-isometry

- U1.P1. [F] Draw/describe Cayley graphs and compute word metrics/growth for Z, Z^2, F_2, D_infinity, and the discrete Heisenberg group with standard generators.
- U1.P2. [T] Prove that word metrics from two finite generating sets are quasi-isometric. Then use the Milnor-Svarc lemma to compare a cocompact group action with the ambient metric space in a concrete example.
- U1.P3. [S] Use growth to prove Z^2 is not quasi-isometric to F_2. Identify a second coarse invariant that distinguishes another pair of groups.

Programme U2: Trees, splittings and growth

- U2.P1. [F] Construct Bass-Serre trees for Z/2 * Z/3 and an HNN extension such as BS(1,2). Compute vertex/edge stabilizers and quotient graph of groups.
- U2.P2. [T] Use normal forms to prove a simple free-product/amalgam statement and explain the Bass-Serre correspondence in that example.
- U2.P3. [R] State Stallings' theorem on ends and Gromov's polynomial-growth theorem precisely. Test both on Z^d, free groups, nilpotent groups, and finite extensions.

Programme U3: Hyperbolic and CAT(0) groups

- U3.P1. [F] Prove trees are 0-hyperbolic and show Z^2 with its word metric is not Gromov hyperbolic by constructing fat geodesic triangles.
- U3.P2. [T] Compute the Gromov boundary of a regular tree/free group and describe the convergence topology. Relate boundary cylinders to infinite reduced words.
- U3.P3. [S] Compare hyperbolic and CAT(0) geometry on trees, hyperbolic plane, Euclidean plane, and product spaces. Give examples showing neither class contains the other in full generality.

Programme U4: Amenability

- U4.P1. [F] Construct Folner sequences for Z^d and prove nonamenability of F_2 via boundary expansion/paradoxical decomposition in a selected formulation.
- U4.P2. [T] Prove equivalence of two standard amenability criteria in the discrete finitely generated setting, such as Folner and invariant mean at theorem-architecture level.
- U4.P3. [S] Analyze simple random walk on Z^d versus F_2: return probabilities/spectral radius qualitatively, and explain how nonamenability manifests spectrally.

Programme U5: Kazhdan property (T)

- U5.P1. [F] Show Z does not have property (T) by exhibiting almost-invariant vectors in suitable one-dimensional/Fourier representations. Show finite groups have property (T).
- U5.P2. [T] Prove one standard equivalence for property (T): almost invariant vectors imply invariant vectors versus isolation of the trivial representation. Work through Kazhdan constants in a finite example.
- U5.P3. [S] Prove that a discrete amenable group with property (T) is finite. Explain the more general locally compact statement (amenable + property (T) forces compactness under the standard hypotheses) and identify which representation/mean argument encodes amenability.

Programme U6: Property (tau), spectral gaps and expanders

- U6.P1. [F] For a finite d-regular Cayley graph, derive the normalized adjacency/random-walk operator and relate eigenvalues to L^2 mixing of the walk.
- U6.P2. [T] Prove one direction of Cheeger's inequality for regular graphs and apply it to cycles, complete graphs, and a candidate expander family.
- U6.P3. [S] Show how property (tau) for a family of finite-index subgroups produces a uniform spectral gap/expander family of finite quotients. Make every identification between representation spaces and graph eigenfunctions explicit.

Cross-unit synthesis (choose one)

- S1. Build a 4-6 page dependency narrative for Geometric Group Theory, Amenability and Property (T): start from “Cayley geometry and quasi-isometry”, pass through “Amenability”, and finish at “Property (tau), spectral gaps and expanders”. Use the named results “Independence of quasi-isometry type from finite generating set” and “Selberg 3/16 theorem as a prototypical arithmetic spectral gap.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct prerequisite for Volume 29 Bourgain-Gamburd/super-approximation and useful for Zimmer rigidity, arithmetic groups and random walks. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Independence of quasi-isometry type from finite generating set; Milnor-Schwarz/Milnor-Svarc lemma; growth type is a quasi-isometry invariant; Morse lemma; thin-triangle characterizations. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Equivalent formulations of property (T); Delorme-Guichardet theorem (T) <-> (FH) for sigma-compact locally compact groups, standard hypotheses; property (T) for SL_n(R), n>=3, and inheritance by lattices; finite generation/finite abelianization consequences for discrete groups; Selberg 3/16 theorem as a prototypical arithmetic spectral gap.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Cayley geometry and quasi-isometry” develops into “Property (tau), spectral gaps and expanders”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 22. Elementary and Analytic Number Theory

Purpose: Build a rigorous classical number-theory spine from congruences and reciprocity through Dirichlet series, zeta/L-functions, the prime number theorem and primes in arithmetic progressions.

Prerequisites: Volumes 1-3, 6 and 8.

Primary and secondary references: Apostol, Introduction to Analytic Number Theory, Chs. 1-13; Davenport, Multiplicative Number Theory, early chapters; Montgomery-Vaughan, Multiplicative Number Theory I, selected early chapters; Hardy-Wright, An Introduction to the Theory of Numbers, selected classical chapters

Quarter output: Produce a 25-page “Euler product to primes in progressions” dossier containing exact statements of every analytic input and at least 60 solved exercises across the quarter.

Research bridges: Foundation for Volumes 23-30, arithmetic geometry, automorphic L-functions and analytic tools in additive combinatorics.

Six two-week study units

Unit 1 (Weeks 1-2): Congruences, reciprocity and continued fractions

Topics: Divisibility; CRT; finite multiplicative groups; quadratic residues; continued fractions; Pell equations.

Key definitions: congruence; reduced residue system; Legendre/Jacobi symbol; primitive root; simple continued fraction; convergent; Pell equation.

Key results to learn:

- Chinese remainder theorem [PROOF]
- Fermat/Euler theorems [PROOF]
- quadratic reciprocity plus supplements [PROOF/ARCH]
- best-approximation property of convergents [PROOF]
- periodic continued fractions for quadratic irrationals [ARCH].

Reading route: Ireland-Rosen or Hardy-Wright classical chapters; Apostol Chs. 1-2 for arithmetic preliminaries.

Warm-up/source exercise focus (secondary): Compute reciprocity examples; solve several Pell equations; prove continued-fraction error estimates.

Week 1: Congruences, reciprocity and continued fractions - definitions and examples. Read: Ireland-Rosen or Hardy-Wright classical chapters; Apostol Chs. 1-2 for arithmetic preliminaries.. Make explicit examples/nonexamples for congruence; reduced residue system; Legendre/Jacobi symbol; primitive root; simple continued fraction; convergent; Pell equation.. Work through Divisibility; CRT; finite multiplicative groups; quadratic residues; continued fractions; Pell equations.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Congruences, reciprocity and continued fractions - theorem and technique pass. Master/audit: Chinese remainder theorem; Fermat/Euler theorems; quadratic reciprocity plus supplements. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Arithmetic functions and convolution

Topics: Multiplicative functions; Dirichlet convolution; Mobius inversion; divisor sums; average orders.

Key definitions: multiplicative/completely multiplicative function; Dirichlet convolution; Mobius function; divisor functions; summatory function.

Key results to learn:

- Mobius inversion [PROOF]
- convolution algebra identities [PROOF]
- Euler product criterion for multiplicative coefficients [PROOF]
- average orders of phi, d(n), mu-related elementary estimates [ARCH].

Reading route: Apostol Chs. 2-4.

Warm-up/source exercise focus (secondary): Derive convolution identities; calculate summatory functions by hyperbola method; prove average-order estimates.

Week 3: Arithmetic functions and convolution - definitions and examples. Read: Apostol Chs. 2-4.. Make explicit examples/nonexamples for multiplicative/completely multiplicative function; Dirichlet convolution; Mobius function; divisor functions; summatory function.. Work through Multiplicative functions; Dirichlet convolution; Mobius inversion; divisor sums; average orders.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Arithmetic functions and convolution - theorem and technique pass. Master/audit: Mobius inversion; convolution algebra identities; Euler product criterion for multiplicative coefficients. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Dirichlet series and zeta

Topics: Dirichlet series; Euler products; partial summation; zeta in Re(s)>1; analytic continuation/functional equation roadmap.

Key definitions: Dirichlet series; abscissa of convergence; Euler product; Mangoldt function; Mellin transform.

Key results to learn:

- Euler product for zeta [PROOF]
- partial/Abel summation [PROOF]
- identities for -zeta'/zeta [PROOF]
- meromorphic continuation and functional equation of zeta [ARCH/USE].

Reading route: Apostol analytic chapters; Davenport Chs. 1-3; complex-analysis cross-reference Volume 8.

Warm-up/source exercise focus (secondary): Manipulate Dirichlet series; derive Euler products; use partial summation to translate counting estimates.

Week 5: Dirichlet series and zeta - definitions and examples. Read: Apostol analytic chapters; Davenport Chs. 1-3; complex-analysis cross-reference Volume 8.. Make explicit examples/nonexamples for Dirichlet series; abscissa of convergence; Euler product; Mangoldt function; Mellin transform.. Work through Dirichlet series; Euler products; partial summation; zeta in Re(s)>1; analytic continuation/functional equation roadmap.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Dirichlet series and zeta - theorem and technique pass. Master/audit: Euler product for zeta; partial/Abel summation; identities for -zeta'/zeta. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Prime number theorem

Topics: Chebyshev functions; zero-free boundary; Tauberian/complex-analytic proof architecture.

Key definitions: pi(x); theta(x); psi(x); zero-free line; explicit formula; Tauberian theorem.

Key results to learn:

- Chebyshev estimates [PROOF/ARCH]
- equivalence formulations of PNT [PROOF]
- zeta(s) !=0 on Re(s)=1 [ARCH]
- prime number theorem psi(x)~x [ARCH]
- Wiener-Ikehara route as an alternate architecture [USE].

Reading route: Davenport PNT chapters; Apostol PNT chapter; optional Newman proof exposition.

Warm-up/source exercise focus (secondary): Prove Chebyshev bounds; write both classical complex and Tauberian proof maps; derive pi(x)~x/log x from psi(x)~x.

Week 7: Prime number theorem - definitions and examples. Read: Davenport PNT chapters; Apostol PNT chapter; optional Newman proof exposition.. Make explicit examples/nonexamples for pi(x); theta(x); psi(x); zero-free line; explicit formula; Tauberian theorem.. Work through Chebyshev functions; zero-free boundary; Tauberian/complex-analytic proof architecture.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Prime number theorem - theorem and technique pass. Master/audit: Chebyshev estimates; equivalence formulations of PNT; zeta(s) !=0 on Re(s)=1. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Characters and Dirichlet L-functions

Topics: Characters mod q; orthogonality; Gauss sums; primitive/imprimitive characters; L-functions.

Key definitions: Dirichlet character; conductor; primitive character; Gauss sum; Dirichlet L-function; generalized Bernoulli number.

Key results to learn:

- Character orthogonality [PROOF]
- |tau(chi)|=sqrt(q) for primitive chi under standard hypotheses [PROOF/ARCH]
- Euler product for L(s,chi) [PROOF]
- analytic continuation and functional equation [ARCH]
- nonvanishing L(1,chi) [ARCH/USE].

Reading route: Davenport character/L-function chapters; Apostol Chs. 6-12 selected.

Warm-up/source exercise focus (secondary): Build character tables; compute Gauss sums; derive Euler products and functional equations in basic parity cases.

Week 9: Characters and Dirichlet L-functions - definitions and examples. Read: Davenport character/L-function chapters; Apostol Chs. 6-12 selected.. Make explicit examples/nonexamples for Dirichlet character; conductor; primitive character; Gauss sum; Dirichlet L-function; generalized Bernoulli number.. Work through Characters mod q; orthogonality; Gauss sums; primitive/imprimitive characters; L-functions.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Characters and Dirichlet L-functions - theorem and technique pass. Master/audit: Character orthogonality; |tau(chi)|=sqrt(q) for primitive chi under standard hypotheses; Euler product for L(s,chi). Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Primes in arithmetic progressions

Topics: Dirichlet theorem; PNT in progressions for fixed modulus; zero-free regions; exceptional zeros concept.

Key definitions: reduced residue class; zero-free region; exceptional/Siegel zero; logarithmic density.

Key results to learn:

- Dirichlet theorem on primes in arithmetic progressions [ARCH]

- PNT for fixed q, pi(x
- q,a)~Li(x)/phi(q) [ARCH/USE]
- classical zero-free region for zeta and Dirichlet L-functions [USE]
- de la Vallee Poussin-type error term [USE].

Reading route: Davenport later classical chapters; Montgomery-Vaughan selected.

Warm-up/source exercise focus (secondary): Derive character decomposition of psi(x;q,a); isolate principal/nonprincipal contributions; write a complete dependency chain from L-function facts to PNT-AP.

Week 11: Primes in arithmetic progressions - definitions and examples. Read: Davenport later classical chapters; Montgomery-Vaughan selected.. Make explicit examples/nonexamples for reduced residue class; zero-free region; exceptional/Siegel zero; logarithmic density.. Work through Dirichlet theorem; PNT in progressions for fixed modulus; zero-free regions; exceptional zeros concept.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Primes in arithmetic progressions - theorem and technique pass. Master/audit: Dirichlet theorem on primes in arithmetic progressions; PNT for fixed q, pi(x; q,a)~Li(x)/phi(q). Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Congruences, reciprocity and continued fractions

U1.P1. [F] Use quadratic reciprocity to determine whether 2,3,5,7 are quadratic residues modulo a curated set of odd primes. Organize the calculations so reciprocity reduces rather than expands the work.

U1.P2. [T] Compute continued fractions of sqrt(2), sqrt(3), sqrt(5), and sqrt(13), derive Pell fundamental solutions, and prove the recurrence generates all positive solutions in one case.

U1.P3. [S] Prove continued-fraction convergents are best approximants in a standard sense and apply the result to bound ||q alpha|| for a quadratic irrational.

Programme U2: Arithmetic functions and convolution

U2.P1. [F] Compute Dirichlet convolutions among 1, mu, phi, tau and selected multiplicative functions; verify multiplicativity from prime-power data.

U2.P2. [T] Prove Mobius inversion in arithmetic-function and summatory forms. Use it to count coprime pairs in [1,N]^2 up to a controllable error.

U2.P3. [S] Derive the Dirichlet hyperbola estimate sum_{n<=x} d(n)=x log x+(2gamma-1)x+O(sqrt x), including the geometric lattice-point interpretation.

Programme U3: Dirichlet series and zeta

U3.P1. [T] Starting from absolute convergence for Re(s)>1, derive the Euler product for zeta and logarithmic derivative -zeta'/zeta=sum Lambda(n)n^{-s}. State every interchange justification.

U3.P2. [F] Prove Euler's divergence of sum_p 1/p using the Euler product/finite products and compare it with divergence of sum 1/n.

U3.P3. [S] Use Mellin transforms/partial summation to translate between summatory arithmetic functions and Dirichlet-series behavior in at least two explicit examples.

Programme U4: Prime number theorem

U4.P1. [F] Prove Chebyshev upper/lower bounds for pi(x) using binomial coefficients or theta/psi estimates. Make constants explicit enough to see pi(x) is of order x/log x.

U4.P2. [T] Prove equivalence psi(x)~x iff pi(x)~x/log x by partial summation and prime-power error estimates.

U4.P3. [R] Reconstruct the complex-analytic PNT chain: meromorphic continuation/zero-free line for zeta, Wiener-Ikehara or classical contour method, psi(x)~x. Prove the elementary steps and audit the analytic input precisely.

Programme U5: Characters and Dirichlet L-functions

U5.P1. [F] Construct all Dirichlet characters modulo 5, 8, and 12; separate primitive/imprimitive characters and verify orthogonality relations.

U5.P2. [T] Compute quadratic Gauss sums and prove |tau(chi)|=sqrt(p) for the Legendre character. Track the role of character orthogonality.

U5.P3. [S] Derive the Euler product and functional-equation ingredients for L(s,chi) in one primitive parity case, making clear which complex-analysis results are imported.

Programme U6: Primes in arithmetic progressions

U6.P1. [T] Derive the character identity for the indicator n=a mod q and use it to reduce prime counting in progressions to estimates for sums chi(n)Lambda(n).

U6.P2. [S] Prove Dirichlet's theorem on primes in arithmetic progressions following a selected proof route through L(1,chi) != 0; identify separately the real and complex character cases.

U6.P3. [R] Compare qualitative Dirichlet, PNT in arithmetic progressions, Siegel-Walfisz, and Bombieri-Vinogradov by writing exact quantifier ranges in q and error strengths.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Elementary and Analytic Number Theory: start from “Congruences, reciprocity and continued fractions”, pass through “Prime number theorem”, and finish at “Primes in arithmetic progressions”. Use the named results “Chinese remainder theorem” and “de la Vallee Poussin-type error term.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Foundation for Volumes 23-30, arithmetic geometry, automorphic L-functions and analytic tools in additive combinatorics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Chinese remainder theorem; Fermat/Euler theorems; quadratic reciprocity plus supplements; best-approximation property of convergents; Mobius inversion. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: nonvanishing L(1,chi); Dirichlet theorem on primes in arithmetic progressions; q,a)~Li(x)/phi(q); classical zero-free region for zeta and Dirichlet L-functions; de la Vallee Poussin-type error term.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Congruences, reciprocity and continued fractions” develops into “Primes in arithmetic progressions”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 23. Advanced Analytic Number Theory

Purpose: Acquire the core research toolkit of sieves, exponential sums, circle method, mean values of L-functions and spectral summation.

Prerequisites: Volumes 9 and 22; Volume 30 is helpful for the final unit but may follow later.

Primary and secondary references: Iwaniec-Kowalski, Analytic Number Theory, selected chapters throughout; Vaughan, The Hardy-Littlewood Method; Friedlander-Iwaniec, Opera de Cribro, selected chapters; Titchmarsh, Theory of the Riemann Zeta-function, selected chapters; Montgomery, Topics in Multiplicative Number Theory, selected chapters

Quarter output: Write a 30-page methods atlas comparing sieve, large sieve, exponential sums, circle method and spectral methods on at least five classical prime/Diophantine problems.

Research bridges: Feeds Bourgain-decoupling/VMVT, automorphic forms, Sato-Tate analytic inputs and modern prime-distribution research.

Six two-week study units

Unit 1 (Weeks 1-2): Combinatorial and Selberg sieves

Topics: Sifting; sieve dimension; upper/lower bound sieves; Selberg weights; fundamental lemma; parity problem.

Key definitions: sifting function; sieve dimension; level of distribution; beta-sieve; Selberg weight; parity obstruction.

Key results to learn:

- Brun sieve upper bounds [ARCH]
- Selberg upper-bound sieve [ARCH]
- fundamental lemma of sieve theory [USE]
- Brun theorem on twin-prime sum/convergence [ARCH]
- parity barrier principle [ARCH/USE].

Reading route: Iwaniec-Kowalski sieve chapters; Opera de Cribro introductory chapters.

Warm-up/source exercise focus (secondary): Build Selberg weights in a model sequence; prove upper bound for almost-prime counts; explain parity obstruction with examples.

Week 1: Combinatorial and Selberg sieves - definitions and examples. Read: Iwaniec-Kowalski sieve chapters; Opera de Cribro introductory chapters.. Make explicit examples/nonexamples for sifting function; sieve dimension; level of distribution; beta-sieve; Selberg weight; parity obstruction.. Work through Sifting; sieve dimension; upper/lower bound sieves; Selberg weights; fundamental lemma; parity problem.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Combinatorial and Selberg sieves - theorem and technique pass. Master/audit: Brun sieve upper bounds; Selberg upper-bound sieve; fundamental lemma of sieve theory. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Large sieve and distribution of primes

Topics: Large-sieve inequalities; duality; Barban-Davenport-Halberstam viewpoint; Bombieri-Vinogradov.

Key definitions: large-sieve constant; dual large sieve; level of distribution; discrepancy in progressions.

Key results to learn:

- Large sieve inequality [PROOF/ARCH]
- duality principle [PROOF]
- Brun-Titchmarsh inequality [ARCH]
- Bombieri-Vinogradov theorem [ARCH/USE].

Reading route: Iwaniec-Kowalski large sieve/distribution chapters; Montgomery-Vaughan.

Warm-up/source exercise focus (secondary): Prove a standard large-sieve form; derive an average result for primes in progressions; compare GRH versus Bombieri-Vinogradov strength.

Week 3: Large sieve and distribution of primes - definitions and examples. Read: Iwaniec-Kowalski large sieve/distribution chapters; Montgomery-Vaughan.. Make explicit examples/nonexamples for large-sieve constant; dual large sieve; level of distribution; discrepancy in progressions.. Work through Large-sieve inequalities; duality; Barban-Davenport-Halberstam viewpoint; Bombieri-Vinogradov.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Large sieve and distribution of primes - theorem and technique pass. Master/audit: Large sieve inequality; duality principle; Brun-Titchmarsh inequality. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Exponential and character sums

Topics: Weyl differencing; van der Corput; completion; character sums; Burgess method.

Key definitions: exponential sum; differencing operator; exponent pair; complete/incomplete character sum.

Key results to learn:

- Weyl inequality [PROOF/ARCH]
- van der Corput differencing inequalities [PROOF/ARCH]
- Polya-Vinogradov [ARCH]
- Burgess character-sum bound in a standard formulation [USE]

- Weil bound for Kloosterman sums as an input [USE].

Reading route: Iwaniec-Kowalski exponential/character sum chapters; Vaughan preliminaries.

Warm-up/source exercise focus (secondary): Estimate polynomial Weyl sums; carry out completion; compare Polya-Vinogradov and Burgess ranges.

Week 5: Exponential and character sums - definitions and examples. Read: Iwaniec-Kowalski exponential/character sum chapters; Vaughan preliminaries.. Make explicit examples/nonexamples for exponential sum; differencing operator; exponent pair; complete/incomplete character sum.. Work through Weyl differencing; van der Corput; completion; character sums; Burgess method.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Exponential and character sums - theorem and technique pass. Master/audit: Weyl inequality; van der Corput differencing inequalities; Polya-Vinogradov. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Circle method and Vinogradov methods

Topics: Major/minor arcs; singular series/integral; Waring; Vinogradov mean values; primes as sums.

Key definitions: major arc; minor arc; singular series; singular integral; Vinogradov mean value J_{s,k}(X).

Key results to learn:

- Hardy-Littlewood circle-method asymptotic in a standard model problem [ARCH]
- Waring-type consequence [ARCH/USE]
- Vinogradov three-primes theorem architecture [USE]
- main conjecture in VMVT [USE], with later decoupling proof cross-reference Volume 35.

Reading route: Vaughan entire central route; Iwaniec-Kowalski circle-method sections.

Warm-up/source exercise focus (secondary): Carry out major/minor arcs for a toy equation; compute a singular series; state sharp VMVT and explain consequences.

Week 7: Circle method and Vinogradov methods - definitions and examples. Read: Vaughan entire central route; Iwaniec-Kowalski circle-method sections.. Make explicit examples/nonexamples for major arc; minor arc; singular series; singular integral; Vinogradov mean value J_{s,k}(X).. Work through Major/minor arcs; singular series/integral; Waring; Vinogradov mean values; primes as sums.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Circle method and Vinogradov methods - theorem and technique pass. Master/audit: Hardy-Littlewood circle-method asymptotic in a standard model problem; Waring-type consequence; Vinogradov three-primes theorem architecture. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Zeta and L-function mean values

Topics: Approximate functional equations; zero density; moments; convexity; zero-free/zero-density techniques.

Key definitions: approximate functional equation; conductor; convexity bound; subconvexity; zero-density estimate; mollifier.

Key results to learn:

- Approximate functional equation for zeta/Dirichlet L [ARCH]
- second moment of zeta on critical line [ARCH]
- convexity bound [ARCH]
- representative zero-density theorem [USE]
- density-to-prime-error mechanisms [USE].

Reading route: Titchmarsh selected; Iwaniec-Kowalski L-function chapters; Montgomery selected.

Warm-up/source exercise focus (secondary): Derive an approximate functional equation in a simple setting; compute second-moment diagonal; map convexity via Phragmen-Lindelof.

Week 9: Zeta and L-function mean values - definitions and examples. Read: Titchmarsh selected; Iwaniec-Kowalski L-function chapters; Montgomery selected.. Make explicit examples/nonexamples for approximate functional equation; conductor; convexity bound; subconvexity; zero-density estimate; mollifier.. Work through Approximate functional equations; zero density; moments; convexity; zero-free/zero-density techniques.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Zeta and L-function mean values - theorem and technique pass. Master/audit: Approximate functional equation for zeta/Dirichlet L; second moment of zeta on critical line; convexity bound. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Spectral methods and trace formulae

Topics: Automorphic spectral decomposition; Kloosterman sums; Petersson/Kuznetsov; Rankin-Selberg; subconvexity philosophy.

Key definitions: Kloosterman sum; Petersson trace; Kuznetsov formula; automorphic conductor; amplifier.

Key results to learn:

- Weil bound for Kloosterman sums [USE]
- Petersson trace formula [ARCH/USE]
- Kuznetsov trace formula [USE]
- Rankin-Selberg mean-value mechanism [USE]
- representative subconvexity theorem and amplification architecture [USE].

Reading route: Iwaniec-Kowalski spectral chapters; cross-read Bump/Goldfeld from Volume 30.

Warm-up/source exercise focus (secondary): Use Petersson formally to estimate a family average; identify geometric/spectral sides of Kuznetsov; prepare a subconvexity method comparison.

Week 11: Spectral methods and trace formulae - definitions and examples. Read: Iwaniec-Kowalski spectral chapters; cross-read Bump/Goldfeld from Volume 30.. Make explicit examples/nonexamples for Kloosterman sum; Petersson trace; Kuznetsov formula; automorphic conductor; amplifier.. Work through Automorphic spectral decomposition; Kloosterman sums; Petersson/Kuznetsov; Rankin-Selberg; subconvexity philosophy.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Spectral methods and trace formulae - theorem and technique pass. Master/audit: Weil bound for Kloosterman sums; Petersson trace formula; Kuznetsov trace formula. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Combinatorial and Selberg sieves

U1.P1. [F] Build the Eratosthenes/Legendre sieve for integers avoiding a finite set of prime divisors and compute exact upper/lower counts for a small z.

U1.P2. [T] Derive Selberg's upper-bound sieve weights for a simplified dimension-1 problem and use them to bound twin-prime-like sifted sets at the correct order of magnitude up to constants/log factors.

U1.P3. [C] Explain the parity problem by constructing a sieve situation where primes and semiprimes are indistinguishable to purely combinatorial inclusion-exclusion data.

Programme U2: Large sieve and distribution of primes

U2.P1. [T] Prove the large sieve inequality in a standard finite form or reconstruct its duality proof. Apply it to a family of character/exponential sums.

U2.P2. [S] Starting from a large-sieve/Bombieri-Vinogradov statement, derive a nontrivial average result for primes in arithmetic progressions over q<=Q. Track the level-of-distribution exponent.

U2.P3. [R] Compare Brun-Titchmarsh, Siegel-Walfisz, Bombieri-Vinogradov, and Elliott-Halberstam by exact ranges and averaging. Mark theorem versus conjecture.

Programme U3: Exponential and character sums

U3.P1. [F] Prove Weyl differencing for a polynomial exponential sum and derive a nontrivial bound for a quadratic phase. Compare with exact Gauss-sum evaluation when available.

U3.P2. [T] Derive van der Corput's A/B process in a selected form and apply it to one incomplete exponential sum.

U3.P3. [S] Prove a Polya-Vinogradov bound for character sums or study Burgess at proof-architecture level; identify what extra idea improves the exponent/range.

Programme U4: Circle method and Vinogradov methods

U4.P1. [F] Write the Hardy-Littlewood circle-method representation for the number of solutions of a Waring-type equation. Separate major/minor arcs and compute the formal singular series/integral in a toy case.

U4.P2. [T] Prove a minor-arc bound using Weyl's inequality sufficient for one classical Waring result in a simplified exponent range.

U4.P3. [R] State Vinogradov mean value theorem in modern sharp form and show algebraically how it improves Weyl-type bounds. Then connect to decoupling in Volume 35.

Programme U5: Zeta and L-function mean values

U5.P1. [F] Derive an approximate functional equation for zeta or a primitive Dirichlet L-function in a simplified symmetric form, explaining the truncation length near the square root of conductor.

U5.P2. [T] Compute the second moment of zeta on the critical line to the main-term level using Dirichlet polynomials and diagonal/off-diagonal separation in a standard route.

U5.P3. [R] Write the hierarchy convexity -> subconvexity -> Lindelof for a family of L-functions with conductor C. Give exact exponents in one GL(1) or GL(2) example and name the method behind a known subconvex result.

Programme U6: Spectral methods and trace formulae

U6.P1. [F] Derive the Petersson trace formula in a finite-dimensional holomorphic cusp-form setting at theorem-application level and compute the diagonal term for a small weight/level example.

U6.P2. [T] State the Kuznetsov formula with the roles of Kloosterman sums and Bessel transforms clearly identified. Use a simplified form to see spectral/geometric duality.

U6.P3. [R] Compare Petersson, Kuznetsov, Selberg trace, and Poisson/Voronoi summation: for each, list the spectral side, arithmetic/geometric side, and a typical analytic-number-theory application.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Advanced Analytic Number Theory: start from “Combinatorial and Selberg sieves”, pass through “Circle method and Vinogradov methods”, and finish at “Spectral methods and trace formulae”. Use the named results “Brun sieve upper bounds” and “representative subconvexity theorem and amplification architecture.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Feeds Bourgain-decoupling/VMVT, automorphic forms, Sato-Tate analytic inputs and modern prime-distribution research. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Large sieve inequality; duality principle; Weyl inequality; van der Corput differencing inequalities. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Weil bound for Kloosterman sums; Petersson trace formula; Kuznetsov trace formula; Rankin-Selberg mean-value mechanism; representative subconvexity theorem and amplification architecture.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Combinatorial and Selberg sieves” develops into “Spectral methods and trace formulae”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 24. Algebraic Number Theory and Local Fields

Purpose: Develop global and local number fields, ideals, class groups, units, p-adic fields, adeles/ideles and class field theory with analytic arithmetic consequences.

Prerequisites: Volumes 3-4, 8, 22; Volume 25 geometry of numbers may be studied concurrently.

Primary and secondary references: Marcus, Number Fields; Neukirch, Algebraic Number Theory, selected Parts I-II; Serre, Local Fields; Milne, Class Field Theory notes; MIT 18.785 Number Theory I and MIT 18.786 Number Theory II/Class Field Theory lecture notes

Quarter output: Build a 30-page local-global dossier for one number field: integral basis, discriminant, splitting, class group, units, selected completions, zeta data, Artin symbols and class-field-theoretic interpretation.

Research bridges: Foundation for Diophantine approximation/transcendence, automorphic forms, arithmetic geometry, Sato-Tate and cryptography.

Six two-week study units

Unit 1 (Weeks 1-2): Number fields and algebraic integers

Topics: Finite extensions of Q; embeddings; trace/norm; algebraic integers; integral bases; discriminant.

Key definitions: number field; ring of integers; embedding; trace; norm; integral basis; discriminant; different.

Key results to learn:

- Algebraic integers form a ring [PROOF]
- trace/norm via embeddings [PROOF]
- finiteness of integral closure as Z-module [ARCH]
- discriminant criterion for ramification roadmap [ARCH].

Reading route: Marcus Chs. 1-3; Neukirch I.1-2; MIT 18.785 early lectures.

Warm-up/source exercise focus (secondary): Compute rings of integers/integral bases in quadratic and selected cubic fields; calculate traces/norms/discriminants.

Week 1: Number fields and algebraic integers - definitions and examples. Read: Marcus Chs. 1-3; Neukirch I.1-2; MIT 18.785 early lectures.. Make explicit examples/nonexamples for number field; ring of integers; embedding; trace; norm; integral basis; discriminant; different.. Work through Finite extensions of Q; embeddings; trace/norm; algebraic integers; integral bases; discriminant.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Number fields and algebraic integers - theorem and technique pass. Master/audit: Algebraic integers form a ring; trace/norm via embeddings; finiteness of integral closure as Z-module. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Dedekind domains and ideal factorization

Topics: Fractional ideals; Dedekind domains; prime factorization; class group; decomposition of rational primes.

Key definitions: fractional ideal; Dedekind domain; ideal norm; class group; ramification index; residue degree; decomposition/inertia group.

Key results to learn:

- Unique factorization of nonzero ideals in Dedekind domains [PROOF/ARCH]
- Dedekind-Kummer factorization criterion under standard index hypotheses [ARCH]
- efg relation sum e_i f_i=[K:Q] [PROOF]
- basic splitting behavior via Frobenius [ARCH].

Reading route: Marcus Chs. 4-6; Neukirch I.3; MIT 18.785 lectures on Dedekind domains/factorization/Frobenius.

Warm-up/source exercise focus (secondary): Factor primes in quadratic/cyclotomic examples; compute ideal classes in small discriminants; verify e-f-g data.

Week 3: Dedekind domains and ideal factorization - definitions and examples. Read: Marcus Chs. 4-6; Neukirch I.3; MIT 18.785 lectures on Dedekind domains/factorization/Frobenius.. Make explicit examples/nonexamples for fractional ideal; Dedekind domain; ideal norm; class group; ramification index; residue degree; decomposition/inertia group.. Work through Fractional ideals; Dedekind domains; prime factorization; class group; decomposition of rational primes.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Dedekind domains and ideal factorization - theorem and technique pass. Master/audit: Unique factorization of nonzero ideals in Dedekind domains; Dedekind-Kummer factorization criterion under standard index hypotheses; efg relation sum e_i f_i=[K:Q]. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Minkowski theory, units and class finiteness

Topics: Minkowski embedding; covolumes; ideal lattices; class number; units/regulator.

Key definitions: Minkowski embedding; ideal lattice; regulator; unit rank; class number.

Key results to learn:

- Minkowski convex-body theorem as arithmetic input [PROOF/ARCH]
- finiteness of class group [PROOF/ARCH]
- Minkowski bound [PROOF]

- Dirichlet unit theorem [ARCH]
- finiteness of roots of unity [PROOF].

Reading route: Marcus geometry chapters; Neukirch I.5; MIT 18.785 geometry-of-numbers and unit lectures; Volume 25 cross-reference.

Warm-up/source exercise focus (secondary): Compute class numbers using Minkowski bounds; find fundamental units in real quadratic fields; calculate regulators in examples.

Week 5: Minkowski theory, units and class finiteness - definitions and examples. Read: Marcus geometry chapters; Neukirch I.5; MIT 18.785 geometry-of-numbers and unit lectures; Volume 25 cross-reference.. Make explicit examples/nonexamples for Minkowski embedding; ideal lattice; regulator; unit rank; class number.. Work through Minkowski embedding; covolumes; ideal lattices; class number; units/regulator.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Minkowski theory, units and class finiteness - theorem and technique pass. Master/audit: Minkowski convex-body theorem as arithmetic input; finiteness of class group; Minkowski bound. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Valuations and local fields

Topics: Nonarchimedean valuations; completions; Q_p; DVRs; Hensel; local extensions; ramification.

Key definitions: valuation; completion; DVR; uniformizer; residue field; ramification index; unramified/tamely/wildly ramified extension.

Key results to learn:

- Ostrowski theorem for Q [ARCH]
- Hensel lemma [PROOF/ARCH]
- classification of finite unramified extensions of local fields [ARCH]
- efg/local degree relation [PROOF]
- structure of unit filtration and ramification groups [ARCH/USE].

Reading route: Serre, Local Fields Chs. I-IV; Neukirch II.1-4; MIT 18.785 local-field lectures.

Warm-up/source exercise focus (secondary): Compute p-adic expansions; apply Hensel to local solvability; classify simple quadratic Q_p extensions; calculate ramification data.

Week 7: Valuations and local fields - definitions and examples. Read: Serre, Local Fields Chs. I-IV; Neukirch II.1-4; MIT 18.785 local-field lectures.. Make explicit examples/nonexamples for valuation; completion; DVR; uniformizer; residue field; ramification index; unramified/tamely/wildly ramified extension.. Work through Nonarchimedean valuations; completions; Q_p; DVRs; Hensel; local extensions; ramification.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Valuations and local fields - theorem and technique pass. Master/audit: Ostrowski theorem for Q; Hensel lemma; classification of finite unramified extensions of local fields. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Adeles, ideles and class field theory

Topics: Restricted products; strong approximation; ideles; local reciprocity; Artin map; global reciprocity.

Key definitions: adele ring; idele group; idele class group; local Artin map; norm-residue symbol; ray class group.

Key results to learn:

- Strong approximation for additive adeles [ARCH]
- local reciprocity theorem [C/B]
- global Artin reciprocity [C/B]
- existence theorem of class field theory [USE]
- Kronecker-Weber theorem [ARCH/USE].

Reading route: Neukirch class-field chapters; Milne CFT; MIT 18.786 complete lecture sequence.

Warm-up/source exercise focus (secondary): Write local/global Artin maps in examples; identify cyclotomic class fields; calculate ray class groups in small cases.

Week 9: Adeles, ideles and class field theory - definitions and examples. Read: Neukirch class-field chapters; Milne CFT; MIT 18.786 complete lecture sequence.. Make explicit examples/nonexamples for adele ring; idele group; idele class group; local Artin map; norm-residue symbol; ray class group.. Work through Restricted products; strong approximation; ideles; local reciprocity; Artin map; global reciprocity.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Adeles, ideles and class field theory - theorem and technique pass. Master/audit: Strong approximation for additive adeles; local reciprocity theorem [C/B]; global Artin reciprocity [C/B]. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Dedekind zeta, class number formula and Chebotarev

Topics: Dedekind zeta; Hecke L-functions; residues; distribution of prime ideals; Frobenius density.

Key definitions: Dedekind zeta function; Hecke character; Frobenius conjugacy class; Chebotarev density.

Key results to learn:

- Euler product and analytic continuation/functional equation of Dedekind zeta [ARCH/USE]
- analytic class number formula [ARCH/USE]
- prime ideal theorem [ARCH/USE]

• Chebotarev density theorem [ARCH/USE].

Reading route: Neukirch analytic chapters; MIT 18.785 zeta/L/class-formula/Chebotarev lectures; Hecke sections as needed.

Warm-up/source exercise focus (secondary): Compute zeta factors for quadratic fields; derive residues from class-number data; use Chebotarev to predict splitting densities.

Week 11: Dedekind zeta, class number formula and Chebotarev - definitions and examples. Read: Neukirch analytic chapters; MIT 18.785 zeta/L/class-formula/Chebotarev lectures; Hecke sections as needed.. Make explicit examples/nonexamples for Dedekind zeta function; Hecke character; Frobenius conjugacy class; Chebotarev density.. Work through Dedekind zeta; Hecke L-functions; residues; distribution of prime ideals; Frobenius density.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Dedekind zeta, class number formula and Chebotarev - theorem and technique pass. Master/audit: Euler product and analytic continuation/functional equation of Dedekind zeta; analytic class number formula; prime ideal theorem. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Number fields and algebraic integers

U1.P1. [F] For Q(sqrt(5)), Q(sqrt(-5)), Q(cuberoot(2)), and Q(zeta_5), compute candidate rings of integers/integral bases where accessible, field discriminants, traces, and norms of selected elements.

U1.P2. [T] Prove the trace/norm/discriminant relations from embeddings and multiplication matrices. Use discriminant congruences to identify the quadratic ring of integers.

U1.P3. [S] Determine splitting of several rational primes in a quadratic field using polynomial factorization modulo p and the discriminant/Legendre symbol.

Programme U2: Dedekind domains and ideal factorization

U2.P1. [F] Factor ideals generated by small rational primes in Z[sqrt(-5)] or another quadratic integer ring. Contrast element factorization with unique ideal factorization.

U2.P2. [T] Prove a selected route that the ring of integers is Dedekind and hence nonzero ideals factor uniquely into prime ideals. Track Noetherian/integrally closed/dimension-one inputs.

U2.P3. [S] Compute the ideal class group of Q(sqrt(-5)) or Q(sqrt(-23)) using a Minkowski bound and explicit prime ideals.

Programme U3: Minkowski theory, units and class finiteness

U3.P1. [T] Derive Minkowski's embedding and covolume formula for O_K in a number field. Verify it numerically for a quadratic field.

U3.P2. [F] Use Minkowski's bound to prove finiteness of the class group and execute the proof in one explicit field far enough to enumerate ideal-class representatives.

U3.P3. [S] Compute units in Q(sqrt(2)) or Q(sqrt(5)) via Pell equations and relate the logarithmic embedding to Dirichlet's unit theorem lattice.

Programme U4: Valuations and local fields

U4.P1. [F] Construct Q_p from a p-adic absolute value/completion and compute p-adic expansions of several rationals. Compare archimedean and nonarchimedean triangle inequalities.

U4.P2. [T] Apply Hensel's lemma to lift roots modulo p to Q_p; determine solvability of selected equations such as x^2=a over Q_p.

U4.P3. [S] For a finite extension of Q_p, compute ramification index/residue degree in a simple unramified or Eisenstein extension and verify e f <= [L:K] with equality in the separable local setting.

Programme U5: Adeles, ideles and class field theory

U5.P1. [F] Write explicit finite adeles/ideles for Q with only a few nonunit components and verify restricted-product conditions. Embed Q diagonally and test discreteness/cocompact quotient ideas.

U5.P2. [T] Prove strong approximation for Q in a selected elementary form (simultaneous approximation at finitely many places) using CRT/weak approximation.

U5.P3. [R] State local and global Artin reciprocity precisely for one standard formulation. Work out the cyclotomic/Q example that recovers Kronecker-Weber/class-field-theoretic reciprocity data.

Programme U6: Dedekind zeta, class number formula and Chebotarev

U6.P1. [F] Derive the Euler product of the Dedekind zeta function from prime-ideal factorization and compute local factors for a quadratic field at split/inert/ramified primes.

U6.P2. [T] State the analytic class number formula and verify all quantities in a quadratic field where class number/regulator/roots of unity are known.

U6.P3. [R] State Chebotarev density theorem precisely and derive Dirichlet's theorem or prime-splitting densities in a Galois extension as consequences.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Algebraic Number Theory and Local Fields: start from “Number fields and algebraic integers”, pass through “Valuations and local fields”, and finish at “Dedekind zeta, class number formula and Chebotarev”. Use the named results “Algebraic integers form a ring” and “Chebotarev density theorem.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Foundation for Diophantine approximation/transcendence, automorphic forms, arithmetic geometry, Sato-Tate and cryptography. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Algebraic integers form a ring; trace/norm via embeddings; Unique factorization of nonzero ideals in Dedekind domains; efg relation sum e_i f_i=[K:Q]; Minkowski convex-body theorem as arithmetic input. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Kronecker-Weber theorem; Euler product and analytic continuation/functional equation of Dedekind zeta; analytic class number formula; prime ideal theorem; Chebotarev density theorem.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Number fields and algebraic integers” develops into “Dedekind zeta, class number formula and Chebotarev”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 25. Geometry of Numbers and Diophantine Approximation

Purpose: Unify lattices, convex geometry, classical and metric Diophantine approximation, Schmidt games and the homogeneous-dynamics dictionary.

Prerequisites: Volumes 2, 6, 14, 21-24.

Primary and secondary references: Cassels, An Introduction to the Geometry of Numbers; Cassels, An Introduction to Diophantine Approximation; Schmidt, Diophantine Approximation; Bugeaud, Approximation by Algebraic Numbers, selected chapters; Kleinbock-Margulis and Kleinbock-Weiss papers/surveys; Beresnevich-Velani surveys

Quarter output: Choose one Diophantine set (badly approximable, very well approximable, or a fractal restriction) and analyze it by three lenses: classical approximation, geometry of numbers and homogeneous dynamics.

Research bridges: Prepares Baker/Schmidt Subspace Theorem, homogeneous dynamics, fractal Diophantine approximation and lattice cryptography.

Six two-week study units

Unit 1 (Weeks 1-2): Lattices and convex bodies

Topics: Lattices in R^n; fundamental domains; determinant/covolume; convex symmetric bodies; lattice-point counting.

Key definitions: lattice; basis; fundamental parallelepiped; determinant/covolume; primitive vector; convex symmetric body.

Key results to learn:

- Blichfeldt lemma [PROOF]
- Minkowski convex-body theorem [PROOF/ARCH]
- Minkowski linear-forms theorem [ARCH]
- basic lattice-point estimates [PROOF/ARCH].

Reading route: Cassels, Geometry of Numbers, Chs. I-III.

Warm-up/source exercise focus (secondary): Compute covolumes; prove Minkowski consequences for linear forms; solve shortest-vector examples in dimensions 2-3.

Week 1: Lattices and convex bodies - definitions and examples. Read: Cassels, Geometry of Numbers, Chs. I-III.. Make explicit examples/nonexamples for lattice; basis; fundamental parallelepiped; determinant/covolume; primitive vector; convex symmetric body.. Work through Lattices in R^n; fundamental domains; determinant/covolume; convex symmetric bodies; lattice-point counting.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Lattices and convex bodies - theorem and technique pass. Master/audit: Blichfeldt lemma; Minkowski convex-body theorem; Minkowski linear-forms theorem. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Successive minima, duality and transference

Topics: Successive minima; polar bodies; dual lattices; covering radius; transference.

Key definitions: successive minimum lambda_i; polar body; dual lattice; covering radius; critical determinant.

Key results to learn:

- Minkowski second theorem [ARCH]
- Mahler transference inequalities [ARCH/USE]
- transference between successive minima of Lambda and Lambda* [ARCH/USE]
- Hermite constant basic bounds [ARCH].

Reading route: Cassels Geometry of Numbers, middle chapters; Gruber-Lekkerkerker selected reference.

Warm-up/source exercise focus (secondary): Compute minima for explicit lattices; verify second theorem bounds; derive elementary transference estimates.

Week 3: Successive minima, duality and transference - definitions and examples. Read: Cassels Geometry of Numbers, middle chapters; Gruber-Lekkerkerker selected reference.. Make explicit examples/nonexamples for successive minimum lambda_i; polar body; dual lattice; covering radius; critical determinant.. Work through Successive minima; polar bodies; dual lattices; covering radius; transference.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Successive minima, duality and transference - theorem and technique pass. Master/audit: Minkowski second theorem; Mahler transference inequalities; transference between successive minima of Lambda and Lambda*. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Classical Diophantine approximation

Topics: Rational approximation; continued fractions revisited; simultaneous/dual approximation; badly approximable systems.

Key definitions: Diophantine exponent; simultaneous approximation; dual approximation; badly approximable vector; singular vector.

Key results to learn:

- Dirichlet approximation theorem in R^n [PROOF]
- best-approximation properties in dimension one [PROOF]
- Khintchine transference principle [ARCH]

• badly approximable numbers iff bounded continued-fraction partial quotients [PROOF].

Reading route: Cassels Diophantine Approximation Chs. I-V; Schmidt early chapters.

Warm-up/source exercise focus (secondary): Prove Dirichlet by pigeonhole and lattices; calculate exponents; translate simultaneous approximation into lattice statements.

Week 5: Classical Diophantine approximation - definitions and examples. Read: Cassels Diophantine Approximation Chs. I-V; Schmidt early chapters.. Make explicit examples/nonexamples for Diophantine exponent; simultaneous approximation; dual approximation; badly approximable vector; singular vector.. Work through Rational approximation; continued fractions revisited; simultaneous/dual approximation; badly approximable systems.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Classical Diophantine approximation - theorem and technique pass. Master/audit: Dirichlet approximation theorem in R^n; best-approximation properties in dimension one; Khintchine transference principle. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Metric Diophantine approximation

Topics: Limsup sets; Lebesgue-null/full laws; Hausdorff dimension; ubiquity/mass transference.

Key definitions: approximating function; limsup set; Khintchine monotonicity condition; Hausdorff measure/dimension; ubiquity.

Key results to learn:

• Khintchine theorem [ARCH]

• Jarnik-Besicovitch theorem [ARCH/USE]

• Duffin-Schaeffer theorem, precise modern statement [USE]

• Mass Transference Principle [USE].

Reading route: Cassels/Schmidt classical metric chapters; Beresnevich-Velani surveys; Koukoulopoulos-Maynard theorem exposition for Duffin-Schaeffer.

Warm-up/source exercise focus (secondary): Apply Borel-Cantelli to convergence cases; compute Hausdorff dimensions of model limsup sets; compare Lebesgue and Hausdorff laws.

Week 7: Metric Diophantine approximation - definitions and examples. Read: Cassels/Schmidt classical metric chapters; Beresnevich-Velani surveys; Koukoulopoulos-Maynard theorem exposition for Duffin-Schaeffer.. Make explicit examples/nonexamples for approximating function; limsup set; Khintchine monotonicity condition; Hausdorff measure/dimension; ubiquity.. Work through Limsup sets; Lebesgue-null/full laws; Hausdorff dimension; ubiquity/mass transference.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Metric Diophantine approximation - theorem and technique pass. Master/audit: Khintchine theorem; Jarnik-Besicovitch theorem; Duffin-Schaeffer theorem, precise modern statement. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Schmidt games and fractal Diophantine approximation

Topics: Winning games; absolute/hyperplane winning; stability; intersections with fractals.

Key definitions: Schmidt game; alpha-winning; absolute winning; hyperplane absolute winning; friendly/decaying measure.

Key results to learn:

• Badly approximable vectors are Schmidt winning [ARCH]

• winning sets have full Hausdorff dimension and stable countable intersections [PROOF/ARCH]

• invariance properties under bi-Lipschitz/C1 maps in relevant variants [ARCH/USE]

• representative fractal winning theorem [USE].

Reading route: Schmidt original chapters; McMullen absolute-winning paper; Kleinbock-Weiss and Fishman-Simmons-Urbański surveys.

Warm-up/source exercise focus (secondary): Play explicit strategies for Bad; prove basic stability properties; work one Cantor-set/fractal restriction example.

Week 9: Schmidt games and fractal Diophantine approximation - definitions and examples. Read: Schmidt original chapters; McMullen absolute-winning paper; Kleinbock-Weiss and Fishman-Simmons-Urbański surveys.. Make explicit examples/nonexamples for Schmidt game; alpha-winning; absolute winning; hyperplane absolute winning; friendly/decaying measure.. Work through Winning games; absolute/hyperplane winning; stability; intersections with fractals.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Schmidt games and fractal Diophantine approximation - theorem and technique pass. Master/audit: Badly approximable vectors are Schmidt winning; winning sets have full Hausdorff dimension and stable countable intersections; invariance properties under bi-Lipschitz/C1 maps in relevant variants. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Homogeneous dynamics correspondence

Topics: Diagonal flows on spaces of lattices; Dani correspondence; quantitative nondivergence; extremality of manifolds/measures.

Key definitions: space of unimodular lattices; diagonal flow; cusp excursion; nondivergence; extremal/strongly extremal manifold.

Key results to learn:

• Dani correspondence between Diophantine properties and bounded/divergent trajectories [ARCH]

• Mahler compactness criterion [PROOF/ARCH]

• Kleinbock-Margulis extremality theorem for nondegenerate manifolds [USE]

• representative friendly/fractal-measure extremality theorem [USE].

Reading route: Kleinbock-Margulis paper/surveys; Einsiedler-Ward selected; Kleinbock-Weiss.

Warm-up/source exercise focus (secondary): Translate inequalities into lattice short-vector conditions; prove one-dimensional Dani correspondence; make a dependency map for quantitative nondivergence.

Week 11: Homogeneous dynamics correspondence - definitions and examples. Read: Kleinbock-Margulis paper/surveys; Einsiedler-Ward selected; Kleinbock-Weiss.. Make explicit examples/nonexamples for space of unimodular lattices; diagonal flow; cusp excursion; nondivergence; extremal/strongly extremal manifold.. Work through Diagonal flows on spaces of lattices; Dani correspondence; quantitative nondivergence; extremality of manifolds/measures.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Homogeneous dynamics correspondence - theorem and technique pass. Master/audit: Dani correspondence between Diophantine properties and bounded/divergent trajectories; Mahler compactness criterion; Kleinbock-Margulis extremality theorem for nondegenerate manifolds. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Lattices and convex bodies

U1.P1. [F] For lattices generated by explicit 2x2 and 3x3 matrices, compute covolume, shortest vectors, fundamental parallelepipeds, and dual lattices. Compare Euclidean length with covolume under shearing.

U1.P2. [T] Prove Blichfeldt's lemma and deduce Minkowski's convex body theorem. Apply it to prove existence of a nonzero lattice point in a carefully chosen ellipse/box with an explicit numerical bound.

U1.P3. [S] Use Minkowski to prove Dirichlet's simultaneous approximation theorem by constructing the appropriate lattice/convex body. Make the dictionary between lattice coordinates and approximation inequalities explicit.

Programme U2: Successive minima, duality and transference

U2.P1. [F] Compute successive minima for Z^2 under several convex bodies and for a highly skew lattice. Verify Minkowski's second theorem numerically in dimension 2.

U2.P2. [T] Prove a selected transference inequality between a lattice and its dual in low dimension, and apply it to simultaneous versus linear-form approximation.

U2.P3. [S] Analyze how an LLL-reduced basis controls successive minima only approximately. Work a 2D/3D example by hand to prepare the cryptographic use later.

Programme U3: Classical Diophantine approximation

U3.P1. [F] Compute convergents and approximation constants for sqrt(2), golden ratio, and one nonquadratic irrational. Verify the sharpness phenomenon behind Hurwitz's theorem for the golden ratio.

U3.P2. [T] Prove Dirichlet's theorem, Hurwitz's theorem in dimension one, and characterize badly approximable reals via bounded continued-fraction partial quotients.

U3.P3. [S] For a quadratic irrational alpha, derive a uniform lower bound q||q alpha|| >= c(alpha)>0 using periodic continued fractions or algebraic arguments.

Programme U4: Metric Diophantine approximation

U4.P1. [F] For psi(q)=q^{-tau}, determine heuristically and then theoremically (where covered) when W(psi) has full/zero Lebesgue measure using the sum of approximation volumes.

U4.P2. [T] Prove Khintchine's theorem in a monotone one-dimensional formulation at proof-architecture level, isolating convergence via Borel-Cantelli and divergence via quasi-independence.

U4.P3. [R] State Duffin-Schaeffer precisely and compare it with Khintchine by identifying monotonicity, coprimality, and the correct divergence sum.

Programme U5: Schmidt games and fractal Diophantine approximation

U5.P1. [F] Prove Schmidt-winning sets are dense and have full Hausdorff dimension in a standard Euclidean setting. Work the strategy for badly approximable numbers explicitly.

U5.P2. [T] Show stability of winning under countable intersections and suitable bi-Lipschitz maps in the version used by your text.

U5.P3. [S] Formulate a fractal/absolute-winning variant relevant to Diophantine approximation on Cantor-type sets and verify the hypotheses in one basic self-similar example.

Programme U6: Homogeneous dynamics correspondence

U6.P1. [F] Show SL_2(R)/SL_2(Z) parametrizes unimodular lattices and write the lattice associated with a real alpha so that short vectors encode good rational approximations.

U6.P2. [T] Prove the Dani correspondence for badly approximable numbers: boundedness of a diagonal orbit iff alpha is badly approximable, with all norm comparisons explicit.

U6.P3. [R] Extend the dictionary to simultaneous approximation in R^n and formulate one higher-dimensional/fractal Diophantine problem as a shrinking-target or nondivergence problem on a homogeneous space.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Geometry of Numbers and Diophantine Approximation: start from “Lattices and convex bodies”, pass through “Metric Diophantine approximation”, and finish at “Homogeneous dynamics correspondence”. Use the named results “Blichfeldt lemma” and “representative friendly/fractal-measure extremality theorem.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Prepares Baker/Schmidt Subspace Theorem, homogeneous dynamics, fractal Diophantine approximation and lattice cryptography. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Blichfeldt lemma; Minkowski convex-body theorem; basic lattice-point estimates; Dirichlet approximation theorem in R^n; best-approximation properties in dimension one. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: invariance properties under bi-Lipschitz/C1 maps in relevant variants; representative fractal winning theorem; Dani correspondence between Diophantine properties and bounded/divergent trajectories; Kleinbock-Margulis extremality theorem for nondegenerate manifolds; representative friendly/fractal-measure extremality theorem.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Lattices and convex bodies” develops into “Homogeneous dynamics correspondence”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 26. Diophantine Equations, Transcendence and Undecidability

Purpose: Develop heights, Roth, linear forms in logarithms, the Subspace Theorem, effective Diophantine equations and Hilbert's tenth problem.

Prerequisites: Volumes 5, 24-25; Volume 31 is helpful for geometric formulations.

Primary and secondary references: Bombieri-Gubler, Heights in Diophantine Geometry, selected chapters; Baker, Transcendental Number Theory; Waldschmidt, Diophantine Approximation on Linear Algebraic Groups, selected chapters; Evertse-Gyory, Unit Equations in Diophantine Number Theory; Schmidt, Diophantine Approximation, Subspace Theorem chapters; Poonen, Hilbert's Tenth Problem and related expository notes

Quarter output: Complete two capstones: (i) an explicit Baker-method case study with all constants tracked; (ii) a 15-page exposition from computability to MRDP and Hilbert's tenth problem.

Research bridges: Directly targets Baker linear forms, Schmidt Subspace Theorem and Hilbert's tenth problem, and prepares arithmetic geometry finiteness theorems.

Six two-week study units

Unit 1 (Weeks 1-2): Heights and finiteness principles

Topics: Absolute values on number fields; product formula; projective heights; height functoriality; Northcott property.

Key definitions: Weil height; logarithmic height; local height; product formula; height of projective point; bounded degree.

Key results to learn:

- Product formula for number fields [PROOF/ARCH]
- height invariance under scaling [PROOF]
- functorial height estimates for rational maps [ARCH]
- Northcott theorem [PROOF/ARCH].

Reading route: Bombieri-Gubler Chs. 1-2; Hindry-Silverman height preliminaries.

Warm-up/source exercise focus (secondary): Compute heights of rationals/algebraic numbers; prove basic height inequalities; enumerate bounded-height examples.

Week 1: Heights and finiteness principles - definitions and examples. Read: Bombieri-Gubler Chs. 1-2; Hindry-Silverman height preliminaries.. Make explicit examples/nonexamples for Weil height; logarithmic height; local height; product formula; height of projective point; bounded degree.. Work through Absolute values on number fields; product formula; projective heights; height functoriality; Northcott property.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Heights and finiteness principles - theorem and technique pass. Master/audit: Product formula for number fields; height invariance under scaling; functorial height estimates for rational maps. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Liouville, Thue-Siegel-Roth

Topics: Approximation to algebraic numbers; irrationality measures; auxiliary polynomials.

Key definitions: irrationality exponent; algebraic approximation; Roth exponent; exceptional approximation.

Key results to learn:

- Liouville inequality [PROOF]
- Thue-Siegel historical improvement, statement [USE]
- Roth theorem: |alpha-p/q|<q^{-2-epsilon} finitely often for algebraic irrational alpha [ARCH/USE]
- Roth as height inequality [ARCH/USE].

Reading route: Bombieri-Gubler approximation chapters; Schmidt; Roth expositions.

Warm-up/source exercise focus (secondary): Derive Liouville bounds; compare exponents; translate Roth into projective-height form and applications.

Week 3: Liouville, Thue-Siegel-Roth - definitions and examples. Read: Bombieri-Gubler approximation chapters; Schmidt; Roth expositions.. Make explicit examples/nonexamples for irrationality exponent; algebraic approximation; Roth exponent; exceptional approximation.. Work through Approximation to algebraic numbers; irrationality measures; auxiliary polynomials.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Liouville, Thue-Siegel-Roth - theorem and technique pass. Master/audit: Liouville inequality; Thue-Siegel historical improvement, statement; Roth theorem: |alpha-p/q|<q^{-2-epsilon} finitely often for algebraic irrational alpha. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Baker theory and linear forms in logarithms

Topics: Logarithms of algebraic numbers; linear forms; explicit lower bounds; effective exponential Diophantine equations.

Key definitions: linear form in logarithms; multiplicative independence; logarithmic height; S-unit.

Key results to learn:

- Baker theorem on nonzero linear forms in logarithms [ARCH/USE]
- Baker-Wustholz theorem, exact quantitative form at reference level [USE]
- Matveev explicit lower bound [USE]
- effective finiteness consequences for exponential Diophantine equations [ARCH/USE].

Reading route: Baker, central chapters; Waldschmidt surveys; Matveev theorem reference.

Warm-up/source exercise focus (secondary): Apply a stated explicit bound to a small exponential equation; work multiplicative-independence examples; build a constant-dependency table.

Week 5: Baker theory and linear forms in logarithms - definitions and examples. Read: Baker, central chapters; Waldschmidt surveys; Matveev theorem reference.. Make explicit examples/nonexamples for linear form in logarithms; multiplicative independence; logarithmic height; S-unit.. Work through Logarithms of algebraic numbers; linear forms; explicit lower bounds; effective exponential Diophantine equations.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Baker theory and linear forms in logarithms - theorem and technique pass. Master/audit: Baker theorem on nonzero linear forms in logarithms; Baker-Wustholz theorem, exact quantitative form at reference level; Matveev explicit lower bound. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Schmidt Subspace Theorem and unit equations

Topics: Simultaneous linear-form inequalities; S-adic absolute values; exceptional subspaces; S-unit equations.

Key definitions: height of vector; S-integer; S-unit; exceptional subspace; decomposable form.

Key results to learn:

- Schmidt Subspace Theorem [ARCH/USE]
- S-adic Subspace Theorem [USE]
- finiteness of S-unit equation x+y=1 [ARCH/USE]
- representative quantitative Subspace Theorem [USE]
- applications to digit/recurrence problems roadmap [USE].

Reading route: Schmidt Subspace chapters; Evertse-Gyory; Bombieri-Gubler selected.

Warm-up/source exercise focus (secondary): Derive Roth from a one-dimensional shadow of Subspace ideas conceptually; solve elementary S-unit examples; map standard consequences.

Week 7: Schmidt Subspace Theorem and unit equations - definitions and examples. Read: Schmidt Subspace chapters; Evertse-Gyory; Bombieri-Gubler selected.. Make explicit examples/nonexamples for height of vector; S-integer; S-unit; exceptional subspace; decomposable form.. Work through Simultaneous linear-form inequalities; S-adic absolute values; exceptional subspaces; S-unit equations.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Schmidt Subspace Theorem and unit equations - theorem and technique pass. Master/audit: Schmidt Subspace Theorem; S-adic Subspace Theorem; finiteness of S-unit equation x+y=1. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Integral points and effective equations

Topics: Binary forms; Thue equations; curves; integral points; effective versus ineffective methods.

Key definitions: Thue equation; Siegel integral point; genus; effective bound; regulator.

Key results to learn:

- Thue finiteness theorem [ARCH]
- Siegel theorem on integral points [ARCH/USE]
- effective resolution of many Thue/Thue-Mahler equations using Baker theory [ARCH/USE]
- Catalan-Mihailescu theorem, statement and method map [USE].

Reading route: Evertse-Gyory; Bombieri-Gubler; selected Baker-method case studies.

Warm-up/source exercise focus (secondary): Solve small Thue equations computationally and justify bounds; compare Siegel and Baker mechanisms; write a finiteness/effectivity chart.

Week 9: Integral points and effective equations - definitions and examples. Read: Evertse-Gyory; Bombieri-Gubler; selected Baker-method case studies.. Make explicit examples/nonexamples for Thue equation; Siegel integral point; genus; effective bound; regulator.. Work through Binary forms; Thue equations; curves; integral points; effective versus ineffective methods.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Integral points and effective equations - theorem and technique pass. Master/audit: Thue finiteness theorem; Siegel theorem on integral points; effective resolution of many Thue/Thue-Mahler equations using Baker theory. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Hilbert's tenth problem and Diophantine undecidability

Topics: Diophantine sets; computably enumerable sets; coding; Pell sequences; exponential growth; MRDP theorem; variants over Q/number fields.

Key definitions: Diophantine set; recursively enumerable/c.e. set; Diophantine representation; universal Diophantine equation.

Key results to learn:

- Davis-Putnam-Robinson reduction framework [USE]
- Matiyasevich theorem that exponentiation is Diophantine via recurrence/Pell-type phenomena [USE]

- MRDP theorem: c.e. sets are Diophantine [ARCH/USE]
- undecidability of Hilbert's tenth problem over Z [ARCH/USE]
- Koymans-Pagano 2025 theorem: H10 is undecidable for every infinite ring finitely generated over Z; know explicitly why this leaves H10(Q) open [USE]
- H10(Q) remains open [PROOF]
- selected number-field status/results [USE].

Reading route: Poonen expository notes; Matiyasevich, Hilbert's Tenth Problem selected; Logic Volume 5 cross-reference.

Warm-up/source exercise focus (secondary): Write a toy Diophantine encoding; analyze a Pell recurrence used for exponential growth; produce a precise logic-to-number-theory dependency graph.

Week 11: Hilbert's tenth problem and Diophantine undecidability - definitions and examples. Read: Poonen expository notes; Matiyasevich, Hilbert's Tenth Problem selected; Logic Volume 5 cross-reference.. Make explicit examples/nonexamples for Diophantine set; recursively enumerable/c.e. set; Diophantine representation; universal Diophantine equation.. Work through Diophantine sets; computably enumerable sets; coding; Pell sequences; exponential growth; MRDP theorem; variants over Q/number fields.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Hilbert's tenth problem and Diophantine undecidability - theorem and technique pass. Master/audit: Davis-Putnam-Robinson reduction framework; Matiyasevich theorem that exponentiation is Diophantine via recurrence/Pell-type phenomena; MRDP theorem: c.e. sets are Diophantine. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Heights and finiteness principles

U1.P1. [F] Compute absolute logarithmic heights of rational numbers, quadratic algebraic numbers, roots of unity, and points in P^n(Q) from local/global definitions.

U1.P2. [T] Prove basic height inequalities h(alpha beta), h(alpha+beta), h(alpha^n), and Northcott finiteness for bounded degree and height in a selected formulation.

U1.P3. [S] Use heights to turn a naive infinite Diophantine search into a finite search once a height bound is given. Work a simple Thue/Mordell-type toy example.

Programme U2: Liouville, Thue-Siegel-Roth

U2.P1. [T] Prove Liouville's inequality for algebraic approximation and derive a transcendence result for a Liouville number.

U2.P2. [S] State Roth's theorem precisely and show how its exponent 2 improves Liouville for algebraic irrationals of degree >2. Test the theorem against continued-fraction examples.

U2.P3. [R] Build the proof-architecture ladder Thue -> Siegel -> Roth, identifying the auxiliary-polynomial principle and what changes in the exponent at each stage.

Programme U3: Baker theory and linear forms in logarithms

U3.P1. [F] For equations such as 2^m-3^n=1 or x^2-Dy^2=1, isolate a candidate nonzero linear form in logarithms whose smallness follows from a hypothetical large solution.

U3.P2. [T] Assuming a standard Baker-Wustholz/Matveev lower bound, combine it with an elementary upper bound to derive an explicit upper bound for an exponent in one exponential Diophantine equation.

U3.P3. [R] Audit one modern explicit linear-form theorem: exact degree/height/logarithm hypotheses, dependence of constants, and which parameters dominate applications. Then compare with a qualitative transcendence statement.

Programme U4: Schmidt Subspace Theorem and unit equations

U4.P1. [F] Rewrite a unit equation x+y=1 in S-units as a projective linear-form problem and compute solutions in one small S for Q if feasible.

U4.P2. [T] State Schmidt's Subspace Theorem in a standard absolute-value/height formulation and derive the finiteness of nondegenerate solutions to a simple S-unit equation at proof-architecture level.

U4.P3. [S] Compare Roth and the Subspace Theorem by embedding rational approximation into a two-variable linear-form inequality. Identify exactly how the latter generalizes the former.

Programme U5: Integral points and effective equations

U5.P1. [F] Solve several classical Thue/Pell/Thue-Mahler equations computationally after deriving a rigorous finite search region from elementary or Baker-type estimates.

U5.P2. [T] State Siegel's theorem on integral points and contrast qualitative finiteness with effective bounds from Baker theory. Give examples where each applies naturally.

U5.P3. [R] Choose one effective Diophantine theorem (Catalan/Mihailescu, S-unit bounds, linear recurrence equation). Trace the reduction from the original equation to heights/logarithmic forms.

Programme U6: Hilbert's tenth problem and Diophantine undecidability

U6.P1. [F] Show that conjunction and finite disjunction of Diophantine conditions can be encoded by polynomial equations over Z. Encode divisibility and selected bounded arithmetic relations explicitly.

U6.P2. [T] Work out Pell/Fibonacci recurrence identities that exhibit exponential growth through polynomial relations, and explain why such growth was the missing ingredient in the Davis-Putnam-Robinson route.

U6.P3. [R] Reconstruct MRDP as the chain c.e. -> exponential-Diophantine -> Diophantine -> undecidability of H10(Z). Then state the Koymans-Pagano 2025 theorem for every infinite ring finitely generated over Z, explain the elliptic-curve/2-descent/additive-combinatorics architecture, and prove from definitions why this theorem does not include Q; finish with the exact open status of H10(Q).

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Diophantine Equations, Transcendence and Undecidability: start from “Heights and finiteness principles”, pass through “Schmidt Subspace Theorem and unit equations”, and finish at “Hilbert's tenth problem and Diophantine undecidability”. Use the named results “Product formula for number fields” and “selected number-field status/results.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Directly targets Baker linear forms, Schmidt Subspace Theorem and Hilbert's tenth problem, and prepares arithmetic geometry finiteness theorems. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Product formula for number fields; height invariance under scaling; Northcott theorem; Liouville inequality; H10(Q) remains open. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Davis-Putnam-Robinson reduction framework; Matiyasevich theorem that exponentiation is Diophantine via recurrence/Pell-type phenomena; MRDP theorem: c.e. sets are Diophantine; undecidability of Hilbert's tenth problem over Z; selected number-field status/results.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Heights and finiteness principles” develops into “Hilbert's tenth problem and Diophantine undecidability”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 27. Finite Fields and Arithmetic over Function Fields

Purpose: Develop finite-field algebra, algorithms, character/exponential sums, curves and zeta functions over finite fields, emphasizing analogies with number fields.

Prerequisites: Volumes 3, 8, 22, 24; Volume 31 useful for the final two units.

Primary and secondary references: Lidl-Niederreiter, Finite Fields; Ireland-Rosen, A Classical Introduction to Modern Number Theory, finite-field chapters; Rosen, Number Theory in Function Fields; Katz, Gauss Sums, Kloosterman Sums, and Monodromy Groups, selected advanced chapters

Quarter output: Create a computational-arithmetic notebook covering finite-field construction, factorization, Gauss sums and curve point counts, accompanied by a 15-page number-field/function-field analogy chart.

Research bridges: Prepares Deligne/Weil theory, arithmetic geometry, coding/cryptography and finite-field additive combinatorics.

Six two-week study units

Unit 1 (Weeks 1-2): Structure of finite fields

Topics: Existence/uniqueness; Frobenius; subfields; cyclic multiplicative groups; trace/norm; normal bases.

Key definitions: finite field F_q; Frobenius automorphism; primitive element; trace; norm; normal basis.

Key results to learn:

- Existence and uniqueness of F_{p^n} [PROOF/ARCH]
- F_q^* cyclic [PROOF/ARCH]
- subfield classification [PROOF]
- Gal(F_{q^n}/F_q) cyclic generated by Frobenius [PROOF]
- normal basis theorem for finite fields [ARCH].

Reading route: Lidl-Niederreiter Chs. 1-2.

Warm-up/source exercise focus (secondary): Construct F_{p^n} explicitly; compute Frobenius orbits/traces/norms; find primitive elements and normal bases.

Week 1: Structure of finite fields - definitions and examples. Read: Lidl-Niederreiter Chs. 1-2.. Make explicit examples/nonexamples for finite field F_q; Frobenius automorphism; primitive element; trace; norm; normal basis.. Work through Existence/uniqueness; Frobenius; subfields; cyclic multiplicative groups; trace/norm; normal bases.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Structure of finite fields - theorem and technique pass. Master/audit: Existence and uniqueness of F_{p^n}; F_q^* cyclic; subfield classification. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Polynomial arithmetic and algorithms

Topics: Irreducibility tests; factorization; square-free/distinct-degree/equal-degree factorization; fast exponentiation.

Key definitions: irreducible polynomial; square-free factorization; Berlekamp subalgebra; distinct-degree factorization.

Key results to learn:

- Criterion x^{q^n}-x product of monic irreducibles with degrees dividing n [PROOF]
- Berlekamp factorization principle [ARCH]
- Cantor-Zassenhaus randomized factorization principle [ARCH]
- polynomial-time factorization over finite fields, complexity-level statement [USE].

Reading route: Lidl-Niederreiter computational chapters; von zur Gathen-Gerhard reference selected.

Warm-up/source exercise focus (secondary): Implement factorization of sample polynomials; prove irreducibility criteria; compare deterministic/randomized methods.

Week 3: Polynomial arithmetic and algorithms - definitions and examples. Read: Lidl-Niederreiter computational chapters; von zur Gathen-Gerhard reference selected.. Make explicit examples/nonexamples for irreducible polynomial; square-free factorization; Berlekamp subalgebra; distinct-degree factorization.. Work through Irreducibility tests; factorization; square-free/distinct-degree/equal-degree factorization; fast exponentiation.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Polynomial arithmetic and algorithms - theorem and technique pass. Master/audit: Criterion x^{q^n}-x product of monic irreducibles with degrees dividing n; Berlekamp factorization principle; Cantor-Zassenhaus randomized factorization principle. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Characters, Gauss and Jacobi sums

Topics: Additive/multiplicative characters; Fourier analysis on F_q; Gauss/Jacobi sums; quadratic characters.

Key definitions: additive character; multiplicative character; Gauss sum; Jacobi sum; quadratic character.

Key results to learn:

- Character orthogonality [PROOF]
- |G(chi)|=sqrt(q) for nontrivial multiplicative chi with nontrivial additive character [PROOF/ARCH]
- Gauss-Jacobi identities [PROOF/ARCH]

- quadratic Gauss sum evaluation up to sign/phase [ARCH].

Reading route: Lidl-Niederreiter character chapters; Ireland-Rosen.

Warm-up/source exercise focus (secondary): Compute Gauss/Jacobi sums for small fields; prove orthogonality; derive point counts for diagonal equations.

Week 5: Characters, Gauss and Jacobi sums - definitions and examples. Read: Lidl-Niederreiter character chapters; Ireland-Rosen.. Make explicit examples/nonexamples for additive character; multiplicative character; Gauss sum; Jacobi sum; quadratic character.. Work through Additive/multiplicative characters; Fourier analysis on F_q; Gauss/Jacobi sums; quadratic characters.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Characters, Gauss and Jacobi sums - theorem and technique pass. Master/audit: Character orthogonality; |G(chi)|=sqrt(q) for nontrivial multiplicative chi with nontrivial additive character; Gauss-Jacobi identities. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Exponential sums and Weil bounds

Topics: Character sums of rational functions; curves; Hasse bound; Kloosterman-type sums.

Key definitions: additive character sum; multiplicative character sum; genus; Kloosterman sum.

Key results to learn:

- Weil bound for multiplicative character sums under standard nonsingularity/non-power hypotheses [ARCH/USE]
- Hasse bound |#E(F_q)-(q+1)|<=2sqrt(q) [ARCH/USE]
- representative Weil bound for additive sums [USE].

Reading route: Lidl-Niederreiter advanced chapters; Katz selected; elliptic curve cross-reference Volume 32.

Warm-up/source exercise focus (secondary): Estimate sums using exact small-field calculations versus Weil bounds; count points on elliptic/diagonal curves.

Week 7: Exponential sums and Weil bounds - definitions and examples. Read: Lidl-Niederreiter advanced chapters; Katz selected; elliptic curve cross-reference Volume 32.. Make explicit examples/nonexamples for additive character sum; multiplicative character sum; genus; Kloosterman sum.. Work through Character sums of rational functions; curves; Hasse bound; Kloosterman-type sums.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Exponential sums and Weil bounds - theorem and technique pass. Master/audit: Weil bound for multiplicative character sums under standard nonsingularity/non-power hypotheses; Hasse bound |#E(F_q)-(q+1)|<=2sqrt(q); representative Weil bound for additive sums [USE].. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Function fields and curves

Topics: Global function fields; places/valuations; divisors; Riemann-Roch; class groups; units/constants.

Key definitions: function field of one variable; place; divisor; degree; principal divisor; genus; Riemann-Roch space.

Key results to learn:

- Product formula for function fields [PROOF/ARCH]
- degree of principal divisor zero [PROOF]
- Riemann-Roch theorem for curves [ARCH/USE]
- finiteness of degree-zero divisor class group over finite constants [ARCH].

Reading route: Rosen Chs. 1-6; Volume 31 curve material.

Warm-up/source exercise focus (secondary): Compute divisors on P1 and elliptic curves; calculate Riemann-Roch spaces in low genus; compare number-field/function-field dictionaries.

Week 9: Function fields and curves - definitions and examples. Read: Rosen Chs. 1-6; Volume 31 curve material.. Make explicit examples/nonexamples for function field of one variable; place; divisor; degree; principal divisor; genus; Riemann-Roch space.. Work through Global function fields; places/valuations; divisors; Riemann-Roch; class groups; units/constants.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Function fields and curves - theorem and technique pass. Master/audit: Product formula for function fields; degree of principal divisor zero; Riemann-Roch theorem for curves. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Zeta functions and Frobenius

Topics: Point counts over extensions; zeta functions; Frobenius eigenvalues; rationality, functional equation and RH for curves.

Key definitions: zeta function Z(C,T); Frobenius endomorphism; reciprocal root; L-polynomial.

Key results to learn:

- Rationality of Z(C,T) [ARCH/USE]
- functional equation [ARCH/USE]
- Riemann hypothesis for curves over finite fields [ARCH/USE]
- elliptic-curve zeta formula [PROOF/ARCH]
- prime polynomial theorem as function-field analogue [ARCH].

Reading route: Rosen zeta chapters; Weil/étale preparation Volume 33.

Warm-up/source exercise focus (secondary): Recover point counts from zeta roots; compute zeta functions for P1 and sample elliptic curves; compare PNT over F_q[t] with Z.

Week 11: Zeta functions and Frobenius - definitions and examples. Read: Rosen zeta chapters; Weil/étale preparation Volume 33.. Make explicit examples/nonexamples for zeta function Z(C,T); Frobenius endomorphism; reciprocal root; L-polynomial.. Work through Point counts over extensions; zeta functions; Frobenius eigenvalues; rationality, functional equation and RH for curves.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Zeta functions and Frobenius - theorem and technique pass. Master/audit: Rationality of Z(C,T); functional equation; Riemann hypothesis for curves over finite fields. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Structure of finite fields

U1.P1. [F] Construct F_4, F_8, F_9, and F_25 explicitly as polynomial quotients. List subfields, Frobenius automorphisms, multiplicative-group orders, traces, and norms.

U1.P2. [T] Prove every finite subgroup of a field's multiplicative group is cyclic and deduce F_q^* is cyclic. Use it to count elements of each possible order.

U1.P3. [S] Prove existence/uniqueness of F_{p^n} up to isomorphism and determine when F_{p^m} embeds in F_{p^n}.

Programme U2: Polynomial arithmetic and algorithms

U2.P1. [F] Implement/execute by hand Euclidean algorithm, modular exponentiation, square-free factorization, and one irreducibility test for polynomials over finite fields.

U2.P2. [T] Use Berlekamp or Cantor-Zassenhaus ideas on a small polynomial, carrying out the linear algebra/factor extraction explicitly.

U2.P3. [S] Compare computational complexity of naive versus fast arithmetic in F_q[x] at a high level, identifying which algebraic operations dominate finite-field cryptography/coding.

Programme U3: Characters, Gauss and Jacobi sums

U3.P1. [F] Construct additive and multiplicative characters of F_p and F_{p^n} using trace. Verify orthogonality identities explicitly.

U3.P2. [T] Prove the quadratic Gauss-sum magnitude and derive relations between Gauss and Jacobi sums. Evaluate a Jacobi sum in a small field.

U3.P3. [S] Use character orthogonality to count solutions of one polynomial equation over F_q, separating the main term and character-sum error.

Programme U4: Exponential sums and Weil bounds

U4.P1. [F] Compute exponential sums for linear and quadratic polynomials exactly and observe cancellation relative to the trivial q bound.

U4.P2. [T] State Weil's bound for additive/multiplicative character sums on curves/polynomials in a precise special case and verify sharp-scale consequences in examples.

U4.P3. [S] Apply a Weil bound to count points or incidences over F_q and derive a nontrivial combinatorial consequence such as existence of a solution once q is large relative to degree.

Programme U5: Function fields and curves

U5.P1. [F] For rational function fields F_q(t) and simple hyperelliptic/elliptic curves, compute places/valuations/divisors of selected functions.

U5.P2. [T] Prove the degree-zero property of principal divisors and use Riemann-Roch in genus 0/1 examples to construct functions with prescribed poles.

U5.P3. [S] Compare number fields and function fields: primes/places, units, zeta functions, Frobenius, and Riemann-Roch/class-group analogies in a detailed table.

Programme U6: Zeta functions and Frobenius

U6.P1. [F] Count #P^1(F_{q^n}) and derive its zeta function directly from the exponential generating definition.

U6.P2. [T] For an elliptic curve over a small finite field, count points over F_q and F_{q^2}, infer Frobenius trace, and check the local zeta numerator relation.

U6.P3. [R] Derive the rationality/formal determinant expression for curve zeta functions at theorem-architecture level, preparing the cohomological Frobenius formulation in Volume 33.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Finite Fields and Arithmetic over Function Fields: start from “Structure of finite fields”, pass through “Exponential sums and Weil bounds”, and finish at “Zeta functions and Frobenius”. Use the named results “Existence and uniqueness of F_{p^n}” and “prime polynomial theorem as function-field analogue.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Prepares Deligne/Weil theory, arithmetic geometry, coding/cryptography and finite-field additive combinatorics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Existence and uniqueness of F_{p^n}; F_q^* cyclic; subfield classification; Gal(F_{q^n}/F_q) cyclic generated by Frobenius; Criterion x^{q^n}-x product of monic irreducibles with degrees dividing n. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: finiteness of degree-zero divisor class group over finite constants; Rationality of Z(C,T); functional equation; Riemann hypothesis for curves over finite fields; prime polynomial theorem as function-field analogue.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Structure of finite fields” develops into “Zeta functions and Frobenius”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 28. Additive Combinatorics, Sum-Product and Approximate Groups

Purpose: Develop structure-versus-randomness methods from sumsets and Fourier analysis through higher-order uniformity, sum-product, growth in groups and approximate-group structure.

Prerequisites: Volumes 3, 9, 21-23, 27.

Primary and secondary references: Tao-Vu, Additive Combinatorics; Nathanson, Additive Number Theory: Inverse Problems and the Geometry of Sumsets; Green, Approximate Groups survey; Breuillard-Green-Tao, The Structure of Approximate Groups; Helfgott, Growth and Generation in SL2(Z/pZ), plus surveys

Quarter output: Write a 25-page “small doubling to group growth” synthesis with worked examples in Z, F_p and SL2(F_p), and present BKT, Helfgott and BGT as a single structural narrative.

Research bridges: Direct prerequisite for Bourgain-Gamburd expansion, modern incidence geometry, additive number theory and entropy inverse methods.

Six two-week study units

Unit 1 (Weeks 1-2): Sumsets and Ruzsa calculus

Topics: Addition sets; doubling; energy; Cauchy-Davenport; Ruzsa distance; covering; Plunnecke.

Key definitions: sumset/difference set; doubling constant; additive energy; Ruzsa distance; Plunnecke graph.

Key results to learn:

- Cauchy-Davenport theorem [PROOF/ARCH]
- Ruzsa triangle inequality [PROOF]
- Ruzsa covering lemma [PROOF/ARCH]
- Plunnecke inequalities [ARCH]
- energy/doubling relations [PROOF].

Reading route: Tao-Vu Chs. 1-2; Nathanson selected.

Warm-up/source exercise focus (secondary): Compute extremal sumsets in Z/pZ; prove Ruzsa inequalities; construct small-doubling examples.

Week 1: Sumsets and Ruzsa calculus - definitions and examples. Read: Tao-Vu Chs. 1-2; Nathanson selected.. Make explicit examples/nonexamples for sumset/difference set; doubling constant; additive energy; Ruzsa distance; Plunnecke graph.. Work through Addition sets; doubling; energy; Cauchy-Davenport; Ruzsa distance; covering; Plunnecke.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Sumsets and Ruzsa calculus - theorem and technique pass. Master/audit: Cauchy-Davenport theorem; Ruzsa triangle inequality; Ruzsa covering lemma. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Inverse sumset theory

Topics: Balog-Szemeredi-Gowers; Freiman homomorphisms; generalized arithmetic progressions; modeling.

Key definitions: Freiman homomorphism/isomorphism; GAP; proper progression; Freiman dimension.

Key results to learn:

- Balog-Szemeredi-Gowers theorem [ARCH]
- Freiman theorem in Z [ARCH/USE]
- Ruzsa modeling lemma [ARCH]
- Freiman-Ruzsa theorem/philosophy in finite abelian groups, standard formulations [USE].

Reading route: Tao-Vu inverse-theorem chapters; Green-Ruzsa papers/surveys.

Warm-up/source exercise focus (secondary): Recover structure from high energy; model finite sets; work explicit GAP examples and dimension bounds.

Week 3: Inverse sumset theory - definitions and examples. Read: Tao-Vu inverse-theorem chapters; Green-Ruzsa papers/surveys.. Make explicit examples/nonexamples for Freiman homomorphism/isomorphism; GAP; proper progression; Freiman dimension.. Work through Balog-Szemeredi-Gowers; Freiman homomorphisms; generalized arithmetic progressions; modeling.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Inverse sumset theory - theorem and technique pass. Master/audit: Balog-Szemeredi-Gowers theorem; Freiman theorem in Z; Ruzsa modeling lemma. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Fourier analysis and Roth

Topics: Fourier transform on finite abelian groups; convolution; spectrum; density increment; 3-term progressions.

Key definitions: Fourier coefficient; large spectrum; Bohr set; density increment; dissociated set.

Key results to learn:

- Parseval/convolution identities [PROOF]

- Roth theorem on 3-term APs in dense sets [ARCH/USE]
- Chang lemma on large spectrum [ARCH]
- Meshulam theorem in F_3^n [ARCH/USE].

Reading route: Tao-Vu Fourier/Roth chapters; Tao-Vu or Green notes.

Warm-up/source exercise focus (secondary): Run Roth density increment in a simplified cyclic model; compute spectra; construct Bohr sets.

Week 5: Fourier analysis and Roth - definitions and examples. Read: Tao-Vu Fourier/Roth chapters; Tao-Vu or Green notes.. Make explicit examples/nonexamples for Fourier coefficient; large spectrum; Bohr set; density increment; dissociated set.. Work through Fourier transform on finite abelian groups; convolution; spectrum; density increment; 3-term progressions.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Fourier analysis and Roth - theorem and technique pass. Master/audit: Parseval/convolution identities; Roth theorem on 3-term APs in dense sets; Chang lemma on large spectrum. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Gowers norms and higher-order structure

Topics: Uniformity norms; generalized von Neumann; inverse theorems; Szemeredi; transference.

Key definitions: Gowers U^k norm; nilsequence; polynomial phase; pseudorandom majorant; transference principle.

Key results to learn:

- Generalized von Neumann inequality [PROOF/ARCH]
- inverse theorem for U^2 [PROOF]
- higher U^k inverse theorem over finite fields/Z_N, representative precise formulation [USE]
- Szemeredi theorem [USE]
- Green-Tao theorem on APs in primes, architecture [USE].

Reading route: Tao-Vu higher-order chapters; Green-Tao/Ziegler surveys; Gowers expositions.

Warm-up/source exercise focus (secondary): Compute U2/U3 for model functions; prove generalized von Neumann in simple systems; map Green-Tao dependencies.

Week 7: Gowers norms and higher-order structure - definitions and examples. Read: Tao-Vu higher-order chapters; Green-Tao/Ziegler surveys; Gowers expositions.. Make explicit examples/nonexamples for Gowers U^k norm; nilsequence; polynomial phase; pseudorandom majorant; transference principle.. Work through Uniformity norms; generalized von Neumann; inverse theorems; Szemeredi; transference.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Gowers norms and higher-order structure - theorem and technique pass. Master/audit: Generalized von Neumann inequality; inverse theorem for U^2; higher U^k inverse theorem over finite fields/Z_N, representative precise formulation. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Sum-product phenomena

Topics: Growth under addition/multiplication; incidence geometry; finite-field expansion; discretized variants.

Key definitions: sum-product set; multiplicative energy; incidence; subfield obstruction.

Key results to learn:

- Erdos-Szemeredi sum-product conjecture: exact open statement and best-known structural philosophy [USE]
- Elekes incidence-based sum-product bound [ARCH]
- Bourgain-Katz-Tao finite-field sum-product theorem under size/subfield conditions [ARCH/USE]
- representative discretized sum-product theorem [USE].

Reading route: Tao-Vu sum-product chapters; Bourgain-Katz-Tao paper; Elekes/Szemeredi-Trotter cross-reference Volume 35.

Warm-up/source exercise focus (secondary): Prove elementary sum-product estimates; derive an incidence-based bound; identify subfield obstructions in F_{p^n}.

Week 9: Sum-product phenomena - definitions and examples. Read: Tao-Vu sum-product chapters; Bourgain-Katz-Tao paper; Elekes/Szemeredi-Trotter cross-reference Volume 35.. Make explicit examples/nonexamples for sum-product set; multiplicative energy; incidence; subfield obstruction.. Work through Growth under addition/multiplication; incidence geometry; finite-field expansion; discretized variants.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Sum-product phenomena - theorem and technique pass. Master/audit: Erdos-Szemeredi sum-product conjecture and best-known philosophy; Elekes incidence-based sum-product bound; Bourgain-Katz-Tao finite-field sum-product theorem under size/subfield conditions. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Growth in groups and approximate groups

Topics: Product sets in nonabelian groups; tripling; approximate groups; product theorems in simple groups.

Key definitions: K-approximate group; tripling constant; control; nilprogression; product growth.

Key results to learn:

- Noncommutative Ruzsa covering/BSG principles [ARCH]
- Helfgott product-growth theorem in SL2(F_p) [ARCH/USE]
- product theorem for finite simple groups of Lie type of bounded rank, representative statement [USE]

• Breuillard-Green-Tao approximate-group structure theorem [USE].

Reading route: BGT paper/survey; Helfgott paper/surveys; Tao noncommutative additive combinatorics notes.

Warm-up/source exercise focus (secondary): Analyze product growth in small matrix groups; verify approximate-group axioms; make an implication map from small tripling to nilpotent structure.

Week 11: Growth in groups and approximate groups - definitions and examples. Read: BGT paper/survey; Helfgott paper/surveys; Tao noncommutative additive combinatorics notes.. Make explicit examples/nonexamples for K-approximate group; tripling constant; control; nilprogression; product growth.. Work through Product sets in nonabelian groups; tripling; approximate groups; product theorems in simple groups.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Growth in groups and approximate groups - theorem and technique pass. Master/audit: Noncommutative Ruzsa covering/BSG principles; Helfgott product-growth theorem in SL2(F_p); product theorem for finite simple groups of Lie type of bounded rank, representative statement. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Sumsets and Ruzsa calculus

U1.P1. [F] For arithmetic progressions, random-looking sets, subgroups, and geometric progressions in Z/pZ, compute |A+A|, |A-A|, and additive energy. Compare structure signatures.

U1.P2. [T] Prove Ruzsa triangle inequality, covering lemma, and a selected Plunnecke inequality. Starting from |A+A|<=K|A|, derive explicit polynomial-in-K bounds for mA-nA.

U1.P3. [S] Derive E(A)>=|A|^4/|A+A| by Cauchy-Schwarz and explain why large energy is the bridge from small sumsets to inverse structure.

Programme U2: Inverse sumset theory

U2.P1. [F] Classify exact/small-doubling examples in Z or Z/pZ for small sizes and compare arithmetic progressions with unions/cosets.

U2.P2. [T] Work through a simplified Balog-Szemeredi-Gowers implication: large additive energy gives a large subset with small doubling. Track losses in all parameters.

U2.P3. [R] State Freiman's theorem in Z and one finite-field/model-group version precisely. For sample sets, identify the generalized arithmetic progression predicted by the theorem.

Programme U3: Fourier analysis and Roth

U3.P1. [F] Compute Fourier transforms of indicators in Z/NZ for intervals, subgroups, and random subsets. Verify Parseval and convolution identities.

U3.P2. [T] Prove Roth's theorem in Z/NZ via a Fourier density-increment argument in a simplified quantitative form. Identify the large Fourier coefficient and progression/subspace increment step.

U3.P3. [S] Compare physical-space BSG/Freiman and Fourier-space Roth methods: given three additive problems, decide which framework is more natural and explain why.

Programme U4: Gowers norms and higher-order structure

U4.P1. [F] Compute U^2 and U^3 norms for characters, random functions, and polynomial phases on finite vector spaces. Verify U^2 controls Fourier coefficients.

U4.P2. [T] Prove the generalized von Neumann inequality for 3-term progressions and formulate the analogous role of U^{k-1} for k-term patterns.

U4.P3. [R] State one inverse theorem for Gowers norms precisely in the setting you study (finite fields or integers), and work several examples of structured functions matching the conclusion.

Programme U5: Sum-product phenomena

U5.P1. [F] Compare A+A and A*A for arithmetic and geometric progressions in R/F_p. Quantify why simultaneous smallness is exceptional.

U5.P2. [T] Prove an elementary sum-product lower bound via incidences/energy in a toy regime, or reconstruct the Bourgain-Katz-Tao finite-field proof architecture with exact non-concentration assumptions.

U5.P3. [S] Convert a sum-product estimate into an expansion statement for a simple set of matrices/Mobius transformations, showing explicitly how additive and multiplicative growth interact.

Programme U6: Growth in groups and approximate groups

U6.P1. [F] Compute A^2 and A^3 for small subsets of S_3, Heisenberg groups, and SL_2(F_p); identify examples with small tripling caused by subgroup/coset structure.

U6.P2. [T] Prove noncommutative Ruzsa covering/triangle lemmas in a standard form and derive a small-tripling consequence.

U6.P3. [R] Compare Helfgott product growth in SL_2(F_p) with the Breuillard-Green-Tao approximate-group theorem: state hypotheses/conclusions and explain the structural obstruction to growth in each.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Additive Combinatorics, Sum-Product and Approximate Groups: start from “Sumsets and Ruzsa calculus”, pass through “Gowers norms and higher-order structure”, and finish at “Growth in groups and approximate groups”. Use the named results “Cauchy-Davenport theorem” and “Breuillard-Green-Tao approximate-group structure theorem.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct prerequisite for Bourgain-Gamburd expansion, modern incidence geometry, additive number theory and entropy inverse methods. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Cauchy-Davenport theorem; Ruzsa triangle inequality; Ruzsa covering lemma; energy/doubling relations; Parseval/convolution identities. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: representative discretized sum-product theorem; Noncommutative Ruzsa covering/BSG principles; Helfgott product-growth theorem in SL2(F_p); product theorem for finite simple groups of Lie type of bounded rank, representative statement; Breuillard-Green-Tao approximate-group structure theorem.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Sumsets and Ruzsa calculus” develops into “Growth in groups and approximate groups”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 29. Expanders, Spectral Gaps and the Bourgain-Gamburd Method

Purpose: Learn spectral graph theory and the modern expansion machine combining quasirandomness, product growth, flattening and escape from subgroups, through super-approximation.

Prerequisites: Volumes 7, 18, 21 and 28; Volume 24 helpful for arithmetic applications.

Primary and secondary references: Hoory-Linial-Wigderson, Expander Graphs and Their Applications; Lubotzky, Discrete Groups, Expanding Graphs and Invariant Measures; Bourgain-Gamburd, Uniform expansion bounds for Cayley graphs of SL2(F_p), and related papers; Helfgott/Breuillard-Green-Tao product-growth references; Salehi Golsefidy-Varju and Bourgain-Varju papers on expansion/super-approximation

Quarter output: Deliver a three-lecture mini-course: (1) spectral expanders, (2) product growth/flattening, (3) Bourgain-Gamburd to super-approximation, with a written 30-page companion.

Research bridges: Connects property (T)/(tau), approximate groups, arithmetic groups, affine sieve, random walks and modern spectral-gap methods.

Six two-week study units

Unit 1 (Weeks 1-2): Spectral graph theory

Topics: Adjacency/Markov/Laplacian operators; conductance; edge expansion; mixing; expander families.

Key definitions: normalized adjacency; normalized Laplacian; spectral gap; conductance/Cheeger constant; expander family.

Key results to learn:

- Cheeger inequalities for regular graphs [PROOF/ARCH]
- expander mixing lemma [PROOF]
- spectral-gap mixing bound for random walks [PROOF/ARCH]
- equivalence of bounded-degree spectral/combinatorial expansion [PROOF/ARCH].

Reading route: HLW survey Secs. 2-4; Lubotzky early chapters.

Warm-up/source exercise focus (secondary): Compute spectra of cycles/complete graphs/Cayley examples; prove expander mixing; estimate mixing times from gaps.

Week 1: Spectral graph theory - definitions and examples. Read: HLW survey Secs. 2-4; Lubotzky early chapters.. Make explicit examples/nonexamples for normalized adjacency; normalized Laplacian; spectral gap; conductance/Cheeger constant; expander family.. Work through Adjacency/Markov/Laplacian operators; conductance; edge expansion; mixing; expander families.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Spectral graph theory - theorem and technique pass. Master/audit: Cheeger inequalities for regular graphs; expander mixing lemma; spectral-gap mixing bound for random walks. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Explicit expanders and representation theory

Topics: Cayley/Schreier graphs; property (T)/(tau); representation multiplicities; Ramanujan graphs.

Key definitions: Cayley graph; Schreier graph; Ramanujan graph; Alon-Boppana bound; property (tau).

Key results to learn:

- Alon-Boppana lower bound [ARCH]
- property (T) -> spectral expansion for finite quotient families [PROOF/ARCH]
- Selberg/property-tau paradigm [USE]
- existence/construction of Ramanujan graph families in classical settings, statement [USE].

Reading route: Lubotzky; HLW; representation-theory cross-reference Volumes 18/21.

Warm-up/source exercise focus (secondary): Relate operator norm to irreducible reps in Cayley graphs; prove easy property-tau implications; compare Ramanujan and generic expanders.

Week 3: Explicit expanders and representation theory - definitions and examples. Read: Lubotzky; HLW; representation-theory cross-reference Volumes 18/21.. Make explicit examples/nonexamples for Cayley graph; Schreier graph; Ramanujan graph; Alon-Boppana bound; property (tau).. Work through Cayley/Schreier graphs; property (T)/(tau); representation multiplicities; Ramanujan graphs.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Explicit expanders and representation theory - theorem and technique pass. Master/audit: Alon-Boppana lower bound; property (T) -> spectral expansion for finite quotient families; Selberg/property-tau paradigm. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Quasirandomness and flattening

Topics: Minimal representation dimension; mixing in quasirandom groups; convolution norms; L2 flattening; noncommutative BSG.

Key definitions: D-quasirandom group; probability measure convolution; L2 flattening; approximate subgroup.

Key results to learn:

- Gowers quasirandom mixing estimates [PROOF/ARCH]
- noncommutative Balog-Szemeredi-Gowers [ARCH/USE]
- L2-flattening lemma in a standard Bourgain-Gamburd form [ARCH/USE].

Reading route: Gowers quasirandom groups paper/surveys; Bourgain-Gamburd expositions; Tao notes.

Warm-up/source exercise focus (secondary): Bound triple-product counts from representation dimensions; compute convolution norms; derive how nonflattening yields approximate structure.

Week 5: Quasirandomness and flattening - definitions and examples. Read: Gowers quasirandom groups paper/surveys; Bourgain-Gamburd expositions; Tao notes.. Make explicit examples/nonexamples for D-quasirandom group; probability measure convolution; L2 flattening; approximate subgroup.. Work through Minimal representation dimension; mixing in quasirandom groups; convolution norms; L2 flattening; noncommutative BSG.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Quasirandomness and flattening - theorem and technique pass. Master/audit: Gowers quasirandom mixing estimates; noncommutative Balog-Szemeredi-Gowers; L2-flattening lemma in a standard Bourgain-Gamburd form [ARCH/USE].. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Escape from subgroups and product growth

Topics: Random-word escape; subgroup classification; product theorem; concentration control.

Key definitions: escape from subgroups; proper algebraic subgroup; girth; nonconcentration.

Key results to learn:

- Helfgott growth theorem in SL2(F_p) [ARCH/USE]
- classification facts on proper subgroups of SL2(F_p) needed by the method [ARCH]
- escape/nonconcentration estimate for admissible generating sets [USE].

Reading route: Helfgott; Bourgain-Gamburd preliminary sections; Dickson subgroup classification reference as needed.

Warm-up/source exercise focus (secondary): Classify model obstructions; trace word growth; verify where sum-product/product growth rules out concentration.

Week 7: Escape from subgroups and product growth - definitions and examples. Read: Helfgott; Bourgain-Gamburd preliminary sections; Dickson subgroup classification reference as needed.. Make explicit examples/nonexamples for escape from subgroups; proper algebraic subgroup; girth; nonconcentration.. Work through Random-word escape; subgroup classification; product theorem; concentration control.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Escape from subgroups and product growth - theorem and technique pass. Master/audit: Helfgott growth theorem in SL2(F_p); classification facts on proper subgroups of SL2(F_p) needed by the method; escape/nonconcentration estimate for admissible generating sets [USE].. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): The Bourgain-Gamburd expansion machine

Topics: Three-input expansion strategy: quasirandomness, product growth/approximate groups, nonconcentration; flattening iteration.

Key definitions: spectral measure; flattening scale; quasirandomness parameter; escape scale.

Key results to learn:

- Bourgain-Gamburd uniform expansion theorem for suitable fixed/Zariski-dense generators reduced mod p in SL2(F_p), precise standard formulation [ARCH/USE]
- flattening-to-spectral-gap implication [ARCH/USE]
- expansion-machine abstraction [USE].

Reading route: Bourgain-Gamburd original paper plus expository surveys.

Warm-up/source exercise focus (secondary): Write every input/output of the machine as lemmas; run a toy flattening iteration numerically; give a 45-minute theorem-chain presentation.

Week 9: The Bourgain-Gamburd expansion machine - definitions and examples. Read: Bourgain-Gamburd original paper plus expository surveys.. Make explicit examples/nonexamples for spectral measure; flattening scale; quasirandomness parameter; escape scale.. Work through Three-input expansion strategy: quasirandomness, product growth/approximate groups, nonconcentration; flattening iteration.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: The Bourgain-Gamburd expansion machine - theorem and technique pass. Master/audit: Bourgain-Gamburd uniform expansion theorem for suitable fixed/Zariski-dense generators reduced mod p in SL2(F_p), precise standard formulation; flattening-to-spectral-gap implication; expansion-machine abstraction [USE].. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Super-approximation and applications

Topics: Expansion modulo composite moduli; Zariski-dense subgroups; perfect identity component; affine sieve/random walks.

Key definitions: super-approximation; congruence quotient; Zariski closure; perfect group; affine sieve.

Key results to learn:

- Bourgain-Varju expansion in SL_d(Z/qZ) for suitable generators, selected precise formulation [USE]
- Salehi Golsefidy-Varju super-approximation theorem in its standard algebraic setting [USE]
- affine-sieve expansion input and representative almost-prime consequence [USE].

Reading route: Bourgain-Varju; Salehi Golsefidy-Varju; Bourgain-Gamburd-Sarnak affine sieve surveys.

Warm-up/source exercise focus (secondary): Check hypotheses for example matrix groups; explain prime versus composite modulus difficulties; map expansion to sieve level of distribution.

Week 11: Super-approximation and applications - definitions and examples. Read: Bourgain-Varju; Salehi Golsefidy-Varju; Bourgain-Gamburd-Sarnak affine sieve surveys.. Make explicit examples/nonexamples for super-approximation; congruence quotient; Zariski closure; perfect group; affine sieve.. Work through Expansion modulo composite moduli; Zariski-dense subgroups; perfect identity component; affine sieve/random walks.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Super-approximation and applications - theorem and technique pass. Master/audit: Bourgain-Varju expansion in SL_d(Z/qZ) for suitable generators, selected precise formulation; Salehi Golsefidy-Varju super-approximation theorem in its standard algebraic setting; affine-sieve expansion input and representative almost-prime consequence [USE].. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Spectral graph theory

- U1.P1. [F] Compute spectra, spectral gaps, conductance, and mixing times for cycles, complete graphs, hypercubes, and small Cayley graphs. Compare normalized versus unnormalized adjacency conventions.
- U1.P2. [T] Prove the variational characterization of lambda_2 and one side of Cheeger's inequality. Use it to convert a combinatorial expansion bound into a spectral bound and vice versa.
- U1.P3. [S] Derive an L^2 and total-variation mixing estimate for a reversible regular graph from the spectral gap, including dependence on the smallest stationary mass.

Programme U2: Explicit expanders and representation theory

- U2.P1. [F] Construct Cayley graphs of finite quotients such as SL_2(F_p) with small generating sets and compute spectra numerically for small p.
- U2.P2. [T] Use representation decomposition of the regular representation to express adjacency eigenvalues of a Cayley graph through irreducible representations.
- U2.P3. [R] Study one explicit expander construction (Margulis/Gabber-Galil, LPS Ramanujan, or property-T quotient family): write the algebraic input and the theorem yielding the spectral gap.

Programme U3: Quasirandomness and flattening

- U3.P1. [F] For probability measures mu on a finite group, compute L^1/L^2 norms of convolution powers in subgroup-supported, uniform, and generating examples.
- U3.P2. [T] Prove a finite-group L^2-flattening lemma in a simplified setting from high multiplicative energy plus BSG/product-growth input. Make the contrapositive structure explicit.
- U3.P3. [S] Explain quasirandomness through minimal dimensions of nontrivial representations and prove how this forces a nearly flat measure to mix once its L^2 norm crosses a threshold.

Programme U4: Escape from subgroups and product growth

- U4.P1. [F] In SL_2(F_p), classify/inspect standard proper subgroup types needed in an expansion proof and compute intersections of a sample generating set with them for small p.
- U4.P2. [T] Prove a toy escape-from-subgroups statement for a free/non-elementary generating set using word maps or random-walk counting in a simplified model.
- U4.P3. [R] Audit Helfgott growth: identify trace amplification, tori/centralizers, escape from subvarieties, and the conclusion |A^3| >= min(|A|^{1+delta},|G|). Mark which ingredients feed Bourgain-Gamburd.

Programme U5: The Bourgain-Gamburd expansion machine

- U5.P1. [S] Starting from (i) non-concentration in proper subgroups, (ii) product growth/flattening, and (iii) quasirandomness, write and prove the abstract three-phase route to a spectral gap for convolution by a generating measure.
- U5.P2. [T] Translate the operator spectral gap into expansion of the Cayley graph and exponential random-walk mixing. Keep normalization constants consistent throughout.
- U5.P3. [R] Reconstruct one Bourgain-Gamburd theorem for SL_2(F_p): state the generating-set assumptions uniformly in p, the expansion conclusion, and where each of the three machine inputs is established.

Programme U6: Super-approximation and applications

- U6.P1. [F] For reductions of a fixed subgroup Gamma<SL_d(Z) modulo primes, identify when the image is large/surjective in elementary examples and what strong approximation would assert generally.
- U6.P2. [R] State a representative super-approximation theorem (Salehi Golsefidy-Varju/Bourgain-Varju style) precisely enough to show the algebraic hypotheses and family of congruence quotients.
- U6.P3. [S] Trace one application - affine sieve, random walks, thin groups, or Diophantine counting - from spectral gap to a quantitative arithmetic conclusion. Explicitly write the inequality where expansion is used.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Expanders, Spectral Gaps and the Bourgain-Gamburd Method: start from “Spectral graph theory”, pass through “Escape from subgroups and product growth”, and finish at “Super-approximation and applications”. Use the named results “Cheeger inequalities for regular graphs” and “affine-sieve expansion input and representative almost-prime consequence.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Connects property (T)/(tau), approximate groups, arithmetic groups, affine sieve, random walks and modern spectral-gap methods. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Cheeger inequalities for regular graphs; expander mixing lemma; spectral-gap mixing bound for random walks; equivalence of bounded-degree spectral/combinatorial expansion; property (T) -> spectral expansion for finite quotient families. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: flattening-to-spectral-gap implication; expansion-machine abstraction; Bourgain-Varju expansion in SL_d(Z/qZ) for suitable generators, selected precise formulation; Salehi Golsefidy-Varju super-approximation theorem in its standard algebraic setting; affine-sieve expansion input and representative almost-prime consequence.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Spectral graph theory” develops into “Super-approximation and applications”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 30. Modular Forms and Automorphic Representations

Purpose: Progress from classical modular forms through Hecke theory and L-functions to Maass forms, adelic GL2, Rankin-Selberg theory and the Langlands viewpoint.

Prerequisites: Volumes 8, 18, 22-24; Volume 23 helpful for spectral methods.

Primary and secondary references: Diamond-Shurman, A First Course in Modular Forms; Miyake, Modular Forms, selected chapters; Bump, Automorphic Forms and Representations; Goldfeld, Automorphic Forms and L-functions for GL(n,R), selected chapters; Gelbart, Automorphic Forms on Adele Groups, selected; Tate, Thesis, in Cassels-Froehlich or modern expositions

Quarter output: Take one normalized newform and produce a 30-page classical-to-adelic dossier: q-expansion, Hecke eigenvalues, classical L-function, local components, Satake parameters, global representation and Rankin-Selberg context.

Research bridges: Foundation for FLT/modularity, Sato-Tate, spectral analytic number theory and modern Langlands/arithmetic geometry.

Six two-week study units

Unit 1 (Weeks 1-2): Classical modular forms

Topics: Upper half-plane; SL2(Z) action; modular/cusp forms; q-expansions; Eisenstein series; discriminant form; modular curves basics.

Key definitions: weight-k modular form; cusp form; slash operator; q-expansion; Eisenstein series; cusp; modular curve.

Key results to learn:

- Valence formula [ARCH]
- dimension formulas for M_k(SL2Z), standard [ARCH]
- graded ring M_*(SL2Z)=C[E4,E6] [PROOF/ARCH]
- q-expansion principle in basic setting [ARCH]
- transformation laws of E_k and Delta [PROOF/ARCH].

Reading route: Diamond-Shurman Chs. 1-3; Serre, A Course in Arithmetic, modular-form chapters.

Warm-up/source exercise focus (secondary): Compute bases in low weights; prove identities by dimensions/q-expansions; calculate first Fourier coefficients.

Week 1: Classical modular forms - definitions and examples. Read: Diamond-Shurman Chs. 1-3; Serre, A Course in Arithmetic, modular-form chapters.. Make explicit examples/nonexamples for weight-k modular form; cusp form; slash operator; q-expansion; Eisenstein series; cusp; modular curve.. Work through Upper half-plane; SL2(Z) action; modular/cusp forms; q-expansions; Eisenstein series; discriminant form; modular curves basics.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Classical modular forms - theorem and technique pass. Master/audit: Valence formula; dimension formulas for M_k(SL2Z), standard; graded ring M_*(SL2Z)=C[E4,E6]. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Hecke operators, eigenforms and newforms

Topics: Hecke correspondences/operators; Petersson product; eigenforms; old/new subspaces; multiplicity one.

Key definitions: Hecke operator T_n; normalized eigenform; newform; oldform; Atkin-Lehner operator.

Key results to learn:

- Hecke algebra commutativity/self-adjointness under standard conditions [PROOF/ARCH]
- Hecke eigenvalue multiplicativity [PROOF]
- Euler product attached to eigenform [PROOF/ARCH]
- newform decomposition [ARCH/USE]
- strong multiplicity-one flavor for GL2 [USE].

Reading route: Diamond-Shurman Hecke/newform chapters; Miyake selected.

Warm-up/source exercise focus (secondary): Compute T_p on q-expansions; diagonalize low-dimensional spaces; derive Euler factors from Hecke relations.

Week 3: Hecke operators, eigenforms and newforms - definitions and examples. Read: Diamond-Shurman Hecke/newform chapters; Miyake selected.. Make explicit examples/nonexamples for Hecke operator T_n; normalized eigenform; newform; oldform; Atkin-Lehner operator.. Work through Hecke correspondences/operators; Petersson product; eigenforms; old/new subspaces; multiplicity one.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Hecke operators, eigenforms and newforms - theorem and technique pass. Master/audit: Hecke algebra commutativity/self-adjointness under standard conditions; Hecke eigenvalue multiplicativity; Euler product attached to eigenform. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Modular L-functions and converse philosophy

Topics: Mellin transforms; completed L-functions; functional equations; twists; converse theorems.

Key definitions: completed L-function; root number; conductor; twist; gamma factor.

Key results to learn:

- Mellin transform gives analytic continuation/functional equation for cusp-form L-functions [PROOF/ARCH]
- Hecke converse theorem in a classical formulation [ARCH/USE]

- Euler product and Ramanujan-Petersson/Deligne bound statement for holomorphic eigenforms [USE].

Reading route: Diamond-Shurman; Iwaniec topics; Bump introductory chapters.

Warm-up/source exercise focus (secondary): Derive functional equation of a modular L-function; compute local Euler factors; compare converse-theorem hypotheses.

Week 5: Modular L-functions and converse philosophy - definitions and examples. Read: Diamond-Shurman; Iwaniec topics; Bump introductory chapters.. Make explicit examples/nonexamples for completed L-function; root number; conductor; twist; gamma factor.. Work through Mellin transforms; completed L-functions; functional equations; twists; converse theorems.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Modular L-functions and converse philosophy - theorem and technique pass. Master/audit: Mellin transform gives analytic continuation/functional equation for cusp-form L-functions; Hecke converse theorem in a classical formulation; Euler product and Ramanujan-Petersson/Deligne bound statement for holomorphic eigenforms [USE].. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Maass forms and spectral theory

Topics: Laplacian on Gamma\H; Maass cusp forms; Eisenstein series; continuous spectrum; Selberg theory.

Key definitions: Maass form; automorphic Laplacian; spectral parameter; Eisenstein series; scattering matrix.

Key results to learn:

- Self-adjoint spectral decomposition of L2(Gamma\H) [ARCH/USE]
- Fourier expansion of Maass forms [ARCH]
- meromorphic continuation/functional equation of Eisenstein series [ARCH/USE]
- Selberg trace formula philosophy and a standard formula [USE].

Reading route: Bump/Goldfeld spectral chapters; Iwaniec, Spectral Methods, selected.

Warm-up/source exercise focus (secondary): Compute Laplacian eigen-equations; derive Fourier-Bessel form; decompose model automorphic functions into spectral pieces conceptually.

Week 7: Maass forms and spectral theory - definitions and examples. Read: Bump/Goldfeld spectral chapters; Iwaniec, Spectral Methods, selected.. Make explicit examples/nonexamples for Maass form; automorphic Laplacian; spectral parameter; Eisenstein series; scattering matrix.. Work through Laplacian on Gamma\H; Maass cusp forms; Eisenstein series; continuous spectrum; Selberg theory.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Maass forms and spectral theory - theorem and technique pass. Master/audit: Self-adjoint spectral decomposition of L2(Gamma\H); Fourier expansion of Maass forms; meromorphic continuation/functional equation of Eisenstein series. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Adeles and automorphic representations of GL2

Topics: Adele group GL2(A); automorphic forms; restricted tensor products; local representations; spherical vectors; Satake parameters.

Key definitions: automorphic representation; restricted tensor product; unramified representation; spherical vector; Satake parameter; local L-factor.

Key results to learn:

- Classical-to-adelic correspondence for modular/newforms [ARCH/USE]
- factorization pi=⊗' pi_v [USE]
- Satake classification for unramified GL2 local representations [ARCH/USE]
- strong multiplicity one for GL2 [USE].

Reading route: Bump central GL2 chapters; Gelbart; Tate thesis background from Volume 24.

Warm-up/source exercise focus (secondary): Adelize a classical newform step by step; identify archimedean/p-adic local components; compute unramified local factors.

Week 9: Adeles and automorphic representations of GL2 - definitions and examples. Read: Bump central GL2 chapters; Gelbart; Tate thesis background from Volume 24.. Make explicit examples/nonexamples for automorphic representation; restricted tensor product; unramified representation; spherical vector; Satake parameter; local L-factor.. Work through Adele group GL2(A); automorphic forms; restricted tensor products; local representations; spherical vectors; Satake parameters.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Adeles and automorphic representations of GL2 - theorem and technique pass. Master/audit: Classical-to-adelic correspondence for modular/newforms; factorization pi=⊗' pi_v; Satake classification for unramified GL2 local representations. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Rankin-Selberg, trace formula and Langlands bridges

Topics: Automorphic L-functions; Rankin-Selberg integrals; local-global Euler products; functoriality/converse frameworks.

Key definitions: Rankin-Selberg convolution; standard L-function; cusp representation; contragredient; functorial lift.

Key results to learn:

- Rankin-Selberg integral representation and analytic continuation for GL2 x GL2 in a standard setting [ARCH/USE]
- Jacquet-Langlands correspondence, statement [USE]
- local/global functional-equation architecture [USE]
- Langlands reciprocity/functoriality principles in precise conjectural form [USE].

Reading route: Bump later chapters; Gelbart; Jacquet-Langlands expository sources; Tate thesis.

Warm-up/source exercise focus (secondary): Unfold a basic Rankin-Selberg integral; match local Euler factors; create a dictionary classical modular form <-> automorphic representation <-> L-function.

Week 11: Rankin-Selberg, trace formula and Langlands bridges - definitions and examples. Read: Bump later chapters; Gelbart; Jacquet-Langlands expository sources; Tate thesis.. Make explicit examples/nonexamples for Rankin-Selberg convolution; standard L-function; cusp representation; contragredient; functorial lift.. Work through Automorphic L-functions; Rankin-Selberg integrals; local-global Euler products; functoriality/converse frameworks.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Rankin-Selberg, trace formula and Langlands bridges - theorem and technique pass. Master/audit: Rankin-Selberg integral representation and analytic continuation for GL2 x GL2 in a standard setting; Jacquet-Langlands correspondence, statement; local/global functional-equation architecture. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Classical modular forms

U1.P1. [F] Compute q-expansions of E4, E6, Delta and verify modular weights/products to enough order to see identities such as E4^3-E6^2 proportional to Delta.

U1.P2. [T] Prove the valence/dimension formula in a selected level-one setting and use it to determine spaces M_k and S_k for several small weights.

U1.P3. [S] Derive Mellin transform of a cusp form and the associated completed L-function functional equation from modular transformation, with all normalizing powers recorded.

Programme U2: Hecke operators, eigenforms and newforms

U2.P1. [F] Compute T_p on q-expansions and verify Hecke relations T_m T_n for coprime m,n and prime powers on selected forms.

U2.P2. [T] Prove simultaneous diagonalization of commuting self-adjoint Hecke operators under the Petersson inner product in a finite-dimensional cusp space.

U2.P3. [S] From Hecke eigenform relations derive multiplicativity of Fourier coefficients and the Euler product of L(s,f), including the quadratic local factor at unramified primes.

Programme U3: Modular L-functions and converse philosophy

U3.P1. [T] Derive the functional equation of L(s,f) for a level-one holomorphic eigenform via Mellin transform. Identify conductor, gamma factors, and center.

U3.P2. [F] Compare Euler products/gamma factors of zeta, Dirichlet L-functions, and a GL_2 cusp-form L-function. Write the local/global data side by side.

U3.P3. [R] State a converse theorem in a simplified GL_2 form and explain what analytic properties of twists are sufficient to recover modularity/automorphy.

Programme U4: Maass forms and spectral theory

U4.P1. [F] Write the hyperbolic Laplacian, verify invariance under SL_2(R), and compute Fourier-Bessel form of a Maass cusp eigenfunction from periodicity plus eigen-equation.

U4.P2. [T] Derive the spectral decomposition of L^2(SL_2(Z)\H) at theorem-architecture level: constants, cusp spectrum, Eisenstein continuous spectrum.

U4.P3. [S] Use unfolding on an Eisenstein/Poincare series to derive one Rankin-Selberg integral identity or Petersson-type formula.

Programme U5: Adeles and automorphic representations of GL2

U5.P1. [F] Construct the adele ring A_Q and the embedding of GL_2(Q) into GL_2(A). Rewrite a classical modular form as an adelic function satisfying left rational invariance and right compact/K-type conditions.

U5.P2. [T] Factor an automorphic representation pi as a restricted tensor product of local pi_v in the unramified almost-everywhere setting and match T_p eigenvalues with Satake parameters.

U5.P3. [S] Starting from an unramified local GL_2 representation, write its local L-factor and recover the classical Euler factor of an eigenform.

Programme U6: Rankin-Selberg, trace formula and Langlands bridges

U6.P1. [T] Unfold a Rankin-Selberg convolution integral and identify the Euler product/local zeta integrals in a standard GL_2 example.

U6.P2. [R] Compare Selberg/Petersson/Kuznetsov trace formulas with the Arthur trace-formula philosophy: list geometric conjugacy/orbital data versus spectral automorphic data without pretending to prove the full formula.

U6.P3. [R] Build a Langlands bridge diagram for GL_1 class field theory and GL_2 modular forms: local parameters, global automorphic representation, L-function, and Galois representation where available.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Modular Forms and Automorphic Representations: start from “Classical modular forms”, pass through “Maass forms and spectral theory”, and finish at “Rankin-Selberg, trace formula and Langlands bridges”. Use the named results “Valence formula” and “Langlands reciprocity/functoriality principles in precise conjectural form.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Foundation for FLT/modularity, Sato-Tate, spectral analytic number theory and modern Langlands/arithmetic geometry. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: graded ring M_*(SL2Z)=C[E4,E6]; transformation laws of E_k and Delta; Hecke algebra commutativity/self-adjointness under standard conditions; Hecke eigenvalue multiplicativity; Euler product attached to eigenform. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: strong multiplicity one for GL2; Rankin-Selberg integral representation and analytic continuation for GL2 x GL2 in a standard setting; Jacquet-Langlands correspondence, statement; local/global functional-equation architecture; Langlands reciprocity/functoriality principles in precise conjectural form.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Classical modular forms” develops into “Rankin-Selberg, trace formula and Langlands bridges”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 31. Algebraic Geometry: Varieties, Schemes and Cohomology

Purpose: Build a working language of schemes, morphisms, divisors, line bundles, sheaf cohomology and intersection theory sufficient for modern arithmetic geometry.

Prerequisites: Volumes 3-4, 8, 11-12; commutative algebra from Volume 4 is essential.

Primary and secondary references: Vakil, The Rising Sea: Foundations of Algebraic Geometry, selected chapters; Hartshorne, Algebraic Geometry, Chapters I-III and selected V; Gortz-Wedhorn, Algebraic Geometry I, selected chapters; Mumford, The Red Book of Varieties and Schemes, selected

Quarter output: Create a 30-page scheme dossier comparing a smooth elliptic curve, a nodal/cuspidal cubic and an arithmetic base change, including local rings, divisors, cohomology and morphism properties.

Research bridges: Essential prerequisite for arithmetic geometry, etale cohomology, Faltings, Weil/Deligne and modern moduli problems.

Six two-week study units

Unit 1 (Weeks 1-2): Affine and projective varieties

Topics: Affine algebraic sets; coordinate rings; morphisms; irreducibility; dimension; projective space and projective varieties.

Key definitions: affine variety; coordinate ring; Zariski topology; irreducible component; dimension; projective variety; homogeneous ideal.

Key results to learn:

- Hilbert Nullstellensatz [PROOF/ARCH]
- algebra-geometry correspondence for affine varieties over algebraically closed fields [PROOF/ARCH]
- dimension equals transcendence degree for irreducible affine varieties [ARCH]
- projective Nullstellensatz in standard form [ARCH].

Reading route: Vakil early chapters; Hartshorne I.1-7.

Warm-up/source exercise focus (secondary): Compute coordinate rings, closures and dimensions; analyze singular plane curves; work projective closures and morphisms.

Week 1: Affine and projective varieties - definitions and examples. Read: Vakil early chapters; Hartshorne I.1-7.. Make explicit examples/nonexamples for affine variety; coordinate ring; Zariski topology; irreducible component; dimension; projective variety; homogeneous ideal.. Work through Affine algebraic sets; coordinate rings; morphisms; irreducibility; dimension; projective space and projective varieties.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Affine and projective varieties - theorem and technique pass. Master/audit: Hilbert Nullstellensatz; algebra-geometry correspondence for affine varieties over algebraically closed fields; dimension equals transcendence degree for irreducible affine varieties. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Schemes, Spec and fiber products

Topics: Prime spectrum; structure sheaf; affine schemes; gluing; functor of points; fiber products/base change.

Key definitions: scheme; affine scheme; local ring; stalk; residue field; morphism of schemes; fiber product; base change.

Key results to learn:

- Anti-equivalence between commutative rings and affine schemes [PROOF/ARCH]
- gluing schemes from affine data [PROOF/ARCH]
- universal property/existence of fiber products [PROOF/ARCH]
- Yoneda viewpoint in standard representable settings [ARCH].

Reading route: Vakil scheme foundations; Hartshorne II.1-3.

Warm-up/source exercise focus (secondary): Compute Spec of Z, DVRs and nonreduced rings; glue P1; calculate representative fiber products and fibers.

Week 3: Schemes, Spec and fiber products - definitions and examples. Read: Vakil scheme foundations; Hartshorne II.1-3.. Make explicit examples/nonexamples for scheme; affine scheme; local ring; stalk; residue field; morphism of schemes; fiber product; base change.. Work through Prime spectrum; structure sheaf; affine schemes; gluing; functor of points; fiber products/base change.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Schemes, Spec and fiber products - theorem and technique pass. Master/audit: Anti-equivalence between commutative rings and affine schemes; gluing schemes from affine data; universal property/existence of fiber products. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Morphisms: proper, flat, smooth and etale

Topics: Geometric properties of morphisms; fibers; dimension; valuative criteria; flatness; smoothness; etaleness.

Key definitions: finite-type/finite morphism; separated; proper; flat; smooth; unramified; etale; geometric fiber.

Key results to learn:

- Valuative criterion for separatedness/properness [ARCH]
- generic flatness [ARCH]

- upper semicontinuity/dimension-of-fibers principles [ARCH/USE]
- smoothness criteria/Jacobian criterion in standard finite-type settings [ARCH]
- basic stability under base change [PROOF/ARCH].

Reading route: Vakil morphism chapters; Hartshorne II.4, III.9-10 selected; Gortz-Wedhorn.

Warm-up/source exercise focus (secondary): Check morphism properties in explicit families; use valuative criteria on A1/P1; analyze singular and smooth fibers.

Week 5: Morphisms: proper, flat, smooth and etale - definitions and examples. Read: Vakil morphism chapters; Hartshorne II.4, III.9-10 selected; Gortz-Wedhorn.. Make explicit examples/nonexamples for finite-type/finite morphism; separated; proper; flat; smooth; unramified; etale; geometric fiber.. Work through Geometric properties of morphisms; fibers; dimension; valuative criteria; flatness; smoothness; etaleness.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Morphisms: proper, flat, smooth and etale - theorem and technique pass. Master/audit: Valuative criterion for separatedness/properness; generic flatness; upper semicontinuity/dimension-of-fibers principles. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Divisors, line bundles and curves

Topics: Cartier/Weil divisors; Picard groups; invertible sheaves; linear systems; nonsingular curves.

Key definitions: Weil divisor; Cartier divisor; line bundle/invertible sheaf; Picard group; degree; canonical divisor; linear system.

Key results to learn:

- Cartier divisors modulo principal divisors correspond to line bundles under standard hypotheses [ARCH]
- Riemann-Roch theorem for smooth projective curves [ARCH/USE]
- genus-degree consequences [PROOF/ARCH]
- Hurwitz formula [ARCH].

Reading route: Hartshorne II.6, IV selected; Vakil divisor/curve chapters.

Warm-up/source exercise focus (secondary): Compute Picard/divisor classes on P1 and elliptic curves; use Riemann-Roch to construct functions/maps; apply Hurwitz to covers.

Week 7: Divisors, line bundles and curves - definitions and examples. Read: Hartshorne II.6, IV selected; Vakil divisor/curve chapters.. Make explicit examples/nonexamples for Weil divisor; Cartier divisor; line bundle/invertible sheaf; Picard group; degree; canonical divisor; linear system.. Work through Cartier/Weil divisors; Picard groups; invertible sheaves; linear systems; nonsingular curves.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Divisors, line bundles and curves - theorem and technique pass. Master/audit: Cartier divisors modulo principal divisors correspond to line bundles under standard hypotheses; Riemann-Roch theorem for smooth projective curves; genus-degree consequences. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Sheaf cohomology and duality

Topics: Derived functors of global sections; Cech comparisons; coherent cohomology; projective space; Serre vanishing/duality.

Key definitions: sheaf cohomology H^i; acyclic sheaf; coherent sheaf; derived functor; Serre twist; dualizing sheaf.

Key results to learn:

- Long exact cohomology sequence [PROOF/ARCH]
- cohomology of O(n) on projective space [ARCH]
- Serre vanishing [ARCH/USE]
- Serre finiteness [ARCH/USE]
- Serre duality for smooth projective varieties/curves [ARCH/USE].

Reading route: Hartshorne III.1-7 selected; Vakil cohomology chapters.

Warm-up/source exercise focus (secondary): Compute H^i(P^n,O(m)) in low cases; use exact sequences for plane curves; verify duality dimensions.

Week 9: Sheaf cohomology and duality - definitions and examples. Read: Hartshorne III.1-7 selected; Vakil cohomology chapters.. Make explicit examples/nonexamples for sheaf cohomology H^i; acyclic sheaf; coherent sheaf; derived functor; Serre twist; dualizing sheaf.. Work through Derived functors of global sections; Cech comparisons; coherent cohomology; projective space; Serre vanishing/duality.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Sheaf cohomology and duality - theorem and technique pass. Master/audit: Long exact cohomology sequence; cohomology of O(n) on projective space; Serre vanishing. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Intersection theory and moduli viewpoint

Topics: Intersection multiplicity; Chow groups; Chern classes; Bezout; families/moduli; Hilbert schemes/stacks orientation.

Key definitions: cycle; rational equivalence; Chow group; intersection product; Chern class; moduli functor; fine/coarse moduli.

Key results to learn:

- Bezout theorem for plane/projective intersections [ARCH]
- basic intersection pairing on smooth surfaces [ARCH/USE]
- Grothendieck-Riemann-Roch, statement [USE]

• representability caveats and fine/coarse moduli distinction [PROOF/ARCH].

Reading route: Fulton, Intersection Theory selected as reference; Vakil later chapters; Hartshorne V selected.

Warm-up/source exercise focus (secondary): Compute plane intersections with multiplicity; work Chow rings of projective space; formulate moduli functors for points/elliptic curves.

Week 11: Intersection theory and moduli viewpoint - definitions and examples. Read: Fulton, Intersection Theory selected as reference; Vakil later chapters; Hartshorne V selected.. Make explicit examples/nonexamples for cycle; rational equivalence; Chow group; intersection product; Chern class; moduli functor; fine/coarse moduli.. Work through Intersection multiplicity; Chow groups; Chern classes; Bezout; families/moduli; Hilbert schemes/stacks orientation.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Intersection theory and moduli viewpoint - theorem and technique pass. Master/audit: Bezout theorem for plane/projective intersections; basic intersection pairing on smooth surfaces; Grothendieck-Riemann-Roch, statement. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Affine and projective varieties

U1.P1. [F] For ideals I in k[x,y] such as (y-x^2), (xy), and (x^2,y^2), compute V(I), radicals, irreducible components, coordinate rings, and dimensions in elementary cases.

U1.P2. [T] Prove the affine/projective Nullstellensatz consequences needed to translate radical ideals and algebraic sets. Apply homogenization/dehomogenization to a plane curve and find points at infinity.

U1.P3. [S] Compute tangent spaces/singular loci of a node, cusp, and smooth conic from Jacobian criteria. Relate singularity to failure of local regularity.

Programme U2: Schemes, Spec and fiber products

U2.P1. [F] Draw/describe Spec Z, Spec k[x], Spec Z[i], and Spec k[x,y]/(xy), listing generic/closed points and specialization relations.

U2.P2. [T] Compute fiber products on affine schemes by tensor products in examples such as Spec Z[i] x_{Spec Z} Spec F_p. Interpret split/inert/ramified fibers.

U2.P3. [S] Glue two affine lines along G_m to construct P^1 and verify the structure sheaf on overlaps. Explain why schemes retain nilpotent/infinitesimal data lost by point sets.

Programme U3: Morphisms: proper, flat, smooth and etale

U3.P1. [F] Test finite-type morphisms in explicit affine examples for smoothness/etaleness using Jacobian/differential criteria; compute fibers and relative dimensions.

U3.P2. [T] Prove base-change stability for one property (flat, smooth, etale, proper) in a selected theorem and verify it in concrete fiber squares.

U3.P3. [C] Construct examples separating flatness, smoothness, etaleness, and properness. For each near-miss identify which algebraic/geometric condition fails.

Programme U4: Divisors, line bundles and curves

U4.P1. [F] Compute Weil/Cartier divisors and Picard groups on P^1, and divisors of rational functions on a smooth projective curve.

U4.P2. [T] Prove Riemann-Roch for P^1 directly and verify the general curve formula on several low-degree divisors where dimensions are known.

U4.P3. [S] Relate line bundles, Cartier divisors, and invertible sheaves by converting one explicit object through all three descriptions.

Programme U5: Sheaf cohomology and duality

U5.P1. [F] Compute Cech cohomology of O(n) on P^1 for several n using the standard two-affine cover. Verify H^0/H^1 dimensions.

U5.P2. [T] Derive the long exact sequence in sheaf cohomology from a short exact sequence at theorem-architecture level and use it in a divisor/line-bundle computation.

U5.P3. [R] State Serre duality precisely for a smooth projective curve and check it numerically on P^1. Explain how it generalizes Riemann-Roch.

Programme U6: Intersection theory and moduli viewpoint

U6.P1. [F] Compute intersection numbers of plane curves in simple transverse and tangential examples, including multiplicity at a point via local algebra when feasible.

U6.P2. [T] Work through Bezout's theorem for selected projective plane curves and identify hypotheses preventing components in common.

U6.P3. [R] Study one elementary moduli problem (elliptic curves, line bundles, points on P^1). Identify why automorphisms obstruct naive fine moduli and motivate stacks without developing full stack theory.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Algebraic Geometry: Varieties, Schemes and Cohomology: start from “Affine and projective varieties”, pass through “Divisors, line bundles and curves”, and finish at “Intersection theory and moduli viewpoint”. Use the named results “Hilbert Nullstellensatz” and “representability caveats and fine/coarse moduli distinction.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Essential prerequisite for arithmetic geometry, etale cohomology, Faltings, Weil/Deligne and modern moduli problems. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Hilbert Nullstellensatz; algebra-geometry correspondence for affine varieties over algebraically closed fields; Anti-equivalence between commutative rings and affine schemes; gluing schemes from affine data; universal property/existence of fiber products. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Serre finiteness; Serre duality for smooth projective varieties/curves; Bezout theorem for plane/projective intersections; basic intersection pairing on smooth surfaces; Grothendieck-Riemann-Roch, statement.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Affine and projective varieties” develops into “Intersection theory and moduli viewpoint”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 32. Arithmetic Geometry, Elliptic Curves and Modularity

Purpose: Develop elliptic curves over global/local/finite fields, heights and Mordell-Weil, Galois representations, modular curves, modularity/FLT, Faltings and Sato-Tate roadmaps.

Prerequisites: Volumes 24, 27, 30-31; Volume 26 supplies heights/Diophantine finiteness.

Primary and secondary references: Silverman, The Arithmetic of Elliptic Curves; Silverman, Advanced Topics in the Arithmetic of Elliptic Curves, selected; Hindry-Silverman, Diophantine Geometry, selected; Cornell-Silverman-Stevens (eds.), Modular Forms and Fermat's Last Theorem, selected; Milne, Elliptic Curves and Arithmetic Duality notes, selected

Quarter output: Prepare three linked dossiers: Mordell-Weil/descent on one curve, the FLT modularity chain, and a Sato-Tate/Faltings theorem dependency map.

Research bridges: Direct target volume for FLT, Faltings and Sato-Tate; feeds etale cohomology and arithmetic statistics.

Six two-week study units

Unit 1 (Weeks 1-2): Elliptic curves and isogenies

Topics: Weierstrass models; group law; discriminant/j-invariant; morphisms/isogenies; torsion.

Key definitions: elliptic curve; Weierstrass equation; discriminant; j-invariant; isogeny; dual isogeny; torsion subgroup.

Key results to learn:

- Chord-tangent group law [PROOF]
- classification by j over algebraic closure, with special cases [PROOF/ARCH]
- degree properties of isogenies [ARCH]
- dual-isogeny relation [ARCH]
- structure of E[n] in characteristic prime to n [PROOF/ARCH].

Reading route: Silverman AEC Chs. III-V selected.

Warm-up/source exercise focus (secondary): Compute group laws and torsion; construct low-degree isogenies; classify reductions of sample curves.

Week 1: Elliptic curves and isogenies - definitions and examples. Read: Silverman AEC Chs. III-V selected.. Make explicit examples/nonexamples for elliptic curve; Weierstrass equation; discriminant; j-invariant; isogeny; dual isogeny; torsion subgroup.. Work through Weierstrass models; group law; discriminant/j-invariant; morphisms/isogenies; torsion.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Elliptic curves and isogenies - theorem and technique pass. Master/audit: Chord-tangent group law; classification by j over algebraic closure, with special cases; degree properties of isogenies. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Finite/local arithmetic and reduction

Topics: Elliptic curves over finite fields and local fields; good/multiplicative/additive reduction; minimal models; formal groups.

Key definitions: good/bad reduction; minimal Weierstrass model; formal group; Tamagawa number; conductor exponent.

Key results to learn:

- Hasse bound [PROOF/ARCH]
- reduction exact sequence in good reduction [ARCH]
- Nagell-Lutz theorem over Q [PROOF/ARCH]
- Neron-Ogg-Shafarevich criterion, statement [ARCH/USE]
- Tate algorithm purpose/output [USE].

Reading route: Silverman AEC Chs. VII; Advanced selected; Washington computational examples.

Warm-up/source exercise focus (secondary): Count points over finite fields; determine good/bad primes; compute reduction types for sample curves using software and theory.

Week 3: Finite/local arithmetic and reduction - definitions and examples. Read: Silverman AEC Chs. VII; Advanced selected; Washington computational examples.. Make explicit examples/nonexamples for good/bad reduction; minimal Weierstrass model; formal group; Tamagawa number; conductor exponent.. Work through Elliptic curves over finite fields and local fields; good/multiplicative/additive reduction; minimal models; formal groups.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Finite/local arithmetic and reduction - theorem and technique pass. Master/audit: Hasse bound; reduction exact sequence in good reduction; Nagell-Lutz theorem over Q. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Heights, descent and Mordell-Weil

Topics: Naive/canonical heights; weak Mordell-Weil; descent; Selmer groups; rank.

Key definitions: canonical/Neron-Tate height; descent map; n-Selmer group; Mordell-Weil rank; Tate-Shafarevich group.

Key results to learn:

- Quadraticity/positivity of canonical height modulo torsion [PROOF/ARCH]
- weak Mordell-Weil theorem [ARCH]

• Mordell-Weil theorem [ARCH/USE]
• descent exact sequences and Selmer finiteness [ARCH/USE].

Reading route: Silverman AEC Ch. VIII-X; Hindry-Silverman selected.

Warm-up/source exercise focus (secondary): Perform 2-descent on simple curves; compute canonical heights numerically; reconstruct the Mordell-Weil proof architecture.

Week 5: Heights, descent and Mordell-Weil - definitions and examples. Read: Silverman AEC Ch. VIII-X; Hindry-Silverman selected.. Make explicit examples/nonexamples for canonical/Neron-Tate height; descent map; n-Selmer group; Mordell-Weil rank; Tate-Shafarevich group.. Work through Naive/canonical heights; weak Mordell-Weil; descent; Selmer groups; rank.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Heights, descent and Mordell-Weil - theorem and technique pass. Master/audit: Quadraticity/positivity of canonical height modulo torsion; weak Mordell-Weil theorem; Mordell-Weil theorem. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Tate modules and Galois representations

Topics: l-adic Tate module; Galois action; Frobenius traces; Weil pairing; l-adic representations.

Key definitions: Tate module T_l(E); Galois representation rho_{E,l}; Frobenius element; Weil pairing; cyclotomic character.

Key results to learn:

• Weil pairing properties [PROOF/ARCH]
• determinant of rho_{E,l} equals cyclotomic character [PROOF/ARCH]
• characteristic polynomial of Frobenius X^2-a_pX+p at good primes [PROOF]
• Serre open-image theorem for non-CM elliptic curves over Q, statement [USE]
• Faltings isogeny theorem interface [USE].

Reading route: Silverman AEC/Advanced; Serre l-adic representation expositions.

Warm-up/source exercise focus (secondary): Compute a_p for primes and compare to Frobenius; verify Weil pairing relations; map local/global Galois data.

Week 7: Tate modules and Galois representations - definitions and examples. Read: Silverman AEC/Advanced; Serre l-adic representation expositions.. Make explicit examples/nonexamples for Tate module T_l(E); Galois representation rho_{E,l}; Frobenius element; Weil pairing; cyclotomic character.. Work through l-adic Tate module; Galois action; Frobenius traces; Weil pairing; l-adic representations.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Tate modules and Galois representations - theorem and technique pass. Master/audit: Weil pairing properties; determinant of rho_{E,l} equals cyclotomic character; characteristic polynomial of Frobenius X^2-a_pX+p at good primes. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Modular curves, modularity and Fermat

Topics: Elliptic curves/modular forms; X_0(N); modular parametrization; Galois reps; level lowering; FLT.

Key definitions: modular elliptic curve; modular parametrization; modular curve X_0(N); residual representation; level lowering.

Key results to learn:

• Modularity theorem for elliptic curves over Q [USE]
• Eichler-Shimura relation, statement [USE]
• Ribet level-lowering theorem in the FLT setting [USE]
• Frey-Serre-Ribet mechanism [ARCH/USE]
• Fermat's Last Theorem deduction from modularity plus level lowering [ARCH/USE].

Reading route: Cornell-Silverman-Stevens selected chapters; Diamond-Shurman cross-reference; Wiles/Taylor-Wiles expositions.

Warm-up/source exercise focus (secondary): Build exact FLT dependency diagram; match elliptic curves to newforms in examples; explain conductor/level behavior in the Frey curve.

Week 9: Modular curves, modularity and Fermat - definitions and examples. Read: Cornell-Silverman-Stevens selected chapters; Diamond-Shurman cross-reference; Wiles/Taylor-Wiles expositions.. Make explicit examples/nonexamples for modular elliptic curve; modular parametrization; modular curve X_0(N); residual representation; level lowering.. Work through Elliptic curves/modular forms; X_0(N); modular parametrization; Galois reps; level lowering; FLT.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Modular curves, modularity and Fermat - theorem and technique pass. Master/audit: Modularity theorem for elliptic curves over Q; Eichler-Shimura relation, statement; Ribet level-lowering theorem in the FLT setting. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Faltings and Sato-Tate roadmaps

Topics: Higher-genus rational points; abelian varieties; isogeny finiteness; Sato-Tate distribution; potential automorphy.

Key definitions: abelian variety; Jacobian; polarization; Tate module; Sato-Tate group; equidistribution.

Key results to learn:

• Faltings theorem/Mordell conjecture: finite K-rational points on genus >1 curves [ARCH/USE]
• Faltings isogeny theorem/Tate conjecture for homomorphisms over number fields [USE]
• Sato-Tate theorem for non-CM elliptic curves over Q [ARCH/USE]

- CM Sato-Tate distribution [ARCH]
- potential automorphy strategy, architecture [USE].

Reading route: Faltings expositions; Hindry-Silverman; Sato-Tate surveys and modularity/automorphy references.

Warm-up/source exercise focus (secondary): Write separate 8-10 page roadmaps for Faltings and Sato-Tate; compute empirical normalized a_p histograms and compare CM/non-CM expectations.

Week 11: Faltings and Sato-Tate roadmaps - definitions and examples. Read: Faltings expositions; Hindry-Silverman; Sato-Tate surveys and modularity/automorphy references.. Make explicit examples/nonexamples for abelian variety; Jacobian; polarization; Tate module; Sato-Tate group; equidistribution.. Work through Higher-genus rational points; abelian varieties; isogeny finiteness; Sato-Tate distribution; potential automorphy.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Faltings and Sato-Tate roadmaps - theorem and technique pass. Master/audit: Faltings theorem/Mordell conjecture: finite K-rational points on genus >1 curves; Faltings isogeny theorem/Tate conjecture for homomorphisms over number fields; Sato-Tate theorem for non-CM elliptic curves over Q. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Elliptic curves and isogenies

U1.P1. [F] Put several cubic curves into Weierstrass form, compute discriminant/j-invariant, add points by chord-tangent law, and identify torsion points in small examples.

U1.P2. [T] Prove the group law is algebraic/associative at theorem-architecture level via divisor/Picard interpretation rather than a brute-force coordinate proof.

U1.P3. [S] Compute degrees/kernels of multiplication-by-n and one explicit isogeny. Verify dual-isogeny relations in a small example where formulas are available.

Programme U2: Finite/local arithmetic and reduction

U2.P1. [F] Count E(F_p) for several small elliptic curves and verify Hasse's bound. Compute Frobenius trace a_p and local Euler factors.

U2.P2. [T] Apply good/multiplicative/additive reduction criteria at several primes for a curve over Q; compute minimal-model data where manageable.

U2.P3. [S] For a local field Q_p and good reduction, describe the exact reduction sequence E_1(Q_p)->E_0(Q_p)->E_tilde(F_p) and use it to analyze local points in a simple case.

Programme U3: Heights, descent and Mordell-Weil

U3.P1. [F] Compute naive/canonical heights numerically for multiples of a rational point on a simple curve and observe quadratic growth hhat(nP)=n^2 hhat(P).

U3.P2. [T] Work through a 2-descent on a curve with accessible rational 2-torsion or use a standard worked example to bound E(Q)/2E(Q) and the rank.

U3.P3. [R] Reconstruct Mordell-Weil theorem architecture: weak Mordell-Weil + height descent + Northcott. Identify precisely where each earlier number-theory result is used.

Programme U4: Tate modules and Galois representations

U4.P1. [F] Compute E[l] over an algebraic closure for l not equal to p and describe the Galois action matrix for simple reductions/examples conceptually.

U4.P2. [T] Define the Tate module T_l(E) and prove its rank-2 Z_l structure in characteristic 0/good settings. Relate determinant of Frobenius to p and trace to a_p.

U4.P3. [S] State the l-adic representation rho_{E,l}:G_Q->GL_2(Z_l), identify ramification properties, and connect its characteristic polynomial at good primes to the Hasse-Weil L-function.

Programme U5: Modular curves, modularity and Fermat

U5.P1. [F] Study X_0(N) moduli interpretation for small N: identify points as elliptic curves plus cyclic subgroup and understand cusps at a basic level.

U5.P2. [T] Translate a normalized newform of weight 2 into an isogeny class of elliptic curves over Q at modularity-theorem statement level; compare Fourier coefficients a_p with point counts.

U5.P3. [R] Build the FLT proof map: hypothetical Fermat solution -> Frey curve -> semistability -> Ribet level lowering/nonmodularity -> modularity theorem -> contradiction. State exactly what each theorem contributes.

Programme U6: Faltings and Sato-Tate roadmaps

U6.P1. [R] State Faltings' theorem (Mordell conjecture) and the isogeny theorem in precise standard formulations. Build a dependency map through abelian varieties, heights, moduli, and finiteness without attempting the full proof.

U6.P2. [F] For an elliptic curve without CM, compute normalized Frobenius traces a_p/(2sqrt p) for many small good primes by code and compare with the Sato-Tate density; for a CM example compare the different behavior.

U6.P3. [R] State a modern Sato-Tate theorem for elliptic curves over Q in a precise non-CM formulation and map the automorphy/potential automorphy plus equidistribution inputs.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Arithmetic Geometry, Elliptic Curves and Modularity: start from "Elliptic curves and isogenies", pass through "Tate modules and Galois representations", and finish at "Faltings and Sato-Tate roadmaps". Use the named results "Chord-tangent group law" and "potential automorphy strategy, architecture." with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct target volume for FLT, Faltings and Sato-Tate; feeds etale cohomology and arithmetic statistics. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Chord-tangent group law; classification by j over algebraic closure, with special cases; structure of E[n] in characteristic prime to n; Hasse bound; Nagell-Lutz theorem over Q. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Faltings theorem/Mordell conjecture: finite K-rational points on genus >1 curves; Faltings isogeny theorem/Tate conjecture for homomorphisms over number fields; Sato-Tate theorem for non-CM elliptic curves over Q; CM Sato-Tate distribution; potential automorphy strategy, architecture.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how "Elliptic curves and isogenies" develops into "Faltings and Sato-Tate roadmaps", naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 33. Etale Cohomology, Weil Conjectures and Modern Finite-Field Arithmetic

Purpose: Learn etale/l-adic cohomology and Frobenius methods far enough to understand the cohomological formulation of zeta functions, Deligne's Weil theorems and modern exponential-sum applications.

Prerequisites: Volumes 4, 27, 31-32; category/derived-functor fluency is essential.

Primary and secondary references: Milne, Etale Cohomology; Milne, Lectures on Etale Cohomology; SGA 4, SGA 4 1/2 and SGA 5, selected reference passages; Deligne, La conjecture de Weil I and II; Katz, Sommes exponentielles and monodromy-related expositions, selected

Quarter output: Write a 30-page “finite fields to Deligne” roadmap, beginning with a concrete exponential sum and ending with its sheaf/cohomology/Frobenius interpretation, plus a theorem map of all Weil conjectures.

Research bridges: Direct target for Deligne/Bombieri and a key bridge between finite-field arithmetic, automorphic forms and modern arithmetic geometry.

Six two-week study units

Unit 1 (Weeks 1-2): Etale morphisms and sites

Topics: Etale maps; Grothendieck topologies; etale site; sheaves; fundamental group; finite etale covers.

Key definitions: etale morphism; site; covering family; etale sheaf; geometric point; etale fundamental group.

Key results to learn:

- Jacobian/infinitesimal criterion for etaleness in standard settings [ARCH]
- equivalence between finite etale covers and finite continuous pi_1-sets [ARCH/USE]
- basic exact sequence of etale fundamental groups [ARCH/USE].

Reading route: Milne Lectures early chapters; SGA 1 selected for pi_1.

Warm-up/source exercise focus (secondary): Classify finite etale covers of basic curves; compute etale pi_1 in elementary cases; compare classical/etale topology.

Week 1: Etale morphisms and sites - definitions and examples. Read: Milne Lectures early chapters; SGA 1 selected for pi_1.. Make explicit examples/nonexamples for etale morphism; site; covering family; etale sheaf; geometric point; etale fundamental group.. Work through Etale maps; Grothendieck topologies; etale site; sheaves; fundamental group; finite etale covers.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Etale morphisms and sites - theorem and technique pass. Master/audit: Jacobian/infinitesimal criterion for etaleness in standard settings; equivalence between finite etale covers and finite continuous pi_1-sets; basic exact sequence of etale fundamental groups [ARCH/USE].. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Etale and l-adic cohomology

Topics: Derived etale cohomology; torsion sheaves; inverse systems; l-adic sheaves; comparison over C.

Key definitions: etale cohomology H^i_et; constructible sheaf; l-adic sheaf; Tate twist; geometric cohomology.

Key results to learn:

- Finite-dimensionality/finiteness theorems for constructible torsion/l-adic cohomology in standard finite-type settings [USE]
- comparison theorem with singular cohomology over C [ARCH/USE]
- Kummer sequence and cohomological interpretations of Picard/Brauer groups [ARCH].

Reading route: Milne Etale Cohomology core chapters.

Warm-up/source exercise focus (secondary): Compute H^1_et via Kummer in simple cases; compare cohomology of P1; track Tate twists and Galois actions.

Week 3: Etale and l-adic cohomology - definitions and examples. Read: Milne Etale Cohomology core chapters.. Make explicit examples/nonexamples for etale cohomology H^i_et; constructible sheaf; l-adic sheaf; Tate twist; geometric cohomology.. Work through Derived etale cohomology; torsion sheaves; inverse systems; l-adic sheaves; comparison over C.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Etale and l-adic cohomology - theorem and technique pass. Master/audit: Finite-dimensionality/finiteness theorems for constructible torsion/l-adic cohomology in standard finite-type settings; comparison theorem with singular cohomology over C; Kummer sequence and cohomological interpretations of Picard/Brauer groups [ARCH].. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Six functors, base change and duality

Topics: Proper/compact-support cohomology; f*, f_*, f_!, f^!; base change; Poincare/Verdier duality.

Key definitions: compact-support cohomology; proper base change; smooth base change; dualizing complex; Poincare duality.

Key results to learn:

- Proper base change theorem [ARCH/USE]
- smooth base change theorem [USE]

- Poincare duality for smooth varieties [ARCH/USE]
- Kunneth formula in etale cohomology [ARCH/USE].

Reading route: Milne; SGA 4/5 reference sections.

Warm-up/source exercise focus (secondary): Apply base change to family examples; compute compact-support cohomology of A1/G_m; check duality dimensions.

Week 5: Six functors, base change and duality - definitions and examples. Read: Milne; SGA 4/5 reference sections.. Make explicit examples/nonexamples for compact-support cohomology; proper base change; smooth base change; dualizing complex; Poincare duality.. Work through Proper/compact-support cohomology; f*, f_*, f_!, f^!; base change; Poincare/Verdier duality.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Six functors, base change and duality - theorem and technique pass. Master/audit: Proper base change theorem; smooth base change theorem; Poincare duality for smooth varieties. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Frobenius and trace formula

Topics: Geometric/arithmetic Frobenius; l-adic action; fixed points; Lefschetz trace; zeta functions.

Key definitions: geometric Frobenius; arithmetic Frobenius; characteristic polynomial; local factor; trace of Frobenius.

Key results to learn:

- Grothendieck-Lefschetz trace formula over finite fields [ARCH/USE]
- cohomological expression Z(X,t)=product det(1-tF|H^i_c)^{(-1)^{i+1}} [PROOF/ARCH]
- rationality of zeta from finite-dimensional cohomology [ARCH].

Reading route: Milne trace-formula chapters; SGA 5 selected.

Warm-up/source exercise focus (secondary): Compute zeta of affine/projective spaces cohomologically; derive point-count sequences from Frobenius eigenvalues.

Week 7: Frobenius and trace formula - definitions and examples. Read: Milne trace-formula chapters; SGA 5 selected.. Make explicit examples/nonexamples for geometric Frobenius; arithmetic Frobenius; characteristic polynomial; local factor; trace of Frobenius.. Work through Geometric/arithmetic Frobenius; l-adic action; fixed points; Lefschetz trace; zeta functions.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Frobenius and trace formula - theorem and technique pass. Master/audit: Grothendieck-Lefschetz trace formula over finite fields; cohomological expression Z(X,t)=product det(1-tF|H^i_c)^{(-1)^{i+1}}; rationality of zeta from finite-dimensional cohomology [ARCH].. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Weil conjectures and Deligne

Topics: Rationality; functional equation; Betti-number interpretation; RH; weights; purity/mixedness.

Key definitions: weight; pure/mixed l-adic sheaf; Weil number; eigenvalue weight; semisimplicity context.

Key results to learn:

- Four Weil conjectures, precise formulations and logical separation of their components [ARCH]
- Grothendieck cohomological proof of rationality framework [ARCH/USE]
- Deligne Weil I theorem proving RH for smooth projective varieties [ARCH/USE]
- Deligne Weil II theory of weights, main purity/mixedness statements [USE].

Reading route: Deligne Weil I/II with Milne/Katz expositions.

Warm-up/source exercise focus (secondary): State each conjecture and identify its cohomological mechanism; compute expected weights/eigenvalue sizes in examples; prepare a 60-minute Deligne roadmap.

Week 9: Weil conjectures and Deligne - definitions and examples. Read: Deligne Weil I/II with Milne/Katz expositions.. Make explicit examples/nonexamples for weight; pure/mixed l-adic sheaf; Weil number; eigenvalue weight; semisimplicity context.. Work through Rationality; functional equation; Betti-number interpretation; RH; weights; purity/mixedness.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Weil conjectures and Deligne - theorem and technique pass. Master/audit: Four Weil conjectures, precise formulations; Grothendieck cohomological proof of rationality framework; Deligne Weil I theorem proving RH for smooth projective varieties. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Bombieri, Katz and exponential sums

Topics: Sheaf-function dictionary; trace functions; monodromy; equidistribution; square-root cancellation.

Key definitions: trace function; lisse sheaf; conductor; geometric monodromy group; equidistribution of Frobenius.

Key results to learn:

- Deligne square-root bounds for suitable trace/exponential sums [USE]
- Stepanov-Bombieri auxiliary-function proof of the Riemann-hypothesis/Hasse-Weil bound for curves; separately, Deligne-Katz sheaf-theoretic square-root bounds for exponential sums [ARCH/USE]
- Katz monodromy-to-equidistribution paradigm and representative theorem [USE]
- Chebotarev over finite fields/function fields in sheaf language [USE].

Reading route: Katz selected monographs/papers; Deligne; Bombieri expositions.

Warm-up/source exercise focus (secondary): Translate classical character sums into sheaf trace functions at a conceptual level; compare elementary/Weil/Deligne bounds; create a monodromy checklist.

Week 11: Bombieri, Katz and exponential sums - definitions and examples. Read: Katz selected monographs/papers; Deligne; Bombieri expositions.. Make explicit examples/nonexamples for trace function; lisse sheaf; conductor; geometric monodromy group; equidistribution of Frobenius.. Work through Sheaf-function dictionary; trace functions; monodromy; equidistribution; square-root cancellation.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Bombieri, Katz and exponential sums - theorem and technique pass. Master/audit: Deligne square-root bounds for suitable trace/exponential sums; Stepanov-Bombieri auxiliary-function proof of the Riemann-hypothesis/Hasse-Weil bound for curves; separately, Deligne-Katz sheaf-theoretic square-root bounds for exponential sums; Katz monodromy-to-equidistribution paradigm and representative theorem. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Etale morphisms and sites

U1.P1. [F] Verify examples of etale morphisms: Spec k[t,t^{-1}]->Spec k[s,s^{-1}] with s=t^n when char(k) does not divide n, and identify failure in bad characteristic.

U1.P2. [T] Compare Zariski and etale covers on a scheme and construct the etale site of a very simple scheme at the level of objects/covers. Explain why ordinary topology is too coarse for finite-cover/Galois information.

U1.P3. [S] Compute the etale fundamental group of Spec F_q and explain its relation to profinite completion of Z generated by Frobenius.

Programme U2: Etale and l-adic cohomology

U2.P1. [F] Compute H^i_et(Spec F_q, Z/lZ) in low degrees using continuous Galois cohomology in a simple coefficient module.

U2.P2. [T] For P^1 over an algebraically closed field, state and verify the expected l-adic cohomology groups and Tate twist in degree 2.

U2.P3. [S] Compare singular cohomology of a complex variety with l-adic etale cohomology via the comparison theorem in an example. Identify which formal properties are being transported.

Programme U3: Six functors, base change and duality

U3.P1. [T] Write the proper base-change theorem and smooth base-change theorem in precise standard forms and test them on a family with easily understood fibers.

U3.P2. [F] Track f^*, f_*, f_!, f^!, tensor, and RHom on a simple finite/proper morphism, stating which adjunctions are visible.

U3.P3. [R] Build a six-functor dependency diagram sufficient for the Grothendieck-Lefschetz trace formula; do not develop the entire derived-category machinery, but identify where each operation enters.

Programme U4: Frobenius and trace formula

U4.P1. [F] For P^1/F_q and an elliptic curve/F_q, express #X(F_{q^n}) in terms of Frobenius eigenvalues and recover the zeta function.

U4.P2. [T] Derive the Grothendieck-Lefschetz trace formula in these examples from known cohomology groups, checking signs and Tate twists.

U4.P3. [S] Show how rationality and functional-equation structure of zeta follows formally from finite-dimensional Frobenius actions plus duality, leaving the Riemann-hypothesis bound as the deep input.

Programme U5: Weil conjectures and Deligne

U5.P1. [F] State all Weil conjectures for a smooth projective variety over F_q and verify them for P^n and curves using known formulas.

U5.P2. [R] Reconstruct Deligne's proof architecture for the Riemann hypothesis over finite fields: weights, Lefschetz pencils/monodromy or the chosen exposition, tensor-power trick, purity conclusion. Mark black-box inputs.

U5.P3. [S] Translate the weight bound |alpha|=q^{i/2} into a square-root cancellation estimate for a point count. Work one explicit error term.

Programme U6: Bombieri, Katz and exponential sums

U6.P1. [T] Use an l-adic sheaf/trace-function viewpoint to reinterpret a classical character or Kloosterman sum. Identify conductor/ramification qualitatively in a simple example.

U6.P2. [R] State a precise Deligne/Katz square-root bound for a selected trace-function or character-sum class, including geometric irreducibility/conductor hypotheses, and derive the numerical O(q^{1/2})-type estimate. Do not call this the Stepanov-Bombieri proof of the curve RH; that is a different method.

U6.P3. [R] Compare two routes side by side: (A) the Stepanov-Bombieri auxiliary-function proof of the Hasse-Weil/RH bound for curves; (B) a Deligne-Katz sheaf/trace-function square-root estimate. State the object, hypotheses, mechanism and scope of each, and explain why route (A) is not a proof of the higher-dimensional Weil conjectures.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Etale Cohomology, Weil Conjectures and Modern Finite-Field Arithmetic: start from “Etale morphisms and sites”, pass through “Frobenius and trace formula”, and finish at “Bombieri, Katz and exponential sums”. Use the named results “Jacobian/infinitesimal criterion for etaleness in standard settings” and “Chebotarev over finite fields/function fields in sheaf language.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct target for Deligne/Bombieri and a key bridge between finite-field arithmetic, automorphic forms and modern arithmetic geometry. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: cohomological expression Z(X,t)=product det(1-tF|H^i_c)^{(-1)^{i+1}}; Four Weil conjectures, precise formulations. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: Deligne Weil II theory of weights, main purity/mixedness statements; Deligne square-root bounds for suitable trace/exponential sums; Stepanov-Bombieri auxiliary-function proof of the Riemann-hypothesis/Hasse-Weil bound for curves; separately, Deligne-Katz sheaf-theoretic square-root bounds for exponential sums; Katz monodromy-to-equidistribution paradigm and representative theorem; Chebotarev over finite fields/function fields in sheaf language.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Etale morphisms and sites” develops into “Bombieri, Katz and exponential sums”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 34. Fractal Geometry, Entropy and Self-Similar Measures

Purpose: Build geometric measure theory and entropy methods through Hochman inverse entropy, Shmerkin inverse Lq theory and modern Bernoulli-convolution arithmetic including Varju.

Prerequisites: Volumes 6, 9, 14, 25 and 28; algebraic-number background Volume 24 is useful for Bernoulli convolutions.

Primary and secondary references: Falconer, Fractal Geometry; Mattila, Geometry of Sets and Measures in Euclidean Spaces; Hochman, On self-similar sets with overlaps and inverse theorems for entropy, plus related papers; Shmerkin, inverse theorems for Lq norms/dimensions and self-similar measures, selected papers; Peres-Solomyak surveys on Bernoulli convolutions; Varju and Breuillard-Varju papers on Bernoulli convolutions/dimension/absolute continuity

Quarter output: Produce a 30-page research-entry guide to Bernoulli convolutions with four technical appendices: entropy at scale, Lq dimension, algebraic separation and Fourier decay.

Research bridges: Direct target for Hochman inverse entropy, Shmerkin inverse Lq and Varju Bernoulli-convolution work; also supports Falconer/Furstenberg questions.

Six two-week study units

Unit 1 (Weeks 1-2): Hausdorff dimension and Frostman theory

Topics: Hausdorff/packing/box dimensions; measures; energy; Frostman measures; mass distribution.

Key definitions: Hausdorff measure/dimension; upper/lower box dimension; Frostman exponent; s-energy.

Key results to learn:

- Frostman lemma [PROOF/ARCH]
- mass distribution principle [PROOF]
- energy criterion for lower dimension [PROOF/ARCH]
- basic dimension inequalities for products/images [ARCH].

Reading route: Falconer early chapters; Mattila Chs. 1-8 selected.

Warm-up/source exercise focus (secondary): Compute dimensions of Cantor/self-similar sets; construct Frostman measures; calculate energies in model cases.

Week 1: Hausdorff dimension and Frostman theory - definitions and examples. Read: Falconer early chapters; Mattila Chs. 1-8 selected.. Make explicit examples/nonexamples for Hausdorff measure/dimension; upper/lower box dimension; Frostman exponent; s-energy.. Work through Hausdorff/packing/box dimensions; measures; energy; Frostman measures; mass distribution.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Hausdorff dimension and Frostman theory - theorem and technique pass. Master/audit: Frostman lemma; mass distribution principle; energy criterion for lower dimension. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Projections, Fourier decay and distance methods

Topics: Orthogonal projections; Fourier transform of measures; energy integrals; Marstrand; Mattila integral.

Key definitions: Fourier dimension; projection measure; radial projection; distance set; Mattila integral.

Key results to learn:

- Marstrand projection theorem [ARCH/USE]
- energy/Fourier identity for Riesz energies [ARCH]
- Mattila integral criterion for distance sets [ARCH/USE]
- Kaufman-type exceptional set estimates, representative [USE].

Reading route: Mattila projection/Fourier chapters; Falconer selected.

Warm-up/source exercise focus (secondary): Use Fourier energy to prove a projection statement in simplified form; compute projections of product Cantor measures.

Week 3: Projections, Fourier decay and distance methods - definitions and examples. Read: Mattila projection/Fourier chapters; Falconer selected.. Make explicit examples/nonexamples for Fourier dimension; projection measure; radial projection; distance set; Mattila integral.. Work through Orthogonal projections; Fourier transform of measures; energy integrals; Marstrand; Mattila integral.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Projections, Fourier decay and distance methods - theorem and technique pass. Master/audit: Marstrand projection theorem; energy/Fourier identity for Riesz energies; Mattila integral criterion for distance sets. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Self-similar measures, overlaps and Bernoulli convolutions

Topics: IFS; open set/strong separation; exact overlaps; dimension formulas; Bernoulli convolutions.

Key definitions: self-similar set/measure; similarity dimension; exact overlap; exponential separation; Bernoulli convolution mu_lambda.

Key results to learn:

- Hutchinson fixed-point theorem [PROOF/ARCH]
- dimension formula under OSC/SSC [PROOF/ARCH]

- Erdos singularity for reciprocal Pisot parameters, statement [ARCH/USE]
- Garsia absolute-continuity theorem for reciprocals of Garsia numbers, with the algebraic-integer norm/conjugate hypotheses stated exactly; keep this distinct from Erdos singularity for reciprocal Pisot parameters [ARCH/USE]
- Solomyak a.e. absolute continuity for lambda in (1/2,1), standard formulation [USE].

Reading route: Falconer self-similar chapters; Peres-Solomyak surveys; original landmark papers as guided reading.

Warm-up/source exercise focus (secondary): Compute exact-overlap examples; approximate Bernoulli densities numerically; compare Fourier transforms for Pisot/non-Pisot parameters.

Week 5: Self-similar measures, overlaps and Bernoulli convolutions - definitions and examples. Read: Falconer self-similar chapters; Peres-Solomyak surveys; original landmark papers as guided reading.. Make explicit examples/nonexamples for self-similar set/measure; similarity dimension; exact overlap; exponential separation; Bernoulli convolution mu_lambda.. Work through IFS; open set/strong separation; exact overlaps; dimension formulas; Bernoulli convolutions.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Self-similar measures, overlaps and Bernoulli convolutions - theorem and technique pass. Master/audit: Hutchinson fixed-point theorem; dimension formula under OSC/SSC; Erdos singularity for reciprocal Pisot parameters, statement. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Entropy at scale and Hochman inverse theory

Topics: Shannon entropy at dyadic/discrete scales; convolution; local entropy averages; inverse theorem; dimension of convolutions/self-similar measures.

Key definitions: entropy H; normalized entropy at scale; entropy dimension; local entropy average; superexponential concentration.

Key results to learn:

- Entropy increases under convolution unless structural concentration occurs, Hochman inverse theorem in a selected precise formulation [USE]
- local entropy averages principle [ARCH/USE]
- Hochman dimension theorem for self-similar measures on R absent superexponential concentration [ARCH/USE]
- algebraic-parameter no-exact-overlap consequence [USE].

Reading route: Hochman Annals paper and expository notes; Hochman-Shmerkin background as needed.

Warm-up/source exercise focus (secondary): Compute discrete entropies of approximations/convolutions; state inverse theorem with alternatives; map it to self-similar dimension.

Week 7: Entropy at scale and Hochman inverse theory - definitions and examples. Read: Hochman Annals paper and expository notes; Hochman-Shmerkin background as needed.. Make explicit examples/nonexamples for entropy H; normalized entropy at scale; entropy dimension; local entropy average; superexponential concentration.. Work through Shannon entropy at dyadic/discrete scales; convolution; local entropy averages; inverse theorem; dimension of convolutions/self-similar measures.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Entropy at scale and Hochman inverse theory - theorem and technique pass. Master/audit: Entropy increases under convolution unless structural concentration occurs, Hochman inverse theorem in a selected precise formulation; local entropy averages principle; Hochman dimension theorem for self-similar measures on R absent superexponential concentration. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Lq dimensions and Shmerkin inverse theory

Topics: Dyadic Lq norms; Renyi entropy; Lq dimension; convolution smoothing; inverse theorem for Lq norm.

Key definitions: Lq dimension; Renyi entropy; flattening; porous set/measure; multifractal spectrum basics.

Key results to learn:

- Submultiplicativity/basic properties of Lq dimensions [PROOF/ARCH]
- Shmerkin inverse theorem for Lq norms of convolutions in a selected precise form [USE]
- consequences for Lq dimension of self-similar measures [USE]
- representative Furstenberg-intersection/dimension consequence [USE].

Reading route: Shmerkin papers and surveys; Falconer/Mattila supporting measure theory.

Warm-up/source exercise focus (secondary): Compute Lq dimensions for Bernoulli/self-similar measures under separation; compare entropy and Lq inverse alternatives.

Week 9: Lq dimensions and Shmerkin inverse theory - definitions and examples. Read: Shmerkin papers and surveys; Falconer/Mattila supporting measure theory.. Make explicit examples/nonexamples for Lq dimension; Renyi entropy; flattening; porous set/measure; multifractal spectrum basics.. Work through Dyadic Lq norms; Renyi entropy; Lq dimension; convolution smoothing; inverse theorem for Lq norm.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Lq dimensions and Shmerkin inverse theory - theorem and technique pass. Master/audit: Submultiplicativity/basic properties of Lq dimensions; Shmerkin inverse theorem for Lq norms of convolutions in a selected precise form; consequences for Lq dimension of self-similar measures. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Varju, algebraic parameters and modern Bernoulli convolutions

Topics: Mahler measure; algebraic approximation; quantitative separation; dimension close to one; absolute continuity criteria.

Key definitions: Mahler measure; algebraic parameter; entropy rate; Fourier decay; exceptional parameter set.

Key results to learn:

- Representative Varju theorem(s) giving dimension/absolute-continuity information for Bernoulli convolutions at algebraic parameters under quantitative assumptions [USE]
- Breuillard-Varju entropy/dimension estimates [USE]
- Shmerkin/Hochman modern zero-dimensional exceptional-set/absolute-continuity results in precise selected formulations [USE].

Reading route: Varju and Breuillard-Varju original papers; Hochman/Shmerkin surveys; algebraic-number cross-reference Volumes 24/26.

Warm-up/source exercise focus (secondary): Track dependence on degree/Mahler measure in one theorem; compare algebraic and transversality mechanisms; prepare an annotated chronology Erdos->Garsia->Solomyak->Hochman->Shmerkin->Varju.

Week 11: Varju, algebraic parameters and modern Bernoulli convolutions - definitions and examples. Read: Varju and Breuillard-Varju original papers; Hochman/Shmerkin surveys; algebraic-number cross-reference Volumes 24/26.. Make explicit examples/nonexamples for Mahler measure; algebraic parameter; entropy rate; Fourier decay; exceptional parameter set.. Work through Mahler measure; algebraic approximation; quantitative separation; dimension close to one; absolute continuity criteria.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Varju, algebraic parameters and modern Bernoulli convolutions - theorem and technique pass. Master/audit: Representative Varju theorem(s) giving dimension/absolute-continuity information for Bernoulli convolutions at algebraic parameters under quantitative assumptions; Breuillard-Varju entropy/dimension estimates; Shmerkin/Hochman modern zero-dimensional exceptional-set/absolute-continuity results in precise selected formulations [USE].. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Hausdorff dimension and Frostman theory

U1.P1. [F] Compute Hausdorff dimension of the middle-third Cantor set, product Cantor sets, and a self-similar set satisfying the open set condition using coverings and mass distribution.

U1.P2. [T] Prove Frostman's lemma in one direction and use a Frostman measure to lower-bound dimension of an explicit fractal.

U1.P3. [S] Convert between energy integrals I_s(mu) and dimension lower bounds in a standard setting. Compute/estimate energy for a simple Cantor measure.

Programme U2: Projections, Fourier decay and distance methods

U2.P1. [F] Compute projections of product/Cantor sets in special directions and identify exceptional directions caused by arithmetic structure.

U2.P2. [T] Prove the Mattila projection theorem/Marstrand theorem at proof-architecture level via energies/Fourier transform in the formulation used by your text.

U2.P3. [S] Derive the Mattila integral criterion for the Falconer distance problem conceptually and test the scaling threshold on a Frostman measure.

Programme U3: Self-similar measures, overlaps and Bernoulli convolutions

U3.P1. [F] For Bernoulli convolution mu_lambda, derive the self-similarity equation and Fourier product formula. Work explicitly with lambda=1/2 and one reciprocal Pisot parameter.

U3.P2. [X] Compute finite Bernoulli-convolution approximations for lambda=0.45,0.5,0.6,1/phi and inspect overlaps/Fourier coefficients. Record phenomena without inferring unproved regularity.

U3.P3. [S] Translate an exact/near overlap into a small value of a polynomial with coefficients in {-1,0,1}. Explain why algebraic properties of lambda become relevant.

Programme U4: Entropy at scale and Hochman inverse theory

U4.P1. [F] Compute dyadic entropy H_n(mu) for a Dirac mass, uniform interval measure, finite uniform sets, and finite Cantor approximations. Normalize by n and compare with dimension.

U4.P2. [T] Prove elementary entropy inequalities under convolution/conditioning and a local-entropy-averages identity in a finite dyadic model.

U4.P3. [R] State Hochman's inverse theorem for entropy growth in one precise finite-scale formulation. Build a toy example for each structural alternative, then trace how the theorem yields the dimension formula for self-similar measures absent superexponential concentration.

Programme U5: Lq dimensions and Shmerkin inverse theory

U5.P1. [F] Compute dyadic L^q sums sum_I mu(I)^q and L^q dimensions for uniform/self-similar measures under separation. Compare q=2 with additive energy intuition.

U5.P2. [T] Prove basic convolution inequalities for discrete L^q norms and construct examples where convolution flattens strongly versus barely changes the norm.

U5.P3. [R] State Shmerkin's inverse theorem for L^q norms of convolutions in the exact formulation chosen from the reference. For a toy multiscale measure, identify the porous/concentrated alternatives and explain the implication for L^q dimension growth.

Programme U6: Varju, algebraic parameters and modern Bernoulli convolutions

U6.P1. [F] For algebraic lambda, compute minimal polynomial, degree, and Mahler measure in several examples; relate polynomial evaluations at lambda to root-separation bounds.

U6.P2. [R] Select one Varju or Breuillard-Varju theorem on Bernoulli convolutions and write its hypotheses exactly, including algebraicity/Mahler-measure or approximation conditions. Trace how the arithmetic estimate feeds entropy/dimension growth.

U6.P3. [S] Prepare a chronological theorem map Erdos -> Garsia -> Solomyak -> Hochman -> Shmerkin -> Breuillard-Varju -> Varju. For each stage give one exact statement and conclusion type (singularity/Fourier decay, absolute continuity, dimension, or Lq regularity). Explicitly separate reciprocal Pisot parameters (Erdos singularity) from reciprocal Garsia-number parameters (Garsia absolute continuity), and distinguish dimension one from absolute continuity.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Fractal Geometry, Entropy and Self-Similar Measures: start from “Hausdorff dimension and Frostman theory”, pass through “Entropy at scale and Hochman inverse theory”, and finish at “Varju, algebraic parameters and modern Bernoulli convolutions”. Use the named results “Frostman lemma” and “Shmerkin/Hochman modern zero-dimensional exceptional-set/absolute-continuity results in precise selected formulations.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct target for Hochman inverse entropy, Shmerkin inverse Lq and Varju Bernoulli-convolution work; also supports Falconer/Furstenberg questions. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Frostman lemma; mass distribution principle; energy criterion for lower dimension; Hutchinson fixed-point theorem; dimension formula under OSC/SSC. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: consequences for Lq dimension of self-similar measures; representative Furstenberg-intersection/dimension consequence; Representative Varju theorem(s) giving dimension/absolute-continuity information for Bernoulli convolutions at algebraic parameters under quantitative assumptions; Breuillard-Varju entropy/dimension estimates; Shmerkin/Hochman modern zero-dimensional exceptional-set/absolute-continuity results in precise selected formulations.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Hausdorff dimension and Frostman theory” develops into “Varju, algebraic parameters and modern Bernoulli convolutions”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 35. Modern Harmonic and Geometric Analysis

Purpose: Unify time-frequency analysis, restriction, Kakeya/incidence geometry, Falconer/Furstenberg problems, polynomial partitioning and decoupling through current research-level theorem statements.

Prerequisites: Volumes 9, 23, 28 and 34; Volumes 6-7 are assumed.

Primary and secondary references: Stein, Harmonic Analysis, selected advanced chapters; Muscalu-Schlag, Classical and Multilinear Harmonic Analysis, selected time-frequency material; Demeter, Fourier Restriction, Decoupling, and Applications; Tao, restriction/Kakeya lecture notes and surveys; Guth, polynomial partitioning papers/notes; Mattila, Fourier Analysis and Hausdorff Dimension, selected; Carleson-Hunt expositions and landmark papers

Quarter output: Write a 35-page “five modern proof machines” guide comparing time-frequency trees, restriction/wave packets, polynomial partitioning, incidence/sum-product and decoupling, with entry maps to Carleson, Kakeya, Falconer, Furstenberg and VMVT.

Research bridges: Direct target for Carleson, Bourgain decoupling, Falconer/Furstenberg/Kakeya and Hong Wang-related geometric analysis.

Six two-week study units

Unit 1 (Weeks 1-2): Time-frequency analysis and Carleson

Topics: Maximal partial Fourier sums; wave packets; tiles; trees; size/density; time-frequency decomposition.

Key definitions: Carleson operator; tile; tree; size; density; wave packet; exceptional set.

Key results to learn:

- Carleson theorem on a.e. convergence of Fourier series for L2 [ARCH/USE]
- Hunt extension to Lp, p>1 [USE]
- weak/strong Carleson operator estimates in standard formulations [USE]
- tree lemma/size-density architecture [ARCH/USE].

Reading route: Selected Carleson theorem lecture notes; Muscalu-Schlag time-frequency chapters; original Carleson/Lacey-Thiele guided reading.

Warm-up/source exercise focus (secondary): Work discrete tile models; prove toy tree estimates; create an end-to-end Carleson proof dependency map.

Week 1: Time-frequency analysis and Carleson - definitions and examples. Read: Selected Carleson theorem lecture notes; Muscalu-Schlag time-frequency chapters; original Carleson/Lacey-Thiele guided reading.. Make explicit examples/nonexamples for Carleson operator; tile; tree; size; density; wave packet; exceptional set.. Work through Maximal partial Fourier sums; wave packets; tiles; trees; size/density; time-frequency decomposition.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Time-frequency analysis and Carleson - theorem and technique pass. Master/audit: Carleson theorem on a.e. convergence of Fourier series for L2; Hunt extension to Lp, p>1; weak/strong Carleson operator estimates in standard formulations. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): Restriction theory

Topics: Fourier extension/restriction; Knapp examples; curvature; bilinear/multilinear restriction.

Key definitions: extension operator; restriction estimate; Knapp example; transversality; wave packet.

Key results to learn:

- Stein-Tomas restriction theorem [PROOF/ARCH]
- Knapp necessary conditions [PROOF]
- bilinear restriction gains under transversality [ARCH/USE]
- Bennett-Carbery-Tao multilinear restriction theorem, statement [USE].

Reading route: Stein; Tao restriction notes; Bennett-Carbery-Tao paper.

Warm-up/source exercise focus (secondary): Derive Stein-Tomas from TT*; construct Knapp examples; prove a toy bilinear transverse estimate.

Week 3: Restriction theory - definitions and examples. Read: Stein; Tao restriction notes; Bennett-Carbery-Tao paper.. Make explicit examples/nonexamples for extension operator; restriction estimate; Knapp example; transversality; wave packet.. Work through Fourier extension/restriction; Knapp examples; curvature; bilinear/multilinear restriction.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: Restriction theory - theorem and technique pass. Master/audit: Stein-Tomas restriction theorem; Knapp necessary conditions; bilinear restriction gains under transversality. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Kakeya, incidence geometry and polynomial method

Topics: Kakeya sets/maximal functions; incidence theorems; polynomial partitioning; broad/narrow analysis.

Key definitions: Kakeya set; Kakeya maximal function; incidence; polynomial partition; broad norm.

Key results to learn:

- Besicovitch existence of measure-zero Kakeya sets [PROOF/ARCH]
- Wolff lower bounds/inequalities, selected [USE]
- Szemeredi-Trotter theorem [PROOF/ARCH]

- Guth-Katz distinct-distances theorem, statement [USE]
- polynomial partitioning theorem [ARCH/USE]; Wang-Zahl 2025 R^3 Kakeya theorem as the current endpoint of the three-dimensional branch [USE].

Reading route: Tao Kakeya notes; Guth papers/book notes; incidence geometry surveys.

Warm-up/source exercise focus (secondary): Construct Kakeya examples; prove Szemeredi-Trotter via crossing-number or polynomial ideas at chosen depth; perform polynomial partition count in a toy setup.

Week 5: Kakeya, incidence geometry and polynomial method - definitions and examples. Read: Tao Kakeya notes; Guth papers/book notes; incidence geometry surveys.. Make explicit examples/nonexamples for Kakeya set; Kakeya maximal function; incidence; polynomial partition; broad norm.. Work through Kakeya sets/maximal functions; incidence theorems; polynomial partitioning; broad/narrow analysis.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Kakeya, incidence geometry and polynomial method - theorem and technique pass. Master/audit: Besicovitch existence of measure-zero Kakeya sets; Wolff lower bounds/inequalities, selected; Szemeredi-Trotter theorem. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Falconer distance problem

Topics: Distance sets; energy/Fourier methods; spherical averages; decoupling/restriction inputs; pinned variants.

Key definitions: distance set; pinned distance set; spherical average; Mattila integral; Frostman measure.

Key results to learn:

- Falconer distance conjecture, exact open statement and scaling threshold [USE]
- Mattila integral criterion [ARCH]
- Wolff planar and Erdogan higher-dimensional threshold theorems [ARCH/USE]
- Guth-Iosevich-Ou-Wang planar pinned-distance theorem above dimension 5/4, in the precise source formulation [USE].

Reading route: Mattila; Wolff/Erdogan papers/expositions; current Hong Wang-related papers as guided research reading.

Warm-up/source exercise focus (secondary): Derive Mattila criterion; identify where restriction enters; reproduce dimensional bookkeeping in one modern paper.

Week 7: Falconer distance problem - definitions and examples. Read: Mattila; Wolff/Erdogan papers/expositions; current Hong Wang-related papers as guided research reading.. Make explicit examples/nonexamples for distance set; pinned distance set; spherical average; Mattila integral; Frostman measure.. Work through Distance sets; energy/Fourier methods; spherical averages; decoupling/restriction inputs; pinned variants.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Falconer distance problem - theorem and technique pass. Master/audit: Falconer conjecture, exact open statement; Mattila integral criterion; Wolff planar and Erdogan higher-dimensional threshold theorems. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Furstenberg sets and modern geometric combinatorics

Topics: Furstenberg s-sets; line-rich sets; incidence estimates; discretization; projection/sum-product connections.

Key definitions: Furstenberg set; (s,t)-Furstenberg set; discretized incidence; two-ends reduction.

Key results to learn:

- Classical Wolff lower bounds for Furstenberg sets [ARCH/USE]
- Ren-Wang sharp planar (s,t)-Furstenberg bound dim_H(E)>=min(s+t,(3s+t)/2,s+1), with parameter ranges [USE]
- projection/incidence principles used in proofs [USE].

Reading route: Wolff expositions; recent Furstenberg-set papers selected for the research track.

Warm-up/source exercise focus (secondary): Compute classical lower-bound exponents; compare constructions to bounds; make a lemma dependency tree for one recent proof.

Week 9: Furstenberg sets and modern geometric combinatorics - definitions and examples. Read: Wolff expositions; recent Furstenberg-set papers selected for the research track.. Make explicit examples/nonexamples for Furstenberg set; (s,t)-Furstenberg set; discretized incidence; two-ends reduction.. Work through Furstenberg s-sets; line-rich sets; incidence estimates; discretization; projection/sum-product connections.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Furstenberg sets and modern geometric combinatorics - theorem and technique pass. Master/audit: Classical Wolff lower bounds for Furstenberg sets; precise modern sharp/near-sharp theorem(s) selected from recent literature, including Hong Wang-related work; projection/incidence principles used in proofs [USE].. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Decoupling and Vinogradov mean values

Topics: l2 decoupling; paraboloid/moment curve; induction on scales; multilinear-to-linear; epsilon removal; number-theory consequences.

Key definitions: decoupling constant; extension operator; frequency cap; wave packet; broad/narrow decomposition; Vinogradov mean value.

Key results to learn:

- Bourgain-Demeter l2 decoupling theorem for the paraboloid [ARCH/USE]
- sharp decoupling for the moment curve [ARCH/USE]
- sharp Vinogradov mean value theorem as consequence [ARCH/USE]
- representative Strichartz/restriction consequences [USE].

Reading route: Demeter monograph; Bourgain-Demeter papers; Wooley VMVT comparison; Volume 23 circle-method cross-reference.

Warm-up/source exercise focus (secondary): Compute scaling/critical exponents; perform low-dimensional decoupling examples; write the decoupling->VMVT chain with all parameter translations.

Week 11: Decoupling and Vinogradov mean values - definitions and examples. Read: Demeter monograph; Bourgain-Demeter papers; Wooley VMVT comparison; Volume 23 circle-method cross-reference.. Make explicit examples/nonexamples for decoupling constant; extension operator; frequency cap; wave packet; broad/narrow decomposition; Vinogradov mean value.. Work through l2 decoupling; paraboloid/moment curve; induction on scales; multilinear-to-linear; epsilon removal; number-theory consequences.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Decoupling and Vinogradov mean values - theorem and technique pass. Master/audit: Bourgain-Demeter l2 decoupling theorem for the paraboloid; sharp decoupling for the moment curve; sharp Vinogradov mean value theorem as consequence. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Time-frequency analysis and Carleson

U1.P1. [F] Starting from Fourier partial sums, derive the maximal partial-sum operator whose L^2 boundedness implies almost-everywhere convergence. Explain why uniform boundedness of individual partial-sum operators is insufficient.

U1.P2. [T] Discretize a model Carleson operator into tiles; define time/frequency intervals, partial order, trees, size, and density. Work a finite toy tile collection completely.

U1.P3. [R] Prove a simplified tree estimate and size decomposition, then write a dependency map showing how orthogonality + tree control + packing yield the Carleson theorem architecture.

Programme U2: Restriction theory

U2.P1. [F] Determine scaling and Knapp necessary conditions for restriction to the parabola/sphere. Construct the cap example and associated physical-space tube explicitly.

U2.P2. [T] Prove the Stein-Tomas theorem by TT* and Fourier decay of surface measure, tracking exponents.

U2.P3. [S] Perform a wave-packet decomposition for a cap-localized function at scale R in a model parabola setting and derive tube dimensions/orientations from uncertainty principle.

Programme U3: Kakeya, incidence geometry and polynomial method

U3.P1. [F] Construct epsilon-neighborhoods of line sets in R^2/R^3 and compute volume lower bounds from elementary direction counting. Contrast true Kakeya sets with finite-scale Kakeya configurations.

U3.P2. [T] Work through a polynomial partitioning toy example for incidences in R^2 or points/lines. Count contributions in cells versus on the zero set.

U3.P3. [R] State the Wang-Zahl 2025 theorem that every Kakeya set in R^3 has Hausdorff and Minkowski dimension 3. Build a proof-architecture map through the non-clustering/sticky reduction, multiscale analysis and volume estimates, using Guth's later outline as a companion; explicitly note that the general Kakeya conjecture in dimensions >=4 is not settled by this theorem.

Programme U4: Falconer distance problem

U4.P1. [F] For a Frostman measure mu, derive the Fourier representation of the distance measure and identify the Mattila integral controlling L^2 density.

U4.P2. [T] Reproduce the classical Falconer threshold from a chosen Fourier/energy estimate in the dimension range covered by your source.

U4.P3. [R] State the planar pinned-distance result of Guth-Iosevich-Ou-Wang in the source formulation (in particular the dim_H(E)>5/4 threshold giving a pinned distance set of positive measure). Then compare the Fourier/decoupling, incidence/polynomial and radial-projection mechanisms that appear in modern Falconer work, marking which conclusions are theorems and which Falconer thresholds remain open.

Programme U5: Furstenberg sets and modern geometric combinatorics

U5.P1. [F] Construct standard Furstenberg set examples and derive basic lower bounds from counting/slicing. Work both continuous and discretized formulations.

U5.P2. [T] Translate a discretized Furstenberg problem into an incidence problem between points and tubes/lines, keeping scale and multiplicity parameters explicit.

U5.P3. [R] State the Ren-Wang sharp planar (s,t)-Furstenberg theorem, including s,t ranges and the bound dim_H(E)>=min(s+t,(3s+t)/2,s+1). Analyze which term is dominant in each parameter region, then reconstruct the multiscale/incidence architecture and one sum-product/projection consequence.

Programme U6: Decoupling and Vinogradov mean values

U6.P1. [F] State l^2 decoupling for the parabola and moment curve with the exact scaling exponent in a standard range. Verify the trivial orthogonality and triangle-inequality bounds for comparison.

U6.P2. [T] Work through one induction-on-scales step in a simplified decoupling proof, separating transverse from nearly parallel pieces and recording the recurrence for the decoupling constant.

U6.P3. [R] Derive the Vinogradov mean-value consequence from decoupling by expressing the relevant moment integral as a count of Diophantine solutions. Check normalization/exponent arithmetic carefully.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Modern Harmonic and Geometric Analysis: start from “Time-frequency analysis and Carleson”, pass through “Falconer distance problem”, and finish at “Decoupling and Vinogradov mean values”. Use the named results “Carleson theorem on a.e. convergence of Fourier series for L2” and “representative Strichartz/restriction consequences.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Direct target for Carleson, Bourgain decoupling, Falconer/Furstenberg/Kakeya and Hong Wang-related geometric analysis. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: Stein-Tomas restriction theorem; Knapp necessary conditions; Besicovitch existence of measure-zero Kakeya sets; Szemeredi-Trotter theorem; Falconer conjecture, exact open statement. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: projection/incidence principles used in proofs; Bourgain-Demeter l2 decoupling theorem for the paraboloid; sharp decoupling for the moment curve; sharp Vinogradov mean value theorem as consequence; representative Strichartz/restriction consequences.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Time-frequency analysis and Carleson” develops into “Decoupling and Vinogradov mean values”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

VOLUME 36. Mathematical Cryptography and Computational Arithmetic

Purpose: Build the mathematical and complexity foundations of modern public-key and post-quantum cryptography, emphasizing reductions and arithmetic structure rather than implementation folklore.

Prerequisites: Volumes 3, 5, 21, 24-27 and 32; algorithms/complexity basics assumed.

Primary and secondary references: Hoffstein-Pipher-Silverman, An Introduction to Mathematical Cryptography; Katz-Lindell, Introduction to Modern Cryptography; Washington, Elliptic Curves: Number Theory and Cryptography; Galbraith, Mathematics of Public Key Cryptography; Micciancio-Goldwasser, Complexity of Lattice Problems; Regev, Learning with Errors lecture notes/papers

Quarter output: Build a mathematical cryptography portfolio containing one formal security proof, one finite-field/ECC implementation, one lattice-reduction experiment and a 15-page comparison of classical versus lattice hardness assumptions.

Research bridges: Integrates finite fields, number theory, elliptic curves, geometry of numbers, logic/complexity and representation/group ideas into computational applications.

Six two-week study units

Unit 1 (Weeks 1-2): Complexity and security definitions

Topics: Polynomial-time algorithms; negligible functions; indistinguishability; one-way functions; semantic security; reductions.

Key definitions: PPT algorithm; negligible function; computational/statistical indistinguishability; one-way function; IND-CPA/IND-CCA; reduction.

Key results to learn:

- Equivalence of semantic security and IND-CPA in standard symmetric/public-key formulations [ARCH/USE]
- hybrid argument principle [PROOF/ARCH]
- Goldwasser-Micali style reduction logic [ARCH].

Reading route: Katz-Lindell Chs. 1-4; HPS preliminaries.

Warm-up/source exercise focus (secondary): Write formal games; prove toy reductions/hybrid bounds; distinguish information-theoretic from computational security.

Week 1: Complexity and security definitions - definitions and examples. Read: Katz-Lindell Chs. 1-4; HPS preliminaries.. Make explicit examples/nonexamples for PPT algorithm; negligible function; computational/statistical indistinguishability; one-way function; IND-CPA/IND-CCA; reduction.. Work through Polynomial-time algorithms; negligible functions; indistinguishability; one-way functions; semantic security; reductions.. Complete U1.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 2: Complexity and security definitions - theorem and technique pass. Master/audit: Equivalence of semantic security and IND-CPA in standard symmetric/public-key formulations; hybrid argument principle; Goldwasser-Micali style reduction logic [ARCH].. Complete U1.P2-U1.P3, writing every hypothesis used. Five to seven days later redo U1.P2 without notes and record the first reconstruction failure, if any.

Unit 2 (Weeks 3-4): RSA, factoring and primality

Topics: Modular arithmetic algorithms; RSA; Chinese remainder acceleration; pseudoprimes; primality testing; factoring algorithms.

Key definitions: RSA modulus; Euler/Carmichael function; strong pseudoprime; Miller-Rabin witness.

Key results to learn:

- RSA correctness [PROOF]
- Miller-Rabin error bound [PROOF/ARCH]
- deterministic polynomial-time primality (AKS), theorem statement [USE]
- Pollard rho/quadratic sieve/NFS complexity philosophies [USE].

Reading route: HPS RSA/primality chapters; Crandall-Pomerance selected reference.

Warm-up/source exercise focus (secondary): Implement RSA/Miller-Rabin; prove error bounds; compare asymptotic factoring methods.

Week 3: RSA, factoring and primality - definitions and examples. Read: HPS RSA/primality chapters; Crandall-Pomerance selected reference.. Make explicit examples/nonexamples for RSA modulus; Euler/Carmichael function; strong pseudoprime; Miller-Rabin witness.. Work through Modular arithmetic algorithms; RSA; Chinese remainder acceleration; pseudoprimes; primality testing; factoring algorithms.. Complete U2.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 4: RSA, factoring and primality - theorem and technique pass. Master/audit: RSA correctness; Miller-Rabin error bound; deterministic polynomial-time primality (AKS), theorem statement. Complete U2.P2-U2.P3, writing every hypothesis used. Five to seven days later redo U2.P2 without notes and record the first reconstruction failure, if any.

Unit 3 (Weeks 5-6): Discrete logarithms in finite fields

Topics: Diffie-Hellman; ElGamal; generic group algorithms; index calculus; hardness assumptions.

Key definitions: DLP; CDH/DDH; generic group; baby-step giant-step; Pollard rho; index calculus.

Key results to learn:

- BSGS O(sqrt(N)) time/memory tradeoff [PROOF]
- Pollard rho expected O(sqrt(N)) generic complexity [ARCH]

- generic lower-bound philosophy (Shoup), statement [USE]
- ElGamal IND-CPA under DDH in standard group model [ARCH/USE].

Reading route: HPS/Galbraith DLP chapters; Katz-Lindell reduction examples.

Warm-up/source exercise focus (secondary): Implement generic DLP algorithms; write a DDH reduction; compare finite-field group choices.

Week 5: Discrete logarithms in finite fields - definitions and examples. Read: HPS/Galbraith DLP chapters; Katz-Lindell reduction examples.. Make explicit examples/nonexamples for DLP; CDH/DDH; generic group; baby-step giant-step; Pollard rho; index calculus.. Work through Diffie-Hellman; ElGamal; generic group algorithms; index calculus; hardness assumptions.. Complete U3.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 6: Discrete logarithms in finite fields - theorem and technique pass. Master/audit: BSGS O(sqrt(N)) time/memory tradeoff; Pollard rho expected O(sqrt(N)) generic complexity; generic lower-bound philosophy (Shoup), statement. Complete U3.P2-U3.P3, writing every hypothesis used. Five to seven days later redo U3.P2 without notes and record the first reconstruction failure, if any.

Unit 4 (Weeks 7-8): Elliptic curves and pairings

Topics: ECC scalar multiplication; ECDLP; curve selection; Weil/Tate pairings; embedding degree; pairing-based reductions.

Key definitions: ECDLP; embedding degree; Weil pairing; Tate pairing; distortion map; pairing-friendly curve.

Key results to learn:

- Hasse bound [PROOF/ARCH]
- MOV reduction principle [ARCH/USE]
- bilinearity/nondegeneracy of Weil/Tate pairings [ARCH]
- security dependence on subgroup/order/embedding degree [ARCH].

Reading route: Washington; Galbraith ECC/pairing chapters; Volume 32 arithmetic cross-reference.

Warm-up/source exercise focus (secondary): Implement EC group law and scalar multiplication; compute pairings in small examples; analyze MOV-vulnerable curves.

Week 7: Elliptic curves and pairings - definitions and examples. Read: Washington; Galbraith ECC/pairing chapters; Volume 32 arithmetic cross-reference.. Make explicit examples/nonexamples for ECDLP; embedding degree; Weil pairing; Tate pairing; distortion map; pairing-friendly curve.. Work through ECC scalar multiplication; ECDLP; curve selection; Weil/Tate pairings; embedding degree; pairing-based reductions.. Complete U4.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 8: Elliptic curves and pairings - theorem and technique pass. Master/audit: Hasse bound; MOV reduction principle; bilinearity/nondegeneracy of Weil/Tate pairings. Complete U4.P2-U4.P3, writing every hypothesis used. Five to seven days later redo U4.P2 without notes and record the first reconstruction failure, if any.

Unit 5 (Weeks 9-10): Lattices, LLL, SIS and LWE

Topics: Euclidean lattices; SVP/CVP; LLL; q-ary lattices; SIS; LWE; trapdoors; worst-case/average-case reductions.

Key definitions: successive minima; SVP/CVP; LLL-reduced basis; SIS; LWE; smoothing parameter; discrete Gaussian.

Key results to learn:

- LLL polynomial-time reduction and approximation guarantee [PROOF/ARCH]
- Ajtai worst-case-to-average-case paradigm for lattice problems, statement [USE]
- Regev quantum reduction from worst-case lattice problems to LWE, standard theorem [USE]
- classical reductions for modern LWE variants, selected [USE].

Reading route: HPS lattice chapters; Micciancio-Goldwasser; Regev notes; Peikert surveys.

Warm-up/source exercise focus (secondary): Run LLL on explicit bases; solve toy SIS/LWE; state reduction hypotheses and parameter mappings exactly.

Week 9: Lattices, LLL, SIS and LWE - definitions and examples. Read: HPS lattice chapters; Micciancio-Goldwasser; Regev notes; Peikert surveys.. Make explicit examples/nonexamples for successive minima; SVP/CVP; LLL-reduced basis; SIS; LWE; smoothing parameter; discrete Gaussian.. Work through Euclidean lattices; SVP/CVP; LLL; q-ary lattices; SIS; LWE; trapdoors; worst-case/average-case reductions.. Complete U5.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 10: Lattices, LLL, SIS and LWE - theorem and technique pass. Master/audit: LLL polynomial-time reduction and approximation guarantee; Ajtai worst-case-to-average-case paradigm for lattice problems, statement; Regev quantum reduction from worst-case lattice problems to LWE, standard theorem. Complete U5.P2-U5.P3, writing every hypothesis used. Five to seven days later redo U5.P2 without notes and record the first reconstruction failure, if any.

Unit 6 (Weeks 11-12): Post-quantum constructions and proof frameworks

Topics: Ring/Module-LWE; KEMs; signatures; Fiat-Shamir; hash/random-oracle models; quantum algorithms and implications.

Key definitions: Ring-LWE; Module-LWE; KEM; signature; random oracle; Fiat-Shamir transform; quantum Fourier transform.

Key results to learn:

- Ring-LWE hardness framework via ideal lattices under standard distributions/parameters [USE]
- Fujisaki-Okamoto transform principle for KEM security [USE]
- Fiat-Shamir security caveats/selected theorems [USE]
- Shor polynomial-time algorithms for factoring/DLP [ARCH/USE].

Reading route: Modern lattice-crypto texts/surveys; Katz-Lindell; Regev/Peikert; standards should be checked separately when implementing.

Warm-up/source exercise focus (secondary): Compare proof structures of KEM/signature schemes; map algebraic structure to attacks; explain exactly why Shor changes RSA/DH/ECC but not known LWE assumptions.

Week 11: Post-quantum constructions and proof frameworks - definitions and examples. Read: Modern lattice-crypto texts/surveys; Katz-Lindell; Regev/Peikert; standards should be checked separately when implementing.. Make explicit examples/nonexamples for Ring-LWE; Module-LWE; KEM; signature; random oracle; Fiat-Shamir transform; quantum Fourier transform.. Work through Ring/Module-LWE; KEMs; signatures; Fiat-Shamir; hash/random-oracle models; quantum algorithms and implications.. Complete U6.P1 plus 3-5 source exercises chosen to expose hypotheses rather than repeat a template.

Week 12: Post-quantum constructions and proof frameworks - theorem and technique pass. Master/audit: Ring-LWE hardness framework via ideal lattices under standard distributions/parameters; Fujisaki-Okamoto transform principle for KEM security; Fiat-Shamir security caveats/selected theorems. Complete U6.P2-U6.P3, writing every hypothesis used. Five to seven days later redo U6.P2 without notes and record the first reconstruction failure, if any.

Precision problem programmes (core)

These 18 problems replace quota-driven generic exercises. They are deliberately linked to the six two-week units. Uj.P1 makes the objects concrete; Uj.P2 develops the central technique/proof; Uj.P3 tests hypotheses, synthesis, computation, or research-paper architecture. Treat multi-part items as one substantial problem.

Programme U1: Complexity and security definitions

U1.P1. [F] Formalize negligible functions, PPT adversaries, indistinguishability, one-wayness, IND-CPA, IND-CCA, and EUF-CMA. For each notion give a concrete game transcript and one construction that obviously fails it.

U1.P2. [T] Prove a simple reduction: semantic-style confidentiality of a one-time-pad-like construction or PRG-based encryption from pseudorandomness, writing the adversary transformation and advantage equation explicitly.

U1.P3. [C] Construct examples showing why correctness, confidentiality, authenticity, and chosen-ciphertext security are distinct. Identify which attack each missing notion permits.

Programme U2: RSA, factoring and primality

U2.P1. [F] Implement modular exponentiation, extended Euclid, CRT, Miller-Rabin, and RSA on toy parameters. Verify decryption both by Euler/Carmichael and CRT decomposition.

U2.P2. [T] Prove RSA correctness with all gcd cases handled and analyze textbook RSA's deterministic-security failure. Explain exactly what randomized padding/security notion must change.

U2.P3. [R] Compare primality testing and factoring: state AKS polynomial-time primality, probabilistic Miller-Rabin, and best-known factoring status without conflating them.

Programme U3: Discrete logarithms in finite fields

U3.P1. [F] Compute discrete logarithms in small F_p^* by brute force, baby-step/giant-step, and Pollard-rho conceptually. Compare time-memory tradeoffs.

U3.P2. [T] Derive Diffie-Hellman/ElGamal correctness and formulate CDH, DDH, and DLP assumptions. Give a group where DDH is easy although DLP may remain hard, explaining the distinction.

U3.P3. [S] Analyze subgroup/confinement attacks in a composite-order multiplicative group and show how group-order validation/cofactor handling changes security.

Programme U4: Elliptic curves and pairings

U4.P1. [F] Implement elliptic-curve addition/doubling over a small finite field and compute group order, scalar multiples, and a toy ECDH exchange.

U4.P2. [T] Explain why generic discrete-log algorithms cost about sqrt(|G|) and compare field versus elliptic-curve parameter sizes. Avoid relying on empirical 'security bits' without assumptions.

U4.P3. [R] For pairings, state bilinearity/nondegeneracy and work a toy algebraic consequence such as DDH testing or identity-based-encryption structure; do not treat insecure tiny examples as deployable crypto.

Programme U5: Lattices, LLL, SIS and LWE

U5.P1. [F] Run LLL by hand on a small integer basis, recording Gram-Schmidt coefficients and Lovasz checks. Compare output vector lengths with successive minima bounds.

U5.P2. [T] Define SVP, CVP, SIS, and LWE precisely. Construct a toy LWE instance and show why solving exact linear equations fails because of noise while a secret still exists.

U5.P3. [R] Trace the worst-case/average-case reduction philosophy for LWE at theorem-architecture level: lattice problem, Gaussian/smoothing parameter, average LWE distribution, cryptographic hardness conclusion.

Programme U6: Post-quantum constructions and proof frameworks

U6.P1. [T] Prove IND-CPA security of a simple LWE/Regev-style encryption scheme from decisional LWE in a clean hybrid sequence, writing each hybrid difference and reduction.

U6.P2. [S] Compare module/ring lattice assumptions with plain LWE: what algebraic structure buys efficiency and what additional assumption/attack surface it introduces.

U6.P3. [R] Choose one standard post-quantum primitive family (lattice KEM/signature, code-based, hash-based). Produce a proof-framework audit: hardness assumption, reduction model, correctness failure probability, and implementation-side issues that are outside the mathematical proof.

Cross-unit synthesis (choose one)

S1. Build a 4-6 page dependency narrative for Mathematical Cryptography and Computational Arithmetic: start from “Complexity and security definitions”, pass through “Elliptic curves and pairings”, and finish at “Post-quantum constructions and proof frameworks”. Use the named results “Equivalence of semantic security and IND-CPA in standard symmetric/public-key formulations” and “Shor polynomial-time algorithms for factoring/DLP.” with every hypothesis checked in at least one explicit example.

S2. Research-bridge audit: Integrates finite fields, number theory, elliptic curves, geometry of numbers, logic/complexity and representation/group ideas into computational applications. Choose one downstream target, identify exactly three results from Units 1-6 that it consumes, and give one counterexample or obstruction showing why a listed hypothesis cannot simply be dropped.

Volume-specific mastery checkpoint

- Full-proof requirement: reproduce without notes the standard proofs of: hybrid argument principle; RSA correctness; Miller-Rabin error bound; BSGS O(sqrt(N)) time/memory tradeoff; Hasse bound. For a compound [PROOF/ARCH] result, prove the core textbook form and state the stronger formulation separately.
- Architecture/use requirement: state all hypotheses and normalizations, then give a dependency graph and one nontrivial application for: classical reductions for modern LWE variants, selected; Ring-LWE hardness framework via ideal lattices under standard distributions/parameters; Fujisaki-Okamoto transform principle for KEM security; Fiat-Shamir security caveats/selected theorems; Shor polynomial-time algorithms for factoring/DLP.
- Problem requirement: complete U1.P1-U6.P3. Full written solutions are mandatory for every [T] problem and at least one [S]/[C]/[R] problem from each half of the volume; any failed delayed P2 reattempt must be corrected.
- Oral synthesis: in 30-45 minutes explain how “Complexity and security definitions” develops into “Post-quantum constructions and proof frameworks”, naming at least four theorem dependencies and one obstruction/counterexample rather than reciting chapter titles.
- Delayed retention: seven days after the oral, reproduce a two-page theorem/definition map and solve one previously unseen tool-selection problem. Passing requires choosing the correct theorem and checking its hypotheses.

PART VI. RESEARCH-TARGET INDEX, EXERCISE POLICY AND HOW TO USE THE 36 VOLUMES

Purpose: This final index converts the volume sequence into routes toward the named research targets. It also fixes a concrete exercise and theorem-review policy so that the document functions as a study programme rather than a bibliography.

A. Research-target dependency index

Target	Recommended volume route	What the route supplies
Carleson theorem	V6 -> V7 -> V9 -> V35	Measure/Fourier foundations; maximal and Calderon-Zygmund tools; time-frequency tiles/trees; Carleson-Hunt.
Bourgain-Demeter decoupling / VMVT	V6 -> V9 -> V23 -> V35	Oscillatory integrals and restriction; circle method/VMVT; l2 decoupling and the moment curve.
Falconer distance problem	V6 -> V9 -> V28 -> V34 -> V35	Frostman/energy; restriction/Fourier methods; incidence/sum-product; fractal measures; modern distance-set machinery.
Furstenberg sets / Kakeya / Hong-Wang-related geometric analysis	V9 -> V28 -> V34 -> V35	Restriction and wave packets; additive/incidence tools; dimension/projection methods; polynomial/geometric analysis.
Baker linear forms in logarithms	V3 -> V22 -> V24 -> V25 -> V26	Algebraic numbers and heights; number fields/local fields; Diophantine approximation; explicit logarithmic-form bounds.
Schmidt Subspace Theorem	V22 -> V24 -> V25 -> V26	Heights/places; geometry of numbers and approximation; S-adic linear forms; S-unit applications.
Hilbert's tenth problem	V5 -> V22 -> V26	Computability and incompleteness background; elementary Diophantine equations; MRDP and Diophantine coding.
Fermat's Last Theorem	V24 -> V30 -> V31 -> V32	Number fields/Galois; modular forms and representations; schemes; elliptic curves, level lowering and modularity.
Faltings / Mordell conjecture	V24 -> V26 -> V31 -> V32	Arithmetic of number fields and heights; Diophantine finiteness; schemes/cohomology; abelian varieties and Faltings.
Sato-Tate	V23 -> V24 -> V30 -> V32 -> V33	L-functions; Galois/Frobenius; automorphic forms; elliptic curves; l-adic/cohomological equidistribution viewpoint.
Weil conjectures / Deligne / Bombieri over finite fields	V27 -> V31 -> V33	Finite-field curves and zeta; scheme/cohomology language; etale cohomology, weights and Frobenius bounds.
Oppenheim / Ratner	V12 -> V18 -> V20	Lie groups/manifolds; representations; unipotent measure classification and homogeneous dynamics.
Einsiedler-Katok-Lindenstrauss	V14 -> V18 -> V19 -> V20	Entropy and measure; Lie representations; smooth hyperbolicity; higher-rank entropy/measure rigidity.
Effective Ratner / Strombergsson / Marklof / Lei Yang	V7 -> V18 -> V20 -> V21 -> V29	Spectral tools; Lie groups; homogeneous dynamics; property (T); quantitative spectral gaps/expansion.
Benoist-Quint	V18 -> V20 -> V21 -> V29	Linear groups/representations; stationary measures; geometry/property (T); random-walk spectral expansion.
Bourgain sum-product	V9 -> V23 -> V27 -> V28	Fourier/incidence background; exponential sums; finite fields; finite-field sum-product.
Approximate groups / Breuillard-Green-Tao	V3 -> V21 -> V28	Group theory; geometric group theory; noncommutative Ruzsa calculus and BGT

Target	Recommended volume route	What the route supplies
		structure theorem.
Bourgain-Gamburd expansion	V18 -> V21 -> V28 -> V29	Representation dimension; property (T)/(tau); product growth; flattening and expansion machine.
Hochman inverse entropy	V6 -> V14 -> V28 -> V34	Measure/probability and entropy; convolution/additive structure; inverse entropy and self-similar measures.
Shmerkin inverse Lq theory	V6 -> V9 -> V28 -> V34	Lp/Fourier tools; convolution; additive inverse ideas; Lq dimensions and inverse theorems.
Varju on Bernoulli convolutions	V24 -> V26 -> V28 -> V29 -> V34	Algebraic numbers/heights; sum-product/expansion; entropy/separation and Bernoulli convolutions.
Poincare conjecture / Ricci flow	V6 -> V10 -> V11 -> V12 -> V13	Analysis/PDE; topology; manifolds; Riemannian curvature and Ricci-flow architecture.
Hauptvermutung	V11 -> V12 -> V13	Simplicial/CW topology; manifolds; PL/topological/smooth category comparisons and counterexamples.
Zimmer programme	V7 -> V18 -> V19 -> V20 -> V21	Unitary reps/property (T); Lie groups; smooth dynamics; homogeneous rigidity; geometric group theory.
Percolation / SLE / Ising	V6 -> V8 -> V9 -> V14 -> V16	Analysis/complex analysis; harmonic ideas; probability; planar models and conformal invariance.
Particle systems -> kinetic equations / Hilbert VI	V10 -> V14 -> V15 -> V17	PDE; probability; stochastic processes; BBGKY, chaos, Boltzmann-Grad and hydrodynamic limits.
Mathematical cryptography	V3 -> V5 -> V24 -> V25 -> V27 -> V32 -> V36	Algebra/complexity; number fields/lattices/finite fields; elliptic curves; reductions and post-quantum arithmetic.

B. Precision problem protocol for every two-week unit

- Core rule: complete Uj.P1-Uj.P3 for the active unit. These are not routine repetitions: each is intended to take roughly 45 minutes to several hours, and research-bridge items may require a short paper-reading session.
- Write-up rule: fully write P2 in every unit. Also fully write either P1 or P3. The remaining core problem may be a concise solution, computation log, or theorem audit, but it must be checkable.
- Source-drill rule: add only 3-8 textbook/lecture-note exercises per unit after the core problems reveal what technique needs practice. If ten near-identical routine problems are already easy, stop and move to a harder problem.
- Hint rule: after 60-90 minutes of serious work, consult only the smallest useful hint (definition, lemma name, or first step), then restart the argument. Record the hint used. Do not read a full solution before writing a failed-attempt summary.
- Hypothesis rule: for every major theorem used in a problem, explicitly check the hypotheses. At least one problem per unit should include a near-miss or counterexample showing why a hypothesis matters.
- Research-level rule: [R] does not normally mean reproduce a 40-page proof. It means state a precise standard formulation from the cited source, build the dependency chain, verify the theorem in examples/toy models, reconstruct one key lemma or mechanism, and derive a real consequence.
- Computational rule: [X] experiments must state the mathematical claim being tested, parameter range, numerical observable, and what the computation cannot establish. Code is evidence for intuition, never a substitute for proof.
- Volume target: 18 core problems + 2 optional cross-unit synthesis tasks, with at least 12 full write-ups, plus roughly 20-40 selected source exercises. This is intentionally smaller and harder than the old 50-80-polished-exercise quota.

C. How to use references efficiently

- Choose one primary source for a linear first pass. Do not read three full books in parallel. The additional references are for alternate explanations, missing prerequisites, exercises and later research consultation.
- For each two-week unit, read approximately 80-140 textbook pages or the equivalent in lecture notes/papers, adjusted downward for technically dense works such as Hartshorne, Serre, Deligne, Ratner or Bourgain-Demeter.

- When the reading route cites a research paper, first read the introduction and theorem statements, then the proof outline/section structure, and only then selected technical sections. For research papers, follow the quarter-specific M3/M4 requirement: reproduce only the named lemmas/mechanisms, but state the cited research theorem and its hypotheses precisely.
- Use lecture notes to bridge books and papers, especially for algebraic number theory/class field theory, representation theory, algebraic geometry, homogeneous dynamics, decoupling and etale cohomology.
- Once a theorem becomes a recurring tool, upgrade it explicitly in your theorem notebook: add a full proof if feasible, at least two nontrivial applications, and a hypothesis/counterexample audit. Do not rely on generic letter grades.

D. The theorem notebook

One page per central theorem: For every major result record: (1) exact statement and hypotheses; (2) two examples satisfying the hypotheses; (3) one near-miss/counterexample showing why an assumption matters; (4) proof architecture in 5-15 lines; (5) three downstream uses; (6) references to the primary source and one alternate exposition; (7) date last reproduced from memory.

Definition notebook: For every definition record the canonical example, a nonexample, an equivalent characterization when available, and the first theorem for which the definition becomes indispensable.

Dependency graph: At the end of each quarter draw a one-page directed graph whose nodes are the 15-30 most important results. This should make prerequisites visible before later research-paper reading.

E. Ten-year execution rule

- Years 1-3: favor breadth and satisfy the explicit M1-M5 targets printed in each quarter. Do not substitute a generic “proof architecture” label for the named proof and application requirements.
- Years 4-6: maintain two active spines at once: an arithmetic/group-theoretic spine and an analysis/probability/dynamics spine. Use the four-quarter yearly structure to alternate concentration.
- Years 7-9: begin original-paper reading every week. For each quarter choose one “anchor theorem” whose original proof you read substantially beyond the syllabus requirement.
- Year 10: do not add more survey material. Use the four synthesis quarters to consolidate one primary specialization, one secondary specialization, and one concrete research problem or expository monograph.
- Every 13th week, or approximately once per quarter if calendar constraints require, perform a closed-notes cumulative examination: definitions, theorem statements, two proof reconstructions, and one synthesis essay.

F. Final standard for research readiness

Research-ready in a subject means: You can read a current paper with the original references beside it; recognize all standard objects and theorem names; reconstruct the dependency chain; verify hypotheses in concrete examples; reproduce the central classical lemmas; explain what is genuinely new in the paper; and identify at least one plausible question or variation to investigate.

What this plan does not require: It does not require memorizing all proofs or reading every reference cover to cover. The objective is exact command of definitions and theorem statements, strong proof competence on the foundational results, and increasingly research-oriented architectural understanding for the deepest landmark theorems.